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PAPERS PRESENTED AT MEETINGS

THIS CALENDAR lists meetings of the Society which have been approved by the Council at which papers may be presented. Programs of Annual Meetings appear in the *Notices* and on the AMS website; programs for sectional meetings appear on the AMS Web pages in the Meetings & Conferences section, and are electronically archived in the *Notices* section on the AMS website.

MEETING #	DATE	PLACE	ABSTRACT DEADLINE	ABSTRACT ISSUE
1159	September 12–13, 2020	El Paso, TX	July 14	Vol 41, No. 3
1160	October 3–4, 2020	State College, PA	August 11	Vol 41, No. 3
1161	October 10–11, 2020	Chattanooga, TN	August 18	Vol 41, No. 4
1162	October 24–25, 2020	Salt Lake City, UT	September 1	Vol 41, No. 4
1163	January 6–9, 2021	Washington, DC	September 8	TBA
1164	March 13–14, 2021	Atlanta, GA	TBA	TBA
1165	March 20–21, 2021	Providence, RI	TBA	TBA
1166	April 17–18, 2021	Cincinnati, OH	TBA	TBA
1167	May 1–2, 2021	San Francisco, CA	TBA	TBA
1168	July 5–9, 2021	Grenoble, France	TBA	TBA
1169	July 19–23, 2021	Buenos Aires, Argentina	TBA	TBA

This issue includes abstracts from the 2020 spring sectional meetings in Charlottesville, VA, Medford, MA, West Lafayette, IN, and Fresno, CA, canceled due to the COVID-19 pandemic. These abstracts were not presented as scheduled and may appear in later issues.

CHARLOTTESVILLE, VA, March 13–15, 2020

Abstracts of the 1155th Meeting.

00 ► General

1155-00-58 **Chad E. L. Kinney*** (ckinney@fit.edu). *Clarifying the Relation Between Geometric and Algebraic Slope on a Non-Homogeneous Coordinate System: Preliminary Report.* Preliminary report.

Zaslavsky et al. (2002) found mathematicians may be challenged or confused by the relation between algebraic and geometric slope when a line is plotted on a non-homogeneous coordinate system (rather than on a Cartesian or homogeneous coordinate system). The researchers also noted that in a non-homogeneous coordinate system, the relation between geometric and algebraic systems loses utility (or meaning) since distances and angles are no longer invariant (whereas in a Cartesian coordinate system, distances and angles are relatively invariant). However, this paper shows how to easily find the angle of inclination for a line on any coordinate system by simply expanding upon the traditional calculation of the tangent inverse of a line's algebraic slope, $\theta = \tan^{-1}(m)$. Through this expanded calculation, $\theta = \tan^{-1}(m * GVQ)$, the angle of inclination for lines across any type of coordinate system can quickly be determined—preserving the isomorphism between algebraic and geometric slope. The term Graphic Variability Quotient (GVQ) is derived by dividing the variability ratio of the y-axis (YVR) by the variability ratio of the x-axis (XVR), and will be explained further, along with some potential applications of this advancement. (Received December 08, 2019)

1155-00-346 **Bojko Bakalov** and **Samantha Kirk*** (slkirk@ncsu.edu), NC State Department of Mathematics, 2108 SAS Hall, Box 8205, Raleigh, NC 27695. *A Vertex Algebra Construction of Representations of Toroidal Lie Algebras.* Preliminary report.

Given a simple finite-dimensional Lie algebra and an automorphism of finite order, one can construct a twisted toroidal Lie algebra. Similar to twisted affine Lie algebras, which are well-studied in the literature, we can create representations of twisted toroidal Lie algebras with the help of vertex algebras. In this talk, we will discuss twisted modules of vertex algebras and we will show how representations of twisted toroidal Lie algebras can be formed from such twisted modules. (Received January 18, 2020)

1155-00-368 **Maira Khan*** (mairakhan@college.harvard.edu). *Evaluating Geometric Methods of Distance as Tools to Determine Racial Imbalance.* Preliminary report.

In this expository talk, I will discuss and evaluate the metrics used to evaluate racial segregation. Specifically, I will explain the mathematical significance of the various weighting structures of each metric, and critically

evaluate the social implications of such metrics with respect to communities. I will then introduce a notion of distance classically used in manifold theory and evaluate its potential use as a means for determining the scaling limit of uniform random plane maps. I will evaluate this method as a means to construct maps that minimize racial imbalance. (Received January 19, 2020)

1155-00-370 **Hannah E Wheelen*** (hwheelen@princeton.edu). *Will math and data ever completely remove humans from the redistricting equation?*

As data collection and technology has advanced over the past decade, the twists and turns of political district boundaries have often been finely tuned to produce specific political outcomes. Meanwhile, mathematics has become a powerful tool for analyzing such political decisions as well as proposing new ways to draw district maps. Practitioners, however, warn that the human element in mapmaking should never be replaced by technology–only aided by it. For instance, the desire for a simple, unified mathematical approach that applies to every state neglects the intricacies of each state's geography and political climate. This talk will look at state specific applications of statistical reasoning where human involvement has been a necessary factor. We will learn from examples in Virginia, North Carolina, and New Jersey, and discuss how we can continue to use math to assure the 2021 redistricting cycle is much more representative of the people than the last. (Received January 19, 2020)

1155-00-393 **Ellen Veomett*** (erv2@stmarys-ca.edu), **Karen Saxe** (kxs@ams.org), **Gowri Ramachandran** (gowri.ramachandran@gmail.com) and **Andrew Chin** (chin@unc.edu). *Panel Discussion - What happens now that partisan gerrymandering has been ruled to be nonjusticiable in federal courts?*

Many in the redistricting reform world were disappointed by the May 2019 Supreme Court decision that partisan gerrymandering is nonjusticiable in federal courts. While this particular door may have been closed, there remain other projects for reformers to pursue. This panel will address some of these issues, including state constitutional challenges, the role of independent redistricting commissions, the consequences for the Voting Rights Act, the impact of the 2020 census on redistricting, and the role that mathematicians can play in these efforts. (Received January 19, 2020)

1155-00-462 **Benjamin Plener Cover*** (bcover@uidaho.edu), 514 W. Jefferson St., Boise, ID 83716. *Gerrymandering as Jerk: Measuring Partisan Fairness Using an Interpolation Function in Indexed Vote Share Space.*

This paper uses a graphical framework and interpolation methods to analyze some of the leading measures of partisan gerrymandering in the simplified case of two parties and constant voter turnout. We index the districts in ascending order of vote share and graphically represent a particular election's vote pattern as a scatterplot in this indexed vote share space. We then use interpolation methods to transform this scatterplot into a continuous and differentiable function so that each party's wasted votes (both lost and excess) correspond to some integral of this function. We demonstrate how leading measures of partisan gerrymandering, including competitiveness measures, the efficiency gap, the declination, and the mean-median difference, can all be expressed as operators on this imputation function. Finally, we propose a new measure of partisan gerrymandering based on the curvature of the interpolation function. This approach suggests that partisan gerrymandering can be understood as a third derivative phenomenon somewhat analogous to the physical concept of jerk. (Received January 20, 2020)

01 ► History and biography

1155-01-349 **Moira Chas***, moira.chas@stonybrook.edu. *Alicia Boole Stott in the fourth dimension.*

The untimely death of George Boole in 1864, left his widow, Mary Everest and his five young daughters in a precarious economic situation. Despite this, all of them, mother and daughters, pursued with remarkable persistence artistic creation or scientific understanding of some sort.

Alicia (1860-1940), the middle daughter, without any formal mathematical training, rediscovered the convex regular four dimensional polytopes (the four dimensional analogues of the Platonic solids in dimension three and regular polygons in dimension two) and studied their sections, that is, intersections with three dimensional planes.

In dimension four, there are six convex regular polytopes: the simplex (whose faces are tetrahedra), the 4D-cube, the 16-cell (dual of the 4D cube, whose faces are tetrahedra), the 24-cell (self-dual, faces are octahedra), the 120-cell (whose faces are dodecahedra) and the 600-cell (dual of the 120-cell, faces are tetrahedra).

In this talk we will discuss Alicia Boole Stott's geometrical ideas, as well as recently discovered facts about her mathematical path. (Received January 18, 2020)

05 ► Combinatorics

1155-05-1 **Moon Duchin*** (moon.duchin@tufts.edu), Moon.Duchin@tufts.edu. *How we divide ourselves up to vote, and why it matters.*

The theory of Markov chains—random walks with no memory, found across pure mathematics and applied statistics—has found a fruitful application in electoral redistricting. I'll discuss some surprisingly simple questions about graphs and geometry that lie at the heart of this impactful intervention that mathematicians are currently making in politics and civil rights. (Received January 21, 2020)

1155-05-18 **Kevin J. McCall*** (mccallkj@vcu.edu). *2-Bootstrap Percolation in Kneser Graphs.*

A Kneser Graph, denoted $K(s,t)$ is a graph whose vertex set is all t-element subsets of a set with s elements. Two vertices u, v are adjacent if their respective sets have no elements in common. r-neighbor boostrap percolation, also known as r-boostrap percolation, is a process in a graph which begins by selecting an initial set of vertices to be infected in the first round. In every subsequent round infected vertices remain infected while uninfected vertices become infected if they have at least r infected neighbors.

In 2-Bootstrap Percolation, we begin by selecting two initial infected vertices and in each round infect every vertex with at least two infected neighbors. A graph is 2-Bootstrap Good (2-BG) if it is possible to select two vertices in the graph so that the infection percolates throughout the entire graph. A natural question is to investigate classes of graphs which are 2-BG. We characterize those Kneser graphs which are 2-BG. (Received October 07, 2019)

1155-05-48 **Daryl R DeFord*** (ddeford@mit.edu). *Multiresolution Redistricting and Municipality Preservation.* Preliminary report.

Geospatial data for political redistricting is reported at several levels of resolution, from census blocks to voting precincts to county boundaries, that are not always compatible. This talk discusses two problems related to this inhomogeneity of scale. First, we examine the impact of resolution choice on the behavior of Markov chain sampling methods for generating ensembles of districting plans. Provided that there are sufficient permissible plans using fixed resolution units, the observed distributions of partisan metrics are similar using the ReCom method, while the boundary flip method returns inconsistent results. Secondly, we analyze municipality preservation rules that enforce rigid county boundaries and describe their impacts on minority representation. In order to accomplish this task, we introduce new algorithmic procedures for generating compliant ensembles. This analysis is supported with empirical results taken from real-world data in states where related legislation has been proposed with concerns for racial and partisan representation. (Received December 01, 2019)

1155-05-122 **Alex Chandler**, **Radmila Sazdanovic**, **Salvatore Stella** and **Martha Yip*** (martha.yip@uky.edu). *Chromatic symmetric homology for graphs: some new developments.*

For any graph G with n vertices, Stanley defined the chromatic symmetric function $X_G(\mathbf{x})$, which is a multivariable lift of the chromatic polynomial $\chi_G(x)$. The symmetric function X_G can be further lifted to a homological setting involving graded $\mathfrak{S}_n$-modules so that its bigraded Frobenius character recovers X_G.

In this talk, we discuss some new results regarding the strength of the chromatic symmetric homology of a graph, and state some conjectures regarding torsion in the homology. This is based on joint work with Chandler, Sazdanovic, and Stella. (Received January 07, 2020)

1155-05-128 **Zachary Hamaker**, **Oliver Pechenik**, **David E Speyer** and **Anna Weigandt*** (weigandt@umich.edu). *Derivatives of Schubert polynomials and proof of a determinant conjecture of Stanley.*

We describe the action of a differential operator when applied to the Schubert basis. We use this to give a short proof of the Macdonald identity. We also discuss an application to the study of the (strong) Sperner property of the weak order on the symmetric group. This property was recently proven by Gaetz and Gao (2018). Our description of the differential operator leads to a proof of a determinant conjecture of Stanley (2017), which also implies the Sperner property. (Received January 07, 2020)

1155-05-129 **Zachary Hamaker** and **Oliver Pechenik***, Department of Mathematics, University of Michigan, Ann Arbor, MI 48109, and **Anna Weigandt**. *Gröbner geometry of Schubert polynomials through ice.*

Knutson and Miller (2005) showed that the multidegree of a matrix Schubert variety X_w is the corresponding Schubert polynomial $\mathfrak{S}_w$. Moreover, after Gröbner degeneration with respect to any antidiagonal term order, the resulting irreducible components are naturally labeled by the pipe dreams for w. In later work with Yong (2009), they used diagonal term orders to obtain alternative combinatorics for certain X_w. We present further results in this direction, with connections to a neglected Schubert polynomial formula of Lascoux (2002) in terms of the square-ice model (recently rediscovered by Lam, Lee, and Shimozono in the guise of "bumpless pipe dreams"). (Received January 07, 2020)

1155-05-145 **John Rhodes** and **Anne Schilling***, Department of Mathematics, University of California, One Shields Avenue, Davis, CA 95616. *Mixing times for finite Markov chains.*

We show that the stationary distribution of a finite Markov chain can be expressed as the sum of certain normal distributions. These normal distributions are associated to planar graphs consisting of a straight line with attached loops. The loops touch only at one vertex either of the straight line or of another attached loop. Our analysis is based on our previous work, which derives the stationary distribution of a finite Markov chain using semaphore codes on the Karnofsky–Rhodes and McCammond expansion of the right Cayley graph of the finite semigroup underlying the Markov chain. This method also provides a way to analyze the mixing time of the Markov chain.

This is joint work with John Rhodes based on arXiv:1902.01042 and arXiv:1711.10689. (Received January 08, 2020)

1155-05-146 **Maria Monks Gillespie*** (maria.gillespie@colostate.edu). *Projective Embeddings of $\overline{M}_{0,n}$ and Parking Functions.*

The moduli space $\overline{M}_{0,n}$ may be embedded into the product of projective spaces $\mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^{n-3}$, using a combination of the Kapranov map $|\psi_n| : \overline{M}_{0,n} \to \mathbb{P}^{n-3}$ and the forgetful maps $\pi_i : \overline{M}_{0,i} \to \overline{M}_{0,i-1}$. We give an explicit combinatorial formula for the multidegree of this embedding in terms of certain parking functions of height $n-3$. We use this combinatorial interpretation to show that the total degree of the embedding (thought of as the projectivization of its cone in $\mathbb{A}^2 \times \mathbb{A}^3 \cdots \times \mathbb{A}^{n-2}$) is equal to $(2(n-3)-1)!! = (2n-7)(2n-9)\cdots(5)(3)(1)$. As a consequence, we also obtain a new combinatorial interpretation for the odd double factorial. (Received January 08, 2020)

1155-05-161 **Brendon Rhoades** and **Andy Wilson*** (andwils2@pdx.edu). *Vandermondes in superspace.*

Superspace of rank n is a $\mathbb{Q}$-algebra with n commuting generators $x_1, \ldots, x_n$ and n anticommuting generators $\theta_1, \ldots, \theta_n$ which arises in mathematical physics and the study of differential forms. We present a family of extensions of the Vandermonde determinant to superspace and use superspace Vandermondes to construct graded representations of the symmetric group. Our constructions specialize to hook-shaped Tanisaki quotients, the coinvariant ring for the Delta Conjecture constructed by Haglund, Rhoades, and Shimozono, and a superspace quotient related to positroids and Chern plethysm constructed by Billey, Rhoades, and Tewari. We also conjecture connections between these representations and the symmetric function $\Delta'_{e_{k-1}} e_n$ from the Delta Conjecture as well as the Hard Lefschetz Theorem. (Received January 10, 2020)

1155-05-173 **Daniel Irving Bernstein**, **Grigoriy Blekherman** and **Kisun Lee*** (klee669@gatech.edu). *Typical ranks in real symmetric matrix completion.*

In this talk, we consider the problem of low rank matrix completion for symmetric matrices. It is well known that the minimum rank of a completion of a generic partially specified symmetric matrix depends only on the location of the specified entries, and not their values, if complex entries are allowed. When we restrict ourselves to have only real entries, this is no longer the case and it has several minimum ranks which are called typical ranks. The Schur complement theory will be firstly discussed as a main tool. We give a combinatorial description of *nimesn* symmetric matrices that have n as a typical rank. Moreover, we also characterize the patterns of entries with low maximal typical ranks. (Received January 11, 2020)

1155-05-174 **Rong Luo*** (rluo@mail.wvu.edu), Morgantown, WV 26505. *Modulo orientations and integer flows of signed graphs.* Preliminary report.

A (signed) graph is modulo-$(2p+1)$-orientable if it has an orientation such that indegree is congruent to outdegree modulo $2p+1$ at each vertex. We will talk about the relations among modulo orientations, integer-valued and

real-values circular flows, and integer flows and study the monotonicity of modulo orientations and circular flows. This is joint work with Miaomiao Han, Jiaao Li, Yongtang Shi and C.-Q. Zhang. (Received January 11, 2020)

1155-05-184 **Laura Colmenarejo**, **Rosa C Orellana*** (rosa.c.orellana@dartmouth.edu), **Franco Saliola**, **Anne Schilling** and **Mike Zabrocki**. *An insertion algorithm on multiset partitions with applications to diagram algebras.*

We generalize the Robinson-Schensted-Knuth algorithm to the insertion of two row arrays of multisets. This generalization leads to new enumerative results that have representation theoretic interpretations as decompositions of centralizer algebras and the spaces they act on. In addition, restrictions on the multisets lead to further identities and representation theory analogues. For instance, we obtain a bijection between words of length k with entries in [n] and pairs of tableaux of the same shape with one being a standard Young tableau of size n and the other being a standard multiset tableau of content [k]. We also obtain an algorithm from partition diagrams to pairs of a standard tableau and a standard multiset tableau of the same shape, which has the remarkable property that it is well-behaved with respect to restricting a representation to a subalgebra. This insertion algorithm matches recent representation-theoretic results of Halverson and Jacobson. This is joint work with Laura Colmenarejo, Rosa Orellana, Franco Saliola, and Mike Zabrocki (Received January 12, 2020)

1155-05-194 **Olya Mandelshtam*** (olya@math.brown.edu). *Compact formulas for Macdonald polynomials via the exclusion process.*

The asymmetric simple exclusion exclusion process (ASEP) is a one-dimensional model of hopping particles that has been extensively studied in statistical mechanics, probability, and combinatorics. It also has remarkable connections with orthogonal symmetric polynomials in many variables such as Macdonald and Koornwinder polynomials.

In this talk we will give an overview of some recent work, and discuss how we can use multiline queues, a new object recently introduced by James Martin to compute stationary probabilities of the ASEP on a circle, to obtain new combinatorial formulas for both the symmetric Macdonald polynomials $P_\lambda(X;q,t)$ and the nonsymmetric Macdonald polynomials $E_\lambda(X;q,t)$, where λ is a partition. Building on this result, we obtain new, compact formulas for modified Macdonald polynomials $H_\lambda(X;q,t)$. Finally, we introduce a new quasisymmetric analogue of Macdonald polynomials, which refine the P_λ's and specialize at $q = t = 0$ to the quasisymmetric Schur polynomials defined by Haglund, Luoto, Mason, and van Willigenburg. (Received January 13, 2020)

1155-05-223 **Jennifer Vandenbussche** (jvandenb@kennesaw.edu) and **Erik E Westlund*** (ewestlun@kennesaw.edu), 370 Paulding Avenue NW, Room 534, MD 1102, Kennesaw, GA 30144. *Extending Matchings to Perfect Matchings in Cartesian Products of Even Cycles.* Preliminary report.

In a bipartite graph G, a set $S \subseteq V(G)$ is *deficient* if $|N(S)| < |S|$. A matching M (with vertex set U) is *k-suitable* if $G - U$ has no deficient set of size less than k. Let C_{2m}^d, where $m \geq 2$, be the d-fold cartesian product of the cycle C_{2m}. We show for any k such that $1 \leq k \leq d$, every k-suitable matching in C_{2m}^d with at most $f(k,d)$ edges extends to a perfect matching in C_{2m}^d, where $f(k,d) = k(2d-k) + \binom{k-1}{2}$. Moreover, for each $1 \leq k \leq d$, we show there exists a k-suitable matching with $f(k,d)+1$ edges that does not extend to a perfect matching in C_{2m}^d. We use a similar approach to that of the first author and West, who solved this problem in 2009 for hypercubes. (Received January 14, 2020)

1155-05-239 **Colleen Robichaux*** (cer2@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801, **Harshit Yadav** (yadav7@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801, and **Alexander Yong** (ayong@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801. *The A·B·C·Ds of Schubert calculus.*

We discuss *Atiyah-Bott Combinatorial Dreams* (A·B·C·Ds) in Schubert calculus. One result relates equivariant structure coefficients for two isotropic flag manifolds, with consequences to the thesis of C. Monical. We contextualize using work of N. Bergeron-F. Sottile, S. Billey-M. Haiman, P. Pragacz, and T. Ikeda-L. Mihalcea-I. Naruse. The relation complements a theorem of A. Kresch-H. Tamvakis in quantum cohomology. Results of A. Buch-V. Ravikumar rule out a similar correspondence in K-theory. This is joint work with H. Yadav and A. Yong. (Received January 14, 2020)

1155-05-246 **Jeanne Clelland*** (jeanne.clelland@colorado.edu), Dept. of Mathematics, 395 UCB, University of Colorado, Boulder, CO 80309-0395, and **Nicholas Bossenbroek**, **Thomas Heckmaster**, **Adam Nelson**, **Peter Rock** and **Jade VanAusdall**. *District compactness in the ReCom sampling method.* Preliminary report.

In recent years, ensemble analysis has emerged as a powerful tool for identifying districting plans that appear to be extreme outliers with regard to partisan outcome of elections, as well as to other election-related metrics. An issue of fundamental mathematical importance is how to choose a sampling method for constructing an ensemble of plans, and how to describe the associated probability distribution on the space of districting plans.

In this talk, we consider the ReCom sampling method developed by the Metric Geometry and Gerrymandering Group. It has been observed that this method tends to favor plans with relatively compact districts. We will describe why this sort of qualitative behavior might be expected, and we will present preliminary experimental evidence of a quantitative relationship between the ReCom sampling distribution and district compactness as measured by the number of cut edges in each plan.

This work was carried out as part of a summer REU program at The University of Colorado-Boulder. (Received January 15, 2020)

1155-05-251 **Brendon Rhoades*** (bprhoades@math.ucsd.edu) and **Andrew Timothy Wilson** (andwils2@pdx.edu). *Vandermondes, superspace, and spanning line configurations.*

Given two positive integers $k \leq n$, let $X_{n,k}$ be the moduli space of n-tuples of lines in $\mathbb{C}^k$ whose span equals $\mathbb{C}^k$. The presenter and Pawlowski introduced $X_{n,k}$ as an extension of the complete flag variety $\mathcal{F}\ell_n$ to the Haglund-Remmel-Wilson *Delta Conjecture*. The space $X_{n,k}$ is not compact; its cohomology ring does not exhibit Poincaré Duality in general. We introduce an extension of the Vandermonde determinant to superspace and use this extension to build multigraded S_n-module extending the cohomology of $X_{n,k}$ which exhibit a 'bigraded' version of Poincaré Duality and (conjecturally) give a representation-theoretic model for the Delta Conjecture. Joint with Andy Wilson. (Received January 15, 2020)

1155-05-253 **Katharine Ahrens*** (kaahrens@ncsu.edu), Raleigh, NC 27606. *Combinatorial Applications of the k-Fibonacci Numbers.*

Fibonacci numbers have fascinated mathematicians for hundreds of years. They continue to appear in surprising places across diverse branches of math.

In this talk, we examine a generalization known as the k-Fibonacci numbers. We explain how they arose organically during the course of our research on a cryptographically-motivated problem in polynomial and matrix theory, and how they neatly answer a question about k-tridiagonal matrices. We also present four other combinatorially interesting objects which the k-Fibonacci numbers count: classes of restricted permutations, integer compositions, binary strings, and subsets. (Received January 16, 2020)

1155-05-255 **M.M.M. Jaradat*** (mmjst4@qu.edu.qa), Department of Mathematics, Statistics and Phy, Qatar University, Doha, 2713, Qatar, and **A Baniabedalruhman**, **M Bataineh** and **A Jaradat**. *The theta-complete graph Ramsey number $r(\theta_k, K_6); k \geq 6$.*

The Ramsey number $r(G, H)$ is the smallest positive integer n such that any graph W of order n contains G as a subgraph or its complement contains H as a subgraph. In this paper, we find the exact value for the Ramsey number $r(\theta_k, K_6); k \geq 6$ where θ_k is a theta graph of order k and K_n is a complete graph of order n. In fact, we prove that $r(\theta_n, K_6) = 5(n-1)+1$, $n \geq 6$. (Received January 15, 2020)

1155-05-260 **Deepak Bal** and **Jonathan Cutler*** (jonathan.cutler@montclair.edu). *Enumerative Nordhaus-Gaddum inequalities.*

Nordhaus and Gaddum proved the following inequalities that give upper and lower bounds on the sum and product of the chromatic number of a graph and its complement.

$$2\sqrt{n} \leq \chi(G) + \chi(\overline{G}) \leq n+1 \quad \text{and} \quad n \leq \chi(G)\chi(\overline{G}) \leq \frac{(n+1)^2}{4}.$$

Inspired by these results, Nordhaus-Gaddum inequalities have been studied for many other graph invariants. Recently, Wagner gave a lower bound on the sum of the number of dominating sets in a graph on its complement. In this talk, we discuss some related results and their connections to well-known areas of study in graph theory. (Received January 15, 2020)

1155-05-301 **Susanna D Fishel*** (sfishel1@asu.edu), Arizona State University, **Jessica Gatica**, University of Talca, **Luc Lapointe**, University of Talca, and **Maria-Elena Pinto**, University of Talca. *Quasisymmetric functions in superspace.*

Symmetric functions in superspace were developed to study the supersymmetric version of the quantum Calogero-Sutherland model of identical particles on a circle. They are a generalization of symmetric functions: we still have the variables $x_1, x_2, \ldots$ and additionally we have anticommuting variables $\theta_1, \theta_2, \ldots$. Superspace analogues of Macdonald, Jack, and Schur polynomials have been defined in a series of papers by Desrosiers and others. I will discuss quasisymmetric functions in superspace. This is work in progress, joint with Jessica Gatica, Luc Lapointe, and Maria-Elena Pinto. (Received January 17, 2020)

1155-05-311 **Linyuan Lu*** (lu@math.sc.edu), Columbia, SC 29208, and **Joshua Thompson** (joshuact@email.sc.edu), Columbia, SC 29208. *Poset Ramsey Numbers for Boolean Lattices.*

A subposet Q' of a poset Q is a *copy of a poset* P if there is a bijection f between elements of P and Q' such that $x \leq y$ in P iff $f(x) \leq f(y)$ in Q'. For posets P, P', let the *poset Ramsey number* $R(P, P')$ be the smallest N such that no matter how the elements of the Boolean lattice Q_N are colored red and blue, there is a copy of P with all red elements or a copy of P' with all blue elements. Axenovich and Walzer introduced this concept in *Order* (2017), where they proved $R(Q_2, Q_n) \leq 2n + 2$ and $R(Q_n, Q_m) \leq mn + n + m$, where Q_n is the Boolean lattice of dimension n. They later proved $2n \leq R(Q_n, Q_n) \leq n^2 + 2n$. Walzer later proved $R(Q_n, Q_n) \leq n^2 + 1$. We provide some improved bounds for $R(Q_n, Q_m)$ for various $n, m \in \mathbb{N}$. In particular, we prove that $R(Q_n, Q_n) \leq n^2 - n + 2$, $R(Q_2, Q_n) \leq \frac{5}{3}n + 2$. (Received January 17, 2020)

1155-05-317 **Rebecca Goldin** and **Brent Gorbutt***, bgorbutt@gmu.edu. *A positive formula for type A Peterson Schubert calculus.*

The S_1-equivariant cohomology of a Peterson variety in type A has a $H^*_{S^1}\{pt\}$-module basis of Peterson Schubert classes. We will present a formula for the product of two of these Peterson Schubert classes. The formula is manifestly positive and integral. We will also present a combinatorial identity we discovered while proving the formula. (Received January 17, 2020)

1155-05-330 **Rosa Orellana** (rosa.c.orellana@gmail.com), Mathematics Department, 6188 Kemeny Hall, Hanover, NH 03755, and **Mike Zabrocki*** (zabrocki@mathstat.yorku.ca), Mathematics & Statistics, 4700 Keele Street, Toronto, ON M3J1P3, Canada. *Representations of a multiset partition algebra and Kronecker coefficients.* Preliminary report.

We define a diagram algebra of multiset partitions which is the centralizer of the the symmetric group of S_n acting on a polynomial ring in k sets of n variables. The irreducible representations of this algebra have dimensions equal to the number of multiset tableaux and the Kronecker coefficients appear as multiplicities in the restriction of irreducible modules. (Received January 18, 2020)

1155-05-344 **Guantao Chen*** (gchen@gsu.edu), Department of Mathematics and Statistics, Georgia State University, Atlanta, GA 30303, **Xing Feng**, Faculty of Science, Jiangxi University of Science and Technology, Ganzhou, Peoples Rep of China, **Fuliang Lu**, School of Mathematics, Minnan Normal University, Zhangzhou, Peoples Rep of China, **Claudio Lucchesi**, Institute of Computing, UNICAMP, Campinas, Brazil, and **Lianzhu Zhang**, School of Mathematial Science, Xiamen University, Xiamen, Peoples Rep of China. **Laminar tight cuts in matching covered graphs**. Preliminary report.

An edge cut C of a graph G is tight if $|C \cap M| = 1$ for every perfect matching M of G. Barrier cuts and 2-separation cuts are called ELP-cuts, which are two important types of tight cuts in matching covered graphs. Edmonds, Lovász, and Pulleyblank proved that if a matching covered graph has a nontrivial tight cut, then it also has a nontrivial ELP-cut. Carvalho, Lucchesi, and Murty made a stronger conjecture: given any nontrivial tight cut C in a matching covered graph G, there exists a nontrivial ELP-cut D in G which does not cross C. In this talk, we will present our recent proof of this conjecture. (Received January 18, 2020)

1155-05-357 **Nicholas A. Loehr** and **Gregory S. Warrington***, Department of Mathematics & Statistics, 82 University Place, Burlington, VT 05405. *Abacus histories and creation operators.*

Creation operators act on symmetric functions to build Schur functions, Hall–Littlewood polynomials, and related symmetric functions one row at a time. Haglund, Morse, Zabrocki, and others have studied more general symmetric functions H_α, C_α, and B_α obtained by applying any sequence of creation operators to 1. We develop

new combinatorial models for the Schur expansions of these and related symmetric functions using objects called abacus-histories. These formulas arise by chaining together smaller abacus-histories that encode the effect of an individual creation operator on a given Schur function. We give a similar treatment for operators such as multiplication by h_m, $h_m^\perp$, ω, etc., which serve as building blocks to construct the creation operators. We use involutions on abacus-histories to give bijective proofs of properties of the Bernstein creation operator and Hall–Littlewood polynomials indexed by three-row partitions. (Received January 18, 2020)

1155-05-376 **Konstantin Matveev*** (kmatveev@math.rutgers.edu), 411 Frelinghuysen Road, Piscataway, NJ 08901. *Positivity for symmetric functions and vertex models.* Preliminary report.

One question is about classifying homomorphisms with positive values on the family of Macdonald symmetric functions. The other is about non-obvious positivity of certain quantities for the stochastic six-vertex model. I will talk about the relation between these two questions and what it has to do with the RSK. (Received January 19, 2020)

1155-05-388 **Yan Cao**, **Guantao Chen** and **Guangming Jing*** (gjing@augusta.edu), 1120 15th Street, Augusta, GA 30912. *A note on the Total Coloring of multigraphs.*

Let $G = (V(G), E(G))$ be a multigraph with maximum degree $\Delta(G)$, chromatic index $\chi'(G)$ and total chromatic number $\chi''(G)$. The Total Coloring conjecture proposed by Behzad and Vizing, independently, states that $\chi''(G) \leq \Delta(G) + \mu(G) + 1$ for a multigraph G, where $\mu(G)$ is the multiplicity of G. Moreover, Goldberg conjectured that $\chi''(G) = \chi'(G)$ if $\chi'(G) \geq \Delta(G) + 3$ and noticed the conjecture holds when G is an edge-chromatic critical graph. We recently showed that $\chi''(G) = \chi'(G)$ if $\chi'(G) \geq \max\{\Delta(G) + 2, |V(G)| + 1\}$, and as a consequence $\chi''(G) = \chi'(G)$ if G contains a critical subgraph of size at least $|V(G)|/2$. (Received January 19, 2020)

1155-05-390 **Nicholas A. Loehr*** (nloehr@vt.edu), Virginia Tech Dept. of Mathematics, 225 Stanger Street, 460 McBryde Hall, Blacksburg, VA 24061-0123. *The combinatorics of evenly spaced binomial coefficients.*

A curious identity for binomial coefficients states that $\sum_k \binom{n}{km} = \frac{1}{m}\sum_{j=0}^{m-1}(1 + e^{2\pi ij/m})^n$. There are similar formulas for the sum of $\binom{n}{a}$ over all a's with a given remainder mod m. This talk undertakes a combinatorial exploration of these formulas emphasizing bijective proofs. Our goal is to find a combinatorial explanation of why these sums are "almost" $2^n/m$. We give a bijective proof that the minimum of the sums $\sum_k \binom{n}{km+r}$ equals $(2^n - \ell(n,m))/m$, where the "error term" $\ell(n,m)$ has an explicit combinatorial interpretation involving words satisfying certain parenthesis-matching conditions. (Received January 19, 2020)

1155-05-391 **Chris Lambie-Hanson*** (cblambiehanso@vcu.edu). *Finite subgraphs of uncountably chromatic graphs.*

A major line of inquiry in infinite graph theory has been around the extent to which one can determine the global structure of an infinite graph simply by looking at its finite subgraphs. An early result in this direction, which is a consequence of the De Bruijn-Erdős compactness theorem, shows that if G is a graph with infinite chromatic number, then G contains finite subgraphs of every possible finite chromatic number. One can then define a function $f_G : \mathbb{N} \to \mathbb{N}$ by letting $f_G(k)$ be the least number of vertices in a subgraph of G with chromatic number k. f_G is an increasing function, and a natural question to ask is how quickly f_G can grow. Results of Erdős show that, for every function $f : \mathbb{N} \to \mathbb{N}$, there is a graph G with *countably* infinite chromatic number such that f_G grows faster than f. In 1982, Erdős, Hajnal, and Szemerédi asked if the analogous statement is true if we moreover require that G have *uncountable* chromatic number. We will answer this question and more broadly discuss other problems involving finite subgraphs of uncountable graphs. (Received January 19, 2020)

1155-05-402 **Louis DeBiasio**, **Ryan Martin** and **Theodore Molla*** (molla@usf.edu). *Powers of Hamiltonian cycles in multipartite graphs.*

The minimum proportional degree of a multipartite graph is the minimum over all vertices v and all parts U not containing v of the number of neighbors of v in U divided by the order of U. In 1963 Moon and Moser proved that every balanced bipartite graph with minimum proportional degree greater than $1/2$ contains a Hamiltonian cycle. In this talk, we will discuss the following related result: For every positive integer r and $\gamma > 0$ there exists n_0 such that if a multipartite graph on $n \geq n_0$ vertices has minimum proportional degree at least $1 - 1/r + \gamma$ and no part has order greater than n/r, then there exists a cyclic ordering of the vertices in which every set of r consecutive vertices is a clique. (Received January 19, 2020)

1155-05-408 **Joshua E Ducey*** (duceyje@jmu.edu), 60 Bluestone Dr, MSC 1911, Harrisonburg, VA 22807. *Critical groups of the Kneser and Grassmann strongly regular graphs.*

The critical group (or sandpile group) of a graph is the abelian group presented by the Smith normal form of a Laplacian matrix of the graph, and is an interesting graph invariant. In this talk methods of representation theory are applied to compute the critical group of the Grassmann graph on 2-dimensional subspaces of a finite vector space. The same basic technique can be applied to compute critical groups of Kneser graphs, using some modular representation theory of the symmetric group. (Received January 19, 2020)

1155-05-419 **Alexander Diaz-Lopez*** (alexander.diaz-lopez@villanova.edu), 800 Lancaster Ave (SAC 305), Villanova, PA 19462. *Peaks and descents of permutations: A story with open questions.*

Peaks and descents of permutations are properties based on when a permutation has the consecutive patterns 132 or 231 for peaks and 21 for descents. They have been studied algebraically for some time (e.g. Solomon algebra, peak algebra) and more recently, from a combinatorial point of view. In this talk, I will give an overview of some important facts about them with open questions intertwined. Everybody is welcome to attend the talk and I will keep the background to a minimum. (Received January 20, 2020)

1155-05-439 **Martha E Precup*** (martha.precup@wustl.edu). *Hessenberg varieties and the Stanley–Stembridge conjecture.*

In 2015, Brosnan and Chow, and independently Guay-Paquet, proved the Shareshian–Wachs conjecture, which links the Stanley–Stembridge conjecture in combinatorics to the geometry of Hessenberg varieties through Tymoczko's permutation group action on the cohomology ring of regular semisimple Hessenberg varieties. This talk will give a brief overview of that story and present a new set of linear relations satisfied by the multiplicities of certain permutation representations in Tymoczko's representation. As an application of these results, we prove an inductive formula for the multiplicity coefficients corresponding to partitions with a maximal number of parts. This is joint work with Megumi Harada. (Received January 20, 2020)

1155-05-451 **Yan Cao**, **Guantao Chen**, **Guangming Jing** and **Songling Shan*** (sshan12@ilstu.edu). *Proof of the Core Conjecture of Hilton and Zhao.* Preliminary report.

Let G be a simple graph with maximum degree Δ. We call G *overfull* if $|E(G)| > \Delta\lfloor |V(G)|/2 \rfloor$. The *core* of G, denoted G_Δ, is the subgraph of G induced by its vertices of degree Δ. A classic result of Vizing shows that $\chi'(G)$, the chromatic index of G, is either Δ or $\Delta + 1$. It is NP-complete to determine the chromatic index for a general graph. However, if G is overfull then $\chi'(G) = \Delta + 1$. Hilton and Zhao in 1996 conjectured that if G is a simple connected graph with $\Delta \geq 3$ and $\Delta(G_\Delta) \leq 2$, then $\chi'(G) = \Delta + 1$ if and only if G is overfull or $G = P^*$, where P^* is obtained from the Petersen graph by deleting a vertex. This conjecture, if true, implies an easy approach for calculating $\chi'(G)$ for graphs G satisfying the conditions. The progress on the conjecture has been slow: it was only confirmed for $\Delta = 3, 4$, respectively, in 2003 and 2017. We confirm this conjecture for all $\Delta \geq 4$. (Received January 20, 2020)

1155-05-452 **Xiaofeng Gu***, University of West Georgia, 1601 Maple St., Carrollton, GA 30118. *Graph rigidity in the Euclidean plane.*

A bar-and-joint framework in Euclidean plane is a pair (G, p), where G is a graph and p is a map from $V(G)$ to the Euclidean plane. Given a framework (G, p), a natural question is to ask, is it rigid? This question has both combinatorial and geometric aspects. A combinatorial characterization of rigidity in the Euclidean plane has been obtained by Laman in 1970, which results the following definition of rigid graphs. A graph G is sparse if $|E(H)| \leq 2|V(H)| - 3$ for every subgraph H of G with $|V(H)| \geq 2$; if in addition $|E(G)| = 2|V(G)| - 3$, then G is minimally rigid. A graph is rigid if it contains a spanning minimally rigid subgraph. It is interesting to find sufficient conditions for graph rigidity. Some old and new results on related topics will be presented in this talk. Joint work with Wei Meng, Martin Rolek, Yue Wang, and Gexin Yu. (Received January 20, 2020)

1155-05-457 **Chris Stephens** and **Dong Ye***, Department of Mathematical Sciences, Middle Tennessee State University, Murfreesboro, TN 37132. *Connectivity for kite-linked graphs.*

For a given graph H, a graph G is *H-linked* if, for every injection $\varphi : V(H) \to V(G)$, the graph G contains a subdivision of H with $\varphi(v)$ corresponding to v, for each $v \in V(H)$. Let $f(H)$ be the minimum integer k such that every k-connected graph is H-linked. Among graphs H with at least four vertices, the exact value $f(H)$ is only know when H is a path with four vertices or a cycle with four vertices. A *kite* is graph obtained from K_4 by deleting two adjacent edges, i.e., a triangle together with a pendant edge. Recently, Liu, Rolek and Yu proved that every 8-connected graph is kite-linked. The exact value of $f(H)$ when H is the kite remains open.

In this talk, we present a recent result which settles this problem by proving that every 7-connected graph is kite-linked. (Received January 20, 2020)

1155-05-481 **James A Long** and **Kevin G Milans*** (milans@math.wvu.edu), Morgantown, WV , and **Andrea Munaro**. *Longest Path Transversals.* Preliminary report.

A *longest path transveral* in a graph G is a set of vertices S such that every longest path in G contains a vertex in S. Let $\mathrm{lpt}(G)$ be the minimum size of a longest path transversal in G. Gallai asked whether $\mathrm{lpt}(G) = 1$ when G is connected. The answer is no; the best known construction is due to Grünbaum (1973), giving a connected graph G with $\mathrm{lpt}(G) = 3$. In 2014, Rautenbach and Sereni showed that $\mathrm{lpt}(G) \leq \left\lceil \frac{n}{4} - \frac{n^{2/3}}{90} \right\rceil$ when G is an n-vertex connected graph. We show that $\mathrm{lpt}(G) \leq O(n^{3/4})$ when G is an n-vertex connected graph. Our results also provide sublinear sets in G which intersect all maximum subdivisions of any fixed graph F. (Received January 20, 2020)

1155-05-483 **Yue Wang*** (150640131750163.com), Williamsburg, VA 23185, and **Gexin Yu** (gyu@wm.edu), Williamsburg, VA 23187. *Enhancing the Erdős-Lo'vasz Tihany conjecture for line graphs of multigraphs.*

In this paper, we prove an enhanced version of the ErdHos-Lov'asz Tihany Conjecture for line graphs of multigraphs. That is, for every graph G whose chromatic number $chi(G)$ is more than its clique number $omega(G)$ and for nonnegative integer ell, any two integers $s, tgeq4(ell1)$ with $st = chi(G)1$, there is a partition (S, T) of the vertex set $V(G)$ such that $chi(G[S])geqs$ and $chi(G[T])geqtell$. In particular, when $ell = 1$, we can obtain the same result just for any $s, tgeq4$. The ErdHos-Lo'vasz Tihany conjecture is a special case when $ell = 0$. (Received January 21, 2020)

1155-05-494 **Allen Knutson*** (allenk@math.cornell.edu) and **Paul Zinn-Justin**. *Puzzles for Segre-Schwarz-MacPherson Schubert calculus.*

The Białynicki-Birula strata $BwT/T \subseteq G/T$ (G complex) are much simpler than those on G/B (the Bruhat cells). To relate them to actual Bruhat cells we follow them under the Grothendieck-Springer degeneration of G/T to $T^*(G/B)$, obtaining Maulik-Okounkov's "stable basis", then formally divide them by the class of the zero section to get *Segre-Schwarz-MacPherson classes.* I will give a rule, joint with Paul Zinn-Justin, for multiplying these classes (on T^* of up to 4-step flag manifolds) using a new formulation of puzzles, closer to pipe dreams. This T^*G/P rule implies the G/P rule up to 3-step, and also lets one compute the Euler characteristic of **positive-dimensional** (transverse) intersections of Bruhat cells. (Received January 20, 2020)

1155-05-514 **Jonah Blasiak*** (jblasiak@gmail.com), **Jennifer Morse**, **Anna Pun** and **Daniel Summers**. *Catalan functions and k-Schur positivity.*

Catalan functions are a family of symmetric functions indexed by pairs consisting of a partition contained in the staircase $(n-1, \ldots, 1, 0)$ and a weight in $\mathbb{Z}^n$. They include the Hall-Littlewood polynomials and their parabolic generalizations. Li-Chung Chen and Mark Haiman conjectured that the k-Schur functions are a subclass of Catalan functions. We settle their conjecture, expose a new miraculous shift invariance property of k-Schur functions, and use this to establish Schur positivity of k-Schur functions.

Our techniques also yield a formula for the k-Schur expansion of the product of a Schur function and a k-Schur function when the indexing partitions concatenate to a partition, thereby describing a new class of Gromov-Witten invariants. (Received January 20, 2020)

1155-05-533 **Luc Lapointe***, Instituto de Matemáticas, Universidad de Talca, Avenida Lircay s/n, Talca, Chile. *Positivity conjectures for non-symmetric Macdonald polynomials.*

We study non-symmetric Macdonald polynomials whose variables $x_{m+1}, x_{m+2}, \ldots$ are symmetrized (using the Hecke symmetrization), which we call m-symmetric Macdonald polynomials (the case $m = 0$ corresponds to the usual Macdonald polynomials). In the space of m-symmetric polynomials, we can define m-symmetric Schur functions (now depending on the parameter t) by certain triangularity conditions. We conjecture that the m-symmetric Macdonald polynomials are positive (after a plethystic substitution) when expanded in the basis of m-symmetric Schur functions and that the corresponding m-(q,t)-Kostka coefficients embed naturally into the $m+1$-(q,t)-Kostka coefficients. As m goes to infinity, the m-symmetric Macdonald polynomials become non-symmetric Macdonald polynomials and the m-(q,t)-Kostka coefficients stabilize. As such, we obtain positivity conjectures for non-symmetric Macdonald polynomials whose corresponding (q,t)-Kostka coefficients contain as special cases the usual (q,t)-Kostka coefficients. We conjecture that natural recursion relations among these (q,t)-Kostka coefficients as well as generalizations of Butler's rule unify this new (q,t)-Kostka table. (Received January 21, 2020)

1155-05-541 **Deepak Bal***, deepak.bal@montclair.edu, and **Louis DeBiasio**. *Size Ramsey number of paths.*

Given a graph H, let $\hat{R}(H)$ be the minimum m such that there exists a graph G with m edges such that in every 2-coloring of the edges G, there is a monochromatic copy of H. Let P_n denote the path on n vertices. To prove that $\hat{R}(P_n) > m$, one must show that every graph on m edges can be 2-colored such that every monochromatic path has order less than n. We show that $\hat{R}(P_n) > (3.75 - o(1))n$ thereby improving the previous best-known lower bound of $(2.5 - o(1))n$ due to Dudek and Pralat. We also discuss some results concerning the r-color version of the problem. This is joint work with Louis DeBiasio. (Received January 21, 2020)

1155-05-577 **Wei Meng*** (mengwei@sxu.edu.cn), **Martin Rolek** (msrolek@wm.edu), **Yue Wang** (m15064013175@163.com) and **Gexin Yu** (gyu@wm.edu). *Linear connectivity for tournaments to be highly linked.*

A digraph is k-linked if for any two disjoint sets of vertices $\{x_1, \ldots, x_k\}$ and $\{y_1, \ldots, y_k\}$ there are vertex disjoint paths $P_1, \ldots, P_k$ such that P_i is directed from x_i to y_i for $i = 1, \ldots, k$. Pokrovskiy in 2015 proved that every strongly $452k$-connected tournament is k-linked. In this paper, we significantly reduce this connectivity bound and show that any $(24k - 19)$-connected tournament is k-linked. (Received January 21, 2020)

1155-05-602 **Daniel P Johnston*** (djohnst1@skidmore.edu) and **Puck Rombach**. *On Rainbow Turán Numbers of Paths and Other Trees.*

For a fixed graph F, we consider the maximum number of edges in a properly edge-colored graph on n vertices which does not contain a rainbow copy of F, that is, a copy of F all of whose edges receive a different color. This maximum, denoted by $ex^*(n; F)$, is the rainbow Turán number of F, and its systematic study was initiated by Keevash, Mubayi, Sudakov and Verstraëte [*Combinatorics, Probability and Computing* **16** (2007)]. In this talk, we look at previous results and explore the rainbow Turán number when F is a path or another tree. (Received January 21, 2020)

1155-05-606 **Wayne Goddard*** (goddard@clemson.edu), Mathematical and Statistical Sciences, Clemson University, Clemson, SC. *Fractional Versions of the Total Domatic and Idomatic Numbers of a Graph.*

The total domatic and idomatic numbers of a graph are the maximum numbers of disjoint total dominating and independent dominating sets. A fractional version of these parameters is introduced. The results focus on planar graphs. For example, every triangular disc (a planar graph with at most one region not a triangle) has fractional total domatic number at least 3/2 and this is best possible. A planar graph has at most 4 disjoint total dominating sets, which is best possible, but its fractional total domatic number can exceed 4. A planar graph of minimum degree at least 2 has fractional idomatic number at least 2 (though there need not be 2 disjoint independent dominating sets) and so its independent domination number is at most half its order, which is sharp. We conclude with some conjectures related to the Four Color Theorem. Joint work with Michael A. Henning. (Received January 21, 2020)

1155-05-614 **Michael Wigal***, School of Mathematics, Atlanta, GA 30332, and **Xingxing Yu**, School of Mathematics, Atlanta, GA 30332. *Large cycles in essentially 4-connected planar graphs.*

Tutte proved that every 4-connected planar graph contains a Hamilton cycle, but there are 3-connected n-vertex graphs whose longest cycles have length $\Theta(n^{\log_3 2})$. On the other hand, Jackson and Wormald proved that an essentially 4-connected n-vertex planar graph contains a cycle of length at least $(2n+4)/5$, which was improved to $5(n+2)/8$ by Fabrici *et al.* We improve this bound to $\lceil (2n+6)/3 \rceil$ for $n \geq 6$ by proving a quantitative version of a result of Thomassen, and the bound is best possible. (Received January 21, 2020)

11 ► Number theory

1155-11-11 **Eva Goedhart*** (eva.goedhart@williams.edu) and **Helen G. Grundman**. *Using Lehmer Numbers to Solve Diophantine Equations.*

After a little history on Diophantus, I'll give a friendly introduction to Diophantine equations and Lehmer numbers. Then applying a well-known result of Bilu, Hanrot, and Voutier (2001), I'll show how Lehmer numbers are used to solve a family of Diophantine equations. (Received September 20, 2019)

1155-11-16 **Dorisa Tabaku*** (dorisatabaku@oakland.edu), 368 Mathematics Science Center, Rochester, MI 48039. *Weil heights in weighted projective spaces.* Preliminary report.

We will give an introduction of Cartier and Weil divisors in the weighted projective spaces and construct the height machinery a-la Weil in such spaces. To illustrate the theory will will provide examples of Weil heights superelliptic Jacobians. (Received September 28, 2019)

1155-11-22 **Sumit Chandra Mishra*** (sumit.chandra.mishra@emory.edu). *Local-global principle for norm one tori over semi-global fields.*

Let K be a complete discretely valued field with residue field κ. Let F be a function field of a smooth, projective, geometrically integral curve over K and $\mathcal{X}$ a regular proper model of F with the reduced special fibre X a union of regular curves with normal crossings. Suppose that the graph associated to $\mathcal{X}$ is a tree (e.g. $F = K(x)$). Let L/F be a Galois extension of degree n such that n coprime to $\mathrm{char}(\kappa)$. Suppose that κ is an algebraically closed field or a finite field containing a primitive n^{th} root of unity. Then we show that the local-global principle holds for the norm one torus corresponding to the extension L/F with respect to discrete valuations on F i.e. an element in $F^\times$ is a norm from the extension L/F if and only if it is a norm from the extensions $L \otimes_F F_\nu/F_\nu$ for all discrete valuations ν of F. (Received October 31, 2019)

1155-11-27 **Suresh Venapally***, suresh.venapally@emory.edu. *Local-global principle principal homogeneous spaces over function fields of p-adic curves.*

Let F be the function field of of p-adic curve and G a semi simple simply connected linear algebraic group over F. Colliot-Thélène, Parimala and Suresh have conjectured that a principal homogeneous space under G over F is trivial if and only is it is trivial over the completion of F at all discrete valuations of F. This conjecture has been proved for many groups of classical type. One of the open cases is $SU(A, \sigma)$, where A is a central simple algebra over F with an involution σ of seconnd kind. In this talk we discuss the validity of this conjecture for $SU(A, \sigma)$. (Received November 09, 2019)

1155-11-32 **Eva Bayer*** (eva.bayer@epfl.ch), EPFL-FSB-MATH, Station 8, Batiment MA, 1025 Lausanne, Switzerland. *Isometries of lattices and Hasse principle.*

Starting from a question of Gross and McMullen concerning characteristic polynomials of isometries of even, unimodular lattices of given signature, we investigate some integral Hasse principles. (Received November 14, 2019)

1155-11-38 **Huy Dang*** (hqd4bz@virginia.edu), 2304 Fontaine Avenue, Charlottesville, VA 22903. *Hurwitz tree and equal characteristic deformation of Artin-Schreier covers.*

An Artin-Schreier curve is a $\mathbb{Z}/p$-branched cover of the projective line over a field of characteristic $p > 0$. A unique aspect of positive characteristic is that there exist flat deformations of a wildly ramified cover that change the number of branch points but fix the genus. In this talk, we introduce the notion of Hurwitz tree. It is a combinatorial-differential object that is endowed with essential degeneration data of a deformation. We then show how the existence of a deformation between two covers with different branching data equates to the existence of a Hurwitz tree with behaviors determined by these branching data. One application of this result is to prove that the moduli space of Artin-Schreier covers of fixed genus g is connected when g is sufficiently large. (Received November 21, 2019)

1155-11-44 **Vishal Arul*** (varul.math@gmail.com). *Torsion points on Fermat quotients of the form $y^n = x^d + 1$.*

We classify torsion points on the curve $y^n = x^d + 1$ over $\mathbf{C}$, where n and d are coprime and satisfy $n, d \geq 2$. When $n + d \geq 8$, we show that the only torsion points on this curve are: (i) those whose x-coordinate is zero, (ii) those whose y-coordinate is zero, (iii) the point at infinity. When $n + d = 7$, there are more torsion points and we classify them all. As an application, we classify all torsion points on a generic superelliptic curve, which generalizes a result of Poonen and Stoll in the hyperelliptic case. (Received November 25, 2019)

1155-11-49 **Daniel Corey**, **Jordan S Ellenberg** and **Wanlin Li*** (wanlinli@mit.edu). *Computations on Ceresa class.*

The Ceresa cycle is an algebraic cycle attached to a smooth algebraic curve, which is trivial when the curve is hyperelliptic, and which has not been proven non-trivial for any non-hyperelliptic curve over a number field. Its image under a certain cycle class map provides a class in etale cohomology called the Ceresa class. This class encodes information of the action of the absolute Galois group on the second term of the lower central series of the pro-l etale fundamental group of the curve. Using an l-adic version of the Johnson morphism, via explicit computation, we determine whether the Ceresa class for curves over C((t)) is rationally trivial. Moreover, we

define a Ceresa invariant for tropical curves corresponding to the special fiber of curves over C[[t]] and show it is always torsion when all vertices have weight 0. (Received December 01, 2019)

1155-11-51 **Anastassia Etropolski**, **Maarten Derickx**, **Mark van Hoeij** and **Jackson Morrow*** (jmorrow4692@gmail.com), 400 Dowman Drive, Atlanta, GA 30307, and **David Zureick-Brown**. *Sporadic Cubic Torsion.*

In Mazur's celebrated 1978 Inventiones paper, he classified the torsion subgroups which can occur in the Mordell-Weil group of an elliptic curve over $\mathbf{Q}$. His result was extended to elliptic curves over quadratic number fields by Kamienny, Kenku, and Momose, with the full classification being completed in 1992. What both of these cases have in common is that each subgroup in the classification occurs for infinitely many elliptic curves; however, this no longer holds for cubic number fields. In 2012, Najman showed that there exists a unique (up to $\overline{\mathbf{Q}}$-isomorphism) elliptic curve whose torsion subgroup over a particular cubic field is $\mathbf{Z}/21\mathbf{Z}$. This curve yielded the first example of a "sporadic" torsion subgroup.

In this talk,I will recall previous literature concerning torsion subgroups of elliptic curves over number fields, introduce new results about sporadic points on the modular curves $X_1(N)$ and $X_1(M,N)$, and discuss some of the tools used in the analysis of cubic points on these modular curves. This is joint work with A. Etropolski, M. Derickx, M. van Hoeij, and D. Zureick-Brown. (Received December 02, 2019)

1155-11-60 **Jon Grantham*** (grantham@super.org) and **Hester Graves** (hkgrave@super.org). *Primes Which Are Values of Cyclotomic Polynomial.*

We will discuss work on computations of prime values of cyclotomic polymonomials $\Phi_n(x)$. For $n = 5$, these are related to a conjecture of Goldbach, which we verify for $x < 2.5 \times 10^{14}$. For $n = 5$, these computations are used to disprove a conjecture of Schott. I will cover a few more conjectures, and the history of such computations. (Received December 09, 2019)

1155-11-63 **Clifford Blakestad*** (blakestad@postech.ac.kr). *On generalizing p-adic sigma functions to Jacobians of curves of genus two.*

The classical Weierstrass sigma function plays an important role in the theory of elliptic curves and modular forms. This function is a complex analytic function defined in terms of the elliptic curve $\mathbb{C}/\Lambda$ with nice transformation properties under translation by points of the given lattice Λ. In the late '80s Mazur and Tate developed a p-adic analog attached to an elliptic curve E defined over the p-adic numbers, when E has ordinary reduction modulo p.

We will discuss a generalization of the Mazur–Tate theory to the Jacobian of a curve C of genus two with a marked Weierstrass point ∞ defined over the p-adic numbers, when C has invertible Hasse–Witt matrix. In doing so, we will be producing a p-adic analog of the two dimensional complex sigma function of Klein. (Received December 10, 2019)

1155-11-74 **Carrie E. Finch-Smith*** (finchc@wlu.edu), Washington and Lee University, Mathematics Department, 204 W. Washington St., Lexington, VA 24416. *Riesel numbers and Sierpiński numbers.*

In 1956, Hans Riesel showed that 509203 has an interesting property - for all $n \in \mathbb{N}$, the integer $509203 \cdot 2^n - 1$ is composite. Furthermore, he showed that there are infinitely many integers with this property. In honor of Riesel's work, an odd positive integer k with the property that $k \cdot 2^n - 1$ is composite for all natural numbers n is known as a Riesel number. Riesel's work showed that there are infinitely many Riesel numebrs. A few years later, Waclaw Sierpiński proved a similar result for positive integers k where $k \cdot 2^n + 1$ is composite for all n, and these integers are known as Sierpiński numbers. In this talk, we give an overview of known results and some new results about these interesting sets of numbers, focusing in particular on the intersection of these sets with other interesting integer sequences. (Received December 18, 2019)

1155-11-87 **Michael Filaseta*** (filaseta@mailbox.sc.edu), Department of Mathematics, 1523 Greene Street, Columbia, SC 29208, and **Jeremiah Southwick** (southwij@math.sc.edu), Department of Mathematics, 1523 Greene Street, Columbia, SC 29208. *Primes that become composite after changing a digit.* Preliminary report.

In 1978, Murray Klamkin asked whether there exist prime numbers such that if any digit is replaced by any other digit, the resulting number is composite. In 1979, several examples were published together with a proof by Paul Erdős that infinitely many such primes exist. Following the terminology of Jackson Hopper and Paul Pollack, we call such primes "digitally delicate." The smallest digitally delicate prime is 294001. In this talk, we discuss some of the history surrounding digitally delicate primes, implications of prior work, and recent work by the speaker with Jeremiah Southwick. (Received December 30, 2019)

1155-11-111 **Preston Wake*** (wakepres@msu.edu). *Tame derivatives and the Eisenstein ideal.* Preliminary report.

As was made famous by Mazur, the mod-5 Galois representation associated to the elliptic curve X_0(11) is reducible. Less famously, but also noted by Mazur, the mod-25 Galois representation is reducible. We'll talk about this kind of extra reducibility phenomenon more generally, for cuspforms of even weight k and prime level. We'll observe that the characters appearing in the reducible representation are related, on one hand, to an algebraic invariant (the 'tame deriviative' of an L-function), and, on the other hand, to an algebraic invariant (the 'tame L-invariant'). This type of 'algebraic=analytic' relation is predicted by a version of the Bloch-Kato conjecture for families of motives formulated by Kato. (Received January 06, 2020)

1155-11-118 **Jordan S Ellenberg**, **Wanlin Li*** (wanlinli@mit.edu) and **Mark Shusterman**. *Nonvanishing of hyperelliptic zeta functions over finite fields.*

Fixing $t \in \mathbb{R}$ and a finite field $\mathbb{F}_q$ of odd characteristic, we give an explicit upper bound on the proportion of genus g hyperelliptic curves over $\mathbb{F}_q$ whose zeta function vanishes at $\frac{1}{2} + it$. Our upper bound is independent of g and tends to 0 as q grows. (Received January 06, 2020)

1155-11-131 **Evan M. O'Dorney***, Fine Hall, Washington Road, Princeton, NJ 08544. *Reflection theorems generalizing the Ohno-Nakagawa identities.*

Scholz's celebrated 1932 reflection principle, relating the 3-torsion in the class groups of $\mathbb{Q}(\sqrt{D})$ and $\mathbb{Q}(\sqrt{-3D})$, can be viewed as an equality among the numbers of cubic fields of different discriminants. In 1997, Y. Ohno discovered (quite by accident) a beautiful reflection identity relating the number of cubic *rings,* or equivalently binary cubic forms, of discriminants D and $-27D$, where D is not necessarily squarefree. In my talk, I will present a new method for proving this identity, based on Poisson summation on adelic cohomology (in the style of Tate's thesis), which generalizes to certain curious results on quadratic and quartic (the latter in progress) rings and forms. (Received January 07, 2020)

1155-11-132 **Jesse Thorner*** (jesse.thorner@gmail.com). *Extremal behavior of class numbers.*

Fix integers $n \geq 3$ and $\ell \geq 2$. Assuming Artin's holomorphy conjecture and the generalized Riemann hypothesis (GRH), Duke proved that there exist number fields K of arbitrarily large discriminant such that $[K : \mathbb{Q}] = n$ and the magnitude of the order of the class group of K is as large as provably possible. Cho proved that Duke's result holds under the strong Artin conjecture without appealing to GRH. We prove that Duke's result holds under a weak form of the Dedekind conjecture. Additionally, we prove that each of these class groups must have nontrivial bounds on their ℓ-torsion that are as strong as GRH predicts. Each number field K that we ensure has large class group with small ℓ-torsion is totally real, and the Galois group of the Galois closure of K (over $\mathbb{Q}$) is isomorphic to the full symmetric group S_n. (Received January 07, 2020)

1155-11-140 **Theresa Anderson***, 150 N. University St., West Lafayette, IN 47907, and **Eyvindur Palsson**. *Sharp bounds for discrete bilinear spherical maximal functions.*

Spherical maximal functions are classical objects of study in harmonic analysis. Discrete variants bring in number theory and satisfy (sometimes strikingly) different bounds than their continuous counterparts. After much investigation, sharp bounds were shown for the continuous bilinear spherical maximal function in recent work of Jeong and Lee. Surprisingly, a modification of their continuous techniques with some number theory leads to the (different) sharp bounds in the discrete setting. We discuss the (short!) proof and mention some related work. This is joint work with Eyvindur Palsson. (Received January 08, 2020)

1155-11-143 **Spencer Hamblen*** (shamblen@mcdaniel.edu), McDaniel College, 2 College Hill, Westminster, MD 21157. *Waring numbers for totally ramified p-adic rings.* Preliminary report.

Given an integer $k \geq 2$, the k-th Waring number, $g(k)$, is the least positive integer such that every positive integer can be expressed as the sum of $g(k)$ k-th powers. For example, Lagrange's Four Squares Theorem – that every positive integer can be written as the sum of 4 squares – implies that $g(2) = 4$. The idea behind Waring numbers can be extended from the positive integers to any ring. We will examine Waring numbers over p-adic integer rings, and note how these numbers change as k varies and in totally ramified extensions. (Received January 08, 2020)

1155-11-147 **Sean Howe*** (sean.howe@utah.edu). *New spaces of p-adic automorphic forms.*

Traditionally, spaces of p-adic automorphic forms are constructed through the p-adic interpolation of algebraic avatars of complex automorphic forms. In this talk, we explain a new perspective, in which p-adic automorphic forms exist on an equal footing with complex automorphic forms, with both occurring as spaces of functions on

Igusa moduli spaces; here the interpolation properties are a secondary consequence of uniformization theorems for Shimura varieties. As an application, we deduce some new results about modular forms. (Received January 08, 2020)

1155-11-164 **Vaidehee Thatte*** (thatte@math.binghamton.edu). *Arbitrary Valuation Rings and Wild Ramification.*

Classical ramification theory deals with extensions of complete discrete valuation rings with perfect residue fields. The study of arbitrary valuation rings with possibly imperfect residue fields and possibly non-discrete valuations of rank ≥ 1 presents many fascinating complications. In particular, the *defect* could be non-trivial when in positive residue characteristic (i.e. we can have a non-trivial extension, such that there is no extension of the residue field or the value group).

In this talk, we will present some new results for Artin-Schreier extensions of arbitrary valuation fields in positive characteristic p. These results relate the "higher ramification ideal" of the extension with the ideal generated by the inverses of Artin-Schreier generators via the norm map. This ideal plays the role of the classical Swan conductor in the arbitrary case. We will also introduce a generalization and further refinement of Kato's refined Swan conductor in this setting. Similar results are true in mixed characteristic $(0, p)$. If time permits, we will briefly discuss the connection between these results and some recent work (joint with K. Kato) on upper ramification groups. (Received January 10, 2020)

1155-11-165 **F Sprung***, 901 S. Palm Walk, Wexler Hall, Tempe, AZ 85287-1804. *Subleading terms of p-adic L-functions.*

The refined Birch and Swinnerton-Dyer conjecture relates the leading term of the Hasse-Weil L-function of an elliptic curve to various algebraic invariants. Recently, Wuthrich has related the subleading (i.e. second non-zero) term to the leading term. Bianchi found a p-adic analogue of this theorem. We report on a chromatic version of this result in terms of integral p-adic L-functions. This is joint work with C. Dion. (Received January 10, 2020)

1155-11-171 **Michael J. Mossinghoff*** (mossinghoff@alumni.stanford.edu), Center for Communications Research, 805 Bunn Dr., Princeton, NJ 08540, and **Timothy S. Trudgian** and **Tomás Oliveira e Silva**. *The Distribution of k-free Numbers.*

For an integer $k \geq 2$, let $Q_k(x)$ denote the number of *k-free integers* in $[1, x]$, that is, integers that have no divisor of the form p^k with p prime. Since the k-free integers have density $1/\zeta(k)$ in $\mathbb{Z}^+$, it is natural to consider the error incurred when estimating $Q_k(x)$ using this density, so let $R_k(x) = Q_k(x) - x/\zeta(k)$. It is well known that $R_k(x) = \Omega(x^{1/2k})$ and $O(x^{1/k})$, and often conjectured that $R_k(x) = O(x^{\frac{1}{2k}+\epsilon})$. While the Riemann hypothesis (and more) would follow if in fact $R_k(x) = O(x^{1/2k})$, we discuss why this bound is often thought to be false, despite experimental investigations that might seem to support it. We also describe some improved lower bounds on $|R_k(x)/x^{1/2k}|$ for various k, deduced by establishing weak linear independence of ordinates of certain zeros of the Riemann zeta function. In addition, we report on some statistics for square-free and cube-free integers obtained by computations up to 10^{18}, including some results on gaps between such numbers. (Received January 11, 2020)

1155-11-175 **Lawrence C Washington*** (lcw@umd.edu), Department of Mathematics, University of Maryland, College Park, MD 20742. *Odd colossally abundant numbers and the Robin-Lagarias criterion for the Riemann Hypothesis.*

In the talk, I'll begin with abundant numbers, superabundant numbers, and colossally abundant numbers, and then discuss applications of this last sequence and variations. Colossally abundant numbers were introduced by Ramanujan in 1915. Robin used them in 1984 to show the Riemann Hypothesis is equivalent to the statement $\sigma(n) < e^\gamma n \log\log n$ for all $n \geq 5041$, where $\sigma(n)$ is the sum of divisors of n and γ is the Euler constant. We consider an odd number version of this result and prove that the Riemann hypothesis is equivalent to the statement $\sigma(n) < \frac{e^\gamma}{2} n \log\log n$ for all odd numbers $n \geq 3^4 \cdot 5^3 \cdot 7^2 \cdot 11 \cdots 67$. Lagarias modified Robin's criterion and proved the Riemann hypothesis is equivalent to $\sigma(n) < H_n + \exp H_n \log H_n$ for all $n \geq 1$, where H_n is the nth harmonic number. We establish an odd number analogue of Lagarias's criterion for the Riemann hypothesis by creating a modified harmonic series $H'_n = 2H_n - H_{2n}$ and demonstrating that the Riemann hypothesis is equivalent to $\sigma(n) \leq \frac{3n}{\log n} + \exp H'_n \log H'_n$ for all odd $n \geq 3$. (This is joint work with Montgomery Blair High School student Ambrose Yang.) (Received January 11, 2020)

1155-11-188 **Antara Mukherjee*** (antara.mukherjee@citadel.edu), 171 Moultrie Street, Department of Mathematical Sciences, The Citadel, Charleston, SC 29409, and **R Florez** and **H Ching**. *Bridging combinatorial triangles with linear algebra.*

Combinatorial triangles like the Pascal triangle maybe used to study properties of various number theoretic sets. For example, the Hosoya (or Fibonacci) triangle has been the source of many geometric, combinatorial, and linear algebraic properties of Fibonacci and by extension generalized Fibonacci numbers. Using the idea of Hosoya triangle, yet another triangle maybe constructed called the determinant Hosoya triangle (See sequence A108038 in www.oeis.org), which has as its entries determinants of specific two-by-two Fibonacci matrices. One of the most interesting features arise from matrices embedded in this triangle mod 2. In fact, certain family of symmetric matrices in this triangle give rise to a family of integral graphs. A graph is integral if its eigenvalues are all integral. The integral graphs are extremely rare and the techniques used to find them are quite complicated. The determinant Hosoya triangle presents a way to uncover infinitely many of these graphs. The techniques used to uncover the integral graphs will be discussed in details in this presentation. (Received January 13, 2020)

1155-11-196 **Alison Beth Miller*** (alimil@umich.edu). *Asymptotics for $\mathrm{Sp}_{2g}(\mathbb{Z})$-orbits on $2g$-ary quadratic forms.*

We use geometry of numbers to give a weighted asymptotic count for the number of orbits of the symmetric square of the standard representation of $\mathrm{Sp}_{2g}(\mathbb{Z})$. That is, we count $2g$-ary quadratic forms up to $\mathrm{Sp}_{2g}(\mathbb{Z})$-equvalence. Our results generalize the classical results of Gauss-Mertens-Siegel on binary quadratic forms, and they can also be generalized further to count integral orbits of the adjoint representation of any semisimple algebraic group. We show that the number of positive definite orbits with invariant height $\leq X$ is asymptotic to $X^{g(2g+1)}$, and give the analogous result weighted by regulators in the indefinite case. (Received January 13, 2020)

1155-11-200 **Keith A Pardue*** (kapardu@nsa.gov). *Ideals and Congruences for the Natural Numbers.* Preliminary report.

In a ring, every congruence – an equivalence relation compatible with addition and multiplication – is given by an ideal, and vice versa. This correspondence breaks in semirings, such as the semiring of natural numbers under ordinary addition and multiplication. I will explain the ideal and congruence theory for the natural numbers, paying special attention to prime ideals and congruences. Aside from prime ideals and congruences coming from the integers, the natural numbers enjoy one extra prime ideal and one extra prime congruence. We will see that congruences for the natural numbers factor uniquely into prime congruences! (Received January 13, 2020)

1155-11-211 **Helen G. Grundman*** (grundman@brynmawr.edu). *An Overview of Sets Arising from Happy Numbers and Their Generalizations.*

In this talk, I will define happy numbers and various types of generalized happy numbers, providing some interesting properties of these sets. I will then define augmented generalized happy functions, and discuss some properties of the corresponding sets of numbers, of the sets of fixed points, and of newly defined sets of integers arising from consideration of these fixed points. My goal is to give an overview of what is known and, in some cases, general ideas behind the proofs. (Received January 13, 2020)

1155-11-224 **Borys Kadets*** (bkadets@mit.edu), MA. *Number of points on abelian varieties over finite fields.*

Given a variety X over a finite field $\mathbb{F}_q$, the Weil conjectures give a good estimate for the size of the set $X(\mathbb{F}_q)$. When X is a particularly nice variety, for example a curve or the Jacobian of a curve, further improvements are possible. I will describe improvements on the Weil bound $(\sqrt{q}-1)^{2g} \leqslant A(\mathbb{F}_q) \leqslant (\sqrt{q}+1)^{2g}$ for a g-dimensional simple abelian variety A. In particular, I will show how to get an exponential lower bound on $A(\mathbb{F}_q)$ for $q < 5$, when the Weil lower bound is trivial. (Received January 14, 2020)

1155-11-225 **Judy A Holdener*** (holdenerj@kenyon.edu). *Perfect Numbers, Descartes Numbers, and Deficient Perfect Numbers.*

In a 1638 letter to Marin Mersenne, René Descartes reported his discovery of the odd "spoof perfect number" $198{,}585{,}576{,}189 = 3^2 7^2 11^2 13^2 (22021) = 3^2 7^2 11^2 13^2 (19^2 61)$, which would be an odd perfect number if only 22021 were prime. A number such as this is now known as a *Descartes number*, and at first glance this perfect spin-off might appear to be a curious fluke. However, a closer look reveals that Descartes did not happen upon this number by chance. In this talk we demonstrate that Descartes' discovery follows naturally from examining the structure of odd perfect numbers. In particular, we prove that every perfect number has a deficient perfect divisor, and one is certain to stumble upon Descartes' well-known number while examining deficient perfect

numbers. In the interest of full disclosure, our approach does rely on the use of a computer, which is something that Descartes did not have! (Received January 14, 2020)

1155-11-227 **Daniel E. Martin*** (daniel.e.martin@colorado.edu). *-15, -19, -20, -23, -24, -31, ...*
Preliminary report.

If O is a ring of integers, the group $SL_n(O)$ is generated by its elementary matrices except when $n = 2$ and O is non-Euclidean, imaginary quadratic. In this special case (when the discriminant belongs to the title sequence), $SL_2(O)$ presentations are computed algorithmically—there is no known explicit and general presentation. We will take a geometric perspective on these groups that produces an explicit generating set and a new algorithm for finding the corresponding relations. (Received January 14, 2020)

1155-11-232 **Charlotte Chan*** (charchan@mit.edu), MIT, Department of Mathematics, Simons Building (Building 2), Room 106, 77 Massachusetts Avenue, Cambridge, MA 02139. *A family of arithmetic automorphic forms on GU(3).*

In the mid-19th century, Kummer proved that the existence of p-torsion of the class group of $Q(\mu_p)$ is equivalent to the p-divisibility of certain Bernoulli numbers. Modern refinements and variations of this connection relate Selmer groups to special values of L-functions, and following an idea of Ribet from the 1970s, one can relate these two objects to congruences of automorphic forms. In this talk, we discuss a construction of an arithmetic automorphic form on GU(3) and its (potential) applications to this picture. This is joint work in progress with Christopher Skinner. (Received January 14, 2020)

1155-11-240 **Daniel J Kriz*** (dkriz@mit.edu), MIT Department of Mathematics, 77 Massachusetts Avenue, Cambridge, MA 02139. *Supersingular main conjectures, Sylvester's conjecture and Goldfeld's conjecture.*

In this talk, I formulate and prove a new Rubin-type Iwasawa main conjecture for imaginary quadratic fields in which p is inert or ramified, as well as a Perrin-Riou type Heegner point main conjecture for certain supersingular CM elliptic curves. These main conjectures and their proofs are related to p-adic L-functions that I have previously constructed, and have applications to two classical problems of arithmetic. First, I prove the 1879 conjecture of Sylvester stating that if $p = 4, 7, 8 \mod 9$, then $x^3 + y^3 = p$ has a solution with x, y rational numbers. Second, combined with previous Selmer distribution results, I show that 100% of squarefree $d = 5, 6, 7 \mod 8$ are congruent numbers, thus establishing Goldfeld's conjecture for the family $y^2 = x^3 - d^2x$, and solving the congruent number problem in 100% of cases. (Received January 14, 2020)

1155-11-242 **Padmavathi Srinivasan*** (padmavathi.srinivasan@uga.edu), **Andrew Obus** and **Kirsten Wickelgren**. *Topological and number-theoretic invariants in algebraic geometry.*

This will be a two part talk. The first half will be the story of conductors and discriminants, which are two closely-related invariants associated to a degenerating family of curves. These invariants show up in many conjectures in number theory, and also have avatars in algebraic geometry and algebraic topology. In this talk, we will introduce a combinatorial refinement of the discriminant, which allows us to prove an inequality between the conductor and the discriminant for families of hyperelliptic curves.

In the second half, we will give a brief introduction to A^1 enumerative geometry. A^1 enumerative geometry is a new field that infuses tools from algebraic topology into algebraic geometry. Using these tools, one can provide "enriched counts" for certain classical enumerative problems in algebraic geometry, valued in quadratic forms. Such counts impose interesting restrictions on the arithmetic of solution sets over non-algebraically closed fields. In this talk, we will explain such an enrichment (joint with Kirsten Wickelgren) for the classic problem of the number of lines meeting four lines in 3-space. (Received January 14, 2020)

1155-11-265 **Nicholas George Triantafillou*** (nicholas.triantafillou@uga.edu), 235 Hill St. Apt #1, Athens, GA 30601. *Restriction of Scalars, Chabauty's Method, and* $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.

For a number field K and a curve C/K, Chabauty's method is a powerful p-adic tool for bounding/enumerating the set $C(K)$. The method typically requires that dimension of the Jacobian J of C is greater than the rank of $J(K)$. Since this condition often fails, especially when $[K : \mathbb{Q}]$ is large, several techniques have been proposed to augment Chabauty's method. For proper curves, Wetherell/Siksek introduced an analogue of Chabauty's method for the restriction of scalars $\mathrm{Res}_{K/\mathbb{Q}}C$ that can succeed when the rank of $J(K)$ is as large as $[K : \mathbb{Q}] \cdot (\dim J - 1)$. Using an analogue of Chabauty's method for restrictions of scalars in the non-proper case, we study the power of this approach together with descent for computing $C = (\mathbb{P}^1 \setminus \{0, 1, \infty\})(\mathcal{O}_{K,S})$. As an application, we show that if 3 splits completely in K and $[K : \mathbb{Q}]$ is prime to 3, then there are no solutions to the unit equation $x + y = 1$ with $x, y \in \mathcal{O}_K^\times$. (Received January 15, 2020)

1155-11-269 **Alexander J Dunn***, 511 West Church Street, Apartment 9, Champaign, IL 61820, and **Alexandru Zaharescu**. *The twisted second moment of modular half integral weight L-functions.*

Given a half-integral weight holomorphic Kohnen newform f on $\Gamma_0(4)$, we prove an asymptotic formula for the second moment of the L-function attached to $f \otimes \chi$, over all primitive χ modulo a prime p. Our result is unconditional, it does not rely on the Ramanujan—Petersson conjecture for the form f. This gives a very sharp Lindelöf on average result for L-series attached to Hecke eigenforms without an Euler product. The Lindelöf hypothesis for such series was originally conjectured by Hoffstein. (Received January 16, 2020)

1155-11-275 **Sharon Anne Garthwaite*** (sag028@bucknell.edu), 1 Dent Dr, Lewisburg, PA 17837-2005, and **Marie Jameson**. *Narrowing the search for beauty: Incongruence properties for generalized Frobenius partitions.*

Let $p(n)$ denote that partition counting function, that is, the number of ways to express n as the sum of non-increasing positive integers. Ramanujan discovered and proved surprising congruence properties for $p(n)$, such as $p(5n+4) \equiv 0 \pmod 5$. Over the last century, many other examples of congruences for the partition function and other counting functions have been proved. In this talk we narrow the search for when such congruences occur – for example, why $p(5n+4)$ and not $p(5n+3)$? – using the theory of automorphic forms. We build on work by C.-S. Radu, S. Ahlgren, B. Kim, N. Andersen, and S. Löbrich that employs the q-expansion principle of P. Deligne and M. Rapoport. We focus on generalized Frobenius partitions to demonstrate this method. This talk aims to be accessible and does not assume familiarity with automorphic form theory. (Received January 16, 2020)

1155-11-290 **Jeffrey C. Lagarias*** (lagarias@umich.edu), 530 Church Street, Ann Arbor, MI 48109-1043. *Arithmetic statistics of some interesting integer sequences.*

This talk will survey various known and new integer sequences related to primes, binomial coefficients and Farey fractions. It will discuss arithmetics statistics of such sequences, which themselves form integer sequences. (Received January 16, 2020)

1155-11-291 **Madeline Locus Dawsey**, **Ken Ono** and **Ian Wagner*** (ian.c.wagner@vanderbilt.edu). *Fields generated by characters of finite groups and a conjecture of John Thompson.*

For a finite group G, let $K(G)$ denote the field generated over $\mathbb{Q}$ by its character values. For alternating groups, G. R. Robinson and J. G. Thompson determined $K(A_n)$ as an explicit multiquadratic field. Confirming a speculation of Thompson, we show that arbitrary suitable multiquadratic fields are similarly generated by the values of A_n-characters restricted to elements whose orders are only divisible by ramified primes. We also extend this result to suitable linear groups and show that cyclotomic fields and subfields are generated by the values of characters restricted to elements with prime power order. (Received January 16, 2020)

1155-11-307 **Soumya Sankar*** (ssankar3@wisc.edu). *Proportion of ordinary curves in families.*

A curve over a finite field of characteristic $p > 0$ is called ordinary if the p torsion of its Jacobian is the largest possible. One can ask, what proportion of curves in a given family are ordinary? We answer this question for some families of curves, namely Artin-Schreier and Superelliptic families. I will talk about these results, as well as some heuristics about what one might expect for random curves. (Received January 17, 2020)

1155-11-308 **Abbey Bourdon*** (bourdoam@wfu.edu), **David Gill**, **Jeremy Rouse** and **Lori D. Watson**. *On Isolated Points of Odd Degree.*

Let C be a curve defined over a number field k, and suppose $C(k)$ is nonempty. We say a closed point $x \in C$ of degree d is isolated if it does not belong to an infinite family of effective divisors of degree d parametrized by the projective line or a positive rank abelian subvariety of the the curve's Jacobian. In this talk we will characterize the non-CM elliptic curves with rational j-invariant which give rise to an isolated point of odd degree on $X_1(N)/\mathbb{Q}$ for some positive integer N. This builds on work of the first author in collaboration with Ejder, Liu, Odumodu, and Viray. (Received January 17, 2020)

1155-11-309 **Chantal David** and **Alexandra Florea***, aflorea@math.columbia.edu, and **Matilde Lalin**. *Non-vanishing of cubic L-functions.*

Chowla conjectured that $L(1/2, \chi)$ never vanishes, for any χ a Dirichlet character. Soundararajan showed that more than 87.5% of the values $L(1/2, \chi_d)$, for χ_d a quadratic character, do not vanish. Much less is known about cubic characters. Baier and Young showed that more than $X^{6/7-\epsilon}$ of $L(1/2, \chi)$ are non-vanishing, for χ a primitive, cubic character of conductor of size up to X. In joint work with C. David and M. Lalin, we show

that a positive proportion of these central L–values are non-vanishing in the function field setting, but the same techniques can be used to prove the analogous result in the number field setting, conditional on the Generalized Riemann Hypothesis. (Received January 17, 2020)

1155-11-312 **Nickolas Andersen*** (nick@math.byu.edu) and **Jesse Thorner**. *Zeros of* GL_2 *L-functions on the critical line.*

We use Levinson's method and the work of Blomer and Harcos on the GL_2 shifted convolution problem to prove a lower bound for the proportion of zeros of the L-function of a holomorphic or Maass cusp form that lie on the critical line. (Received January 17, 2020)

1155-11-355 **Riad Masri** and **Wei-Lun Tsai*** (wltsai@math.tamu.edu), Department of Mathematics, Mailstop 3368, Texas A&M University, College Station, TX 77843-3368. *The distribution of Fourier coefficients of weak Maass forms.*

In this talk, I will discuss recent work in which we show that the normalized Fourier coefficients of weak Maass forms of prime level p become equidistributed on $[-1, 1]$ as p approaches infinity. For integral weight forms, these coefficients are equidistributed with respect to the Sato-Tate measure, while for half-integral weight forms, these coefficients are equidistributed with respect to the arc length measure. The proofs involve blend of geometric and analytic methods. This is joint work with Riad Masri. (Received January 18, 2020)

1155-11-367 **Amy Feaver*** (amy.feaver@kingsu.ca), T5W 4C4, Canada. *Supprime Fibonacci Numbers.*

Richard Guy et. al. was inspired by Conway to study subprime Fibonacci numbers. Motivated by the work of R. Guy, T. Khovanova and J. Salazar, we consider subprime Fibonacci sequences: that is, if we modify the Fibonacci recurrence relation so that the result is (1) left alone if the sum is prime or (2) divided by the largest prime factor if the sum is composite, then what will the resulting sequence be? In this talk, we look at the possible cycles that can result if we start with any two initial terms, and create a sequence using this recurrence relation. These are joint results with K. Hammer, K. Lasola, M. Neguisse and V. Yasa. (Received January 19, 2020)

1155-11-380 **Sarah Arpin**, **Richard Griffon**, **Libby Taylor*** (lt691@stanford.edu) and **Nicholas Triantafillou**. *On the arithmetic of a twisted constant family of superelliptic curves.*

We consider the family of superelliptic curves over a function field with affine equation $y^b = x^a + t^q - t$. We compute the L-functions of the Jacobians of such curves and compute various arithmetic invariants of the Jacobian, such as the regulator, Tate-Shafarevich group, and local Tamagawa numbers. We use this family to produce simple abelian varieties of arbitrary dimension with unbounded algebraic and analytic rank, and prove that these Jacobians satisfy BSD. (Received January 19, 2020)

1155-11-409 **Jinbo Ren***, 141 Cabell Dr, Kerchof Hall, Charlottesville, VA 22903. *Mathematical logic and its applications in number theory.*

A large family of classical problems in number theory such as:

(a) Finding rational solutions of the so-called trigonometric Diophantine equation $F(\cos 2\pi x_i, \sin 2\pi x_i) = 0$, where F is an irreducible multivariate polynomial with rational coefficients;

(b) Determining all $\lambda \in \mathbb{C}$ such that $(2, \sqrt{2(2-\lambda)})$ and $(3, \sqrt{6(3-\lambda)})$ are both torsion points of the elliptic curve $y^2 = x(x-1)(x-\lambda)$;

can be regarded as special cases of the Zilber-Pink conjecture in Diophantine geometry. In this short talk, I will explain how we use tools from mathematical logic to attack this conjecture. In particular, I will present a series partial results toward the Zilber-Pink conjecture, including those proved by Christopher Daw and myself. (Received January 19, 2020)

1155-11-414 **Michael J Griffin*** (mjgriffin@math.byu.edu), 320 TMCB, Brigham Young University, Provo, UT 84602, and **Ken Ono**. *Elliptic curves and lower bounds for class numbers.*

Ideal class pairings map the rational points of rank $r \geq 1$ elliptic curves $E/\mathbb{Q}$ to the ideal class groups $\mathrm{CL}(-D)$ of certain imaginary quadratic fields. These pairings imply that

$$h(-D) \geq (c(E) - \varepsilon)(\log D)^{\frac{r}{2}}$$

for sufficiently large discriminants $-D$ in certain families. These bounds are effective, and they offer improvements to known lower bounds for many discriminants. (Received January 19, 2020)

1155-11-416 **Victor H Moll*** (vhm@tulane.edu), 324 Pine St, New Orleans, LA 70118. *Valuations of integer sequences.* Preliminary report.

Given a prime p and an integer x, the p-adic valuation $\nu_p(x)$ is the highest power of p dividing x. In this talk we will describe a variety of phenomena encountered in the process of trying to describe the valuation of sequences specific sequences. Examples include valuations of integers generated from a polynomial, the valuation of sequences appearing in a combinatorial context (Stirling and Fibonacci Numbers) and sequences produced in the process of evaluating definite integrals. (Received January 19, 2020)

1155-11-417 **Irene Bouw**, **Ozlem Ejder*** (ejder@math.colostate.edu) and **Valentijn Karemaker**. *Arboreal Galois Representations of Dynamical Belyi Maps.*

A dynamical Belyi map is a finite morphism $f : \mathbb{P}^1_{\mathbb{C}} \to \mathbb{P}^1_{\mathbb{C}}$ defined over $\mathbb{C}$ which is branched exactly at the three ordered points $0, 1, \infty$ such that $f(\{0, 1, \infty\}) \subseteq \{0, 1, \infty\}$ which makes all the iterates f^n also Belyi maps. Given a dynamical Belyi map defined over a number field K and a non-preperiodic point $\alpha \in K$, one can construct a tree of preimages of α. This construction leads to the phenomena: one has a tower of fields $K = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$ where $K_n := K\left(\phi^{-n}(\alpha)\right)$. One also has a natural Galois representation on the tree of preimages, the so-called Arboreal Galois representation of the function f. In this talk, we describe the Arboreal Galois representations and the monodromy groups of iterations of a large class of dynamical Belyi maps. Studying these Galois groups has applications in the study of the density of prime divisors of elements of dynamical sequences. If time allows, I will mention some applications as well. (Received January 20, 2020)

1155-11-528 **Frank G. Garvan*** (fgarvan@ufl.edu). *A new approach to Dyson's rank conjectures.* Preliminary report.

In 1944 Dyson defined the rank of a partition as the largest part minus the number of parts, and conjectured that the residue of the rank mod 5 divides the partitions of $5n + 4$ into five equal classes. This gave a combinatorial explanation of Ramanujan's famous partition congruence mod 5. He made an analogous conjecture for the rank mod 7 and the partitions of $7n + 5$. In 1954 Atkin and Swinnerton-Dyer proved Dyson's rank conjectures by constructing several Lambert-series identities basically using the theory of elliptic functions. In 2016 the author gave another proof using the theory of weak harmonic Maass forms. In this talk we describe a new and more elementary approach using Hecke-Rogers series. (Received January 21, 2020)

1155-11-531 **Larry Rolen*** (larry.rolen@vanderbilt.edu), Department of Mathematics, 1420 Stevenson Center, Vanderbilt University, Nashville, TN 37240. *Riemann Hypothesis for Period Polynomials and their Analogues.*

Recently, a number of authors, including El-Guindy and Raji, Conrey, Farmer, and Imamoglu, and Jin, Ma, Ono, and Soundararajan (JMOS) have studied the roots of period polynomials. These polynomials have a distinguished history, and arise naturally from Eichler-Shimura theory and the study of critical L-values of modular forms. JMOS proved the so-called "Riemann Hypothesis for Period Polynomials"(RHPP), which states that all roots of these polynomials, in the case of Hecke eigenforms, line up on the natural line of symmetry, namely, the unit circle. In this talk, I will discuss further analogues, conjectural and proven, for this phenomenon. In particular, describing joint work with Diamantis, I will describe a recent probe into period polynomials for "higher L-derivative" period polynomials which we have described a cohomological theory for and formulated several conjectures on. I will also discuss forthcoming work with Babei and Wagner, where we prove that the RHPP holds for a large class of Hilbert modular forms using a newly formulated notion of period polynomials. Time permitting, we will discuss further experimental data on a much broader phenomenon, which will serve as jumping off points for future investigations. (Received January 21, 2020)

1155-11-559 **Ananth Shankar***, ananths@mit.edu. *Isogenies between abelian varieties in positive and mixed characteristic.*

I will speak about various results pertaining to isogenies between abelian varieties, including the Hecke orbit conjecture and the question of finding abelian varieties not isogenous to Jacobians. This is based on joint work with Tsimerman, Maulik-Tang, and Kisin-Lam-Srinivasan. (Received January 21, 2020)

1155-11-572 **Joel Specter***, 3400 N. Charles Street, Baltimore, MD 21218-3529. *Producing surjective arboreal Galois representations.*

For most degree n polynomials defined over a number field, the Galois action on the roots is as unrestricted as possible - the splitting field is an S_n extension. In 1985, Robert Odoni found the same is true for polynomial iterates. For each integer k, the Galois group of the splitting field of the k-th iterate of a polynomial of degree n is most often the k-fold iterated wreath product of S_n - the largest group possible for an extension obtained via iteration.

While Odoni's result shows that for every integer k the Galois group of the k-th iterate of most polynomials is as large as possible, his methods do not guarantee the existence of any polynomial that has maximal Galois groups for all iterates, simultaneously. He conjectures that such polynomials exist. In this talk, I will discuss the proof of this conjecture. (Received January 21, 2020)

1155-11-574 **Nathan G McNew***, 8000 York Road, Department of Mathematics, Towson, MD 21252. *Counting primitive subsets and other statistics of the divisor graph of* $\{1, 2, ..., n\}$.

A set of integers is primitive if no element of the set divides another. Let $Q(n)$ denote the count of the primitive subsets of the integers $\{1, 2 \ldots, n\}$. Erdős and Cameron conjectured in 1990 that $Q(n) = c^{n+o(n)}$ for some constant c. This conjecture was proven in 2018 by Angelo, however his proof was not effective in that it gave no information as to the value of the constant c. We give a new proof of the fact that $Q(n) = c^{n+o(n)}$ which provides an algorithm to compute the value of this constant c to arbitrary precision, and also gives a much stronger error term than $o(n)$ in the exponent. The method developed can be applied to various related problems which can be stated in terms of the divisor graph. In particular it dramatically improves the error term in the estimates of Mazet and Chadozeau for the path cover number of the divisor graph of the integers from 1 to n. (Received January 21, 2020)

1155-11-584 **Leo Goldmakher*** (lg5@williams.edu) and **Greg Martin**. *Some refinements of Artin's conjecture.*

In 1927, Artin gave a heuristic argument that 2 is a primitive root (mod p) approximately 37% of the time. No one has been able to make his argument rigorous, and even the weaker problem of showing that 2 is a primitive root (mod p) for infinitely many p remains open.

Artin's initial heuristic has been generalized, giving rise to conjectures on the proportion of primes p for which any given integer is a primitive root (mod p); the most general form of this is now known as Artin's conjecture. In this talk I will describe several new conjectures (joint with Greg Martin, UBC) on the proportion of the time a given integer is "almost" a primitive root (mod p). Our conjectures subsume Artin's conjecture, and are borne out in computations. I'll also prove that our conjectures hold on average, and derive some consequences of this. For example, we obtain a new proof that Artin's conjecture holds on average, a result originally due to Goldfeld. (Received January 21, 2020)

12 ► Field theory and polynomials

1155-12-65 **David J Saltman*** (saltman@idaccr.org), 805 Bunn Dr, Princeton, NJ 08540. *Two generalizations of Artin-Schreier polynomials.* Preliminary report.

Inspired by a deep paper by Suresh Venapally, we can define a generic polynomial which gives a short proof of local/global results in mixed characteristic for cyclic Galois extensions. Along the way we discovered a cute generalization of the Artin-Schreier polynomial and Galois action. A second part of the talk will revisit an old characteristic p inverse Galois result and give an improved proof. Renewed interest in these questions was spurred by the questions and research of Catherine Ray in the theory of homotopy groups of spheres. (Received December 13, 2019)

1155-12-631 **David Harbater** and **Daniel Krashen***, Department of Mathematics, Rutgers University, Piscataway, NJ 08854, and **Alena Pirutka**. *Relative local-global principles.*

In various contexts, the Hasse principle can be used to transfer questions of rational points and triviality of Galois cohomology classes from global fields to local fields. Some such results have been extended, for example, in the work of Kato, Bayer-Flukiger-Parimala, Parimala-Preeti, Parimala-Sujatha, to apply to function fields over global fields. In this talk, I will discuss recent joint work with David Harbater and Alena Pirutka in which we examine to what extent local-global principles for one field extend to local-global principles for a function field over this field. We focus particularly on the case where one starts with a semiglobal field (a function fields over discretely valued fields). (Received January 22, 2020)

13 ► Commutative rings and algebras

1155-13-28 **Alexander S Merkurjev*** (merkurev@math.ucla.edu), CA. *On a pairing for algebraic tori.*

There is a pairing between the groups of torsors for a torus and its dual with values in the third Galois cohomology group with coefficients in $Q/Z(2)$. We study the kernel of this pairing. (Received November 09, 2019)

1155-13-68 **Simplice Tchamna*** (simplice.tchamna@gcsu.edu), Milledgeville, GA 31061. *Some proprieties of cancellation ideals of a ring extension.*

We study properties of cancellation ideals of ring extensions. Let $R \subseteq S$ be a ring extension. A nonzero S-regular ideal I of R is called a *(quasi)-cancellation ideal* of the ring extension $R \subseteq S$ if whenever $IB = IC$ for two S-regular (finitely generated) R-submodules B and C of S, then $B = C$. We show that a finitely generated ideal I is a cancellation ideal of the ring extension $R \subseteq S$ if and only if I is S-invertible. (Received December 16, 2019)

1155-13-84 **Saeed Nasseh*** (snasseh@georgiasouthern.edu), Department of Mathematical Sciences, Georgia Southern University, Statesboro, GA 30460. *Deformations and vanishing of homology.* Preliminary report.

A commutative noetherian local ring R is homologically persistent if every finitely generated R-module M with $\operatorname{Tor}^R_{i\gg 0}(M, M) = 0$ has finite projective dimension. In this talk, we discuss some of the recent progress on tracking the homologically persistence property along deformations with some of its applications. This talk is based on an in-progress joint work with L. Avramov, S. Iyengar, and S. Sather-Wagstaff. (Received December 27, 2019)

1155-13-109 **Daniel Duarte** and **Luis Núñez-Betancourt*** (luisnub@cimat.mx), Guanajuato, Mexico. *Nash blowup in prime characteristic.* Preliminary report.

The Nash blowup is a natural modification of algebraic varieties that replace singular points by limits of certain vector spaces associated to the variety at non-singular points. For several decades, It has been studied whether it is possible to resolve singularities of algebraic varieties by iterating Nash blowups. This problem has mostly been treated in characteristic zero due to an example given by Nobile. In this talk, we will discuss a new approach in prime characteristic using differential operators. (Received January 06, 2020)

1155-13-121 **Craig Huneke*** (huneke@virginia.edu). *Expected Resurgence for ideals defining Gorenstein rings.* Preliminary report.

This talk will be a report on work in progress, joint with E. Grifo and V. Mukundan, that proves the following theorem: if I is a reduced ideal in a regular ring containing a field and defining a Gorenstein quotient, then the resurgence of I is strictly less than its codimension. As a consequence, the ideal satisfies the stable Harbourne conjecture. (Received January 07, 2020)

1155-13-127 **F. C. Yhee*** (fyhee@umich.edu). *Ulrich modules do not always exist.*

Ulrich modules were introduced by Bernd Ulrich in 1984 and has since been a very active area of research. The existence of Ulrich modules for complete local domains have powerful applications. For example, existence implies Lech's conjecture: given a flat local map of local rings from R to S, the Hilbert-Samuel multiplicity of S is at least the Hilbert-Samuel multiplicity of R. Until recently, it was unknown if there were any counterexamples to the existence of Ulrich modules for local domains. In this talk, I will introduce the notion of Ulrich modules and discuss implications of their existence. Then I will give a counterexample to the existence of Ulrich modules for complete local domains. (Received January 07, 2020)

1155-13-151 **Monica Ann Lewis*** (malewi@umich.edu). *The local cohomology of a parameter ideal with respect to an arbitrary ideal.*

Let S be a complete intersection presented as R/J for R a regular ring and J a parameter ideal in R. Let I be an ideal containing J. It is well known that the set of associated primes of $H^i_I(S)$ can be infinite, but far less is known about the set of minimal primes. In 2017, Hochster and Núñez-Betancourt showed that if R has prime characteristic $p > 0$, then the finiteness of $\operatorname{Ass} H^i_I(J)$ implies the finiteness of $\operatorname{Min} H^{i-1}_I(S)$, raising the following question: is $\operatorname{Ass} H^i_I(J)$ always finite? We give a positive answer when $i = 2$ but provide a counterexample when $i = 3$. The counterexample crucially requires $\operatorname{Ass} H^2_I(S)$ to be infinite. The following question, to the best of our knowledge, is open: (under suitable hypotheses on R) does the finiteness of $\operatorname{Ass} H^{i-1}_I(S)$ imply the finiteness of $\operatorname{Ass} H^i_I(J)$? When S is a domain, we give a positive answer when $i = 3$. When S is locally factorial, we extend

this to $i = 4$. Finally, if R has prime characteristic $p > 0$ and S is regular, we give a complete answer by showing that Ass $H^i_I(J)$ is finite for all $i \geq 0$. (Received January 09, 2020)

1155-13-169 **Rachel N Diethorn*** (rngettin@syr.edu). *Koszul homology of quotients by edge ideals.*
We show that the Koszul homology algebra of a quotient by the edge ideal of a forest is generated by the lowest linear strand. This provides an answer, for such rings, to a question of Avramov about the Koszul homology algebra of Koszul algebras. We also recover a result of Roth and Van Tuyl on the graded Betti numbers of quotients of edge ideals of trees. (Received January 11, 2020)

1155-13-237 **Craig Huneke** and **Daniel Katz*** (dlk53@ku.edu). *Uniform symbolic topologies and hypersurfaces.*
We show that complete local normal domains satisfy the uniform symbolic topology property if local hypersurface rings have uniform symbolic topologies at regular primes. We also discuss two types of bootstrapping results related to the uniform symbolic topology property. (Received January 14, 2020)

1155-13-284 **Sarasij Maitra** (sm3vg@virginia.edu). *Berger's Conjecture from the viewpoint of an Invariant of the Module of Differentials.*
Let R be a commutative 1 dimensional domain whose integral closure is a DVR. For the first part of this talk, we assume further that R is a complete k-algebra where k is any field. Under this hypothesis, there is a long standing conjecture of R.W. Berger which states that R is regular if and only if the (universally finite) differential module $\Omega_{R/k}$ is torsionfree. We introduce an invariant of $\Omega_{R/k}$ and discuss some lower bounds on it which would give some new cases of this conjecture. In particular, we can generalize the quasihomogeneous case proved by Scheja. Finally, we explore some relations between this invariant and the colength of the conductor. (Received January 16, 2020)

1155-13-314 **John Eagon**, **Ezra Miller** and **Erika Ordog***, erika.ordog@duke.edu. *Minimal resolutions of monomial ideals.*
It has been an open problem since the 1960s to construct closed-form, canonical, combinatorial minimal free resolutions of arbitrary monomial ideals in polynomial rings. This talk explains how to solve the problem, in characteristic 0 and almost all positive characteristics, using sums over lattice paths of combinatorial data from simplicial complexes, one simplicial complex for each lattice point. Any minimal free resolution of any monomial ideal must – either implicitly or explicitly – produce homomorphisms between various homology groups of these simplicial complexes. Therefore an important aspect of the solution is an explicit way to write down canonical homomorphisms between these homology groups without choosing bases. (Received January 17, 2020)

1155-13-315 **Rebecca R.G.*** (rrebhuhn@gmu.edu), **Neil Epstein** and **Janet C. Vassilev**. *Closure-interior duality over complete local rings.* Preliminary report.
The relationship between tight closure and its test ideal can be described in terms of a duality between a closure operation and a corresponding interior operation. We discuss a duality between closure operations, interior operations, and test ideals over complete local rings, and give applications of this technique to trace ideals and the core of an ideal with respect to certain closure operations. (Received January 17, 2020)

1155-13-324 **Uwe Nagel***, University of Kentucky, Department of Mathematics, Lexington, KY 40506. *Stabilization in Sequences of Symmetric Ideals.*
Ideals for studying algebraic-statistical models often have a rich algebraic structure, reflected, for example, in a large symmetry group. It is natural to expect that the properties of the ideals remain somewhat invariant when one preserves the structure of the model, but changes one of its parameters. We discuss an algebraic framework in order to capture such phenomena. Related questions arise in representation theory and topology. A starting point are ascending chains of symmetric ideals in more and more variables. Finiteness results in a suitable category lead to a uniform description of properties of all but finitely many ideals in such a chain.

This is mainly based on joint work with Tim Römer. (Received January 17, 2020)

1155-13-359 **Benjamin Briggs*** (briggs@math.utah.edu) and **Josh Pollitz**. *Asymptotic properties of Tor modules over complete intersections.* Preliminary report.
In 2000 Avramov and Buchweitz introduced a support theory for complete intersections, and used it to establish some remarkable asymptotic properties of Ext modules over such rings.

If M and N are finitely generated modules over a local complete intersection R, then the minimal number of generators of $\operatorname{Ext}^i_R(M, N)$ grows polynomially with i, and the order of this growth is known as the complexity

$\operatorname{cx}\operatorname{Ext}_R(M,N)$. Avramov and Buchweitz showed that $\operatorname{cx}\operatorname{Ext}_R(M,N) = \operatorname{cx}\operatorname{Ext}_R(N,M)$, by connecting complexity to the dimension of their support varieties. They also showed that the $\operatorname{Ext}^i_R(M,N)$ eventually vanish if and only if the $\operatorname{Tor}^R_i(M,N)$ do the same.

Nevertheless, the question of whether $\operatorname{Ext}_R(M,N)$ and $\operatorname{Tor}^R(M,N)$ have the same growth in general has remained open. Dao made partial progress on the problem in 2007. We establish that $\operatorname{cx}\operatorname{Tor}^R(M,N) = \operatorname{cx}\operatorname{Ext}_R(M,N)$ always holds.

One of the steps is interesting in its own right: the set of associated primes of Tor modules (for odd and even degrees separately) eventually stabilise.

Depending on time, I'll talk about some generalisations, and some related developments on the homological behaviour of complete intersections. (Received January 18, 2020)

1155-13-384 **Haydee Lindo***, 179 Park St, Apt 2, Williamstown, MA 01267. *Trace Modules in Commutative Algebra.*

A trace module is the homomorphic image of one module in another. In this talk, I will discuss some historical uses for and recent developments in the theory of trace modules over commutative Noetherian rings. This will include applications of trace modules in understanding free summands, endomorphism rings, and special classes of rings. (Received January 19, 2020)

1155-13-422 **Justin Lyle*** (justin.lyle@ku.edu) and **Jonathan Montaño**. *Extremal growth of Betti numbers and rigidity of (co)homology.*

Let $(R,\mathfrak{m},k)$ be a Cohen-Macaulay local ring and let M,N be finitely generated R-modules. We consider the vanishing of $\operatorname{Tor}^R_i(M,N)$ or $\operatorname{Ext}^i_R(M,N)$ for $i \gg 0$. In particular, we establish obstructions to this vanishing using various numerics of M, N, and R. Our methods allow us to settle the Auslander-Reiten conjecture in several new cases, as well as establish new cases of a dual version of the Auslander-Reiten conjecture introduced by Şega. (Received January 20, 2020)

1155-13-435 **Zhan Jiang*** (zoeng@umich.edu), Ann Arbor, MI 48105. *Size of an ideal.*

Hochster and Huneke defined quasilength for any I-torsion modules, generalizing the notion of length to non-maximal ideals I. Based on quasilength, we develop a new numerical invariant for ideals, called "size". It is invariant up to taking radicals and bounded between the arithmetic rank and height of the ideal. We will present some results in low dimensions and discuss a lot of open questions related to this new notion "size". (Received January 20, 2020)

1155-13-445 **Andrew Kustin**, **Rebecca R.G.** and **Adela Vraciu*** (vraciu@math.sc.edu). *The resolution of $R/(x^N,y^N,z^N,w^N)$ over $R=k[x,y,z,w]/(x^n+y^n+z^n+w^n)$.* Preliminary report.

We study the resolution of $R/(x^N,y^N,z^N,w^N)$ over the hypersurface ring $R=k[x,y,z,w]/(x^n+y^n+z^n+w^n)$, and the related resolution of $P/(x^N,y^N,z^N,w^N):(x^n+y^n+z^n+w^n)$, where $P=k[x,y,z,w]$ and k is a field of characteristic zero. (Received January 20, 2020)

1155-13-446 **Hugh R Geller*** (hgeller@clemson.edu). *DG-Algebra Resolutions for Products of Ideals.* Preliminary report.

Avramov, Iyengar, Nasseh, and Sather-Wagstaff show how the existence of a DG-algebra structure on a minimal resolution can be used to prove a strong Tor-rigidity result. We give an explicit construction of DG algebra resolutions for certain products of ideals. Moreover, we give sufficient conditions for this construction to be minimal and, in other cases, show how the construction can be reduced to a minimal DG algebra resolution. (Received January 20, 2020)

1155-13-479 **Haydee Lindo***, 179 Park St, Apt 2, williamstown, MA. *Endomorphism Invariance and Ring Classifications.*

I will speak on modules which are invariant under endomorphisms of their envelopes. This will include connections to the general theory of trace modules with some preliminary applications to ring classifications and conjectures involving modules with no self-extensions. (Received January 20, 2020)

1155-13-480 **Vivek Mukundan*** (vm6y@virginia.edu), **Jugal Verma** and **Kriti Goel**. *ON THE VANISHING OF THE NORMAL HILBERT COEFFICIENTS OF IDEALS.*

Using vanishing of graded components of local cohomology modules of the Rees algebra of the normal filtration of an ideal, we give bounds on the normal reduction number. This helps to get necessary and sufficient conditions

in Cohen-Macaulay local rings of dimension $d \geq 3$, for the vanishing of the normal Hilbert coefficients $\overline{e}_k(I)$ for $k \leq d$, in terms of the normal reduction number. (Received January 20, 2020)

1155-13-498 **Claudia Miller*** (clamille@syr.edu), **Hamid Rahmati** and **Rebecca R.G.**. *Betti numbers of the Frobenius powers of the maximal ideal over general hypersurfaces in 3 variables.*

We discuss the conjectured curious behavior of the Betti numbers of the Frobenius powers of the maximal ideal in hypersurfaces $R = k[x, y, z]/(f)$, where k is a field of positive characteristic p, as described by Kustin and Ulrich. We prove that if f is chosen to be very general (with k algebraically closed) and of degree the opposite parity to p then high enough Frobenius powers of the maximal ideal have identical high graded Betti numbers up to explicit shifts, and we find some structural features of the resolution and its associated matrix factorization. We also find the Hilbert-Kunz function of R. The proof is based on our constructions of free resolutions of relatively compressed Artinian graded algebras. (Received January 20, 2020)

1155-13-537 **James Edward Gossell*** (jgossel@clemson.edu), 155 Anderson Hwy apt. 823, Clemson, SC 29631. *Gorenstein Injective Modules under Flat Base Change.* Preliminary report.

For $A \to B$ a flat homomorphism of commutative noetherian rings and M an injective A-module, we can calculate the injective dimension of $M \otimes_A B$ as a B-module by calculating the injective dimensions of the fibers $F(\mathfrak{p}) = B_{\mathfrak{p}}/\mathfrak{p}$ for each $\mathfrak{p} \in \operatorname{Ass}(M)$ by the formula $\operatorname{id}_B(M \otimes_A B) = \sup_{\mathfrak{p} \in \operatorname{Ass}(M)} \operatorname{id}_{F(\mathfrak{p})} F(\mathfrak{p})$ (Foxby, 1975). This formula can be used to recover the fact that injective modules remain injective under certain flat base changes, such as a localization. We work towards a generalization of Foxby's Theorem to calculate the Gorenstein injective dimension of the base change of a Gorenstein-injective module. (Received January 21, 2020)

1155-13-561 **David A. Jorgensen**, **Liana M. Şega*** (segal@umkc.edu) and **Peder Thompson**. *Laurent series and asymptotic behavior of Ext over graded complete intersections.* Preliminary report.

Let R be a graded commutative ring generated over a field by finitely many homogeneous elements of positive degree. Work of Avramov, Buchweitz and Sally shows that, under certain assumptions, the Laurent series expansions around 1 of the Hilbert series of two graded R-modules and those of some of their Ext modules are related. When R is a complete intersection, we show that such relations are connected to the asymptotic behavior of the Ext modules. (Received January 21, 2020)

1155-13-610 **Jessica Sidman** (jsidman@mtholyoke.edu), **Will Traves** (traves@usna.edu) and **Ashley Wheeler*** (awheeler@mtholyoke.edu). *Geometric equations for matroid varieties.*

Each point x in $\operatorname{Gr}(r, n)$ corresponds to an $r \times n$ matrix A_x which gives rise to a matroid M_x on its columns. Gel'fand, Goresky, MacPherson, and Serganova showed that the sets $\{y \in \operatorname{Gr}(r, n) | M_y = M_x\}$ form a stratification of $\operatorname{Gr}(r, n)$ with many beautiful properties. However, results of Mnëv and Sturmfels show that these strata can be quite complicated, and in particular may have arbitrary singularities. We study the ideals I_x of matroid varieties, the Zariski closures of these strata. We construct several classes of examples based on theorems from projective geometry and describe how the Grassmann-Cayley algebra may be used to derive non-trivial elements of I_x geometrically when the combinatorics of the matroid is sufficiently rich. (Received January 21, 2020)

14 ► Algebraic geometry

1155-14-13 **Tony Shaska** (shaska@oakland.edu), 546 Mathematics Science Center, Rochester, MI. *Weighted greatest common divisors and weighted heights.*

We introduce the weighted greatest common divisor of a tuple of integers and explore some of its basic properties. Furthermore, for a set of weights $\mathfrak{w} = (q_0, \ldots, q_n)$, we use the concept of the weighted greatest common divisor to define a height $\mathfrak{h}(\mathfrak{p})$ on weighted projective spaces $\mathbb{WP}^n_{\mathfrak{w}}(k)$, over a field k. We prove some of the basic properties of this weighted height, including an analogue of the Northcott's theorem for heights on projective spaces. (Received January 14, 2020)

1155-14-14 **Lubjana Beshaj*** (lubjanabeshaj@gmail.com), 606 Thayer Road, West Point, NY 10966, and **Artur Elezi** and **Tony Shaska**. *Isogenous components of Jacobian surfaces.*

Let $\mathcal{X}$ be a genus 2 curve defined over a field K, $\operatorname{char} K = p \geq 0$, and $\operatorname{Jac}(\mathcal{X}, \iota)$ its Jacobian, where ι is the principal polarization of $\operatorname{Jac}(\mathcal{X})$ attached to $\mathcal{X}$. Assume that $\operatorname{Jac}(\mathcal{X})$ is (n, n)-geometrically reducible with E_1 and E_2 its elliptic components. We prove that there are only finitely many curves $\mathcal{X}$ (up to isomorphism)

defined over K such that E_1 and E_2 are N-isogenous for $n = 2$ and $N = 2, 3, 5, 7$ with $\mathrm{Aut}(\mathrm{Jac}\mathcal{X}) \cong V_4$ or $n = 2$, $N = 3, 5, 7$ with $\mathrm{Aut}(\mathrm{Jac}\mathcal{X}) \cong D_4$. The same holds if $n = 3$ and $N = 5$. Furthermore, we determine the Kummer and the Shioda-Inose surfaces for the above $\mathrm{Jac}\mathcal{X}$ and show how such results in positive characteristic $p > 2$ suggest nice applications in cryptography. (Received January 17, 2020)

1155-14-21 **Garritt Johns** and **Zach Teitler** (zteitler@boisestate.edu), Department of Mathematics, Boise State University, 1910 University Drive, Boise, ID 83725-1555. *An exponentially improved upper bound for the Waring rank of the determinant.* Preliminary report.

The Waring rank of the generic $d \times d$ determinant has Waring rank bounded above by $d \cdot d!$. This exponentially improves previous upper bounds, which were of the form an exponential times the factorial. (Received January 13, 2020)

1155-14-33 **Srimathy Srinivasan*** (srimathy.srinivasan@colorado.edu), Boulder, CO 80301. *Good Reduction of Unitary Groups of Quaternionic Skew-Hermitian Forms.*

Given a field K equipped with a set of discrete valuations V, we develop a general theory to relate ramification of skew-hermitian forms over a quaternion K-algebra Q to ramification of quadratic forms over the function field $K(Q)$ obtained via Morita equivalence. Using this we show that if (K, V) satisfies certain conditions, then the number of K-isomorphism classes of special unitary groups of quaternionic skew-hermitian forms that have good reduction at all valuations in V is finite and bounded by a value that depends on size of a quotient of the Picard group of V and the size of the kernel and cokernel of residue maps in Galois cohomology of K with finite coefficients. As a corollary we prove a conjecture of Chernousov, Rapinchuk, Rapinchuk for groups of this type. (Received November 15, 2019)

1155-14-34 **Charlotte Ure*** (cu9da@virginia.edu) and **Rajesh Kulkarni**. *An Element in the Brauer group of an Elliptic Curve.*

The Clifford algebra associated to a quadratic form is a classical algebraic object with many applications. Its definition can be generalized to forms of higher degree. In the case of a binary cubic form $f(u, v)$, the associated Clifford algebra gives an element in the Brauer group of an elliptic curve – the Jacobian associated to the curve $w^3 - f(u, v)$. In this talk, we will explore this phenomenon in families and show that the Brauer class associated to the binary cubic generic Clifford algebra gives a non trivial Brauer class over its center. This is joint work in progress with Rajesh Kulkarni. (Received November 15, 2019)

1155-14-43 **Hara Charalambous** (hara@math.auth.gr), Department of Mathematics, Aristotle University of Thessaloniki, GR-54124 Thessaloniki, Greece, **Kostas Karagiannis*** (konstantinos.v.karagiannis@gmail.com), Department of Mathematics, Aristotle University of Thessaloniki, GR-54124 Thessaloniki, Greece, and **Aristides Kontogeorgis** (kontogar@math.uoa.gr), Department of Mathematics, National and Kapodistrian University of Athen, GR-15784 Athens, Greece. *The Relative Canonical Ideal of the Kummer - Artin Schreier - Witt family of curves.*

We study the *relative canonical embedding* of the flat family of curves obtained from lifting an Artin-Schreier curve to a Kummer curve. Combining elements of Gröbner theory with deformation-theoretic arguments we will give an explicit set of generators for the *relative canonical ideal*, obtaining in the process both a relative version of Petri's Theorem and a combinatorial criterion on whether a set of polynomials generates the canonical ideal. Finally, if time permits, we will present how our results are related to the Galois module structure of n-differentials, the tangent space to the deformation functor of curves with automorphisms and a characteristic p version of Green's conjecture. The talk will be based on ongoing joint work with Hara Charalambous and Aristides Kontogeorgis. (Received November 24, 2019)

1155-14-62 **Andrew Kobin*** (ak5ah@virginia.edu). *Stacky curves in characteristic p.*

As stacks continue to become an essential part of a modern algebraic geometer's toolbox, researchers look to their local structure as a guide to their nature. Over the complex numbers, this local structure is called a complex orbifold, or 'orbit space of a manifold' under a cyclic group action. In this talk, I will survey the classification of stacky curves in characteristic 0 and introduce a new construction, called an Artin–Schreier root stack, which allows for similar classification results in characteristic p. (Received December 10, 2019)

1155-14-85 **Atticus Christensen*** (atticusc@mit.edu). *Specialization of Neron-Severi groups in characteristic p.*

We will present two results on the specialization of Neron–Severi groups in characteristic $p > 0$ for arbitrary varieties then will explain how these results specialize to results about the specialization of endormorphism algebras of abelian varieties. The first result bounds the exponent of p which kills the cokernel of specialization of Neron–Severi groups coming from any family of algebraic varieties. The second result is analogous to a result by André and Maulik–Poonen in mixed characteristic and establishes that given a smooth projective family of algebraic varietiesover an algebraically closed field K of characteristic $p > 0$ not isomorphic to $\bar{F}_p$, there exists a K point outside of the Noether-Lefschetz locus. (Received December 28, 2019)

1155-14-97 **Andrew Obus*** (andrewobus@gmail.com) and **Tony Shaska** (shaska@oakland.edu). *Superelliptic curves: many automorphisms vs complex multiplication.*

We consider the class of superelliptic curves over $\mathbb{C}$ with affine equation given by $y^n = f(x)$, where $f(x)$ is a separable polynomial. Suppose that such a curve has *many automorphisms*, i.e., it has more automorphisms than any nearby curve in the moduli space of curves. Does the Jacobian of such a curve have complex multiplication? In general the answer is no. We give a complete classification of those superelliptic curves as above with many automorphisms where the answer is yes. (Received January 02, 2020)

1155-14-99 **Jesse Leo Kass***, Department of Mathematics, University of South Carolina, University of South Carolina, Columbia, SC 29208, and **Candace Bethea**. *How to count curves using equivariant homotopy theory?*

A major theme in modern algebraic geometry is that it is often productive to analyze a counting problem ("how many singular curves?") by identifying the desired count as an invariant in algebraic topology and then studying it using tools from topology. One direction for current research is apply equivariant tools to study counting problems in a way that records interesting information about symmetries.

In this talk, I will explain Candace Bethea's dissertation work on this topic. Bethea describes a count of the singular elements of a general pencil of plane conics with group action that is valued in the Burnside-ring. She does not directly use methods from algebraic topology, but explaining some unexpected results she found provides a challenge for topologists. (Received January 03, 2020)

1155-14-124 **Rebecca Goldin** and **Changlong Zhong*** (czhong@albany.edu), 1400 washington ave, albany, NY 12222. *Structure constants of equivariant oriented cohomology of flag varieties.*

Equivariant oriented cohomology theory generalizes singular cohomology/Chow group and K-theory. It can be characterized by the formal group law associated to the first Chern class. In this talk I will talk about a formula for the structure constants of equivariant oriented cohomology of flag varieties. We will use the method of formal affine Demazure algebra. It generalizes the result of Goldin-Knutson for Schubert classes, and that of Su for cohomology stable basis. This is joint work with R. Goldin. (Received January 07, 2020)

1155-14-150 **Daoji Huang***, 310 Malott Hall, Cornell University, Ithaca, NY 14853. *Bruhat Atlases in Coordinates.*

A stratified smooth variety admits a Bruhat atlas if it can be covered by open charts isomorphic to (opposite) Bruhat cells in some Kac-Moody flag manifold via stratified isomorphisms. In this talk, I will present a Bruhat atlas on the wonderful compactification of a symmetric space where the modelling flag manifold is finite type, which allows explicit computation in coordinates using Bott-Samelson maps and other familiar techniques in studying geometry of flag varieties. (Received January 09, 2020)

1155-14-160 **Lara B. Anderson*** (lara.anderson@vt.edu), Department of Physics, 850 West Campus Dr, Blacksburg, VA 24061. *New aspects of heterotic geometry.* Preliminary report.

In this talk I will describe new aspects of the total geometry of heterotic string compactifications – consisting of holomorphic, poly-stable vector bundles over Calabi-Yau manifolds. In particular, we will consider the case that the bundle/manifold geometry arises from a (0,2) Gauged Linear Sigma Model (GLSM). We demonstrate that Calabi-Yau manifolds arising as complete intersections in toric varieties and bundles defined via the monad construction over them have a simple combined toric description. We use this description to try to shed new light on (0,2) heterotic dualities. (Received January 10, 2020)

1155-14-168 **Swastik Kopparty**, **Guy Moshkovitz** and **Jeroen Zuiddam*** (jzuiddam@ias.edu), 1 Einstein Drive, Princeton, NJ 08540. *Geometric rank of tensors and the subrank of matrix multiplication.* Preliminary report.

Motivated by problems in algebraic complexity theory, extremal combinatorics and quantum information theory, and inspired by the analytic rank of tensors, we introduce and study a new tensor parameter called the geometric rank. We use the geometric rank to answer a question that was left open by Strassen in 1987. (Received January 11, 2020)

1155-14-172 **Hamza Jaffali** (hamza.jaffali@utbm.fr), 90010 Belfort, France, and **Luke Oeding*** (oeding@auburn.edu), Department of Mathematics and Statistics, Auburn, AL 36849. *Learning Algebraic Models of Quantum Entanglement.*

We give a thorough overview of supervised learning and network design for learning membership on algebraic varieties via deep neural networks. We show how artificial neural networks can be trained to predict the entanglement type for quantum states. We give examples for detecting degenerate states, as well as border rank classification for up to 5 binary qubits and 3 qutrits (ternary qubits). (Received January 11, 2020)

1155-14-206 **Dean Bisogno*** (dbisogno@rams.colostate.edu). *Abhyankar's Inertia Conjecture for Some Sporadic Groups.*

Fix a wildly ramified Galois cover of curves in positive characteristic. Abhyankar's Inertia Conjecture indicates which inertia groups potentially occur at ramified points. We verify Abhyankar's Inertia Conjecture for a collection of sporadic groups in various characteristics. For a larger set of sporadic groups, all but finitely many possible ramification invariants are shown to occur. (Received January 13, 2020)

1155-14-295 **Yun Shi***, 5 Cowperthwaite Street, Apt 512, Cambridge, MA 02138. *On motivic Donaldson-Thomas theory on the local projective plane.*

Donaldson-Thomas (DT) theory is an enumerative theory which counts ideal sheaves of 1-dimensional subschemes on a Calabi-Yau 3-fold. Motivic Donaldson-Thomas theory is a categorification of the DT theory. This categorification contains more refined information of the moduli space. In this talk, I will give a brief introduction to motivic DT theory. I will also discuss some results on this theory for moduli spaces of sheaves on the local projective plane. (Received January 16, 2020)

1155-14-296 **Paolo Aluffi**, **Leonardo Mihalcea*** (lmihalce@math.vt.edu), **Jorg Schurmann** and **Changjian Su**. *Positivity of characteristic classes.*

Motivated by the theory of stable envelopes of Okounkov and collaborators, Cotangent Schubert Calculus is a generalization of Schubert Calculus where the ordinary Schubert classes are replaced by characteristic classes of Schubert cells. A study of these classes leads to many positivity and log concavity properties, most of them conjectural. I will survey some of these statements, and explain the ideas behind the proofs of the known results. Our main examples are the positivity of the expansion of Chern-Schwartz-MacPherson classes into Schubert classes, and the sign-alternating property of the Schubert expansion of the Segre-MacPherson classes. Joint work with P. Aluffi, J. Schurmann, C. Su. (Received January 17, 2020)

1155-14-297 **Jean-Louis Colliot-Thélène*** (jlct@math.u-psud.fr), Orsay, France. *On the integral Tate conjecture for 1-cycles on the product of a curve and a surface over a finite field.* Preliminary report.

Let X be the product of a smooth projective curve C and a smooth projective surface S over a finite field $\mathbf{F}$. Assume the Chow group of zero-cycles on S is just $\mathbf{Z}$ over any algebraically closed field extension of $\mathbf{F}$ (example : Enriques surface). We give a simple condition on C and S which ensures that the integral ℓ-adic Tate conjecture holds for 1-cycles on X. An equivalent formulation is a vanishing result for unramified cohomology of degree 3. This generalizes a result of A. Pirutka (2016).

It is a joint work with Federico Scavia (UBC, Vancouver). (Received January 17, 2020)

1155-14-325 **Han-Bom Moon*** (hmoon8@fordham.edu) and **Sang-Bum Yoo**. *Finite generation of the algebra of conformal blocks via birational geometry.*

A Conformal block is a representation-theoretic object constructed as an example of two-dimensional chiral conformal field theory (WZW model). For each simple Lie algebra $\mathfrak{g}$ and a stable curve, the set of conformal blocks naturally form an algebra structure. We prove, by employing birational geometric techniques, finite generation of the algebra of conformal blocks for $\mathfrak{g} = \mathfrak{sl}_r$. This is joint work with Sang-Bum Yoo. (Received January 17, 2020)

1155-14-440 **Mark R Bly*** (mbly@coastal.edu), PO Box 261954, Conway, SC 29528. *An Automorphic Classification of Real Cubic Curves.* Preliminary report.

The action of ring automorphisms of the polynomial ring in two variables over the real numbers on real plane curves is considered. The orbits containing degree-three polynomials have been computed, with one representative per orbit being selected. In addition, this talk will include preliminary results related to a classification of real cubic curves with respect to affine coordinate ring isomorphism that include a two-dimensional moduli space of polynomials with equal j-invariant whose associated real normal domains are each pairwise non-isomorphic. (Received January 20, 2020)

1155-14-442 **Sabrina Pauli*** (sabrinp@math.uio.no). *Lines on a Quintic Threefold.*

A smooth cubic surface contains 27 complex lines. There are two types of real lines on a smooth cubic surface called hyperbolic and elliptic, and the number of real hyperbolic lines minus the number of real elliptic lines on a real smooth cubic surface is equal to 3. Motivated by Morel's Brouwer degree which takes values in the Grothendieck-Witt ring $GW(k)$, Kass and Wickelgren define the type of a line on a smooth cubic surface defined over a field k as an element in $GW(k)$ and show that the sum over the types of the lines on a cubic surface equals $15\langle 1\rangle + 12\langle -1\rangle \in GW(k)$ which recovers both the complex (take the rank) and real count (take the signature).

In my talk I will define the type of a line on a quintic threefold in $GW(k)$. There are infinitely many lines on the Fermat Quintic threefold $X = \{F = X_0^5 + X_1^5 + X_2^5 + X_3^5 + X_4^5 = 0\} \subset \mathbb{P}^4$. However, a generic deformation X_t of X contains only finitely many lines. I will compute the types of the finitely many deformed lines and show that they do not depend on the chosen deformation. Summing up all those types one gets $1445\langle 1\rangle + 1430\langle -1\rangle \in GW(k)$ which again recovers the complex and real count of lines on quintic threefolds. (Received January 20, 2020)

1155-14-475 **Rares Rasdeaconu*** (rares.rasdeaconu@vanderbilt.edu), Department of Mathematics, Vanderbilt University, 1326 Stevenson Center, Nashville, TN 37240, and **Viatcheslav Kharlamov**. *Moduli spaces of stable rank one torsion free sheaves on real curves.*

The counting of rational curves representing primitive homology classes on complex or real K3 surfaces is governed by the Yau- Zaslow formula and its real analog, respectively. An approach to extend such formulae to the non-primitive case has been suggested by Jun Li and requires the computation of the Euler characteristic of moduli spaces of stable, rank one sheaves on curves which are possibly reducible and non-reduced. Recent developments in this direction will be presented. (Received January 20, 2020)

1155-14-491 **James Phillips*** (jp100@wellesley.edu). *Deformations of wildly ramified one-point covers.*

Abhyankar's Conjecture gives explicit conditions for the existence of a G-cover of a given curve in characteristic p with a given number of branch points. It does not, however, give a description of the possible ramification groups above the branch points. In this talk, I will generalize techniques used by Pries to study the conductors of one-point covers of $\mathbb{P}^1$ when G has a cyclic Sylow p-subgroup. These techniques involve the study of deformations of covers over equicharacteristic discretely valued rings. (Received January 20, 2020)

1155-14-506 **Andrew Kobin*** (ak5ah@virginia.edu) and **Libby Taylor** (lt691@stanford.edu). *$\mathbb{A}^1$-Local Degree via Stacks.*

Using a root stack construction, we extend results of Kass and Wickelgren to define an Euler class for a non-orientable vector bundle on a smooth scheme, valued in the Grothendieck–Witt group $GW(k)$ of the ground field. This allows one to apply Kass–Wickelgren's techniques for arithmetic enrichment of enumerative geometry problems to a larger class of problems. For example, our construction produces a new arithmetic count of the number of lines meeting 6 general hyperplanes in $\mathbb{P}^4$. (Received January 20, 2020)

1155-14-515 **Dave Anderson**, **Linda Chen*** (lchen@swarthmore.edu) and **Nicola Tarasca**. *Degeneracy loci, Brill-Noether theory, and tableau formulas.*

Degeneracy loci of morphisms between vector bundles have been used in a wide range of situations, including classical approaches to the Brill-Noether theory of special divisors on curves. I will describe these connections and recent developments in Schubert calculus and Brill-Noether theory, including K-theoretic formulas which yield formulas for K-classes of Brill-Noether loci in terms of set-valued tableaux as well as formulas for Chern-Schwarz-MacPherson classes. (Received January 20, 2020)

1155-14-556 **C Damiolini** (chiarad@princeton.edu), Princeton University, Department of Mathematics, Fine Hall, Washington Road, Princeton, NJ 08544-1000, **Angela Gibney*** (angela.gibney@gmail.com), Department of Mathematics, Rutgers University, Hill Center for the Mathematical Sciences, 110 Frelinghuysen Rd., Piscataway, NJ 08854-8019, and **Nicola Tarasca** (tarascan@vcu.edu), Virginia Commonwealth University, Department of Mathematics & Applied Mathemati, 1015 Floyd Avenue — Box 842014, Richmond, VA 23284-2014. *Vertex algebras of CohFT- type.*

Representations of vertex algebras of CohFT-type can be used to construct vector bundles of coinvariants and conformal blocks on moduli spaces of stable pointed curves. As I'll explain in the talk, the name comes from the fact that such bundles define semisimple cohomological field theories. I'll present motivation for why one may be interested in such bundles and their classes, as well as a few examples. (Received January 21, 2020)

1155-14-566 **Aaron Pixton*** (pixton@umich.edu). *Boundary splitting and socle evaluations.*

Cohomological field theories are families of cohomology classes on the moduli space of stable curves that satisfy various compatibility axioms. I will discuss some families of cohomology classes that only satisfy a single boundary splitting axiom but otherwise have many nice features (such as being of pure cohomological degree). I will conclude by giving a conjecture relating one of these families to a socle (top-degree) evaluation in the tautological ring of the moduli space of bridgeless curves. (Received January 21, 2020)

1155-14-573 **Judson Everett Beougher*** (judson.beougher@westpoint.edu). *Algebraic and Geometric Definitions of Addition for Superelliptic Curves.*

Shaska and Kopeliovich in a recent paper define geometric addition for the Jacobians of Superelliptic Curves. We seek to use their definition to derive an algebraic equivalent while also exploring various examples of this addition. We use the definitions of addition for elliptic and hyperelliptic curves to inform our search and help to illuminate any potential patterns. (Received January 21, 2020)

1155-14-583 **Jean-Louis Colliot-Thélène**, **David Harbater**, **Julia Hartmann*** (hartmann@math.upenn.edu), **Daniel Krashen**, **R. Parimala** and **V. Suresh**. *Local-global principles for tori over arithmetic curves.*

We study local-global principles for tori over arithmetic curves, which are curves over complete discretely valued fields. For tori defined over the underlying discrete valuation ring, the obstruction to such local-global principles can be described using techniques from patching. We give a vanishing criterion for this obstruction in terms of the reduction graph of a model of the curve. (Received January 21, 2020)

1155-14-627 **Julianna Tymoczko*** (jtymoczko@smith.edu), Department of Mathematics, Smith College, Northampton, MA 01063. *Springer fibers and webs.*

This talk describes the combinatorics and representation theory related to a family of subvarieties of the flag variety called *Springer fibers*. Recall that the flag variety is the collection of nested vector subspaces $V_1 \subseteq V_2 \subseteq \cdots V_{n-1} \subseteq \mathbb{C}^n$ of a fixed complex vector space. The Springer fiber of a linear operator X consists of those flags that are fixed by X in the sense that $XV_i \subseteq V_i$ for all i. Springer fibers are well-known as 1) the fundamental example of a geometric representation (their cohomology carries an action of the symmetric group S_n) and 2) a family of varieties whose geometry is encoded by subtle combinatorics (they inherit and refine some of the permutation statistics of Schubert varieties in the flag variety). In addition, many geometric characteristics of Springer fibers can be encoded by *webs*, a kind of planar graph that encodes quantum representations of $\mathfrak{sl}_t$. We give a quick overview of the geometric structure of Springer fibers and then describe their connection to webs in more detail, including both recent results and open questions. (Received January 22, 2020)

15 ► Linear and multilinear algebra; matrix theory

1155-15-19 **Alessandro De Paris*** (deparis@unina.it), Dipartimento di Matematica e Applicazioni, "Renato Caccioppoli", via Cintia, Monte S. Angelo, I-80126 Napoli, NA, Italy. *A short talk on lengthy symmetric tensor decompositions.*

To date, a systematic knowledge on CP decompositions and tensor rank is still under construction, and several related problems of basic nature are still to be solved. For instance, the maximum rank for tensors of given size is unknown, apart from special cases. The same problem for the symmetric rank is unsolved as well, in spite of the fact that it is equivalent to a classical Waring problem on polynomials, which had already been considered in a geometric framework nearly a century ago.

We discuss some techniques that have led to the best known upper bound on the Waring rank of ternary forms of given degree. To go further and detect the actual maximum, it is reasonable to investigate the structure of the variety of all decompositions of length equal to the expected answer: we illustrate some progress in that direction. (Received October 14, 2019)

1155-15-116 **Shmuel Friedland*** (friedlan@uic.edy), Department of Mathematics, Statistics and Com, University of Illinois at Chicago, 851 S. Morgan Street , 322 SEO, Chicago, IL 60607-7045. *Tensors, Entanglement, Computations and Complexity.*

Tensors are most suitable to describe multipartite quantum systems. In this talk we will address a number of functions on tensors as: rank, spectral and nuclear norm, nuclear rank – for general, symmetric, skew symmetric and bihermitian tensors. We will emphasize the connections to some concepts of quantum physics as entanglement and inseparability. We will discuss the numerical and complexity aspects of computing these functions. (Received January 06, 2020)

1155-15-142 **Jarosław Buczyński*** (jabu@mimuw.edu.pl), Institute of Mathematics, Polish Academy of Sciences, Śniadeckich 8, 02-632 Warsaw, Poland. *On Strassen's rank additivity for small three-way tensors.*

We will describe the problem of the additivity of the tensor rank. That is, for two independent tensors we study if the rank of their direct sum is equal to the sum of their individual ranks. A positive answer to this problem was previously known as Strassen's conjecture until recent counterexamples were proposed by Shitov. The latter are not very explicit, and they are only known to exist asymptotically for very large tensor spaces. In this talk I will report on a proof that for some small three-way tensors the additivity holds. For instance, if the rank of one of the tensors is at most 6, then the additivity holds. Or, if one of the tensors lives in $\mathbb{C}^k \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$ for any k, then the additivity also holds. More generally, if one of the tensors is concise and its rank is at most 2 more than the dimension of one of the linear spaces, then additivity holds.

Based on a joint work with Elisa Postinghel and Filip Rupniewski. (Received January 08, 2020)

1155-15-193 **Kevin Pratt***, 5700 Ellsworth Ave, Apt. A4, Pittsburgh, PA 15232. *Waring Rank, Parameterized and Exact Algorithms.*

Given nonnegative integers n and d, where $n \gg d$, what is the minimum number r such that there exist linear forms $\ell_1, \ldots, \ell_r \in \mathbb{C}[x_1, \ldots, x_n]$ so that $\ell_1^d + \cdots + \ell_r^d$ is supported exactly on the set of all degree-d multilinear monomials in $x_1, \ldots, x_n$? We show that this and related questions have intimate connections to the areas of parameterized and exact algorithms in computer science, generalizing earlier methods and providing a concrete approach to obtain faster approximate counting and deterministic decision algorithms. As an application application we give faster algorithm for approximately counting subgraphs of bounded treewidth, improving on earlier work of Alon et al. (Received January 13, 2020)

1155-15-336 **Jonathan Shi*** (jonathan.shi@unibocconi.it), **Sam Hopkins** and **Tselil Schramm**. *Fast generically robust tensor rank decomposition through convex relaxation.*

We overview a family of techniques for finding the symmetric rank decompositions of possibly overcomplete tensors. The techniques are united by a common view of a convex relaxation of the original problem, although in some cases they are implemented using only basic linear algebraic operations, in nearly the time to compute a singular value decomposition of a matrix reshaping of the input.

A common feature of these algorithms is that they are provably robust to adversarial perturbations of bounded spectral norm. In particular, generically decomposing overcomplete order-4 tensors is done faster than previously known possible even without said robustness guarantee. (Received January 22, 2020)

1155-15-340 **Alana Huszar*** (huszara@umich.edu) and **Harm Derksen** (hderksen@umich.edu). *Non-commutative Rank and Stability of Quiver Representations.* Preliminary report.

We start with $n \times n$ matrices $A_1, \ldots, A_m$, and consider the matrix $A = x_1 A_1 + \ldots + x_m A_m$. We would like to determine the rank of A, where $x_1, ..., x_m$ are viewed as non-commuting generators of a free skew-field. This is the non-commutative rank of A. Finding this rank is related to the stability of a representation of a Kronecker quiver. We generalize this non-commutative rank to any acyclic quiver representation. (Received January 18, 2020)

1155-15-382 **Harm Derksen*** (hderksen@umich.edu), **Zhi Jiang** (zhijiang@umich.edu), **Jonathan Gryak** (gryakj@umich.edu) and **Kayvan Najarian** (kayvan@umich.edu). *The G-stable rank for tensors and the cap set problem.*

We introduce the G-stable rank for tensors, a new notion related to stability in geometric invariant theory. We will compare it to other rank notions, such as the tensor rank, border rank and the slice rank. There are applications in data science such as dimension reduction and noise removal. Applied to the cap set problem in additive number theory, one can slightly improve upon the Ellenberg-Gijswijt upper bound for the cardinality of cap sets. (Received January 19, 2020)

1155-15-387 **Shmuel Friedland**, **Zehua Lai** and **Lek-Heng Lim*** (lekheng@uchicago.edu). *Tensor norms and approximability.* Preliminary report.

It is common knowledge that tensor ranks play a key role in the *complexity* of various computational problems; we will see that tensor norms play an analogous role in the *approximability*. We show that for some important NP-hard problems, the obstruction to polynomial-time approximability is often given by an injective norm of an appropriate 3-tensor. Examples include the Goemans–Williamson constant in the approximation of the MAXCUT of a graph and the Grothendieck constant in the approximation of the $(\infty, 1)$-norm of a matrix. Among other things, the Unique Games Conjecture can be decided if we know the exact value of a certain norm of a specific 3-tensor. (Received January 19, 2020)

1155-15-470 **Mohsen Aliabadi*** (maliab2@uic.edu), Chicago, IL , and **Shmuel Friedland**. *On the complexity of finding tensor ranks.*

It is a classical result that the rank of a matrix is polynomially computable, using the Gauss elimination. However for tensors, for finite fields the complexity of computation of the tensor rank is NP-complete. For general fields it is NP-hard. It is well understood that matrices are closely related to linear transformations, while tensors are closely related to polynomial maps. The purpose of this talk is to estimate the number of steps to find if a given linear system over integers or rational numbers is solvable in precise arithmetic. We provide a good answer to the problem of "finding an upper bound on complexity to find a rank of a tensor of integers or rationals". (Received January 20, 2020)

1155-15-489 **Christina Pospisil*** (pospisil.christina@gmx.de). *Generalization Theory of Linear Algebra II.*

The talk continues the first part of the project Generalization Theory of Linear Algera I (first part was presented at JMM 2019) and presents inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be applications to physics and other natural sciences be explored. (Received January 20, 2020)

1155-15-512 **Austin Conner***, connerad@math.tamu.edu, and **Fulvio Gesmundo**, **Joseph M. Landsberg** and **Emanuele Ventura**. *Kronecker powers of tensors and the complexity of matrix multiplication.*

Determining the complexity of matrix multiplication has been a central problem ever since Strassen showed in 1968 that the usual algorithm for matrix multiplication is not optimal. Efforts to obtain better algorithms have essentially stalled since the work of Coppersmith and Winograd in 1989. I will present both positive and negative results with regard to ways forward on the question of upper bounds on the complexity of matrix multiplication. (Received January 20, 2020)

16 ► Associative rings and algebras

1155-16-26 **Sergey V. Tikhonov*** (tikhonovsv@bsu.by), Belarusian State University, Department of Higher Algebra, Nezavisimosti Ave., 4, 220030 Minsk, Belarus. *On genus of division algebras.*

The genus $\mathbf{gen}(D)$ of a finite-dimensional central division algebra D over a field F is defined as the collection of classes $[D'] \in Br(F)$, where D' is a central division F-algebra having the same maximal subfields as D.

In [2], we show that the fact that quaternion division algebras D and D' have the same maximal subfields does not imply that the matrix algebras $M_l(D)$ and $M_l(D')$ have the same maximal subfields for $l > 1$. Moreover, for any $n > 1$, we construct a field L such that there are two quaternion division L-algebras D and D' and a central division L-algebra C of degree and exponent n such that $\mathbf{gen}(D) = \mathbf{gen}(D')$, but the algebras $D \otimes C$ and $D' \otimes C$ do not have the same maximal subfields.

In particular, this gives negative answers to questions formulated in [1].

References

1. V.I. Chernousov, A.S. Rapinchuk, I.A. Rapinchuk, *The finiteness of the genus of a finite-dimensional division algebra, and some generalizations*, arXiv:1802.00299.

2. S.V. Tikhonov, On genus of division algebras, arXiv:1904.03933. (Received November 09, 2019)

1155-16-107 **Joshua Sussan*** (joshuasussan@gmail.com), CUNY Medgar Evers, 1638 Bedford Avenue, Brooklyn, NY 11225. *p-DG structures in higher representation theory.*

We will show an approach to categorification of representations of quantum sl(2) at a prime root of unity. (Received January 06, 2020)

1155-16-369 **Mee Seong Im*** (meeseongim@gmail.com), Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996. *On the category of finite-dimensional representations of periplectic Lie superalgebras.*

Translation functors on the category of finite-dimensional representations of periplectic Lie superalgebras $\mathfrak{p}(n)$ using the endomorphism of the endofunctor $-\otimes V$ will be defined. I will then define the actions of translation functors on thick and thin Kac modules, which categorically lift the Temperley–Lieb relations associated to the infinite symmetric group. Next, I will define the notion of weight diagrams for dominant weights and explain the associated combinatorics of the actions of translation functors on standard and costandard objects in terms of the diagrams. This involves moving a shaded ball left or right, depending on the translation functor and the weight. I will also explain the duality for simple modules in terms of weight diagrams.

This is joint with Martina Balagovic, Zajj Daugherty, Inna Entova-Aizenbud, Iva Halacheva, Johanna Hennig, Gail Letzter, Emily Norton, Vera Serganova, and Catharina Stroppel. (Received January 19, 2020)

1155-16-403 **Ethan Kowalenko** and **Carl Mautner*** (carlm@ucr.edu). *Category $\mathcal{O}$ for oriented matroids.* Preliminary report.

Braden-Licata-Proudfoot-Webster defined a class of finite-dimensional graded algebras associated to linear programs, meaning real hyperplane arrangements with a choice of linear functional. The representation theory of these algebras behaves much like integral blocks of Bernstein-Gelfand-Gelfand's category $\mathcal{O}$. For example the algebras are quasi-hereditary and Koszul.

I will review the set-up and then discuss joint work with Ethan Kowalenko, in which we extend the definition of Braden-Licata-Proudfoot-Webster to the more general combinatorial setting of oriented matroid programs. Our results are part of a larger program of the speaker with Jens Eberhardt to categorify matroidal Schur algebras, introduced in joint work with Tom Braden. (Received January 19, 2020)

1155-16-477 **Peter Samuelson*** (psamuels@ucr.edu). *The BMW skein algebra of the torus.*

The Homflypt skein relations are linear relations satisfied by the R-matrix associated to $\mathfrak{gl}_n$ for any n, and they are the relations describing the kernel of the map from the group algebra of the braid group to type A Hecke algebra. There are similar-looking skein relations in type BCD (the Kauffman skein relations) which describe the Birman-Murakami-Wenzl algebra. In this talk we describe the "skein algebra of the torus" in both the type A and type BCD settings. (This talk is based on joint works with Morton and Morton-Pokorny.) (Received January 20, 2020)

1155-16-492 **Stephen Doty*** (doty@math.luc.edu), Mathematics and Statistics, 1032 W. Sheridan Rd, Chicago, IL 60660. *Integral Schur-Weyl duality for partition algebras.*

Partition algebras are finite-dimensional "diagram" algebras with a combinatorial basis given by set partitions, with multiplication defined by stacking diagrams. They were (independently) discovered in the 1990s by Vaughan Jones and Paul P. Martin in connection with the Potts model in physics. Partition algebras can be regarded as generic centralizers of the natural action of the Weyl group W of GL(V) on the tensor algebra of V, where V is a finite-dimensional complex vector space. By construction, partition algebras satisfy a Schur-Weyl duality with the group algebra of W, at least over a field of characteristic zero, in which case the group algebra is semisimple. I will try to explain why Schur-Weyl duality still holds, even when the underlying field is replaced by an arbitrary ring (of any characteristic). In particular, it holds over the integers. This vastly extends a result of Peter M. Gibson (1980) on generalised doubly-stochastic matrices. The result is joint work with Chris Bowman and Stuart Martin. (Received January 20, 2020)

1155-16-500 **Huanchen Bao** and **Thomas Sale*** (tws2mb@virginia.edu). *Quantum symmetric pairs at roots of* 1.

A quantum symmetric pair $(\mathbf{U}, \mathbf{U}^{\mathbf{i}})$ consists of a quantum group $\mathbf{U}$ and a coideal subalgebra $\mathbf{U}^{\mathbf{i}}$ (called ıquantum group), which quantizes a symmetric pair $(\mathfrak{g}, \mathfrak{g}^{\theta})$, where θ is an involution of $\mathfrak{g}$. When specializing the quantum parameter q to a root of unity, we establish the compatibility of $\mathbf{U}^{\mathbf{i}}$ and the Frobenius-Lusztig homomorphism on (the integral form of) $\mathbf{U}$, and formulate a small ıquantum group. (Received January 20, 2020)

1155-16-543 **James F Clark*** (jfclark@albany.edu), 1400 Washington Ave, ES 151, Albany, NY 12222. *Further Remarks of the Unrolled Quantum Group of* $\mathfrak{sl}_2$. Preliminary report.

In this talk we will be expanding on the representation theory of the unrolled quantum group $\overline{U}_q^H(\mathfrak{sl}_2)$ where q is any primitive root of unity. We will talk about its center, how the category of finite-dimensional $\overline{U}_q^H(\mathfrak{sl}_2)$-modules forms a ribbon category, the simple and projective modules forms a subcategory, logarithmic modules, and results that have applications to topological quantum field theory (TQFT). (Received January 21, 2020)

1155-16-580 **Robert W Muth*** (rmuth@washjeff.edu), Washington & Jefferson College, 60 S Lincoln St, Washington, PA 15301. *Superalgebra deformations of Schur algebras and related categories.* Preliminary report.

Given a superalgebra A, one may construct certain A-deformations of Schur algebras and related diagrammatic categories. I will discuss the connection between these objects and some structural results regarding bases and double centralizer properties. These deformations are motivated by some conjectures in the block theory of finite groups and Schur algebras, as a number of important objects arise in this way; for instance, RoCK blocks of Hecke algebras were shown by Evseev and Kleshchev to be Morita equivalent to zigzag-deformations of Schur algebras. This talk describes joint work with Alexander Kleshchev, as well as work with Nicholas Davidson, Jonathan Kujawa, and Jieru Zhu. (Received January 21, 2020)

1155-16-587 **Angelo Bianchi** and **Evan Wilson***, ewilson@jtcc.edu. *Groebner-Shirshov bases for Weyl modules of the current algebra of* $sl(2)$.

The theory of the theory of Gröbner-Shirshov bases have many useful properties. Recently, in joint work with Angelo Bianchi, they have been used to construct bases for graded local Weyl modules for the current algebras of $\mathfrak{sl}_2$ in positive characteristic. In the characteristic zero setting, the method provides a different construction of Chari-Pressley, Kus-Littelmann, and Chari-Venkatesh bases for local Weyl modules. In this talk, we discuss these results and Grobner basis methods in general. (Received January 21, 2020)

1155-16-593 **Zajj B Daugherty*** (zdaugherty@gc.cuny.edu), The City College of New York, Department of Mathematics, 160 Convent Ave, New York, NY 10031. *Centralizer algebras for orthogonal and symplectic Lie algebras and their generalizations.* Preliminary report.

Brauer algebras, BMW algebras, and their generalizations arise as tensor power centralizer algebras for orthogonal and symplectic Lie algebras and quantum groups. In this talk, we will explore some of the newer generalizations, building upon combinatorial and algebraic structure of these families of diagram algebras. (Received January 21, 2020)

1155-16-595 **Zajj B Daugherty*** (zdaugherty@gc.cuny.edu), The City College of New York, Department of Mathematics, 160 Convent Ave, New York, NY 10031. *Two-boundary diagram algebras.* Preliminary report.

Two-boundary Temperley-Lieb algebras began in mathematical physics related to solving certain lattice models, but have since inspired the study of other two-boundary diagram algebras that arise as natural generalizations from a Schur-Weyl duality perspective. In this talk, we will explore some interesting examples of these algebras and their connections to multiplicity-free tensor products of classical Lie algebra and Quantum group modules and the combinatorics thereof. (Received January 21, 2020)

17 ► Nonassociative rings and algebras

1155-17-72 **Christopher M Drupieski*** (c.drupieski@depaul.edu) and **Jonathan R Kujawa** (kujawa@math.ou.edu). *Support varieties and modules of finite projective dimension for modular Lie superalgebras.*

Let $\mathfrak{g}$ be a finite-dimensional Lie superalgebra over an algebraically closed field k of characteristic $p \geq 3$. Using the cohomology ring of $\mathfrak{g}$, one can define for each finite-dimensional $\mathfrak{g}$-supermodule M a corresponding affine algebraic variety $X_{\mathfrak{g}}(M)$, called the cohomological support variety of M. In this talk I'll discuss recent work with Jonathan

Kujawa in which we give an explicit 'rank variety' description for $X_{\mathfrak{g}}(M)$ as a subset of $X_{\mathfrak{g}} = \{x \in \mathfrak{g}_{\overline{1}} : [x, x] = 0\}$, the odd nullcone of $\mathfrak{g}$. A key tool in our approach is the Clifford filtration on $\mathfrak{g}$, which helps us to compare $X_{\mathfrak{g}}(M)$ to the support variety of a certain related unipotent finite supergroup scheme. As corollaries of our main result, I'll discuss a characterization of when a finite-dimensional $\mathfrak{g}$-supermodule is of finite projective dimension, and a positive characteristic analogue of a theorem of Bøgvad concerning a criterion for when the enveloping algebra of $\mathfrak{g}$ is of finite global dimension. (Received December 19, 2019)

1155-17-135 **Kailash C. Misra***, 2311 Stinson Dr., Raleigh, NC 27695-8205. $D_6^{(1)}$- *Geometric Crystal and its ultra-discretization.*

Let $\mathfrak{g}$ be an affine Lie algebra with index set $I = \{0, 1, 2, \cdots, n\}$ and $\mathfrak{g}^L$ be its Langlands dual. It is conjectured that for each Dynkin node $i \in I \setminus \{0\}$ the affine Lie algebra $\mathfrak{g}$ has a positive geometric crystal whose ultra-discretization is isomorphic to the limit of certain coherent family of perfect crystals for $\mathfrak{g}^L$. In this talk we construct a positive geometric crystal $\mathcal{V}(D_6^{(1)})$ in the level zero fundamental spin $D_6^{(1)}$- module $W(\varpi_6)$. Then we define explicit 0-action on the level ℓ known $D_6^{(1)}$- perfect crystal $B^{6,l}$ and show that $\{B^{6,l}\}_{l\geq 1}$ is a coherent family of perfect crystals with limit $B^{6,\infty}$. Finally we show that the ultra-discretization of $\mathcal{V}(D_6^{(1)})$ is isomorphic to $B^{6,\infty}$ as crystals which proves the conjecture in this case. This is joint work with Suchada Pongprasert. (Received January 08, 2020)

1155-17-167 **Jonathan Brundan**, **Alistair Savage*** (alistair.savage@uottawa.ca) and **Ben Webster**. *Heisenberg and Kac-Moody categorification.*

The Heisenberg category and Kac-Moody 2-categories categorify the Heisenberg algebra and Kac-Moody algebras, respectively. We show how module categories over the Heisenberg category may be viewed as 2-representations of Kac-Moody 2-categories, and vice versa. This gives a way to construct Kac-Moody actions in many representation-theoretic settings. (Received January 11, 2020)

1155-17-271 **Brian Boe**, Dept. of Math, University of Georgia, Athens, GA 30602, and **Jonathan Kujawa***, Dept. of Math, University of Oklahoma, Norman, OK 73069. *Complexity and support varieties for Lie superalgebras of type P.*

Using a mixture of combinatorial, algebraic, and geometric arguments we compute the complexity and support varieties for the (thick) Kac supermodules for the Lie superalgebras of type P. (Received January 16, 2020)

1155-17-272 **Jonathan Kujawa***, Dept. of Math, University of Oklahoma, Norman, OK 73069. *Clifford supercategories.*

A Clifford supercategory is one in which simple objects can admit nontrivial involutive automorphisms. We will discuss several examples of these categories and how they naturally arise in representation theory (e.g. for the type Q Lie superalgebras). (Received January 16, 2020)

1155-17-282 **Alex J Feingold*** (alex@math.binghamton.edu), Dept of Math Sciences, Whitney Hall, Binghamton University, 4400 Vestal Parkway East, Binghamton, NY 13902-6000. *Hyperbolic Kac-Moody algebras and groups.* Preliminary report.

Several approaches to the study of hyperbolic Kac-Moody (KM) algebras, g, and their groups, G, will be discussed. (1) New insights of Carbone-Feingold-Freyn come from the embedding of the twin building for G in the union of Tits cones in the real compact form of g. (2) New work with Kleinschmidt-Nicolai concerns imaginary root groups in hyperbolic KM groups. (3) Decomposition of g with respect to a subalgebra k provides information about the structure of g using the representation theory of k. Feingold-Frenkel (1983) studied the rank 3 hyperbolic g with Weyl group $PGL(2, \mathbb{Z})$ and affine k of type $A_1^{(1)}$. That g contains all rank 2 hyperbolic KM algebras with symmetric Cartan matrices (Feingold-Nicolai 2004). The case when k is the rank 2 "Fibonacci" hyperbolic was studied in Penta's 2016 dissertation. In addition to k itself and infinitely many irreducible highest and lowest weight k-modules, Penta found four other integrable k-modules in the decomposition of g which are neither highest nor lowest weight modules. Weight multiplicity formulas for these "non-standard" modules are not yet known, but a recursive algorithm for computing a basis provided numerical data suggestive of a Racah-Speiser type multiplicity formula. (Received January 16, 2020)

1155-17-320 **Chun-Ju Lai***, Boyd Graduate Studies Research Center, University of Georgia, Athens, GA 30602, and **Daniel K Nakano** and **Ziqing Xiang**. *q-Schur duality of classical type, coordinate coalgebras, and categorification.*

In this talk we investigate the q-Schur algebras corresponding to the Hecke algebras of classical type, which are closely related to quantum symmetric pair coideal subalgebras for the type A quantum groups. We present a coordinate coalgebra construction that allows us to realize these q-Schur algebras as the duals of the d-th

graded components of certain graded coalgebras. Such a concrete realization leads to an isomorphism theorem between q-Schur algebras of classical type. Furthermore, we obtain a 1-faithful quasi-hereditary cover of the Hecke algebra which identifies the category O for certain rational Cherednik algebra. A potential categorification of our coordinate coalgebras will be discussed. (Received January 17, 2020)

1155-17-392 **Matt Szczesny*** (szczesny@math.bu.edu), 111 Cummington Mall, Boston, MA 02215, and **Brian Williams** and **Jackson Walters**. *Holomorphic fibrations, factorization algebras, and toroidal vertex algebras.*

Let X be a complex manifold, and $\pi : E \to X$ a locally trivial holomorphic fibration with fiber F. We associate to this data a holomorphic prefactorization algebra $\mathcal{F}_{\pi,\mathfrak{g}}$ in the formalism of Costello-Gwilliam. When $X = \mathbb{C}$, $\mathfrak{g}$ is simple, and F has the structure of a smooth affine variety, we extract from $\mathcal{F}_{\pi,\mathfrak{g}}$ a vertex algebra which is a vacuum module for the universal central extension of $\mathfrak{g} \otimes H^0(F, \mathcal{O}F)[z, z^{-1}]$. When F is an n-dimensional torus, we get a vertex algebra naturally associated to an $(n+1)$-toroidal algebra. (Received January 19, 2020)

1155-17-399 **Neil Saunders** and **Arik Wilbert*** (arik.wilbert@uga.edu). *From exotic Springer fibers to highest weight categories.*

Parabolic versions of category $\mathcal{O}$ provide important examples of highest weight categories in the sense of Cline, Parshall and Scott. For a parabolic subalgebra $\mathfrak{p} \subseteq \mathfrak{gl}_n$ with Levi subalgebra $\mathfrak{gl}_k \times \mathfrak{gl}_{n-k}$, it is known that two-block Springer fibers can be used to geometrically construct a convolution algebra whose category of finite-dimensional modules is equivalent to the parabolic category $\mathcal{O}^{\mathfrak{p}}$. It turns out that the diagrammatic tools that describe the geometry of the Springer fibers can also be used to combinatorially describe the highest weight category structure of $\mathcal{O}^{\mathfrak{p}}$. In this talk, we discuss the geometry of a certain family of exotic Springer fibers from an explicit, diagrammatic point of view. For the exotic Springer fibers, it remains an open problem if the corresponding convolution algebras can be used to construct new examples of interesting highest weight categories. (Received January 19, 2020)

1155-17-484 **Jude L. Quintero*** (jlquintero@randolphcollege.edu), 2500 Riveermont Ave., Box 612, Lynchburg, VA 24503. *Finite Orbifolds of the Rank 2 Heisenberg Vertex Algebra $\mathcal{H}(2)$.*

It is well known that the full automorphism group of the Heisenberg Vertex Algebra $\mathcal{H}(n)$ is the orthogonal group, and the finite subgroups of the two-dimensional orthogonal group are either cyclic or dihedral. In this talk we calculate a minimal strong generating set of the rank 2 Heisenberg Vertex Algebra $\mathcal{H}(2)$ with respect to all finite subgroups of its own automorphism group. Our proof relies on constructing families of decoupling relations of all weights. (Received January 20, 2020)

1155-17-505 **Michael Reeks*** (mar055@bucknell.edu) and **Alistair Savage**. *Trace of the Frobenius Heisenberg category.*

The Heisenberg algebra plays an important role in many areas of mathematics and physics. Khovanov constructed a categorical analogue of this algebra which emphasizes its connections to representation theory and combinatorics. While the Grothendieck group of this category is the Heisenberg algebra, applying another decategorification functor called the trace to the Heisenberg category yields a richer structure: a W-algebra, an infinite-dimensional Lie algebra related to conformal field theory. In this talk, we will describe recent work extending this latter result to a generalized version of the Heisenberg category associated to an arbitrary graded Frobenius algebra F. We describe a basis of the trace of this Frobenius Heisenberg category and discuss progress towards computing its algebra structure. (Received January 20, 2020)

1155-17-519 **Christopher K Chung*** (cc2wn@virginia.edu), 110 Kerchof Hall, 141 Cabell Drive, Charlottesville, VA 22904. *Serre presentation and canonical basis for $\imath$-quantum covering groups.* Preliminary report.

In 2016, Bao and Wang developed a theory of canonical basis for quantum symmetric pairs $(\mathbf{U}, \mathbf{U}^\imath)$, generalizing the canonical basis of Lusztig-Kashiwara for quantum groups. Soon after, Chen, Lu and Wang established a presentation for the $\imath$-quantum group $\mathbf{U}^\imath$ of the quasi-split quantum symmetric pair of Kac-Moody type, explicitly in terms of Serre relations. In a series of papers, Clark, Hill and Wang developed a theory of quantum covering algebras, which generalizes both the Lusztig quantum group and quantum supergroups. In this setting, we will describe how the results on the Serre presentation and canonical basis generalize to the quantum covering setting. (Received January 21, 2020)

1155-17-538 **James Lepowsky*** (lepowsky@math.rutgers.edu). *Logarithmic tensor category theory – a very brief survey.* Preliminary report.

I will very briefly sketch some highlights of logarithmic tensor category theory for vertex operator algebras (Y.-Z. Huang, J. Lepowsky and L. Zhang), and I'll mention some recent results. (Received January 21, 2020)

1155-17-553 **Nick Davidson*** (njd@reed.edu), **Jonathan Kujawa** and **Robert Muth**. *Webs for the Type P Lie Superalgebra.* Preliminary report.

Webs are combinatorially defined diagrams originally introduced by Kuperberg to study homomorphisms between representations of rank 2 Lie algebras (e.g $\mathfrak{sl}_3$). Since then, they have been used to study representations of other Lie (super)algebras. I will discuss webs associated to the type P Lie superalgebra $\mathfrak{p}_n$. (Received January 21, 2020)

1155-17-589 **Juan Jose Villarreal*** (jjvillarrealm@vcu.edu), 1709 Hanover avenue, Richmond, VA 23220, and **Marco Aldi**. *Non local generalizations of vertex algebras involving logarithmic fields.* Preliminary report.

In this work we want to describe some non local generalizations of vertex algebras. We consider a set of axioms on a vector space which describe the algebraic properties of logarithmic fields with algebraic relations given by logarithmic singularities. Additionally, we consider other generalizations involving the lambda bracket formalisms. (Received January 21, 2020)

18 ► Category theory; homological algebra

1155-18-54 **Theo Johnson-Freyd***, Perimeter Institute for Theoretical Physics, 31 Caroline St N, Waterloo, ON N2L2Y5, Canada, and **Davide Gaiotto**, Perimeter Institute for Theoretical Physics, 31 Caroline St N, Waterloo, ON N2L2Y5, Canada. *A secondary invariant of the elliptic genus.*

The elliptic genus is a powerful deformation invariant of 2d SQFTs: if it is nonzero, then it protects the SQFT from admitting a deformation to one with spontaneous supersymmetry breaking. This talk will describe a "secondary" invariant that goes beyond the elliptic genus, protecting SQFTs with vanishing elliptic genus. Our invariant relates directly to mock modularity. It furthermore supports the hypothesis that the space of minimally supersymmetric 2d SQFTs provides a geometric model for the spectrum of topological modular forms. (Received December 03, 2019)

1155-18-166 **Jonathan Brundan**, **Alistair Savage*** (alistair.savage@uottawa.ca) and **Ben Webster**. *Foundations of the Frobenius Heisenberg category.*

Frobenius Heisenberg categories are monoidal categories depending on a Frobenius algebra. We describe bases for the morphism spaces of these categories, proving several open conjectures. We also prove, in a more general setting than previous work, that these categories categorify lattice Heisenberg algebras. (Received January 11, 2020)

1155-18-339 **Zongzhu Lin*** (zlin@ksu.edu), Department of Mathematics, Kansas State University, Cardwell 138, Manhattan, KS 66506. *Realizing Hall algebras as Grothendieck rings.* Preliminary report.

Alperin studied the fusion properties of subgroups of a Sylow p-subgroup of finite group in 1967, which lead to later development of fusion systems. Fusion system is regarded as a local group theory. In the attempted to characterize fusion systems in terms of homotopy theory, the concept of localities and partial groups were introduced. Around Kriz and May also introduced partial algebras in an attempt to formalize the intersection multiplication of motives. It terms out that both concepts are pretty much the same in categorical setting, where multiplication needed to be defined as they should. To do so one has to define the concepts of domains in both settings. The concept of partial categories can be defined in a similar fashion. Partial category can be used as framework to bridge several different mathematical structures. In this talk I will focus on realizing Hall algebras of an abelian category as the Grothendieck group of a partial category attached to the abelian category. (Received January 18, 2020)

1155-18-343 **Ian Coley*** (iacoley@math.rutgers.edu), Hill Center for the Mathematics Sciences, 110 Frelinghuysen Road, Piscataway, NJ 08854. *Stabilization of homotopy theories via derivators.*

Derivators, introduced variously by Franke, Heller, and Grothendieck in the 1990s, are a means by which abstract homotopy theories can be studied in a general fashion. Under mild hypotheses, any derivator can be stabilized by passing through unstable spectrum objects, and the stabilized derivator satisfies a nice 2-categorical universal property. The proof scheme is similar to but more general than work of Lurie in Higher Algebra (§1.4). This talk aims to present an outline of the method and introduce some key diagrammatic techniques in the realm of derivators. (Received January 18, 2020)

1155-18-464 **Michael Horst*** (horst.59@osu.edu). *A Categorification of Group Cohomology.* Preliminary report.

We present a categorification of group cohomology by way of Picard categories, which are to be thought of as categories that behave like Abelian groups. After reviewing background information on Picard categories, we will present a framework for a cohomology of Picard categories, including categorical modules and derived 2-functors. We will close by using free Picard categories to form projective resolutions and compute various specific examples. (Received January 20, 2020)

1155-18-511 **Benjamin Cooper***, University of Iowa, Department of Mathematics, 14 MacLean Hall, Iowa City, IA 52242. *Hall algebras of surfaces.*

I'll discuss ongoing work on computing the Hall algebras of Fukaya categories of surfaces. I'll talk about presentations for these algebras and discuss the role they play in low dimensional topology. Some of this work is joint with Peter Samuelson and Fabian Haiden. (Received January 20, 2020)

1155-18-591 **Peter Webb*** (webb@math.umn.edu), School of Mathematics, University of Minnesota, Minneapolis, MN 55455. *Biset functors for categories.* Preliminary report.

In the context of group theory, biset functors have been useful in various ways: in computing the values of group cohomology, and providing fundamental constructions such as the (torsion free part of) the Dade group. Biset functors can also be done for categories in general, not just groups, with similar goals in mind. We describe the basics of this theory, paying attention to the role and structure of the Burnside ring functor for categories. We then show that the cohomology of a category is a biset functor, provided that a condition is imposed on the bisets. In the case of groups, it is that the bisets are free on one side, and we show how to extend this condition to categories. The approach provides a solution to the problem of defining restriction and corestriction on the homology of categories. Prior approaches to this usually require induction and restriction functors to be adjoint on both sides, and we avoid this by using the construction by Bouc and Keller of a map on Hochschild homology associated to a bimodule, and the realization by Xu of category cohomology as a summand of Hochschild cohomology. (Received January 21, 2020)

1155-18-612 **Nicolle Gonzalez*** (nicolle@math.ucla.edu) and **Matt Hogancamp**. *A skein-theoreric formulation of the $A_{q,t}$ algebra.* Preliminary report.

The $A_{q,t}$ algebra was originally introduced by Carlsson and Mellit in their proof to the shuffle conjecture. In this talk I will discuss ongoing work with Matt Hogancamp entailing a skein-theoretic formulation of the $A_{q,t}$ algebra specialized at $t = q^{-1}$ that recovers the original algebraic formulation and provides the prime structure for categorification. (Received January 21, 2020)

19 ► K-theory

1155-19-82 **Sherry Gong*** (sgongli@math.ucla.edu). *The Novikov conjecture, the group of volume preserving diffeomorphisms, and Hilbert-Hadamard spaces.*

We discuss the Novikov conjecture and explain that a version of it holds for manifolds whose fundamental groups admit proper and isometric actions on certain non-positively curved infinite dimensional spaces called Hilbert-Hadamard spaces. This talk is based on joint work with Jianchao Wu and Guoliang Yu (arxiv1811.02086). (Received December 22, 2019)

1155-19-259 **Rufus Willett*** (rufus@math.hawaii.edu), 2565 McCarthy Mall, Keller 401A, Honolulu, HI 968222234. *Approximate ideals and K-theory.*

If a C*-algebra A admits ideals I and J so that A=I+J, then a Mayer-Vietoris sequence connecting the K-theory of A to the K-theories of I, J, and the intersection of I and J can be constructed. This allows one to deduce

information about the K-theory of A from information about the K-theories of (one hopes) simpler objects, just as in classical algebraic topology.

I'll discuss decompositions into approximate ideals, and how they can be used to compute K-theory in a similar way. Such approximate ideals exist quite often even when the ambient C*-algebra is simple, and are connected to the theory of nuclear dimension; I'll try to sketch some of this. (Received January 15, 2020)

1155-19-560 **Jesse Wolfson*** (wolfson@uci.edu), University of California, Irvine. *Derived motivic integration.* Preliminary report.

The Grothendieck ring of varieties is a universal source for "cut and paste invariants" of varieties such as Euler characteristics, Hodge-Deligne polynomials, zeta functions and more. Recent work of Zakharevich and Campbell allows us to realize the Grothendieck ring as π_0 of a K-theory spectrum, and we are still in the early stages of understanding what information this higher K-theory encodes. Motivic integration was introduced by Kontsevich in order to prove that the Hodge numbers of birational smooth Calabi-Yaus are equal, and it remains one of the most powerful tools to analyze the class of a variety in the Grothendieck ring. In this talk, I will discuss joint work-in-progress to lift motivic integration to a map of spectra with the goal of constructing new invariants of birational automorphisms of smooth Calabi-Yaus. This is joint work with Oliver Braunling, Michael Groechenig and Inna Zakharevich. (Received January 21, 2020)

20 ► Group theory and generalizations

1155-20-20 **Ted Chinburg*** (ted@math.upenn.edu), 209 S. 33rd Street, Philadelphia, PA 19104, and **Frauke Bleher** and **Alex Lubotzky**. *Linear representations of profinite groups and of the Grothendieck-Teichmuller group.* Preliminary report.

This paper is about a version for profinite free groups of results of Lubotzky and Grunewald that construct linear representations of finite index subgroups of free groups. We will apply this to construct representations of the Grothendieck-Teichmuller group. (Received October 16, 2019)

1155-20-24 **Amrita Acharyya**, **Jon M Corson** and **Bikash C Das*** (bikash.das@ung.edu). *Varieties of Profinite Graphs.*

Given a category C of finite graphs, we say that a graph Γ is a $Pro-C$ graph, if it is the inverse limit of an inverse system of $C-$graphs and surjective $C-$maps. Initially, we investigate some $Pro-C$ graphs satisfying certain nice properties. Then, we specialize to a specific category that arises naturally when considering end point compactifications (viewed as profinite graphs in a certain way) of abstract connected graphs. In the analogous situation for groups, classes of finite groups H that satisfy various closure properties are of interest and a $Pro-H$ group is defined to be an inverse limit of an inverse system of H-groups and surjective homomorphisms. (Received November 04, 2019)

1155-20-25 **Vladimir Chernousov*** (vladimir@ualberta.ca), Math. and Statistical Sciences Departmenent, University of Alberta, Edmonton, Alberta T6G 2G1, Canada. *R-triviality of groups of type* F_4 *arising from the first Tits construction.*

Any group of type F_4 is obtained as the automorphism group of an Albert algebra. We prove that such a group is R-trivial whenever the Albert algebra is obtained from the first Tits construction. Our proof uses cohomological techniques and the corresponding result on the structure group of such Albert algebras. Joint work with S. Alsaody and A. Pianzola. (Received November 04, 2019)

1155-20-31 **Boris Kunyavskii*** (kunyav@gmail.com). *Word maps on linear algebraic groups over special fields.*

Given a group word $w = w(x_1, \dots, x_d)$ on d letters and a group G, one can consider the evaluation map $w\colon G^d \to G$ induced by w. We will discuss the case where $G = \mathcal{G}(K)$ is the group of K-points of a linear algebraic K-group $\mathcal{G}$. Our focus is on the special ground fields K (complex, real, p-adic, or close to such). We will present a survey of various generalizations of a seminal theorem of Borel, which states that the map w is dominant for any non-identity word w whenever G is semisimple, and discuss several open problems. (Received November 14, 2019)

1155-20-39 **M. Hunnell** (hunnellm@wssu.edu), **J. Hutchens*** (hutchensjd@wssu.edu) and **N. Schwartz** (gnat79@gmail.com). *Involutions of the Orthogonal Group over a field of characteristic 2.* Preliminary report.

Symmetric k-varieties are defined to be the homogeneous spaces $G(k)/H(k)$ where G is a reductive algebraic group defined over a field k and H is the fixed point group of an automorphism of order 2. Here we begin classifying these automorphisms of order 2 up to conjugacy where $G(k) = \mathrm{O}(V, \mathrm{q}, k)$, the orthogonal group for a vector space V and quadratic form q over a field of characteristic 2. There is a full classification in the case that (V, q) is non-defective, and further results with certain constraints on $\mathrm{rad}(V)$. We also classify the fixed point groups for the conjugacy classes of these involutions. (Received November 22, 2019)

1155-20-144 **Daniel K Nakano*** (nakano@math.uga.edu), Department of Mathematics, University of Georgia, Athens, GA 30602. *A Tribute to the Work of Cline, Parshall and Scott (CPS).*

For over 35 years, the triple author collaboration of Edward Cline, Brian Parshall and Leonard Scott was a leading force in the area of the representation theory of algebraic groups. The Cline-Parshall-Scott collaboration (also known as "CPS") was the longest active triple authored collaboration in the history of mathematics. Many of the CPS results such as Mackey Imprimitivity and their foundation work on the induction functor were key to the early development of the subject.

In this talk I will survey three of their most important contributions, (i) the cohomology of finite groups of Lie type, (ii) rational and generic cohomology and (iii) highest weight categories. In the process, I will provide examples of how CPS's work has influenced my research (with my collaborators) during the past 30 years. (Received January 08, 2020)

1155-20-176 **Leonard L. Scott*** (lls2l@virginia.edu). *Variations on a theme of Cline and Donkin.*

The starting point for my discussion is a paper with Brian Parshall, Variations on a theme of Cline and Donkin, Algebras and Representation Theory (3013) 16, 793-817. The paper's content is related to the Humphreys-Verma conjecture on projective indecomposable modules, but uses an approach that goes back to a 1972 paper by Ed Cline and related work. This mathematician, the Cline mentioned in the title, was a collaborator of Parshall and Scott for over 40 years. He unfortunately was too ill to collaborate on the paper in the title, and passed away shortly after its on-line publication in 2012. I intend to use the paper and its origin as a jumping-off point to discuss some aspects of Ed Cline's career and his collaboration with Parshall and myself. (Received January 12, 2020)

1155-20-177 **Marissa Kawehi Loving*** (mloving6@gatech.edu). *Symmetries of Surfaces.*

There are many ways to study surfaces: geometrically, number theoretically, algebraically, and combinatorially, just to name a few. We will touch on some of the motivation for studying surfaces and their associated mapping class groups, which is the collection of symmetries of a surface. We will also describe a few of the ways that these different perspectives for studying surfaces appear in practice and how they often overlap in beautiful ways. (Received January 12, 2020)

1155-20-183 **Rosa C Orellana***, 6188 Kemeny Hall, Hanover, NH 03755, and **Mike Zabrocki** (zabrocki@mathstat.yorku.ca). *Kronecker products and multiset tableaux.*

In a recent article Zabrocki and I introduced two new bases of symmetric functions that have the property that they evaluate to characters of the symmetric group when evaluated at eigenvalues of permutation matrices. One of the basis evaluates to the irreducible characters, the structure coefficients of this basis are the stable Kronecker coefficients. A byproduct of this work was the introduction of multiset tableaux in the study of the Kronecker product. In this talk I will discuss several combinatorial descriptions of Kronecker coefficients in terms of multiset tableaux.

This is joint work with Mike Zabrocki (Received January 12, 2020)

1155-20-202 **Rachel Skipper*** (skipper.26@osu.edu) and **Benjamin Steinberg**. *Generating lamplighter groups with bireversible automata.*

We use the language of formal power series to construct finite state automata generating groups of the form A wr Z, where A is the additive group of a finite commutative ring and Z is the integers. We then provide conditions on the units of the ring and the power series which make automata bireversible. (Received January 13, 2020)

1155-20-204 **David C Vella*** (dvella@skidmore.edu), Dept. Mathematics and Statistics, Skidmore College, 815 N. Broadway, Saratoga Springs, NY 12866. *Subgroups of Algebraic Groups with Property F.*

Let G be a connected, reductive algebraic group over an algebraically closed field k. A closed subgroup H of G is said to have property F if whenever V is a finite dimensional rational H-module, then the induced module $V|_H^G$ is also finite dimensional. We outline the historical interest in this property and discuss connections between subgroups with this property and certain other classes of subgroups, such as parabolic subgroups and epimorphic subgroups. (Received January 13, 2020)

1155-20-234 **Christopher P. Bendel*** (bendelc@uwstout.edu), Department of Math, Stats, and Comp Sci, Jarvis Hall Science Wing 231, University of Wisconsin-Stout, Menomonie, WI 54751, and **Daniel K. Nakano**, **Cornelius Pillen** and **Paul Sobaje**. *New results on the Tilting Module Conjecture.* Preliminary report.

Let G be a simple, simply connected algebraic group scheme over an algebraically closed field of prime characteristic p, and let G_r denote the rth Frobenius kernel of G. Further, let $Q_r(\lambda)$ denote the injective hull of the simple G_r-module of highest weight λ. For large primes, each $Q_r(\lambda)$ admits the structure of a G-module, and this is conjectured to be true in general. In the 1990s, Donkin made a more precise conjecture, that each $Q_r(\lambda)$ lifts to a certain indecomposable tilting G-module. While also known to be true for large p, it was recently shown by the authors that this stronger conjecture fails to hold for a group of type G_2 when the prime is 2. In this talk, we review the status of the conjecture and present some new strategies for verifying the conjecture, doing so in some new small rank cases. (Received January 14, 2020)

1155-20-236 **Christopher P Bendel**, **Daniel K Nakano**, **Cornelius Pillen*** (pillen@southalabama.edu) and **Paul Sobaje**. *On Donkin's tilting module conjecture for groups of type G_2.*

Tilting modules for reductive algebraic groups over fields with positive characteristics feature prominently in new approaches towards finding character formulas for simple modules. A long-standing conjecture by Donkin says that certain tilting modules are isomorphic to indecomposable injective modules when restricted to the Frobenius kernel of the algebraic group. It was recently shown that the conjecture does not hold for a group of type G_2 and characteristic 2. In this talk we verify Donkin's conjecture for G_2 and characteristic 3. (Received January 14, 2020)

1155-20-264 **Caglar Uyanik***, caglar.uyanik@yale.edu. *Large deviations for Teichmuller geodesics.*

Chaika and Eskin proved that the conclusions of Birkhoff's and Oseledets' theorems for the Teichmuller geodesic flow hold at every translation surface in almost every direction. We extend their results by showing that the Hausdorff dimension of the set of directions exhibiting a definite amount of deviation from the correct limit is not full. This is joint work with Al-Saqban, Apisa, Erchenko,Khalil and Mirzadeh. (Received January 15, 2020)

1155-20-279 **George Imad Domat*** (domat@math.utah.edu). *Big Pure Mapping Class Groups are Never Perfect.*

We prove that pure mapping class groups of infinite type surfaces are never perfect, i.e., they have non-trivial abelianization. This is in contrast to the result of Powell that pure mapping class groups of finite type surfaces are perfect if the surfaces have genus at least 3. In fact, we show that the abelianization of a particular closed subgroup always contains uncountable direct sums of rationals. To accomplish this we make use of the projection complex machinery of Bestvina, Bromberg, and Fujiwara in order to build specific quasimorphisms on finite type mapping class groups. (Received January 16, 2020)

1155-20-281 **Brian J Parshall*** (bjp8w@virginia.edu). *On some unpublished work on Edward Cline.*

This talk opens with a discussion of collaborative work done by Ed Cline and Roger Carter in Charlottesville that resulted in a joint paper in 1976. There, they computed the lattice of submodules of Weyl modules for G=$SL(2,k)$, where k is algebraically closed of positive characteristic. Using this result, Cline published a determination, in 1979, of all $Ext^1_G(M,N)$ for arbitrary simple G modules M,N. Then, in an unpublished paper, based on work he did at the Institute for Advanced Study in 1979, Cline gave an alternate construction of the lattice of submodules and explained certain regularities that had been observed earlier. He used these results to give an easy inductive determination of the composition factors and the structure graph of a dual Weyl module. In my talk I will provide some detail of this work. (Received January 16, 2020)

1155-20-294 **Aaron Calderon*** (aaron.calderon@yale.edu). *Mapping class groups, vector fields, and holomorphic differentials.*

The mapping class group of a surface is the set of its topological symmetries. When the surface is equipped with a vector field, one would like to understand which symmetries preserve (the isotopy class of) this vector field. Despite the fundamental nature of this question, these "framed mapping class groups" remain poorly understood. In this talk I will explain the discovery, joint with Nick Salter, of explicit, finite generating sets for certain framed mapping class groups. I will also discuss the implications of this result for the topology of moduli spaces of holomorphic 1–forms. (Received January 16, 2020)

1155-20-321 **Jean Pierre Mutanguha*** (jpmutang@uark.edu). *Irreducible endomorphisms of F_n are hyperbolic.*

Mapping tori of irreducible free group automorphisms are well-understood. In this talk, I will sketch the reason why the mapping torus of an irreducible-nonsurjective free group endomorphism is word-hyperbolic. The proof uses various standard tools that are important in the study of low-dimensional dynamics and geometry: laminations on graphs, the Culler-Vogtmann outer space, and the Bestvina-Feighn combination theorem. (Received January 17, 2020)

1155-20-334 **Viveka Erlandsson**, **Juan Souto** and **Jing Tao*** (jing@ou.edu). *Genericity of pseudo-Anosov mapping classes.*

We will show that pseudo-Anosov mapping classes are generic with respect to certain notions of genericity reflecting that we are dealing with mapping classes. (Received January 18, 2020)

1155-20-337 **Rose Morris-Wright*** (rmorriswright@brandeis.edu). *An analogy of the curve complex for FC type Artin groups.*

The curve complex is an important geometric construction for studying the mapping class group of a topological surface. This talk will construct an analogous simplicial complex for Artin groups of FC type. Artin groups are a generalization of braid groups that provide a rich field of examples and counter-examples for many algebraic, geometric, and topological properties. The complex of parabolic subgroups generalizes the curve complex for braid groups and the extension graph for right angled Artin groups. I will define the complex of parabolic subgroups, as well as discuss properties that the curve complex and the complex of parabolic subgroups share. (Received January 18, 2020)

1155-20-345 **Ian Runnels*** (iir4pk@virginia.edu). *Effectively Generating RAAGs in MCGs.*

A now classical theorem in the theory of mapping class groups, due to Ivanov and McCarthy, states that given a collection of independent pseudo-Anosov mapping classes f_1, ..., f_m, there is some (possibly large) power R such that $\langle f_i^R, ..., f_m^R \rangle$ is free of rank m. If one includes reducible mapping classes, this no longer holds, as mapping classes supported on disjoint subsurfaces necessarily commute. However, Koberda has shown that there is again some power R such that f_1^R, ..., f_m^R generate a *right-angled Artin group*. His proof relies on a choice of hyperbolic metric on the surface and playing ping-pong with geodesic laminations. We give an alternate proof using the so-called Masur-Minsky machinery, which relies only on the data of the mapping classes in question, and moreover provides an explicit upper bound for R in terms of this data. (Received January 18, 2020)

1155-20-385 **Gordon Rojas Kirby*** (girkirby@gmail.com), University of California, Santa Barbara, Department of Mathematics, South Hall 6607, Santa Barbara, CA 93106. *A new approach to understanding the structure of Artin groups.*

I would like to give a short talk about the word problem for Artin groups, a class of groups which are a far-reaching generalization of the much better-known braid groups. In particular, Artin groups can be realized as the fundamental group of a quotient of a complexified hyperplane complement, and the word problem reduces to understanding the universal cover of this space. My work centers on a combinatorially defined cell complex that is homotopy equivalent to an intermediate covering space and decomposing it into understandable pieces. In this talk, I will reduce the word problem to a simpler problem that can be stated about compact, combinatorially defined, zonotopal cell complexes, give brief illustrations of some work in my dissertation, and give concrete solutions to word problem for specific Euclidean Artin groups. Lastly, I will discuss why my work shows promise of generalizing to general Artin groups. (Received January 19, 2020)

1155-20-386 **Kasia Jankiewicz***, kasia@math.uchicago.edu. *Residual finiteness of certain Artin groups.*

A group is residually finite if for every non-identity element there is a finite quotient of the group where the given element is nontrivial. In this talk I will discuss the residual finiteness of many 3-generator Artin groups. (Received January 19, 2020)

1155-20-396 **Radhika Gupta**, **Kasia Jankiewicz** and **Thomas Ng*** (thomas.ng@temple.edu). *Using cubical geometry to construct free semigroups.* Preliminary report.

The strong Tits alternative for groups acting properly on CAT(0) cube complexes states that all finitely generated subgroups are either virtually abelian or contain a free subgroup. Verifying when pairs of elements generate a free subgroup, however, is well-known to be quite challenging. In this talk, I will describe joint work with Radhika Gupta and Kasia Jankiewicz explicitly constructing of free semigroups in arbitrary generating sets that rely on combinatorial characterizations of the dynamics of group actions on cube complexes as well as applications to quantifying the strong Tits alternative and uniform exponential growth. (Received January 19, 2020)

1155-20-411 **Jinkui Wan*** (wjk302@hotmail.com) and **Weiqiang Wang** (ww9c@virginia.edu). *Stability of the centers of integral group algebras of classical groups over finite fields.*

The center of the integral group algebra of the general linear group $GL(n,q)$ over a finite field admits a filtration with respect to the reflection length. We show that the structure constants of the associated graded algebras are independent of n and this stability leads to a universal stable center with positive integer structure constants. Various structure constants of the stable center are computed. Furthermore, as applications we extend the stability property to the centers of the integral group algebras of other classical finite groups including affine groups $P(n,q)$, symplectic groups $Sp(2n,q)$ and unitary groups $U(n,q^2)$ with respect to the filtrations defined via reflections belonging to these groups, respectively.This is partially based on joint work with Weiqiang Wang. (Received January 19, 2020)

1155-20-436 **Paul Sobaje*** (psobaje@georgiasouthern.edu). *Steinberg quotients of projective G_1T-modules.*

For a reductive group G in positive characteristic, the character of a projective G_1T-module is divisible by the character of the Steinberg module. Computing the resulting "Steinberg quotient" is equivalent to computing the composition multiplicities of baby Verma modules for G_1T, the latter a well-studied problem. However, in the form of Steinberg quotients some of the patterns in these multiplicities are more readily seen. We will discuss some of these patterns, with particular attention to the problem of understanding when these characters exhibit a generic pattern (which holds for very large p), and what happens in smaller characteristics. (Received January 20, 2020)

1155-20-453 **Spencer Dowdall***, Department of Mathematics, Vanderbilt University, Station B 407807, Nashville, TN 37240, and **Ilya Kapovich** and **Christopher J Leininger**. *Universal trees and canonical polynomials for free-by-cyclic groups.* Preliminary report.

A free-by-cyclic group $G = F \rtimes_\varphi \mathbb{Z}$ can often be expressed as a free-by-cyclic $G \cong F_u \rtimes_{\varphi_u} \mathbb{Z}$ in infinitely many ways. Supposing the initial monodromy φ is fully irreducible and atoroidal, we construct a universal real tree T equipped with a G–action and a canonical polynomial invariant $p \in \mathbb{Z}[G/G']$ built from the arcs in T. Together, the tree and polynomial encode a wealth of information the free-by-cyclic splittings of G. Specifically, the polynomial calculates both a component of the BNS-invariant of G and the stretch factor for each monodromy $\varphi_u \in \mathrm{Out}(F_u)$ in this component; meanwhile the topological tree T may be given an $\mathbb{R}$–tree metric for each such monodromy that yields the unique attracting fixed point of φ_u in compactified Outer space. (Received January 20, 2020)

1155-20-455 **Talia Fernos***, UNCG Dept. of Math and Stats, Petty 116, PO Box 26170, Greensboro, NC 27402, and **Jean Lecureux** and **Frederic Matheus**. *Random Walks and CAT(0) Cube Complexes.*

Let G be a group acting on a finite dimensional CAT(0) cube complex X. By studying equivariant maps from the Furstenberg-Poisson boundary to the Roller boundary, we deduce a variety of phenomena concerning the push-forward of the random walk from G to an orbit in X. Under mild and natural assumptions, we deduce positivity of the drift and sublinear tracking. This is joint work with Jean Lécureux and Frédéric Mathéus. (Received January 20, 2020)

1155-20-463 **Yulan Qing*** (yulan.qing@gmail.com), 648 Markham Street, Toronto, Ontario M6G 2L9, Canada, and **Ilya Gekhtman**, **Kasra Rafi** and **Abdul Zalloum**. *the sublinearly Morse boundary of CAT(0) Cube Complexes.*

To every Gromov hyperbolic space X one can associate a space at infinity called the Gromov boundary of X. This boundary is a fundamental tool for studying hyperbolic groups and hyperbolic 3-manifolds. Similarly, the sublinearly Morse boundary for any given CAT(0) space is QI-invariant and metrizable. In this talk we discuss the combinatorial characterization of this boundary in the setting of CAT(0) cube complexes. We also show that the sublinearly Morse boundary for a CAT(0) space with cocompact action is a metrical model for the associated Poisson boundaries. This talk is based on joint projects with Abdul Zalloum and also with Ilya Gekhtman and Kasra Rafi. (Received January 20, 2020)

1155-20-471 **Terrell L. Hodge*** (terrell.hodge@wmich.edu), Department of Mathematics, 1908 Western Michigan Avenue, Western Michigan University, Kalamazoo, MI 49008-5248, **Paramasamy Karrapuchamy**, Department of Mathematics, University of Toledo, Toledo, OH 43606, and **Leonard L. Scott**, Department of Mathematics, University of Virginia, Charlottesville, VA 22903. *Truncation and the Induction Theorem.*

A key result in a 2004 paper by S. Arkhipov, R. Bezrukavnikov, and V. Ginzburg compares the bounded derived category $D^b(block(\mathbb{U}))$ of finite dimensional modules for the principal block of a Lusztig quantum enveloping algebra $\mathbb{U}$ at an ℓth root of unity, with a special full subcategory $D_{triv}(\mathbb{B})$ of the bounded derived category of integrable type 1 modules for a Borel part $\mathbb{B} \subset \mathbb{U}$. With their work as motivation, we consider characteristic $p > 0$ representations of algebraic groups, and an equivalence $\mathrm{RInd}_B^G : D_{triv}(B) \xrightarrow{\sim} D^b(block(G))$, relating an analog of $D_{triv}(\mathbb{B})$, defined using a Borel subgroup B of a simply connected semisimple algebraic group G, to the bounded derived category of the principal block of finite dimensional rational G-modules, for $p > h$, h the Coxeter number. Under the assumption $p > 2h-2$, we show the equivalence behaves well with respect to certain weight poset "truncations". This means, in particular, that the equivalence above can be reformulated in terms of derived categories of finite dimensional quasi-hereditary algebras. We expect that a similar result holds in the quantum case. (Received January 20, 2020)

1155-20-502 **Aaron Lauve*** (lauve@math.luc.edu), Dept. of Math. and Stats., 1032 W. Sheridan Road, Chicago, IL 60660-1537, and **Anthony A Lazzeroni** (alazzeroni@luc.edu). *Powersums in the Hopf Algebra of Quasisymmetric Functions.* Preliminary report.

The classical (commutative) powersums wear a number different hats. To name a few, they: demonstrate the Cartier–Milnor–Moore isomorphism for the ring SYM of symmetric functions; provide for a recursive construction of character tables for the symmetric group; provide elegant combinatorial links to other classical bases of SYM; .... After a brief survey, we'll discuss the main problem motivating our work: the limitless sea of generalizations to the Hopf algebra NSYM of noncommutative symmetric functions (and, by duality, to the Hopf algebra QSYM of quasisymmetric functions). In this talk, I offer yet another generalization to QSYM that maintains an elegant combinatorial link to other bases. I'll close with a discussion of open questions and possible applications. (A report on work in progress.) (Received January 20, 2020)

1155-20-509 **Pramod N. Achar*** (pramod@math.lsu.edu), **William Hardesty** and **Simon Riche**. *Around the Humphreys conjecture.*

Let G be a reductive algebraic group over a field of characteristic $p > h$. The "G_1-cohomology construction" associates to any G-module M a certain coherent sheaf on the nilpotent variety of G. This construction seems to be especially rich when M is a tilting module. The Humphreys conjecture (now a theorem if p is large, or if G is of type A) predicts the support of this coherent sheaf; more recent conjectures propose finer descriptions of the coherent sheaves arising in this way. I will explain these conjectures and their connections to other topics (e.g., tensor ideals of tilting modules), with a special focus on the case of GL_n. (Received January 20, 2020)

1155-20-525 **Carolyn Abbott*** (abbott@math.columbia.edu) and **Michael Hull**. *Free products and random walks in acylindrically hyperbolic groups.*

The properties of a random walk on a group which acts on a hyperbolic metric space have been well-studied in recent years. In this talk, I will focus on random walks on acylindrically hyperbolic groups, a class of groups which includes mapping class groups, $\mathrm{Out}(F_n)$, and right-angled Artin and Coxeter groups, among many others. I will discuss how a random element of such a group interacts with fixed subgroups, including convex cocompact subgroups of mapping class groups and $\mathrm{Out}(F_n)$. In particular, I will discuss when the subgroup generated by a random element and a fixed subgroup is a free product, and I will also describe some of the geometric properties of that free product. This is joint work with Michael Hull. (Received January 21, 2020)

22 ► Topological groups, Lie groups

1155-22-104 **Feng Zhu*** (zhufeng@umich.edu). *Relatively dominated representations.*
Convex cocompact subgroups of rank-one semisimple Lie groups such as PSL(2,R) form a structurally stable class of quasi-isometrically embedded discrete subgroups which are naturally associated to negatively-curved geometric structures. Anosov representations give a higher-rank analogue of convex cocompactness which shares many of its good geometric and dynamical properties, and have become important objects of study in higher Teichmüller theory. I will introduce a relativization of Anosov representations, or in other words a higher-rank analogue of geometric finiteness—a controlled weakening of convex cocompactness to allow for isolated failures of hyperbolicity. (Received January 06, 2020)

1155-22-276 **Jiuzu Hong** (jiuzu@email.unc.edu), Department of Mathematics, 329 Phillips Hall, Chapel Hill, NC 27599. *Twisted conformal blocks and theta functions.*
The theory of conformal blocks is important in 2d conformal field theory. It is defined via Wess-Witten-Zumino, more precisely in terms of Kac-Moody theory. It is related to the geometry of moduli space of algebraic curves. Moreover, conformal blocks can be identified with the theta functions on the moduli stack of principle G-bundles.

I will talk about a twisted theory of conformal blocks attached to Galois covers of algebraic curves, where twisted Kac-Moody algebra will play key roles. I will also talk about the identification between twisted conformal blocks and the theta functions on the moduli stack of torsors over parahoric Bruhat-Tits group schemes over curves. This talk is based on the joint work with Shrawan Kumar. (Received January 21, 2020)

1155-22-493 **Iva Halacheva*** (i.halacheva@northeastern.edu), **Allen Knutson** and **Paul Zinn-Justin**. *Restriction of Maulik-Okounkov classes from type A to type C Grassmannians.* Preliminary report.
One of the classical questions in Schubert calculus studies the expansion of the product of two Schubert classes in the Schubert basis, under the pullback of the diagonal inclusion of the Grassmannian. Knutson and Tao gave a positive puzzle formula in 2003 for computing this expansion in equivariant cohomology. We study a similar pullback, of the inclusion of the symplectic Grassmannian into the usual Grassmannian. I will present a puzzle formula describing the expansion of Schubert classes under this map, using self-dual puzzles. I will then discuss a generalization using Maulik-Okounkov classes and a more geometric context for puzzles involving Lagrangian correspondences. This is joint work with Allen Knutson and Paul Zinn-Justin. (Received January 20, 2020)

1155-22-508 **J. Matthew Douglass*** (mdouglas@nsf.gov). *Equivariant coherent sheaves on a point and Kazhdan-Lusztig bases.* Preliminary report.
Suppose G is a connected, reductive, complex, algebraic group with Lie algebra $\mathfrak{g}$. In this talk I will discuss (1) a refined version of a question of Ostrik that describes the $G \times \mathbb{C}^*$-equivariant coherent sheaves on the nilpotent cone of $\mathfrak{g}$ that are supported at $\{0\}$ in terms of the Kazhdan-Lusztig basis of the extended, affine Hecke algebra, and (2) how this description is connected with the computation of the $G \times \mathbb{C}^*$-equivariant K-theory of some generalized Steinberg varieties. (Received January 20, 2020)

26 ► Real functions

1155-26-415 **Caleb Ashley*** (ashley.cj@umich.edu), **Naiomi Cameron**, **Edray Goins**, **Emille Lawrence**, **Theo McKenzie** and **Karoline Pershell**. *On Computing Monodromy Groups for Compositions of Belyi Maps.* Preliminary report.
Among all hyperbolic structures on a Riemann surface S_g of fixed genus $g \geq 2$, those that are also arithmetic are special. Grothendieck's famous CNRS research proposal "Esquisse d'un Programme" outlines a link between combinatorial data, so-called "dessin d' enfants," the complex geometry of algebraic curves, and their corresponding algebraic number fields of coefficients. In this talk we describe some fundamental objects and results which outline this theory. Our main goal is to describe $\mathrm{Mon}(\beta \circ \alpha)$, the monodromy group of the composition of two dynamical Belyi maps α and β of a certain broad subclass in a straightforward manner. This work departs from the Ph. D. thesis of Jacob Bond and is joint with N. Cameron, E. Goins, E. Lawrence, T. McKenzie, K. Pershell.) (Received January 19, 2020)

28 ► Measure and integration

1155-28-117 **Ben Jaye*** (bjaye@clemson.edu). *Multi-scale analysis of Jordan curves.*
In this talk we will describe how one can detect regularity in Jordan curves through analysis of associated geometric square functions. We will particularly focus on the resolution to a conjecture of L. Carleson. Joint work with Xavier Tolsa and Michele Villa (https://arxiv.org/abs/1909.08581). (Received January 06, 2020)

1155-28-257 **Vyron Vellis***, Ayres Hall, University of Tennessee, 1403 Circle Dr, Knoxville, TN 37916. *Hölder parameterization of IFS.*
The rectifiability problem, one of the most important problems in geometric measure theory, asks conditions under which a metric space is contained in a Lipschitz or Hölder curve. A theorem of Remes from 1998 asserts that any connected, self-similar set in Euclidean space that satisfies the open set condition is the image of a $(1/s)$-Hölder curve, where s is the similarity dimension. In this talk we generalize Remes' theorem to connected iterated function systems in complete metric spaces. As a special case, we discuss Bedford-McMullet carpets and determine the sharp Hölder exponent in this case. This is joint work with Matthew Badger. (Received January 15, 2020)

1155-28-375 **Matthew Badger*** (matthew.badger@uconn.edu). *Anisotropic Linear Approximation, Square Functions, and Rectifiability of Measures.*
In geometric measure theory, we are interested in understanding the structure of a measure on a space through its interactions with canonical lower dimensional sets such as rectifiable curves or Lipschitz graphs. An important dichotomy arises between the rectifiable measures, which give all of their mass to lower dimensional sets, and purely unrectifiable measures, which vanish on every lower dimensional set. In recent years, the infusion of ideas from harmonic analysis has led to a renaissance of results by several groups of mathematicians, including a complete characterization of locally finite measures carried by or singular to rectifiable curves in joint work by myself and Raanan Schul. I will give an overview of these developments with an emphasis on ideas from harmonic analysis and discuss future challenges. (Received January 19, 2020)

33 ► Special functions

1155-33-299 **Chris Jennings-Shaffer** and **Antun Milas*** (antun.milas@gmail.com). *q-series identities from logarithmic CFT and 3d/4d supersymmetric gauge theory.* Preliminary report.
I will present several new families of q-series identities that are motivated by research on super logarithmic conformal field theory and 4d supersymmetric gauge theory. We shall also discuss related identities for the classical and quantum dilogarithm functions. (Received January 17, 2020)

34 ► Ordinary differential equations

1155-34-9 **Joseph Paullet*** (jep7@psu.edu), School of Science, Penn State Behrend, 4205 College Drive, Erie, PA 16563, and **Joseph Previte**, School of Science, Penn State Behrend, 4205 College Drive, Erie, PA. *Analysis of Nanofluid Flow Past a Permeable Stretching/Shrinking Sheet.*
In this talk we analyze a recently proposed model for boundary layer flow of a nanofluid past a permeable stretching/shrinking sheet. The boundary value problem (BVP) governing fluid velocity reduces to:

$$f'''(\eta) + f(\eta)f''(\eta) - f'(\eta)^2 = 0, \quad 0 < \eta < \infty,$$

subject to

$$f(0) = S, \quad f'(0) = \lambda, \quad f'(\infty) = 0,$$

where λ measures the stretching ($\lambda > 0$) or shrinking ($\lambda < 0$) of the sheet, and S controls the suction ($S > 0$) or injection ($S < 0$) of fluid through the sheet.

For $\lambda \geq 0$ and $S \in \mathbb{R}$, we present a closed-form solution to the BVP and prove that this solution is unique. For $\lambda < 0$ and $S < 2\sqrt{-\lambda}$ we prove no solution exists. For $\lambda < 0$ and $S = 2\sqrt{-\lambda}$ we present a closed-form solution to the BVP and prove that it is unique. For $\lambda < 0$ and $S > 2\sqrt{-\lambda}$ we present two closed-form solutions to the BVP and prove the existence of an infinite number of solutions in this parameter range. The analytical results proved here differ from the numerical results reported in the literature. We discuss the mathematical aspects of the

problem that lead to the difficulty in obtaining accurate numerical approximations to the solutions. (Received January 06, 2020)

1155-34-55 **Jeffrey T Neugebauer*** (jeffrey.neugebauer@eku.edu) and **Johnny Henderson** (johnny_henderson@baylor.edu). *First Extremal Point Comparison for a Fractional Boundary Value Problem with a Fractional Boundary Condition.*

Let $n \geq 2$ be a natural number, and let $n-1 < \alpha \leq n$ and $0 < \gamma \leq \alpha - 1$ be real numbers. Let $\beta > 0$ and $b \in (0, \beta]$. We compare first extremal points of the differential equations $D^{\alpha}_{0+}u + p(t)u = 0$, $D^{\alpha}_{0+}u + q(t)u = 0$, $t \in (0, \beta)$, each of which satisfies the boundary conditions $u^{(i)}(0) = 0$, $i = 0, 1, \ldots, n-2$, $D^{\gamma}_{0+}u(b) = 0$. While it is assumed that q is nonnegative, no sign restrictions are put on p. The fact that the associated Green's function $G(b; t, s)$ is nonnegative and increasing with respect to b plays an important role in the analysis. (Received December 05, 2019)

1155-34-67 **James Sochacki*** (sochacjs@jmu.edu), **Roger Thelwell** and **Anthony Tongen**. *Polynomial Methods for the Single Pendulum - Including Planar Motion and External Forces.* Preliminary report.

The differential equations describing the motion of a bob on a rod with a moving pivot in a plane with internally and externally applied forces is converted to polynomial form. A conserved quantity for these polynomial differential equations and a decoupled form of these polynomial differential equations is presented. We present both analytical and numerical solutions for these differential equations for several interesting models including a circular motion of the pivot, a motion to a vertical pendulum and the motion of the pendulum under unbounded forces. The power series method used for the numerical solutions is outlined. (Received December 16, 2019)

1155-34-69 **Andres Larrain-Hubach*** (alarrainhubach1@udayton.edu), University of Dayton, Department of Mathematics, 300 College Park, Dayton, OH 45469. *An ODE Index Theorem related to Instantons on Taub-NUT Space.*

An *Instanton* is a special solution to *Yang-Mills Equations* (YM) defined on a four-manifold. YM is a difficult system of PDE's and it is rare to find instantons explicitly. In the case of *Taub-NUT space* (TN), a process known as *Nahm Transform* produces instantons by solving a related system of ODE's and linear equations organized in a *Bow Representation*. In joint work with Sergey Cherkis and Mark Stern, we proved that every generic instanton on TN can be obtained from a bow representation. One important step in this proof is the computation of the index of certain Fredholm differential operator defined using the bow representation. In this talk, I will explain the general setup of the Nahm transform and the index calculation. (Received December 16, 2019)

1155-34-73 **Abdul-Aziz Yakubu** and **Eric Ngang Che Fnu***, 10203 Birdie Ln, Upper Marlboro, MD 20774, and **Yeona Kang**. *Risk Structured Model of Cholera Infections in Cameroon.*

Since 1991, Cameroon, a cholera endemic African country, has been experiencing large cholera outbreaks and cholera related deaths. In this talk, we use a "fitted" demographic equation (disease-free equation) to capture the total population of Cameroon, and then use a fitted low-high risk structured cholera differential equation model to study reported cholera cases in Cameroon from 1987-2004. For simplicity, our model has no spatial structure. The basic reproduction number of our fitted cholera model, $\mathcal{R}_0$, is bigger than 1 and our model predicted cholera endemicity in Cameroon. In addition, the fitted risk structured model predicted a decreasing trend from 1987 to 1994 and an increasing trend from 1995 to 2004 in the pre-intervention reported number of cholera cases in Cameroon from 1987 to 2004. Using the fitted risk structured cholera model, we study the impact of vaccination, treatment and improved sanitation on the number of cholera infections in Cameroon from 2004 to 2022. Finally, using sensitivity analysis, we study the impact of our model parameters on the demographic threshold, basic reproduction number, effective reproduction number and on the total number of our model's predicted cholera cases. (Received December 18, 2019)

1155-34-88 **Stephen K Lucas*** (lucassk@jmu.edu), Department of Mathematics & Statistics, James Madison University, Roop Hall, MSC 1911, 60 Blue Stone Drive, Harrisonburg, VA 22807. *High Precision Solutions for Ordinary Differential Equations.*

There are a variety of approaches to numerically approximating the solution to initial value ordinary differential equations, typically some form of Runge-Kutta method. While fast and efficient for modest accuracy requirements, standard order four/five methods require absurdly small time steps for high precision solutions. Here, we will briefly outline the Power Series Method (PSM), which recasts ordinary differential equations in polynomial form in the dependent variables and can be made at most quadratic in order. Then Taylor series of arbitrary order can be found for each time step. We show how this approach can be used to develop Taylor series with

hundreds of coefficients at any time step and find solutions accurate to hundreds or thousands of digits accuracy. We shall see that arbitrarily large orders are not necessarily the best choice for all problems, and a finite radius of convergence for the unknown functions can be a limiting factor. (Received December 30, 2019)

1155-34-90 **Muhammad N Islam*** (mislam1@udayton.edu) and **Jeff Neugebauer**. *Fractional Differential Equation of Riemann-Liouville Type.* Preliminary report.

An initial value problem for a Riemann-Liouville fractional differential equation of order $q, 0 < q < 1$, is studied. First, the existence of a continuous solution on an interval $(0, T], T > 0$ is obtained. Then it is shown that if a continuous solution exists on $(0, \infty)$, then it is absolutely integrable on the same interval. (Received January 01, 2020)

1155-34-123 **JING JIAO*** (jing.jiao@cos.name), 1122 Volunteer Blvd., Suite 106, Knoxville, TN 37996-3410. *Host metapopulation, disease epidemiology and host evolution.*

While most evolutionary studies of host-pathogen dynamics consider pathogen evolution alone or host-pathogen coevolution, there is evidence that hosts can evolve more rapidly than their pathogen during initial outbreaks after disease introduction. In this talk, we will discuss the spatial, temporal, and epidemiological factors may drive the evolutionary dynamics of the host population. We consider a simplified system of two host genotypes that trade off factors of disease robustness and spatial mobility or growth. For diseases that infect hosts for life, we find that migration and disease-driven mortality can have either antagonistic or synergistic effects on host composition, depending on disease starting status. For diseases that allow hosts to recover with immunity, we explore the conditions under which the disease dies out, becomes endemic, or has periodic outbreaks, and show how these dynamics relate to the relative success of the robust and wild type hosts in the population over time. Overall, we will discuss how combinations of host spatial structure, demography, and epidemiology of infectious disease can significantly influence host evolution and disease prevalence. We will conclude with some profound implications for wildlife conservation and zoonotic disease control. (Received January 07, 2020)

1155-34-180 **Gampola Waduge Nalin Fonseka*** (g_fonsek@uncg.edu), **Ratnasingham Shivaji**, **Jerome Goddard**, **James Cronin** and **Jackson Leonard**. *Modeling the effects of density dependent emigration, weak Allee effects, and matrix hostility on patch-level population persistence.*

The relationship between conspecific density and the probability of emigrating from a patch can play an essential role in determining the population-dynamic consequences of an Allee effect. In this paper, we model a population that inside a patch is diffusing and growing according to a weak Allee effect per-capita growth rate, but the emigration probability is dependent on conspecific density. The habitat patch is one-dimensional and is surrounded by a tuneable hostile matrix. We consider five different forms of density dependent emigration (DDE) that have been noted in previous empirical studies. Our models predict that at the patch-level, DDE forms that have a positive slope will counteract Allee effects, whereas, DDE forms with a negative slope will enhance them. Also, DDE can have profound effects on the dynamics of a population, including producing very complicated population dynamics with multiple steady states whose density profile can be either symmetric or asymmetric about the center of the patch. Our results are obtained mathematically through the method of sub-super solutions, time map analysis, and numerical computations using Wolfram Mathematica. (Received January 12, 2020)

1155-34-285 **Dan Neugebauer*** (neugebauerd1@udayton.edu), 1261 Laura Lane, Mason, OH 45040. *Perturbations of a periodic-like boundary value problem at resonance.*

Let q denote a continuous function of t, periodic with period 1. Let $h > 0$. Assume $\lambda \in \mathbb{R}$ and assume g is continuous on $[0, 1]$. We consider a boundary value problem

$$x''(t) + (\lambda + q(t))x(t) = g(t), \quad t \in (0, 1), \tag{1}$$

$$x(0) = hx(1), \quad x'(0) = x'(1). \tag{2}$$

If $h = 1$, the boundary value problem (1), (2) is called a periodic boundary value problem.

Let λ_0 denote the smallest eigenvalue of the boundary value problem,

$$x''(t) + (\lambda + q(t))x(t) = 0, \quad t \in (0, 1),$$

with the boundary conditions (2). Then the boundary value problem (1), (2) is at resonance for $\lambda = \lambda_0$ and no Green's function exists. We obtain conditions on $\lambda < \lambda_0$ or $\lambda > \lambda_0$ such that the boundary value problem (1), (2) satisfies a maximum principle or the boundary value problem (1), (2) satisfies an antimaximum principle.

We extend results that are known in the case $h = 1$. (Received January 16, 2020)

1155-34-363 **Nsoki Mavinga*** (nmaving1@swarthmore.edu) and **Yusuf Qaddura**. *Delay Differential Equations with Applications to the Analysis of the Spread of Vector-Borne Diseases.*

We are concerned with the stability of equilibrium solutions for a two-lag delay differential equation which models the spread of vector-borne diseases where the lags are incubation periods in humans and vectors. We show that there are some values of the transmission and recovery rates for which either the disease dies out or it spreads into an endemic state. The approach is based on the linearization method and the analysis of roots of transcendental equations. (Received January 18, 2020)

1155-34-450 **Ming Zhong*** (mzhong5@jhu.edu), 500 W. University Parkway, Apt. 10P, Baltimore, MD 21210, and **Jason Miller** and **Mauro Maggioni**. *Data-driven Discovery of Emergent Behaviors in Collective Dynamics.*

Agent-based systems are a ubiquitous modeling tool in many disciplines. We consider the fundamental problem of inferring interaction kernels from observations of agent-based collective dynamical systems exhibiting various kinds of emergent behaviors. As extension of [1], we provide extensive numerical evidence that the estimators provide faithful approximations to the interaction kernels, and provide accurate predictions for trajectories started at new initial conditions, both throughout the "training" time interval in which the observations were made, and often much beyond. We demonstrate these features on a number of prototypical systems. Our experiments also suggest that our estimated systems can display the same emergent behaviors of the observed systems. Finally, in the case of families of systems governed by a parameterized family of interaction kernels, we introduce novel estimators that estimate the parameterized family of kernels. We demonstrate this in the case of gravity, by learning both the "common component $1/r^2$" and the dependency on mass, without any a priori knowledge of either one.

[1] F. Lu, M. Zhong, S. Tang, M. Maggioni. Nonparametric inference of interaction laws in systems of agents from trajectory data, PNAS, 116 (3), 14424 - 14433, 2019. (Received January 20, 2020)

1155-34-527 **Zachariah Sinkala*** (zachariah.sinkala@mtsu.edu), Department of Mathematical Sciences, Middle Tennessee State University, Murfreesboro, TN 37132. *Application of the one-step method to parameter estimation in Glioblastoma ode model.*

In this paper, we study the glioblastoma ordinary differential equations model. We use the so-called one-step method(for example, Le Cam's) for parameter estimation. The most popular method for parameter estimation is nonlinear least squares estimator, which requires the use of a multi-step iterative algorithm and repetitive numerical integration of the ODE system. The one-step method starts from a preliminary a consistent estimator of the parameter of interest and next into an asymptotic equivalent of the least-squares estimator through a numerically straightforward procedure. We demonstrate the performance of the one-step estimator compared with traditional technique(nonlinear least estimator) using real glioblastoma data. The method does parameter estimate values and parameters intervals estimate. The preliminary consistent estimator that we use depends on nonparametric smoothing. It is also data dependent for choosing its tuning parameter. The advantage of this approach, it is easy to implement practically. (Received January 21, 2020)

35 ► Partial differential equations

1155-35-17 **Alexander S Pankov*** (alexander.pankov@morgan.edu), 1700 E Cold Spring Lane, Morgan State University, Mathematics Department, Baltimore, MD 21251. *Standing waves in discrete nonlinear Schrödinger equations.*

In this talk we present results on the existence of standing solitary waves in discrete nonlinear Schrödinger equations. We consider two cases: (a) infinitely growing linear potential, and (b) periodic linear and nonlinear potential. The results are based on various critical point theorems for smooth functionals. (Received September 30, 2019)

1155-35-35 **Michael P Presho*** (mpresho@semo.edu), Southeast Missouri State University, Department of Mathematics, Cape Girardeau, MO 63701. *An Accurate Multiscale Method for Modeling Two-Phase Flow with Capillary Pressure.*

In this presentation we investigate a numerical method used for solving a two-phase flow model within the context of a physics-based two-level operator splitting scheme. The governing equations involve an elliptic pressure equation coupled with a convection-dominated parabolic equation. The underlying permeability of the porous medium is assumed to exhibit high variation over many length scales which leads to poor approximation properties in standard multiscale finite element spaces. In order to effectively solve the system we construct an

approach that combines a mass-conservative generalized multiscale finite element method with an operator splitting scheme. In particular, we split the coupled system into three subsystems: the elliptic, the hyperbolic, and the parabolic pieces. To solve the elliptic equation we propose the use of a mass-conservative generalized multiscale finite element method (GMsFEM-FV), for the parabolic equation we use a mass-conservative multiscale finite element method (MsFEM-FV), and for the hyperbolic equation we use a standard non-oscillatory finite-volume scheme. The performance of the methodology is validated through a variety of numerical experiments. (Received November 17, 2019)

1155-35-56 **W. E. Fitzgibbon***, Department of Mathematics, Houston, TX 77204, and **J. J. Morgan**, **G. F. Webb** and **Y. Wu**. *The Role of Advection and Diffusion for Modeling the Spatial Spread of Disease.*

There is a growing literature regarding the use of reaction diffusion equations for modeling the spatio-temporal spread of diseases. Recently we have been working on models that incorporate both diffusion and advection as mechanisms of dispersal for either hosts, vectors, or pathogens. Dispersion can result from the movement of hosts and/or vectors, it can be wind driven, or result from the flow of surface or subsurface water. The modelling may couple systems of partial differential equations across fragmented domains, different domains or across domains of different dimension. The domains may be bounded or unbounded or both. In this talk, we will discuss our modeling and analysis of a variety of scenarios for the advective-diffusive spread of infectious disease. (Received December 06, 2019)

1155-35-70 **Joseph Daniel Rudmin*** (rudminjd@jmu.edu), JMU ISAT MSC4106, 801 Carrier Dr., Harrisonburg, VA 22807. *A Topological Approach to the Parker Sochacki Method for Solving Differential Equations.*

A general all-order topological approach to the Parker Sochacki Method of solving differential equations applied to a Yang Mills field with a non-trivial interaction generates homotopic solutions at all energy scales. The shift operator extends domains of convergence of resulting Taylor Series. Properties of the first-order polynomial form of differential equations guides one in optimizing initial conditions to find bounded periodic states. While a simple solution to the Yang Mills equations of motion has the form of a tower of states, the simplicity of the Yang Mills equations of motion admits arbitrary scaling of the energy spacing of the tower of states. (Received December 17, 2019)

1155-35-76 **Gro Hovhannisyan*** (ghovhann@kent.edu), 6000 Frank ave NW, North Canton, OH 44720. *3 soliton solution to Sine-Gordon equation on a space scale.*

Using Hirota's direct approach we construct the three-soliton solutions to Sine-Gordon equation on a space scale. We derive also determinant form of these solutions. (Received December 19, 2019)

1155-35-120 **Ruiwen Shu*** (rshu@cscamm.umd.edu). *Equilibration of aggregation-diffusion equations with weak interaction forces.*

I will discuss some recent results on the large time behavior of aggregation-diffusion equations. The talk will focus on proving the convergence to the unique steady state as time goes to infinity, in the case where the confining effect of the interaction force is weak. (Received January 07, 2020)

1155-35-139 **Zebang Shen** and **Zhenfu Wang*** (zwang423@math.upenn.edu). *Interacting Particle System to the Barycenter Problem.*

We propose a new non-parametric method to approximate the barycenter of a set of probability measures. This method is based on constructing an interacting particle system. The mean field limit is given by a gradient flow in the probability measure space and the long time limit of the gradient flow corresponds to the target barycenter. Joint work with Zebang Shen. (Received January 08, 2020)

1155-35-178 **Gampola Waduge Nalin Fonseka*** (g_fonsek@uncg.edu), Greensboro, Greensboro, 27402, and **Ratnasingham Shivaji**, **Byungjae Son** and **Keri Spetzer**. *Classes of reaction diffusion equations where a parameter influences the equation as well as the boundary condition.*

We study positive solutions to steady state reaction diffusion equations of the form:

$$\begin{cases} -\Delta u = \lambda f(u);\ \Omega, \\ \frac{\partial u}{\partial \eta} + \mu(\lambda) u = 0;\ \partial\Omega, \end{cases}$$

where $\lambda > 0$, Ω is a bounded domain in $\mathbb{R}^N$; $N \geq 1$ with smooth boundary $\partial\Omega$, $\frac{\partial u}{\partial \eta}$ is the outward normal derivative of u, $\mu \in C([0,\infty))$ is strictly increasing such that $\mu(0) \geq 0$ and $f \in C^2([0, r_0))$ with $0 < r_0 \leq \infty$. If $r_0 < \infty$ we assume $f \in C^2([0, r_0])$ with $f(r_0) = 0$ and $f(s) \leq 0$ for $s \in (r_0, \infty)$, and if $r_0 = \infty$ we assume

$\lim_{s\to\infty} f(s) > 0$ and $\lim_{s\to\infty} \frac{f(s)}{s} = 0$ (sublinear at ∞). Note here that the parameter λ influences both the equation and the boundary condition. We discuss existence, nonexistence, multiplicity and uniqueness results for the cases when (A) $f(0) = 0$, (B) $f(0) < 0$, and (C) $f(0) > 0$. We obtain existence and multiplicity results by the method of sub-super solutions and uniqueness results by comparison principles and a priori estimates. (Received January 12, 2020)

1155-35-179 **K.D. Chu**, **D.D. Hai** and **R Shivaji***, Department of Mathematics and Statistics, University of North Carolina at Greensboro, Greensboro, NC 27455. *Uniqueness of positive radial solutions for a class of infinite semipositone p-Laplacian problems in a ball.*

We prove uniqueness of positive radial solutions to the p-Laplacian problem

$$\begin{cases} -\Delta_p u = \lambda f(u) \text{ in } \Omega, \\ u = 0 \text{ on } \partial\Omega, \end{cases}$$

where $\Delta_p u = div(|\nabla u|^{p-2}\nabla u),\ p \geq 2,\ \Omega$ is the open unit ball in $R^N, N > 1,\ f : (0,\infty) \to \mathbb{R}$ is concave, $p-$ sublinear at ∞ with infinite semipositone structure at 0, and λ is a large parameter. (Received January 12, 2020)

1155-35-192 **Jaemin Park*** (jpark776@gatech.edu), 686 Cherry St, NW, Atlanta, GA 30332. *Symmetry in Steady and Stationary Solutions of Active Scalar Equations.*

We study whether a stationary solution to 2D Euler equation has to be radially symmetric if the vorticity is compactly supported or has some decay at infinity. Our main results are the following: (1) On the one hand, we are able to show that for any non-negative smooth stationary vorticity that is compactly supported, it must be radially symmetric up to a translation. (2) On the other hand, if we allow vorticity to change sign, then by applying bifurcation arguments to sign-changing radial patches, we are able to show that there exists a compactly- supported, sign-changing smooth stationary vorticity that is non-radial. We have also obtained some symmetry results for uniformly-rotating solutions for 2D Euler equation, as well as stationary/rotating solutions for the SQG equation. The symmetry results are mainly obtained by calculus of variations and elliptic equation techniques. This is a joint work with Javier Gomez-Serrano, Jia Shi and Yao Yao. (Received January 13, 2020)

1155-35-217 **Fei Wang*** (fwang256@umd.edu), 4176 Campus Drive - William E. Kirwan Hall, College Park, MD 20740, and **Igor Kukavica** and **Vlad Vicol**. *The inviscid limit for the Navier-Stokes equations with data analytic only near the boundary.*

We address the inviscid limit for the Navier-Stokes equations in a half space, with the initial datum that is analytic only close to the boundary of the domain, and has finite Sobolev regularity in the complement. We prove that for such data the solution of the Navier-Stokes equations converges in the vanishing viscosity limit to the solution of the Euler equation, on a constant time interval. (Received January 13, 2020)

1155-35-238 **John K Hunter**, **Jingyang Shu** and **Qingtian Zhang*** (qingtian.zhang@mail.wvu.edu). *SOLUTIONS OF GSQG FRONT PROBLEMS.*

We consider a family of patch-like solutions of generalized surface quasi-geostrophic (GSQG) equation, where the patch may be unbounded. We derive the equations of the contour dynamics under different geometrical situations and prove that the initial value problems have unique local smooth solutions. Under a smallness assumption on the initial data, with the help of the dispersive estimate, we are able to prove the global existence of the solutions for SQG front problem. This is a joint work with John Hunter and Jingyang Shu. (Received January 14, 2020)

1155-35-241 **Trinh Nguyen*** (txn5114@psu.edu), Penn State. *Vanishing viscosity limit for the vortex-wave system.*

We discuss the vanishing viscosity limit of 2D Navier-Stokes when the initial vorticity is a sum of point vortices and a smooth regular part that is compactly supported away from the point vortices. We prove that for a time interval independent of viscosity, the solution of Navier-Stokes converges to the vortex-wave system, which was derived by Marchioro and Pulvirenti in the early 90s. (Received January 17, 2020)

1155-35-245 **Wei-Jian Bo*** (bowj13@lzu.edu.cn), **Guo Lin** and **Yuan-Wei Qi**. *Propagation dynamics of a time periodic diffusion equation with degenerate nonlinearity.* Preliminary report.

This paper is on study of traveling wave solutions and asymptotic spreading of a class of time periodic diffusion equations with degenerate nonlinearity. The asymptotic behavior of traveling wave solutions is investigated by using auxiliary equations and a limit process. In addition, the monotonicity and uniqueness, up to translation, of

traveling wave solution with critical speed are determined by sliding method. Finally, combining super and sub-solutions and the stability of steady states, some sufficient conditions on asymptotic spreading are given, which indicates that the success or failure of asymptotic spreading are dependent on the degeneracy of nonlinearity as well as the size of compact support of initial value. (Received January 20, 2020)

1155-35-266 **Yu Deng**, **Andrea Nahmod** and **Haitian Yue*** (haitiany@usc.edu). *Invariant Gibbs measures and global strong solutions for 2D nonlinear Schrödinger equations.*

In this talk, I'll present our result of the long-standing problem of proving almost sure global well-posedness (i.e. existence with uniqueness) for the periodic nonlinear Schrödinger equation (NLS) in 2D on the support of the Gibbs measure, for any (defocusing and renormalized) odd power nonlinearity. Consequently, we get the invariance of the Gibbs measure. This is achieved by a new method we call *random averaging operators* method, which precisely captures the implicit randomness structure of the high-low interactions. This is work with Yu Deng (USC) and Andrea Nahmod (UMass Amherst). (Received January 16, 2020)

1155-35-274 **Jeff Borggaard**, **Nathan Glatt-Holtz** and **Justin Krometis*** (jkrometis@vt.edu), Wright House, RM 202, Virginia Tech, 765 West Campus Dr, Blacksburg, VA 24061. *On Bayesian Consistency for Flows Observed Through a Passive Scalar.*

We consider the statistical inverse problem of estimating a background fluid flow field $\mathbf{v}$ from the partial, noisy observations of the concentration θ of a substance passively advected by the fluid, so that θ is governed by the partial differential equation

$$\frac{\partial}{\partial t}\theta = -\mathbf{v}\cdot\nabla\theta + \kappa\Delta\theta$$

on $[0,T]\times\mathbb{T}^2$. The initial condition and diffusion coefficient κ are assumed to be known and the data consist of point observations of θ corrupted by additive, i.i.d. Gaussian noise. We adopt a Bayesian approach to this estimation problem and establish that the inference is consistent, i.e., that the posterior measure identifies the true background flow as the number of scalar observations grows large. Since the inverse map is ill-defined for some classes of problems even for perfect, infinite measurements, multiple experiments (initial conditions) are required to resolve the true fluid flow. Under this assumption, suitable conditions on the observation points, and given support and tail conditions on the prior measure, we show that the posterior measure converges to a Dirac measure centered on the true flow as the number of observations goes to infinity. (Received January 16, 2020)

1155-35-277 **Wesley Hamilton*** (wham@live.unc.edu), **Jeremy L. Marzuola** and **Hau-tieng Wu**. *On the behavior of 1-Laplacian Ratio Cuts on nearly rectangular domains.*

Given a connected set $\Omega_0 \subset \mathbb{R}^2$, define a sequence of sets $(\Omega_n)_{n=0}^{\infty}$ where Ω_{n+1} is the subset of Ω_n where the first eigenfunction of the (properly normalized) Neumann $p-$Laplacian $-\Delta^{(p)}\phi = \lambda_1|\phi|^{p-2}\phi$ is positive (or negative). For $p=1$, this is also referred to as the Ratio Cut of the domain. We conjecture that, unless Ω_0 is an isosceles right triangle, these sets converge to the set of rectangles with eccentricity bounded by 2 in the Gromov-Hausdorff distance as long as they have a certain distance to the boundary $\partial\Omega_0$. We establish some aspects of this conjecture for $p=1$ where we prove that (1) the 1-Laplacian spectral cut of domains sufficiently close to rectangles of a given aspect ratio is a circular arc that is closer to flat than the original domain (leading eventually to quadrilaterals) and (2) quadrilaterals close to a rectangle of aspect ratio 2 stay close to quadrilaterals and move closer to rectangles in a suitable metric. We also discuss some numerical aspects and pose many open questions. (Received January 16, 2020)

1155-35-280 **Yuan Gao*** (yg86@duke.edu). *Nolocal equation/systems: dislocation dynamics and Dyson Brownian motion.*

Dislocations are important line defects in crystalline materials. In this talk, I will first talk about the static Peierls-Nabarro (PN) models for a single straight/curved dislocation line, which can be reduced to a Ginzburg-Landau equation/systems involving "anisotropic half-Laplacian". The existence of global minimizers and the exponential relaxation of dynamic solutions to the global minimizer (uniquely determined) will be discussed. Second, the mean field limit of N straight dislocations (with global interactions between each other) leads to the Dyson equation, which is the trace equation of the complex Burgers equation with a force term on the upper half complex plane. The global weak solution and the convergence (after space-time rescaling) to the steady dislocation density function satisfying Wigner's semicircle law will be discussed. (Received January 16, 2020)

1155-35-287 **Mickaël D. Chekroun**, **Honghu Liu*** (hhliu@vt.edu), **James C. McWilliams** and **Shouhong Wang**. *From approximation of random invariant manifolds to closure of stochastic PDEs.*

The modeling of physical phenomena oftentimes leads to PDEs that are usually nonlinear and can also be subject to various uncertainties. Solutions of such equations typically involve multiple spatial and temporal scales, which can be numerically expensive to fully resolve. On the other hand, for many applications, it is often some large-scale features of the solutions that are of interest. The closure problem of a given PDE system seeks essentially for a smaller system that governs to a certain degree the evolution of such large-scale features, in which the small-scale effects are modeled through various parameterization schemes.

In this talk, we discuss an approach for this parameterization problem by adopting a variational framework. We will show that efficient parameterizations can be explicitly determined as parametric deformations of invariant manifolds. The minimizers are objects, called the optimal parameterizing manifolds, that are intimately tied to the conditional expectation of the original system. We will highlight a simple semi-analytic approach to determine such manifolds based on backward-forward auxiliary systems. The approach will be illustrated on the Kuramoto-Sivashinsky equation and a stochastic Burgers equation. (Received January 16, 2020)

1155-35-352 **Adam Nowak**, **Peter Sjögren** and **Tomasz Z. Szarek*** (tzs7@math.rutgers.edu). *Sharp estimates of the spherical heat kernel.*

The classical spherical heat kernel is an important object in analysis, probability and physics, among other fields. It is the integral kernel of the spherical heat semigroup and thus provides solutions to the heat equation based on the Laplace-Beltrami operator on the sphere. It is also a transition probability density of the spherical Brownian motion.

In this talk we prove sharp two-sided global estimates for the heat kernel associated with a Euclidean sphere of arbitrary dimension. If time permits, we will present a generalization of this result to the compact rank-one symmetric spaces.

The talk is based on joint papers with Adam Nowak and Peter Sjögren. (Received January 18, 2020)

1155-35-353 **J Goddard**, **A Muthunayake** and **R Shivaji*** (shivaji@uncg.edu), Dept of Mathematics & Statistics, University of North Carolina at Greensboro, Greensboro, NC 27412. *Modeling the effects of interaction-mediated dispersal on coexistence.*

We analyse positive solutions to the steady state reaction diffusion equation:

$$\begin{cases} -\Delta u = \lambda u(1-u);\ \Omega \\ -\Delta v = \lambda r v(1-v);\ \Omega \\ \frac{\partial u}{\partial \eta} + \sqrt{\lambda} g(v) u = 0;\ \partial\Omega \\ \frac{\partial v}{\partial \eta} + \sqrt{\lambda} h(u) v = 0;\ \partial\Omega \end{cases}$$

where λ and r are positive parameters, Ω is a bounded domain with smooth boundary $\partial\Omega$ and $g, h \in C^1(\mathbb{R}, \mathbb{R}^+)$ are decreasing functions. We will discuss existence results via the method of sub-supersolutions. (Received January 18, 2020)

1155-35-394 **Logan Stokols*** (lstokols@math.utexas.edu) and **Alexis Vasseur**. *SQG on Bounded Domains.*

The SQG equation on $\mathbb{R}^2$ was shown in the late '00s to be well posed with smooth solutions. Recently, Constantin and Ignatova proposed a model for SQG on bounded open subsets of $\mathbb{R}^2$, defined in terms of the Dirichlet Laplacian. This model is particularly complex because it involves a nonlocal operator on a bounded domain.

We will discuss this model, including existence and interior regularity, and our own recent proof (co-authored with Alexis Vasseur) of continuity at the boundary. (Received January 19, 2020)

1155-35-443 **Chenjie Fan*** (cjfanpku@gmail.com), University of Chicago, Math Department, 5734 S University Ave, Chicago, IL 60615, and **Weijun Xu**. *On wellposedness of stochastic NLS.*

We will discuss some recent progress on the wellposedness for the mass critical stochastic NLS. Joint work with Weijun Xu (Oxford). (Received January 20, 2020)

1155-35-444 **Yongki Lee***, Department of Mathematical Sciences, Statesboro, GA 30458. *Wave breaking conditions for the traffic flow models and the Whitham-type equation.*

In this talk, we discuss threshold conditions for wave breaking in a class of non-local conservation laws. We review various nonlocal traffic flow models with look-ahead interactions and the Whitham-type equation. We are interested in the local and global well-posedness of these models. The goal is to understand whether smooth solutions persist in all time, or there is a finite time singularity formation. We discuss a sharp critical threshold

condition on the initial data which distinguishes the global smooth solutions and finite time wave break-down. Some work in this talk is a joint work with Dr. Changhui Tan(University of South Carolina). (Received January 20, 2020)

37 ► Dynamical systems and ergodic theory

1155-37-37 **Vaibhav Gadre** and **Joseph Maher*** (joseph.maher@csi.cuny.edu). *Random mapping classes have generic foliations.*

A pseudo-Anosov element of the mapping class group determines a quadratic differential, which lies in the principal stratum of Teichmuller space if all zeroes are simple, equivalently, if the corresponding foliations have trivalent singularities. We show that this occurs with asymptotic probability one for random walks on the mapping class group, and furthermore, the hitting measure on the boundary gives weight zero to foliations with saddle connections. This is joint work with Vaibhav Gadre. (Received November 19, 2019)

1155-37-263 **CUNY-Hunter R College***, 695 Park Ave, Department of Mathematics & Statistics, New York, NY 10065. *Unique ergodicity and exponential mixing for the damped-driven stochastic KdV equation.*

We discuss a proof for the uniqueness and regularity of the invariant measure for the damped-driven stochastic Korteweg-de Vries equation, where the noise is additive and sufficiently non-degenerate. It is shown that a simple, but versatile control strategy, typically employed to establish exponential mixing for strongly dissipative systems, can nevertheless be applied in this weakly dissipative setting to obtain unique ergodicity, albeit without mixing rates. Under the assumption of large damping, however, exponential mixing can be recovered. Time permitting, application of this approach to other weakly dissipative systems, such as the damped-driven Nonlinear Schrodinger equation will also be discussed. This is joint work with Nathan Glatt-Holtz and Geordie Richards. (Received January 15, 2020)

1155-37-342 **Jane Wang*** (wangjan@iu.edu) and **Sunrose Shrestha**. *Statistics of Square-Tiled Surfaces: Symmetry and Short Loops.*

Square-tiled surfaces are an important class of translation surfaces that are formed by gluing together unit-area squares. The square torus is the simplest square-tiled surface and has many nice symmetry properties that sometimes also occur for other square-tiled surfaces. In this talk, we will introduce some of these nice symmetry properties and discuss combinatorial problems relating to how prevalent these properties are among various strata of square-tiled surfaces. (Received January 18, 2020)

1155-37-398 **James Belk**, **Justin Lanier*** (jlanier8@gatech.edu), **Dan Margalit** and **Rebecca R. Winarski**. *Two pullback maps on Teichmüller spaces.*

Thurston proved a classification theorem for post-critically finite branched covers of spheres by defining and analyzing a certain pullback map on Teichmüller spaces. In recent joint work with Belk, Margalit, and Winarski, we define and analyze a different pullback map on Teichmüller spaces that respects a certain simplicial structure and which yields a recognition algorithm for the class of branched covers called topological polynomials. I will discuss our approach and results and will in particular highlight these different pullback maps and some questions that arise in comparing them. (Received January 19, 2020)

1155-37-521 **Abdullah M Abu-Rqayiq*** (abdullah.aburqayiq@tamucc.edu), 6300 Ocean DR, Texas A&M University-Corpus Christi, Corpus Christi, TX 78412. *Dimorphisms of a Lotka-Volterra Model by Singularity Theory.*

In this talk, we present to classify the singularity of a Lotka-Volterra competitive model with a Gaussian competition function and non-Gaussian carrying capacity functions. These functions need not be completely different to affect the adaptive dynamics of the model. For instance, it will be seen how ostensibly similar models can actually give rise to quite different behaviors due to their properties under unfolding. the use of Gaussian-like carrying capacity functions can also show the sensitivity of the model to assumptions on the carrying capacity function's shapes. The classification is achieved using singularity theory of fitness functions under dimorphism equivalence. We also investigate the effect of the presence of unfolding and bifurcation parameters on the evolution of the system near its singular points. Particularly, we study the adaptive dynamics of the system near the singularity by focusing on ESS and CvSS types, and dimorphisms. Mutual invasibility plots are used to show regions of coexistence. (Received January 21, 2020)

39 ► Difference and functional equations

1155-39-71 **Murat Adivar*** (madivar@uncfsu.edu), 1200 Murchison Rd, Fayetteville, NC 28301, and **Youssef N. Raffoul** (yraffoul1@udayton.edu), Department of Mathematics, University of Dayton, Dayton, OH 45469. *Continuation and Boundedness of Volterra Integral Equations on time scales.*

In this study, we apply the concept of resolvent that was developed in our paper [Adıvar, M. and Raffoul, Y. N., Appl. Math. Comput., (273), 258–266, 2016] for vector Volterra integral dynamic equations on time scales and show the existence and boundedness of solutions. The resolvent is an abstract term that makes it difficult, if not impossible, to make efficient use of it. However, with the help of Lyapunov functionals and variation of parameters, we will be able to verify all the conditions that are related to the resolvent. This paper does not only provide generalized results over the time scales but also completely new results even for well-known particular time scales. All results of this paper are new for the summation equations while some results are new for integral equations in the continuous case. (Received December 18, 2019)

1155-39-197 **Candace M Kent** (cmkent@vcu.edu) and **David M Chan*** (dmchan@vcu.edu). *Parameter Exploration of an Epilepsy Model.* Preliminary report.

In this talk we will examine a model for neuron excitation and its role in epilepsy. In particular we explore the model's parameter space for complex behavior. We construct bifurcation diagrams and phase portraits in which seizures are frequently initiated and terminated. (Received January 13, 2020)

1155-39-300 **John Dale Luebking*** (jluebking1@udayton.edu), 1512 Town Hall Rd, Beavercreek, OH 45432-2552. *Neutral Volterra Difference Equations of Advanced Type.*

The inversion of a perturbed difference operator may yield the sum of a contraction and a compact operator. In this talk, we consider a neutral difference equation, we add and subtract a linear term and do the proper inversion to get what we call, Neutral Volterra Difference Equations of Advanced Type. Then we use Krasnoselskii fixed point theorem to study existence of solutions. (Received January 17, 2020)

1155-39-410 **Kimberly E. Howard*** (khowar43@students.kennesaw.edu). *Existence of solutions for a second order discrete boundary value problem with mixed periodic boundary conditions.*

The main result of this talk is to show a second order mixed periodic boundary value problem with inputs from a Banach space admits multiple solutions. We obtain this conclusion by using critical point theory along with functional analysis and linear algebra. Then we close with a quick example that can be applied to the boundary value problem to show the main result is possible. (Received January 19, 2020)

42 ► Fourier analysis

1155-42-23 **Betsy Stovall*** (stovall@math.wisc.edu). *An inverse problems approach to some questions arising in harmonic analysis.*

One strategy for developing a proof of a claimed theorem is to start by understanding what a counter-example should look like. In this talk, we will discuss a few recent results in harmonic analysis that utilize a quantitative version of this approach. A key step is the solution of an inverse problem with the following flavor. Let $T : X \to Y$ be a bounded linear operator and let $0 < a \leq \|T\|$. What can we say about those functions $f \in X$ obeying the reverse inequality $\|Tf\|_Y \geq a\|f\|_X$? (Received November 01, 2019)

1155-42-137 **Kasso A Okoudjou*** (okoudjou@umd.edu), Massachusetts Institute of Technology, Department of Mathematics, 77 Massachusetts Avenue, Cambridge, MA 02139. *An introduction to Gabor analysis.*

In 1946, Dennis Gabor claimed that any square Lebesgue integrable function can be written as an infinite linear combination of time and frequency shifts of the standard Gaussian. Since then, decomposition methods for larger classes of functions or distributions in terms of various elementary building blocs have lead to an impressive body of work in harmonic analysis. For example, Gabor analysis which originated from Gabor's claim is concerned with both the theory and the applications of the approximation properties of sets of time and frequency shifts of a given function. It re-emerged with the advent of wavelets at the end of the last century and is now at the intersection of many fields of mathematics, applied mathematics, engineering, and science. In this talk, I will introduce the fundamental of the theory highlighting some applications and open problems. (Received January 08, 2020)

1155-42-185 **Shaoming Guo** (shaomingguo@math.wisc.edu), **Joris Roos** (jroos@math.wisc.edu), **Andreas Seeger*** (seeger@math.wisc.edu) and **Po-Lam Yung** (polam.yung@anu.edu.au). *A maximal function for families of Hilbert transforms along homogeneous curves.*

Let $H^{(u)}$ be the Hilbert transform along the parabola (t, ut^2) where $u \in \mathbb{R}$. For a set U of positive numbers consider the maximal function $\mathcal{H}^U f = \sup\{H^{(u)} f : u \in U\}$. We obtain (essentially) optimal results for the L^p operator norm of $\mathcal{H}^U$ when $2 < p < \infty$ and close to optimal results when $1 < p \leq 2$. The results are proved for families of Hilbert transforms along more general non-flat homogeneous curves. (Received January 12, 2020)

1155-42-220 **Gennady Uraltsev*** (guraltsev@math.cornell.edu), Malott Hall - Dept of Mathematics, Cornell University, Ithaca, NY 14853. *Uniform Bounds for the Bilinear Hilbert Transform.*

The Bilinear Hilbert Transform is a prototypical modulation invariant multi-linear singular operator.

$$\mathrm{BHT}_\alpha(f_1, f_2)(x) := \int_{\mathbb{R}} \int_{\mathbb{R}} \widehat{f_1}(\xi_1) \widehat{f_2}(\xi_2) 1_{[0,+\infty)}(\xi_1 - \alpha\xi_2) e^{2\pi i(\xi_1+\xi_2)x} d\xi_1 \xi_2$$

It arises in many contexts including Cauchy integrals along Lipschitz curves and in the study of Calderón commutators relevant for semi-linear non-constant-coefficient t ellyptic PDEs.. The bounds

$$\|\mathrm{BHT}_\alpha(f_1, f_2)(x)\|_{L^{p'_3}(\mathbb{R})} \leq C_{p_1,p_2,\alpha} \|f_1\|_{L^{p_1}(\mathbb{R})} \|f_2\|_{L^{p_2}(\mathbb{R})}$$

hold for any $p_{1,2,3} \in (1, +\infty)$ with $\sum_{i=1}^3 p_i^{-1} = 1$.

A longstanding open problem is weather the above bounds holds with a uniform constant in the parameter α. The dyadic analog of this problem has been solved in Oberlin, Thiele '10.

We will give a general overview of the topic. We will then sketch how the framework of outer measure L^p spaces in time-scale-frequency space plays a crucial role in proving the result and how the problem gives useful insights into embeddings of functions into times-scale-frequency space. (Received January 14, 2020)

1155-42-310 **David Beltran*** (dbeltran@math.wisc.edu), **Shaoming Guo**, **Jonathan Hickman** and **Andreas Seeger**. *The circular maximal operator on Heisenberg radial functions.*

A variant of the Stein spherical maximal function defined on the Heisenberg group $\mathbb{H}^n$ was introduced by Nevo and Thangavelu in 1997. Whilst the sharp Lebesgue space estimates for this object are known for $n \geq 2$ (independently obtained by Müller and Seeger and Narayanan and Thangavelu), the case of $\mathbb{H}^1$ remains open and it is currently unknown if the circular maximal function is bounded in $L^p(\mathbb{H}^1)$ for any finite p.

In this talk, we present sharp L^p estimates for this object when restricted to a class of Heisenberg radial functions. Under this assumption, the problem reduces to studying a variable-coefficient version of Bourgain's circular maximal operator on the Euclidean plane which presents a number of interesting singularities: it is associated to a non-smooth curve distribution and, furthermore, fails both the usual rotational curvature and cinematic curvature conditions.

This is joint work with Shaoming Guo, Jonathan Hickman and Andreas Seeger. (Received January 17, 2020)

1155-42-323 **Rui Han**, **Vjekoslav Kovac**, **Michael Lacey**, **Jose Madrid*** (jmadrid@math.ucla.edu) and **Fan Yang**. *Improving estimates for discrete polynomial averages.*

For a polynomial P mapping the integers into the integers, define an averaging operator $A_N f(x) := \frac{1}{N} \sum_{k=1}^N f(x + P(k))$ acting on functions on the integers. We prove sufficient conditions for the ℓ^p-improving inequality

$$\|A_N f\|_{\ell^q(\mathbb{Z})} \lesssim_{P,p,q} N^{-d(\frac{1}{p}-\frac{1}{q})} \|f\|_{\ell^p(\mathbb{Z})}, \qquad N \in \mathbb{N},$$

where $1 \leq p \leq q \leq \infty$. For a range of quadratic polynomials, the inequalities established are sharp, up to the boundary of the allowed pairs of (p, q). For degree three and higher, the inequalities are close to being sharp. In the quadratic case, we appeal to discrete fractional integrals as studied by Stein and Wainger. In the higher degree case, we appeal to the Vinogradov Mean Value Theorem, recently established by Bourgain, Demeter, and Guth. (Received January 17, 2020)

1155-42-338 **Eyvindur A. Palsson** and **Sean R. Sovine*** (sovine5@gmail.com), 460 McBryde Hall, Virginia Tech, 225 Stanger Street, Blacksburg, VA 24061-1026. *The Triangle Averaging Operator in Higher Dimensions.*

Greenleaf and Iosevich introduced the bilinear operator that averages a pair of functions over the set of equilateral triangles in the plane with one vertex at a given point. We study the generalization of this operator that applies to functions on higher dimensional Euclidean space. We obtain non-trivial Lp bounds for this operator. (Received January 18, 2020)

1155-42-397 **Rachel Greenfeld*** (greenfeld@math.ucla.edu). *Translational Tilings.*

We say that a domain F in some group G *tiles by translations* if it is possible to cover (almost) every point in G exactly once by translates of F. The study of translational tilings is over 100 years old and is still very active nowadays.

Suppose F tiles by translations. What can be said about the structure of the tiling? What can be said about the geometric and analytic properties of F? We will discuss these questions, and present some recent developments as well as some open problems. (Received January 19, 2020)

1155-42-405 **Evangelos Dimou*** (ed8bg@virginia.edu), 141 Cabell Dr, Kerchof Hall, Charlottesville, VA 22904, and **Andreas Seeger** (seeger@math.wisc.edu). *On pointwise convergence of Schrodinger means.*

For $a > 0$, we consider the IVP

$$i\partial_t u(x,t) + (-\partial_{xx})^{a/2} u(x,t) = 0, \qquad u(x,0) = f(x); \quad (x,t) \in \mathbb{R} \times \mathbb{R}.$$

The solution is given by the generalized Schrödinger means operator

$$S^a f(x,t) = (2\pi)^{-1} \int_{\mathbb{R}} e^{i(x\xi + t|\xi|^a)} \hat{f}(\xi)\, d\xi.$$

It is a well known result that if $a > 1$, $\lim_{t\to 0} S^a f(x,t) = f(x)$ a.e. for every f in the Sobolev space H^s, if and only if $s \geq 1/4$. However it is conceivable that if the limit is taken along a sequence $t_n \to 0$, the space $H^{1/4}$ can be replaced with a bigger one. We will show that while this is true for some sequences $\{t_n\}$, it fails for others. More precisely, given $s > 0$, we provide a characterization of decreasing, convex sequences $\{t_n\}$ for which

$$\lim_{n\to\infty} S^a f(x,t_n) = f(x) \quad \text{a.e.}$$

holds for all $f \in H^s$. Some related results may be discussed if time permits. (Received January 19, 2020)

1155-42-434 **Neil Lyall*** (lyall@math.uga.edu). *Finite Point Configurations in both the Continuous and Discrete Settings.*

We will discuss some results on pertaining to finite point configurations in both the continuous and discrete settings. (Received January 20, 2020)

1155-42-437 **Alex Iosevich***, 145 Dunrovin Ln, Rochester, NY 14618-4815, and **Akos Magyar** (iosevich@gmail.com), Alex Iosevich, Rochester, NY 14618-4815. *Simplexes in thin sets.*

We will show that in dimensions three and higher, there exists a non-trivial threshold such that all sets of Hausdorff dimension greater than this threshold contains vertices of a non-degenerate simplex with the number of vertices smaller than the ambient dimension. (Received January 20, 2020)

1155-42-438 **Eyvindur Ari Palsson*** (palsson@vt.edu), Department of Mathematics, 225 Stanger Street, Blacksburg, VA 24061-1026, and **Joeun Jung**. *Bounds for singular integral operators motivated by Calderón's commutators.*

The commutators first studied by Calderón

$$\mathcal{C}_A^{(k)} f(x) = p.v. \int_{\mathbb{R}} \frac{1}{x-y} \left(\frac{A(x) - A(y)}{x-y} \right)^k f(y) dy,$$

for $k = 1, 2, \ldots$, were important in the study of the Cauchy integral on Lipschitz curves. By dropping an average in the first commutator one obtains the bilinear Hilbert transform. Finding L^p estimates for the bilinear Hilbert transform was an important open question in harmonic analysis until Lacey and Thiele showed a wide range of L^p estimates in two Annals papers in 1997 and 1999. Dropping two averages in the second commutator yields the trilinear Hilbert transform for which no L^p estimates are known. In this talk we will discuss both L^p bounds and sparse domination for operators obtained by dropping a single average in the Calderón commutators. (Received January 20, 2020)

1155-42-456 **Polona Durcik** and **Vjekoslav Kovac*** (vjekovac@math.hr), Department of Mathematics, Faculty of Science, Bijenicka cesta 30, 10000 Zagreb, Croatia. *A Szemerédi-type theorem for subsets of the unit cube.*

We are interested in arithmetic progressions in positive measure subsets of $[0,1]^d$. After a counterexample by Bourgain, it seemed as if nothing could be said about the longest interval formed by sizes of their gaps. However, Cook, Magyar, and Pramanik gave a positive result for 3-term progressions if their gaps are measured in the ℓ^p-norm for $p \neq 1, 2, \infty$ and the dimension d is large enough. We establish an appropriate generalization of their result to longer progressions. The main difficulty lies in handling a class of multilinear singular integrals associated with arithmetic progressions that includes the well-known multilinear Hilbert transforms, bounds for which still constitute an open problem. As a substitute, we use the previous work of the authors with Thiele,

on power-type cancellation of those transforms, which was, in turn, motivated by a desire to quantify the results of Tao and Zorin-Kranich. The basic outline of our approach is a modification of the one by Cook, Magyar, and Pramanik. It allows us to be sufficiently quantitative and promises useful future applications to related problems. (Received January 20, 2020)

1155-42-467 **Jeremy M Schwend*** (jschwend@math.wisc.edu), 2921 Fish Hatchery Rd Apt 214, Fitchburg, WI 53713. *Optimal $L^p \to L^q$ Estimates for Euclidean Averages Over Prototypical Hypersurfaces in $\mathbb{R}^3$.*

We look at a case study on the effects of a manifold having vanishing curvature, through the convolution operator. We find the precise range of $(\frac{1}{p}, \frac{1}{q})$ for which local averages along graphs of a class of two-variable polynomials in $\mathbb{R}^3$ are bounded (at least in the restricted-weak sense) from L^p to L^q, given the hypersurfaces have Euclidean surface measure. We derive these results for a model class of polynomials bearing a strong connection to the general real-analytic cases. Techniques and ideas used here have since shown promise being applied to other operators, such as the Fourier Restriction operator. (Received January 20, 2020)

1155-42-485 **Shaoming Guo**, **Changkeun Oh**, **Joris Roos*** (jroos@math.wisc.edu), **Po-Lam Yung** and **Pavel Zorin-Kranich**. *Decoupling for two quadratic forms in three variables: a complete characterization.*

The talk will be about a recent work, where we prove decoupling inequalities for all degenerate surfaces of codimension two in R5 given by two quadratic forms in three variables. Together with previous work by Demeter, Guo, and Shi in the non-degenerate case, this provides a classification of decoupling inequalities for pairs of quadratic forms in three variables. (Received January 20, 2020)

1155-42-504 **Xiumin Du***, xdu@umd.edu. *Special Schroedinger solutions.*

I'll present some special Schroedinger solutions. Then discuss what these examples tell about Fourier decay rates of fractal measures and Falconer's distance set problem. (Received January 20, 2020)

1155-42-523 **Polona Durcik*** (durcik@caltech.edu). *On some multi-parameter singular integrals.* Preliminary report.

We discuss L^p bounds for some multi-parameter fiber-wise paraproducts, which generalize the twisted paraproduct studied by Kovač to the multi-parameter setting. Based on joint work in progress with Frédéric Bernicot. (Received January 21, 2020)

1155-42-569 **Dominique Kemp*** (dekemp@iu.edu). *Extending Known $\ell^2(L^p)$ and $\ell^p(L^p)$ Decoupling Theory.*

The celebrated ℓ^2 decoupling theorem of Jean Bourgain and Ciprian Demeter presented a new perspective on a range of problems related to hypersurfaces with nonzero Gaussian curvature, such as exponential sum estimates, additive energy estimates, local smoothing, and counting solutions to Diophantine inequalities. The same authors also extended their theory to the n-dimensional cone. Following their steps, we prove optimal ℓ^2 and ℓ^p decoupling results for a new class of surfaces which have some vanishing curvature. (Received January 21, 2020)

1155-42-576 **Shahaf Nitzan*** (shahaf.nitzan@math.gatech.edu), **Michael Northington** and **Alex Powell**. *Uncertainty principles for Fourier multipliers.*

We study the possible regularity properties of a function w, which has a zero on the d dimensional torus, and who's reciprocal $1/w$ is a $(2, q)$ Fourier multiplier. Several extensions of this problem are also discussed.

We then relate this question to problems of the uncertainty type, determining the best possible time-frequency localization of generators of certain Gabor systems, and generators of certain shift invariant spaces. (Received January 21, 2020)

1155-42-578 **Larry Guth**, **Hong Wang** and **Ruixiang Zhang*** (rzhang347@wisc.edu). *Local smoothing for the wave equation in $2+1$ dimensions.*

Sogge's local smoothing conjecture for the wave equation predicts that the solution to this equation gets smoother when averaged over time. Jointly with Larry Guth and Hong Wang, we recently proved the conjecture in $\mathbb{R}^{2+1}$. I will talk about a sharp square function estimate we proved which implies the local smoothing conjecture in dimensions $2+1$. A key ingredient in the proof is an incidence type theorem. (Received January 21, 2020)

1155-42-608 **Bartosz Langowski*** (balango@iu.edu), Indiana University, Department of Mathematics, 831 East 3rd St., Bloomington, IN 47405. *Spherical maximal function along thin subsets of integers.*

Let us define the sets of the form

$$\mathbb{N}_h = \{\lfloor h(n) \rfloor : n \in \mathbb{N}\},$$

with the function

$$h(x) = x^c L(x),$$

where $L(x)$ is a suitable slowly varying function and $c \geq 1$, one can think that $L(x) = \log x$.

In our joint project with A. Iosevich, M. Mirek and T.Z. Szarek we study the variant of the spherical maximal function along $\mathbb{N}_h$. Let σ_N be a characteristic function of the sphere

$$\sigma_N = \chi_{\{(n_1,n_2,n_3)\in \mathbb{N}_{h_1}\times\mathbb{N}_{h_2}\times\mathbb{N}_{h_3} : n_1+n_2+n_3=N\}}$$

and

$$r(N) = \#\{(n_1, n_2, n_3) \in \mathbb{N}_{h_1} \times \mathbb{N}_{h_2} \times \mathbb{N}_{h_3} : n_1 + n_2 + n_3 = N\}.$$

We define the maximal spherical operator

$$M_\star f(x) = \sup_{N\in\mathbb{N}} \left| \frac{\sigma_N}{r(N)} * f(x) \right|, \qquad x \in \mathbb{Z}^3.$$

As our main result we obtain the estimate of the form

$$\|M_\star f\|_{\ell^p} \lesssim \|f\|_{\ell^p},$$

for the suitable range of p. (Received January 21, 2020)

1155-42-618 **Victor D Lie***, 1932 18th Str., Apt J 204, Bellingham, WA 98225. *Interplay between modulation invariance and curvature.*

In this talk we outline several themes within the harmonic analysis area that rely on two key concepts: modulation invariance and curvature. We investigate various representative problems and develop a new unifying method in approaching them. (Received January 21, 2020)

43 ► Abstract harmonic analysis

1155-43-490 **Mahya Ghandehari*** (mahya@udel.edu), 15 Orchard Rd, Newark, DE 19716, and **Yemon Choi**. *Derivations on the Fourier algebras of non-Abelian groups.*

Fourier transform and its analogues are the cornerstone of classical harmonic analysis. To generalize the concept of Fourier transform to non-Abelian groups, the modern field of non-commutative harmonic analysis was initiated. The broad philosophy here is to employ group representations and operator theory to capture the non-Abelian nature of a group. A major trend in non-commutative harmonic analysis is to investigate function algebras related to Fourier analysis (and representation theory) of non-Abelian groups. The Fourier algebra, which is associated with the regular representation of the ambient group, is a fundamental example of such function algebras. In this talk, we investigate Banach algebraic behavior, in particular derivation theory, of the Fourier algebra for various classes of locally compact (Lie) groups, and show that these function algebras encode the properties of the underlying groups in various ways. (Received January 20, 2020)

45 ► Integral equations

1155-45-326 **Colleen Kirk*** (ckirk@calpoly.edu) and **W. Edward Olmstead**. *Thermal blow-up in a reactive-diffusive medium with superdiffusive properties.*

We investigate the problem of a high-energy source localized within a one-dimensional superdiffusive medium of finite length. The problem is modeled by a fractional diffusion equation with a nonlinear source term. For the boundary conditions, we consider both the case of homogeneous Dirichlet conditions and the case of homogeneous Neumann conditions. We investigate this model to determine whether or not blow-up occurs. It is demonstrated that a blow-up may or may not occur for the Dirichlet case. On the other hand, a blow-up is unavoidable for the Neumann case. (Received January 17, 2020)

46 ► Functional analysis

1155-46-79 **Scott A Atkinson*** (scott.atkinson@ucr.edu) and **Srivatsav Kunnawalkam Elayavalli**. *Ultraproduct embeddings and amenability for tracial von Neumann algebras.*

The separably acting hyperfinite/amenable II_1-factor R has the property that any two embeddings into its own ultrapower are unitarily conjugate. Jung showed in 2009 that the converse holds modulo the Connes Embedding Problem. In this talk we will discuss the recent result that amenability for tracial von Neumann algebras satisfying the Connes Embedding Problem is characterized by the weaker property that any two embeddings into an ultrapower of R are conjugate by unital completely positive maps. Time permitting, we will discuss other recent characterizations of amenability within this context. (Received December 20, 2019)

1155-46-138 **Tim Ferguson**, **Tao Mei*** (tao_mei@baylor.edu) and **Brian Simanek**. *Monotonicity of Markov semigroups and H^∞-calculus.*

I plan to explain a recent discovery of a monotonicity that is satisfied by all subordinated Markov semigroup of operators. For the classical Poisson semigroup P_t generated by the square root of Laplacian, this property simply means that $P_t - \frac{t}{s}P_s$ is positive preserving for any pair of parameters $s > t > 0$. I will explain how this monotonicity helps us to prove a bounded H^∞ -calculus for general "Laplacian" operators on BMO. (Received January 08, 2020)

1155-46-218 **Don Hadwin*** (don@unh.edu), **Qihui Li** and **Junhao Shen**. *A note on amalgamated free products of C*-algebras.* Preliminary report.

We prove a family of results on properties of amalgamated free products of C*-algebras. For example, if A and B are strongly quasidiagonal C*-algebras with a common nuclear subalgebra D such that each of A and B is generated by D and its relative commutant, then the amalgamated free product is quasidiagonal. Similar results hold for the properties: inner quasidiagonal, MF, and RFD. This extends a result of T. Shulman and K. Courtney, where D is assumed to be contained in the centers of A and B. (Received January 14, 2020)

1155-46-248 **Roy Araiza*** (raraiza@purdue.edu), **Adam Dor-On** and **Thomas Sinclair**. *Tensor Products and Categorical Constructions of Matrix Convex Sets.*

Beginning with the work of Choi-Effros, Wittstock, and Effros-Winkler it was realized that there was an intimate relationship between operator systems (self-adjoint unital subspaces of C*-algebras) and compact matrix convex sets. Motivated by the work of Kavruk et al. in the development of the tensor product theory of operator systems, I will present the foundations for a theory of tensor products of matrix convex sets and relate this to the study of non-commutative Choquet theory. (Received January 15, 2020)

1155-46-250 **Mahya Ghandehari*** (mahya@udel.edu), 15 Orchard Rd, Room 501, Newark, DE 19716, and **Hun Hee Lee**, **Jean Ludwig**, **Nico Spronk** and **Ludmyla Turowska**. *Gelfand spectrum of weighted Fourier algebras.*

Weighted Fourier algebras, *a.k.a.* Beurling-Fourier algebras, are analogues of the Beurling algebra in the non-commutative setting. These Banach algebras for general locally compact groups were defined as the predual of certain weighted von Neumann algebras, where a weight on $\widehat{G}$ is defined to be a suitable unbounded operator affiliated with the group von Neumann algebra. In this talk, we present the general definition of a Beurling-Fourier algebra, and discuss how their spectra can be identified. In particular, we determine the Gelfand spectrum of Beurling-Fourier algebras for some representative examples of Lie groups, such as $\mathrm{SU}(n)$ and the Heisenberg group, representing spectra in terms of the complexification of the underlying Lie groups. This talk is based on joint work with Lee, Ludwig, Spronk, and Turowska. (Received January 15, 2020)

1155-46-254 **Zead Mustafa*** (zead@qu.edu.qa), Dept. of Mathematics, Statistics and Physics, Qatar University, Doha, 2713, Qatar, and **Vahid Parvaneh**, **Mohammed Jaradat** and **Zoran Kadelburg**. *Extended Rectangular b-Metric Spaces and Some Fixed Point Theorems for Contractive Mappings.*

In this paper, we introduce the class of extended rectangular b-metric spaces as a generalization of both rectangular metric and rectangular b-metric spaces. In addition, some fixed point results connected with certain contractions are obtained and examples are given to illustrate these results. (Received January 15, 2020)

1155-46-298 **Rajula Srivastava*** (rsrivastava9@wisc.edu), Madison, WI 53715. *Wavelet Systems as Unconditional Bases in Sobolev Spaces.* Preliminary report.

In this talk, we consider for various wavelet systems, the following problem: on which Sobolev spaces does the system form an unconditional basis? We also consider the analogous problem on Triebel-Lizorkin spaces. This

builds upon previous work of Seeger and Ullrich, which established a precise range in the case of the Haar system. (Received January 21, 2020)

1155-46-447 **Kathryn McCormick*** (kamccorm@umn.edu). *Some conditions for a C^*-Morita context to descend to a subcontext.* Preliminary report.

In (Blecher, Muhly, Paulsen, 2000), the authors define two notions of Morita equivalence for (possibly non-selfadjoint) operator algebras, which they call strong Morita equivalence and *cb*-Morita equivalence. The latter equivalence is strictly weaker. In some situations, a *cb*-Morita equivalence can be factored into a strong Morita equivalence composed with a similarity; this is true for an equivalence containing a separable C^*-algebra, for instance. We will describe a few cases in which we can use a *cb*-Morita equivalence of C^*-algebras to more precisely pin down a *cb*-Morita equivalence of their subalgebras. (Received January 20, 2020)

1155-46-448 **Sara Azzali**, **Sarah Browne**, **Maria Paula Gomez Aparicio**, **Lauren C. Ruth*** (lauren.c.ruth@vanderbilt.edu) and **Hang Wang**. *The Baum-Connes correspondence for the pure braid group on 4 strands.* Preliminary report.

We calculate the left-hand side and the right-hand side of the Baum-Connes correspondence for the pure braid group on 4 strands, each side relying on different techniques. (Received January 20, 2020)

1155-46-449 **Ben Hayes**, **David Jekel**, **Brent Nelson*** (brent@math.msu.edu) and **Thomas Sinclair**. *A random matrix approach to absorption in free products.*

Consider a tracial free product von Neumann algebra $\mathcal{M} = \mathcal{P} * \mathcal{Q}$, where $\mathcal{P}$ and $\mathcal{Q}$ can be embedded into $\mathcal{R}^\omega$. Using random matrix theory, one can show that $\mathcal{P}$ contains any von Neumann subalgebra $\mathcal{N} \subset \mathcal{M}$ which intersects $\mathcal{P}$ diffusely and has 1-bounded entropy zero. Here the 1-bounded entropy of $\mathcal{N}$ is a modification of the free entropy dimension that—roughly speaking—measures how many matrix approximations $\mathcal{N}$ has. This quantity is known to be a von Neumann algebra invariant and in particular is zero for all amenable algebras. Consequently, as a corollary of the aforementioned absorption result, one obtains a novel proof of Popa's famous theorem that the generator MASA in a free group factor is maximal amenable. In this talk I will discuss this result, some aspects of its proof, and a few of its consequences. This is based on joint work with Ben Hayes, David Jekel, and Thomas Sinclair. (Received January 20, 2020)

1155-46-520 **Thomas Sinclair*** (tsincla@purdue.edu). *On approximative commutation of matrices.* Preliminary report.

We discuss a possible new approach and experimental results related to the main result in von Neumann's "Approximative Properties of Matrices of High Finite Order." This work is joint with Garrett Mulcahy. (Received January 21, 2020)

1155-46-544 **Christopher Schafhauser*** (cschafhauser2@unl.edu). *On the classification of simple nuclear C*-algebras.*

I will give an overview of some of the recent progress in the classification of simple, nuclear C*-algebras and some applications of these techniques. Much of the talk will be based on joint work with Jose Carrion, James Gabe, Aaron Tikuisis, and Stuart White. (Received January 21, 2020)

1155-46-547 **Rolando de Santiago***, rdesantiago@math.ucla.edu, and **Ben Hayes**, **Dan Hoff** and **Thomas Sinclair**. *Maximal rigid subalgebras of deformations and L^2-cohomology.*

The classification program for von Neumann algebras witnessed remarkable progress which is in large part due to Popa's Deformation/Rigidity theory. Proceeding from where Ben Hayes ended in Part I, we expand on the implications of the existence of maximal rigid algebras, provide concrete examples in the group setting, and describe applications when considering families of deformations as in L^2 rigidity; the latter result lends further support to a conjecture of Peterson and Thom. (Received January 21, 2020)

1155-46-579 **Michael J Nasta*** (michael.nasta@westpoint.edu). *Quantum correlations generated from Jones-Wenzl Projectors.* Preliminary report.

Formulations of Topological Quantum Computing (TQC) often involve the aid of a graphical calculus to simplify complex calculations that would otherwise be computed on Hilbert Spaces. This graphical calculus specifically is derived from the Temperley-Lieb algebra, wherein nonzero idempotent elements known as Jones-Wenzl projectors are an important part of many computations. In this project, we explore the sets of quantum correlations generated by Jones-Wenzl projectors to assess their utility in quantum information theory. Particularly, we consider tensor products of generalized Jones-Wenzl projectors and their relationships to physical systems in the

form of entangled quasiparticles known as anyons in order to demonstrate their utility in quantum information transmission. (Received January 21, 2020)

47 ► Operator theory

1155-47-365 **Min He*** (mhe@kent.edu). *A Smoothness Property of Classical Solutions of Cauchy Problems with Non-Densely Defined Operators.*

This work investigates the differentiability with respect to parameters of the classical solutions of abstract Cauchy problems, and studies the case that the operators of the equations are not densely defined by employing the theory of integrated semigroup. (Received January 19, 2020)

1155-47-507 **Chaoyuan Liu*** (mary.liu@eku.edu), Eastern Kentucky University, Department of Mathematics and Statistics, Richmond, KY 40475. *Operator Spaces in Probabilistic Normed Spaces.*

In M-PN space with a probabilistic bounded set N, we introduce a norm. Under the norm, we prove that the continuous operator spaces, the completely continuous operator space, the strong bounded operator space, and the sub-strong operator space are M-PN spaces. We show that the completely continuous operator space is the closed subspace of the continuous operator space. We also prove that sub-strong bounded operator space is subspace of continuous operator space and the strong bounded operator space is the subspace of sub-strong bounded operator space. (Received January 20, 2020)

1155-47-558 **Anna Duwenig**, **Elizabeth Gillaspy**, **Rachael Norton** and **Sarah A Reznikoff***, sarahrez@ksu.edu, and **Sarah Wright**. *Cartan subalgebras of non-principal twisted groupoid C*-algebras.*

The study of Cartan C*-subalgebras, initiated by Kumjian and Renault several decades ago, has led to numerous advances in the classification and representation theory of C*-algebras. In particular, Renault showed in 2008 that the presence of a Cartan subalgebra guarantees that a C*-algebra has a dynamical representation via a topologically principal "twisted" groupoid. We identify a class of non-topologically principal groupoids that give rise to twisted groupoid C*-algebras admitting Cartan subalgebras—namely subalgebras generated by certain abelian subgroupoids. By Renault's theorem, these algebras have alternative, topologically principal, twisted groupoid models. In the special case that the original groupoid is a discrete group G, we further describe the relationship between G and the groupoid arising from application of Renault's theorem. (Received January 21, 2020)

51 ► Geometry

1155-51-15 **Artur Elezi*** (aelezi@american.edu), Department of Mathematics and Statistics, American University, 4400 Mass Ave. NW, Washington, DC 20016, and **Tony Shaska** (shaska@oakland.edu), 546 Mathematics Science Center, 146 Library Drive, Oakland University, Rochester, MI 48309. *Reduction of Binary Forms Via the Hyperbolic Centroid.*

In this paper we introduce a reduction theory based on the hy- perbolic center of mass, which is different from the reduction introduced by Julia (1917). We show that the zero map via the Julia quadratic is different than the hyperbolic center of mass. Moreover, we discover some interesting formulas for computing the hyperbolic centroid. (Received September 28, 2019)

1155-51-103 **Sunrose Thapa Shrestha*** (sunrose.shrestha@gmail.com). *Statistics of Random Square-tiled Surfaces.* Preliminary report.

Square-tiled surfaces are finite branched covers of the standard square torus, with branching over exactly one point. These can be also thought of as surfaces obtained by taking finitely many Euclidean unit squares, and gluing the edges in parallel pairs. In this talk we will see topological and geometric statistics of random square-tiled surfaces using a simple combinatorial model. (Received January 05, 2020)

1155-51-148 **Lorenzo Ruffoni*** (lruffoni@fsu.edu) and **Stefano Francaviglia**. *Local deformations of branched projective structures: Schiffer variations and the Teichmüller map.*

We consider CP1-structures on closed surfaces of genus at least 2, which are geometric structures locally modelled on the geometry of Möbius transformations of the Riemann sphere. Every Riemann surface admits a CP1-structure by uniformization, and vice versa a CP1-structure induces a complex structure on the surface. In the

unbranched case a CP1-structure is uniquely determined by the underlying complex structure together with a holonomy representation of the fundamental group into PSL(2,C). If we allow branch points, then these structures admit non-trivial holonomy-preserving deformations, and in this talk we consider the problem of understanding if the induced deformation of the underlying complex structure is trivial (per se, or at least infinitesimally), in terms of analytic conditions on the collection of branch points. This is joint work with S. Francaviglia. (Received January 08, 2020)

1155-51-198 **James Gray***, Department of Physics MC0435, 850 West Campus Drive, Blacksburg, VA 24061. *Chern-Simons and Heterotic Superpotentials.*

Compactifications of Heterotic theories on Calabi-Yau manifolds have a Chern-Simons like contribution to their four dimensional superpotential. In this talk I will discuss some aspects of computations and general theory behind understanding this often ignored contribution to the effective theory of such constructions. (Received January 13, 2020)

1155-51-199 **Wei Cui***, Department of Physics MC0435, 850 West Campus Drive, Blacksburg, VA 24061. *Numerical Ricci-Flat Metrics and Curvature Hierarchies in Calabi-Yau Manifolds.*

We will discuss the application of numerical approximations to Ricci-flat metrics in searching for hierarchies of curvature scales on Calabi-Yau manifolds. Knowledge of such hierarchies is central to understanding the regimes of validity of string effective theories. We show that numerical methods can be effective at studying these issues and illustrate our discussion with some explicit examples. (Received January 13, 2020)

1155-51-293 **Yang Xiao*** (yang_xiao@brown.edu), 151 Thayer St, Providence, RI 02912. *Characterizing Covers by Simple Closed Curves.*

Given a finite degree regular cover S' of a surface S, we characterize the covering map by studying the simple closed curves on S' that come from lifts of closed curves on S. Our result shows that given two such covers, suppose any closed curve on S lifts to a simple closed curve on one cover if and only if it does to another, then the covers are equivalent. The proof uses some Teichmuller theory and the curve complex. This represents joint work with Tarik Aougab, Max Lahn, and Marissa Loving. (Received January 16, 2020)

1155-51-362 **Konstantinos Tsouvalas*** (tsouvkon@umich.edu), Department of Mathematics, 2074 East Hall, 530 Church Street, Ann Arbor, MI 48109. *Characterizations of Benoist representations.*

Benoist representations into PSL(n,R) (n>2) is a large class of Anosov representations whose domain groups are of maximum (virtual) cohomological dimension. In this talk, we are going to provide a characterization of Benoist representations in terms of the existence of limit maps. This is joint work with Richard Canary. (Received January 18, 2020)

1155-51-488 **Sara Tukachinsky***, School of Mathematics, IAS, 1 Einstein Drive, Princeton, NJ 08540. *Defining and calculating disk counting invariants via A_∞ algebras.*

Genus zero open Gromov-Witten invariants should count pseudoholomorphic maps from the disk to a symplectic manifold, with boundary conditions in a totally real submanifold and with various constraints on boundary and interior marked points. The presence of boundary leads to bubbling phenomena that pose a fundamental obstacle to invariance. A Fukaya A_∞ algebra associated with the boundary condition L is a deformation of the dg algebra of smooth differential forms on L by pseudoholomorphic disks, with coefficient in a module over $\mathbb{R}$. This language turns out to be useful in open Gromov-Witten theory, as demonstrated in joint works with Jake Solomon. In particular, we develop a way of defining numeric invariants and show that they satisfy some general properties. For the projective space $\mathbb{C}P^n$ with $L = \mathbb{R}P^n$, these properties are sufficient to calculate all invariants explicitly. No prior knowledge of the subject matter will be assumed. (Received January 20, 2020)

52 ► Convex and discrete geometry

1155-52-64 **Mokshay Madiman*** (madiman@udel.edu). *Concentration of information for convex measures.*

We present work with Matthieu Fradelizi and Jiange Li that obtains sharp bounds on the concentration of information content for random vectors drawn from sufficiently convex measures. We also discuss various consequences of these bounds, including non-sharp but much more usable corollaries that clarify the role of dimension and convexity parameter in the strength of concentration. (Received December 13, 2019)

1155-52-86 **Alexander Koldobsky*** (koldobskiya@missouri.edu), **Grigoris Paouris** and **Artem Zvavitch**. *Measure comparison and distance inequalities for convex bodies.*

We prove new versions of the isomorphic Busemann-Petty problem for two different measures and show how these results can be used to recover slicing and distance inequalities. (Received December 29, 2019)

1155-52-113 **Galyna V Livshyts*** (glivshyts6@math.gatech.edu), 686 Cherry st NW, Atlanta, GA 30097. *Maximal perimeter with respect to probability measures.*

We discuss estimates on maximal perimeter of a convex set with respect to probability measures. We show that any unimodule probability measure can be brought to a position such that the maximal perimeter of a convex set is bounded from above by Cn. We also show that the maximal perimeter of a convex set with respect to an isotropic log-convave measure is bounded by Cn^2. Lastly, we provide a very general estimate from below on the maximal perimeter of a convex set, in terms if the thin shell parameter of the measure, and as a corollary, deduce that for any log-convave isotropic measure, there exists a convex set with perimeter of order at least $n^{1/8}$ view of the best currently known bound which is due Lee and Vempala)/ (Received January 06, 2020)

1155-52-119 **Galyna Livshyts*** (glivshyts6@math.gatech.edu), 686 Cherry st NW, Atlanta, GA 30097, and **Alexander Kolesnikov** and **John Hosle**. *On the log-brunn-minkowski conjecture and related questions.*

We discuss the Log-Brunn-Minkowski conjecture of Boroczky, Lutwak, Yang and Zhang, and related questions, and prove several new estimates. Joint work with Hosle and Kolesnikov. (Received January 06, 2020)

1155-52-158 **Matthias Reitzner**, University of Osnabrueck, Osnabrueck, Germany, **Carsten Schuett*** (schuett@math.uni-kiel.de), 2517 Overlook Rd, Apt 18, Cleveland, OH 44106, and **Elisabeth M Werner**. *The convex hull of random points on the boundary of a simple polytope.*

The convex hull of N independent random points chosen on the boundary of a simple polytope in $\mathbb{R}^n$ is investigated. Asymptotic formulas for the expected number of vertices and facets, and for the expectation of the volume difference are derived. This is the first successful attempt of investigations which lead to rigorous results for random polytopes which are neither simple nor simplicial. The results contrast existing results when points are chosen in the interior of a convex set. (Received January 10, 2020)

1155-52-228 **Yiting Cao*** (yxc827@case.edu), 10900 Euclid Ave, Cleveland, OH 44106, and **Victor Glasgo**, **Andreas Kreuml** and **Elisabeth Werner**. *Gravitational illumination bodies (Part II).*

We give a precise description of the gravitational illumination bodies for n-dimensional Euclidean balls and for 2- dimensional polytopes. We determine the behaviour of the volume difference of the Euclidean ball and its gravitational illumination ball. Surprising phenomena happen in dimensions 2 and 3 in the case of the ball.

Based on joint work with Victor Glasgo, Andreas Kreuml and Elisabeth M. Werner. (Received January 14, 2020)

1155-52-244 **Victor Glasgo*** (victor.glasgo@case.edu). *Gravitational illumination bodies.* Preliminary report.

We introduce a new class of convex bodies, the gravitational illumination bodies. We show some of their properties and explore their relation to affine surface area of convex bodies.

Based on joint work with Yiting Cao, Andreas Kreuml and Elisabeth M. Werner. (Received January 14, 2020)

1155-52-261 **Matthieu Fradelizi**, **Mokshay Madiman** and **Artem Zvavitch*** (zvavitch@math.kent.edu), Department of Mathematical Sciences, Kent State University, Kent, OH 44242. *Links between different inequalities on mixed volumes of convex bodies.* Preliminary report.

The notion of mixed volumes and the inequalities involving them play a central role in the modern convex geometry, and have many connections to various other areas of mathematics. The most classical inequalities includes Brunn-Minkowski inequality and more general Alexandrov-Fenchel inequality.

In this talk we will discuss a local version of Alexandrov-Fenchel ineqaulity and its connection to the search for the best constant in a number of geometric inequalities including inequality on the volume of orthogonal projections of convex bodies; isomorphic version of Bezout inequality; approximate submodularity of Minkowski sum; as well as to the property of distribution of roots of Steiner polynomial. (Received January 15, 2020)

1155-52-306 **Giorgos Chasapis*** (gchasap1@kent.edu). *Norms of weighted sums of log-concave random vectors.*

Given two centrally symmetric convex bodies C and K of volume 1 in $\mathbb{R}^n$, we study the multi-integral expression

$$\|\mathbf{t}\|_{C^s,K} := \int_C \dots \int_C \left\| \sum_{j=1}^{s} t_j x_j \right\|_K dx_1 \dots dx_s,$$

where $\mathbf{t} = (t_1, \dots, t_s) \in \mathbb{R}^s$. An old question by V. Milman in the pursuit of a generalized Khintchine-type inequality is whether $\|\mathbf{t}\|_{K^s,K}$ is actually equivalent to the Euclidean norm $\|\mathbf{t}\|_2$, up to a term which is logarithmic in the dimension. We discuss general upper bounds for $\|\mathbf{t}\|_{C^s,K}$ in the case that C is isotropic, that can be further tightened when $C = K$ under certain uniform convexity or cotype assumptions on $(\mathbb{R}^n, \|\cdot\|_K)$. Our approach also yields a new proof of the sharp lower bound on $\|\mathbf{t}\|_{C^s,K}$ provided in an earlier work of Gluskin and Milman. Based on joint work with A. Giannopoulos and N. Skarmogiannis. (Received January 17, 2020)

1155-52-350 **Sergii Myroshnychenko*** (myroshny@ualberta.ca), University of Alberta, CAB 632, Edmonton, Alberta T6G 2G1, Canada. *On visual shapes and non-central sections.*

Assume that Earth is made out of a transparent glass and contains a convex body K in its interior. Let K be seen as a disk from every point on the planet's surface, possibly of different radii. Can one conclude that K is a Euclidean ball? What if it is seen as an ellipse or a polygon?

We discuss related open problems, provide known and recent results that answer all of the questions above, as well as their dual counterparts for non-central sections of convex bodies. (Received January 18, 2020)

1155-52-351 **Sergii Myroshnychenko**, **Kateryna Tatarko*** (tatarko@ualberta.ca) and **Vladyslav Yaskin**. *On a characterization of ellipsoids.*

Many interesting problems in convex geometry are concerned with a unique determination of convex and star bodies. We present an approach to uniquely determine ellipsoids by a set of parameters from dual Brunn-Minkowski theory. This is joint work with S. Myroshnychenko and V. Yaskin. (Received January 18, 2020)

1155-52-371 **Florian Besau*** (florian.besau@tuwien.ac.at) and **Christoph Thäle**. *Asymptotic normality for random polytopes in non-Euclidean geometries.*

In this talk I will present our results that show that the centered and normalized volume of a random polytope, that is, the convex hull of random points (independent and identically distributed) inside a fixed strictly convex and compact set with smooth boundary, converges to a standard normal random variable. This was known to be true in the Euclidean space, and our results now show that it holds true also in non-Euclidean spaces, such as the unit sphere or hyperbolic space.

Our results follow from a central limit theorem for weighted random polytopes in Euclidean space and can also be applied in Lutwak's dual Brunn–Minkowski theory to show asymptotic normality of the dual volumes of random polytopes.

The proof relies on probabilistic tools, in particular a generalization of Stein's method due to Chatterjee, and some geometric estimates for the weighted floating body. (Based on joint work together with Christoph Thäle, arXiv:1909.05607.) (Received January 19, 2020)

1155-52-378 **Ferenc Fodor*** (fodorf@math.u-szeged.hu). *Approximations by generalized random polytopes.*

In the classical model, the convex hull of n i.i.d. uniform random points in a convex body K is a random polytope whose geometric properties have been investigated extensively. We will consider certain generalizations of this model. For example, if one takes the intersection of all closed unit balls containing the n points, then one obtains a random convex ball-polytope in K. The asymptotic approximation properties such random ball-polytopes have recently been studied both in the plane and in arbitrary dimension revealing phenomena that cannot be observed in the classical model. A further natural generalization is when one takes the intersection of all translates of a suitable fixed convex body L that contain the random points. Such an object is a random L-polytope. We will consider some asymptotic properties of such objects as the number of generating points tends to infinity under different geometric assumptions on K and L.

Some of the results in this talk are joint with D. Papvári (University of Szeged, Hungary) and V. Vígh (University Szeged, Hungary). (Received January 19, 2020)

1155-52-381 **Gil Kur*** (gilkur@mit.edu), Cambridge, MA 02139. *Estimation of convex sets in high dimensions.*

We are going to give a simple proof, based on both geometrical and statistical methods, that the task of ϵ-estimating a convex set $K \subset \mathbb{R}^d$ with volume one. That is given $X_1, \dots, X_n$ i.i.d. samples from a convex set of volume one, find a set such that $|\hat{K} \Delta K| \leq \epsilon$.

We show that for $\epsilon > 10^{-4}$, it requires that $n \geq c_1 \epsilon^{(d+1)/2}$. When $\epsilon = 10^{-4}$, This results improve an earlier result of L. Rademacher that shows based on a information theoretic argument that shows a weaker lower bound of $2^{c\sqrt{d}}$, where this bound gives $2^{c_1 d}$. (Received January 19, 2020)

1155-52-499 **Han Huang**, **Boaz Slomka**, **Tomasz Tkocz** and **Beatrice-Helen Vritsiou*** (vritsiou@ualberta.ca). *On Hadwiger's covering conjecture.*

A central problem in Discrete Geometry, Hadwiger's covering conjecture, asks to find the smallest integer $N(n)$ with the property that every convex body in $\mathbb{R}^n$ can be covered by at most $N(n)$ translates of its interior.

We will discuss connections with Asymptotic Convex Geometry and measure concentration, as well as with entropic methods, that allow for at least a subexponential improvement to the long-standing general upper bounds of Rogers for $N(n)$.

This is joint work with Han Huang, Boaz Slomka and Tomasz Tkocz. (Received January 20, 2020)

1155-52-522 **Andrea Colesanti**, **Michael Roysdon*** (mroysdon@kent.edu), **Sudan Xing** and **Christos Saroglou**. *A functional version of the L_p-addition and some related inequalities.* Preliminary report.

In the 1960's Firey proposed an extension of the usual Minkowski following sense: given any convex bodies K, L, each containing the origin, $\alpha, \beta > 0$, and any $p \geq 1$, define the L_p-sum of K and L to be hp

$$h^p_{\alpha \cdot_p K +_p \beta \cdot_p L} = \alpha h^p_K + \beta h^p_L,$$

where h_K and h_L denote the support functions of K and L, respectively. In the early 2010s Böröczky, Lutwak, Yang and Zhang extended Firey's addition to the setting $0 \leq p < 1$ by describing it in terms of the following Wülff shape:

$$\alpha \cdot_p K +_p \beta \cdot_p L = \bigcap_{y \in \mathbb{R}^n} \{x \colon x \cdot y \leq (\alpha h_K(y)^p + \beta h_L(y)^p)^{1/p}\}.$$

From these objects many important questions have arisen including the famous *log*-Brunn-Minkowski conjecture.

In this talk we propose an extension of the L_p-addition to a class of s-concave function that naturally embeds the L_p-Minkowski structure and describe some questions that arise as a result. This is part of an ongoing work A. Colesanti, C. Saroglou, and S. Xing. In the case, $p = 1$, we also describe some functional Rogers-Shephard type inequalities that arose as part of a joint work with D. Alonso, M. Cifre, J. Yepes, and A. Zvavitch. (Received January 21, 2020)

53 ► Differential geometry

1155-53-230 **Falk Hassler*** (falk@fhassler.de). *Generalised Quotients.*

Generalised Scherk-Schwarz reductions are a powerful tool to construct consistent truncations in generalised geometry. Recently, it turned out that they are also closely related to Poisson-Lie T-duality. However, the most general form of Poisson-Lie T-duality, the dressing coset construction, can not be implemented in terms of a generalised Scherk-Schwarz ansatz. I will show that implementing it in generalised geometry leads to a natural extension of the generalised Scherk-Schwarz ansatz which comes with many new features: 1) Partial or full breaking of SUSY which allows to find many new examples of generalised Kaehler or Calabi-Yau Manifolds. 2) Singular backgrounds with localised sources. 3) Localised vector multiplets while still resulting in consistent truncations. (Received January 14, 2020)

1155-53-428 **Daniele Alessandrini*** (daniele.alessandrini@gmail.com), Columbia University, 2990 Broadway, New York, NY 10027. *Non commutative cluster coordinates for Higher Teichmuller Spaces.*

Fock-Goncharov found a beautiful structure of cluster variety on the decorated Hitchin components of punctured surfaces, generalizing Penner's decorated Teichmüller Theory. This is an algebraic theory based on the notion of positivity. Hitchin components are an example of Higher Teichmüller Spaces, and the spaces of Maximal Representations are another example. In this latter case, we found new coordinates on these Higher Teichmüller Spaces that give them a structure of non-commutative cluster varieties, in the sense defined by Berenstein-Rethak. This is joint work with Guichard, Rogozinnikov and Wienhard. (Received January 20, 2020)

1155-53-617 **Nathaniel Eldredge*** (neldredge@unco.edu), **Maria Gordina** and **Laurent Saloff-Coste**. *Uniform estimates for left-invariant Brownian motion on compact Lie groups via volume doubling.*

A classical connection between geometry and probability arises on a compact Riemannian manifold (M, g), where geometric properties (e.g. Ricci curvature) influence the probabilistic properties of the Brownian motion B_t on M (e.g. its rate of convergence to equilibrium). If the metric g is allowed to vary, then the quantitative properties of B_t vary as well, but maybe not as much as one might think. We study the case where $M = K$ is a fixed compact Lie group and the metric g is left-invariant, and consider when certain properties of the corresponding Brownian motions and their heat kernels may be controlled uniformly over all such g, despite the failure of uniform curvature bounds. The key tool is a notion of *uniform volume doubling*. We show that this works on the three-dimensional special unitary group $K = \mathrm{SU}(2)$, and discuss a conjecture, motivated by ideas from sub-Riemannian geometry, that this phenomenon may hold much more broadly. (Received January 21, 2020)

54 ► General topology

1155-54-110 **Eleni Panagiotou***, 415 EMCS Building, 615 McCallie Ave, Chattanooga, TN 37403, and **Louis H. Kauffman**. *Knot polynomials of open and closed curves.*

In this manuscript we introduce a method to measure entanglement of curves in 3-space that extends the notion of knot and link polynomials to open curves. We define the bracket polynomial of curves in 3-space and show that it has real coefficients and is a continuous function of the chain coordinates. This is used to define the Jones polynomial in a way that it is applicable to both open and closed curves in 3-space. For open curves, it has real coefficients and it is a continuous function of the chain coordinates and as the endpoints of the curve tend to coincide, the Jones polynomial of the open curve tends to that of the resulting knot. For closed curves, it is a topological invariant, as the classical Jones polynomial. We show how these measures attain a simpler expression for polygonal chains and provide a finite form for their computation in the case of chains of 3 and 4 edges. (Received January 06, 2020)

1155-54-331 **Hannah Turner***, hannahturner@math.utexas.edu. *Orderability, knots, and SL(2,R) representations.*

Some groups can be given an ordering which is respected by the group operation. The question of when a 3-manifold group is orderable is intimately related to topological properties of the manifold. Three-manifold groups with a SL(2,R) representation which has trivial Euler class are orderable. We exploit this fact and techniques of Hu and Tran to determine for any branched cyclic cover of a family of hyperbolic knots, whether its fundamental group is orderable. (Received January 18, 2020)

1155-54-421 **Courtney M Thatcher*** (cthatcher@pugetsound.edu), Department of Math and CS, University of Puget Sound. *Math, Geography, and Districting: Using Topological Data Analysis to Assess Voting Districts.* Preliminary report.

In this talk will discuss some of the work done by the Spatial Model and Electoral Districting REU housed at University of Washington Tacoma and University of Puget Sound. Students worked on two projects related to the construction of districts based upon how individuals experience, come to know, and move through the spaces in which they live. One of the projects used a tool from topological data analysis (TDA) to determine communities of interest, and the other used travel-time to assess outliers in current districts and to create new districts. We will discuss both projects briefly and then dive deeper into the first one. In particular, the mapper algorithm from TDA provides a method for obtaining clusters not found using traditional clustering techniques. The mapper algorithm and its benefits will be described, and some of its implications for districting will be discussed. (Received January 20, 2020)

55 ► Algebraic topology

1155-55-94 **Nitu Kitchloo***, Johns Hopkins University, Baltimore, MD. *Link Homologies for Homotopy Theorists.*

In my talk, I will describe a geometric equivariant stable homotopy type that recovers several known link homology theories (Khovanov homology, sl(n)-link homology, Khovanov-Rozansky homology etc) when one applies suitable equivariant cohomology theories to the homotopy type. In addition, most of the algebraic structure known in the context of link homologies (like spectral sequences connecting different homology theories) will be

shown to arise in a natural manner that is very familiar to homotopy theorists. The geometry alluded to above refers to the framework of symmetry breaking for principal bundles on a circle with marked points. More on that if time permits. (Received January 02, 2020)

1155-55-95 **Duncan A Clark*** (clark.1843@osu.edu), 231 W 18th Ave, Columbus, OH 43210.
Homotopy coherent operads and the Goodwillie derivatives of the identity functor.

By work of Ching it is known that $\partial_* I$, the symmetric sequence of Goodwillie derivatives of the identity functor in spaces, is an operad. In this talk, we offer a new approach to this result by showing that $\partial_* I$ is a homotopy coherent operad: the structure of which we encode via a certain $\mathbf{N}$-colored operad which is a "fattened-up" version of the colored operad whose algebras are operads, and also offer new algebraic descriptions for both as $\mathbf{N}$-colored operads *with levels*. In addition, we show that $\partial_* I$ naturally acts on the suspension spectrum associated to the completion of a 1-connected space with respect to the $(\Sigma^\infty, \Omega^\infty)$-adjunction.

One benefit of our approach is that our constructions carry over to the category of algebras over a reduced operad, $\mathcal{O}$, in spectra. Using these techniques, we answer positively the conjecture that $\partial_* I_{\mathsf{Alg}_\mathcal{O}}$ should be weakly equivalent to $\mathcal{O}$ (as homotopy-coherent operads), as well as show that the TQ-completion of a 0-connected $\mathcal{O}$-algebra inherits an action of $\partial_* I_{\mathsf{Alg}_\mathcal{O}}$. (Received January 21, 2020)

1155-55-201 **Nikolas Schonsheck*** (schonsheck.2@osu.edu). TQ*-completion of certain fibration sequences.*

By considering algebras over an operad $\mathcal{O}$ in one's preferred category of spectra, we can encode various flavors of algebraic structure (e.g. commutative ring spectra). Drawing intuition from singular homology of spaces and Quillen homology of rings, topological Quillen (TQ) homology is a naturally occurring notion of homology for these objects with analogies to both singular homology and stabilization of spaces.

For a given $\mathcal{O}$-algebra X, there is a canonical way (following Bousfield-Kan) to "glue together" iterates $\mathsf{TQ}^n(X)$ of the TQ-homology spectrum of X to construct "the part of X that TQ-homology sees," namely its TQ-completion. We then ask, "When can X be 'recovered from' $\mathsf{TQ}(X)$ in this way?"

Bousfield-Kan consider the analogous question in spaces and conclude that all nilpotent spaces are weakly equivalent to their homology completion. The key technical maneuver of their proof involves showing that certain fibration sequences are preserved by completion. In this talk, we will review the construction of TQ-homology and discuss certain types of fibration sequences of $\mathcal{O}$-algebras which are preserved by TQ-completion. (Received January 21, 2020)

1155-55-303 **Christopher J. R. Lloyd*** (cjl8zf@virginia.edu). *Chromatic fixed point theory, and the Balmer spectrum of D_8-equivariant stable homotopy.*

Given a finite group G with subgroup H, one can ask for what pairs (n, m) does there exist a finite G-spectrum with geometric G-fixed points of chromatic type n and geometric H-fixed points of chromatic type m (localized at a prime p). P. Balmer and B. Sanders reduced understanding the spectrum of the tensor triangulated ideals of the equivariant stable homotopy category of a fixed group G to classifying such pairs. Recently, this question was completely answered when G is abelian by T. Barthel, M. Hausmann, N. Naumann, T. Nikolaus, J. Noel, and N. Stapleton. In my talk, I will show how to completely classify these pairs when G is the dihedral group of order 8, using a new technique for constructing examples. This is joint with Nick Kuhn. (Received January 17, 2020)

1155-55-313 **Anna Marie Bohmann*** (am.bohmann@vanderbilt.edu), Department of Mathematics, 1326 Stevenson Center, Nashville, TN 37240, and **Christy Hazel**, **Jocelyne Ishak**, **Magdalena Kedziorek** and **Clover May**. *Rational equivariant K-theory.* Preliminary report.

Since early work of Serre, we've understood that rationalization "algebraicizes" topology. One manifestation of this is that many categories of rational spectra have an algebraic model—a model category built out of algebraic data to which the category is Quillen equivalent. For example, rational spectra are equivalent to rational chain complexes. Recent work by Barnes–Greenlees–Kedziorek gives an algebraic model for naive-commutative G-equivariant ring spectra when G is a finite group. We use this model to understand rational equivariant K-theory as a naive-commutative ring spectrum and in particular prove a uniqueness result. Time permitting, we may address further levels of commutativity using recent work of Wimmer. (Received January 17, 2020)

1155-55-347 **Jocelyne Ishak*** (jocelyne.ishak@vanderbilt.edu). *Rigidity of the $K(1)$-local stable homotopy category.*

In some cases, it is sufficient to work in the homotopy category $\mathrm{Ho}(\mathcal{C})$ associated to a model category $\mathcal{C}$, but looking at the homotopy level alone does not provide us with higher order structure information. Therefore,

we investigate the question of rigidity: If we just had the structure of the homotopy category, how much of the underlying model structure can we recover? This question has been investigated during the last decade, and some examples have been studied, but there are still a lot of open questions regarding this subject. In this talk, we investigate a new case of rigidity, which is the localisation of spectra with respect to the Morava K-theory $K(1)$, at $p = 2$. (Received January 18, 2020)

1155-55-404 **Rebecca E Field***, MSC 1911, James Madison University, 800 South Main Street, Harrisonburg, VA 22807. *A gentle introduction to equivariant cohomology.*

What does it mean for an object to have symmetries? I'll explain what a classifying space is, do some equivariant cohomology, and elucidate some of the more subtle things that we can mean by 'the shape of a group'. (Received January 19, 2020)

1155-55-407 **Rebecca E Field*** (fieldre@jmu.edu), MSC 1911, James Madison University, 800 South Main Street, Harrisonburg, VA 22807. *BG2 again.*

We revisit a computation of Morava K-theory of the classifying of the compact group G2 describing some old and hopefully new results. (Received January 19, 2020)

1155-55-430 **John D. Berman*** (jberman@math.utexas.edu). *THH and enriched categories.*

THH is an invariant of a category generalizing the trace of linear algebra. It is generated by symbols tr(f) subject to relations tr(fg)=tr(gf). Using this perspective, we will see that THH is controlled by a single universal object, a certain circle bundle over a graph (in the sense of combinatorics). Along the way, we will describe a new and hopefully intuitive model for enriched higher categories. (Received January 20, 2020)

1155-55-441 **Shane W. Clark*** (shane.clark@uky.edu), 719 Patterson Office Tower, Lexington, KY 40517. *Periodic Points: Vanishing and Realizability.*

The Reidemeister trace of an endomorphism of a CW complex is a lower bound for the number of fixed points (up to homotopy) of that endomorphism. For an endomorphism f, the Reidemeister trace of f^n is a lower bound for the number of fixed points of f^n, however it can be a far from an optimal lower bound.

There are many related invariants that refine the Reidemeister trace and have different strengths and weaknesses in regards to computablity and realizability. In this talk we will describe a classes of spaces where these invariants can be computed and realized. (Received January 20, 2020)

1155-55-473 **Dominic Culver*** (dculver@illinois.edu), Urbana, IL 61801, and **JD Quigley**. *On the $\mathbb{C}$-motivic Adams spectral sequence based on Hermitian K-theory.*

In the 1980s, Mahowald initiated the study of the Adams spectral sequence based on connective real K-theory. Using this spectral sequence, he was able to give a complete description of all of the so-called v_1-periodic elements in the stable homotopy groups of spheres. In this talk, I will describe joint work with JD Quigley where we developed Mahowald's program for $\mathbb{C}$-motivic spectra. In particular, I will describe our calculation of the v_1-periodic elements in the stable homotopy groups of the $\mathbb{C}$-motivic sphere. Given time, I will pose a motivic analog of the telescope conjecture. (Received January 20, 2020)

1155-55-474 **Prasit Bhattacharya*** (pb9wh@virginia.edu), 1507 Broad Ave, Apt B, Charlottesville, VA , and **Hood Chatham**. *On the EO-orientation order of complex vector bundles.*

Let p be any prime. When $n = (p-1)k$, the n-th Morava stabilizer group contains C_p as a subgroup. $EO_{n,p}$, the homotopy fixed point of height n prime p Morava E-theory $E_{n,p}$, can be thought of as the higher height analogues of real K-theory KO. Indeed, $E_{1,2}$ is 2-completed KU and $EO_{1,2}$ is 2-completed KO. It is a classical result that the tautological line bundle γ_1 over CP^∞ is not KO-orientable, but $2\gamma_1$ is, meaning KO-orientation order of γ_1 is 2. In this talk, we generalize this result to show that $EO_{n,p}$-orientation order of γ_1 divides $o(n,p) = p^{p^k-1}$. Consequently, we conclude that $o(n,p)$-fold direct sum of any complex vector bundle is $EO_{n,p}$-orientable. This is a joint work with Hood Chatham. (Received January 20, 2020)

1155-55-482 **Alexander Berglund**, **Gregory Michel**, **TriThang Tran** and **Craig Westerland*** (cwesterl@umn.edu). *Second order terms in arithmetic statistics.* Preliminary report.

It's an old conjecture that the the cardinality of the set of number fields of degree n and discriminant bounded by X grows asymptotically linearly in X. This is classical when $n = 2$, due to Davenport-Heilbronn for $n = 3$, and Bhargava for $n = 4, 5$. Work of Taniguchi-Thorne and Bhargava-Shankar-Tsimerman establishes a second-order term for cubic extensions.

In this talk, we will report on work in progress aiming to establish a second order term in the function field setting, as well as giving a general heuristic for what we believe the degree of the second-order term to be. Our

tools are homological algebraic, coming from the Koszul complex of the Nichols algebra associated to the Galois data; this gives refined information on the cohomology of the Hurwitz moduli spaces of branched covers with this Galois data. (Received January 20, 2020)

1155-55-487 **Dominic Leon Culver** (dculver@illinois.edu) and **J.D. Quigley*** (jdq27@cornell.edu), 580 Malott Hall, Ithaca, NY 14853. *The motivic kq-resolution over general base fields.* Preliminary report.

In previous work with Dominic Culver, we analyzed the complex motivic Adams spectral sequence based on the very effective cover of Hermitian K-theory, kq. In this talk, I will report on our progress over other base fields. I plan to highlight an application to low-dimensional Milnor-Witt stems, and if time permits, I will discuss generalizations of some classical results of Brown–Gitler, Brown–Comenetz, and Priddy. (Received January 20, 2020)

1155-55-495 **Maru Sarazola*** (mesarazola@gmail.com). *Cotorsion pairs and a K-theory localization theorem.*

Cotorsion pairs were introduced in the '70s as a generalization of projective and injective objects in an abelian category, and were mainly used in the context of representation theory. In 2002, Hovey showed a remarkable correspondence between compatible cotorsion pairs on an abelian category $\mathcal{A}$ and abelian model structures one can define on $\mathcal{A}$.

In this talk, we turn our attention to Waldhausen categories, and explain how cotorsion pairs can be used to construct different Waldhausen structures on an exact category, with the usual class of admissible monomorphisms as cofibrations, and some freedom to choose the class of desired trivial objects. This allows us to prove a new version of Quillen's Localization Theorem, relating the K-theory of exact categories $\mathcal{A} \subseteq \mathcal{B}$ to that of a cofiber, constructed through a cotorsion pair. Notably, we do not require $\mathcal{A}$ to be a Serre subcategory, which produces new examples. (Received January 20, 2020)

1155-55-496 **Luis Pereira*** (luisalexandrepereira@outlook.com), Office 034, Physics Building, Duke University, Durham, NC 27708. *On models for equivariant operads with norm operations.*

Work of Cisinski-Moerdijk has established an equivalence between the homotopy theory of colored simplicial operads and the homotopy theory of ∞-operads, which can be regarded as "operads where composition is only defined up to homotopy". At the center of their work is the identification of a category Ω of trees, whose objects are tree diagrams that encode "composition of operations in an operad". When given a group G and working in the context of G-equivariant operads, some caution is required when identifying the "correct" notion of weak equivalence, since equivalences need to detect an additional piece of structure called "norm operations". In joint work with Peter Bonventre, we identified a category Ω_G of G-trees, a non-trivial generalization of Ω, whose objects are diagrams that encode "composition of norm operations in a G-operad". This talk will discuss alternative models for the homotopy theory G-equivariant simplicial colored operads suggested by Ω_G, namely the homotopy theory of G-∞-operads, which can be regarded as "G-operads where composition of norm operations is only defined up to homotopy". (Received January 20, 2020)

1155-55-503 **Manuel L Rivera*** (manuelr@purdue.edu) and **Mahmoud Zeinalian**. *The coalgebra of singular chains and the fundamental group.*

I will explain the sense in which the natural algebraic structure of the singular chains on a path-connected space determines its fundamental group. An example of a corollary of our statement is the following extension of a classical theorem of Whitehead: a continuous map between path-connected pointed topological spaces is a weak homotopy equivalence if and only if the induced map between the differential graded coalgebras of singular chains is a Koszul weak equivalence (i.e. a quasi-isomorphism after applying the cobar functor). A deeper implication, which is work in progress, is that this allows us to give a complete description of infinity groupoids in terms of homological data.

There are three main ingredients that come into play in order to give a precise formulation and proof of our main statement: 1) we extend a classical result of F. Adams from 1956 regarding the "cobar construction" as an algebraic model for the based loop space of a simply connected space, 2) we make use of the homotopical symmetry of the chain approximations to the diagonal map on a space, and 3) we apply a duality theory for algebraic structures known as Koszul duality. (Received January 20, 2020)

1155-55-516 **Candace Bethea**, **Niny Arcila Maya**, **Morgan Opie*** (opie.morgan@gmail.com), **Kirsten Wickelgren** and **Inna Zakharevich**. *A compactly supported motivic Euler characteristic via the Hochschild complex.* Preliminary report.

We give a compactly supported motivic Euler characteristic for smooth schemes over a ground field k. Our invariant, which is defined using the Hochschild complex, maps out of $K_0(\mathrm{Var}_k)$ and takes values in the Grothendieck–Witt group of k. As time allows, we will outline work in progress to obtain a spectrum-level version of this construction. (Received January 20, 2020)

1155-55-517 **Matthew Feller*** (feller@virginia.edu). *Excision and unital higher Segal spaces.* Preliminary report.

The notion of an excisive functor comes from Goodwillie calculus, where a functor is k-excisive if it sends certain $(k+1)$-dimensional cubes in the domain category to limit cubes in the codomain category. Meanwhile, a higher Segal space, an idea due to Dyckerhoff and Kapranov, is a contravariant functor whose domain is the simplex category Δ, subject to the condition that certain diagrams in Δ are sent to limit cones. While these conditions at first seem very different, Walde showed in a recent preprint that the higher Segal conditions can actually be characterized in terms of weak excision. In this talk, we'll explore this situation and push beyond Walde's results. In particular, we'll investigate in what situations Walde's weak excision implies full excision, and consider the role played by the extra condition of unitality. (Received January 20, 2020)

1155-55-554 **Niny Arcila-Maya*** (ninyam@math.ubc.ca). *Decomposition of Topological Azumaya Algebras.* Preliminary report.

An Azumaya algebra of degree n over X is a bundle of algebras over X that is locally isomorphic to the matrix algebra. The tensor product of algebras can be extended to Azumaya algebras by performing the operation fiberwise. In topology, we can give conditions for positive integers m and n and the space X such that a topological Azumaya algebra of degree mn can be decomposed as the tensor product of Azumaya algebras of degrees m and n, but no such conditions are known in algebra. We will explain the topological proof and indicate the difficulties in extending these to the algebraic context. (Received January 21, 2020)

1155-55-564 **Agnes Beaudry**, **Irina Bobkova*** (ibobkova@math.tamu.edu), **Michael Hill** and **Vesna Stojanoska**. *Invertible modules at chromatic height 2 and $p = 2$.*

For Morava E-theory E_n and a finite subgroup F of the Morava stabilizer group, the spectrum E_n^{hF} is periodic and the Picard group of the category of modules over the ring spectrum E_n^{hF} contains the cyclic subgroup generated by ΣE_n^{hF}. In most known examples, the Picard group is found to be precisely this cyclic group. However, at chromatic height $n = 2$ and $p = 2$, the Picard group of the category of $E_2^{hC_4}$-modules is not cyclic and contains an extra element of order 2. I will describe the tools we use to compute this Picard group: a group homomorphism from $RO(C_4)$ to it and the Picard spectral sequence. This talk is based on joint work with A. Beaudry, M. Hill and V. Stojanoska. (Received January 21, 2020)

1155-55-571 **Kyle Ormsby*** (ormsbyk@reed.edu), **Maxine Calle**, **Sam Ginnett**, **Harry Chen** and **Xinling Chen**. *Tambara generators for the trace ideal.*

The Dress homomorphism links the Burnside ring of a Galois group to the Grothendieck-Witt ring of the base field in a manner compatible with restriction, transfer, and norm maps. We use this additional "Tambara functoriality" to produce new results on the trace ideal, i.e., the kernel of the Dress map. The stable Galois correspondence between equivariant and motivic spectra induces the Dress map (on endomorphism rings of unit objects), so this furthers our understanding of the connection between these fields. As an application, we compute the multiplicative norms in the Grothendieck-Witt ring of a finite field. (Received January 21, 2020)

1155-55-590 **Peter J Bonventre*** (peterbonventre@uky.edu), **Bert J Guillou** and **Nat J Stapleton**. *Power operations and transfers in equivariant cohomology.*

Power operations are an analogue of the m-th power map, found in suitably multiplicative cohomology theories. In general, these are only ring maps after modding out by a certain transfer ideal. Suitably multiplicative equivariant cohomology theories are instead valued in Green functors. In this talk, we investigate the interaction of power operations with this additional structure. In particular, we explicitly describe a natural, minimal ideal so that the reduced power operation is a map of Green functors. This work yields similar results for ultra-commutative global cohomology theories. (Received January 21, 2020)

1155-55-600 **Bogdan Krstic*** (bk2fh@virginia.edu). *Bispans in quasicategories and global Tambara fucntors.*

Given a suitable (e.g. locally Cartesian closed) quasicategory $\mathcal{C}$, we construct an associated quasicategory whose 1-simplices are bispans in $\mathcal{C}$, and whose n-simplices encode n-fold composites of bispans. This enables us to rigorously construct a theory of global Tambara functors. These are algebraic structures indexed on finite groups with operations of additive and multiplicative induction along subgroup inclusions along with restrictions, and a twisted distributive law governing the relationship between the induction maps. These structures bear a close resemblance to Schwede's global power functors, which encode the power operations on highly-structured global commutative ring spectra, and exploring this relationship is the aim of future work. (Received January 21, 2020)

57 ► Manifolds and cell complexes

1155-57-46 **Ciprian Manolescu**, **Marco Marengon**, **Sucharit Sarkar** and **Michael Willis*** (msw188@g.ucla.edu), 520 Portola Plaza, Los Angeles, CA 90095-0001. *Generalising Rasmussen's s-invariant. Part 1: Definition and basic properties.*

Previous work of Rozansky and of the author led to the construction of a Khovanov complex for links in connected sums of copies of $S^1 \times S^2$. We introduce a Lee deformation of this complex, allowing for a definition of an s-invariant for null-homologous links in such 3-manifolds. We discuss the behavior of the s-invariant under cobordisms. We also introduce a re-formulation in $S^1 \times S^2$, closer in spirit to Rozansky's original definition of the Khovanov complex, which allows us to make an important computation for a specific set of links. This is joint work with Ciprian Manolescu, Marco Marengon, and Sucharit Sarkar. (Received November 25, 2019)

1155-57-47 **Ciprian Manolescu**, **Marco Marengon*** (marengon@math.ucla.edu), **Sucharit Sarkar** and **Michael Willis**. *Generalising Rasmussen's s-invariant. Part 2: Applications.*

We use a computation of the s-invariant for the set of links discussed in part 1 to deduce an adjunction inequality for null-homologous surfaces in a (punctured) connected sum of $\overline{\mathbb{CP}^2}$. This inequality has several consequences: first, the s-invariant of a link in S^3 does not increase under the operation of adding a full twist. Second, the s-invariant cannot be used to distinguish S^4 from homotopy 4-spheres obtained by Gluck twist on S^4. We also prove a connected sum formula for the s-invariant of links in S^3, improving a previous result of Beliakova and Wehrli. We define two s-invariants for links in $\#^r(S^1 \times S^2)$. One of them gives a lower bound to the slice genus in $\natural^r(S^1 \times B^3)$ and the other one to the slice genus in $\natural^r(D^2 \times S^2)$. Lastly, we give a combinatorial proof of the slice Bennequin inequality in $\#^r(S^1 \times S^2)$. This is joint work with Ciprian Manolescu, Sucharit Sarkar, and Michael Willis. (Received November 25, 2019)

1155-57-75 **Scott A Taylor***, sataylor@colby.edu. *Lower bounds on the tunnel number of composite spatial θ-curves.*

The tunnel number of a graph embedded in a 3-dimensional manifold is the fewest number of arcs needed so that the union of the graph with the arcs has handlebody exterior. The behavior of tunnel number with respect to connected sum of knots can vary dramatically, depending on the knots involved. However, a classical theorem of Scharlemann and Schultens says that the tunnel number of a composite knot is at least the number of factors. For theta graphs, trivalent vertex sum is the operation which most closely resembles the connected sum of knots. The analogous theorem of Scharlemann and Schultens no longer holds, however. I will provide a sharp lower bound for the tunnel number of composite theta graphs, using recent work on a new knot invariant which is additive under connected sum and trivalent vertex sum. This is joint work with Maggy Tomova. (Received December 19, 2019)

1155-57-89 **Charles Frohman*** (charles-frohman@uiowa.edu), Professor, and **Adam Sikora** (asikora@buffalo.edu). *Classifying SL_3-webs in finite type surfaces up to isotopy.*

An SL_3-web is a trivalent graph whose edges have been oriented so that each vertex is a source or sink, embedded in a finite type surface F. A web is non-elliptic if none of its components is a simple closed curve bounding a disk or contains a bigon, or quadrigon that bounds a disk. Nonelliptic SL_3-webs are a basis for the coordinate ring of the $SL_3\mathbb{C}$-character variety of the fundamental group of the underlying surface. Given an ideal triangulation of the surface F we find coordinates for the set of isotopy classes of non-elliptic SL_3-webs in F. The coordinates consist of two counting numbers for each edge of the ideal triangulation on one integer for each ideal triangle.

The SL_3-skein algebra is a noncommutative algebra having the same basis as the coordinate ring of the SL_3-character variety where the multiplication comes from stacking webs in $F \times [0, 1]$. Using these coordinates

we prove the the SL_3-skein algebra of a finite type surface is finitely generated (affine), by imitating the proof of the Hilbert basis theorem. (Received December 31, 2019)

1155-57-100 **Eugene Gorsky***, University of California, Davis, Department of Mathematics, One Shields Avenue, Davis, CA 95616, and **Tye Lidman**, **Beibei Liu** and **Allison Moore**. *Triple linking numbers and Heegaard Floer homology.* Preliminary report.

We establish some new relationships between higher linking invariants (such as Sato-Levine invariant and triple linking number) and Heegaard Floer homology. We give explicit formulas for these invariants in terms of Heegaard Floer homology and apply these results to study Dehn surgeries along multi-component links. (Received January 03, 2020)

1155-57-105 **Aliakbar Daemi** and **Christopher Scaduto*** (c.scaduto@math.miami.edu). *Equivariant aspects of singular instanton Floer homology.*

We associate several invariants to a knot in an integer homology 3-sphere using SU(2) singular instanton gauge theory. There is a space of framed singular connections for such a knot, equipped with a circle action and an equivariant Chern-Simons functional, and our constructions are morally derived from the associated equivariant Morse chain complexes. In particular, we construct a triad of groups analogous to the knot Floer homology package in Heegaard Floer homology, several Froyshov-type invariants which are concordance invariants, and more. (Received January 06, 2020)

1155-57-108 **Colin Adams*** (cadams@williams.edu), Bascom Hall, 33 Stetson Ct., Williamstown, Williamstown, MA 01267, and **Judah Devadoss**, **Mohamed Elhamdadi** and **Alireza Mashaghi**. *Quandles and Bondles for Distinguishing Knotted Proteins.*

Proteins are linear molecular chains that often fold to function. The topology of folding is widely believed to define its properties and function, and knot theory has been applied to study protein structure and its implications. More that 97% of proteins are, however, classified as unknots when intra-chain interactions are ignored. This raises the question as to whether knot theory can be extended to include intra-chain interactions and thus be able to categorize topology of the proteins that are otherwise classified as unknotted. Here, we develop knot theory for folded linear molecular chains and apply it to proteins.

In particular, we explore the possibility of applying the algebraic structure of quandles to distinguish the topologies of proteins. Because of the presence of bonds, we extend the theory to define bondles, a type of quandle particularly adapted to distinguishing the topological types of proteins. (Received January 06, 2020)

1155-57-125 **Thomas D. Eddy** and **Clayton Shonkwiler***, Department of Mathematics, Colorado State University, clay@shonkwiler.org. *New Stick Number Bounds from Random Sampling of Confined Polygons.*

The stick number of a knot is the minimum number of segments needed to build a polygonal version of the knot. Despite its elementary definition and relevance to physical knots, the stick number is poorly understood: for most knots we only know bounds on the stick number. We adopt a Monte Carlo approach to finding better bounds, producing very large ensembles of random polygons in tight confinement to look for new examples of knots constructed from few segments. We generated a total of 220 billion random polygons, yielding either the exact stick number or an improved upper bound for more than 40% of the knots with 10 or fewer crossings for which the stick number was not previously known, as well as determining for the first time the exact stick number of a non-torus prime knot with more than 9 crossings. (Received January 07, 2020)

1155-57-130 **Kenneth Bromberg**, **Autumn Kent*** (kent@math.wisc.edu) and **Yair Minsky**. *New theorems on Dehn filling, drilling, and geometric inflexibility.* Preliminary report.

Thurston's Hyperbolic Dehn Filling Theorem tells us that all but finitely many Dehn fillings of a cusp on a hyperbolic manifold are hyperbolic. The universal hyperbolic Dehn Filling Theorem of Hodgson and Kerckhoff tells us that if we normalize the cusp cross section to be a unit area flat torus, then any filling slope of length greater than 8 will produce a hyperbolic Dehn filling. The Drilling Theorem of Brock and Bromberg further tells us that these hyperbolic fillings don't disturb the geometry much away from the cusp.

We prove a new universal hyperbolic Dehn Filling Theorem that retains the geometric control of Brock–Bromberg but allows for filling slopes of small normalized length. We trade Hodgson and Kerckhoff's normalized length constraint for constraints on the actual length and the volume growth of the cusp.

The geometric inflexibility theorems of McMullen and Brock–Bromberg tell us that the effect of a quasiconformal deformation of a geometrically finite Kleinian group decays exponentially as you move deeper into the convex core. These theorems are very useful in the convex cocompact setting, but don't get a lot of mileage in

the presence of rank-1 cusps. Our Filling Theorem provides new inflexibility theorems that are insensitive to this presence. (Received January 07, 2020)

1155-57-133 **Kenneth C Millett*** (millett@mathl.ucsb.edu), Department of Mathematics, UCSB, Santa Barbara, CA 93106. *Knots in Proteins.* Preliminary report.

Since 2000, knots in proteins have attracted the attention of biologists and mathematicians. Following a brief historical inventory, we will explore the recent discoveries of instances of knots in specific proteins, their mathematical character, and reflect on what they tell us about their biological and functional consequences. (Received January 07, 2020)

1155-57-134 **Hugh N Howards***, Math Dept, WFU, Winston Salem, NC 27109, and **Jiong Li** and **Xiotian Liu**. *Detecting Flypes and Hexagonal Mosaics.*

We introduce a new tool which makes it easier to systematically recognize when two knots differ by a sequence of Flypes (for example, giving a process to recognize that the Perko Pair were in fact the same knot). We then apply it to hexagonal mosaics to find an infinite family of knots which do not achieve their hexagonal mosaic number while also in a a projection which achieves their crossing number, extending a result of Lew Ludwig et al. (Received January 07, 2020)

1155-57-152 **Bahar Acu*** (baharacu@northwestern.edu), 2033 Sheridan Road, Evanston, IL 60208, and **Orsola Capovilla-Searle**, **Agnes Gadbled**, **Aleksandra Marinkovic**, **Emmy Murphy**, **Laura Starkston** and **Angela Wu**. *Weinstein Handlebodies for Complements of Smoothed Toric Divisors.*

Toric manifolds together with their divisors are of great importance in four-dimensional symplectic geometry. These divisors initially have "normal crossing singularities" and are important classes of symplectic divisors. It is, moreover, known that the complement of the normal crossing divisor is symplectically T^*T^2. Any of the normal crossing singularities can be smoothed so the divisor has fewer singularities. This smoothing operation changes the topology of the complement. In this work, the fruit of the meeting at the 2019 Research Collaboration Conference for Women in Symplectic and Contact Geometry and Topology, we develop a method to draw a Weinstein handlebody decomposition for the complement of any smoothed toric divisor. (Received January 09, 2020)

1155-57-153 **Miriam Kuzbary***, 686 Cherry St NW, Atlanta, GA 30313. *Pure Braids and Link Concordance.*

The set of knots in the 3-sphere with the connected sum operation forms a group when its quotient is taken by a 4-dimensional equivalence relation called concordance. This knot concordance group, defined by Fox and Milnor in 1966, can be contextualized as organizing problems about 3- and 4-dimensional spaces and the relationships between them. However, one of its limitations is that every 3-manifold is surgery on a link, not necessarily a knot, and thus it may be useful to understand such a group for links. Unfortunately, the connected sum of links is not well defined; in 1988, Le Dimet constructed the string link concordance group for links with n components to get around this problem. Notably, the knot concordance group is abelian while, for each n, the string link concordance group on n strands is non-abelian as it contains the pure braid group on n strands as a subgroup. In this talk, I will discuss my recent result that even the quotient of each string link concordance group by its pure braid subgroup is non-abelian. (Received January 09, 2020)

1155-57-155 **Aaron Calderon** and **James Farre*** (james.farre@yale.edu). *Shear-Shape Coordinates for Teichmüller Space and Applications.* Preliminary report.

We introduce a family of coordinate systems for the Teichmüller space of a closed surface adapted to an arbitrary (non-maximal) geodesic lamination, generalizing the shearing coordinates of Bonahon and Thurston. When the lamination carries a transverse measure μ of full support, the coordinates also parameterize quadratic differentials whose vertical foliation is measure equivalent to μ. Following a strategy of Mirzakhani, we then exhibit dynamically natural mappings between strata of quadratic differentials and earthquake invariant subsets of the bundle of measured laminations over Teichmüller space. (Received January 09, 2020)

1155-57-156 **Honghao Gao*** (gaohongh@msu.edu), 619 Red Cedar Road, Wells Hall C212, Department of Mathematics, East Lansing, MI 48823. *Legendrian links with infinitely many Lagrangian fillings.*

A filling is an oriented surface bounding a link. Lagrangian fillings can be constructed via local moves in finite steps, but it was unknown whether a Legendrian link could admit infinitely many Lagrangian fillings. In this talk, I will show that Legendrian torus links other than (2,m), (3,3), (3,4), (3,5) indeed have infinitely many

fillings. These fillings are constructed using Legendrian loops, and proven to be distinct using the microlocal theory of sheaves and the theory of cluster algebras. This is a joint work with Roger Casals. (Received January 19, 2020)

1155-57-157 **Erica Flapan*** (ericaflapan@gmail.com). *Generalized Simon Invariants for Embeddings of Complete Graphs in S^3.* Preliminary report.

We define invariants of certain spatial graphs which are a generalization of the Simon invariant for embeddings of K_5 and $K_{3,3}$ in S^3. Our invariants are easy to compute and can be used to prove that K_7, all Möbius ladders with an odd number of rungs, and the Heawood graph each have the property that all of their embeddings in S^3 are chiral. They also give lower bounds for the minimal crossing number of particular embeddings of graphs in S^3. This talk will include past results as well as work in progress to define generalized Simon invariants for all complete graphs. (Received January 10, 2020)

1155-57-163 **Yuanan Diao***, 11908 Three Vistas Ct, Charlotte, NC 28277. *Braid Index Bounds Ropelength From Below.* Preliminary report.

For an un-oriented link K, let $L(K)$ be the ropelength of K. It is known that in general $L(K)$ is at least of the order $O((Cr(K))^{3/4})$, and at most of the order $O(Cr(K)\ln^5(Cr(K))$ where $Cr(K)$ is the minimum crossing number of K. Furthermore, it is known that there exist families of (infinitely many) links with the property $L(K) = O(Cr(K))$. A long standing open conjecture states that if K is alternating, then $L(K)$ is at least of the order $O(Cr(K))$. In this paper, we show that the braid index of a link also gives a lower bound of its ropelength. More specifically, let $B(K)$ be the largest braid index among all braid indices corresponding to all possible orientation assignments of the components of K, we show that there exists a constant $a > 0$ such that $L(K) \geq aB(K)$ for any K. Consequently, $L(K) \geq O(Cr(K))$ for any link K whose largest braid index is proportional to its crossing number. In the case of alternating links, the largest braid indices for many of them are proportional to their crossing numbers hence the above conjecture holds for these alternating links. (Received January 10, 2020)

1155-57-190 **Kristen Hendricks*** (kristen.hendricks@rutgers.edu), **Jennifer Hom**, **Matthew Stoffregen** and **Ian Zemke**. *A surgery triangle in involutive Heegaard Floer homology.*

We discuss work-in-progress with J. Hom, M. Stoffregen, and I. Zemke on a surgery exact triangle in involutive Heegaard Floer homology. This construction is based on a model for the involution using doubling Heegaard diagrams. We show how in certain nice cases it gives a "surgery formula" which may be used to compute the local equivalence class of the involutive Heegaard Floer homology of certain integer homology spheres. (Received January 13, 2020)

1155-57-207 **Ramon Flores**, **Delaram Kahrobaei** and **Thomas Koberda*** (thomas.koberda@gmail.com). *An application of 3–manifold topology to cryptography.*

I will present an application of 3–manifold topology to cryptography. Namely, I will discuss a symmetric key cryptographic scheme based on the Thurston norm of a hyperbolic 3-manifold. (Received January 13, 2020)

1155-57-216 **Paolo Aceto**, **Jeffrey Meier**, **Allison Miller**, **Maggie Miller** and **JungHwan Park*** (junghwan.park@math.gatech.edu), 686 Cherry Street, Atlanta, GA 30332, and **Andras Stipsicz**. *Branched covers bounding rational homology balls.*

Prime-power-fold cyclic branched covers along smoothly slice knots all bound rational homology balls. This phenomenon, however, does not characterize slice knots: In this talk, we give examples of non-slice knots that have the above property. This is joint work with Aceto, Meier, A. Miller, M. Miller, and Stipsicz. (Received January 13, 2020)

1155-57-219 **Mariano Echeverria*** (mariano.echeverria@rutgers.edu). *A Generalization of the Tristram-Levine Knot Signatures as a Singular Furuta-Ohta Invariant for Tor.*

Given a knot K inside an integer homology sphere Y , the Casson-Lin-Herald invariant can be interpreted as a signed count of conjugacy classes of irreducible representations of the knot complement into SU(2) which map the meridian of the knot to a fixed conjugacy class. It has the interesting feature that it determines the Tristram-Levine signature of the knot associated to the conjugacy class chosen. Turning things around, given a 4-manifold X with the integral homology of S1 × S3, and an embedded torus which is homologically non trivial, we define a signed count of conjugacy classes of irreducible representations of the torus complement into SU(2) which satisfy an analogous fixed conjugacy class condition to the one mentioned above for the knot case. Our count recovers the Casson-Lin-Herald invariant of the knot in the product case, thus it can be regarded as implicitly defining a

Tristram-Levine signature for tori. This count can also be considered as a singular Furuta-Ohta invariant, and it is a special case of a larger family of Donaldson invariants which we also define. (Received January 14, 2020)

1155-57-221 **Matt Hedden** and **Katherine Raoux*** (raouxkat@msu.edu), 619 Red Cedar Road, East Lansing, MI 48824. *Knot Floer homology and relative adjunction inequalities.* Preliminary report.

In this talk, we present a relative adjunction inequality for 4-manifolds with boundary. We begin by constructing generalized Heegaard Floer tau-invariants associated to a knot in a 3-manifold and a nontrivial Floer class. Given a 4-manifold with boundary, the invariant associated to a Floer class provides a lower bound for the genus of a properly embedded surface provided that the Floer class is in the image of the cobordism map induced by the 4-manifold. We will also discuss several applications to links and contact manifolds.

This is joint work with Matt Hedden. (Received January 14, 2020)

1155-57-229 **Claus Ernst*** (claus.ernst@wku.edu), Department of Mathematics, Western Kentucky University, Bowling Green, KY 42101, and **Donald Price** (donald.price077@topper.wku.edu), Department of Mathematics, Western Kentucky University, Bowling Green, KY 42101. *Generating all Minimal Diagrams of Alternating Knots.*

By the Taite Flyping Conjecture all all possible minimal diagrams of an alternating knot are related by a finite sequence of flypes. We we use this fact to generate all possible minimal diagrams of an alternating knot. Starting with any given minimal diagram our algorithm performs flypes exhaustively until all possible diagrams have been produced. We display our results for all distinct alternating knot diagrams up to 16 crossings. We discuss knots families that have only a small number of distinct minimal diagrams and knot families that have a large number of minimal diagrams. (Received January 14, 2020)

1155-57-231 **Eric J Rawdon*** (ejrawdon@stthomas.edu). *Open linking.* Preliminary report.

In the last 10 years, there has been increasing interest in measuring entanglement in open curves, e.g. knotting in proteins. We present a simple technique for measuring linking between two or more open curves. (Received January 14, 2020)

1155-57-235 **Carson Rogers*** (carson.rogers@bc.edu). *Cosmetic twisting operations on fibered knots.*

One important open problem in knot theory is the Cosmetic Crossing Conjecture. In this talk, we will examine a generalization of this problem to the operation of an *n-tangle replacement* on a knot, where n is a nonzero integer. This consists of adding $|n|$ half-twists of the same sign to two oppositely-oriented strands of the knot. We will cover two related results which restrict the n-tangle replacements on a fibered knot which may produce an isotopic knot. The first shows that if n is even, then a non-nugatory n-tangle replacement on a fibered knot K in a rational homology sphere M will never do so. This generalizes a theorem of Kalfagianni. Our second result shows that if a non-nugatory n-tangle replacement on such K can be realized by twisting its fiber surface along a separating arc, then the resulting knot may only be isotopic to K if $n = \pm 1$.

After indicating how these theorems are proven, we will exhibit simple examples to show that the second theorem cannot be improved. If time allows, we'll conclude by discussing some special features of the case $n = \pm 1$, which is that of non-coherent band surgery. (Received January 14, 2020)

1155-57-262 **Samuel A Ballas***, 208 Love Building, 1017 Academic Way, Tallahassee, FL 32303, and **Alex Casella**. *Gluing equations for projective structures on 3-manifolds.*

One of Thurston's many amazing ideas are his hyperbolic gluing equations. Roughly speaking, given an ideally triangulated 3-manifold M, one can construct a set of complex polynomial equations whose solutions correspond to hyperbolic structures on M. In this talk I will describe some recent work generalizing these equations in the context of real projective structures. Specifically, we construct a system of equations whose solutions can be used to construct a projective structure on M. Furthermore, solutions to our equations detect several interesting classes of projective structures including hyperbolic structures, anti-de Sitter structures, and convex projective structures. (Received January 15, 2020)

1155-57-270 **William M. Goldman*** (wmg@umd.edu). *Dynamics and moduli of geometries on surfaces.* Preliminary report.

We describe dynamical systems arising from the classification of locally homogeneous geometric structures and flat connections. Their classification mimics that of Riemann surfaces by the Riemann moduli space, the quotient of Teichmueller space by the mapping class group. However, unlike Riemann surfaces, the mapping class group

actions on character varieties may be chaotic, leading to dynamic complexity. A striking elementary example is Baues's theorem that the deformation space of complete affine structures on the 2-torus is the plane with the usual linear action of GL(2,Z) (the mapping class group of the torus). We discuss specific examples of these dynamics for some simple surfaces, where the relative character varieties appear as cubic surfaces in 3-space. Complicated dynamics seems to accompany complicated topology, and we interpret them in terms of hyperbolic structures on surfaces, with or without singularities. (Received January 16, 2020)

1155-57-278 **Steve J Trettel*** (trettel@stanford.edu). *Riemannian Limits of the Product Geometries H2xR and S2xR.*

As the curvature k goes to zero, the 1-parameter family of hyperbolic spaces of curvature k converges to Euclidean space. This fact was already noted by Klein at the end of the 19th century, who produced an explicit model of this transition as a conjugacy limit in projective space. As understanding the limiting behaviors of a geometry informs the study of degenerations of geometric structures, the classification of such limits is of interest to geometric topology. And in dimension three, geometrization suggests a natural project: classifying the limits of the Thurston geometries. More precisely - when can a Thurston geometry X be deformed in projective space so that it degenerates to some geometry containing some other Thurston geometry Y? Work of Cooper, Danciger, and Wienhard nearly completed this classification, resolving 62 of the 64 possible cases. In this talk I will discuss the remaining two, and show the product geometries H2xR and S2xR do not limit to Nil. (Received January 16, 2020)

1155-57-283 **Anar Akhmedov**, **Mohan Bhupal** and **Nur Saglam***, nur77@vt.edu. *Constructions of Lefschetz fibrations using cyclic group actions.*

We construct families of Lefschetz fibrations over $\mathbb{S}^2$ using finite order cyclic group actions on the product manifolds $\Sigma_g \times \Sigma_g$ for $g > 0$. In some cases, we obtain singular fibers which we can apply rational blow-down operation. (Received January 16, 2020)

1155-57-286 **Yvon Verberne*** (yvon.verberne@mail.utoronto.ca), Department of Mathematics, University of Toronto, Room 6290, 40 St. George Street, Toronto, Ontario M5S 2E4, Canada. *Constructing pseudo-Anosov homeomorphisms using positive twists.*

Thurston obtained the first construction of pseudo-Anosov homeomorphisms by showing the product of two parabolic elements is pseudo-Anosov. A related construction by Penner involved constructing whole semigroups of pseudo-Anosov homeomorphisms by taking appropriate compositions of Dehn twists along certain families of curves. In this talk, I will present a new construction of pseudo-Anosov homeomorphisms and discuss some of the unique properties associated to maps obtained through this new construction. (Received January 16, 2020)

1155-57-319 **Sandy Ganzell*** (sganzell@smcm.edu), St. Mary's College of Maryland, 18952 E Fisher Rd, St. Marys City, MD 20686, and **Allison K Henrich**. *Virtual Mosaic Knot Theory.* Preliminary report.

We define a theory of virtual mosaic knots. Using the eleven standard mosaic tiles, we create a virtual knot mosaic by identifying boundary edges of a standard mosaic and viewing virtual knots as equivalence classes of knot diagrams on surfaces. The equivalence of virtual mosaics is given by a set of moves, including isotopy, stabilization and injection (into a larger mosaic), as well as Reidemeister moves. We also define the virtual mosaic number of a virtual knot or link L as the smallest n for which L can be realized on an $n \times n$ virtual mosaic. Virtual mosaic numbers are calculated for classical and virtual knots of small crossing number. (Received January 17, 2020)

1155-57-322 **Aliakbar Daemi**, **Tye Lidman**, **David Shea Vela-Vick** and **C.-M. Michael Wong*** (cmmwong@lsu.edu). *Ribbon homology cobordisms.*

A cobordism between 3-manifolds is ribbon if it has no 3-handles. Such cobordisms arise naturally from several different topological and geometric contexts. In this talk, we describe a few obstructions to their existence, from Thurston geometries, character varieties, and instanton and Heegaard Floer homologies, and some applications. (Received January 17, 2020)

1155-57-329 **Jason Parsley*** (parslerj@wfu.edu) and **Eric Rawdon**. *Census of Petal Knots.* Preliminary report.

This is joint work with Eric Rawdon (and many students). A petal diagram for a knot resembles a flower with n petals, symmetrically placed around a central point. Petal diagrams were introduced by Colin Adams and his students, who showed that every knot has a petal diagram, indexed by a permutation in S_n. Our work builds a census of all petal knots with 13 or fewer petals – over 480 million diagrams, classified by knot type. We

study the properties of these petal knots and draw connections to both classical and modern knot invariants. (Received January 18, 2020)

1155-57-333 **Radmila Sazdanovic*** (rsazdan@ncsu.edu), Department of Mathematics NC State University, Raleigh, NC 27695, and **Adam Lowrance**. *Bounds on torsion in Khovanov and chromatic homology.* Preliminary report.

We prove an upper bound on the order of the torsion part of Khovanov homology in terms of the crossing number of the link. We similarly prove an upper bound on the order of the torsion part of the chromatic homology of a graph in terms of the number of vertices and edges in the graph. Both bounds follow from Hadamard's inequality and an analysis of the boundary in the respective complexes. (Received January 18, 2020)

1155-57-335 **Uta Ziegler*** (uta.ziegler@wku.edu). *Topological and geometric measures of knots in extreme confinement.* Preliminary report.

Nearly-equilateral polygons were generated under extreme confinement. The knot types and some geometric quantities were computed for each polygon. The objective of this investigation is to compare the topological and geometric measures from these extremely confined polygons with previously reported measures of equilateral polygons in more moderate confinement. This work is in collaboration with Claus Ernst, Eric Rawdon, and Yuanan Diao. (Received January 18, 2020)

1155-57-341 **Tarik Aougab***, Mathematics Department, Haverford College, Haverford, PA 19041, and **Jonah Gaster**. *Statistics for random curves on surfaces.*

We study random walks on surface groups and the combinatorial properties of the curves that arise along them. For instance, we show that with probability converging to 1 as n goes to infinity, the curve appearing at the nth step has self-intersection number on the order of n^2. As an application, we show that the log of the dilatation of a random point-pushing pseudo-Anosov is comparable to the square root of the self-intersection number of the underlying filling curve. This is joint work with Jonah Gaster. (Received January 18, 2020)

1155-57-356 **Moshe Cohen*** (cohenm@newpaltz.edu). *The genus of a random 2-bridge knot using Chebyshev billiard table diagrams.* Preliminary report.

In previous work with Krishnan and then with Even-Zohar and Krishnan, the author develops a model for random 2-bridge knots. In the present work, the author restricts this model to obtain a set of alternating diagrams. Analyses on these alternating diagrams give estimations of genera. (Received January 18, 2020)

1155-57-358 **Moshe Cohen*** (cohenm@newpaltz.edu) and **Adam Lowrance**. *The signature of a random 2-bridge knot using Chebyshev billiard table diagrams.* Preliminary report.

In previous work with Krishnan and then with Even-Zohar and Krishnan, the first author develops a model for random 2-bridge knots. In newer work, the first author restricts this model to obtain a set of alternating diagrams, on which analyses are performed to yield estimations of genera. In the present work, the authors exploit these results to perform computational estimations of signature. (Received January 18, 2020)

1155-57-360 **Alexander James Rasmussen***, 503 Orange Street, New Haven, CT 06511. *Nonseparating curve graphs and infinite type surfaces.*

The curve graph of a finite type surface is a crucial tool for understanding the algebra and geometry of the corresponding mapping class group. Recently, researchers have studied mapping class groups of infinite type surfaces using related graphs such as the ray graph and the nonseparating curve graph. In this talk I will discuss the combinatorial machinery of bicorns, first introduced by Przytycki-Sisto, and their applications to the geometry of the nonseparating curve graphs. These applications include results on hyperbolicity, quasiconvex subgraphs, and boundaries. (Received January 18, 2020)

1155-57-383 **Spencer Dowdall**, **Matthew Durham** and **Christopher Leininger*** (clein@illinois.edu), 1409 W. Green St., Urbana, IL 61820, and **Alessandro Sisto**. *Surface bundles over Teichmuller curves.* Preliminary report.

In this talk I will discuss hyperbolic geometric aspects of the fundamental group of the canonical surface bundle over a Teichmuller curve: a totally geodesic surface in the moduli space of Riemann surfaces. These examples are meant to serve as motivating examples that set the stage for a notion of geometric finiteness in the mapping class group. (Received January 19, 2020)

1155-57-389 **Rylee Alanza Lyman*** (rylee.lyman@tufts.edu). *Detecting pseudo-Anosov orbifold braids in automorphisms of free products.* Preliminary report.

Zieschang in 1966 proved that the mapping class group of a punctured surface is isomorphic to the subgroup of outer automorphisms of its (free) fundamental group permuting the conjugacy classes corresponding to simple loops around the punctures. In 1992, Bestvina and Handel constructed *train track maps* on graphs representing *irreducible* elements of $\mathrm{Out}(F_n)$, the outer automorphism group of a free group of rank n. Irreducible elements with irreducible powers are the $\mathrm{Out}(F_n)$ analogue of pseudo-Anosov mapping classes, and Bestvina–Handel additionally show that if an irreducible outer automorphism preserves a nontrivial conjugacy class, then it may be represented as a pseudo-Anosov homeomorphism of a once-punctured surface. The author has recently extended the train track theory to automorphisms of free products. Here the analogy is to braid groups; we show that irreducible automorphisms of free products of finite subgroups of $SO(2)$ preserving a nonperipheral conjugacy class may be represented as psesudo-Anosov braid on a 2-orbifold. (Received January 19, 2020)

1155-57-395 **Joshua Greene** and **John Luecke*** (luecke@math.utexas.edu), Math Department, 1 University Station, University of Texas at Austin, Austin, TX 78712. *Fibered Simple Knots.*

We prove that a simple knot in the lens space $L(p, q)$ fibers if and only if its order in homology does not divide any remainder occurring in the Euclidean algorithm applied to the pair (p, q). One corollary is that if $p = m^2$ is a perfect square, then any simple knot of order m fibers, answering a question of Cebanu. More generally, we compute the leading coefficient of the Alexander polynomial of a simple knot, and we describe how to construct a minimum complexity Seifert surface for one. (Received January 19, 2020)

1155-57-400 **Richard D Canary*** (canary@umich.edu), Department of Mathematics, University of Michigan, Ann Arbor, MI 48109, and **Konstantinos Tsouvalas** (tsouvkon@umich.edu), Department of Mathematics, University of Michigan. *Topological restrictions on Anosov representations.*

We will discuss topological restrictions on groups which admit Anosov representations into SL(n,R) with a focus on restrictions on cohomological dimension. (Joint work with Kostas Tsouvalas.) (Received January 19, 2020)

1155-57-461 **Radmila Sazdanovic*** (rsazdan@ncsu.edu), Department of Mathematics NC State University, Raleigh, NC 27695, and **Vladimir Baranovsky**. *On factorization and chromatic graph homology.* Preliminary report.

Factorization homology, introduced by Ayala, Francis, and Tanaka, generalizes Hochschild homology. Helme-Guizon and Rong's chromatic homology, a Khovanov-type homology for graphs, approximates Hochschild homology when applied to cycle graphs. We provide alternative construction for the chromatic homology of any graph that allows us to view it as an approximation of factorization homology. (Received January 20, 2020)

1155-57-465 **Stanislav Jabuka*** (jabuka@unr.edu), Reno, NV 89557. *Concordance crosscap numbers and rational Witt classes.*

The concordance crosscap number of a knot is the smallest crosscap number among all knots concordant to it, and it can be seen as interpolating between the classical 3-dimensional crosscap number and its 4-dimensional counterpart.

To date only two results have appeared on computing this invariant. The first by Gengyu Zhang relies on the knot signature and determinant, while the second, due to Charles Livingston, uses the Alexander polynomial. We present in this talk a third way of thinking about the concordance crosscap number, one that uses the rational Witt class of a knot. We give examples to show that our methods give obstructions even when Zhang's and Livingston's do not. (Received January 20, 2020)

1155-57-472 **Nicolas Petit*** (petitnicola@gmail.com), Oxford, GA 30054. *The multi-variable affine index polynomial.*

We will be going over a recent generalization of the affine index polynomial to the case of virtual links. We will present the invariant and discuss some properties related to it. (Received January 20, 2020)

1155-57-518 **Krzysztof K. Putyra** and **Alexander N. Shumakovitch*** (shurik@gwu.edu), 801 22nd St. NW, Phillips Hall, Suite 739, Department of Mathematics, The George Washington University, Washington, DC 20052. *Unified Khovanov homology and its properties.* Preliminary report.

Unified Khovanov homology combines even and odd Khovanov homology theories into a single algebraic object that carries the structure of a module over the group ring $\mathbb{Z}\mathbb{Z}_2$. It was previously shown by the authors that the unified Khovanov homology is often a stronger knot invariant than the even and odd Khovanov homology

combined. In this talk, we present evidence that two knots with the same unified Khovanov homology might nonetheless have different even and/or odd Khovanov homology. This is remarkable since one can easily obtain even and odd Khovanov chain complexes from the unified one. (Received January 20, 2020)

1155-57-529 **Siddhi Krishna***, siddhi.krishna@bc.edu. *Taut Foliations and Positive Braids.* Preliminary report.

I will describe a construction for building taut foliations in manifolds obtained by Dehn surgery along positive braid knots (i.e. knots obtained by taking the closure of a positive braid). Namely, we'll discuss how we can reduce building foliations to a combinatorial problem. This has implications towards the L-space conjecture, which predicts that every non-L-space admits a taut foliation. No background in Heegaard Floer homology or foliation theory will be assumed. (Received January 21, 2020)

1155-57-534 **Anastasiia Tsvietkova*** (a.tsviet@rutgers.edu). *The number of surfaces of fixed genus embedded in a 3-manifold.*

It was noticed before that presence of embedded essential surfaces in a 3-manifold can give information about that manifold. However to construct, classify or count such surfaces is a non-trivial task. If 3-manifold is complement of an alternating link with n crossings in a 3-sphere, we previously showed that the number of genus-g closed surfaces is bounded by a polynomial in n. This was the first polynomial bound. We then extended the result to spanning surfaces, and to 3-manifolds obtained as Dehn fillings of alternating links. This was joint work with Joel Hass and Abigail Thompson. In the talk, I will discuss its generalizations. One generalization is a joint work with Jessica Purcell. It concerns any cusped 3-manifold that is complements of a link alternating on some embedded surface in an arbitrary 3-manifold. Another one is joint work with Marc Lackenby. There, we prove that for any closed hyperbolic 3-manifold, there are polynomially many genus-g surfaces in terms of hyperbolic volume of the manifold. (Received January 21, 2020)

1155-57-542 **Dorothy Buck***, Dept of Math, Duke University, Durham, NC 27705. *Entanglement in Biopolymers.*

We discuss several recent results where characterizing entanglement, of knots, knotoids or spatial graphs, lead to improved biological characterization of both DNA and proteins. (Received January 21, 2020)

1155-57-562 **Alexander Ellis** and **Ina Petkova***, ina.petkova@dartmouth.edu, and **Vera Vertesi**. *Knot Floer homology and the gl(1—1) link invariant.*

The Reshetikhin-Turaev construction for the standard representation of the quantum group gl(1—1) sends tangles to C(q)-linear maps in such a way that a knot is sent to its Alexander polynomial. After a brief review of this construction, I will give an introduction to tangle Floer homology — a combinatorial generalization of knot Floer homology which sends tangles to (homotopy equivalence classes of) bigraded dg bimodules. Finally, I will discuss how to see tangle Floer homology as a categorification of the Reshetikhin-Turaev invariant. This is joint work with Alexander Ellis and Vera Vertesi. (Received January 21, 2020)

58 ► Global analysis, analysis on manifolds

1155-58-186 **Karamatou Yacoubou Djima*** (kyacouboudjima@amherst.edu), **Linda Ness**, **Melanie Weber** and **Patricia Medina**. *Heuristic Framework for Multi-Scale Testing of the Multi-Manifold Hypothesis.*

Global linear models often overestimate the number of parameters required to analyze or efficiently represent datasets, for example when a data set in sampled from a manifold of lower dimension than the ambient space. The manifold hypothesis consists in asking whether the data lies on or near a d-dimensional manifold or is sampled from a distribution supported on a manifold. In this talk, we outline a heuristic framework for a hypothesis test suitable for computation and empirical data analysis. We consider two datasets made of multiple manifolds and test our manifold hypothesis on a set of spline-interpolated manifolds constructed based variance-based intrinsic dimensions computed from the data. (Received January 13, 2020)

60 ► Probability theory and stochastic processes

1155-60-12 **Christopher Corley*** (christopher-corley@utc.edu), 415 EMCS Building (Dept. 6956), 615 McCallie Avenue, Chattanooga, TN 37403, and **Andrew Ledoan** (andrew-ledoan@utc.edu), 415 EMCS Building (Dept. 6956), 615 McCallie Avenue, Chattanooga, TN 37403. *The density of complex zeros of random sums.* Preliminary report.

Let $\{\eta_j\}_{j=0}^N$ be a sequence of independent, identically distributed random complex Gaussian variables, and let $\{f_j\}_{j=0}^N$ be a sequence of given analytic functions that are real-valued on the real number line. In this talk, I will present an exact formula for the expected density of the distribution of complex zeros of the equation $\sum_{j=0}^N \eta_j f_j(z) = \mathbf{K}$, where $\mathbf{K} \in \mathbb{C}$. The method of proof employs a formula for the expected absolute value of quadratic forms of Gaussian random variables. In addition, I will discuss the limiting behavior of the density function as N tends to infinity and the numerical computations for the density function and empirical distributions for certain functions f_j, such as polynomials orthogonal on the real line and the unit circle. (Received September 24, 2019)

1155-60-61 **Indrajit Jana***, 1027 Wachman Hall, Department of Mathematics, 1805 North Broad Street, Philadelphia, PA 19122. *CLT for non-Hermitian random band matrices with variance profiles.*

We show that the fluctuations of the linear eigenvalue statistics of a non-Hermitian random band matrix of bandwidth b_n with a continuous variance profile $w_\nu(x)$ converges to a $N(0, \sigma_f^2(\nu))$, where $\nu = \lim_{n\to\infty}(2b_n/n)$. We obtain an explicit formula for $\sigma_f^2(\nu)$, which depends on the test function, and w_ν. When $\nu = 1$, the formula is consistent with Rider, and Silverstein (2006). We also compute an explicit formula for $\sigma_f^2(0)$. We show that $\sigma_f^2(\nu) \to \sigma_f^2(0)$ as $\nu \downarrow 0$. (Received December 09, 2019)

1155-60-77 **Ahmad Reza Soltani*** (ar.soltani@ku.edu.kw), DEPARTMENT OF STATISTICS and OR, College of Science, Kuwait University., 13060 Safat, Kuwait, Kuwait. *Exponential Function and Cauchy Distribution: Characterizations via Random Weight Averages.*

In this note, we look into a function ϕ that satisfies the integral equation $\phi(x) = \int_0^1 \phi(ux)\phi((1-u)x)du$. We prove that $\phi(x) = e^{cx}I_{(-\infty,0)}(x) + e^{bx}I_{[0,+\infty)}(x)$, c, b constants, are the only solutions to the integral equation. This in turn gives the result of [Van Assche, W. (1987), A random variable uniformly distributed between two independent random variables. *Sankhya A*, 49, 207-211] that for i.i.d X and Y, $UX + (1-U)Y \stackrel{d}{=} X$ if and only if X has a the Cauchy distribution. Moreover, we prove that for any n, the random weight mean $U_{(1)}X_1 + (U_{(2)} - U_{(1)})X_2 + \cdots + (1 - U_{(n-1)})X_n$ has a Cauchy distribution if and only if X_1 has a Cauchy distribution; $X_1, ..., X_n$ i.i.d, the random weights are the cuts of $(0, 1)$ by a uniform sample. Funded by Kuwait University, Research Administration, SS02/19. (Received December 20, 2019)

1155-60-98 **Marianna Russkikh***, russkikh@mit.edu. *Dimers and embeddings.*

One of the main questions in the context of the universality and conformal invariance of a critical 2D lattice model is to find an embedding which geometrically encodes the weights of the model and that admits "nice" discretizations of Laplace and Cauchy-Riemann operators. We establish a correspondence between dimer models on a bipartite graph and circle patterns with the combinatorics of that graph. We describe how to construct a 't-embedding' (or a circle pattern) of a dimer planar graph using its Kasteleyn weights, and develop a relevant theory of discrete holomorphic functions on t-embeddings; this theory unifies Kenyon's holomorphic functions on T-graphs and s-holomorphic functions coming from the Ising model. We discuss a concept of 'perfect t-embeddings' of weighted bipartite planar graphs. We believe that these embeddings always exist and that they are good candidates to recover the complex structure of big bipartite planar graphs carrying a dimer model.

Based on: joint work with R. Kenyon, W. Lam, S. Ramassamy; and joint work with D. Chelkak, B. Laslier. (Received January 02, 2020)

1155-60-102 **Le Chen*** (le.chen@emory.edu), Department of Mathematics, 400 Dorman Drive, Atlanta, GA 30322, and **Axel Saenz**. *Second moments for the stochastic heat equation on a ring.*

In this paper, we study the stochastic heat equation with multiplicative space-time white noise on both discrete ring and on a continuous torus. This model is also called the *parabolic Anderson model* in the literature. When the underlying space is the whole space, namely, either $\mathbb{Z}$ or $\mathbb{R}$, the formulas for the second moments are known; see Chen & Dalang 2014 and Borodin & Corwin 2012. In this paper, we obtain corresponding formulas when the underlying space is a discrete ring or a continuous torus. As an application, we obtain the precise second Lyapunov exponents, which converge to those on the whole space when the size of the ring or torus goes to infinity. (Received January 05, 2020)

1155-60-112 **Oanh Nguyen*** (onguyen@princeton.edu), 120 Prospect Ave, Apt G3, Princeton, NJ 08540. *Real roots of random functions.*

Random functions are linear combinations of deterministic functions using independent random coefficients. These innocent-looking objects appear naturally in physics and approximation theory and remain mysterious despite decades of intensive research. We will discuss recent progress on the study of random functions and present our approach via the local universality method to study questions on the real roots. We will also discuss the mean, variance, and the limiting distribution of the number of real roots for several classes of random functions. This talk is based on several joint papers with Mei-Chu Chang, Yen Do, Hoi Nguyen, and Van Vu. (Received January 06, 2020)

1155-60-149 **Elizabeth Meckes*** (ese3@case.edu) and **Kathryn Lockwood Stewart**. *On the eigenvalues of truncations of random unitary matrices.*

We consider the eigenvalues of the top-left block of a large Haar distributed unitary matrix. Using a new Coulomb transport inequality due to Chafaii, Hardy and Maida, we give a parametrized family of concentration inequalities for the spectral measures as a function of the proportion of the matrix which is retained. We also show that individual bulk eigenvalues are concentrated at predicted locations, so that the eigenvalue rigidity phenomenon present in many classical ensembles holds in this context as well. (Received January 09, 2020)

1155-60-154 **Mathav Murugan***, 1984 Mathematics Road, Vancouver, BC V6T 1Z2, Canada. *Singularity and absolute continuity of energy measures.*

For a symmetric diffusion, we show that the two-sided sub-Gaussian heat kernel estimate implies singularity of energy measures with respect to the reference measure. This gives an affirmative answer to a conjecture of Barlow (2003). This is joint work with Naotaka Kajino. (Received January 09, 2020)

1155-60-181 **Robert Buckingham*** (buckinrt@uc.edu), Department of Mathematical Sciences, University of Cincinnati, Cincinnati, OH 45221. *Nonintersecting Brownian bridges on the unit circle with drift and generalized Hastings-McLeod functions.*

Nonintersecting Brownian bridges on the unit circle form a determinantal stochastic process exhibiting random matrix statistics for large numbers of walkers. We investigate the effect of adding a drift term to walkers on the circle conditioned to start and end at the same position. We use Riemann-Hilbert analysis of a family of discrete orthogonal polynomials with a complex weight to compute the asymptotic distribution of total winding numbers in the scaling regime in which the expected total winding is finite. Furthermore, we show that an appropriate double scaling of the drift and return time leads to a generalization of the tacnode process expressed in terms of generalized Hastings-McLeod functions. We further investigate the large-degree asymptotic behavior of the generalized Hastings-McLeod functions, which have arisen in a variety of integrable probability contexts. This is joint work with Karl Liechty (DePaul University) and Kurt Schmidt (University of Cincinnati). (Received January 12, 2020)

1155-60-187 **R Banuelos*** (banuelos@purdue.edu), Depart of Mathematics, Purdue University, West Lafayette, IN 47907, **D Kim**, IN , and **M Kwaśnicki**, IN. *Martingale Transforms and and discrete singular integrals.* Preliminary report.

We discuss a class of discrete singular integral that arise from conditional expectations of martingale transforms of Doob-h processes in the upper half-space of R^d, $d \geq 1$. The goal is to obtain sharp ℓ^p norm inequalities for these operator from the corresponding inequalities for martingales. For $d = 1$ these singular integral include the discrete Hilbert transform. This case was treated in a 2018 paper of Bañuelos and Kwaśnicki proving a long-standing conjecture concerning its ℓ^p norm. (Received January 13, 2020)

1155-60-209 **Erin Crossen Brown**, **Sevak Mkrtchyan*** (sevak.mkrtchyan@rochester.edu) and **Jonathan Pakianathan**. *Asymptotic completeness of random iid gap sequences.*

Various notions of completeness for increasing sequences of positive integers have been studied. For example, Lagrange showed that the sequence of integer squares $1, 4, 9, \ldots, n^2, \ldots$ is weakly $\leq$ 4-complete, while the Goldbach conjecture claims that the sequence of prime numbers is asymptotically weakly $\leq$ 3-complete. In this work we study asymptotic completeness of random weakly increasing sequences of integers which have iid gaps. We show that in general, with probability one, such a sequence is asymptotically complete. We further show that given some conditions on the gap distribution, with probability one, a random iid gap sequence of weights is an asymptotically k-complete sequence for every fixed $k \geq 2$. (Received January 13, 2020)

1155-60-226 **Mirjana Vuletic*** (mirjana.vuletic@umb.edu), 100 William T. Morrissey, Boston, MA 02125. *Free boundary Schur process- models and asymptotic behavior.*

The Schur process, introduced by Okounkov and Reshetikhin, is one of the fundamental models in integrable probability. It encompasses several combinatorial random models such as plane partitions, domino tilings, and last passage percolation. The Schur process is a class of measures on sequences of partitions where the first and last partitions are required to be empty. In this talk we consider the free boundary Schur process, which is a generalization of the original process where the end partitions are free. We present some of these models and some new asymptotic results. The limiting behavior is given by new deformations of universal distributions of Schur processes, such as Airy kernel or Tracy-Widom distribution. This is a joint work with D. Betea, J. Bouttier, and P. Nejjar. (Received January 14, 2020)

1155-60-252 **Grigoris Paouris**, **Konstantin Tikhomirov** and **Petros Valettas*** (valettasp@missouri.edu). *Hypercontractivity, and lower deviation estimates in normed spaces.*

We consider the problem of estimating probabilities of lower deviation $\mathbb{P}\{\|G\| \leq \delta \mathbb{E}\|G\|\}$ in normed spaces with respect to the Gaussian measure. These estimates occupy central role in the probabilistic study of high-dimensional structures. It has been confirmed in several concrete situations, using ad hoc methods, that lower deviations exhibit very different and more complex behavior than the corresponding upper estimates. In this work we develop a general method for proving small ball estimates for norms. In the case of 1-unconditional norms our bounds are best possible up to numerical constants. We also study the lower small deviation estimates for both 1-unconditional and general norms and we obtain optimal results. In all regimes, $\|G\|_\infty$ arises (at the computational level) as an extremal case in the problem. The proofs exploit the convexity and hypercontractive properties of the Gaussian measure. (Received January 15, 2020)

1155-60-256 **Mark Rychnovsky*** (mr3623@columbia.edu) and **Evgeni Dimitrov**. *the GUE corners limit of an exactly solvable Markov process.*

We study an exactly solvable Markov process, valued in six vertex model configurations, which originates from the higher spin stochastic six vertex model. We use a steep descent analysis of integral formulas for the higher spin stochastic six vertex model to show that at large time, this Markov process converges to the GUE corners process. (Received January 15, 2020)

1155-60-267 **Fabirce Baudoin**, **Maria Gordina*** (maria.gordina@uconn.edu) and **David Herzog**. *Ergodicity for Langevin dynamics with singular potentials.*

We discuss Langevin dynamics of N particles on $\mathbb{R}^d$ interacting through a singular repulsive potential, such as the Lennard-Jones potential, and show that the system converges to the unique invariant Gibbs measure exponentially fast in a weighted Sobolev norm. The proof relies on an explicit construction of a Lyapunov function using a modified Gamma calculus. In contrast to previous results for such systems, our results imply geometric convergence to equilibrium starting from an essentially optimal family of initial distributions. This is based on the joint work with F.Baudoin and D.Herzog. (Received January 16, 2020)

1155-60-292 **Gregory J Herschlag***, 120 Science Dr, 117 Physics Building, Durham, NC 27708, and **Jonathan C Mattingly**. *Advances in Sampling Techniques for Planar Graph Partitions.*

Gerrymandering is the manipulation of political districts to advantage a particular group. To understand whether districts have been egregiously manipulated, one must model potential district outcomes in the absence of unfair redistricting in order to obtain a comparative baseline. There is a growing consensus to establish such baselines by sampling a representative collection of alternative and neutral redistricting plans. Sampling the space of redistricting plans may be recast as sampling the space of partitions on a (mostly) planar graph. Although methods to procure representative samples are still in their infancy, several research groups have made a number of compelling recent advances. In this talk I will discuss several novel sampling techniques developed by our research group; the new approaches are empirically more efficient at sampling graph partitions than the first generation of samplers. (Received January 16, 2020)

1155-60-302 **Rodrigo Bañuelos** (banuelos@purdue.edu), **Phanuel Mariano*** (pmariano@newhaven.edu) and **Jing Wang** (jingwang@purdue.edu). *Extremal Problems for the Expected Lifetime of Brownian motion and the Fundamental Frequency of a Drum.*

I will speak about the relationship between the fundamental frequency of a drum and probability. This connection will be through an inequality involving the fundamental frequency of a drumhead $D \subset \mathbb{R}^d$ and the maximum expected lifetime of Brownian motion started inside a domain $D \subset \mathbb{R}^d$. For larger drums the fundamental frequency becomes lower while the maximum expected lifetime increases. We improve on the constants of the

known inequality and prove a new asymptotically sharp inequality involving the moments of the expected lifetime of Brownian motion. We discuss conjectures about the sharp inequality and present our partial results about the extremal domains and sharp constants. (Received January 17, 2020)

1155-60-316 **Fabrice Baudoin** (fabrice.baudoin@uconn.edu), **Nizar Demni** (nizar.demni@univ-rennes1.fr) and **Jing Wang*** (jingwang@purdue.edu). *Quaternionic Brownian windings.*

We study the 3-dimensional winding processes along the Brownian paths in the quaternionic Euclidean, projective, and hyperbolic spaces, and obtain their asymptotic laws in large time. In the flat and spherical spaces the asymptotic laws are shown to be Gaussians, while in the hyperbolic space they are related to Cauchy relativistic distributions. (Received January 17, 2020)

1155-60-318 **Karl E Liechty***, kliechty@depaul.edu, and **Gia Bao Nguyen** and **Daniel Remenik**. *A deformation of the Tracy-Widom GOE distribution.* Preliminary report.

In random matrix theory the Tracy-Widom GOE distribution describes the location of the largest eigenvalue of large real symmetric random matrices. It also describes height fluctuations certain random growth models in the KPZ universality class, and can be characterized as the maximum value of the Airy process minus a parabola. We use this characterization to study a natural transition between the Tracy-Widom GOE distribution and the Gaussian distribution. Namely we study the maximum value of the Airy process with wanderers whose marginal distributions correspond to spiked complex Hermitian random matrix models like GUE. We present both Fredholm determinant and Painleve formulas for the distribution of this maximum. (Received January 17, 2020)

1155-60-332 **Promit Ghosal*** (pg2475@columbia.edu), 520 West 122nd Street, New York, NY 10027, and **Guillaume Remy**, **Xin Sun** and **Yi Sun**. *Probabilistic Conformal Blocks for Liouvlle CFT on the Torus.*

We give a probabilistic expression for the 1-point torus conformal block of Liouvile conformal field theory. Our expression involves an expectation over a Gaussian multiplicative chaos on the unit interval and specializes to known Dotsenko-Fateev type integral expressions for the conformal block at integer values of parameters. Our method uses BPZ equations and operator product expansions for a deformation of our GMC expression for the conformal block to prove certain shift equations hold for its series coefficients. We combine these equations with expressions for the block as a Dotsenko-Fateev integral for integer parameters and as the Nekrasov partition function coming from the AGT correspondence to obtain the desired equality. (Received January 18, 2020)

1155-60-348 **Hao Shen*** (pkushenhao@gmail.com), 309 West Washington Avenue Apt 601, Madison, WI 53703, and **Jian Song** and **Rongfeng Sun**. *SPDE limit of a directed polymer in Poisson random walks.*

We consider a model of a directed polymer in an environment of Poisson random walks. We prove that under diffusive scaling, as temperature goes to infinity, the partition function converges to a stochastic heat equation with a multiplicative noise, whose covariance is long-range and singular (more precisely, the heat kernel). (Received January 18, 2020)

1155-60-364 **Guang Yang*** (guang.yang@uconn.edu). *A version of Hörmander's theorem for semimartingales.*

In this talk, we study the properties of the solution of a rough differential equation driven by semimartingale. In particular, we will give an explicit formula for the Malliavin derivative of the solution and prove the solution has a smooth density when the vector fields satisfy the Hörmander's condition. A main difficulty is to give estimations to the Malliavin matrix. (Received January 18, 2020)

1155-60-366 **Peter Nandori*** (peter.nandori@yu.edu). *Global observables for random walks: law of large numbers.*

We consider the sums $T_N = \sum_{n=1}^{N} F(S_n)$, where S_n is a random walk on $\mathbb{Z}^d$ and $F : \mathbb{Z}^d \to \mathbb{R}$ is a global observable, that is, a bounded function, which admits an average value when averaged over large cubes. We show that T_N always satisfies the weak Law of Large Numbers but the strong law fails in general except for one dimensional walks with drift. Under additional regularity assumptions on F, we obtain the Strong Law of Large Numbers and estimate the rate of convergence. The growth exponents which we obtain are optimal in two special cases: for quasiperiodic observables and for random walks in random scenery. The talk is based on a joint work with Dmitry Dolgopyat and Marco Lenci. (Received January 19, 2020)

1155-60-374 **Mark W Meckes***, 10900 Euclid Ave, Cleveland, OH 44106-1712. *Bounding distances between determinantal point processes.*

I will describe a method for bounding distributional distances between determinantal point processes in terms of their kernels, which was originally developed by E. Meckes and myself to prove a rigorous version of a self-similarity conjecture for eigenvalues of random unitary matrices stated by Coram and Diaconis. I will also describe further developments and applications due to my student Kyle Taljan. (Received January 19, 2020)

1155-60-401 **Nan Chen*** (chennan@math.wisc.edu), 480 Lincoln Dr., Madison, WI 53706, and **Andrew J Majda** and **Xin Tong**. *Beating the curse of dimension with accurate statistics for the Fokker–Planck equation in complex turbulent systems.* Preliminary report.

Solving the Fokker–Planck equation for high-dimensional complex dynamical systems is an important issue. The basic algorithm involves a hybrid strategy with a small number of samples L, where a conditional Gaussian mixture in a high-dimensional subspace via an extremely efficient parametric method is combined with a judicious Gaussian kernel density estimation in the remaining low-dimensional subspace. Then effective strategies are developed and incorporated into efficient statistically accurate algorithms for solving the Fokker–Planck equations associated with a rich class of high-dimensional nonlinear turbulent dynamical systems with strong non-Gaussian features. These effective strategies exploit a judicious block decomposition of high-dimensional conditional covariance matrices and statistical symmetry to facilitate an extremely efficient parallel computation and a significant reduction of sample numbers. The resulting algorithms can efficiently solve the Fokker–Planck equation in much higher dimensions even with orders in the millions and thus beat the curse of dimension. Skillful behavior of the algorithms is illustrated for highly non-Gaussian systems in excitable media and geophysical turbulence (Received January 19, 2020)

1155-60-406 **Alexander Moll*** (a.moll@northeastern.edu). *Gaussian Fields in Geometric Quantization.*

The study of path properties of fractional Gaussian fields – such as their level sets and extreme values – often requires a regularization such as discretization of the domain. In this talk, we will show that the Segal-Bargmann construction of Fock-Sobolev spaces in geometric quantization provides a distinguished regularization of these fields on arbitrary Riemannian manifolds with boundary. As applications, (I) we derive limit shapes and global Gaussian fluctuations for a large class of point processes associated to classical Hamiltonian systems that are not necessarily Liouville integrable and also (II) realize the asymptotics of cycles of large uniform random permutations within our regularization of log-correlated Gaussian fields in 1 dimension. (Received January 19, 2020)

1155-60-413 **Liangbing Luo*** (liangbing.luo@uconn.edu), 341 Mansfield Rd, Storrs, CT 06268. *Logarithmic Sobolev Inequalities on Non-isotropic Heisenberg Groups.*

Heisenberg group is the simplest non-trivial example of a sub-Riemannian manifold. In this talk, we will discuss the dimension (in)dependence of the constants in logarithmic Sobolev inequalities on non-isotropic Heisenberg groups. In this setting, a natural Laplacian is not an elliptic but a hypoelliptic operator. The argument relies on viewing the logarithmic Sobolev inequality as a limiting case of Sobolev inequalities and tensorization of the logarithmic Sobolev inequalities. (Received January 19, 2020)

1155-60-424 **Cecilia F Mondaini*** (cf823@drexel.edu) and **Nathan Glatt-Holtz**. *Rates of convergence to statistical equilibrium: a general approach and applications.*

This talk focuses on the study of convergence/mixing rates for stochastic dynamical systems towards statistical equilibrium. Our approach uses the weak Harris theorem combined with a generalized coupling technique to obtain such rates for infinite-dimensional stochastic systems in a suitable Wasserstein distance. In particular, we provide two application scenarios. First, to show that Markov kernels constructed from a suitable numerical discretization of the 2D stochastic Navier-Stokes equations converge towards the invariant measure of the continuous system. This depends crucially on a spectral gap result for the discrete Markov kernel that is independent of the level of discretization. Second, to approximate the posterior measure obtained via a Bayesian approach to inverse PDE problems, particularly when applied to advection-diffusion type PDEs. In this latter case, the Markov transition kernel is constructed with an exact preconditioned Hamiltonian Monte Carlo algorithm in infinite dimensions. A rigorous proof of mixing rates for such algorithm was an open problem until quite recently. Our approach provides an alternative and flexible methodology to establish mixing rates for other Markov Chain Monte Carlo algorithms. This is a joint work with N. Glatt-Holtz (Tulane U). (Received January 20, 2020)

1155-60-425 **Souvik Dhara**, **Debankur Mukherjee** and **Kavita Ramanan*** (kavita_ramanan@brown.edu), Box F, Brown University, 182 George Street, Providence, RI 02912. *Scaling Limits of r-to-p norms of non-negative random matrices.* Preliminary report.

The study of r-to-p operator norms of large random matrices is a problem of importance in many areas; for example, when $r = p = 2$, this reduces to the question of finding the largest singular value of the matrix, whereas when p is the Hölder conjugate of r, then this problem reduces to the famous Grothendiek ℓ_p problem, which is of relevance in optimization. When $1 < p \leq r < \infty$, we analyze this norm for a class of sequences of non-negative random matrices with dimension growing to infinity, and show that it satisfies a somewhat non-standard central limit theorem. Along the way, we obtain sharp approximations for the maximizing vector, which may be of independent interest. (Received January 20, 2020)

1155-60-426 **David P Herzog***, Carver Hall, 411 Morrill Rd., Ames, IA 50011. *On the large-time behavior of singular stochastic Hamiltonian systems.*

We discuss the problem of convergence to equilibrium in two stochastic differential equations used in statistical sampling algorithms. In each system, the equilibrium probability distribution has an explicit density which is known up to a normalization constant. Moreover, each density is of the Boltzmann-Gibbs form. In the context of the algorithm, this form is exploited in order to take samples from a wide array of probability distributions by running the stochastic dynamics "long enough" started from conveniently chosen prior distributions. However, outside of a particular class of target distributions, very little is known about how fast the stochastic dynamics converges to this equilibrium. This talk will cover joint work with my collaborators to bridge this gap, ultimately resolving a challenge posed by Denis Talay at an AIM conference in 2007 about convergence to equilibrium for the singular, Lennard-Jones potential. (Received January 20, 2020)

1155-60-429 **Israel Klich*** (isklich@gmail.com) and **Rafael N Alexander**. *From Exact Tensor networks to a new representation of TASEP.*

The use and interest in tensor network methods have rapidly grown in recent years and represents a development of major numerical and analytical tools for the study of quantum states. In spite of an extensive amount of work, until now there have been no demonstration of an exact correspondence between networks of finite dimension (finite dimensionality of the involved tensors) and an actual ground state (or even excited state) of a reasonable (i.e., local) quantum critical Hamiltonian in the thermodynamic limit (such exact examples have only been presented for gapped systems before). In this talk I will describe a tensor network construction of the exact ground state of the quantum Motzkin spin chain. This tensor network exhibits a built in scale invariance, and has a finite bond dimension. I will show that a suitable modification of the network results in a novel representation of the well known steady state measure describing the totally asymmetric exclusion process. (Received January 20, 2020)

1155-60-497 **Yen Q Do*** (yendo@virginia.edu), Charlottesville, VA 22904. *Real roots of random polynomials.*

Finding the real roots for a polynomial is often a challenging talk, even numerically, especially when the polynomial has high degree. In this talk I will describe some recent work (by myself and also with coauthors, H. Nguyen, O. Nguyen, V. Vu) about the distribution of the real roots of random polynomials. I will also give a brief survey of the history of the subject. (Received January 20, 2020)

1155-60-536 **Robert Neel*** (robert.neel@lehigh.edu). *Geometric and Martin boundaries on Cartan-Hadamard manifolds.*

We discuss results on the solvability of the Dirichlet problem at infinity (DPI) and the identification of the geometric and Martin boundaries for Cartan-Hadamard manifolds, by both stochastic and non-stochastic methods. We emphasize the situation in the two-dimensional case, which is fundamentally different than the higher-dimensional case, via studying the long-time behavior of Brownian motion. (Received January 21, 2020)

1155-60-546 **Sam Punshon-Smith***, punshs@brown.edu, Providence, RI. *Uniform Scalar Mixing and Scalar Turbulence in Stochastic Fluid Mechanics.*

I will discuss recent work on the advection of a passive scalar by the stochastic Navier-Stokes equations. Using techniques from the theory of Lyapunov exponents for random dynamical systems and the Harris-type geometric ergodic theory for Markov chains, one can prove that the scalar almost surely mixes exponentially fast, uniformly in the diffusivity. This can be used to prove fundamental results in passive scalar turbulence, specifically the first rigorous proof of version of Batchelor's 1/k law on the power spectrum. This, in turn allows one to pass

the zero diffusivity limit in the stochastically forced scalar equation and obtain a non-trivial stationary process which relies on a cascade from the mixing to dissipate the input from the noise. (Received January 21, 2020)

1155-60-581 **Abdelmalek Abdesselam*** (malek@virginia.edu). *Conformal scaling limits on the boundary of an infinite tree.*

I will consider random Schwartz distributions on the boundary of an infinite regular tree and explain what it means for their probability distribution to be conformally invariant. I will then present a candiate for such an object which is a non-Gaussian scaling limit of a hierarchical model of Euclidean quantum field theory. I will report on the results obtained so far, jointly with Ajay chandra and Gianluca Guadagni, on this model: scale invariance and strong universality. (Received January 21, 2020)

1155-60-586 **Yanjmaa Jutmaan***, 4948 Sunburst ln, Charlotte, NC 28213, and **Stanislav Molchanov**, Charlotte, NC 28223. *Branching process in the random environment: the solvable models.*

We study the behavior of the branching process in a random environment on trees in the critical, subcritical and supercritical case. We are interested in the case when both the branching and the step transition parameters are random quantities and binary splitting of non-homogeneous Galton-Watson process. We present quenched and annealed classifications for such processes and determine some limit theorems in the supercritical quenched case. Corollaries cover the percolation problem on the random trees. The main tools are the compositions of the random generating functions. Second solvable model is branching process where moment generating function is canonical representation of the fractional linear functions. (Received January 21, 2020)

1155-60-597 **Tai Melcher***, 141 Cabell Drive, Kerchof Hall, Charlotttesville, VA 22904. *Smoothness of measures associated to Brownian motions.*

Smoothness is a fundamental principle in the study of measures on infinite-dimensional spaces. Smoothness properties on such measure spaces has allowed, for example, the development of a calculus which has become an invaluable tool in the analysis of stochastic processes and their applications. Canonical examples of smooth measures include those induced by a Brownian motion, both its end point distribution and as a real-valued path. We'll consider measures induced by Brownian motions in different infinite-dimensional geometric settings, and discuss the standard types of results of interest, including some motivation for their study. (Received January 21, 2020)

1155-60-598 **Fabrice Baudoin**, **Maria Gordina** and **Tai Melcher*** (melcher@virginia.edu). *Quasi-invariance for generalized Kolmogorov diffusions in infinite dimensions.*

The Kolmogorov diffusion stands as one of the simplest and most natural examples of hypoelliptic diffusions, that is, diffusions whose end point distribution is smooth with respect to Lebesgue measure. We consider different infinite-dimensional versions of these diffusions and show that they retain smoothness properties. These results are attained by establishing uniform control over coefficients appearing in certain functional analytic inequalities for approximating finite-dimensional diffusions. This is joint work with Fabrice Baudoin and Maria Gordina. (Received January 21, 2020)

1155-60-599 **Elizabeth Meckes**, **Tai Melcher*** (melcher@virginia.edu) and **Jing Wang**. *Improved log Sobolev coefficients for compact Lie groups.*

It is a now classical result in stochastic analysis that a lower bound on the Ricci curvature of a Riemannian manifold M implies that a log Sobolev inequality holds for the heat kernel measure on M. In particular, if the Ricci curvature is non-negative, the log Sobolev coefficient is linear. Using basic properties of Lie groups and Brownian motion, we improve this standard estimate for log Sobolev coefficients of the heat kernel measure on compact Lie groups equipped with a bi-invariant metric (and thus of non-negative curvature). This is joint work with Elizabeth Meckes and Jing Wang. (Received January 21, 2020)

1155-60-609 **Petr Plechac** and **Gideon Simpson*** (grs53@drexel.edu), Philadelphia, PA 19104. *Uncertainty Quantification for a Model Problem in Fracture Mechanics.* Preliminary report.

Many important materials, including glasses and ceramics, are of the linearly elastic/brittle variety, obeying a classical Hookean law until an applied load reaches a yield stress, at which point they break into some number of fragments. Predicting the behavior of such materials, through modelling and simulation, is of obvious value to a variety of applications. In this work, we add randomness in material properties into the problem, modelling them via random fields. We then apply tools from uncertainty quantification to investigate the robustness to deformation of materials in the presence of this randomness. (Received January 21, 2020)

1155-60-613 **Jonathan C Mattingly*** (jonm@math.duke.edu) and **David Herzog**. *A perspective on Hypocoercivity through averaging.*

I will discuss some relationships between averaging and hypocoercivity and the construction of some lyapunov functions. In particular I will discuss some applications to Langevin dynamics with with singular potentials. (Received January 21, 2020)

1155-60-615 **Jonathan C Mattingly*** (jonm@math.duke.edu). *Ergodicity in infinite dimensions.* Preliminary report.

I will discuss some problems arising in the ergodicity of infinite dimensional systems and some of their resolutions. (Received January 21, 2020)

1155-60-616 **Sandra Cerrai***, University of Maryland, Department of Mathematics, College Park, MD 20742. *Incompressible viscous fluids in the plane and SPDEs on graphs.*

We present some results on the asymptotic behavior of a class of stochastic reaction-diffusion-advection equations in the plane. We show that as the divergence-free advection term becomes larger and larger, the solutions of such equations converge to the solution of a suitable stochastic PDE defined on the graph associated with the Hamiltonian. Joint works with Mark Freidlin and Guangyu Xi. (Received January 21, 2020)

1155-60-624 **Marta Lewicka*** (lewicka@pitt.edu). *Regularity of Laplace eigenfunctions on random geometric graphs.*

Motivated by applications to semi-supervised learning on graphs, we discuss regularity questions related to eigenfunctions of Laplace-Beltrami operator posed on a random geometric graph on a compact manifold.

Work in collaboration with Jeff Calder and Nicolas Garcia Trillos. (Received January 21, 2020)

1155-60-625 **Gautam Iyer***, Dept. of Math. Sci. WEH # 6113, Pittsburgh, PA 15206. *A stochastic oscillator driven by fractional Brownian motion.*

We study stochastically forced nonlinear oscillators. When the forcing is Brownian, it is easy to see that the long time behaviour is governed by a diffusion driven by Brownian motion. Our aim is to study this problem when the forcing has a memory effect. In a certain parameter space, we show that when the oscillator is driven by fractional Brownian motion, the long time behaviour does not see the memory effect, and is still governed by a diffusion driven by standard Brownian motion. (Received January 21, 2020)

62 ► Statistics

1155-62-431 **Sui Tang***, 3400 North Charles St, Baltimore, MD 21218, and **Fei Lu**, **Mauro Maggioni** and **Ming Zhong**. *Learning interaction kernels in agent based systems.*

Interacting agent-based systems are ubiquitous in science, from the modeling of particles in Physics to prey-predator in Biology, to opinion dynamics in economics and social sciences, where the interaction law between agents yields a rich variety of collective dynamics. Inferring the interaction laws between agents from observational trajectory data is a fundamental task for modeling and prediction.

Given abundant data sampled from multiple trajectories, we use tools from statistical/machine learning to construct estimators for interaction kernels with provably good statistical and computational properties, under the minimal assumptions that the interaction kernels only depend on pairwise distance. In particular, we show that despite the high-dimensionality of the systems, optimal learning rates can still be achieved, equal to that of a one-dimensional regression problem. Numerical simulations on a variety of examples suggest the learnability of kernels in models used in practice, and that our estimators are robust to noise, and produced accurate predictions of collective dynamics in relative large time intervals, even when they are learned from data collected in short time intervals. This talk is based on the joint work with Mauro Maggioni, Fei Lu and Ming Zhong. (Received January 20, 2020)

1155-62-433 **Katherine Daftari***, 3250 Phillips Hall #3, Chapel Hilil, NC 27599. *A statistically robust classifier for passive particle tracking microrheology.* Preliminary report.

Particle tracking microrheology is a widely used technique to experimentally estimate otherwise inaccessible viscoelastic properties of soft complex fluids. Extracting trustworthy information from experimental data presents the challenge of understanding the various sources of error that distort the position time series of observed particles, and consequently, distort the inference of the elastic and viscous moduli. Two significant types of error are presented in this analysis: high frequency (camera) error, and low frequency drift of the sample and embedded beads. Soon-to-be-released software mitigates these error sources while in parallel fitting the thermal

fluctuations of the beads by a fractional Brownian motion (FBM) model. This statistically robust model of the particles' inherent subdiffusive behavior plus the error sources allows us to extract a two-parameter classifier of the thermal fluctuations, which in turn is transformed to estimate local viscoelastic properties around each bead. For FBM, the mean-squared-displacement (MSD) exhibits a power law: $MSD(\tau) \sim D_\alpha \tau^\alpha$. Likewise, the frequency-dependent storage ($G'(\omega)$) and loss ($G''(\omega)$) moduli exhibit power law behavior over the frequency spectrum of the measurements. (Received January 20, 2020)

65 ► Numerical analysis

1155-65-29 **Yuan Chen*** (yuan_chen@gwu.edu), Department of Statistics, George Washington University, Rome Hall, 801 22nd St NW, 7th Floor, Washington, DC 20052, **Songming Hou** (shou@latech.edu), Department of Mathematics&Statistics, P.O. Box 10348, Louisiana Tech University, Ruston, LA 71272, and **Xu Zhang** (xzhang@okstate.edu), Department of Mathematics, Oklahoma State University, 401 Mathematical Sciences Building, Stillwater, OK 74078. *Immersed Finite Element Methods for Interface Problems with Multi-Domains and Triple-Junction Points.*

In the talk, we introduce a class of immersed finite element (IFE) methods for elliptic interface problems with multiple-domain and triple-junction points. Our methods do not require the mesh to fit the interface, therefore structured triangular meshes or Cartesian meshes can be used. On interface elements involving more than one interfaces or with a triple junction point, we construct new IFE basis functions. To handle the non-homogeneous flux jump, we enrich the local approximation spaces by adding up to three flux basis functions. Our numerical results indicate that the interpolation and numerical solutions possess the optimal convergence rates in both L_2 and semi-H_1 norms for multi-domain interface problems. This work is collaborated with Songming Hou and Xu Zhang. (Received November 12, 2019)

1155-65-36 **Shukai Du***, 1324 Gibbs Ave, Falcon Heights, MN 55108. *New analysis techniques of HDG methods.*

For some hybridizable discontinuous Galerkin (HDG) methods, suitably devised projections make their error analysis simple and concise. However, this simple way of analysis only works for a limited class of HDG methods. This limitation makes it difficult to analyze many other important classes of HDG methods and severely hamper their applications to more general problems. In this talk, we present new analytical tools to solve this limitation in a systematical way. We introduce the so called *generalized projection-based error analysis*, which is a framework that enables to analyze vastly more classes of HDG methods in a very concise way. In addition, this new analysis framework also sheds light upon the connections between several different classes of HDG methods. We plan to first introduce these tools for a steady-state diffusion problem for the sake of demonstration. After that, we discuss the applications to elastic waves and Maxwell equations. (Received November 19, 2019)

1155-65-80 **Qiao Zhuang***, 225 Stanger Street, Blacksburg, VA 24061, and **Slimane Adjerid** and **Tao Lin**. *Immersed finite element methods for second order hyperbolic equations in inhomogeneous media.*

In this presentation, we reanalyze the fully discrete PPIFE method presented in the previous article. Utilizing the error bounds given recently for elliptic interface problems, we are able to derive optimal a-priori error bounds for this PPIFE method not only in the energy norm but also in L^2 norm under the standard piecewise H^2 regularity assumption in the space variable of the exact solution, rather than the excessive piecewise H^3 regularity. Numerical simulations for standing and travelling waves are presented, which corroboratively confirm the reported error analysis. (Received December 20, 2019)

1155-65-93 **Xu Zhang*** (xzhang@okstate.edu), Stillwater, OK 74078, and **Cuiyu He** (cuiyu.he@uga.edu), Department of Mathematic, University of Georgia, Athens, 30602. *A posteriori error estimation of an immersed finite element method.*

Interface problems arise in many applications in science and engineering. Partial differential equations (PDEs) are often used to model interface problems. Solutions to these PDE interface problems often involves kinks, singularities, discontinuities, and other non-smooth behaviors. The immersed finite element method (IFEM) is a class of numerical methods for solving PDE interface problems with unfitted meshes. In this talk, we introduce a residual-based a posteriori error estimation for IFEM. We show that the error estimator is globally reliable and locally efficient, with the reliability and efficiency constants independent of the interface location. This error estimator can provide accurate assessment of the numerical approximation and can also be used as a guidance for

adaptive mesh refinement for IFEM. Numerical results are provided to demonstrate the features of the adaptive IFEM. This is a joined work with Cuiyu He. (Received January 02, 2020)

1155-65-106 **Thomas L. Lewis*** (tllewis3@uncg.edu), 116 Petty Building, PO Box 26170, Greensboro, NC 27402. *Admissibility and Stability Analysis of a Convergent Narrow-Stencil Finite Difference Method for Fully Nonlinear Elliptic Boundary Value Problems.*

This talk will introduce a new convergent narrow-stencil finite difference method for approximating the viscosity solution of second order fully nonlinear elliptic partial differential equations including Hamilton-Jacobi-Bellman equations. The proposed finite difference method naturally extends the Lax-Friedrichs method for first order problems to second order problems by introducing a key stabilization term called a numerical moment. We will outline the analytic techniques for proving the admissibility and stability of the proposed scheme which are based on a new discrete Sobolev embedding result instead of the traditional ℓ^∞ analysis associated with monotone schemes. The convergence result for the new method will also be mentioned and some numerical tests will be provided. (Received January 06, 2020)

1155-65-115 **Zheng Sun** and **Yulong Xing*** (xing.205@osu.edu). *On structure-preserving discontinuous Galerkin methods for Hamiltonian partial differential equations: Energy conservation and multi-symplecticity.*

Many partial differential equations (PDEs) can be written as a multi-symplectic Hamiltonian system, which obeys the multi-symplectic conservation law. In this presentation, we present and study semi-discrete discontinuous Galerkin (DG) methods for one-dimensional multi-symplectic Hamiltonian PDEs, and show that the proposed DG methods can simultaneously preserve the multi-symplectic structure and energy conservation with a general class of numerical fluxes, which includes the well-known central and alternating fluxes. Applications to the wave equation, the Benjamin–Bona–Mahony equation, the Camassa–Holm equation, the Korteweg–de Vries equation and the nonlinear Schrodinger equation are discussed. Some numerical results are provided to demonstrate the accuracy and long time behavior of the proposed methods. (Received January 06, 2020)

1155-65-213 **Jenna C. Guenther*** (guenthjc@dukes.jmu.edu). *An Adaptive, Highly Accurate and Efficient, Parker-Sochacki Algorithm for Numerical Solutions to Large Scale Dynamical Systems.*

For nonlinear systems of differential equations, an explicit adaptive procedure using a foundation of the Parker-Sochacki Method (PSM) produces better accuracy in less time with significantly fewer steps when contrasted with many standard adaptive algorithms that use a Runge-Kutta (RK) foundation. First, two simple PSM functions are developed, illustrating a class of functions that represent the backbone of a future PSM tool for the scientific community. At each step across the domain, combinations of these functions efficiently and recursively generate the coefficients of the Taylor polynomial of the solution to the ODE system. An adaptive stepping algorithm is derived which provides a simple way to either increase or decrease the order of the method during computation. PSM Adaptive is first developed theoretically and then demonstrated on several examples, including a 2 degrees of freedom system related to missile defense. Results are compared against standard RK adaptive algorithms including 4th/5th order Dormand Prince, the algorithm which serves as the foundation of MATLAB's renowned ODE45 solver. It is noted in the 2 degrees of freedom example that PSM Adaptive takes roughly two orders of magnitude fewer steps and runs an order of magnitude faster for similar accuracy. (Received January 13, 2020)

1155-65-243 **Ruchi Guo*** (guo.1778@osu.edu) and **Tao Lin**. *An Immersed Finite Element Method For Elliptic Interface Problems In Three Dimensions.*

We present an immersed finite element (IFE) method for solving the typical three-dimensional second order elliptic interface problem with an interface-independent Cartesian mesh. The local IFE space on each interface element consists of piecewise trilinear polynomials. In this space, the IFE shape functions with the Lagrange degrees of freedom can always be constructed regardless of interface location and discontinuous coefficients. Detailed geometric analysis is given for surface approximation and its effect on the construction of IFE functions. The proposed IFE space is proven to have the optimal approximation capabilities to the functions satisfying the jump conditions. (Received January 14, 2020)

1155-65-289 **Jennifer K Ryan*** (jkryan@mines.edu), Applied Math & Statistics, Colorado School of Mines, 1800 Illinois St, Golden, CO 80401. *Utilizing Superconvergence in Numerical Approximations.*

Slimane Adjerid is the leading expert in identifying superconvergent points for discontinuous Galerkin approximations to PDEs. The property of superconvergence leads to the questions of where did it come from and how to exploit this information in a useful manner. In this talk, we discuss the relation between the superconvergent

points and the accurate wave propagation properties. Further, we show that by applying Smoothness-Increasing Accuracy-Conserving (SIAC) filters superconvergence can be achieved throughout the domain. This was joint work with Daniel Frean (UEA). (Received January 17, 2020)

1155-65-361 **SLIMANE ADJERID*** (adjerids@math.vt.edu), Department of Mathematics, 460 McBryde Hall, Blacksburg, VA 24061, and **Haroun Meghaichi** (haroun@vt.edu). *An Immersed DG Method for Acoustic-Elastic Wave Propagataion Problems.*

We present a new Immersed DG method for solving acoustic-elastic wave propagation problems in inhomogeneous media. The method consists of a stable weighted DG formulation combined with a specially designed finite element spaces to coupled acoustic and elastic waves. We show how to construct the IFE spaces and present a stability analysis. We present several numerical examples to show the performance of the proposed method and discuss future research problems. (Received January 18, 2020)

1155-65-373 **Tao Lin***, Department of Mathematics, 460 McBryde Hall, Virginia Tech, 225 Stanger Street, Blacksburg, VA 24061-1026. *Immersed Finite Elements for Some Interface Inverse Problems.*

This talk is about a fixed mesh method for solving interface inverse problems for a diffusion equation whose diffusivity is a piecewise constant representing different materials in the simulation domain. These inverse problems are to determine the interface that separates materials by measurements about the solution to the diffusion equation, which can be either on the boundary and/or in the interior of the domain. These interface inverse problems are formulated as shape optimization problems whose objective functionals depend on the interface. Both the governing partial differential equations and objective functionals in this shape optimization are discretized accurately by an immersed finite element (IFE) method on a fixed mesh independent of the interface. Formulas for the shape sensitivities of the discretized objective functions are derived within the IFE framework that can be computed accurately and efficiently with the discretized adjoint method. Features of this IFE-based shape optimization algorithm are demonstrated by a group of representative interface inverse/design problems. (Received January 19, 2020)

1155-65-466 **Jerry L. Bona**, **Hongqiu Chen** and **Ohannes A. Karakashian*** (okarakas@utk.edu), Knoxville, TN 37919, and **Michael Wise**. *Finite Element Methods for a System of Non linear Dispersive Equations.*

We contruct a priori and a posteriori error estimates for a system of KdV-KdV equations coupled through the non linear terms. We present numerical experiments validating the theoretical results. In particular we study the stability of solitary wave solutions and point out differences with the behaviour of solutions of the (single) KdV equation. (Received January 20, 2020)

1155-65-539 **Burak Aksoylu** (fceliker@gmail.com) and **Fatih Celiker*** (celiker@wayne.edu), Department of Mathematics, Detroit, MI 48202, and **George Gazonas**. *Convergence and Asymptotic Compatibility of Higher Order Collocation Methods for Nonlocal Problems.*

We study the convergence and asymptotic compatibility of higher order collocation methods for nonlocal operators inspired by peridynamics, a nonlocal formulation of continuum mechanics. We prove that the methods are optimally convergent with respect to the polynomial degree of the approximation. A numerical method is said to be asymptotically compatible if the sequence of approximate solutions of the nonlocal problem converges to the solution of the corresponding local problem as the horizon and the grid sizes simultaneously approach zero. We carry out a calibration process via Taylor series expansions and a scaling of the nonlocal operator via a strain energy density argument to ensure that the resulting collocation methods are asymptotically compatible. We find that, for polynomial degrees greater than or equal to two, there exists a calibration constant independent of the horizon size and the grid size such that the resulting collocation methods for nonlocal diffusion are asymptotically compatible. We verify these findings through extensive numerical experiments. (Received January 21, 2020)

1155-65-626 **Mahboub Baccouch*** (mbaccouch@unomaha.edu), DSC 233, 6001 Dodge st, Omaha, NE 68182. *A superconvergent local discontinuous Galerkin method for two-dimensional nonlinear elliptic equations on Cartesian grids.*

In this talk, we present a superconvergent local discontinuous Galerkin (LDG) method for two-dimensional nonlinear elliptic equations on Cartesian grids. Optimal a priori error estimates for the LDG solution and for the auxiliary variable that approximate its gradient are derived in the L^2-norm. The order of convergence is proved to be $p+1$, when tensor product polynomials of degree at most p are used. Moreover, we present several superconvergence results for the LDG solution and its gradient. These results allow us to construct a posteriori

LDG error estimates. The proposed a posteriori error estimates are computationally simple and are obtained by solving a local problem with no boundary conditions on each element. Finally, we show that, for smooth solutions, these a posteriori error estimates converge to the exact spatial errors in the L^2-norm under mesh refinement. Several numerical experiments are provided to validate the theoretical results. (Received January 22, 2020)

1155-65-630 **Xiaobing Feng*** (xfeng@math.utk.edu), Department of Mathematics, 1403 Circle Drive, Knoxville, TN 37996, and **Liet Anh Vo** (lvo6@vols.utk.edu), Department of Mathematics, 1403 Circle Drive, Knoxville, TN 37996. *Recent developments in numerical methods for stochastic Stokes equations with multiplicative noise.*

Besides the mathematical interests, stochastic Stokes (and Navier-Stokes) equations have been proposed to study turbulence flow under random forcing. Even in the simplest setting, their PDE solutions have very low regularity in time, which then poses a significant challenge for developing efficient and convergent numerical methods for the stochastic Stokes and (Navier-Stokes) equations. In particular, the most natural and popular class of numerical methods for those equations, namely mixed finite element methods, had not been proven to work. In this talk we shall present some recent developments in mixed finite element and splitting methods for the Stokes equations with multiplicative noise. We shall highlight the continuous and discrete stochastic inf-sup conditions and the strong convergence for both the velocity and pressure approximations, as well as the new analysis techniques used to obtain these results. Numerical experiments will also be presented to validate the theoretical results. (Received January 22, 2020)

68 ► Computer science

1155-68-5 **Josh Alman*** (jalman@mit.edu), 32 Vassar Street, 32G-634, Cambridge, MA 02139. *Limits on the Universal Method for Matrix Multiplication.*

In this work, we prove limitations on the known methods for designing matrix multiplication algorithms. Alman and Vassilevska Williams recently defined the *Universal Method*, which substantially generalizes all the known approaches including Strassen's Laser Method and Cohn and Umans' Group Theoretic Method. We prove concrete lower bounds on the algorithms one can design by applying the Universal Method to many different tensors. Our proofs use new tools for upper bounding the *asymptotic slice rank* of a wide range of tensors. Our main result is that the Universal Method applied to any Coppersmith-Winograd tensor CW_q cannot yield a bound on ω, the exponent of matrix multiplication, better than 2.16805. By comparison, it was previously only known that the weaker 'Galactic Method' applied to CW_q could not achieve an exponent of 2. (Received August 03, 2019)

1155-68-91 **Moon Duchin**, **Tom Needham** and **Thomas Weighill*** (thomas.weighill@tufts.edu). *The Topology of Redistricting.*

A recent trend in redistricting research is the use of large computer-generated ensembles of maps to evaluate proposed or enacted redistricting plans. These ensembles can help us understand the redistricting baseline for a state when combined with election and demographic data. Even though these ensembles represent tiny samples of the intractable space of all possible plans, they themselves contain a huge amount of geographic variation. In this talk, we will use persistent homology, a technique from topological data analysis, to cut through this complexity to capture essential partisan and geographic features of an ensemble of maps. We demonstrate this technique by using it to examine how district scale and election year affect representation in Pennsylvania, as well as to analyze two enacted plans for North Carolina. (Received January 02, 2020)

1155-68-222 **Yusu Wang*** (yusu@cse.ohio-state.edu). *Geometric and topological methods for graph analysis.*

Graphs form one of the most important types of data in various applications across science and engineering. They could be geometric in nature, such as road networks in GIS, or relational and abstract, such as protein-protein interaction networks. In this talk, I will give some examples where topological and geometric ideas can be used to analyze various forms of graph data. In particular, we will discuss the reconstruction of hidden geometric graphs from density data, the recovery of graph metrics from noisy observations, and the comparison of metric graphs. In each of the case, we aim to also provide theoretical understanding and guarantees. Through these topics, I hope to illustrate how geometric and topological ideas can be very useful in the (qualitative) analysis of graphs. (Received January 14, 2020)

1155-68-377 **Austin Conner**, Department of Mathematics, Texas A&M University, Mailstop 3368, College Station, TX 77843-3368, **Alicia Harper**, Department of Mathematics, Texas A&M University, Mailstop 3368, College Station, TX 77843-3368, and **Joseph M Landsberg*** (jml@math.tamu.edu), Department of Mathematics, Texas A&M University, Mailstop 3368, College Station, TX 77843-3368. *The punctual Hilbert scheme and matrix multiplication.*

In 1968 V. Strassen discovered the usual way we multiply matrices is not the most efficient one. This raised the question as to just how efficiently matrices can be multiplied, and led to the astounding conjecture that for large matrices, it is almost as easy to multiply them as to add them. After giving a brief history of the problem, I will explain new uses of algebraic geometry and representation theory that have advanced lower bounds in this central question in computer science. (Received January 20, 2020)

1155-68-468 **Peter Bürgisser**, **Ankit Garg**, **W. Cole Franks*** (franks@mit.edu), **Rafael Oliveira**, **Michael Walter** and **Avi Wigderson**. *Geodesically convex optimization for invariants and moment polytopes.*

Moment polytopes are convex bodies associated to certain group actions on manifolds. When the manifold is a projective variety invariant under the action of a reductive Lie group, such as the general linear group acting on the space of 3-tensors, the moment polytope encodes asymptotic information about the the invariant polynomials and the representation theory of the coordinate ring. This talk concerns the computational complexity of deciding moment polytope membership. This question finds surprising applications ranging from algebraic complexity to real analysis. Before this work, methods included enumerating the (potentially exponentially many) facets of the polytope and evaluating highest weight polynomials (of potentially exponential degree). We will discuss new optimization-based algorithms, initially inspired by Sinkhorn's famous algorithm for matrix scaling, that do not seem to face the same hurdles. These algorithms are notable for their simplicity, their applicability to arbitrary representations, and their ability to compute preimages under the moment map, a problem of considerable practical interest. The approach goes via the Kempf-Ness criterion and connects to the exciting area of geodesically convex optimization. (Received January 20, 2020)

1155-68-585 **Joshua D. Priest*** (jdp8jb@virginia.edu), **Madhav V. Marathe** (mmarathe66@gmail.com), **S. S. Ravi** (ssravi0@gmail.com), **Daniel J. Rosenkrantz** (drosenkrantz@gmail.com) and **Richard E. Stearns** (thestearns2@gmail.com). *Using Boolean Satisfiability to Solve Decision Problems for Synchronous Dynamical Systems.*

A discrete dynamical system over $\{0,1\}$ consists of an undirected graph in which each node has a time varying state value from $\{0,1\}$. Each node v is associated with a local transition function f_v whose inputs are the current state of v and those of its neighbours, and whose output is in $\{0,1\}$. A **Synchronous Dynamical System** (SyDS) has an update mechanism that computes and updates every node's state value *simultaneously*. The **configuration** of a system at a time t is the vector of state values of the nodes at that time. SyDS **trajectories** are sequences of successive configurations over time. Such trajectories can model contagion processes in social networks as well as cascading failures in a networked infrastructure system (such as an electrical power system).

Many decision problems for SyDSs have been studied in the literature. We show that that such decision problems can be reduced to instances of the Boolean Satisfiability Problem (SAT) and solved in practice with open source SAT solvers. These solvers can produce one, several or all solutions to a given decision problem, and under certain circumstances, can do so extremely quickly even when the given SyDS is large. (Received January 21, 2020)

74 ► Mechanics of deformable solids

1155-74-622 **Ben Vadala-Roth***, benLvr@live.unc.edu, and **Boyce E Griffith**. *Effect of fiber dispersion in fluid-structure interaction models of bicuspid aortic heart valves.*

The aortic valve in the human heart usually consists of three symmetric leaflets. One of the most common cardiac defects is bicuspid aortic valves (BAV), in which two leaflets have fused together. One of the two remaining leaflets closely resembles one from a typical trifleat valve, whereas the other consists of two cusps and a fibrous ridge-like region, called a raphe, in between the cusps. Many mathematical models of the dynamics BAV's, use the same constitutive laws and material parameters of normal tricuspid valves. However, it has been reported that the underlying fiber architecture of BAV's are less organized near the raphe region. We study the effect of varying levels of fiber dispersion in an invariant based material law for the solid mechanical behavior of a BAV leaflet. This material law is used to define the structural response in an immersed boundary (IB) simulation of fluid-structure interaction. (Received January 21, 2020)

76 ► Fluid mechanics

1155-76-126 **Ebrahim M. Kolahdouz*** (ebrahimk@email.unc.edu), 220 E Cameron Ave, 396 Philips Hall, Mathematics Department, Chapel Hill, NC 27514, and **Brent A. Craven** and **Boyce E. Griffith**. *An Immersed Lagrangian Eulerian Method for Biological Fluid Structure Interaction.* Preliminary report.

The interaction of incompressible viscous flows with immersed objects is ubiquitously found in biology and medicine. An immersed Lagrangian Eulerian coupling strategy is presented within the framework of the immersed interface method that allows fluid and solid subproblems to be solved in a partitioned manner and coupled through interface conditions. The present fluid-structure interaction (FSI) approach allows for general complex geometries with discrete surfaces while retaining sharp resolution of stresses at the fluid-solid interface. In the coupling of the fluid to the solid, the interfacial fluid stresses drive the solid motion, and a penalty method is used to ensure that the fluid satisfies the no-slip condition along the fluid-solid interface. The method is extensively tested against benchmark computational and experimental studies in two and three dimensions including geometries with non-smooth features. In particular, with no FSI sub-iterations for problems with so called artificial added-mass instabilities, the method is shown to be stable for simulations involving low, nearly equal, equal and high solid-fluid density ratios. Applications of this method to the simulation of deformable blood clots inside inferior vena cava will be presented. (Received January 07, 2020)

1155-76-182 **Yuanyuan Feng*** (yzf58@psu.edu), State College, PA 16802, and **Gautam Iyer**, **Jean-Luc Thiffeault** and **Yu Feng**. *Dissipation enhancement by mixing and suppression of phase separation.*

We quantitatively study the interaction between diffusion and mixing in both the continuous, and discrete time setting. In discrete time, we consider a mixing dynamical system interposed with diffusion. In continuous time, we consider the advection diffusion equation where the advecting vector field is assumed to be sufficiently mixing. We then study the addictive Cahn-Hilliard equation. We prove that when the flow is sufficiently mixed, phase separation will be prevented. (Received January 12, 2020)

1155-76-189 **Rafal Komendarczyk*** (rako@tulane.edu), 6823 St. Charles Ave., New Orleans, LA 70002. *On Wojtier's force-free minimizers and Moffatt's magnetic relaxation.*

We will exhibit a situation where a stationary state of the Moffatt's magnetic relaxation problem is different than the corresponding force-free minimizer of the Wojtier's variational principle under the assumption of only a weak convergence of the minimizing sequence. Such examples have been envisioned in Moffatt's seminal work on the subject and involve divergence-free vector fields supported on collections of essentially linked magnetic tubes and are valid under an assumption of the strong convergence of the minimizing sequence. (Received January 13, 2020)

1155-76-249 **Saad Qadeer*** (saadq@email.unc.edu) and **Boyce E Griffith**. *The Smooth Forcing Extension Method.*

Numerical techniques for solving fluid problems in irregular domains are typically low-order accurate, making them unsuitable for complex fluid computations since the stresses may fail to converge pointwise. The Immersed Boundary Smooth Extension method was introduced to remedy this; however, its structure rendered it ill-conditioned and unfeasible for time-dependent problems. In this talk, we shall present a similarly inspired technique, known as the Smooth Forcing Extension method. Our approach allows for convergence at arbitrarily high orders and can be easily extended to higher dimensions and complicated geometries. We shall demonstrate its effectiveness on a variety of test problems, including the time-dependent setting, before outlining strategies for further exploration (Received January 15, 2020)

1155-76-327 **Wei Zhang*** (wz7ap@virginia.edu), Department or Mechanical and Aerospace Engine, University of Virginia, Charlottesville, VA 22904, **Junshi Wang** (jw9un@virginia.edu), Department or Mechanical and Aerospace Engine, University of Virginia, Charlottesville, VA 22904, and **Haibo Dong** (hd6q@virginia.edu), Department or Mechanical and Aerospace Engine, University of Virginia, Charlottesville, VA 22904. *Application of sharp-interface immersed boundary method with block-based adaptive mesh refinement on a bio-inspired flow.*

Simulations on the bio-inspired flows with moderate Reynolds numbers, complex geometry require dense and/or large scale mesh, which is a challenge for the current numerical methods. Based on the 3D sharp interface immersed boundary method with multigrid acceleration (Mittal, et al. JCP 2008), an improved method is developed in the current work, with a block-wise Adaptive Mesh Refinement (AMR), an MPI/OpenMP hybrid

parallelization method compatible to the AMR, and a Level-set boundary reconstruction. With the block-wise AMR, a moving-block technique is developed to make it possible to numerically study a body moving in a big domain; and a multi-block technique is developed to simulate multiple bodies in a schooling distribution. The current method provides a highly efficient and extensible way to simulate the bio-inspired problems with dense mesh in a big domain. The newly developed is used to study the airflows and the resultant aerodynamic pressure or force in the pharyngeal airway to understanding the pathophysiology of snoring and sleep apnea. The flow is analyzed for the pressure fluctuations in the pharyngeal airway and found the vortex formation, aerodynamic pressure, and pharyngeal wall force was significantly affected by the width of the pharyngeal airway. (Received January 18, 2020)

1155-76-412 **Pengtao Yue*** (ptyue@vt.edu), 460 McBryde Hall, 225 Stanger Street, Blacksburg, VA 24061, and **Jiaqi Zhang** (zjiaqi@vt.edu), 460 McBryde Hall, 225 Stanger Street, Blacksburg, VA 24061. *Phase-field modeling of contact angle hysteresis and its application in drop impact dynamics.*

In reality, most solid surfaces are intrinsically rough or chemically heterogeneous. As a consequence, the contact line may get pinned at topological or chemical defects, which is known as contact angle hysteresis. In this talk, we introduce a thermodynamically consistent phase-field model that accurately captures contact angle hysteresis. The formulation satisfies a dissipative energy law, based on which we develop a finite-element discretization that is at least unconditionally energy-stable in some simple cases. In the end, we use this new method to simulate the transient process of a drop impacting a solid surface, where the drop may deposit, rebound, or partially rebound depending on the drop impact velocity and the surface wetting property. By tuning the hysteresis window, our numerical results can achieve very good agreement with the experimental measurements, which also manifests the importance of hysteresis in contact line dynamics. (Received January 19, 2020)

1155-76-423 **H. Reed Ogrosky*** (hrogrosky@vcu.edu). *Linear stability, nonlinear dynamics, and predicting plug formation for film flows inside a tube in the presence of surfactant.*

Viscous film flows inside a tube arise in numerous scientific applications. If such films are thick enough, they may pinch off and form liquid bridges, or plugs. The presence of an insoluble surfactant at the film's free surface has been previously shown to delay plug formation and in this talk we explore this phenomenon using a long-wave model. Linear stability analysis identifies two modes; in the absence of base flow, the 'interface' mode is the only unstable mode. The growth rates of this mode serve as an accurate predictor of how surfactant concentration increases plug formation time, and the effects of film thickness on this increase are quantified. In the presence of base flow, both the interface mode and 'surfactant' mode may be unstable. In the absence of surfactant, turning points in traveling wave solution families have previously been shown to be a good indicator of h_c, the critical thickness past which plugs may form. Here, in the presence of surfactant, turning points in solution families that arise from an unstable surfactant mode give an estimate of h_c, if the interface mode is linearly stable. When both modes are unstable, interpretation of these turning points as they relate to plug formation is more complicated. (Received January 20, 2020)

1155-76-545 **Alexander P Hoover***, 1318 W 89 St, Cleveland, OH 44102, and **Ricardo Cortez**. *A Force Doublet Approach to the Motion and Behavior of Microscopic Swimmers.*

In the world of bacterial locomotion, motile organisms generate wakes and eddies that affect their local fluid environment. When many of them are present around each other, they interact with the flows generated by other organisms and form active suspensions whose complex flow structures play a significant role in fluid transport and mixing. In this talk, we present a reduced model for bacterial locomotion that describes their self-propelled motion in a low Reynolds number viscous incompressible fluid. The model is based on a particular limit of regularized Stokeslets with built-in asymmetry in order to produce a swimming direction. The result is a single-particle model of a swimmer that does not require special treatment of the self velocity due to the regularization, while allowing us to efficiently study the collective motion of bacteria. With this model, we are able to model pusher and puller organisms in a straightforward manner for both freespace and periodic domains. We will characterize the particle dynamics and discuss the diffusion of these particles as a function of the concentration density. We will then take advantage of the regularized Stokeslets framework to understand how active suspensions interact with viscoelastic structures, such as biofilms. (Received January 21, 2020)

1155-76-565 **D. Michael Senter*** (dmsenter@live.unc.edu). *Tools and Techniques for Semi-Automated Generation of Finite Difference Based Meshes for Use with Immersed Boundary Solvers.* Preliminary report.

Numerous fluid-structure interaction problems in biology have been investigated using the immersed boundary method. The advantage of this method is that the complex geometries frequently encountered in biological applications, e.g., internal or external morphology, can easily be handled without the need to generate matching grids for both the fluid and the structure. Consequently, the difficulty of modeling the structure lies often in discretizing the boundary of the complex geometry (morphology). While tools for finite element methods have long been established, immersed boundary solvers like IB2d are based on a finite difference approximation. We will present a Python package that leverages computer vision and machine learning techniques to facilitate the semi-automated generation of the necessary finite difference meshes based on digital images of the morphology of interest. (Received January 21, 2020)

1155-76-611 **Mark S Schwitzerlett*** (msschwitzerl@vcu.edu), 1015 Floyd Ave, Richmond, VA 23220, and **H Reed Ogrosky** and **Ihsan A Topaloglu**. *Effects of slip on highly viscous thin-film flows inside a tube.* Preliminary report.

Viscous liquid film flows in a tube arise in numerous industrial and biological applications. Previous modeling studies have typically used no-slip boundary conditions, but in some applications the effects of slip at the boundary may not be negligible. We derive a long-wave model based on lubrication theory which allows for slippage along the boundary. Linear stability analysis verifies the impact of slip-length on the speed, growth rate, and wavelength of the most unstable mode. Nonlinear simulations demonstrate the impact of slip-length on plug formation and wave dynamics. These simulations are conducted for flows driven by gravity, core flow, or a combination of the two. (Received January 21, 2020)

1155-76-620 **MD MONIR HOSSAIN*** (mdmoh81@vt.edu) and **Anne Staples** (staplesa@vt.edu). *Evaluating canopy flow closure models for coral reefs using fully resolved 3d flow field data.*

Much research has been performed to understand the hydrodynamics of coral reefs above the canopy, but detailed flow dynamics inside the reef have largely remained unexplored. To make the analysis tractable, the Navier-Stokes equations are double-averaged (in space and time), resulting in a canopy flow model. The momentum equation obtained by the averaging approach generates spatially averaged Reynolds and dispersive stresses that require closure modeling. Most of these closure models are developed for simple homogenous geometries. In addition, we still depend on analytical models or semi-empirical data for prediction of the drag force on the coral reef. To determine these parameters directly, three-dimensional simulations of the flow-through a P. meandrina colony were performed using the immersed boundary method implemented in the large eddy simulation framework. The same methods were then used to perform the first simulation of the flow inside a model coral reef consisting of 24 P. Meandrina colonies. In this talk we will show how obtaining this fully resolved internal flow field will allow us to i) evaluate existing closure models used in the canopy flow theory, and ii) quantify the exact drag force that develops on the individual reef elements for the first time. (Received January 21, 2020)

1155-76-623 **Junshi Wang*** (jw9un@virginia.edu), 412 Seymour Rd Apt 1, Charlottesville, VA 22903, and **Yan Ren**, **George V. Lauder** and **Haibo Dong**. *Hydrodynamics and Low Dimensional Analysis of Fishlike Forward Swimming.*

Highly flexible body undulations are commonly observed in fish swimming. However, quantifying hydrodynamic advantages of body flexion remains unexplored in live fish swimming. In this work, a combined experimental and computational approach will be introduced to study the hydrodynamic role of body flexibility in a trout's steady swimming. A high-speed photogrammetry system and 3D model reconstruction techniques are used together to measure the kinematics of the body and fins with extraordinary details. A singular value decomposition (SVD)-based model reduction tool is developed to extract the dominant kinematical components of the entire fish for kinematics analysis and computational modeling. An immersed-boundary-method (IBM)-based computational fluid dynamics solver is then used to simulate the corresponding unsteady flows in all their complexity. Vortex dynamics and hydrodynamic benefits of different kinematical components are then studied. The methods and resulted findings from this work are expected to bring new insights into the design of next-generation bio-inspired autonomous underwater systems. (Received January 21, 2020)

81 ► Quantum theory

1155-81-40 **David J Bertucci*** (davidjbertucci@gmail.com), Physics Department, Shippensburg University, 1871 Old Main Dr, Shippensburg, PA 17257, and **A Abdurrahman** (ababdu@ship.edu), Physics Department, Shippensburg University, 1871 Old Main Dr, Shippensburg, PA 17257. *Infinite matrices in the half-string field theory of the open bosonic string.* Preliminary report.

In the half-string operator formulation of the open bosonic string proposed by Witten, one encounters infinite dimensional matrices. These infinite dimensional matrices correspond to the physical string states. For example, the interaction at the level of vertices can be computed in terms of product and trace of three infinite dimensional matrices. Moreover, the form of the interaction vertex in the half-string oscillator modes is given by an inverse of infinite dimensional matrix. However, it is not easy to invert these infinite dimensional matrices due to the lack of any formal procedure in matrix algebra to invert such matrices. In this paper, we shall achieve that by using the so-called Neumann function techniques. (Received November 22, 2019)

1155-81-41 **Andrew Charles Mueller*** (am2209@ship.edu), Physics Department, Shippensburg University, 1871 Old Main Drive, Shippensburg, PA 17257, and **A Abdurrahman** (ababdu@ship.edu), Physics Department, Shippensburg University, 1871 Old Main Drive, Shippensburg, PA 17257. *Infinite sums of the first and second kind.* Preliminary report.

In the generalization of the Comma 3-vertex of the open bosonic string to N strings, one needs to generalize the Neumann coefficients to an N-number of strings. In the process, one encounters infinite sums called sums of the first kind, the second kind and the third kind. These sums are highly nontrivial as they involve summing over infinite dimension matrices and the Taylor coefficient of multivalued functions. In this paper, we show that these sums obey certain recursion relations. We use the recursion relation to derive various identities, which enable us to perform the sums. Here we give a complete treatment of the first and second kind. (Received November 22, 2019)

1155-81-42 **Aeron Liss*** (al7874@ship.edu), Shippensburg University, 1871 Old Main Drive, Physics department, Shippensburg, PA 17257, and **A Abdurrahman** (ababdu@ship.edu), Shippensburg University, 1871 Old Main Drive, Physics department, Shippensburg, PA. *Infinite sums of the third kind.* Preliminary report.

In formulating the quantum field theory of N bosonic open bosonic strings using the half-string oscillator approach, the N-vertex is described in terms of degrees of freedom on two halves of a split string. In solving for the N-interaction vertex, one encounters Neumann coefficient. The explicit form of the Neumann coefficients contain three classes of infinite sums over infinite dimensional matrices. These sums can be classified into three groups, called sums of the first, second and third kinds. The first and second kinds of sums had been considered in another paper. Here we give various mathematical identities required to perform these sums and work out all the sums of the third kind. The form of sums we find suggests an analogous class of sums in the original theory of Witten, which is an essential ingredient in the proof of the equivalence of Witten's theory and the comma theory of the open bosonic string. (Received November 22, 2019)

1155-81-45 **Abdelkrim Kallich*** (ak2206@ship.edu), Physics Depatement, 1871 Old Main Drive, Shippensburg, PA 17257, and **Abdulmajeed Abdurrahman** (ababdu@ship.edu), Physics Department, Shippensburg, PA 17257. *Neumann Coefficients in the Comma theory of Witten's open bosonic string.* Preliminary report.

In the Comma approach to Witten's string theory of the open bosonic string one uses the modes on the right and left halves of the string to formulate the theory in half string oscillator modes. In the comma approach as it is the case in the full string, the central role in the formalism is played by the 3-interaction vertex, which is represented as exponentials of quadratic forms in the creation operators with concrete matrix coefficients (known as the Neumann coefficients). Using Neumann and conformal mapping techniques we obtain their explicit form and show that they are precisely the ones derived in the full string oscillator approach derived by Gross, et al. Hence establishing the equivalence of the 3-interaction vertex in both approaches to Witten's theory of the open bosonic string. (Received November 25, 2019)

1155-81-52 **Anatoly Dymarsky***, 505 Rose Street, Lexington, KY 40506. *Quantum KdV hierarchy at large central charge.*

I will review recent progress of "solving" quantum KdV hierarchy at large central charge, which includes finding the spectrum, the transfer matrix, and the generalized character. I will also discuss how explicit answers in the

large central charge limit can shed light on modular properties of the generalized 2d CFT partition function and be used to establish Eigenstate Thermalization Hypothesis. (Received December 02, 2019)

1155-81-208 **Hao Zou*** (hzou@vt.edu), Department of Physics, 850 West Campus Dr, Blacksburg, VA 24061, and **Eric Sharpe** and **Wei Gu**. *Gauged Linear Sigma Models for Symplectic Grassmannians.*

In this talk, I will discuss two-dimensional gauged linear sigma models realizing symplectic Grassmannians. After describing the construction of the GLSMs themselves, I will explain how the Coulomb branch computations in such GLSMs reproduce known ordinary and equivariant quantum cohomology ring relations. Similar constructions and analysis can also be used for more general isotropic spaces, for example, orthogonal Grassmannians and the G2 flag manifold. (Received January 13, 2020)

1155-81-305 **Florencia Orosz Hunziker*** (orosz@math.harvard.edu), 1 Oxford Street, Cambridge, MA 02138, and **Thomas Creutzig**, **Cuipo Jiang**, **David Ridout** and **Jinwei Yang**. *Tensor categories arising from the Virasoro algebra.*

We show that there is a braided tensor category structure on the category of C_1-cofinite modules for the Virasoro vertex operator algebra of arbitrary central charge. (Received January 17, 2020)

1155-81-535 **David E. V. Rose*** (davidrose@unc.edu) and **Logan Tatham**. *On the quantum type C spider.* Preliminary report.

Following Kuperberg, a spider is a generators-and-relations description of the category of representations of a quantum group, given in terms of trivalent graphs called webs. It is an open problem (outside of type A and rank 2) to find such a description for quantum groups associated to simple complex Lie algebras. We will discuss work that provides a conjectural formulation of the type C spider in rank 3. Further, we prove that our construction shares a number of properties with the category of finite-dimensional representations of quantum sp(6); in particular, we provide a purely combinatorial/diagrammatic formulation for the quantum sp(6) link invariant. Time permitting, we'll also discuss the general sp(2n) case, and implications for categorification. (Received January 21, 2020)

1155-81-563 **Du Pei***, 20 Garden St, Cambridge, MA 02138. *BPS States and Invariants of 3-Manifolds.*

String theory predicts a class of new homological invariants of 3-manifolds that generalizes the Gopakumar-Vafa and Ooguri-Vafa invariant. I will review recent progress in understanding these new invariants, focusing on their "decategorifications" that take the form of q-series with integer coefficients. (Received January 21, 2020)

1155-81-592 **Marco A Aldi*** (maldi2@vcu.edu). *What do generalized complex branes form?* Preliminary report.

In light of brane quantization, we discuss a possible approach to the category of generalized complex branes on tori. (Received January 21, 2020)

83 ► Relativity and gravitational theory

1155-83-170 **William D Linch III*** (wdlinch3@gmail.com). *Off-shell Supersymmetry and the M-theory Effective Action.*

The leading part of the M-theory low-energy effective action has been conjectured to be the minimal supersymmetric completion of a certain curvature-to-the-fourth correction to eleven-dimensional supergravity. This idea is difficult to exploit, because a component Noether procedure would have an infinite number of steps of ever-higher order in the curvature expansion. In an off-shell superspace, the supersymmetry closes independently of the interactions, but any 11D, N=1 superspace would put the theory on-shell. In this talk, I will describe an ongoing program to construct an 11D, N = 1/8 supergeometry that is potentially "sufficiently off shell" to supersymmetrize the leading M-theory correction to the eleven-dimensional supergravity action. (Received January 11, 2020)

90 ► Operations research, mathematical programming

1155-90-30 **Hamidreza Validi**, **Austin Buchanan*** (buchanan@okstate.edu) and **Eugene Lykhovyd**. *Imposing contiguity constraints in political districting models.*

Beginning in the 1960s, techniques from operations research began to be used to generate political districting maps. A classical example is the integer programming model of Hess et al. (*Operations Research* 13(6):998–1006, 1965). Due to the model's compactness-seeking objective, it tends to generate contiguous or nearly-contiguous districts, although none of the model's constraints explicitly impose contiguity. Consequently, Hess et al. had to manually adjust their solutions to make them contiguous. Since then, there have been several attempts to adjust the Hess model and other models so that contiguity is explicitly ensured. In this talk, we review two existing models for imposing contiguity, propose two new ones, and analytically compare them in terms of their strength and size. We conduct an extensive set of numerical experiments to evaluate their performance. While many believe that contiguity constraints are particularly difficult to deal with, we find that the problem does not become harder when contiguity is imposed. In fact, a branch-and-cut implementation of a cut-based model generates, for the first time, optimally compact districting plans for 21 different US states at the census tract level (under the compactness objective proposed by Hess et al.). (Received November 13, 2019)

1155-90-268 **Paul L. Goethals*** (paul.goethals@westpoint.edu), West Point, NY 10996. *Cybersecurity Attributes: Trends in Modeling and Application.*

In generating solutions to cybersecurity problems, many researchers will attempt to measure characteristics that seem less tangible in nature, such as resilience, trust, loyalty, and detectability. Developing a methodology or process for evaluating these attributes can be extremely challenging – a number of assumptions and factors may diminish the value of any one metric. The difficulty in integrating or applying proposed solutions in a security environment is also apparent and may contribute to the complexity of the problem. In this talk, trends in establishing models for cybersecurity attributes are presented. Future areas of work where research in these characteristics may be beneficial are identified. And, analogies to existing models unrelated to cybersecurity in the literature are also offered to promote further work in this research space. (Received January 16, 2020)

91 ► Game theory, economics, social and behavioral sciences

1155-91-8 **Kristopher Tapp*** (ktapp@sju.edu). *Clustering and Expected Seat-Share for District Maps.*

In the context of modern sampling methods for redistricting, we define the *clustering* of a political party (or of a geographic region), and we study how clustering affects the expected election outcome. (Received August 15, 2019)

1155-91-195 **Daryl Deford** and **Nicholas Eubank*** (nick@nickeubank.com), 1014 Hale Street, Durham, NC 27705, and **Jonathan Rodden**. *Partisan Dislocation: A Local Measure of District Naturalness and Gerrymandering.*

In this paper, we introduce a fine-grained local measure of the extent to which electoral districts combine and split local communities of co-partisans in unnatural ways. Our indicator – which we term *partisan dislocation* – is a measure of the difference between the partisan composition of a voter's geographic nearest neighbors and that of her assigned district. We show that our measure is a good local (and global) indicator of district manipulation, easily identifying instances in which gerrymanderers carve up clusters of co-partisans (cracking) or combining them in unnatural ways (packing). Using districting simulations, we examine the degree to which minimizing dislocation can 1) place limits on the ability of district drawers to create gerrymandered districts, and 2) reduce bias in the transformation of votes to seats. Our measure holds promise not only as a tool for detecting gerrymandering, but also for helping commissions and demographers who wish to draw better districts. (Received January 13, 2020)

1155-91-354 **Jeffrey S. Buzas** and **Gregory S. Warrington***, Department of Mathematics & Statistics, University of Vermont, 82 University Place, Burlington, VT 05405. *Simulated Packing and Cracking.*

A number of metrics have been proposed for evaluating partisan asymmetry in district plans. Many rely not on the shapes of districts, but only the distribution of votes. There is little agreement on which of these metrics are the most effective. There are many reasons for this lack of clarity, but one is certainly the difficulty of interpreting

analyses based on historical election data. In this talk we describe simulated packing and cracking as an *in vitro* technique for making such analyses. We assess how faithfully four metrics (efficiency gap, declination, partisan bias and mean-median difference) record the primary means by which partisan gerrymanders are effected: packing and cracking. Such determinations are especially important given the use of such metrics in conjunction with outlier analyses based on computer-generated district plans. (Received January 18, 2020)

1155-91-605 **Haley Colgate** and **Beth Malmskog*** (bmalmskog@coloradocollege.edu). *Finding the Baseline: A Colorado Case Study in Ensemble Analysis.*

Ensemble analysis is the process of generating a large ensemble of random, legal districting plans for a region and using real election data to model what political outcomes would have been for the random plans in real elections. In recent years, Ensemble analysis has become an essential tool in identifying and fighting egregious partisan gerrymanders, but the applications are broader than that. The technique can be used to create a baseline, i.e. to understand the range of outcomes that are reasonable for a state/region's unique physical, human, and political geography. In 2018, Colorado voters passed ballot measures that banned partisan gerrymandering and required maps to be as competitive as possible, as well as put redistricting in the hands of independent commissions of voters and created a framework for public input to the process. In this context, our group has created a data rich map of Colorado which we have used to generate a large ensemble of maps that satisfy Colorado's redistricting criteria. This analysis allows to paint a more complete picture of Colorado's electoral landscape, and to quantify possible unforeseen effects of competitiveness constraints in the state. (Received January 21, 2020)

92 ► Biology and other natural sciences

1155-92-59 **Yoichi Enatsu*** (yenatsu@rs.tus.ac.jp), 1-3 Kagurazaka, Shinjuku, Tokyo 162-8601, Japan, and **Takeo Ushijima** and **Emiko Ishiwata**. *Traveling wave for epidemic models with a free boundary.*

Free boundary problems are recently used to model phenomena of biological invasion for species such as migration into a new habitat (see, e.g., Du and Lin (2010) and references therein). These ideas are also applied to epidemic models. Kim et al. (2013) proposed an SIR epidemic model with free boundary to investigate the front motion of infected individuals spreading into a region where disease is not prevalent. In this presentation, we consider a simple diffusive epidemic model. Hosono and Ilyas (1995) and Kaellen (1984) proved the existence of traveling wave solutions of the model. We extend the results in Kaellen (1984) to the model with free boundary, namely we prove the existence of a semi wave solution (solutions propagating with the same profile and the same speed on a half space). We numerically observe the semi wave and the front motion of this model with free boundary. This is a joint work with Takeo Ushijima and Emiko Ishiwata. (Received December 09, 2019)

1155-92-81 **Tyrus Hunter Berry*** (tberry@gmu.edu), 4400 University Dr., Fairfax, VA 22030, and **Steven Schiff** and **Timothy Sauer**. *Spatiotemporal modeling and control of neonatal sepsis and hydrocephalus in Africa.* Preliminary report.

Neonatal sepsis is a significant contributor to neonatal and infant mortality in sub-Saharan Africa. Of those that recover, a significant number of infants go on develop hydrocephalus as a long-term consequence of the infection. Our goal is to model the disease across all of Uganda and find optimal strategies for prevention and treatment using ideas from control theory. We start by extending the Susceptible-Infected-Recovered (SIR) model to a new SIRH model by adding a Hydrocephalic class. Using publicly available statistics, we obtain coarse estimates of parameters. We then extend the model spatially and develop a method to estimate spatial fields for infection rates and rates of developing hydrocephalus. Finally, we incorporate the spatiotemporal model into a control theory framework by considering prevention and treatment as competing for health care resources. (Received December 21, 2019)

1155-92-83 **Xi-Chao Duan**, **Xue-Zhi Li** and **Maia Martcheva*** (maia@ufl.edu), Department of Mathematics, University of Florida, Gainesville, FL 32611. *Coinfection dynamics of heroin transmission and HIV infection in a single population.* Preliminary report.

We propose a model of a joint spread of heroin use and HIV infection. The unique disease-free equilibrium always exists and it is stable if the basic reproduction numbers of heroin use and HIV infection are both less than 1. The semi-trivial equilibrium of HIV infection (heroin use) exists if the basic reproduction number of HIV infection (heroin use) is larger than 1 and it is locally stable if and only if the invasion number of heroin use (HIV infection) is less than 1. When both semi-trivial equilibria lose their stability, a coexistence equilibrium occurs, which may not be unique. We compare the model to US data on heroin use and HIV transmission.

We conclude that the two diseases in the US are in a coexistence regime. Elasticities of the invasion numbers suggest two foci for control measures: targeting the drug abuse epidemic and reducing HIV risk in drug-users. (Received December 25, 2019)

1155-92-136 **Jin Wang*** (jin-wang02@utc.edu). *Basic reproduction numbers for a class of reaction-convection-diffusion epidemic models.* Preliminary report.

A numerical approach is presented to compute the basic reproduction numbers for a class of reaction-convection-diffusion epidemic models. Meanwhile, the numerical formulation provides useful insight into the characterization of those basic reproduction numbers. Using matrix analysis, we show that the basic reproduction numbers are the same for such PDE models and their associated ODE models in several important cases that include zero convection, uniform convection and diffusion, and partial convection and diffusion. (Received January 08, 2020)

1155-92-141 **Sivan Leviyang*** (sivan.leviyang@georgetown.edu), **Nate Strawn** and **Igor Griva**. *Constructing regulatory modules using a sparse regression approach with application to interferon stimulated genes at homeostasis.*

Interferon stimulated genes (ISGs) are a collection of genes important in the early innate immune response. There are 100s of ISGs and their regulations is complex, involving dozens of regulatory factors and varying across cell types. The regulation of ISGs has been mostly studied in cells exposed to significant interferon stimulation, while less is known about ISG regulation in homeostatic regimes in which extracellular interferon levels are low. In order to better understand homeostatic ISG regulation, we collected expression datasets across multiple cell types from the NCBI GEO database. Using a novel approach based on sparse linear regression, we constructed regulatory modules which grouped subsets of ISGs with a small number of correlated regulation factors. Application of our approach to the GEO datasets reveals significant levels of ISG regulation at homeostatis. (Received January 08, 2020)

1155-92-159 **Guido Felipe España*** (guido.espana@nd.edu) and **Alex Perkins**. *Projecting the impact of the CYD-TDV vaccination against dengue in Puerto Rico.*

Dengue is the fastest spreading vector-borne disease worldwide causing around 100 million cases every year. Almost 3 million cases were reported in Latin America in 2019. Even though a vaccine (CYD-TDV) has been developed against dengue, its efficacy is heterogeneous in the population. Hence, the World Health Organization has recommended to use serological screening prior to vaccination to administer the vaccine only to individuals that would benefit from it. However, serological screening with poor accuracy could result in misclassification of individuals, reducing the benefits of vaccination. We developed a stochastic, agent-based model to project the impact of routine vaccination strategies in a setting representative of Puerto Rico. We used our model to project the benefits of vaccination strategies with different levels of the accuracy of serological screening, and estimated the cost-effectiveness of these interventions. Our results suggest that this vaccination strategy could benefit Puerto Rico, assuming a moderate level of transmission. From a public payer perspective, the cost-effectiveness of vaccination strategies increased with high screening sensitivity. From an individual perspective, high specificity of serological screening reduced the risk of disease. (Received January 10, 2020)

1155-92-162 **Baoling Ma*** (baoling.ma@millersville.edu), **Chuan Li** and **Jack Warner**. *Structured mathematical models to investigate the interactions between malaria parasites and host immune response.*

Malaria infection has posed a major health threat for hundreds of years in human history. Yet, due to the complex interactions between a host immune response and the parasite, few mathematical models exist to study its dynamics. In this talk, a system of structured partial differential equations are established to account for the dependence of red blood cell infectivity on maturation level. These equations are coupled with another set of differential equations for investigating the population dynamics of Plasmodium falciparum and its interaction with red blood cells and cells of the immune system. A finite difference scheme is developed to solve the system. Numerical simulations are applied to investigate the interplay between the host immune response and the parasite dynamics. The disease dynamics in acute infection is also investigated. (Received January 10, 2020)

1155-92-203 **Hollee A McGinnis**, **Michael D Broda**, **Claire Luce**, **Cameron Tomlinson**, **Joy Ma**, **Jeremy Winslow** and **David M Chan*** (dmchan@vcu.edu). *Preliminary results of the impact of undergraduate student social and support networks on education and mental health.*

In this talk I will present preliminary data collected in the fall of 2019 from undergraduate students who were interviewed and described their social and support networks. These networks produce egocentric graphs that

are then studied. Our main interests include how their support network affects their mental health as well as their education. (Received January 13, 2020)

1155-92-205 **Matthew Mills**, **David M Chan*** (dmchan@vcu.edu), **James Forsberg**, **Michelle Chabot**, **Sarah Yimam** and **Ben Ramage**. *Effects of disturbance on species-specific feedbacks, and the implications for forest diversity.* Preliminary report.

In this talk I will present some preliminary work on an agent-based model that investigates the interactions between heterospecific and conspecific effects between tree species and different types of disturbances (e.g. hurricanes, forest fires, clear cutting, etc.). In particular we are interested in the effect of these interactions on forest biodiversity. (Received January 13, 2020)

1155-92-215 **Xiaoyuan Chang*** (smilingchang@126.com), **Junping Shi** and **Hao Wang**. *Analysis of a stoichiometric bacteria-grazer reaction-diffusion model for organic matter biodegradation.*

The dynamical behaviors of a bacteria-carbon-nitrogen reaction-diffusion system with grazers and without grazers are investigated in this paper. It is shown that the bacteria death rate and the grazer death rate play a decisive role in the dynamics of the system. The existence, stability and persistence of the solutions are proved by applying the theorem of existence of eigenvalue, the local and global bifurcation theory, and the abstract persistence theory. Through numerical simulations, the transient dynamic and the oscillation phenomenon are obtained. (Received January 13, 2020)

1155-92-233 **Stanca Ciupe***, 460 McBryde Hall, Blacksburg, VA 24060, and **Sarah Kadelka** and **Harel Dahari**. *Understanding the antiviral effects of RNAi-based therapy on chronic hepatitis B infection.* Preliminary report.

Reaching functional cure following chronic hepatitis B virus infections is hindered difficult by the presence of large numbers of HBsAg in the blood of infected patients. Therapies with the RNA interference drug ARC-520, which silence viral translation, together with daily administration of the nucleoside analogue drug entecavir have showed reduction in the overall levels of serum HBsAg in HBeAg-positive, treatment naive patients. Understanding the relative effects of ARC-520 alone, and in combination with entecavir, is particularly important in informing the development of new generation anti-HBsAg drugs. A mathematical model describing the mechanistic interactions between HBV DNA, HBsAg, and HBeAg in the presence of ARC-520 and entecavir has been developed. We fitted the model to patient data and investigated the long term dynamics of the virus and viral protein titers under entecavir alone and under combination therapy. We showed that ARC-520 reduces HBsAg to new positive equilibrium corresponding to its production from integrated viral genomes.We run in silico boosting experiments and used them to determine whether and when RNA interference therapies can lead to functional cure and what is the tradeoff between cure and drug induced toxicity. Such results can inform policy. (Received January 14, 2020)

1155-92-273 **Punit Gandhi*** (gandhipr@vcu.edu), **Sarah Iams**, **Sara Bonnetti**, **Amilcare Porporato** and **Mary Silber**. *A Fast-Slow Model of Banded Vegetation Pattern Formation in Drylands.*

From infiltration of water into the soil during rainstorms to seasonal plant growth and death, the ecohydrological processes that are thought to be relevant to the formation of banded vegetation patterns in drylands occur across multiple timescales. On gently sloped terrain, the patterns often appear as bands of vegetation growth alternating with bare soil. The vegetation bands can be tens of meters thick with spacing on the order of a hundred meters, and form a regular striped pattern that often occupy tens of square kilometers on the landscape. We propose a fast-slow switching model in order to capture the relevant processes on appropriate timescales within a conceptual modeling framework based on reaction-advection-diffusion equations. We explore the fast-slow model through numerical simulation on a one-dimensional hillslope, and find agreement with certain observations about the phenomenon, including band spacing and upslope colonization rates. We also find that the predicted soil moisture dynamics is consistent with time series data that has been collected at a banded vegetation site. (Received January 16, 2020)

1155-92-288 **Keng Deng***, Department of Mathematics, University of Louisiana at Lafayette, Lafayette, LA 70504, and **Qihua Huang**. *A hybrid parabolic and hyperbolic equation model for a population with separate dispersal and stationary stages.*

In this talk, we introduce a hybrid parabolic and hyperbolic equation model, in which a reaction-diffusion equation governs the random movement and settlement of dispersal individuals, while a first-order hyperbolic equation describes the growth of stationary individuals with age structure. We prove the existence and uniqueness of the

solution of the model using the monotone method based on a comparison principle. We study the population persistence criteria in terms of three related measures. (Received January 16, 2020)

1155-92-304 **Eric Forgoston*** (eric.forgoston@montclair.edu), Dept. of Applied Mathematics and Statistics, Montclair State University, 1 Normal Avenue, Montclair, NJ 07043. *Invasion and Extinction in Stochastic Epidemic Models.*

Noise plays a fundamental role in a wide variety of physical and biological dynamical systems. The noise may be internal or external - internal noise is inherent to the system itself and arises due to the random interactions of a finite number of agents in the system, while external noise arises from a source outside of the system. In recent years, it has been shown that even weak noise can induce a large fluctuation that leads to population extinction, switching between metastable states in ecological systems, or the escape of a particle from a potential well. After providing an overview of the relevant stochastic theory, we will show the development and application of new techniques to better understand the outbreak vulnerability and extinction dynamics for stochastic disease models. (Received January 17, 2020)

1155-92-328 **Jae Ho Lee*** (jaeholee@live.unc.edu), **Robert Hunt** (huntrl@live.unc.edu), **Margaret Anne Smith** (masmath@live.unc.edu) and **Boyce E. Griffith** (boyceg@email.unc.edu). *Fluid-Structure Interaction Models of Bioprosthetic Heart Valves to Study Leaflet Dynamics.*

The incidence of valvular heart disease increases rapidly with age, and the only treatment for severe valvular disease remains repair or replacement. Pulse duplicator systems are widely used in academia and industry and by regulators to design and assess prosthetic heart valves. This study leverages a fluid-structure interaction model of bioprosthetic valve dynamics in a ViVitro pulse duplicator system to study the role of valve geometry and leaflet flexibility on the leaflet dynamics. Our model is based on a version of the immersed boundary method that allows for the use of general hyperelastic material models for the immersed structures. Using this simulation framework, we present direct comparisons between model predictions and bulk flow properties (flow rates, pressure differences) and measurements of leaflet dynamics (valve area) obtained from the experimental pulse duplicator system as well as initial work towards comparing flow profiles to experimental flow profiles obtained using particle image velocimetry (PIV) in the ViVitro pulse duplicator. We study the role of leaflet geometry and elasticity on the dynamics of the device that may be relevant to leaflet durability. Our results suggest that a fully coupled FSI approach is needed to capture the full leaflet dynamics. (Received January 18, 2020)

1155-92-372 **Angela M Reynolds*** (areynolds2@vcu.edu), 1015 Floyd Ave, Richmond, VA 23284. *Modeling the Innate Immune Cells.*

During an inflammatory response there is a complex cascade of reactions, which may lead to health or sustained inflammation during many diseases and processes, such as wound healing and infections. In order to understand how the immune cells involved in the inflammatory response contribute to the disease progression, we have developed various models for the immune cell dynamics. In this talk we will start by analyzing a reduced model and then adapting this model to various diseases by increasing the complexity of the immune cells in the model to gain more biological insight. Using bifurcations, parameter estimation, and sensitivity analysis, we will explore how modulating the immune response dynamics can alter patient outcome. (Received January 19, 2020)

1155-92-379 **Zhilan Feng**, **Katharine Gurski*** (kgurski@howard.edu), **Margaret Grogan**, **Olivia Prosper** and **Miranda Teboh-Ewungkem**. *A vector-borne disease model with non-exponentially distributed infection and treatment stages.*

Most models for vector-borne disease assume exponentially distributed residence times in disease stages in order to simplify the model formulation and analysis. However, tackling drug resistance in malaria causing parasites requires an accurate description of the interaction between drug concentration and parasite load within hosts. For example, how long a human host has been infected is likely to influence their parasite load and their ability to transmit the parasite to a mosquito, especially among the immunologically naive individuals. The onset of clinical symptoms is likely to dictate when treatment begins. How long an individual has been undergoing treatment for an infection will determine their current blood-drug concentration and parasite load, and therefore, their susceptibility to re-infection with drug resistant parasites, along with their ability to transmit. Thus, we formulate a model by considering arbitrarily distributed sojourn for various disease stages. The model formulation is presented using integral equations. When general distributions are replaced by gamma distributions, the system of integral equations can be reduced to a system of ODEs, which has some non-trivial characteristics which are only captured by non-exponential distributions for disease stages. (Received January 19, 2020)

1155-92-420 **Maria Emelianenko***, memelian@gmu.edu, and **Dan Anderson**, **Fatah Kashanchi**, **Catherine DeMarino**, **Tin Phan** and **Yang Kuang**. *Three-state mathematical HIV model reveals differences in transcriptional dynamics between T-cells and macrophages.*

HIV-1 viral transcription persists in patients despite antiretroviral treatment, potentially due to intermittent HIV-1 LTR activation. While several mathematical models have been explored in the context of LTR-protein interactions, in this work for the first time HIV-1 LTR model featuring repressed, intermediate, and activated LTR states is integrated with generation of long (env) and short (TAR) RNAs and proteins (Tat, Pr55, and p24) in T-cells and macrophages using both cell lines and primary cell infection. This type of extended modeling framework allows us to compare and contrast behavior of these two cell types. We demonstrate that they exhibit unique LTR dynamics, which ultimately results in differences in the magnitude of viral products generated. Incorporation of a transcription inhibitor (F07#13) into the governing equations demonstrates how the model can be used to assess drug efficacy. Theoretical results concerning model stability and numerical sensitivity analysis on the model parameters are presented. (Received January 20, 2020)

1155-92-432 **Sarah B. Minucci*** (minuccisb@vcu.edu), 1015 Floyd Ave, Box 842014, Richmond, VA 23284, and **Rebecca L. Heise**, **Michael S. Valentine** and **Angela M. Reynolds**. *The Effects of Mechanical Ventilation on Macrophage Activation: Mathematical Model and Parameter Estimation.*

Mechanical ventilation is used to provide support to the lungs for patients with severe breathing issues, but as the air is pushed into the alveolar space it can trigger an immune response leading to ventilator-induced lung injury (VILI). A key component of the immune response is recruitment of macrophages, immune cells that differentiate into phenotypes with unique pro- and/or anti-inflammatory roles. An imbalance in pro- and anti-inflammatory responses can have deleterious effects on the individual's health. To develop a greater understanding of the mechanisms of the immune response to VILI and the sensitivity of post-ventilation outcomes, we develop a mathematical model of interactions between the immune system and site of damage, accounting for macrophage polarization. Through Latin Hypercube Sampling and available data, we generate a virtual cohort of patients with biologically feasible dynamics. We use a variety of methods to analyze the results, including a random forest decision tree algorithm and parameter sensitivity with eFAST. Analysis shows that parameters and properties related to epithelial repair and M1 activation and de-activation best predict outcome. We hypothesize interventions and use these treatment strategies to modulate damage in select virtual patients. (Received January 20, 2020)

1155-92-454 **Kyle Dahlin*** (kdahlin@purdue.edu), 150 N University Street, West Lafayette, IN 47907. *Investigating the effect of the development of host tolerance in the transmission of mosquito-borne diseases.* Preliminary report.

We develop a general model of sensitive, tolerant, and resistant sub-populations of a host population affected by a mosquito-borne disease. Sensitive hosts experience disease-induced mortality while tolerant hosts do not and are less effective at transmitting infectious agents. The impact of the evolution of tolerance in a sub-population of hosts on the overall host population is investigated. The creation of this reservoir of competent hosts has a non-linear effect on the overall population size. We consider the impact of translocation of tolerant hosts to a totally sensitive population and examine disease control measures that mitigate the potential harmful effects of translocation. We discuss applications of this model to particular systems of mosquito-borne viral or parasitic diseases in human or animal populations. (Received January 20, 2020)

1155-92-458 **Jonathan Forde***, 300 Pulteney St., Geneva, NY 14456. *Models of Chronic Hepatitis B Infection.*

Chronic hepatitis B infection is a major public health burden, leading to more than 750,000 deaths per year as result of cirrhosis and hepatocellular carcinoma. Understanding the interplay between viral factors and the host immune reaction is essential for designing interventions. We develop a mathematical model of chronic HBV infection, incorporating long-term effects such as gradual exhaustion of the immune response. Interacting feedback loops on differing time scales help explain the eventual collapse of immune control. (Received January 20, 2020)

1155-92-469 **Jessica M Conway*** (jmconway@psu.edu), Department of Mathematics, Pennsylvania State University, University Park, PA 16802. *Stochastic time-inhomogeneous HIV dynamics following treatment interruption.*

Antiretroviral therapy (ART) effectively controls HIV infection, suppressing HIV viral loads. Typically suspension of therapy is rapidly followed by rebound of viral loads to high, pre-therapy levels. However, recent

studies suggest that approximately 10% of study participants undergoing ART treatment interruption show viral rebound only months or years after interruption, while some may be controlling infection permanently. We will first define what we mean by viral rebound and describe model-supported hypotheses of HIV viral rebound and control. We will then discuss our branching process model to gain broad insight into these post-treatment dynamics. Specifically we provide theory that explains both short- and long-term viral rebounds, and post-treatment control, via a branching process with time-inhomogeneous rates, validated with data from Li et al. (2016). We will discuss the associated biological interpretation and implications. Finally, treatment interruption clinical trials are used to test efficacy of drug or other interventions to delay or prevent viral rebound; we will discuss how our modeling can be used to guide and inform such clinical trials. (Received January 20, 2020)

1155-92-476 **Michael A Robert*** (m.robert@usciences.edu), 600 S 43rd St, Philadelphia, PA 19104, **Rachel J Sippy**, Gainseville, FL, **Anna M Stewart-Ibarra**, Montevideo, Uruguay, **Rebecca C Christofferson**, Baton Rouge, LA, **Helen J Wearing**, Albuquerque, NM , and **Elizabet L Estallo**, Cordoba, Argentina. *Modeling meteorological influences on emergence of dengue in the temperate city of Cordoba, Argentina.* Preliminary report.

Dengue, a tropical virus transmitted by the mosquito species *Aedes aegypti*, has been spreading to populations in temperate regions across the globe. This spread has been driven by numerous factors including higher temperatures and more erratic precipitation patterns caused by global climate change. Temperature and/or precipitation impact various parts of the dengue transmission cycle, including mosquito development and survival and the incubation period of the virus in the mosquito, and with the continuing threat of climate change, it is critical that we develop a better understanding of meteorological influences on the spread of dengue. In this work, we expand a classic vector-host epidemiological ordinary differential equations model to include seasonal impacts of temperature and precipitation to investigate the potential impacts of each on dengue transmission in temperate regions. We specifically investigate the influence of climate on recent dengue emergence in the city of Córdoba, Argentina, which experienced its first dengue outbreak in 2009 and has since experienced yearly transmission and three other large outbreaks. We discuss the potential implications of our results for dengue mitigation strategies in Córdoba and other temperate cities. (Received January 20, 2020)

1155-92-501 **Tracey G Oellerich*** (toelleri@gmu.edu), **Maria Emelianenko**, **Robyn Araujo**, **Lance Liotta**, **Alessandra Luchini** and **Abdulaziz Alaraini**. *Adaptability Conditions in Biological Networks.*

In this talk we extend adaptability conditions for biological networks to include singular systems with non-hyperbolic equilibria. The proposed theoretical extension is compatible with the notions of homeostasis and robust perfect adaptation (RPA) and clarifies the relationship between the two. The new condition is derived using the notion of Moore-Penrose pseudoinverse and is implemented using a numerically efficient algorithm. The proposed approach is tested on several synthetic systems that are shown to exhibit homeostatic behavior yet lie outside of the scope of earlier work. (Received January 20, 2020)

1155-92-510 **Rebecca Segal***, 1015 Floyd Ave, Richmond, VA 23284, and **Wanda Strychalski**, **Sarah Bryant**, **Baasansuren Jadamba**, **Eirini Kilikian**, **Xiulan Lai**, **Leili Shahriyari**, **Ning Wei** and **Laura Miller**. *Fluid Dynamics of Nematocyst Prey Capture.* Preliminary report.

A nematocyst is a specialized organelle within cells of jellyfish and other Cnidarians that sting. Nematocysts are also present in some single-celled protists. They contain a barbed, venomous thread that accelerates faster than almost anything else in the animal kingdom. Here we simulate the fluid–structure interaction of the barbed thread accelerating through water to puncture its prey using the 2D immersed boundary method. For simplicity, our model describes the discharge of a single barb harpooning a single-celled organism, as in the case of dinoflagellates. One aspect of this project that is particularly interesting is that the micron-sized barbed thread reaches Reynolds numbers above one, where inertial effects become important. At this scale, even small changes in speed and shape can have dramatic effects on the local flow field. This suggests that the large variety of sizes and shapes of nematocysts may have important fluid dynamic consequences. We find that reaching the inertial regime is critical for hitting prey over short distances since the large boundary layers surrounding the barb characteristic of viscous dominated flows effectively push the prey out of the way. (Received January 20, 2020)

1155-92-513 **Shanshan Chen**, **Junping Shi**, **Zhisheng Shuai** and **Yixiang Wu*** (yixiang.wu@mtsu.edu). *Asymptotic profiles of the steady states for an SIS epidemic patch model with asymmetric connectivity matrix.*

The dynamics of an SIS epidemic patch model with asymmetric connectivity matrix is analyzed. It is shown that the basic reproduction number R_0 is strictly decreasing with respect to the dispersal rate of the infected individuals, and the model has a unique endemic equilibrium if $R_0 > 1$. The asymptotic profiles of the endemic equilibrium for small dispersal rates are characterized. In particular, it is shown that the endemic equilibrium converges to a limiting disease-free equilibrium as the dispersal rate of susceptible individuals tends to zero, and the limiting disease-free equilibrium has a positive number of susceptible individuals on each low-risk patch. Moreover a sufficient and necessary condition is found to guarantee that the limiting disease-free equilibrium has no positive number of susceptible individuals on each high-risk patch. (Received January 20, 2020)

1155-92-524 **Glenn F Webb*** (glenn.f.webb@vanderbilt.edu), Mathematics Department, Vanderbilt University, NASHVILLE, TN 37343, and **Pierre Magal** and **Yixiang Wu**. *An Environmental Model of Honey Bee Colony Collapse Due to Pesticide Contamination.*

A model of honey bee colony collapse is developed based on the contamination of forager bees in environmental regions contaminated with pesticides. An important feature of the model is the daily homing capacity each day of foragers bees. The model consists of difference equations describing the daily homing of uncontaminated and contaminated forager bees, with an increased homing failure of contaminated bees. The model quantifies colony collapse in terms of the fraction of contaminated bees subject to this increased homing failure. If the fraction is sufficiently high, then the hive falls below a viability threshold population size that leads to rapid disintegration. If the fraction is sufficiently low, then the hive can rise above the viability threshold and attain a stable population level. (Received January 21, 2020)

1155-92-530 **Libin Rong*** (libinrong@ufl.edu), Department of Mathematics, University of Florida, Gainesville, FL 32611. *Modeling the role of macrophages in HIV persistence.* Preliminary report.

HIV preferentially infects activated CD4+ T cells. Current antiretroviral therapy cannot eradicate the virus. Viral infection of other cells such as macrophages may contribute to viral persistence during antiretroviral therapy. In this talk, we will study the dynamics of macrophage infection. In addition to cell-free virus infection, macrophages can also get infected during the engulfment of infected CD4+ T cells as innate immune sentinels. I will also introduce a new HIV model that includes the infection of CD4+ T cells and macrophages via cell-free virus infection and cell-to-cell viral transmission. We will show the analysis of the model and discuss the roles of macrophages in HIV persistence during treatment. (Received January 21, 2020)

1155-92-532 **Libin Rong*** (libinrong@ufl.edu), Department of Mathematics, University of Florida, Gainesville, FL 32611. *Dynamics of a state-dependent HIV model with the activation status of infected cells.* Preliminary report.

The activation status can dictate the fate of an HIV-infected CD4+ T cell. Infected cells with a low level of activation stay in the latent state and do not produce virus, while cells with a higher level of activation are more productive and thus likely to transfer more virions to uninfected cells. How the activation status of infected cells affects HIV dynamics under antiretroviral therapy remains unclear. In this talk, I will introduce a new mathematical model that structures the population of infected cells continuously according to their activation state. The effectiveness of antiretroviral drugs in blocking cell-to-cell viral transmission is assumed to decrease as the level of activation of infected cells increases because the more virions transferred from infected to uninfected cells, the less effectively the treatment is able to inhibit the transmission. Asymptotic analysis of the model will be provided. We will also show that this model provides a new framework to study the observations, such as the low viral load persistence, extremely slow decay of latently infected cells and transient viral load measurements above the detection limit, in HIV-infected patients during long-term therapy. (Received January 21, 2020)

1155-92-540 **Andrew Nevai*** (math.anevai@gmail.com), Department of Mathematics, University of Central Florida, Orlando, FL 32816, and **Wichanon Sae-jie**. *A model of island community dynamics incorporating niches and species rank.*

We describe a model for species migrating between islands. Each island has a certain number of niches that species can occupy. A species can colonize an island provided that there are niches available. If there is no available niche, then it can replace a species already on the island provided that the colonizing species has a higher competitive ability than at least one existing species. Some results will be presented connecting colonization and extinction. (Received January 21, 2020)

1155-92-548 **Suzanne Robertson*** (srobertson7@vcu.edu). *Controlling vector-borne disease outbreaks on a neighborhood scale.*

Outbreaks of vector-borne disease such as Zika virus can occur when an infected individual introduces the virus to a residential neighborhood after traveling. Management strategies for controlling vector-borne disease typically involve large-scale application of larvicide or adulticide by truck or plane, as well as door-to-door control efforts that require obtaining permission to access private property. The efficacy of the latter efforts depend highly on the compliance of local residents. We present a model for vector-borne disease transmission in a neighborhood, considering a network of houses connected via mosquito dispersal. We use this model to compare the effectiveness of various control strategies and determine how optimal use of door-to-door control and aerial spraying depends on the level of resident compliance as well as mosquito movement. This is joint work with Jeffery Demers, Sharon Bewick, Folashade Agusto, Kevin Caillouet, and Bill Fagan. (Received January 21, 2020)

1155-92-549 **Cameron Browne*** (cambrowne@louisiana.edu). *Predator-prey dynamics of evolving HIV/SIV and immune response within a host.*

The dynamics of virus and immune response within a host can be viewed as a complex and evolving ecological system. For example, during HIV infection, an array of CD8+ T-cell immune response populations effectively target different HIV epitopes (viral proteins), however the virus can rapidly evolve resistance at distinct epitopes. In recent work, we analyzed stability and equilibria in a general model for the prey-predator network of multiple virus and immune response populations in HIV. In a parallel interdisciplinary study, we found significant cross-correlation between several immune cell populations and viral effective population sizes (Ne) sampled from experiments of the simian immunodeficiency virus (SIV)-infected macaque model of HIV infection. Connecting the theoretical modeling with this data, we observe that simulations of extended versions of the virus-immune network model mimic the observed large-scale oscillations in Ne. The results suggest that HIV diversity within the host is governed by a dynamic and complex predator-prey relationship, wherein the virus sequentially adapts to a series of weakening immune responses with accumulating viral diversity. (Received January 21, 2020)

1155-92-550 **Nicholas A Battista*** (battistn@tcnj.edu), 2000 Pennington Road, Ewing Township, NJ 08628. *Whose more sensitive: Jellyfish or Eels?* Preliminary report.

Aquatic organisms use diverse mechanisms for locomotion depending on their shape and size. Jellyfish have been deemed as one of the most efficient swimmers and use propulsatory methods via their bell contractions to swim. Their morphology and kinematics place them within the intermediate Reynolds Number, Re, regime. Many other animals also fall within this regime, such as larval fish, eels, or lamprey, which demonstrate anguilliform modes of locomotion. The effectiveness of both of these modes are highly dependent on a number of parameters, such as the frequency and kinematics of undulation and fluid scale. In this work we quantify the sensitivity of swimming performance (forward speed and cost of transport) to these parameters for both of these swimming modes using fluid-structure interaction tools. (Received January 21, 2020)

1155-92-551 **Igor V. Erovenko***, igor@uncg.edu, and **Johann Bauer**, **Mark Broom**, **Karan Pattni** and **Jan Rychtar**. *The effect of network topology on optimal exploration strategies and the evolution of cooperation in a mobile population.*

We model a mobile population interacting over an underlying spatial structure using a Markov movement model. Interactions take the form of public goods games, and can feature an arbitrary group size. Individuals choose strategically to remain at their current location or to move to a neighboring location, depending upon their exploration strategy and the current composition of their group. This builds upon previous work where the underlying structure was a complete graph (i.e., there was effectively no structure). Here, we consider alternative network structures and a wider variety of, mainly larger, populations. Previously, we had found when cooperation could evolve, depending upon the values of a range of population parameters. In our current work, we see that the complete graph considered before promotes stability, with populations of cooperators or defectors being relatively hard to replace. By contrast, the star graph promotes instability, and often neither type of population can resist replacement. We discuss potential reasons for this in terms of network topology. (Received January 21, 2020)

1155-92-557 **Necibe Tuncer***, ntuncer@fau.edu, and **Kristof Nemeth**, knemeth1@fau.edu. *Identifiability Analysis in Mutli-Scale Models.* Preliminary report.

We study the structural and practical identifiability of multi-scale models with multi-scale data. For multi-scale model, we consider nested within-host between-host models, where the within-host model is the immune response of the infected host, and between-host model is the time-since-infection structured SIR outbreak model. The epidemiological and immunological model are nested by linking the epidemiological parameters such as

transmission and recovery rates to infected host's immune response. For mutli-scale data, we use CDC 2018-2019 Influenza H1N1data combined with H1N1 viremia data within-infected host. We present the structural and practical identifiability analysis of the multi-scale model. (Received January 21, 2020)

1155-92-575 **Laura Ellwein Fix*** (lellwein@vcu.edu). *Dynamics of spontaneous breathing in the extremely preterm infant.*

The extremely preterm infant is at risk of developing chronic lung disease despite common treatments such as surfactant replacement therapy and non-invasive ventilation. Progressive lung collapse observed clinically in these infants has been attributed in part to their undermineralized rib cage and consequent high chest wall compliance, but the underlying mechanisms are not well understood and difficult to assess in this fragile population. To study the impact of variable chest wall compliance on dynamic changes in lung volume, we developed a computational model of nonlinear lung mechanics depicting dynamic airflow and esophageal pressure during spontaneous breathing in an idealized 1 kg infant. Results to date have demonstrated a faster loss of lung volume under high chest wall compliance conditions. We present additional findings from a parameter identifiability study using pseudo-data generated from simulated tracings and published clinical data, and further discuss simulation of thoracoabdominal asynchrony. (Received January 21, 2020)

1155-92-582 **Asma Alshehri**, **John Ford** and **Rachel Leander*** (rachel.leander@mtsu.edu). *The impact of maturation time distributions on the structure and growth of cellular populations.*

We study how the structure and growth of a cellular population vary with the distribution of maturation times from each stage. We consider two cell cycle stages. The first represents early G1. The second includes late G1, S, G2, and mitosis. Passage between the two reflects passage of an important cell cycle checkpoint known as the restriction point. We model the population as a system of partial differential equations in order to characterize the maturation rates in terms of the distributions of maturation times and derive the steady-state age and stage distributions as well as the asymptotic growth rates for models with exponential and inverse Gaussian maturation times. We find that the stable age and stage distributions, transient dynamics, and asymptotic growth rates are substantially different for these two maturation models. We conclude that researchers modeling cellular populations should take care when choosing a maturation time distribution, as the population growth rate and stage structure can be heavily impacted by this choice. (Received January 21, 2020)

1155-92-588 **Ryan Nikin-Beers*** (rnikinbeers@ufl.edu), **Julie Blackwood**, **Lauren Childs** and **Stanca Ciupe**. *Unraveling within-host signatures of dengue infection at the population level.*

Dengue virus causes worldwide concern with nearly 100 million infected cases reported annually. The within-host dynamics differ between primary and secondary infections, where secondary infections with a different virus serotype typically induce more severe disease. We build upon the variable within-host viral dynamics during infections resulting in mild dengue fever and severe dengue hemorrhagic fever. We couple these within-host viral dynamics to a population-level model through a system of partial differential equations. The resulting multiscale model examines the dynamics of between-host infections in the presence of two circulating virus strains. We determine thresholds of disease persistence analytically, while simulating the population dynamics numerically.

Using a similar numerical scheme, we develop and analyze a multiscale model which includes four virus strains. We focus on how the viral dynamics affect both the rate of transmissibility of infection and the rate of susceptibility to infection. We also determine how the viral dynamics contribute to the difference between disease stability and oscillatory behavior at the population level. (Received January 21, 2020)

1155-92-594 **Azmy Ackleh**, LA, and **Karyn Sutton**, **Tingting Tang*** (ttang2@sdsu.edu) and **Lihong Zhao**. *A Second Order Finite Difference Scheme for a Structured Model of Mycobacterium marinum Dynamics in Aquatic Animals.*

In this talk, we present a novel second order finite difference scheme for a modified version of a model, to study the transmission dynamics of a human TB-like bacterium, Mycobacterium marinum, affecting aquatic animals on a similar scale. Convergence of the finite difference approximation to the unique weak solution of the model is shown. Numerical results confirming the second order convergence of the scheme are presented. The computational advantages of using this high-resolution scheme in comparison to the first order scheme is illustrated (Received January 21, 2020)

1155-92-596 **Dane Patey**, **Jennifer Forbey**, **Steve Kern** and **Rongsong Liu***, Dept. of Math. and Stat., University of Wyoming, Laramie, WY 82071. *Using pharmacokinetic models to predict bioavailability of toxins in vertebrate herbivores.*

In this project, compartmental pharmacokinetic models are built to predict the concentration of toxic phytochemical in the gastrointestinal tract and blood following oral intake by an individual vertebrate herbivore. The existing single and multiple dose pharmacokinetic models are extended by inclusion of impulsive differential equations which account for an excretion factor whereby unchanged toxins are excreted in the feces due to gastrointestinal mobility. An index α is defined to measure the fraction of bioavailability attributed to the excretion factor of gastrointestinal motility. Sensitivity analysis was conducted and suggests, for any toxin, the bioavailability index α depends mostly on absorption rate of toxin from gastrointestinal tract into the blood, frequency of elimination due to gastrointestinal motility, and the frequency of toxin intake, under the model assumptions. (Received January 21, 2020)

1155-92-601 **Hayriye Gulbudak*** (hayriye.gulbudak@louisiana.edu), Lafayette, LA. *An Immuno-Epidemiological Vector-Host Model with Within-Vector Viral Kinetics.* Preliminary report.

A current challenge for disease modeling and public health is understanding pathogen dynamics across scales since their ecology and evolution ultimately operate on several coupled scales. This is particularly true for vector-borne diseases, where within-vector, within-host, and between vector-host populations all play crucial roles in diversity and distribution of the pathogen. Here we formulate an age-since-infection structured epidemic model coupled to nonlinear ordinary differential equations describing within-host immune-virus dynamics and within-vector viral kinetics, with feedbacks across these scales. We first define the *within-host viral-immune response and within-vector viral kinetics dependent* basic reproduction number $\mathcal{R}_0$, and then investigate the threshold dynamics of the system. Furthermore, we numerically show that within-vector-viral kinetics and dynamic inoculum size may play a substantial role in epidemic spread. Finally, we address how the model can be utilized to better predict the success of control strategies such as vaccination and drug treatment. (Received January 21, 2020)

1155-92-603 **Yanyu Xiao*** (yanyu.xiao@uc.edu), PBOX210025, cincinnati, OH 45221. *Dynamics of an HIV-1 virus model with both virus-to-cell and cell-to-cell transmissions.*

In this paper, we incorporate both virus-to-cell and cell-to-cell transmissions into an HIV-1 virus model with nonlinear general incidence rate, intracellular delay, and cytotoxic T lymphocyte (CTL) immune responses. By Lyapunov functionals and LaSalle invariance principle, it is verified that global threshold dynamics of three equilibria, infection-free, CTL-inactivated and CTL-activated, can be explicitly described by the basic reproduction numbers. (Received January 21, 2020)

1155-92-604 **Hayriye Gulbudak*** (hayriye.gulbudak@louisiana.edu), Lafayette, LA , and **Cameron Browne** (cambrowne@louisiana.edu), Lafayette, LA. *Infection severity across scales in multi-strain immuno-epidemiological Dengue model structured by host antibody level.*

Infection by distinct Dengue virus serotypes and host immunity are intricately linked. In particular, certain levels of cross-reactive antibodies in the host may actually enhance infection severity leading to Dengue hemorrhagic fever (DHF). The coupled immunological and epidemiological dynamics of Dengue calls for a multi-scale modeling approach. In this work, we formulate a within-host model which mechanistically recapitulates characteristics of antibody-dependent enhancement (ADE) in Dengue infection. The within-host scale is then linked to epidemiological spread by a vector-host partial differential equation model structured by host antibody level. The coupling allows for dynamic population-wide antibody levels to be tracked through primary and secondary infections by distinct Dengue strains, along with waning of cross-protective immunity after primary infection. Analysis of both the within-host and between-host systems are conducted. Stability results in the epidemic model are formulated via basic and invasion reproduction numbers as a function of immunological variables. Additionally, we develop numerical methods in order to simulate the multi-scale model and assess the influence of parameters on disease spread and DHF prevalence in the population. (Received January 21, 2020)

1155-92-607 **Cameron Browne*** (cambrowne@louisiana.edu), **Xuejun Pan**, **Hongying Shu** and **Xiang-Sheng Wang**. *Resonance of periodic combination antiviral therapy and intracellular delays in virus model.*

There is substantial interest in detailed models of viral infection and antiviral drug kinetics in order to optimize treatment against viruses such as HIV. In this paper, we study within-viral dynamics under general intracellular distributed delays and periodic combination antiviral therapy. The basic reproduction number R_0 is established

as a global threshold determining extinction versus persistence, and spectral methods are utilized for analytical and numerical computation of R_0. We derive the critical maturation delay for virus and optimal phase difference between sinusoidally varying drug efficacies under various intracellular delays. Furthermore, numerical simulations are conducted utilizing realistic pharmacokinetics and gamma-distributed viral production delays for HIV. Our results demonstrate that the relative timing of the key viral replication cycle steps and periodic antiviral treatment schedule involving distinct drugs all can interact to critically affect the overall viral dynamics. (Received January 21, 2020)

1155-92-619 **Max R. Mikel-Stites*** (mmikelst@vt.edu) and **Anne E. Staples**. *Small asymmetries produce big gains: tympanal asymmetry in a parasitoid fly.* Preliminary report.

Ormia ochracea is parasitoid fly famous for its ability to localize its hosts phonotactically with a precision of 2° in the azimuthal plane, a level of precision equal to that of humans. This is despite their small size, which should prohibit this degree of accuracy due to fundamental constraints imposed by the physics of sound propagation (Mason *et al., Nature*, 2001). The enhanced precision is due to the mechanical coupling of the fly's tympanal membranes. This coupling allows *O. ochracea* to resolve nanosecond time differences in incoming sound waves by increasing the interaural time delay (ITD) and interaural amplitude difference (IAD) between the fly's two ears. The tympanal system can be approximated by a simple mechanical model whose physics are well represented by two coupled ODEs (Miles *et al., J Acoust Soc Am*, 1995). Here, we modify this mathematical model by introducing an asymmetry in tympanal surface area between the two membranes. We also measured 38 *O. ochracea* tympanal membranes and found an average tympanal area asymmetry of approximately 5.6%. We show that, in the modified model, an asymmetry of just 5% between the left and right tympanal areas can result in more than a 22-fold gain in ITD and IAD. (Received January 21, 2020)

1155-92-621 **Hao Kang** and **Xi Huo*** (x.huo@math.miami.edu), 1365 Memorial Drive, University of Miami, Miami, FL 33156, and **Shigui Ruan**. *A population model with two physiological structures.* Preliminary report.

We consider a scalar population model with two physiological structures and study its fundamental properties and dynamical behaviors. It is well-known for a long time that the age-structure of a population affects the nonlinear dynamics of the species in ecology and the transmission dynamics of infectious diseases in epidemiology. In modeling specific diseases, the age could be chronological age, infection age, recovery age, class age, etc. Other physiological conditions or physical characteristics such as size, location, and status have also been taken in consideration in population dynamical models. Recently there are some studies taking into account the combined effects of two physiological characteristics (such as age-age, age-size, age-maturation, age-stage), and we here present our results on the abstract linear version of such problems. This is a joint work with Hao Kang and Shigui Ruan, University of Miami. (Received January 21, 2020)

1155-92-628 **Laura A Miller*** (lam9@unc.edu), CB 3280 Coker Hall, Department of Biology, Chapel Hill, NC 27599. *Flow through the hairy appendages of small animals: The leaky rake to solid plate transition.*

Numerous small organisms that swim, fly, smell, or feed in flows at the mesoscale where inertial and viscous forces are balanced, rely on using branched, bristled and hairy structures. Such mesoscale structures can augment underlying biological function (e.g., particle capture) by moving in a manner to transition from acting as solid surfaces to leaky/porous rakes at Reynolds number close to one. Although mesoscale flows have been studied in many organisms, the fluid dynamic mechanisms underlying the leaky rake to solid plate transition remain unclear, and robust mathematical models are unavailable. I will present results from numerical simulations and experiments that demonstrate how these transitions affect locomotion and exchange. Examples will include filtering flows through upside down jellyfish oral arms, airflows through bristled insect wings, exchange flows through corals, and sampling flows in crabs. In particular, the immersed boundary method will be used to resolve the flow through moving, flexible and porous structures. (Received January 22, 2020)

94 ► Information and communication, circuits

1155-94-212 **Gretchen L. Matthews*** (gmatthews@vt.edu), Department of Mathematics, Virginia Tech, Blacksburg, VA 24061. *Linear algebra and code-based cryptography.*

Quantum computing poses a threat to current public-key encryption schemes, such as RSA and elliptic curve cryptography. For this reason, there is much interest in cryptosystems which are more resilient in the face

of quantum algorithms, such as the McEliece cryptosystem. The McEliece cryptosystem is based on error-correcting codes and can be explained using undergraduate linear algebra. In this talk, we focus on code-based cryptography from this point of view. (Received January 13, 2020)

1155-94-555 **Bianca Sosnovski*** (bsosnovski@qcc.cuny.edu), 222-05 56th Avenue, Bayside, NY 11364. *Cayley Hash Functions and Open Problems.*

Cayley hash functions are cryptographic hashes constructed from Cayley graphs of non-Abelian groups. Although most of constructions of Cayley hash functions have been proved insecure, the algorithms used to break Cayley hash functions target specific vulnerabilities of each underlying group used and don't seem to invalidate the generic scheme of these functions. We review some of the latest Cayley hash constructions and related open problems. (Received January 21, 2020)

97 ► Mathematics education

1155-97-78 **Barry J. Griffiths*** (barry.griffiths@ucf.edu), Department of Mathematics, University of Central Florida, 4000 Central Florida Boulevard, Orlando, FL 32816-1364. *Cryptography in Undergraduate Education: Perceptions of Postgraduate Students.*

This talk will describe the experience of twenty students from sixteen different countries who have made the transition from being an undergraduate student to pursuing a postgraduate degree in cryptography. In particular, it considers their motivation for studying cryptography, the training that they received as undergraduates, and how it can be improved.

The collected data suggest that most of the participants decided on a career in cryptography midway through their undergraduate program, after gaining a broader knowledge of mathematics or computer science. They view the field as both important and exciting, but cite the value of a strong mathematical background, which could be enhanced by stronger undergraduate programming requirements. Most of the participants plan to stay in academia once their postgraduate program ends, and offer a range of ideas to help promote the field of cryptography so that younger students are aware of it. (Received December 20, 2019)

1155-97-92 **Tony Shaska*** (shaska@oakland.edu), MI. *Educating the "Good" Citizen: An algebraic viewpoint.*

Often the role of algebra in general education is debated in last decade. In this extended talk we will argue the fundamental role of algebra throughout history and its mission in modern education. We will discuss teaching techniques old and new of how to fulfil this mission. The talk will be especially directed to mathematics elementary and high school teachers and instructors of four year colleges. (Received January 02, 2020)

1155-97-486 **Andrea J Pretorian*** (andreap@bitira.com). *Educational Obstacles for Women in Crypto & STEM.*

A high barrier to entry into technology—for women and minorities in particular—has been documented as a consistent phenomenon. In fact, the educational community has been rallying to render STEM subjects more accessible. In this talk, we will review the statistics, what's known to be contributing, and solutions for overcoming these barriers. What can educators do in order to inspire and promote resilience and grit in their students? (Received January 20, 2020)

1155-97-552 **Anna K Vinnedge*** (anna.vinnedge@westpoint.edu). *Cyber Readiness of West Point Cadets.*

The Cyber Branch of the United States Army is the newest and fastest growing branch of the Army as cyber security and cyber warfare continues to grow in relevance to military readiness. The mission of the U.S. Army Cyber Command is to "integrate and conduct full-spectrum cyberspace operations, electronic warfare, and information operations, ensuring freedom of action for friendly forces in and through the cyber domain and the information environment, while denying the same to our adversaries." Cadets at the United States Military Academy from all academic disciplines are given the opportunity to commission as a cyber officer upon graduation, most of whom study computer science or cyber in preparation for their career. This research analyzes the readiness levels of West Point graduates who branch cyber in terms of understanding and utilizing cryptosystems for security purposes in an applied way. (Received January 21, 2020)

00 ► General

1156-00-30 **Eli Barkai***, Physics Dept., Bar-Ilan University, Ramat-Gan, Israel. *Infinite densities for large deviations of Levy walks.*

As shown in experiments conducted by Daphne Weihs and coworkers active transport in the cell breaks breaks the mono-scaling paradigm. In the presence of ATP, tracer particles exhibit strong anomalous diffusion where moments $\langle |x|^q \rangle \sim t^{q\nu(q)}$ increase either super-diffusively $\nu(q) \simeq 0.7$ or quasi ballistically $\nu(q) \sim 1$ for values of q below or above a critical value q_c respectively. Similar dual nature of the transport is found in many systems including deterministic models like the Lorentz gas and stochastic approaches like the Lévy walk. Given that standard diffusion equations and the central limit theorem fail to describe this widely observed bi-scaling we investigate the problem using the Lévy walk model. We show that non-normalised infinite densities are complementary to the central limit theorem in the statistical description of the process. Using a moment generating function formalism to these large deviations we investigate a model of diffusion of atoms in optical lattices, showing how infinite density concepts are related to physical processes involving laser particle interactions. (Received December 19, 2019)

1156-00-313 **Guy Wolf*** (guy.wolf@umontreal.ca), Montréal, Québec , Canada. *Graph Signal Processing Approach to Data Exploration.*

Big data are becoming increasingly common in many fields, especially in biomedical data science. Together with limited annotation, these introduce several challenges in unsupervised data processing aiming to identify meaningful patterns at the initial exploratory stage of the analysis. Here, I will discuss several tools that approach these challenges by leveraging graph signal processing and harmonic analysis on manifolds. A common thread in these tools is the construction of data-driven diffusion geometry that captures intrinsic structure in data and provides a generalization of Fourier harmonics. Further, this allows the separation of intrinsic geometry from data distribution, which is often biased by collection artifacts and data availability while hindering the identification of rare subpopulations and sparse transitions that are of great interest, e.g., in biomedical applications. The extracted harmonics are then used to process data features along the data geometry for denoising and generative purposes, which will be presented within the context of genomic & proteomic data exploration. Finally, I will relate the presented approach to the emerging field of geometric deep learning via the construction of deep graph filter banks known as the geometric scattering transform. (Received January 27, 2020)

1156-00-352 **Alexander Mazel**, **Izabella Stuhl*** (ius68@psu.edu) and **Yuri Suhov**. *The hard-core model on 2D graphs.*

We will report on results about high-density Gibbs measures in the hard-core model on discrete 2D structures: a unit triangular lattice $\mathbb{A}_2$, square lattice $\mathbb{Z}_2$ and a hexagonal tiling $\mathbb{H}_2$. We describe the extreme Gibbs measures for a general value of allowed Euclidean hard-core diameter D by referring to number-theoretical properties of the (integer) number D^2. The main tool is the well-known Pirogov-Sinai theory. In particular, we address the issue of sliding arising for models on $\mathbb{Z}^2$ and $\mathbb{H}_2$. (Received January 27, 2020)

01 ► History and biography

1156-01-35 **Anne Mahoney*** (anne.mahoney@tufts.edu). *An Indian Puzzle Exploiting Place-Value Notation.* Preliminary report.

The 12th-century Indian mathematician Bhāskara, in his *Līlavatī,* introduces what he calls the *aṅkapāśa* or "net of digits," in the context of a discussion of combinatorics. The net is the set of all numbers that can be produced by permuting the digits of a given number, for example {617, 671, 167, 176, 716, 761}. Bhāskara poses and solves the problem of summing up all the numbers in the net. This is apparently the first mathematical puzzle in history that depends on place-value notation.

The title of the text, *Līlavatī,* means "playful one." Because it is feminine, and because the book is addressed to an unnamed young woman, the conventional story is that the book is named for and addressed to Bhāskara's daughter, though there is no independent evidence that he ever had a daughter, by this name or any other.

The book is also presented in the form of a series of puzzles, so it is at least possible that the name refers to mathematical play.

Beyond Bhāskara's work, one can pose other puzzles: find a net all of whose members are prime, find a net whose members include multiples of the first N primes, find a net all of whose members are prime in 2 different bases, and so on. (Received December 24, 2019)

05 ► Combinatorics

1156-05-20 **Zhongyang Li*** (zhongyang.li@uconn.edu). *Exact Recovery of Community Detection in k-partite Graph Models.*

We study the vertex classification problem on a graph in which the vertices are in k different groups, and edges are only allowed between vertices in distinct groups. The observation is the weighted adjacency matrix, perturbed by a Gaussian Orthogonal Ensemble (GOE), or Gaussian Unitary Ensemble (GUE) matrix. Different from the standard symmetric Z^2-synchronization in which there are 2 communities with equal number of vertices, we do not assume that the number of vertices in each group is the same. For the exact recovery of the maximum likelihood estimation (MLE) with various weighted adjacency matrix, we prove a sharp phase transition result with respect to the intensity of the Gaussian perturbation. These weighted adjacency matrices may be considered as natural models for the electric network. In contrast to the well-studied Z^2-sychronization, a new complex version of the semi-definite programming (SDP) is designed to efficiently implement the community detection problem when the number of communities k is greater than 2, and a common region for σ such that SDP exactly recovers the true classification is obtained, which is independent of k. (Received December 10, 2019)

1156-05-29 **Emilio Bujalance**, **Marston Conder**, **Antonio F. Costa** and **Milagros Izquierdo*** (milagros.izquierdo@liu.se), Department of Mthematics, Linköping University, 58183 Linköping, Sweden. *Regular maps with 4g automorphisms and a curve of Wiman.*

n 1896, Wiman gave two (smooth, irreducible) complex algebraic curves for each genus $g \geq 2$: one admitting an automorphism of order $4g + 2$, and another admitting an automorphism of order $4g$. These curves are known as Wiman's curves of type I and II respectively. These curves are determined by regular maps with large automorphism group. Wiman's curves of type II have exactly $8g$ automorphisms, except in the case $g = 2$, when the curve has 48 automorphisms (and it provides the regular map of genus 2 having the maximum number of automorphisms).

Wiman's curve of type I provides the only (regular) map of genus g with $4g + 2$ orientation-preserving automorphisms. In the same way, for all integer values of $g \geq 2$ other than 3, 6, 12 and 30, there are exactly two regular dessins d'enfant of genus g with orientation-preserving automorphism group of order $4g$. Moreover, the regular map $\mathcal{W}_g$ with orientation-preserving automorphism group of order $8g$ corresponding to Wiman's curve of type II can be obtained as a medial subdivision of each of the two non-sporadic dessins with $4g$ orientation-preserving automorphisms. (Received December 19, 2019)

1156-05-31 **Gareth A. Jones*** (g.a.jones@maths.soton.ac.uk), Mathematical Sciences, University of Southampton, Highfield, Southampton, SO17 1BJ, United Kingdom. *Paley graphs and the random graph.* Preliminary report.

In 1963 Erdős and Rényi published a paper with the title Asymmetric Graphs. (A graph is symmetric or asymmetric as it has or does not have a non-identity automorphism.) They proved the remarkable result that while most finite graphs are asymmetric, most countably infinite graphs are symmetric. (Needless to say, the vague word 'most' was made completely precise in both cases.)

To illustrate symmetry properties of finite graphs, they defined what are now called the *Paley graphs*, based on quadratic residues in finite fields. In the case of countably infinite graphs, their construction shows that a randomly chosen graph is, with probability 1, isomorphic to a single graph, Rado's *random graph R*, and that this graph is highly symmetric.

I will show you that these very different graphs (one class finite and deterministically defined, the other infinite and probabilistically defined) are in fact closely related, through infinite but locally finite generalisations of the Paley graphs. A surprising link between them is a character sum estimate by Weil, which he used in proving the Riemann hypothesis for curves over finite fields. (Received December 20, 2019)

1156-05-42 **Leah Meissner**, South Orange, NJ 07079-2646, and **John T Saccoman*** (saccomjt@shu.edu), Department of Mathematics and CS, 400 South Orange Ave, South Orange, NJ 07079. *Strongly regular multigraphs.*

Strongly regular graphs are defined by 4 parameters, n, k, a, and c where n is the number of nodes, k is the degree of each node, a is the number of common neighbors for every adjacent pair of nodes, and c is the number of common neighbors for every nonadjacent pair of nodes. A multigraph is a graph that has no self-loops, but may have multiple edges and is formally defined by specifying a graph G and assigning a multiplicity to each edge of G. We examine underlying strongly regular multigraphs in order to further clarify their properties, specifically with regard to combinatorial configurations.

Keywords: strongly regular graphs, multigraphs, algebraic combinatorics (Received December 31, 2019)

1156-05-60 **Ila Fiete***, fiete@mit.edu, and **Manyi Yim**, **Lorenzo Sadun** and **Thibaud Taillefumier**. *Where can a place cell put its fields? Let us count the ways.*

A hippocampal place cell exhibits multiple firing fields within and across environments. What factors determine the configuration of these fields, and could they be set down in arbitrary locations? We conceptualize place cells as performing evidence combination across many inputs and selecting a threshold to fire. Thus, mathematically they are perceptrons, except that they act on geometrically organized inputs in the form of multiscale periodic grid-cell drive, and external cues. We analytically count which field arrangements a place cell can realize with structured grid inputs, to show that many more place-field arrangements are realizable with grid-like than one-hot coded inputs. However, the arrangements have a rigid structure, defining an underlying response scaffold. We show that the "separating capacity" or spatial range over which all potential field arrangements are realizable equals the rank of the grid-like input matrix, which in turn equals the sum of distinct grid periods, a small fraction of the unique grid-cell coding range. However, compared to random inputs over the same range, grid-structured inputs generate larger margins, conferring relative robustness to place fields when grid input weights are fixed. (Received January 07, 2020)

1156-05-96 **Colleen Robichaux*** (cer2@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801, **Harshit Yadav** (yadav7@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801, and **Alexander Yong** (ayong@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801. *Equivariant cohomology, Schubert calculus, and Edge labeled tableaux.*

This talk concerns edge labeled Young tableaux, introduced by H. Thomas and A. Yong. It is used to model equivariant Schubert calculus of Grassmannians. We report on a new shifted analogue of edge labeled tableaux. Conjecturally, this gives a Littlewood-Richardson rule for the structure constants of the D. Anderson W. Fulton ring, which is related to the equivariant cohomology of isotropic Grassmannians. This is joint work with H. Yadav and A. Yong. (Received January 14, 2020)

1156-05-100 **Donghyun Kim*** (donghyun_kim@berkeley.edu), Science center, 421a, Cambridge, MA 02138. *Combinatorial formulas for the coefficients of the Al-Salam-Chihara polynomials.*

In the last few decades, there has been a lot of work on finding combinatorial formulas for moments of orthogonal polynomials, particularly when they are polynomials with positive coefficients. The Al-Salam-Chihara polynomials are an important family of orthogonal polynomials in one variable x depending on 3 parameters α, β and q. They are closely connected to a model from statistical mechanics called the partially asymmetric simple exclusion process (PASEP) and they can be obtained as a specialization of the Askey-Wilson polynomials. We give two different combinatorial formulas for the coefficients of the (transformed) Al-Salam-Chihara polynomials. Our formulas make manifest the fact that the coefficients are polynomials in α, β and q with positive coefficients. (Received January 14, 2020)

1156-05-116 **Rebecca F. Goldin*** (rgoldin@gmu.edu), 4400 University Drive, Fairfax, VA 22039, and **Brent Gorbutt**. *Positivity of Peterson Schubert calculus.*

We discuss the S^1-equivariant Schubert calculus of the Peterson variety Y in Type A. We present an explicit, positive combinatorial formula for structure constants occurring in a Schubert-like basis of $H^*_{S^1}(Y)$. An immediate corollary is that the structure constants satisfy a natural notion of positivity, proving a conjecture of Harada-Tymoczko and extending their positive Monk formula to the general case. This is joint work with Brent Gorbutt. (Received January 16, 2020)

1156-05-121 **Anthony Weaver*** (anthony.weaver@bcc.cuny.edu). *Topological conjugacy of group actions on surfaces via the (not-) Burnside lemma.* Preliminary report.

Sorting finite group actions on compact surfaces of fixed genus into topological equivalence classes gives a stratification of the moduli space. We show how to accomplish the task for fully ramified actions of the rank 2 elementary abelian p-group, p an odd prime, in arbitrary genus. Actions are represented as zero-sum sets of column vectors from the 2-dimensional vector space $V_2(F_p)$ over the field F_p with p elements. Topological equivalence classes are the $GL_2(F_p)$-orbits on this collection of column sets. The action is by column-wise left multiplication, and is not faithful, since column sets which are merely reordered are the same. We use the lemma that is (not) Burnside's, and several delicate results of elementary number theory. This is work in progress. (Received January 17, 2020)

1156-05-130 **Lenore J Cowen*** (cowen@cs.tufts.edu), Department of Computer Science, Tufts University, Medford, MA 02155. *Random Walk Methods in the Analysis of Biological Networks.*

Many of the computational methods that seek to leverage high-throughput information currently being generated about genes and genetic modules driving disease deal with a representation of some of that information in the form of a massive graph, where there are vertices for each gene or protein, and two genes or proteins are connected by an edge if there is evidence that there is some form of association. For example, in the PPI network, vertices are proteins, and two proteins are connected by an edge if there is experimental evidence that they physically bind in the cell. Applications that perform inference or other general machine learning tasks to mine such graph data, typically encode the similarity (or disimilarity, in the case of a distance metric), according to some measure of graph proximity. Finding the right network-based measure of dissimilarity, customized for the application at hand, can lead to valuable new insights. We survey some of the methods based on random walks on graphs that have been applied to biological network applications, and discuss in particular applications and open problems concerning our novel Diffusion State Distance (DSD) metric. (Received January 19, 2020)

1156-05-132 **Nathan Sun*** (nathan99sun@gmail.com), 20 Main Street, Exeter, NH 03833, and **Aman Agrawal** (amanagrawal@iisc.ac.in) and **Caroline Choi** (cchoi1@stanford.edu). *On Permutation Weights and q-Eulerian Polynomials.*

Weights of permutations were originally introduced by Dugan, Glennon, Gunnells, and Steingrímsson in their study of the combinatorics of tiered trees. Given a permutation σ viewed as a sequence of integers, computing the weight of σ involves recursively counting descents of certain subpermutations of σ. Using this weight function, one can define a q-analog $E_n(x,q)$ of the Eulerian polynomials. We prove two main results regarding weights of permutations and the polynomials $E_n(x,q)$. First, we show that the coefficients of $E_n(x,q)$ stabilize as n goes to infinity, which was conjectured by Dugan, Glennon, Gunnells, and Steingrímsson, and enables the definition of the formal power series $W_d(t)$, which has interesting combinatorial properties. Second, we derive a recurrence relation for $E_n(x,q)$, similar to the known recurrence for the classical Eulerian polynomials $A_n(x)$. Finally, we give a recursive formula for the numbers of certain integer partitions and, based on that, conjecture a recursive formula for the stabilized coefficients mentioned above. (Received January 20, 2020)

1156-05-136 **Mark A Adler*** (adler@brandeis.edu). *A Master Kernel and Universality in Dimer Models.*

The talk will focus on universality in dimer tilings, exhibiting a number of models and providing a Master- kernel which captures these models. These models will be illuminated through pictures and simulations and finally we will discuss how one solves these models. (Received January 20, 2020)

1156-05-167 **Deborah Oliveros*** (debolivero@matem.unam.mx), Instituto de Matematicas, Boulevard Juriquilla 3001., Queretaro, Queretaro, Mexico. *The geometry and the combinatorics of piercing line segments hypergraphs.*

An *r-segment hypergraph* H is a hypergraph whose edges consist of r consecutive integer points on line segments in $\mathbb{R}^2$. In this talk, we will discuss bounds of the chromatic number $\chi(H)$ and covering number $\tau(H)$ of hypergraphs in this family, uncovering several interesting geometric properties in the process. (joint work with Christopher O'Neill and Shira Zerbib) (Received January 22, 2020)

1156-05-182 **Reuven Hodges*** (rhodges@illinois.edu) and **Alexander Yong**. *Decompositions of Demazure modules.* Preliminary report.

In this talk we will investigate the action of Levi subgroups on Schubert varieties in the full flag variety, and the induced action on their Demazure modules. I will introduce the notion of split-symmetric Schur polynomials and then give a combinatorial rule for writing the Demazure character as a sum of these split-symmetric Schur

polynomials. This is equivalent to providing a decomposition of the Demazure module into irreducible Levi representations. The combinatorial rule can then be used to give criterion for when these decompositions are multiplicity free. This has implications for when a Schubert variety is a spherical variety for the action of the Levi subgroup. (Received January 22, 2020)

1156-05-186 **He Guo*** (he.guo@gatech.edu), 686 Cherry Street, School of Mathematics, Georgia Institute of Technology, Atlanta, GA 30332, and **Lutz Warnke** (warnke@math.gatech.edu). *Bounds on Ramsey Games via Alterations.*

In this talk we introduce a refined alteration approach for constructing H-free graphs: we show that removing *all* edges in H-copies of the binomial random graph does not significantly change the independence number (for suitable edge-probabilities); previous alteration approaches of Erdős and Krivelevich remove only a *subset* of these edges. We present two applications to online graph Ramsey games of recent interest, deriving new bounds for Ramsey, Paper, Scissors games and online Ramsey numbers (each time extending recent results of Fox–He–Wigderson and Conlon–Fox–Grinshpun–He). (Received January 22, 2020)

1156-05-188 **Brigitte Servatius** (bservat@wpi.edu), Math Sci, WPI, Worcester, MA 01609. *On the stressability of multidimensional frameworks.*

We prove an equilibrium stressability criterium for trivalent multidimensional tensegrities. The criterium appears in different languages: (1) in terms of stress monodromies, (2) in terms of surgeries, (3) in terms of exact discrete 1-forms, and (4) in Cayley algebra terms. This is joint work with Oleg Karpenkov, Christian Mueller, Gaiane Panina, Herman Servatius, and Dirk Siersma. (Received January 23, 2020)

1156-05-191 **Christopher M Langdon*** (langdon@cshl.edu). *Combinatorial Geometry of Threshold-Linear Networks.* Preliminary report.

We study the bifurcation structure of a threshold-linear network with architecture constrained by a directed graph. Such networks are piecewise-linear dynamical systems frequently used in modeling of neural circuits. We show that bifurcations arising from fixed points crossing boundaries between linear systems correspond to mutations of a chirotope and use this perspective to obtain constraints on the possible bifurcations over a fixed architecture. As an illustration of this perspective we exhibit the bifurcation structure of all sixteen non-isomorphic directed graphs on three nodes. (Received January 23, 2020)

1156-05-193 **Joel H Spencer*** (spencer@cims.nyu.edu). *Four Discrepancies.*

In Combinatorial Descrepancy Paul selects objects and Carole balances them into two piles as evenly as possible.

In this talk the objects will be n n-dimensional vectors with coefficients ± 1. Carole assigns to each vector ± 1 (the two piles) and the payoff (which Carole tries to minimize) is the L^∞ norm of the signed sum.

We consider four variants of this problem. Paul may play randomly (in which case Carole tries to minimize expectation) or adversarially. Carole may play On-Line (selecting each sign immediately after seeing its vector) or Off-Line (seeing all vectors and then selecting all signs).

The arguments (for both Paul and Carole) involve a combination of appropriate weight functions and probabilistic (Carole selects ± 1 randomly!) thinking. (Received January 23, 2020)

1156-05-203 **Adam Logan**, **Mike Molloy** and **Pawel Pralat*** (pralat@ryerson.ca), Department of Mathematics, Ryerson University, Toronto, ON M5B 2K3, Canada. *A variant of the Erdős-Rényi random graph process.*

We consider a natural variant of the Erdős-Rényi random graph process in which k vertices are special and are never put into the same connected component. The model is natural and interesting on its own, but is actually inspired by the combinatorial data fusion problem that itself is connected to a number of important problems in graph theory. We will show that a phase transition occurs when the number of special vertices is roughly $n^{1/3}$, where n is the number of vertices. (Received January 23, 2020)

1156-05-210 **Alan Frieze** and **Tomasz Tkocz***, Carnegie Mellon University, Wean Hall, Pittsburgh, PA 15213. *A randomly weighted minimum spanning tree with a random cost constraint.*

We shall present a constrained version of the minimum spanning tree problem on the complete graph whose edges have random weights and costs. The constraint is that the total cost of a spanning tree does not exceed a given value. Using the dual problem, we establish the asymptotic value of the minimal weight. Based on joint work with A. Frieze. (Received January 23, 2020)

1156-05-218 **Elisenda Feliu** and **Nidhi Kaihnsa*** (nidhi_kaihnsa@brown.edu), Division of Applied Mathematics, 182 George Street, Providence, RI 02912, and **Timo de Wolff** and **Oguzhan Yürük**. *The Kinetic Space of Multistationarity in Dual Phosphorylation.*

Multistationarity in molecular systems underlies switch-like responses in cellular decision making. Determining whether and when a system displays multistationarity is in general a difficult problem. In a joint work with Elisenda Feliu, Timo de Wolff, and Oğuzhan Yürük we completely determine the set of kinetic parameters that enable multistationarity in a ubiquitous motif involved in cell signaling, namely a dual phosphorylation cycle. We employ a suite of techniques from (real) algebraic geometry, which in particular concerns the study of the signs of a multivariate polynomial over the positive orthant and sums of nonnegative circuit polynomials. (Received January 24, 2020)

1156-05-221 **Alice Paul*** (apaul@olin.edu), Needham, MA 02806. *Iterative Methods for Semidefinite Programming Relaxations.* Preliminary report.

We consider the process of fixed-point iteration with a map that optimizes a linear function over a convex domain. We show this process always converges and characterize the fixed points of the process in different convex sets. We consider in particular the fixed points in spectrahedra that arise in semide nite programming relaxations. In this case we see that fixed-point iteration provides a practical deterministic method for rounding the solution of a semidefi nite programming relaxation. This leads to a new clustering method based on the classical semidefi nite programming relaxation of k-way Max-Cut. (Received January 24, 2020)

1156-05-227 **Thomas Lam***, 77 Massachusetts Ave., Cambridge, MA 02139, and **Seung Jin Lee** and **Mark Shimozono**. *Coproduct formulae for affine Schubert classes.*

The cohomology ring of the affine flag variety has a coaction by the cohomology ring of the affine Grassmannian. I will discuss a positive formula for the coaction of a Schubert class, and an application to monomial formulae for affine Schubert polynomials. (Received January 24, 2020)

1156-05-228 **Rebecca F. Goldin*** (rgoldin@gmu.edu), 4400 University Drive. MS 3F2, Fairfax, VA 22039, and **Allen Knutson**. *On the equivariant K-theory of G/B.*

Bernstein, Gelfand, & Gelfand and Demazure introduced divided difference operators that act on the (equivariant) cohomology ring of G/B. They can be used to obtain a geometrically motivated basis, the Schubert basis, for the equivariant cohomology (or equivariant K-theory) of G/B as a module over the cohomology of a point. The underlying geometry implies that intersections are oriented, and hence can be measured with *positive* values. The corresponding ring statement is that the structure constants of multiplication of equivariant Schubert classes are positive in an appropriate sense. We introduce new operators whose coefficients compute the structure constants in a manifestly polynomial but not positive way, resulting in a formula that generalizes the positive Andersen-Jantzen-Soergel/Billey and Graham/Willems formulæ for the restriction of classes to fixed points. Our proof involves Bott-Samelson manifolds, and in particular, the (K-)cohomology basis dual to the (K-)homology basis consisting of classes of sub-Bott-Samelson manifolds. This is joint work with Allen Knutson. (Received January 24, 2020)

1156-05-229 **Andrzej Dudek*** (andrzej.dudek@wmich.edu), Department of Mathematics, Western Michigan University, Kalamazoo, MI 49008. *Variations on twins in permutations.*

Let π be a permutation of the set $[n] = \{1, 2, \ldots, n\}$. Two disjoint order-isomorphic subsequences of π are called *twins*. How long twins are contained in every permutation? The well known Erdős-Szekeres theorem implies that there is always a pair of twins of length $\Omega(\sqrt{n})$. On the other hand, by a simple probabilistic argument Gawron proved that for every $n \geqslant 1$ there exist permutations with no twins of length greater than $O(n^{2/3})$. His conjecture states that the latter bound is the correct size of the longest twins guaranteed in every permutation. In this talk we show that asymptotically almost surely a random permutation contains twins of length at least $\Omega(n^{2/3}/\log^{1/3} n)$, which supports this conjecture. We also discuss several variants of the problem with diverse restrictions imposed on the twins.

This is a joint work with Jarosław Grytczuk and Andrzej Ruciński. (Received January 24, 2020)

1156-05-249 **Gweneth McKinley*** (gweneth@mit.edu), **Asaf Ferber** and **Kyle Luh**. *Resilience of the Rank of Random Matrices.*

For an $n \times m$ matrix M of independent Rademacher (± 1) random variables, it is well known that if $n \leq m$, then M is of full rank with high probability. We show that this property is resilient to adversarial changes to M. More precisely, if $m \geq n + n^{1-\varepsilon/6}$, then even after changing the sign of $(1-\varepsilon)m/2$ entries, M is still of full rank with high probability. Note that this is asymptotically best possible as one can easily make any two

rows proportional with at most $m/2$ changes. Moreover, this theorem gives an asymptotic solution to a slightly weakened version of a conjecture made by Van Vu in 2008. (Received January 25, 2020)

1156-05-250 **Andrea I Lincoln*** (andreali@mit.edu), **Virginia Vassilevska Williams**, **Josh Wang** and **Ryan Williams**. *Deterministic Time-Space Tradeoffs for k-SUM.*

Abstract: Given a set of numbers, the k-SUM problem asks for a subset of k numbers that sums to zero. When the numbers are integers, the time and space complexity of k-SUM is generally studied in the word-RAM model; when the numbers are reals, the complexity is studied in the real-RAM model, and space is measured by the number of reals held in memory at any point.

We present a time and space efficient deterministic self-reduction for the k-SUM problem which holds for both models, and has many interesting consequences. To illustrate: 3-SUM is in deterministic time $O(n^2 lglg(n)/lg(n))$ and space $O(\sqrt{nlg(n)/lglg(n)})$, extending the results of Gold and Sharir. 3-SUM is in deterministic time $O(n^2)$ and space $O(\sqrt{n})$, derandomizing an algorithm of Wang. A popular conjecture states that 3-SUM requires $n^{2-o(1)}$ time on the word-RAM. We show that the 3-SUM Conjecture is in fact equivalent to the (seemingly weaker) conjecture that every $O(n^{.51})$-space algorithm for 3-SUM requires at least $n^{2-o(1)}$ time on the word-RAM Joint work with Virginia Vassilevska Williams, Josh Wang and Ryan Williams (Received January 25, 2020)

1156-05-253 **Huseyin Acan*** (ha627@drexel.edu), **Alan Frieze** (alan@random.math.cmu.edu) and **Boris Pittel** (bgp@math.ohio-state.edu). *Descendant sets in preferential attachment graphs.*

In a preferential attachment graph, vertices arrive one at a time. Upon its arrival at time t, vertex v_t chooses m random neighbors for itself from the existing vertices, where these choices depend on the degrees of the current graph and a parameter δ. In this graph, we are interested in the set of descendants of a given vertex. (If we think that the edges are directed towards the earlier vertices, v_t is a descendant of v_r if there is a directed path from v_t to v_r.) For $m = 1$ and for a given vertex, we show that the size of the descendant set, scaled by the number of vertices in the graph, converges almost surely to a limit whose distribution is a mixture of two beta distributions. When $m > 1$, the same quantity converges almost surely to 1.

Joint work with Alan Frieze and Boris Pittel. (Received January 25, 2020)

1156-05-255 **Jenna Rajchgot**, **Yi Ren**, **Colleen Robichaux**, **Avery St. Dizier** and **Anna Weigandt*** (weigandt@umich.edu). *Degrees of symmetric Grothendieck polynomials and Castelnuovo-Mumford regularity.*

We give an explicit, combinatorial formula for the degrees of symmetric Grothendieck polynomials. We use this formula to study the Castelnuovo-Mumford regularity of open patches of Grassmannian Schubert varieties. In particular, we provide a counterexample to a conjecture of Kummini-Lakshmibai-Sastry-Seshadri and show that our work gives rise to an alternate formula in these cases. (Received January 25, 2020)

1156-05-268 **Iva Halacheva*** (i.halacheva@northeastern.edu), **Allen Knutson** and **Paul Zinn-Justin**. *Branching in Schubert calculus: from type A to type C Grassmannians.*

Triangular puzzles in Schubert calculus are used to describe the product of two Schubert classes of the Grassmannian of type A, expanded back into Schubert classes. A related question is to study the inclusion of the type C Grassmannian into the type A Grassmannian, and the corresponding pullback in equivariant cohomology. I will describe a puzzle rule, using self-dual puzzles, for the expansion of the image of a Schubert class under this pullback. I will also outline a generalization to cotangent bundles which allows a further geometric interpretation of the puzzles in the context of Lagrangian correspondences. This is joint work with Allen Knutson and Paul Zinn-Justin. (Received January 26, 2020)

1156-05-278 **Ted Dobson*** (ted.dobson@upr.si) and **Rachel Barber** (rvb41@msstate.edu). *On Automorphisms of Haar graphs.* Preliminary report.

Let G be a group and $S \subseteq G$. A Haar graph of G with connection set S has vertex set $\mathbb{Z}_2 \times G$ and edge set $\{(0, g)(1, gs) : g \in G \text{ and } s \in S\}$. Haar graphs are then natural bipartite analogues of Cayley digraphs. We report on progress to generalize an earlier result characterizing the automorphism groups of Haar graphs of abelian groups to Haar graphs of any group. We mainly discuss the case for Haar graphs of any group G corresponding to the case of Haar graphs of abelian groups where the graph is connected and can be written as a nontrivial wreath product. It turns out that for general Haar graphs the corresponding operation is not the wreath product, but a generalization of the wreath product called an X-join of a family of graphs. We refine a result of Hemminger which gives necessary and sufficient conditions for the automorphism group of the X-join to the "complete set of natural automorphisms", in the special case when X is bipartite and the family of graphs

are all empty graphs. This allows us to finish this case, which should allow us to finish the characterization. All of these results hold in the more general setting of bi-coset graphs. This is joint work with Rachel Barber of Mississippi State University. (Received January 26, 2020)

1156-05-281 **Lutz Warnke*** (warnke@math.gatech.edu), Georgia Institute of Technology, Atlanta, and **Matas Šileikis**, Czech Academy of Sciences, Prague, Czech Rep. *Counting extensions revisited.*

We consider rooted subgraphs in random graphs, i.e., extension counts such as (i) the number of triangles containing a given 'root' vertex or (ii) the number of paths of length three connecting two given 'root' vertices. In 1989, Spencer gave sufficient conditions for the event that, with high probability, these extension counts are asymptotically equal for all choices of the root vertices (for vertex degrees this is true when np grows faster than log n). For the important strictly balanced case, Spencer also raised the fundamental question whether these conditions are necessary. We answer this question by a careful second moment argument, and discuss some intriguing problems that remain open; see arXiv:1911.03012 (Received January 26, 2020)

1156-05-286 **Anton Bernshteyn*** (abernsht@math.cmu.edu) and **Clinton T. Conley**. *Equitable colorings of graphs on probability spaces.*

A proper k-coloring of a finite graph G is called equitable if every two color classes differ in size at most by one. In particular, if G has n vertices and k divides n, then in an equitable k-coloring of G every color class has size exactly n/k. There is a natural way to extend this definition to infinite graphs on probability spaces. Namely, if G is a graph whose vertex set $V(G)$ is a probability space, then a proper k-coloring of G is equitable when every color class has measure $1/k$. In this talk we will discuss extensions of some classical results about equitable colorings to this setting, including an infinite version of the Hajnal–Szemerédi theorem on equitable k-colorings for $k \geq \Delta(G) + 1$. (Received January 26, 2020)

1156-05-292 **Hao Li*** (hli29@albany.edu), Albany, NY 12206, and **Changlong Zhong**. *On equivariant oriented cohomology of Bott-Samelson varieties.*

Given a flag variety G/B, we have a Bott-Samelson resolution via the canonical map $p_{I_w} : \hat{X}_{I_w} \to G/B$ where $\hat{X}_{I_w}$ is the Bott-Samelson variety. we can study $T-$equivarient cohomology of $\hat{X}_{I_w}$ by using T-fixed points. In present paper, We get the restriction formula for images of basis elements in $h_T(\hat{X}_{I_w})$ under the restriction map from $h_T(\hat{X}_{I_w})$ to $h_T((\hat{X}_{I_w})^T)$ where $(\hat{X}_{I_w})^T$ is the set of all T fixed points on $\hat{X}_{I_w}$. We also characterize the image of restriction map. Furthermore, we study the pushforward map $(p_{I_W})_*$. As a byproduct, we obtain an interesting formula about root system. (Received January 26, 2020)

1156-05-295 **Nima Anari**, **Kuikui Liu**, **Shayan Oveis Gharan** and **Cynthia Vinzant*** (clvinzan@ncsu.edu). *Matroids, log-concavity, and expanders.*

Matroids are combinatorial objects that model various types of independence. They appear several fields mathematics, including graph theory, combinatorial optimization, and algebraic geometry. In this talk, I will introduce the theory of matroids along with the closely related class of polynomials called strongly log-concave polynomials. Strong log-concavity is a functional property of a real multivariate polynomial that translates to useful conditions on its coefficients. Discrete probability distributions defined by these coefficients inherit several of these nice properties. I will discuss the beautiful real and combinatorial geometry underlying these polynomials and describe applications to random walks on the faces of simplicial complexes. Consequences include proofs of Mason's conjecture that the sequence of numbers of independent sets of a matroid is ultra log-concave and the Mihail-Vazirani conjecture that the basis exchange graph of a matroid has expansion at least one. (Received January 26, 2020)

1156-05-300 **Patrick Bennett*** (patrick.bennett@wmich.edu), **Ryan Cushman** and **Andrzej Dudek**. *On The Number of Alternating Paths in Random Graphs.*

In the noisy channel model from coding theory, we wish to detect errors introduced during transmission by optimizing various parameters of the code. Bennett, Dudek, and LaForge framed a variation of this problem in the language of alternating paths in edge-colored complete bipartite graphs in 2016. Here, we extend this problem to the random graph $G(n,p)$. We seek the *alternating connectivity*, $\kappa_{r,\ell}(G)$, which is the maximum t such that there is an r-edge-coloring of G such that any pair of vertices is connected by t internally disjoint and alternating (i.e. no consecutive edges of the same color) paths of length ℓ. (Received January 27, 2020)

1156-05-301 **Sylvester Gates*** (sylvester_gates@brown.edu). *Representation Theory Advances For Space-Time SUSY: Adinkra Networks, Branching Rules, Plethysm, Young Tableaux, & Dynkin Labels.*

In this talk we present new results for supersymmetric representation theory recently derived for superspaces related to superstrings, and M-Theory. (Received January 27, 2020)

1156-05-303 **John C Urschel*** (urschel@mit.edu). *Old and New Results in Force-Directed Graph Drawing.*

Force-directed graph drawing is a general class of techniques used to embed a graph in some space (typically $\mathbb{R}^n$, $n = 2, 3$) in some way that represents the dynamics of the graph itself. This is done by assigning some type of forces among the vertices and edges based on their locations in the space, and then minimizing energy.

This class of techniques originated with the work of Tutte, who proved that, for a three-connected planar graph, if any face of the graph is fixed as the complement of a convex region in the plane, and all other vertices are placed at the mass center of their neighbors, then this results in a unique embedding, and this embedding is planar. This result is commonly referred to as Tutte's spring embedding theorem.

Since Tutte's work, there have been a number of different force-based techniques employed to obtain drawings of graphs, most notably stress majorization algorithms (i.e., the SMACOF algorithm) and the Kamada-Kawaii algorithm. In this talk, a review of the established results and techniques of this area will be presented, as well as some new results. (Received January 27, 2020)

1156-05-306 **Pedro Felzenszwalb*** (pff@brown.edu), **Caroline Klivans** and **Alice Paul**. *Fixed-point Iteration with convex optimization.*

I will discuss the fixed points of a map that optimizes a linear function over a convex domain. (Received January 27, 2020)

1156-05-310 **Zachary Hamaker**, **Oliver Pechenik** and **Anna Weigandt*** (weigandt@umich.edu). *Diagonal Gröbner geometry of matrix Schubert varieties.* Preliminary report.

Knutson and Miller (2005) studied the Gröbner geometry of matrix Schubert varieties. They showed that the multidegree of a matrix Schubert variety is its corresponding Schubert polynomial. Furthermore, when degenerating with respect to any antidiagonal term order, the components of the limit are naturally labeled by pipe dreams. In subsequent work, Knutson, Miller, and Yong (2009) considered diagonal term orders. In the vexillary case, components of the limit are in bijection with flagged tableaux. We present further results and conjectures outside of the vexillary case. (Received January 27, 2020)

1156-05-315 **Wesley Pegden*** (wes@math.cmu.edu), Carnegie Mellon University, Pittsburgh. *Random volumes in d-dimensional polytopes.*

Consider a convex body X in R^d. Suppose we choose random points from this body and take their convex hull. How many points should we choose so that the resulting convex hull is nearly as large as X itself? It is known that if X is the hypercube, then exponentially many (in d) points suffice, while the Euclidean ball requires super-exponentially many points. We will discuss this problem in other cases; e.g., when X is a simplicial polytope. (Received January 27, 2020)

1156-05-318 **Xavier Perez-Gimenez** and **Abigail Raz*** (araz2@unl.edu). *Perfect matchings in the random bipartite geometric graph.* Preliminary report.

We consider the standard random bipartite geometric graph process in which n red vertices and n blue vertices are placed at random on the unit d-dimensional cube and edges are added sequentially, between vertices of different colors, in increasing order of edge-length. A natural question is to ask whether the first edge in the process that results in the minimum degree being at least one coincides, with high probability, with the first edge that creates a perfect matching. While this was already known to be false when $d = 2$, as the thresholds are not even of the same order, we are able to positively answer it for $d \geq 3$.
Joint work with Xavier Pérez-Giménez (Received January 27, 2020)

1156-05-325 **Martha Precup** and **Edward Richmond*** (edward.richmond@okstate.edu). *An equivariant monomial basis for the cohomology of Springer fibers.* Preliminary report.

Springer fibers are subvarieties of the flag variety that play an important role in combinatorics and geometric representation theory. In this talk, we analyze the equivariant cohomology of Springer fibers for $GL_n(\mathbb{C})$ using results of Kumar and Procesi that describe this equivariant cohomology as a quotient ring. We define a basis for the equivariant cohomology of a Springer fiber, generalizing a monomial basis of the ordinary cohomology defined by De Concini and Procesi and studied by Garsia and Procesi. Our construction yields a combinatorial framework

with which to study the equivariant and ordinary cohomology rings of Springer fibers. As an application, we identify an explicit collection of Schubert classes whose images in the cohomology ring of a given Springer fiber form a basis. (Received January 27, 2020)

1156-05-329 **Jim Lawrence*** (lawrence@gmu.edu), Dept of Mathematical Sciences, 4400 University Dr MS 3F2, Fairfax, VA 22030-4444. *On the Equation $\mathcal{J}(\mathcal{P}) \cong \mathcal{P}$.*

Here P represents a strongly locally finite poset and $\mathcal{J}(\mathcal{P})$ is its poset of nonempty intervals. The solutions to the equation will be examined by making use of the notion of "parity representations" of posets. The equation itself represents a certain kind of symmetry, and among the many solutions are some that correspond to certain affine Coxeter arrangements. (Received January 27, 2020)

1156-05-339 **Oliver Pechenik*** (pechenik@umich.edu), **David E Speyer** and **Anna Weigandt**. *Castelnuovo-Mumford regularity of matrix Schubert varieties.* Preliminary report.

Jenna Rajchgot observed that the Castelnuovo-Mumford regularity of matrix Schubert varieties is computed by the degrees of the corresponding Grothendieck polynomials. We give a formula for these degrees. Indeed, we compute the leading terms of the top degree pieces of Grothendieck polynomials and give a complete description of when two Grothendieck polynomials have the same top degree piece (up to scalars). Our formulas rely on some new facts about major index of permutations. (Received January 27, 2020)

1156-05-343 **Kevin Iga*** (kiga@pepperdine.edu), Brown University, Box S, 340 Brook St., Providence, RI 02912. *Adinkras: Graphs that describe supersymmetry.*

An Adinkra is a kind of decorated graph, motivated by the supersymmetry algebra in particle physics. For physicists, it provides a visual and accessible way to understand the subject. But independently of this, Adinkras are interesting diagrams to study and play with. The study of Adinkras has surprising connections with other areas of math, like error correcting codes, algebraic topology, and combinatorics. Most recently, Adinkras have been seen to naturally embed in surfaces, and this not only sheds light on Adinkras, but also provides new ways to construct interesting surfaces. (Received January 27, 2020)

1156-05-345 **JD Nir*** (jnir@huskers.unl.edu) and **Xavier Pérez Giménez**. *The Chromatic Number of Random Lifts of Regular Graphs.*

Graph coloring is one of several constraint satisfaction problems studied on random structures. The problem at the heart of this talk is to bound the chromatic number of a random d-regular graph. Rather than use the uniform model, we take inspiration from a question of Linial and generate our regular graph by taking random lifts of a regular host graph. This method produces a family of noncontiguous models of random graphs. We provide an upper bound for the chromatic number of such lifts. In the case where the host graph is complete, we also provide a nearly matching lower bound. Our proof contains several techniques with broad applications in random structures, including Laplace summation, the saddle point method and the small subgraph conditioning method. (Received January 27, 2020)

1156-05-359 **Paul Balister*** (pbalistr@memphis.edu), **Béla Bollobás**, **Robert Morris**, **Julian Sahasrabudhe** and **Marius Tiba**. *Flat Littlewood polynomials exist.*

A Littlewood polynomial is a polynomial with all coefficients equal to $+1$ or -1. Erdős (1957) and Littlewood (1966) asked if one could find $\delta, \Delta > 0$ such that for all large n there exist Littlewood polynomials $P(z)$ of degree n such that $\delta\sqrt{n} \leq |P(z)| \leq \Delta\sqrt{n}$ for all complex z with $|z| = 1$. In this talk I will describe our recent proof of this conjecture, which uses recent results in discrepancy theory. (Received January 27, 2020)

1156-05-363 **R Karpman*** (karpman1@otterbein.edu) and **Érika Roldán Roa**. *Isotopies of Latin tableaux.*

Latin tableaux are a generalization of Latin squares. They were first introduced in 2002 by Chow, Fan, Goemans, and Vondrak. Chow and his collaborators hoped to use Latin tableaux as a tool to prove Rota's basis conjecture– a question in linear algebra which has been open since the early 1990's. Their approach was foiled, however, because Latin tableaux are surprisingly mysterious objects. Nearly twenty years later, many natural questions about Latin tableaux remain unexplored. In this talk, we extend the notion of *isotopy*, a permutation group action, from Latin squares to Latin tableaux. We define *isotopy graphs* for Latin tableaux, which encode the structure of orbits under the isotopy action, and investigate the relationship between the shape of a Latin tableau and the structure of its isotopy graph. We show, for example, that for any positive integer d, there is a Latin tableau whose isotopy graph is a d-dimensional cube. (Received January 27, 2020)

1156-05-372 **Ross Berkowitz*** (ross.berkowitz@yale.edu) and **Patrick Devlin**. *Bribing three voters in majority dynamics suffices.*

Let V be a set of N vertices, half of which are initially colored red and half are colored blue. At each time step, all of the vertices update their color to match the majority opinion among their neighbors. We examine the evolution of this process when the underlying graph is chosen from $G(N, p)$ and show that giving one color (say blue) a lead of size $O(1/p)$ can lead to every vertex being blue within four steps. (Received January 27, 2020)

1156-05-373 **Anton Dochtermann*** (dochtermann@txstate.edu). *Parking functions, skeleta, and syzygies of monomial ideals.* Preliminary report.

Associated to a graph G on vertex set $[n+1]$, Postnikov and Shapiro introduced an artinian monomial ideal M_G called the 'G-parking function ideal'. The dimension of M_G is given by the number of spanning trees of G, and a (possibly redundant) list of generators of M_G is indexed by the nonempty subsets of $[n]$. The ideal M_G is a certain initial ideal of the binomial toppling ideal, which encodes the linear equivalence of divisors on G. A cellular resolution of M_G is supported on the barycentric subdivision of a simplex, and the Betti numbers of M_G are encoded by data coming from the flats of the underlying matroid of G.

Motivated by Backman's notion of 'restricted set chip-firing' we study the 'k-skeleton' ideals $M_G^{(k)}$, subideals of M_G generated by subsets of $[n]$ of size at most $k+1$. Here we focus on homological properties. For $G = K_{n+1}$ the complete graph, we use discrete Morse theory to describe a minimal cellular resolution of $M_G^{(k)}$. For the case $k = 1$ we show how, for a large class of graphs G, deformations of certain fan arrangements lead to interpretations of the Betti numbers of the ideal $M_G^{(1)}$. Parts of this are joint work with Patrik Norén. (Received January 28, 2020)

1156-05-375 **Giancarlo LaCamera*** (giancarlo.lacamera@stonybrook.edu). *Cortical computations via metastable activity.*

Metastable brain dynamics are characterized by abrupt, jump-like modulations so that the neural activity in single trials appears to unfold as a sequence of discrete, quasi-stationary 'states'. Metastable activity occurs both in response to an external stimulus and during ongoing, self-generated activity. These spontaneous metastable states are increasingly found to subserve internal representations that are not locked to external triggers, including states of deliberations, attention and expectation. So far, in both data and models, it appears that sequences of metastable states result from a combination of a limited number of internal states. Moreover, such sequences can emerge spontaneously in a deterministic network of spiking neurons with a clustered architecture. In this talk, I will present recent theoretical results on the mechanistic origin, characterization, and potential roles of metastable representations. (Received January 28, 2020)

1156-05-381 **William Graham*** (wag@uga.edu) and **Victor Kreiman**. *Rationally indecomposable weights of tangent spaces of Kazhdan-Lusztig varieties.*

We study the tangent space to a Kazhdan-Lusztig variety at a T-fixed point x, and the subspace of the tangent space spanned by tangent lines to T-invariant curves through x. We show that these two spaces have the same rationally indecomposable T-weights, where a weight is rationally indecomposable if it cannot be expressed as a positive rational linear combination of other weights of a certain ambient space. We also study other properties of rationally indecomposable weights. (Received January 28, 2020)

1156-05-389 **J Elias Mochan*** (jaime.mochan@im.unam.mx) and **Isabel Hubard**. *2-Orbit Polytopes.*

Pellicer, Potočnik and Toledo proved that for rank greater or equal than 5 there exist 2-orbit maniplexes with any possible symmetry type. They do this by giving a concrete voltage assignment to every possible symmetry type graph with two vertices.

We have proved a theorem that allows us to know what intersection properties must a voltage group satisfy so that the derived graph from an admissible graph and a voltage assignment is the flag graph of a polytope. By applying this theorem to the voltage assignment constructed by Pellicer, Potočnik and Toledo, we attempt to prove that in some cases the derived graph is actually the flag graph of a polytope, showing that the corresponding graph is the symmetry type graph of a polytope.

This is a work in progress. (Received January 28, 2020)

1156-05-398 **Alexander Postnikov*** (apost@math.mit.edu), Department of Mathematics, MIT, Room 2-367, 77 Massachusetts Ave, Cambridge, MA 02139. *From Schubert calculus to positive Grassmannian.*

We will discuss the positroid decomposition of the Grassmannian, which refines the Schubert decomposition. Its cells are described using interesting combinatorial structures such as plabic graphs and Grassmann necklaces. This construction has interesting links with other areas: cluster algebras, theory of symmetric functions, physics (scattering amplitudes, solitons), convex geometry. We'll talk about some recent advances in this study. (Received January 28, 2020)

1156-05-403 **Keith Frankston*** (keith.frankston@math.rutgers.edu), **J Kahn**, **B Narayanan** and **J Park**. *Thresholds Versus Fractional Expectation-Thresholds.*

Given an increasing family F of subsets of a finite set X, its measure according to μ_p increases and often exhibits a threshold behavior, growing quickly as p increases from near 0 to near 1 around a specific value p_c. These thresholds have been a central focus of the study of random discrete structures, with estimation of thresholds for specific properties the subject of some of the most challenging work in the area.

In 2006, Kahn and Kalai conjectured that a natural (and often easy to calculate) lower bound $q(F)$ (which we call the "expectation-threshold") for p_c is never far from its actual value. We prove a fractional version (proposed by Talagrand) of this so called "expectation-threshold" conjecture showing that for any increasing family we have $p_c(F) = O(q_f(F) \log \ell(F))$, where q_f is the "fractional expectation-threshold" and $\ell(F)$ is the maximum size of a minimal element of F.

This result easily implies previously difficult results in probabilistic combinatorics such as thresholds for perfect hypergraph matchings (Johansson-Kahn-Vu) and bounded-degree spanning trees (Montgomery). Our approach builds on a recent breakthrough of Alweiss, Lovett, Wu and Zhang on the Erdős-Rado "Sunflower Conjecture." (Received January 28, 2020)

1156-05-423 **Julianna Tymoczko*** (jtymoczko@smith.edu), Department of Mathematics, Smith College, Northampton, MA 01063. *A brief introduction to GKM theory.*

GKM theory is the name often given to an algebraic-combinatorial way to compute equivariant cohomology. In this survey, we give a brief introduction to GKM theory, a sketch of how it has been used in the literature, and some questions and directions for future research. (Received January 29, 2020)

1156-05-424 **Meera Nadeem** (mnadeem@smith.edu), Department of Mathematics, Smith College, Northampton, MA 01063, and **Garcia Sun*** (gsun@smith.edu), Department of Mathematics, Smith College, Northampton, MA 01063. *The (n, n) Springer fiber and noncrossing matchings.*

This talk describes the combinatorics and geometry related to a family of subvarieties of the flag variety called *Springer fibers*. Recall that the flag variety is the collection of nested vector subspaces $V_1 \subseteq V_2 \subseteq \cdots V_{n-1} \subseteq \mathbb{C}^n$ of a fixed complex vector space. The Springer fiber of a linear operator X consists of those flags that are fixed by X in the sense that $XV_i \subseteq V_i$ for all i. If X has two Jordan blocks of the same size, many geometric characteristics of its Springer fibers can be encoded by noncrossing matchings. We give a quick overview of the cell structure of Springer fibers and then describe recent results relating the topology of the cells to noncrossing matchings. (Received January 29, 2020)

1156-05-425 **Julianna Tymoczko*** (jtymoczko@smith.edu), Department of Mathematics, Smith College, Northampton, MA 01063. *Hessenberg Schubert polynomials.*

We describe a set of polynomials that generalize Schubert polynomials. Geometrically, these polynomials are connected to Hessenberg varieties: a family of subvarieties of the full flag variety defined by the choice of a linear operator X and a nondecreasing function $h : \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$. Hessenberg varieties can be paved by an appropriate choice of Schubert cells, inducing an additive "Hessenberg Schubert basis" in the homology of each Hessenberg variety. We give open questions as well as some results. This is joint work with Harada, Horiguchi, Murai, and Precup. (Received January 29, 2020)

1156-05-427 **Julianna Tymoczko*** (jtymoczko@smith.edu), Department of Mathematics, Smith College, Northampton, MA 01063. *Springer fibers and webs.* Preliminary report.

The Springer fiber of a linear operator X is the collection of flags $V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n$ that are fixed by X in the sense that $XV_i \subseteq V_i$ for all i. We describe geometric and representation-theoretic connections between Springer fibers and a class of planar graphs with boundary called webs. Time permitting, we will describe a combinatorial extension of webs and some properties. (Received January 29, 2020)

11 ► Number theory

1156-11-13 **Abdulmajeed Abdurrahman*** (ababdu@ship.edu), 1871 Old Main Drive, Shippensburg, PA 17257. *CM Method and Expansion of Numbers.*

We show that an iterative method for computing the center of mass (CM) of q units of mass, placed on a unit interval [0, 1] along the x- axis, give rise to a simple procedure for expanding rational numbers less than unity in powers of r/s < 1, with r, s, integers larger than 0. The method is then extended to all numbers, real or complex, though the procedure for none rational numbers is more time consuming. We also show how our method provides a natural way to generalize Jacobsthal numbers. Moreover, the method provides a way to generate infinitely many sequences of numbers, of which many play an important rule in mathematical sciences and engineering, to name few, Jacobsthal sequence, Fibonacci sequence, and Pell sequence. (Received November 26, 2019)

1156-11-43 **Steven H. Weintraub*** (shw2@lehigh.edu), Dept. of Mathematics, Lehigh University, Bethlehem, PA 18015. *Some questions and answers in the arithmetic theory of generalized continued fractions.*

Let N be an arbitrary positive integer. Let x_0 be a positive real number. We consider continued fraction expansions of x_0 of the form

$$x_0 = a_0 + \cfrac{N}{a_1 + \cfrac{N}{a_2 + \cfrac{N}{a_3 + \cdots}}},$$

with a_0 a nonnegative integer and $a_1, a_2, a_3, \ldots$ positive integers, and refer to them as cf_N expansions. There is a natural notion of the "best" cf_N expansion of x_0, which we denote by $[[x_0]]_N$. We investigate these expansions, with particular attention to the question of the periodicity of $[[x_0]]_N$. It is well-known that classical continued fractions (i.e., the case $N = 1$) have a close connection with Pell's equation, and we investigate the relationship between cf_N expansions and Pell's equation, and Pell-like equations, as well. (Received December 31, 2019)

1156-11-53 **Patrick Ingram*** (pingram@yorku.ca), 4700 Keele St., Toronto, Ontario , Canada. *"Unicritical" polynomials in several variables.* Preliminary report.

We will propose a generalization of the unicritical polynomials to several variables, and make some observations about the dynamics of their critical loci. In particular, we will demonstrate estimates relating the critical height to the moduli height (although not quite the ones one would want). (Received January 06, 2020)

1156-11-169 **Irene Bouw**, **Ozlem Ejder*** (ejder@math.colostate.edu) and **Valentijn Karemaker**. *Arboreal Representations of Dynamical Belyi Maps.*

A dynamical Belyi map is a finite morphism $f : \mathbb{P}^1_{\mathbb{C}} \to \mathbb{P}^1_{\mathbb{C}}$ defined over $\mathbb{C}$ which is branched exactly at the three ordered points $0, 1, \infty$ such that $f(\{0, 1, \infty\}) \subseteq \{0, 1, \infty\}$ which makes all the iterates f^n also Belyi maps. Given a dynamical Belyi map defined over a number field K and a non-preperiodic point $\alpha \in K$, one can construct a tree of preimages of α. This construction leads to the phenomena: one has a tower of fields $K = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$ where $K_n := K\left(\phi^{-n}(\alpha)\right)$. One also has a natural Galois representation on the tree of preimages, the so-called Arboreal Galois representation of the function f. In this talk, we describe the Arboreal Galois representations and the monodromy groups of iterations of a large class of dynamical Belyi maps. Studying these Galois groups has applications in the study of the density of prime divisors of elements of dynamical sequences. If time allows, I will mention some applications as well. (Received January 22, 2020)

1156-11-254 **Charlotte Chan*** (charchan@mit.edu), Department of Mathematics, Simons Building (Building 2), Room 106, 77 Massachusetts Avenue, Cambridge, MA 02139. *Geometric L-packets of toral supercuspidal representations.*

In 2001, Yu gave an algebraic construction of supercuspidal representations of padic groups which is now known to be exhaustive when the residual characteristic is sufficiently large (Kim, Fintzen). There has since been a lot of progress towards explicitly constructing the local Langlands correspondence: DeBacker-Reeder and Kazhdan-Varshavsky (depth zero), Reeder and DeBacker-Spice (unramified toral), and Kaletha (regular supercuspidals). In this talk, we present recent and ongoing work with Masao Oi investigating a geometric counterpart to this story. (Received January 25, 2020)

1156-11-261 **Alexander J Carney*** (alexanderjcarney@rochester.edu), **Ruthi Hortsch** and **Michael Zieve**. *Rational polynomials are (almost) always injective.*

Any polynomial $f \in \mathbb{Q}[X]$ induces a map $\mathbb{Q} \to \mathbb{Q}$. We show that this map is usually 1-to-1 and always at most 6-to-1 over all but finitely many values. Analogous bounds hold over every number field K. If we interpret

Mazur and Merel's theorems on rational torsion of elliptic curves as bounding the rational N-to-1 behavior for morphisms between genus one curves, then our result can be seen as a parallel which doesn't require the structure of an abelian variety. We formulate a conjecture about morphisms between arbitrary varieties which implies both our result and the uniform boundedness conjecture for rational torsion on abelian varieties, and discuss implications and obstructions to proving it. (Received January 25, 2020)

1156-11-360 **Samuel S. Schiavone***, sam.schiavone@gmail.com, and **John Voight** and **Michael Musty**. *Computing, Cataloguing, and Representing Belyi Maps.* Preliminary report.

In this talk, we present our work in computing Belyi maps and creating an exhaustive database in low degree. We also discuss some preliminary work on finding "small" equations in an endeavor to produce particularly simple models for Belyi maps. (Received January 27, 2020)

13 ► Commutative rings and algebras

1156-13-39 **Saeed Nasseh*** (snasseh@georgiasouthern.edu), Department of Mathematical Sciences, Georgia Southern University, Statesboro, GA 30460. *Deformations and vanishing of homology.* Preliminary report.

A commutative noetherian local ring R is homologically persistent if every finitely generated R-module M with $\operatorname{Tor}_{i\gg 0}^R(M,M) = 0$ has finite projective dimension. In this talk, we discuss some of the recent progress on tracking the homologically persistence property along deformations with some of its applications. This talk is based on an in-progress joint work with L. Avramov, S. Iyengar, and S. Sather-Wagstaff. (Received December 27, 2019)

1156-13-94 **Susan Cooper**, **Sabine El Khoury**, **Sara Faridi*** (faridi@dal.ca), **Sarah Mayes-Tang**, **Susan Morey**, **Liana Sega** and **Sandra Spiroff**. *Simplicial resolutions of powers of monomial ideals.* Preliminary report.

The thesis of Diana Taylor in 1966 established a way to construct a free resolution of an ideal I generated by p monomials using the simplicial chain maps of a simplex on p vertices. Work of Bayer, Peeva and Sturmfels in 1998 showed that Taylor's theorem can be extended to all simplicial complexes on p vertices: as long as such a simplicial complex satisfies certain homological conditions, it can "support" a free resolution of I.

The purpose of this talk is to construct a combinatorial class of simplicial complexes indexed by the positive integers, where the n-th complex in this class supports a free resolution of the n-th power I^n. (Received January 14, 2020)

1156-13-111 **Huy Tai Ha*** (tha@tulane.edu). *Containment between powers of ideals and optimal solution to linear programming problems.*

We discuss an interesting connection between containment of powers of monomial ideals and gap estimates for optimal solutions to linear programming problems. (Received January 15, 2020)

1156-13-118 **Ayah Almousa*** (aka66@cornell.edu), **Gunnar Fløystad** and **Henning Lohne**. *Polarizations of Powers of Graded Maximal Ideals.*

We give a complete combinatorial characterization of all possible polarizations of powers of the graded maximal ideal $(x_1, x_2, \ldots, x_m)^n$ of a polynomial ring in m variables. We also give a combinatorial description of the Alexander duals of such polarizations. In the three variable case $m = 3$ and also in the power two case $n = 2$ the descriptions are easily visualized and we show that every polarization defines a (shellable) simplicial ball. We conjecture that any polarization of an Artinian monomial ideal defines a simplicial ball. (Received January 16, 2020)

1156-13-131 **Rachel N. Diethorn*** (rngettin@syr.edu). *Generators of Koszul homology and an application.*

One approach to understanding Koszul homology is to find, explicitly, its generators. In the first part of this talk, I will provide explicit formulas for the generators of Koszul homology on the minimal generators of an ideal J with coefficients in what we call a J-weak complete intersection module. In the second part of this talk, I will demonstrate the utility of having such an explicit understanding of Koszul homology. In particular, I will discuss an application to the Koszul homology algebra of quotients by certain edge ideals, and I will give an answer, for such rings, to a question of Avramov about the Koszul homology algebra of Koszul algebras. (Received January 19, 2020)

1156-13-145 **Claudia Miller*** (clamille@syr.edu) and **Hamid Rahmati**. *Transferring algebra structures on complexes, part 2.*

We discuss how to use the homotopy on the tautological Koszul complex given by the weighted de Rham map to build a concrete permutation invariant dg-algebra structure on a well-known resolution. This is joint work with Hamid Rahmati. (Received January 20, 2020)

1156-13-146 **Claudia Miller** and **Hamid Rahmati*** (hrahmati2@unl.edu). *Transferring algebra structures on complexes, part 1.*

We discuss a homological method for transferring algebra structures on complexes along suitably nice homotopy equivalences. We also show how one can get such homotopy equivalencies, from old ones, using a homological tool called the perturbation lemma. This is joint work with Claudia Miller. (Received January 20, 2020)

1156-13-172 **Jonathan L. O'Rourke*** (jorourk2@tulane.edu). *Local Cohomology and Degree Complexes of Monomial Ideals.*

Studying the local cohomology modules of quotient rings is an area of increasing interest in recent years. Understanding these modules gives insight into the computational complexity of the ideal. Even in the case of powers of squarefree monomial ideals, little is known about these local cohomology modules. In this case, due to a Hochster-like formula of Takayama, the dimensions of the graded local cohomology modules are encoded in the homology of a simplicial complex called the degree complex. I will introduce the construction of the degree complex of a monomial ideal. I will then present results on how this degree complex can be decomposed when the ideal can be written as powers of a sum or fiber product of squarefree monomial ideals, and how the cohomological information can then be recovered. (Received January 22, 2020)

1156-13-178 **Ela Celikbas** and **Jai Laxmi*** (jai.laxmi@uconn.edu), Department of Mathematics, University of Connecticut, Storrs, Storrs, CT 06269, and **Jerzy Weyman**. *Spinor structures on free resolutions of codimension four Gorenstein ideals.*

The problem of classifying Gorenstein ideals of codimension four was around ever since Buchsbaum-Eisenbud classified Gorenstein ideals of codimension three. The first result related to the structure of codimension four case were obtained by Kustin and Miller. We analyze the structure of spinor coordinates on resolutions of Gorenstein ideals of codimension four. As an application we produce a family of such ideals with seven generators which are not specializations of Kustin-Miller model. (Received January 22, 2020)

1156-13-205 **Huy Tai Ha** (morey@txstate.edu) and **Susan Morey*** (morey@txstate.edu), Texas State University, 601 University Dr., San Marcos, TX 78666. *An Application of Rees Algebras to Graph Theory.*

In 1995 Rafael Villarreal determined the generators of the ideal of defining equations of the Rees algebra of the edge ideal of a graph in terms of the structure of the graph. In particular, the non-linear generators of the ideal of equations can be determined from the even closed walks of the graph. Detecting and enumerating cycles and circuits, both even and odd, is an interesting problem in graph theory. In this talk, an alternate description of the equations of the Rees algebra of an edge ideal will be given in terms of minors of a Jacobian dual matrix. Minor variations of this result provide an algorithm for enumerating all circuits (or cycles) of even length in a graph. An application to detecting directed cycles in digraphs will be given. (Received January 23, 2020)

1156-13-220 **Selvi Kara Beyarslan*** (selvi@southalabama.edu), 411 North University Blvd, Mobile, AL 36606. *Weighted Stanley-Reisner Correspondence.* Preliminary report.

In this talk, we will discuss relations between algebraic invariants of the Stanley-Reisner ideals of weighted and unweighted simplicial complexes. (Received January 24, 2020)

1156-13-230 **Luchezar L. Avramov**, **Nicholas R. Packauskas*** (nicholas.packauskas@cortland.edu) and **Mark E. Walker**. *Quasi-Polynomial Growth of Betti Sequences over Complete Intersections.* Preliminary report.

Abstract: Let Q be a regular local ring and I an ideal generated by a regular sequence of c elements in the square of the maximal ideal. It is known that over the complete intersection $R = Q/I$ that any finitely generated module M has Betti numbers eventually given by quasi-polynomial of degree less than c. That is, there are integer-valued polynomial functions p_+^M and p_-^M with the same leading term such that $\beta_{2i}^R(M) = p_+^M(2i)$ and $\beta_{2i+1}^R(M) = p_-^M(2i+1)$ for i sufficiently large. We will show that if q is the height of the ideal generated by the quadratic initial forms of I in the associated graded ring of Q, then the degree of $p_+^M - p_-^M$ is less than $c-q-1$. (Received January 24, 2020)

1156-13-237 **Sema Gunturkun*** (sgunturkun@amherst.edu) and **Mel Hochster**. *On the Eisenbud-Green-Harris Conjecture.*

Given a homogeneous ideal in $K[x_1, \ldots, x_n]$, Eisenbud, Green and Harris conjectured a finer bound on its Hilbert function than Macaulay's bound depending on the ideal's structure. Although there has been some progress on the conjecture, it remains widely open since 90s. In this talk, we focus on the ideals whose generators containing a regular sequence of quadrics, and talk about the recent progress on this conjecture. This is a joint work with Mel Hochster. (Received January 24, 2020)

1156-13-248 **Federico Galetto*** (f.galetto@csuohio.edu). *Finite group actions on free resolutions.* Preliminary report.

Many ideals inside polynomial rings come naturally equipped with group actions. Under mild assumptions, such actions extend to minimal free resolutions of the ideals, providing interesting interpretations of the associated Betti numbers. While describing these actions can in general be a difficult problem, I will present an algorithm that makes the computational approach a viable solution for finite groups. (Received January 25, 2020)

1156-13-252 **Anthony Iarrobino**, **Chris McDaniel** and **Alexandra Seceleanu*** (aseceleanu@unl.edu). *Lefschetz properties for connected sums of rings.*

A standard way to create new varieties from old is by forming connected sums. The cohomology ring of a connected sum of varieties can be obtained from the cohomology rings of the summands by means of an algebraic operation on Artinian Gorenstein rings also called a connected sum.

Cohomology rings of smooth complex algebraic varieties satisfy the Hard Lefschetz Theorem. The conclusion of this theorem has been abstracted in the language of commutative algebra under the name of Strong Lefschetz Property. This property is closely related to the Jordan type of multiplication by a general linear form. It is natural to ask whether all algebraic connected sums of graded Artinian Gorenstein rings enjoy the Lefschetz property in the absence of any underlying topological information. We investigate this question as well as the analogous question concerning a closely related construction, the fibered product. (Received January 25, 2020)

1156-13-270 **Abu C Thomas*** (athoma17@tulane.edu). *Estimation of Resurgence for various classes of Homogeneous Ideals.*

Motivated by Chudnovsky's Conjecture and containment problems, we study the invariants of a homogeneous ideal in a polynomial ring such as its Resurgence, Asymptotic Resurgence etc. A lot of algebraic properties of the ideal and in some cases geometric properties of the Scheme defined by the ideal can be derived from the measure of containment and non containment among symbolic and ordinary powers of ideal. In this aspect, computation of Resurgence, Asymptotic Resurgence which measures the containment/non containment of ideals can be very useful.

Following the joint work (with S. Bisui, T. Hà, A.V Jayanthan) of estimating the Resurgence, Asymptotic Resurgence of ideals corresponding to schemes in a bi-projective space $\mathbb{P}^m \times_k \mathbb{P}^n$, we will look at the estimation of various other classes of homogeneous ideals. (Received January 26, 2020)

1156-13-272 **Lindsey Hill*** (hill178@purdue.edu) and **Rachel Lynn**. *Specialization of Integral Closures of Ideals by General Elements.* Preliminary report.

J. Hong and B. Ulrich proved that the integral closure of an ideal of height at least 2 is compatible with specialization by *generic elements*, allowing one to use induction on the height of the ideal to prove results about integrally closed ideals. This technique's applications are limited by the fact that one must extend the base ring to a polynomial ring or localization thereof in order to define generic elements. We have shown that the integral closure of an ideal of height at least 2 is compatible with specialization by *general elements*, allowing one to induct on the height of the ideal without extending the ring. This is based on joint work with Rachel Lynn. (Received January 26, 2020)

1156-13-297 **Anthony Iarrobino** and **Pedro Macias Marques*** (pmm@uevora.pt), D. Matemática, CIMA, ECT, Universidade de Évora, Rua Romão Ramalho, 69, 7000-671 Évora, Portugal. *Using Jordan type to distinguish families of algebras.* Preliminary report.

We study Artinian Gorentein (AG) algebras and consider the set of Jordan types of elements of the maximal ideal, i.e. the partition giving the Jordan blocks of the respective multiplication map.

In a joint work with Tony Iarrobino, we explore the possible obstructions that the symmetric decomposition of the associated graded algebra of an AG algebra A may impose on the Jordan type of A. We use these obstructions to distinguish different irreducible components of the family of AG algebras of fixed Hilbert function. (Received January 27, 2020)

1156-13-330 **Selvi K. Beyarslan**, **Jennifer Biermann*** (biermann@hws.edu), **Kuei-Nuan Lin** and **Augustine O'Keefe**. *Algebraic invariants of weighted oriented graphs.*

Let $\mathcal{D}$ be a weighted oriented graph and let $I(\mathcal{D})$ be its edge ideal in a polynomial ring R. We give a formula for the Castelnuovo-Mumford regularity of $R/I(\mathcal{D})$ when $\mathcal{D}$ is a weighted oriented path or cycle such that edges of $\mathcal{D}$ are oriented in one direction. Additionally, we compute the projective dimension for this class of graphs. (Received January 27, 2020)

1156-13-332 **Sara Faridi*** (faridi@dal.ca), Halifax, Canada, and **Navid Nemati**, , France. *Weak Lefschetz properties of square-free Gotzmann ideals.* Preliminary report.

A Gotzmann ideal is an ideal whose growth in every degree is the same as the lex ideal which shares its Hilbert function. This talk is about which Gotzmann square-free monomial ideals have the Weak Lefschetz Property. The main tool we will use is the 2012 characterization of Gotzmann square-free monomial ideals by Mermin and Hoefel. (Received January 27, 2020)

1156-13-341 **Ashley Wheeler*** (awheeler@mtholyoke.edu), **Jessica Sidman** (jsidman@mtholyoke.edu) and **Will Traves** (traves@usna.edu). *Geometric equations for matroid varieties.*

Each point x in $\mathrm{Gr}(r,n)$ corresponds to an $r \times n$ matrix A_x which gives rise to a matroid M_x on its columns. Gel'fand, Goresky, MacPherson, and Serganova showed that the sets $\{y \in \{textGr(r,n)|M_y = M_x\}$ form a stratification of $\mathrm{Gr}(r,n)$ with many beautiful properties. However, results of Mnëv and Sturmfels show that these strata can be quite complicated, and in particular may have arbitrary singularities. We study the ideals I_x of matroid varieties, the Zariski closures of these strata. We construct several classes of examples based on theorems from projective geometry and describe how the Grassmann-Cayley algebra may be used to derive non-trivial elements of I_x geometrically when the combinatorics of the matroid is sufficiently rich. (Received January 27, 2020)

1156-13-357 **John Myers*** (john.myers@oswego.edu). *Three algebras and three definitions of Koszulness.* Preliminary report.

Let R be a standard graded commutative algebra over a field k, let K be the Koszul complex on a minimal set of generators of the irrelevant ideal of R, and let H be the homology of K. Recall that R is said to be *Koszul* if k has a linear free resolution over R. We adapt this definition to apply to K (viewed as a DG algebra) and then to H (viewed as a bigraded algebra). We describe how these three Koszul properties transfer back and forth between the three algebras R, K, and H, and we give several examples of classes of algebras R for which H is Koszul. (Received January 27, 2020)

1156-13-361 **Gabriel E Sosa Castillo*** (gsosacastillo@colgate.edu), Hamilton, NY 13346. *A survey of Rees and multi-Rees algebras of strongly stable ideals.*

In this talk we will present methods and results, along with open questions, regarding the defining equations of Rees and multi-Rees algebras of strongly stable ideals. (Received January 27, 2020)

1156-13-364 **Sara L Malec*** (malec@hood.edu), 401 Rosemont Ave, Frederick, MD 21701. *Intersection Algebras, Semigroups from Continued Fractions, and Other Observations.* Preliminary report.

Given two ideals in a Noetherian ring R, the intersection algebra is an object that captures some information on the relationships between those two ideals. This talk discusses a connection between these algebras and classes of semigroups coming from particular continued fraction decompositions, and explores other interesting properties. (Received January 27, 2020)

1156-13-371 **Lars W Christensen*** (lars.w.christensen@ttu.edu) and **Srikanth B Iyengar**. *Krull dimension of finite free complexes.* Preliminary report.

Let R be a commutative noetherian local ring. A simple way to extend invariants from the realm of modules to complexes is to apply them to the homology modules. Foxby's definition of the Krull dimension of a complex is an example:

$$\dim_R X = \sup\{\dim_R \mathrm{H}_n(X) - n \mid n \in \mathbb{Z}\}\,.$$

To apply this definition to compute the dimension even just of a bounded complex of finitely generated free modules, one needs to determine the annihilators of the homology modules. I will discuss an alternative way to compute the dimension of such a complex, a way that only requires one to determine certain Fitting invariants of the differential. (Received January 27, 2020)

1156-13-378 **Claudiu Raicu*** (craicu@nd.edu) and **Steven V Sam**. *Bigraded Koszul modules.*
I will describe the basic theory of bigraded Koszul modules, and present a sharp vanishing theorem for their graded components in the finite length case. This statement has implications to the vanishing of syzygy modules of K3 carpets, which arise as double structures on a rational normal surface scroll. It implies the Canonical Ribbon Conjecture of Bayer and Eisenbud, and extends its validity to positive characteristic. Joint work with Steven Sam. (Received January 28, 2020)

1156-13-380 **Francesca Gandini***, 1200 Academy St, Kalamazoo, MI 49006. *Equivariant Hilbert series of subspace arrangements.*
To an hyperplane arrangement we can associate a combinatorial object, a matroid. Similarly, to a subspace arrangement we associate a polymatroid. Each subspace in the arrangement can be viewed algebraically as a linear ideal. We can study the product of these linear ideals by using the combinatorial data of the polymatroid. In particular, by tensoring the ambient vector space with an n-dimensional vector space, we introduce an action of GL_n on the subspace arrangement and consequently on the product of linear ideals. The structure of the equivariant resolution of the product ideal is determined by the combinatorial data. More precisely, the equivariant Hilbert series of the product of the linear ideals can be recursively constructed from the polymatroid of the subspace arrangement. Furthermore, we will introduce a functor on the category of polynomial representations that produces an ideal in the exterior algebra associated to the product ideal of the subspace arrangement. The corresponding effect of this functor on the equivariant Hilbert series is the involution that maps the Schur symmetric function s_λ to $s_{\lambda'}$, the function indexed by the transpose of the original partition. (Received January 28, 2020)

1156-13-382 **Laura E. Ballard***, lballard@syr.edu. *Properties of the Toric Ring of a Chordal Bipartite Family of Graphs.*
This project concerns the classification and study of a group of Koszul algebras coming from the toric ideals of a chordal bipartite infinite family of graphs (alternately, these rings may be interpreted as coming from determinants of subsets of certain matrices). I have determined a system of parameters for this family of rings, have developed a proof for the regularity of the rings and explicitly determined the Hilbert series for the Artinian reduction of these modulo a linear system of parameters, yielding, in particular, the multiplicity of the original rings. I will also report on progress on the nonadditive derived homological algebra of this family of algebras. (Received January 28, 2020)

1156-13-383 **Francesca Gandini***, 1200 Academy St, Kalamazoo, MI 49006. *Castelnuovo-Mumford regularity of ideals associated to subspace arrangements.*
Given a collection of t subspaces in an n-dimensional vector space W we can associate to them t linear ideals in the symmetric algebra $\mathcal{S}(W^*)$. Conca and Herzog showed that the Castelnuovo-Mumford regularity of the product of t linear ideals is equal to t. Derksen and Sidman showed that the Castelnuovo-Mumford regularity of the intersection of t linear ideals is at most t. In this talk we show that analogous results hold when we work over the exterior algebra $\bigwedge(W^*)$ (and work over a field of characteristic 0). To prove these results we rely on the functoriality of equivariant free resolutions and construct a functor Ω from the category of polynomial functors to itself. The functor Ω transforms resolutions of polynomial functors associated to subspace arrangements over the symmetric algebra to resolutions over the exterior algebra. In particular, the functor Ω preserves regularity bounds. Therefore, in the exterior algebra, ideals constructed from t linear ideals will also have regularity at most t. (Received January 28, 2020)

1156-13-391 **Jennifer Biermann**, **Hernan De Alba Casillas**, **Federico Galetto**, **Satoshi Murai**, **Uwe Nagel**, **Augustine O'Keefe**, **Tim Romer** and **Alexandra Seceleanu*** (aseceleanu@unl.edu). *Betti numbers of symmetric shifted ideals.*
Say an ideal of a polynomial ring is symmetric if it is fixed by the natural action of the symmetric group. We introduce a new class of monomial ideals, which we call symmetric shifted ideals, and which can be considered as an analogue of stable monomial ideals within the class of symmetric ideals. The homological properties of symmetric shifted ideals and their applications will be discussed. (Received January 28, 2020)

1156-13-397 **Branden Stone*** (bstone@hamilton.edu) and **Courtney R Gibbons** (crgibbon@hamilton.edu). *Module decompositions over short and super-stretched rings.* Preliminary report.
The concept of stretched Cohen-Macaulay rings was first introduced by J.D. Sally. In this talk we will explore the relationship between super-stretched (a refinement of stretched) and short rings with dimension at most

one. We are interested in characterizing decompositions of modules based on their numerical invariants (e.g. h-vectors). (Received January 28, 2020)

14 ► Algebraic geometry

1156-14-8 **Tony Shaska*** (shaska@oakland.edu). *The addition on Jacobian varieties from a geometric viewpoint.*

We give a geometric interpretation of the group law for Jacobian varieties by extending the geometric construction of chords and tangents on an elliptic curve to Jac $\mathcal{X}$. For any given curve $\mathcal{X}$ and reduced divisors D_1 and D_2 in Jac $\mathcal{X}$, we define a new curve $\mathcal{Y}$, such that the intersection divisor of $\mathcal{X}$ and $\mathcal{Y}$ determines the divisor $(D_1 + D_2)$. We give exact formulas for $\mathcal{Y}$ for superelliptic Jacobians. (Received January 19, 2020)

1156-14-22 **Angel Carocca** (angel.carocca@ufrontera.cl), Temuco, Chile, and **Sebastián Reyes-Carocca*** (sebastian.reyes@ufrontera.cl), Temuco, Chile. *Riemann surfaces of genus $1 + q^2$ with $3q^2$ automorphisms.*

In this talk we discuss a classification of compact Riemann surfaces of genus $1+q^2$ with a group of automorphisms of order $3q^2$, where q is a prime number. We also study decompositions of the corresponding Jacobian varieties. (Received December 11, 2019)

1156-14-26 **Hyuk Jun Kweon*** (kweon@mit.edu). *Bounds on the Torsion Subgroups of Néron-Severi Groups.*

Let $X \hookrightarrow \mathbb{P}^r$ be a smooth projective variety defined by homogeneous polynomials of degree $\leq d$. We give explicit upper bounds on the order of the torsion subgroup $(\mathrm{NS}\, X)_{\mathrm{tor}}$ of the Néron–Severi group of X. The bounds are derived from an explicit upper bound on the number of irreducible components of either the Hilbert scheme $\mathbf{Hilb}_Q\, X$ or the scheme $\mathbf{CDiv}_n\, X$ parametrizing the effective Cartier divisors of degree n on X. As an application we can bound the order of the torsion subgroups of the second étale cohomology groups of X. We also give a uniform upper bound on the number of generators of $(\mathrm{NS}\, X)[\ell^\infty]$ uniform as ℓ varies. (Received December 15, 2019)

1156-14-28 **Milagros Izquierdo*** (milagros.izquierdo@liu.se), Department of Mathematics, Linköping University, 58183 Linköping, Sweden, and **Sebastián Reyes-Carocca**. *On Large Groups of Automorphisms of Riemann Surfaces.*

Belolipetsky and Jones classified those compact Riemann surfaces of genus g admitting a large group of automorphisms of order $k(g-1)$, for each $k > 6$, under the assumption that $g-1$ is a prime number. Here we study the remaining large cases: we classify Riemann surfaces admitting $5(g-1)$ and $6(g-1)$ automorphisms, with $g-1$ a prime number. As a consequence, we obtain that any Riemann surface de genus g, $g-1$ a prime, admitting $3(g-1)$ automorphisms has $6(g-1)$ automorphisms. (Received December 19, 2019)

1156-14-33 **Rahul Pandharipande** and **Hsian-Hua Tseng*** (hhtseng@math.ohio-state.edu). *descendant Hilb/Sym correspondence for the plane.*

Let S be a nonsingular surface. A version of the crepant resolution conjecture predicts that the descendant Gromov-Witten theory of $Hilb^n(S)$, the Hilbert scheme of n points on S, is equivalent to the descendant Gromov-Witten theory of $Sym^n(S)$, the n-fold symmetric product of S. In this talk we discuss how this works when S is $\mathbb{C}^2$. We explicitly identify a symplectic transformation equating the two descendant Gromov-Witten theories. We also establish a relationship between this symplectic transformation and the Fourier-Mukai transformation which identifies the (torus-equivariant) K-groups of $Hilb^n(\mathbb{C}^2)$ and $Sym^n(\mathbb{C}^2)$. (Received December 23, 2019)

1156-14-34 **Bong Lian***, 415 South St, Waltham, MA 02454. *Mirror Symmetry for Singular Calabi-Yau Varieties.*

I will report on a recent discovery of a new class of mirror pairs of Calabi-Yau Varieties, and discuss their general constructions and consequences, including topological and enumerative applications. This is based on recent joint work with S. Hosono, T.J. Lee, H. Hiromichi Takagi, and S.-T. Yau. (Received December 23, 2019)

1156-14-37 **Emma Previato*** (ep@bu.edu), Dept of Mathematics & Statistics, Boston University, Boston, MA 02215. *Trigonal Cyclic Curves: Modular Aspects.*

The classical relevance of cyclic curves was expanded at the end of the last century by their relevance in mathematical physics, witness the 2011 Springer monograph by H.M. Farkas, S. Zemel, "Generalizations of Thomae's Formula for Z_n Curves"; little is known about the pertaining modular functions. This talk focuses on Z_3 curves

over a family with a nodal singular fiber. Although the Riemann theta function does not have a finite limit, we extend analytically the sigma function for the singular curve. We pose the question of completing the family to a fibration, by inserting an incomplete lattice and exploiting the modular properties of the sigma function. We will discuss a generalization of these constructions for Z_n curves. This work is the result of collaboration with K. Aomoto, Yu. Fedorov, J. Komeda, S. Matsutani. (Received December 25, 2019)

1156-14-46 **Kirill Zainoulline*** (kirill@uottawa.ca), Department of Mathematics and Statistics, University of Ottawa, 150 Louis-Pasteur, Ottawa, Ontario K1N6N5, Canada. *Localized operations on T-equivariant oriented cohomology of projective homogeneous varieties.*

In the present talk we provide a general algorithm to compute multiplicative cohomological operations on algebraic oriented cohomology of projective homogeneous G-varieties, where G is a split reductive algebraic group over a field k of characteristic 0.

More precisely, we extend such operations to the respective T-equivariant (T is a maximal split torus of G) oriented theories, and then compute them using equivariant Schubert calculus techniques. This generalizes an approach suggested by Garibaldi-Petrov-Semenov for Steenrod operations. We also show that operations on the theories of additive type commute with classical push-pull operators up to a twist. (Received January 03, 2020)

1156-14-55 **Gerhard Rosenberger*** (gerhard.rosenberger@math.uni-hamburg.de), Fachbereich Mathematik, Universität Hamburg, Bundesstrasse 55, 20146 Hamburg, Germany, and **Martin Kreuzer**. *On the numbers of the form $x^2 + 11y^2$.* Preliminary report.

Numbers of the form x^2+Ny^2 have been studied for a long time, in particular when N is one of Euler's convenient numbers. Here we consider the first inconvenient number $N = 11$. We use the structure of the modular class group of level 11 to study the set of all numbers having a representation $n = x^2 + 11y^2$. (Received January 06, 2020)

1156-14-56 **Dawei Chen*** (dawei.chen@bc.edu), Department of Mathematics, Boston College, Chestnut Hill, MA 02467, and **Martin Moeller** and **Adrien Sauvaget**. *Volume and Segre class of the cotangent bundle of the moduli space of curves.*

The cotangent bundle of the moduli space of curves can be identified with the moduli space of quadratic differentials, which carries a natural volume form. In this talk I will explain how to compute this volume via the top Segre class of the cotangent bundle and certain linear Hodge integrals. (Received January 06, 2020)

1156-14-97 **Qile Chen*** (qile.chen@bc.edu), Maloney Hall, Fifth Floor, Department of Mathematics, Boston College, Chestnut Hill, MA 02467. *The Logarithmic Gauged Linear Sigma Model.*

We introduce the notion of log R-maps generalizing the notion of stable maps with p-fields, and develop a proper moduli stack of stable log R-maps in the case of a hybrid gauged linear sigma model. Two virtual cycles (canonical and reduced) are constructed for these moduli stacks. The main results are two comparison theorems relating the reduced virtual cycle to the Kiem-Li cosection localized virtual cycle, as well as the reduced virtual cycle to the canonical virtual cycle. This is part of a project aiming at the foundation of a new technique for computing higher genus Gromov–Witten invariants of complete intersections.

This is a joint work with Felix Janda and Yongbin Ruan. (Received January 14, 2020)

1156-14-106 **Allen Knutson***, allenk@math.cornell.edu, and **Paul Zinn-Justin**. *Puzzles, quiver varieties, and Segre-Schwarz-MacPherson classes.*

The "10-10-10 puzzle piece" in Grassmannian Schubert calculus puzzles plays a strange role. It is forbidden in ordinary (equivariant) cohomology, allowed right side up when multiplying K-classes of structure sheaves, and allowed upside down when multiplying K-classes of ideal sheaves. When allowed both ways, we show the resulting puzzles compute products of Segre-Schwarz-MacPherson classes (a close relative of CSM classes, likewise living most naturally on the cotangent bundle). To do this, we implement multiplication by convolution with the graph of diagonal inclusion, soup that graph up to its conormal bundle (a Lagrangian correspondence between the cotangent bundles), factor that Lagrangian correspondence through a quiver variety, and see how puzzles naturally arise on that intermediate space. (Received January 15, 2020)

1156-14-120 **Iulia Gheorghita*** (gheorgiu@bc.edu). *Effective divisors in the projectivization of the Hodge bundle.*

In this talk I will explain how to compute the classes of many effective divisors in the projectivization of the Hodge bundle. These divisors include the counterparts of the pointed Brill-Noether divisors in the moduli space of curves with one marked point. This is joint work with Nicola Tarasca. (Received January 17, 2020)

1156-14-123 **Elana Grace Kalashnikov*** (kalashnikov@math.harvard.edu). *Finding mirrors for Fano quiver flag zero loci.*

One interesting feature of the classification of smooth Fano varieties up to dimension three is that they can all be described as certain subvarieties in GIT quotients; in particular, they are all either toric complete intersections (subvarieties of toric varieties) or quiver flag zero loci (subvarieties of quiver flag varieties). Fano varieties are expected to mirror certain Laurent polynomials; given such a Fano toric complete intersection, one can produce a Laurent polynomial via the Landau-Ginzburg model. In this talk, I'll discuss finding mirrors of four dimensional Fano quiver flag zero loci via finding degenerations of the ambient quiver flag varieties. These degenerations generalise the Gelfand-Cetlin degeneration, which in the Grassmannian case has an important role in the cluster structure of its coordinate ring. (Received January 17, 2020)

1156-14-148 **S. Allen Broughton*** (brought@rose-hulman.edu). *Equivalence of Finite Group Actions on Riemann Surfaces and Algebraic Curves.* Preliminary report.

We consider conformal actions of the finite group G on a closed Riemann surface, as well as algebraic actions of G on smooth, complete algebraic curves over an arbitrary, algebraically closed field. There are several notions of equivalence of actions, the most studied of which is topological equivalence, presumably because of its close relationship to the branch locus of the moduli space. However, the notion of topological equivalence does not work well for positive characteristic. We shall discuss a replacement for topological equivalence, which we dub *equisymmetry*, that may be applied in all characteristics. The relation is induced by families of curves with G-action. After giving an overview of the various equivalence relations (conformal/algebraic, topological, q-differentials, equisymmetry) we focus on equisymmetry and its relationship to rotation constants and q-differentials. (Received January 20, 2020)

1156-14-149 **Tony Shaska*** (shaska@oakland.edu) and **Dorisa Tabaku** (dorisatabaku@oakland.edu). *Computation aspects of moduli spaces: an introduction to weighted moduli heights.*

Rational points in the moduli space of curves are of fundamental interest in arithmetic geometry. Is there a way to count them? We introduce weighted moduli heights and show how they can be used to count rational points in the hyperelliptic moduli. (Received January 21, 2020)

1156-14-152 **John Lesieutre***, jdl@psu.edu. *A new rational surface automorphism with no periodic curves.*

I will describe a new example of a positive entropy automorphism of a rational surface which has no periodic curves. The construction is geometric, with a very classical feel, using some old and new facts about del Pezzo surfaces, Coble surfaces, etc. (Received January 21, 2020)

1156-14-158 **Renzo Cavalieri***, 1874 campus mail, Fort Collins, CO 80523. *A tour of hyperelliptic loci.* Preliminary report.

Hyperelliptic loci are subvarieties of the moduli spaces of curves which parameterize curves that admit a degree two map to the projective line. They are one of the basic examples of tautological classes described by the geometric properties that the curves parameterize satisfy. This talk will survey what we know and what we would like to know about these classes, and highlight the similarities and differences between the collection of all hyperelliptic loci and the datum of a CohFT. Collaborative work with Tarasca and Schmitt will be presented. (Received January 21, 2020)

1156-14-166 **Tina Kanstrup*** (tkanstrup@math.umass.edu). *Link homologies and Hilbert schemes via representation theory.* Preliminary report.

In knot theory one of the main ways to tell if two links are isotopic is to calculate link invariants. One of the most famous ones is Khovanov-Rozansky triply graded link homology. The definition is completely algebraic but it has been conjectured by Gorsky, Negut and Rasmussen that it can also be calculated geometrically in terms of cohomolgy of shaves on Hilbert schemes. A result of a similar nature but in terms of matrix factorization was proved by Oblomkov and Rozansky. In this talk we'll use representation theory to relate the two, and show that their construction is natural from the point of view of representation theory. (Received January 22, 2020)

1156-14-189 **Ewa Kozłowska-Walania***, Bazynskiego 8, 80-309 Gdansk, Poland. *Extremal Riemann surfaces and their properties.*

A Riemann surface shall be called *extremal* if it admits the maximal possible number of symmetries or if it admits the maximal number of ovals for a set of non-conjugate symmetries with ovals. These cases are to be called s-extremal and o-extremal respectively. We shall present a variety of recent results concerning such surfaces, in particular we show the structure of the automorphism group in both s- and o-extremal case and find all

the possible topological types of commuting symmetries together with their defining equations in an o-extremal configuration. Special attention shall be paid to the surfaces of even genera and in particular the so-called *button-like* surfaces, being unique both s- and o-extremal surfaces of genus $g = 4k, k \geq 1$ with a non-abelian automorphism group. (Received January 23, 2020)

1156-14-192 **Joachim Jelisiejew** and **Klemen Šivic*** (klemen.sivic@fmf.uni-lj.si). *Irreducible components of commuting varieties.*

Let $C_n(M_d)$ denote the variety of all n-tuples of pairwise commuting $d \times d$ matrices. These varieties are closely related to Quot schemes, and in particular to Hilbert schemes. While $C_2(M_d)$ is always irreducible, $C_n(M_d)$ is reducible for $n, d \geq 4$. However, very few irreducible components are known. In the talk we characterize irreducible components of $C_n(M_d)$ for all n if $d \leq 8$. In particular, our results show that $C_4(M_8)$, viewed as a scheme defined by the quadratic commutativity relations, has a generically nonreduced component. Using torus actions we also prove existence of other irreducible components for higher d and n. (Received January 23, 2020)

1156-14-194 **Alessio Caminata**, **Noah Giansiracusa**, **Han-Bom Moon*** (hmoon8@fordham.edu) and **Luca Schaffler**. *Point configurations and metric trees.* Preliminary report.

Consider the parameter space of n points in a d-dimensional projective space that lie on a rational normal curve. What are the defining equations of the parameter space? I will explain a connection between the question and the reconstruction problem for metric trees via tropical geometry. This is joint work with Alessio Caminata, Noah Giansiracusa, and Luca Schaffler. (Received January 23, 2020)

1156-14-202 **Yusuf Mustopa***, Department of Mathematics, UMass Boston, Wheatley Hall, 100 William T Morrissey Blvd, Boston, MA 02125, and **Montserrat Teixidor i Bigas**, Department of Mathematics, Tufts University, 503 Boston Avenue, Medford, MA 02155. *Rational Curves on Moduli Spaces of Vector Bundles.*

Given that Fano varieties are rationally connected, the Hilbert schemes parametrizing their rational curves encode fundamental aspects of their geometry. Among the best-known Fano varieties is the moduli space $\mathrm{SU}_C(r, L)$ parametrizing $S-$equivalence classes of semistable vector bundles on a smooth projective curve C which have rank r and determinant L. The irreducible components of each Hilbert scheme of rational curves on $\mathrm{SU}_C(r, L)$ in the case $r = 2, c_1(L) = 1$ were previously classified by Castravet. In this talk, I will report on joint work with Montserrat Teixidor i Bigas which addresses the case of higher rank and arbitrary determinant. (Received January 23, 2020)

1156-14-204 **Rachel Davis*** (rachel.davis@wisc.edu). *Johnson morphisms: definitions and related open questions.* Preliminary report.

Let X be a smooth curve of genus g. Johnson, Morita, Perron, and others have studied the mapping class group of X and its Torelli subgroup using Johnson homomorphisms. I will define and give examples of the Johnson morphisms and explain how work of Hain-Matsumoto, Corey-Ellenberg-Li, and Bisogno-Li-Litt-Srinivasan relates open questions about the Ceresa class of non-hyperelliptic curves to the Johnson morphisms. (Received January 23, 2020)

1156-14-236 **Allen Knutson*** (allenk@math.cornell.edu). *A degeneration of the commuting scheme.*

Is the space of pairs (X, Y) of matrices, satisfying $XY = YX$, a reduced scheme? This natural-looking question (from at least Hochster '84) remains unanswered. I'll give a degeneration of these equations, and obtain from them a stronger conjecture, that the "upper-upper scheme" $\{XY, YX$ upper triangular with reversed diagonals$\}$ is reduced. As time permits I'll talk about related schemes arising in geometric representation theory. (Received January 24, 2020)

1156-14-246 **Paolo Aluffi** and **Leonardo Mihalcea*** (lmihalce@math.vt.edu), 225 Stanger St, Dept. of Mathematics, 460 McBryde Hall, Virginia Tech, Blacksburg, VA 24061, and **Jorg Schurmann** and **Changjian Su**. *Cotangent Schubert Calculus.*

A natural question with roots in representation theory is to find good analogues of Schubert classes in the intersection rings of the cotangent bundle of a flag manifold. One answer is given in terms of the Maulik and Okounkov theory of stable envelopes. These turn out to be equivalent to certain characteristic classes of singular varieties in the flag manifold: the Chern-Schwartz-MacPherson classes in cohomology, and motivic Chern classes in K theory. I will explain how familiar properties of the usual Schubert classes, such as being determined by divided difference operators, or satisfying positivity properties, hold, or are conjectured to hold, in the cotangent situation. Joint work with P. Aluffi, J. Schurmann, C. Su. (Received January 25, 2020)

1156-14-256 **Boris Tsvelikhovskiy*** (tsvelikhovskiy.b@husky.neu.edu), 30 Stearns Rd, Boston, MA 02446. *On G-Hilbert scheme for 1/n(1,...,1,-1,...,-1).* Preliminary report.

Let G be a finite subgroup of $GL_n(\mathbb{C})$ and consider the quotient $X = \mathbb{C}^n/G$. If G is contained in $SL_{2,3}(\mathbb{C})$, the G-Hilbert scheme (the fine moduli space of G-clusters in $\mathbb{C}^n$) is a crepant resolution of X. Moreover, one observes the following phenomenon: the number of nontrivial irreducible representations of G coincides with the number of irreducible components in the central fiber of the resolution. This usually goes under the name of McKay correspondence. The more modern formulation of McKay correspondence is the derived equivalence $D^G(\mathbb{C}^n) \to D(Y)$, where Y is a crepant resolution of X.

The plan for the talk is as follows. After giving the necessary definitions and explaining the McKay correspondence on the example of cyclic subgroups $\mathbb{Z}_n \subset SL_2(\mathbb{C})$, we state the results of the work in progress, namely, that the G-Hilbert scheme for $\frac{1}{n}(1,\dots,1,-1,\dots,-1)$ is smooth and the number of nontrivial irreducible representations of G coincides with the number of irreducible components in the central fiber of the resolution. (Received January 25, 2020)

1156-14-264 **Richard Rimanyi*** (rimanyi@email.unc.edu). *3d mirror symmetry through elliptic Schubert classes.*

The elliptic class of a Schubert variety depends on several sets of parameters: topological, equivariant, and dynamical ones, as well as $\hbar$. If we switch the roles of equivariant and dynamical variables, and replace $\hbar$ with $\hbar^{-1}$ then we should obtain the elliptic classes of certain varieties in the mirror dual. We will explore consequences of this prediction, some in elementary combinatorics, some in the theory of theta functions. (Received January 26, 2020)

1156-14-274 **Kostiantyn Timchenko*** (ktimchen@nd.edu). *Characteristic cycles for K-orbits on Grassmannians.*

Let K be the fixed set of an involution of the complex group GL(n). K acts on the Grassmannian with a finite number of orbits. We compute the characteristic cycle of the intersection cohomology sheaf of the orbit closure (with respect to the trivial local system). The cycles are irreducible for K = GL(p)xGL(q) and Sp(2n). When K is SO(n), roughly half of them are irreducible. (Received January 26, 2020)

1156-14-277 **Rahul Singh***, rahul.sharpeye@gmail.com. *A Robinson-Schensted correspondence for partial permutations.* Preliminary report.

The classical Robinson-Schensted correspondence, a bijection from pairs of Young tableaux to the group of permutations, was interpreted geometrically by Steinberg counting the conormal varieties of Schubert varieties. Steinberg's work has seen many generalizations, built around studying the conormal varieties of various objects.

We undertake a similar project, studying the conormal varieties of matrix Schubert varieties to obtain a bijection from the set of partial permutations of size (p, q) to triples (P, Q, L); here L is a signed Young diagram indexing an orbital variety of the 2 vertex loop quiver, and P, Q are Young tableaux of size p, q respectively.

The key idea is to use a natural compactification of matrix Schubert varieties, and van Leeuwen's geometric interpretations of various combinatorial operations on Young tableaux, to reduce the combinatorics to a simple variation of the Robinson-Schensted-Knuth correspondence. (Received January 26, 2020)

1156-14-284 **Daniel Joo** and **Kalina Mincheva*** (kalina.mincheva@yale.edu), 10 Hillhouse Ave, Department of Mathematics, New Haven, CT 06511. *Prime tropical ideals.*

In tropical geometry most algebraic computations are done on the classical side - using the algebra of the original variety. The theory developed so far has explored the geometric aspect of tropical varieties as opposed to the underlying (semiring) algebra and there are still many commutative algebra tools and notions without a tropical analogue. In the recent years, there has been a lot of effort dedicated to developing the necessary tools for commutative algebra using different frameworks, among which prime congruences, tropical ideals, tropical schemes. These approaches allows for the exploration of the properties of tropicalized spaces without tying them up to the original varieties and working with geometric structures inherently defined in characteristic one (that is, additively idempotent) semifields. In this talk we explore the relationship between tropical ideals and congruences to conclude that the variety of a non-zero prime tropical ideal is either empty or consists of a single point. (Received January 26, 2020)

1156-14-290 **Jean-Louis Colliot-Thélène**, **David Harbater**, **Julia Hartmann*** (hartmann@math.upenn.edu), **Daniel Krashen**, **R. Parimala** and **V. Suresh**. *Local-Global principles for tori over arithmetic function fields.*

We study local-global principles for tori over arithmetic function fields (i.e., one variable function fields over complete discretely valued fields). We show that for tori defined over the underlying discrete valuation ring, the

obstruction to such local-global principles can be described using techniques from patching. As a consequence, we obtain a vanishing criterion as well as a rather explicit description in interesting cases where the obstruction is not trivial. A key ingredient is the use of flasque tori and their cohomological properties. (Received January 26, 2020)

1156-14-291 **Martha Precup*** (martha.precup@wustl.edu). *Springer fibers and divisible tableaux.* Preliminary report.

Springer fibers are subvarieties of the flag variety that play an important role in geometric representation theory. In the type A case, it is a well known result of Spaltenstein that the irreducible components of Springer fibers are indexed by standard tableaux. In this talk, we consider the irreducible components corresponding to so-called divisible tableaux. We show that these components have surprising inductive features which yield new insights into their geometric structure. This is joint work with W. Graham and A. Russell. (Received January 26, 2020)

1156-14-293 **Martha Precup*** (martha.precup@wustl.edu). *Filtrations and bases for the cohomology of regular nilpotent Hessenberg varieties.*

Hessenberg varieties are subvarieties of the flag variety with connections to representation theory, algebraic geometry, and combinatorics. In 2015, Brosnan and Chow proved the Shareshian–Wachs conjecture, linking the Stanley–Stembridge conjecture in combinatorics to the geometry of Hessenberg varieties through Tymoczko's symmetric group action on the cohomology of regular semisimple Hessenberg varieties. The S_n-invariant subspace of this representation is isomorphic to the cohomology of a regular nilpotent Hessenberg variety. This talk will give a brief overview of that story and present a new filtration for the cohomology rings of regular nilpotent Hessenberg varieties. As an application, we obtain a monomial basis for this ring. This is joint work with M. Harada, T. Horiguchi, S. Murai, and J. Tymoczko. (Received January 26, 2020)

1156-14-299 **Shiyue Li*** (shiyue_li@brown.edu), 151 Thayer Street, Providence, RI 02912, **Dagan Karp** (dagan.karp@hmc.edu), 301 Platt Blvd, claremont, CA 91711, and **Siddarth Kannan**, 151 Thayer Street, Providence, RI 02912. *Tropical Derivation of Cohomology of Heavy/Light Hassett Spaces.*

We compute the Chow ring of an arbitrary *heavy/light Hassett space* $\overline{M}_{0,w}$. These spaces are moduli spaces of weighted pointed stable rational curves, where the associated weight vector w consists of only *heavy* and *light* weights. Work of Cavalieri et al. exhibits these spaces as tropical compactifications of hyperplane arrangement complements. The computation of the Chow ring then reduces to intersection theory on the toric variety of the Bergman fan of a graphic matroid. Keel has calculated the Chow ring $A^*(\overline{M}_{0,n})$ of the moduli space $\overline{M}_{0,n}$ of stable nodal n-marked rational curves; his presentation is in terms of divisor classes of stable trees of $\mathbb{P}^1$'s having one nodal singularity. Our presentation of the ideal of relations for the Chow ring $A^*(\overline{M}_{0,w})$ is analogous. We show that pulling back under Hassett's birational reduction morphism $\rho_w : \overline{M}_{0,n} \to \overline{M}_{0,w}$ identifies the Chow ring $A^*(\overline{M}_{0,w})$ with the subring of $A^*(\overline{M}_{0,n})$ generated by divisors of *w-stable trees*, which are those trees which remain stable in $\overline{M}_{0,w}$. (Received January 27, 2020)

1156-14-311 **Andrew Harder*** (anh318@lehigh.edu). *Mirror symmetry and the weight filtration.*

I will describe how mirror symmetry for log Calabi-Yau manifolds should relate Deligne's weight filtration on the cohomology of a log Calabi-Yau manifold to a flag filtration on the cohomology of its mirror dual. I will focus on the case of the rational elliptic surface and its mirror, and mirror symmetry will be taken to mean SYZ mirror symmetry. (Received January 27, 2020)

1156-14-331 **Yohsuke Matsuzawa***, 151 Thayer street, Providence, RI 02912. *On non-density of rational points with small arithmetic degree.*

Kawaguchi-Silverman conjecture is a conjecture that claims the geometric complexity of dynamical systems, dynamical degree, is equal to the arithmetic complexity, arithmetic degree. In a joint work with S. Meng, T. Shibata, and D.-Q. Zhang, we study a generalization of this conjecture, which predicts the non-density of rational points with arithmetic degree strictly smaller than the dynamical degree. I will talk about some results on this generalized conjecture and related problems. (Received January 27, 2020)

1156-14-348 **Rahul Singh***, rahul.sharpeye@gmail.com. *The Conormal Variety of a Schubert Divisor.*

Let X be a compact homogeneous space for some simple group G. The Schubert divisors in X are indexed by the simple roots of G. For X_α a divisor corresponding to a *long root* α, we show that the conormal variety of X_α with respect to X admits a compactification as a Schubert variety corresponding to the loop group of G. We also briefly discuss the implications of this observation for the geometry of the conormal variety. (Received January 27, 2020)

1156-14-358 **Lawrence J Barrott***, Boston College, Chestnut Hill, MA 02467, and **Navid Nabijou**. *Logarithmic Gromov-Witten Theory via degenerations.*

A recent flurry of activity has occurred around the problem of counting curves on $\mathbb{P}^2$, tangent to a fixed elliptic curve E at a single point to maximal order. A natural question is to ask what happens as one degenerates E to the toric boundary.

In this talk I will outline what occurs and how one can apply toric localisation in the non-toric setting. I will then make precise the connection between the invariants we construct and the work on scattering diagrams of Bousseau and Gabele. (Received January 27, 2020)

1156-14-369 **Dori Bejleri*** (bejleri@math.harvard.edu), Science Center 525, Harvard University, One Oxford Street, Cambridge, MA 02138. *Hilbert schemes of points on singular curves.* Preliminary report.

Hilbert schemes of points on singular curves are at the center of a beautiful story connecting algebraic geometry, representation theory and algebraic knot invariants. This story is by now well known for curves with reduced locally planar singularities. After recalling the reduced locally planar case, I will explain both some results and open questions about the non-reduced and non-planar cases. This is based on previous joint work with Ranganathan and Vakil, as well as work in progress, partially with Hung-Hsun Yu. (Received January 27, 2020)

1156-14-387 **He Yun*** (he_yun@brown.edu), Box 1917, 151 Thayer Street, Providence, RI 02912. *The S_n-equivariant rational homology of the tropical moduli space $\Delta_{2,n}$.*

The tropical moduli space $\Delta_{g,n}$ is a topological space that parametrizes isomorphism classes of n-marked stable tropical curves of genus g with total volume 1. In this talk, we focus on $\Delta_{2,n}$ and compute the characters of the S_n-representations afforded by the rational homology of $\Delta_{2,n}$ induced by permuting markings. We use that $\Delta_{2,n}$ is a symmetric Δ-complex as introduced by Chan, Glatius, and Payne. The computation is done in SageMath. (Received January 28, 2020)

1156-14-392 **Aaron D. Wootton*** (wootton@up.edu), 5000 N Willamette Blvd, Portland, OR 97203. *Cyclic surfaces with multiple defining equations.*

A compact Riemann surface S is said to be cyclic n-gonal if it has a defining equation of the form $y^n = q(x)$ for some polynomial $q(x)$, or equivalently, if it admits a conformal automorphism τ of order n such that the quotient space $S/\langle\tau\rangle$ has genus 0. For a given n-gonal surface S, one natural question to ask is whether the morphism τ is unique. That is, does there exists a conformal automorphism λ of S of order m (not necessarily equal to n) with $\lambda \notin \langle\tau\rangle$ and $S/\langle\lambda\rangle$ of genus 0? Such a surface would then admit an equation of the form $y^m = r(x)$ which might be significantly different to the equation $y^n = q(x)$. In this talk we shall discuss the classification of all such surfaces under the assumption that n and m are prime numbers. We shall also consider why the problem becomes much harder when for non-primes. (Received January 28, 2020)

1156-14-395 **Laura Rider*** (laurajoy@uga.edu), **Kei Yuen Chan** and **Paul Sobaje**. *Exotic t-structure for coherent sheaves on a partial resolution of the nilpotent cone.* Preliminary report.

In this talk, I will explain how to define an exotic t-structure for coherent sheaves on a partial resolution of the nilpotent cone. Along the way, I'll mention the special cases of exotic sheaves on the Springer resolution and perverse coherent sheaves on the nilpotent cone, along with some of their applications to geometric representation theory. This is joint work with Paul Sobaje and Kei Yuen Chan. (Received January 28, 2020)

1156-14-404 **Anders Buch*** (asbuch.math@gmail.com), 1, and **Pierre-Emmanuel Chaput**, **Leonardo Mihalcea** and **Nicolas Perrin**. *The Seidel representation on quantum K-theory.* Preliminary report.

The Seidel representation of a flag manifold X = G/P is an action of the fundamental group of the automorphism group of X on the quantum cohomology ring with all deformation parameters specialized to 1. I will speak about a version of the Seidel representation for the quantum K-theory of cominuscule flag manifolds, as well as consequences. This is joint work with P.-E. Chaput, L. Mihalcea, and N. Perrin. (Received January 28, 2020)

15 ► Linear and multilinear algebra; matrix theory

1156-15-71 **Madeleine Udell*** (udell@cornell.edu), NY , and **Alex Townsend**. *Big Data is Low Rank.*

Matrices of low rank are pervasive in big data, appearing in recommender systems, movie preferences, topic models, medical records, and genomics. While there is a vast literature on how to exploit low rank structure in these datasets, there is less attention on explaining why low rank structure appears in the first place. In this talk, we explain the abundance of low rank matrices in big data by proving that certain latent variable models associated to piecewise analytic functions are of log-rank. Any large matrix from such a latent variable model can be approximated, up to a small error, by a low rank matrix. Armed with this theorem, we show how to use a low rank modeling framework to exploit low rank structure even for datasets that are not numeric, with applications in the social sciences, medicine, and automated machine learning. (Received January 10, 2020)

1156-15-144 **Christina Pospisil*** (pospisil.christina@gmx.de). *Generalization Theory of Linear Algebra II.*

The talk continues the first part of the project Generalization Theory of Linear Algebra I (first part was presented at JMM 2019) and presents inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be applications to physics and other natural sciences be explored. (Received January 20, 2020)

1156-15-242 **Anastasiia Minenkova***, 341 Mansfield Road U1009, Storrs, CT 06269. *A Theorem of Joseph-Alfred Serret and its Relation to Perfect Quantum State Transfer.* Preliminary report.

In this talk we recast the Serret theorem about a characterization of palindromic continued fractions in the context of polynomial continued fractions. Then, using the relation between symmetric tridiagonal matrices and polynomial continued fractions we give a quick exposition of the mathematical aspect of the perfect quantum state transfer problem. (Received January 25, 2020)

1156-15-273 **Vladimir Druskin*** (vdruskin@wpi.edu), **Liliana Borcea**, **Alexander Mamonov**, **Mikhail Zaslavsky** and **Jorn Zimmerling**. *Embedding properties of data-driven dissipative reduced order models computed via polynomial recursions.* Preliminary report.

Realizations of reduced order models of passive SISO or MIMO LTE problems can be transformed to tridiagonal and block-tridiagonal forms, respectively, via different modifications of the Stieltjes/Lanczos algorithm. Generally, such realizations can be interpreted as ladder resistor-capacitor-inductor (RCL) networks. In his seminal 1950s works Krein realized that in addition to their compressing properties, network realizations can be used to embed the data back into the state space of the underlying continuum problems. In more recent works of the first author and his collaborators Krein's ideas gave rise to so-called finite-difference Gaussian quadrature rules (FDGQR), allowing to approximately map the data-driven ROM state-space representation to its full order continuum counterpart on a judicially chosen grid. Thus, the state variables can be accessed directly from the transfer function without solving the full problem and even explicit knowledge of the PDE coefficients in the interior, i.e., the FDGQR directly "learns"" the problem from its transfer function. This embedding property found applications in PDE solvers, inverse problems and machine learning. Here we show a generalization of this approach to dissipative systems beyond Stieltjes-Krein theory. (Received January 26, 2020)

16 ► Associative rings and algebras

1156-16-21 **Ajda Fosner** and **Wu Jing*** (wjing@uncfsu.edu), Department of Mathematics & Computer Science, Fayetteville State University, Fayetteville, NC 28301. *Jordan Derivations on Rings.*

We prove that every Jordan derivation on a triangular ring is a derivation. Moreover, we show that, under some conditions, every Jordan derivation on a 2-torsion free ring is a derivation. (Received December 10, 2019)

1156-16-233 **Luigi Ferraro*** (ferrarl@wfu.edu), Winston-Salem, NC 27106. *The Taylor Resolution over a Skew Polynomial Ring.*

Let I be a monomial ideal in the polynomial ring $R = k[x_1, \ldots, x_n]$ over a field k. In her thesis, Taylor introduced a complex which provides a multi-graded free resolution for R/I as an R-module. Later, Gemeda provided a differential graded structure on this complex while Avramov showed that this DG algebra admits a divided power

structure. We generalize these results to monomial ideals J in a skew polynomial ring S. As an application we show that if one fixes the number of generators of the ideal J, then there are finitely many isomorphism classes for $\pi^{\geq 2}(S/J)$, where $\pi(S/J)$ is the homotopy color Lie algebra of S/J, an invariant which was introduced and studied by the first and last author in a different work. As a result it follows that there are finitely many possibilities for the Poincaré series of k over S/J, if the number of generators of J is fixed. (Received January 24, 2020)

1156-16-334 **Alexander Garver*** (garver@umich.edu), **Rebecca Patrias** and **Hugh Thomas**. *Promotion on reverse plane partitions and representations of quivers.*

We start from a Dynkin quiver Q and a choice of minuscule vertex m. There is a bijection between isomorphism classes of representations of Q each whose indecomposable summands is supported at m and reverse plane partitions of whose shape is the minuscule poset corresponding to the chosen vertex m. We study an action on these reverse plane partitions called promotion, which is a sequence of piecewise-linear transformations performed on the entries of a reverse plane partition. Using a modification of the bijection described above we show that the order of this action is the Coxeter number of the associated root system. (Received January 27, 2020)

17 ► Nonassociative rings and algebras

1156-17-27 **Christopher M Drupieski*** (c.drupieski@depaul.edu) and **Jonathan R Kujawa** (kujawa@math.ou.edu). *Support varieties and modules of finite projective dimension for modular Lie superalgebras.*

Let $\mathfrak{g}$ be a finite-dimensional Lie superalgebra over an algebraically closed field k of characteristic $p \geq 3$. Using the cohomology ring of $\mathfrak{g}$, one can define for each finite-dimensional $\mathfrak{g}$-supermodule M a corresponding affine algebraic variety $X_{\mathfrak{g}}(M)$, called the cohomological support variety of M. In this talk I'll discuss recent work with Jonathan Kujawa in which we give an explicit 'rank variety' description for $X_{\mathfrak{g}}(M)$ as a subset of $X_{\mathfrak{g}} = \{x \in \mathfrak{g}_{\overline{1}} : [x, x] = 0\}$, the odd nullcone of $\mathfrak{g}$. A key tool in our approach is the Clifford filtration on $\mathfrak{g}$, which helps us to compare $X_{\mathfrak{g}}(M)$ to the support variety of a certain related unipotent finite supergroup scheme. As corollaries of our main result, I'll discuss a characterization of when a finite-dimensional $\mathfrak{g}$-supermodule is of finite projective dimension, and a positive characteristic analogue of a theorem of Bøgvad concerning a criterion for when the enveloping algebra of $\mathfrak{g}$ is of finite global dimension. (Received December 18, 2019)

1156-17-58 **Yiqiang Li*** (yiqiang@buffalo.edu), 244 Mathematics Building, University at Buffalo, The State University of New York, Buffalo, NY 14051. *Quasi-split symmetric pairs of sl_n and Steinberg varieties of classical type.*

We shall present a Lagrangian construction of the fixed-point subalgebra in a quasi-split symmetric pair of sl_n and its representations via the geometry of the n-step Steinberg varieties of classical type. (Received January 07, 2020)

18 ► Category theory; homological algebra

1156-18-83 **Keller VandeBogert*** (kellerlv@math.sc.edu), LeConte College, 1523 Greene Street, Office 300A, Columbia, SC 29208. *Trimming Complexes and Applications to the Structure of Ideals Defining Compressed Rings.*

In this talk, we introduce so-called Trimming Complexes. Given ideals $I = (\phi_1, \ldots, \phi_n)$ and J of some ring R, these complexes can be used to construct a resolution of the ideal $(\phi_1, \ldots, \phi_{n-1}) + J\phi_n$ (a more general iterated version can also be constructed). In particular, these complexes can be used to deduce the structure of ideals $I \subseteq k[x, y, z]$ defining compressed rings with socle $k(-s)^{\ell} \oplus k(-2s + 1)$ for some integers $s \geq 3$, $\ell \geq 1$. We also discuss additional applications, such as resolving subsets of a generating set for an ideal; in particular, we deduce graded Betti numbers for certain subsets of the generating set of the ideal of maximal minors of a generic matrix. (Received January 13, 2020)

1156-18-180 **Pinches Dirnfeld***, 155 S 1400 E, Room 233, salt lake city, UT 84108. *Base Change Along the Frobenius Endomorphism and the Gorenstein Property.*

Let R be a local ring of positive characteristic and X a complex with nonzero finitely generated homology and finite injective dimension. We prove that if derived base change of X via the Frobenius (or more generally, via a contracting) endomorphism has finite injective dimension then R is Gorenstein. (Received January 22, 2020)

20 ► Group theory and generalizations

1156-20-9 **John C O'Neill***, joneill@siena.edu. *Test elements in the solvable Baumslag–Solitar groups.*

A test element of a group G is an element g with the property that if an endomorphism $\varphi : G \to G$ with $\varphi(g) = g$ then φ is an automorphism. The solvable Baumslag–Solitar Groups are a class of two-generator, one relator groups with presentation $BS_{1,n} = \langle a, t | tat^{-1} = a^n \rangle$ In this talk, normal forms will be established and test elements for this class of groups will be comprehensively classified. In particular, test elements in groups of this type are identified as elements whose total exponent on one of the generators is 0. It will also be shown that Turner's Retract Theorem does not hold for this class of groups. (Received October 16, 2019)

1156-20-32 **Gareth A. Jones*** (g.a.jones@maths.soton.ac.uk), Mathematical Sciences, University of Southampton, Highfield, Southampton, SO17 1BJ, United Kingdom. *A short proof of Greenberg's Theorem.*

In 1960 Greenberg proved that every countable group A is isomorphic to the automorphism group of a non-compact Riemann surface, which can be chosen to have a finitely generated fundamental group if A is finite. His proof is quite complicated, using notions of N-equivalence and N-maximality which he introduced and developed in a separate paper that year. In 1973 he proved that every finite group is isomorphic to the automorphism group of a compact Riemann surface. His proof of this depends on a delicate construction of maximal Fuchsian groups with a given signature. I will give a short and constructive algebraic proof of these results in the case of finitely generated groups A, based on well-known properties of triangle groups and their finite quotient groups. The method is part of a more general construction for realising countable groups as automorphism groups of maps and hypermaps of a given hyperbolic type. (Received December 20, 2019)

1156-20-68 **Yulan Qing*** (yulan.qing@gmail.com), **Kasra Rafi** (rafi@math.toronto.edu) and **Giulio Tiozzo** (tiozzo@math.utoronto.ca). *Boundaries of non-positively curved groups and spaces.*

To every Gromov hyperbolic space X one can associate a space at infinity called the Gromov boundary of X. This boundary is a fundamental tool for studying hyperbolic groups and hyperbolic 3-manifolds. As shown by Gromov, quasi-isometries of hyperbolic metric spaces induce homeomorphisms on their boundaries, thus giving rise to a well-defined notion of the boundary of a hyperbolic group. However, the visual boundary of a CAT(0) space is in general not well-defined. For any sublinear function, we consider a subset of the visual boundary called the sublinearly Morse boundary and show that it is a QI-invariant. In the case of Right-angled Artin group, we show that the Poisson boundary of random walks on groups is naturally identified with the $(\sqrt{t \log t})$-boundary. If time permits we will discuss recent new development of the sublinearly Morse boundary. (Received January 10, 2020)

1156-20-102 **Marston D E Conder*** (m.conder@auckland.ac.nz), Mathematics Department, University of Auckland, 38 Princes Street, Auckland, 1010, New Zealand. *Group actions on pseudo-real surfaces.* Preliminary report.

A Riemann surface is *pseudo-real* if it admits anti-conformal automorphisms, but no anti-conformal automorphism of order 2. In the moduli space of compact Riemann surfaces of given genus, pseudo-real surfaces represent the points that have real moduli, but are not definable over the reals. In this talk I'll describe a number of recent discoveries about pseudo-real surfaces, including analogues of theorems of Wiman and Maclachlan on the maximum order of a cyclic or abelian group of automorphisms. Also I'll discuss the *pseudo-real genus* $\psi(G)$ and the *strong pseudo-real genus* $\psi^*(G)$ of a finite group G. The former is the smallest genus of those pseudo-real surfaces on which G acts faithfully as a group of automorphisms, while the latter is the same but under the assumption that half the elements of G reverse orientation ... when such a surface exists. In particular, I'll show that for every $g > 1$ there is some G with $\psi(G) = \psi^*(G) = g$, so that the range of each of the functions ψ and ψ^* is complete. Finally, I'll give an example of a group G for which $\psi^*(G)$ is defined but $\psi(G) < \psi^*(G)$.

This is joint work with Emilio Bujalance, Javier Cirre and Stephen Lo. (Received January 14, 2020)

1156-20-103 **Marston D E Conder*** (m.conder@auckland.ac.nz), Mathematics Department, University of Auckland, 38 Princes Street, Auckland, 1010, New Zealand. *Chiral polytopes with soluble automorphism group.*

Regular abstract polytopes are relatively well understood and easy to construct. In contrast, chiral polytopes (which admit no reflections but have maximal symmetry by rotation) are much more challenging. In fact it became known only about 10 years ago that chiral polytopes of rank d exist for all $d \geq 3$. Even now, relative

few infinite families of examples of chiral polytopes are known for ranks 4 and higher. Furthermore, many constructions involve simple or almost-simple groups, and yet the vast majority of regular and chiral polytopes of small order have a *soluble* automorphism group. In this talk I will explain how to construct two infinite families of locally toroidal chiral 4-polytopes of type $\{4,4,4\}$, with members of these families having soluble automorphism groups of orders $1024m^2$ and $2048m^2$ for every positive integer m, respectively. Note that these are 2-groups when m is a power of 2.

This is joint work with Yan-Quan Feng and Dong-Dong Hou (Beijing). (Received January 14, 2020)

1156-20-117 **Jane Gilman*** (gilman@rutgers.edu), NJ. *Extending Harvey's Surface Kernel Maps.* Preliminary report.

Let S and S_0 be Riemann surfaces of finite type with S a finite regular branched cover of S_0. Let G be a finite group of conformal automorphisms of S with $S_0 = S/G$. If U denotes the unit disc, let Γ and Γ_0 be uniformizing groups so that $S = U/\Gamma$ and $S_0 = U/\Gamma_0$. There is a group homomorphism of Γ_0 onto G with kernel Γ and this is termed a *surface kernel map*. Two surface kernel maps are equivalent if they differ by an automorphism of Γ_0. In his 1971 paper Bill Harvey showed that when G is a cyclic group, there is a unique simplest representative for this equivalence class. His result has played an important role in establishing subsequent results about conformal automorphism groups of surfaces. We extend his result to surface kernels of maps onto arbitrary finite groups. (Received January 16, 2020)

1156-20-126 **Rose Morris-Wright*** (rmorriswright@brandeis.edu), MA. *An analog of the curve complex in FC type Artin groups.*

The curve complex is an important geometric construction for studying the mapping class group of a topological surface. This talk will construct an analogous simplicial complex for Artin groups of FC type. Artin groups are a generalization of braid groups that provide a rich field of examples and counter-examples for many algebraic, geometric, and topological properties. The complex of parabolic subgroups generalizes the curve complex for braid groups and the extension graph for right-angled Artin groups. I will define the complex of parabolic subgroups, as well as discuss properties that the curve complex and the complex of parabolic subgroups share. (Received January 18, 2020)

1156-20-128 **Jon F. Carlson**, **Jesper Grodal**, **Nadia Mazza** and **Daniel K. Nakano*** (nakano@math.uga.edu). *Torsion Free Endotrivial Modules for Finite Groups of Lie Type.*

In this talk I will give a description of the torsion free rank for the endotrivial group for finite groups of Lie type, and explain how important structural results for simple algebraic groups are utilized in this process. The results encompass endotrivial modules in both the describing and non-describing characteristic cases. As a byproduct, the number of components of the orbit space for the poset of elementary abelian subgroups of rank at least two in a finite group of Lie type is described. (Received January 18, 2020)

1156-20-137 **David Vogan*** (dav@math.mit.edu). *What's special about special?* Preliminary report.

Around 1978, Lusztig's investigations of the representations of finite Chevalley groups led him to single out a class of Weyl group representations he called *special*. The corresponding set of unipotent classes is also called *special*. These unipotent classes arise also in representation theory over any local field. The main result over $\mathbb{R}$ says that *any representation of a real reductive group having integral infinitesimal character gives rise to a special unipotent class*.

In 2018 Meinolf Geck gave a conjectural characterization of special unipotent classes by an integrality criterion, which I will recall. Geck's conjecture was established in 2019 by Junbin Dong and Gao Yang. The point of this talk is to pose the question: can the result above (connecting integral representation theory over $\mathbb{R}$ to special unipotent elements) be directly established using Geck's conjectural characterization of special? (Received January 20, 2020)

1156-20-159 **Thomas Tucker*** (ttucker@colgate.edu), 406 Williston Rd, Box 163, Sagamore Beach, MA 02562, and **Marie Carvacho**, **Jennifer Paulhus** and **Aaron Wootton**. *Finite groups acting with almost all signatures.* Preliminary report.

A finite group G acts preserving orientation on a Riemann surface with signature $(g; m_1, \dots m_k)$, if there is a set of generators $a_1, b_1, \dots a_g, b_g \in G$ and $c_1, \dots c_k \in G$, where c_j has order m_j, such that $\Pi_{i=1}^{g}[a_i, b_i]\Pi_j c_j = id_G$. We say G acts with *almost all signatures* (AAS) if, when we restrict $m_j \in \mathcal{O}(G)$ (the orders of elements of G), there is an action of G for all but finitely many such signatures. We have shown that G is AAS if and only G is generated by elements of order m and $[G, G]$ contains an element of order m, for all $m \in \mathcal{O}(G)$. It follows that all simple groups are AAS and that every AAS group is either perfect or a p-group for some prime p. There are

AAS groups that are not simple or p-groups, and the direct sum of n copies of an AAS group is AAS. In this talk, we consider similar concepts for orientation-reversing actions or actions on non-orientable surfaces (NEC group quotients). Since most such actions require $|G|$ to be even, the only possibilities are 2-groups and perfect groups. (Received January 25, 2020)

1156-20-163 **Ignat Soroko** (ignat.soroko@gmail.com). *Ranks of joins and intersections in free groups.* The famous Hanna Neumann Conjecture (now the Friedman–Mineyev Theorem) stipulates that for the ranks of arbitrary subgroups H and K of a nonabelian free group we have:

$$\operatorname{rank} H \cap K - 1 \leq (\operatorname{rank} H - 1)(\operatorname{rank} K - 1).$$

It is an interesting open question to quantify this bound with respect to the rank of $H \vee K$, the subgroup generated by H and K. We describe a set of realizable values $(\operatorname{rank} H \vee K, \operatorname{rank} H \cap K)$ for arbitrary H, K, and conjecture that this locus is complete. Using the approach developed by Dicks in the context of his Amalgamated Graph Conjecture, we prove that the locus is complete when $\operatorname{rank} H = 2$ and show that the region

$$\operatorname{rank} H \vee K \geq \operatorname{rank} H + \operatorname{rank} K - 3 \quad \& \quad \operatorname{rank} H \cap K \geq 4$$

consists of non-realizable values, thus resolving the remaining open case of R. Guzman's conjecture in the affirmative and obtaining applications to 3-dimensional topology. (Received January 28, 2020)

1156-20-168 **Coy L. May** and **Jay Zimmerman*** (jzimmerman@towson.edu). *The Largest Group Actions on Riemann Surfaces of Genus g.*

A natural problem is to determine, for each value of the integer $g \geq 2$, the largest order of a group that acts on a Riemann surface of genus g. First, let $N(g)$ (respectively $M(g)$) be the largest order of a group of automorphisms of a Riemann surface of genus $g \geq 2$ preserving the orientation (respectively possibly reversing the orientation) of the surface.

The basic inequalities comparing $N(g)$ and $M(g)$ are $N(g) \leq M(g) \leq 2N(g)$. It is easy to see that the upper bound is satisfied for an infinite number of integers g. We also show that there are an infinite number of positive integers g, such that $N(g) = M(g)$.

Specifically, if p is a prime satisfying $p \equiv 1 \pmod 6$ and $g = 3p + 1$, there is a group of order $24(g - 1)$ that acts on a surface of genus g preserving the orientation of the surface. For all such values of $g = 3p + 1$ larger than a fixed constant, there are no groups with order larger than $24(g - 1)$ that act on a surface of genus g. (Received January 22, 2020)

1156-20-173 **Thomas Tucker*** (ttucker@colgate.edu), 406 Williston Rd, Box 163, Sagamore Beach, MA 02562, and **Marston Conder**, **Coy May** and **Jay Zimmerman**. *Group Order Density of Triangle Group Quotients.* Preliminary report.

A set S of non-negative integers has *density* d if $f(n)/n \to d$ where $f(n) = |\{s \in S : s \leq n\}|$. Using a theorem of Bertram (1976), we show that for any r, s, t, the set of orders of all finite triangle group quotients $\{x, y : x^r = y^s = (xy)^t, \dots\}$ have density 0. It follows that the orders of groups G such that $0 < \sigma^o(G) - 1 < |G|/12$ have density 0, where $\sigma^o(G)$ is the smallest genus of a closed orientable surface on which G acts non-trivially preserving orientation. This generalizes a result of May and Zimmerman [2019] for groups of odd order. On the other hand, for odd n, the groups $G = D_n \times C_4$ have $0 < \sigma^o(G) - 1 < |G|/8$ (again by May and Zimmerman [2003]) and the density of their orders $8n$ is non-zero. (Received January 22, 2020)

1156-20-176 **Hoang Thanh Nguyen** and **Hung Cong Tran*** (hung.c.tran-1@ou.edu), 1111 Oak Tree Ave, Apt #321, Norman, OK 73019, and **Wenyuan Yang**. *Strong quasiconvexity and almost malnormality.*

Strong quasiconvexity generalizes quasiconvexity in hyperbolic groups and it is preserved under quasi-isometry. It is well-known that quasiconvex subgroups of hyperbolic groups are finitely generated and have finite height (a generalization of almost malnormality). I prove that strongly quasiconvex subgroups of arbitrary finitely generated groups are also finitely generated and have finite height. Therefore, it may be interesting to ask if a finitely generated, finite height (or even almost malnormal) subgroup is strongly quasiconvex. I will talk about the answer to this question in 3–manifold groups. A part of my talk is joint work with Hoang Thanh Nguyen and Wenyuan Yang. (Received January 22, 2020)

1156-20-215 **Francesca Bartolucci** and **Filippo De Mari*** (demari@dima.unige.it), Dipartimento di Matematica, via Dodecaneso, 45, Genova, Italy, and **Matteo Monti**. *Unitarization of the horocyclic Radon transform on homogeneous trees.*

Let X be a q-homogeneous tree and Ξ be the set of horocycles. The classical horocyclic Radon transform $\mathcal{R}$ maps $C_C(X)$ into the space $L^2(\Xi)^\flat$ of the square integrable functions on Ξ whose average on the boundary is

even. We exhibit the existence of a pseudo-differential operator Λ such that $\Lambda\mathcal{R}$ extends to a unitary operator $\mathcal{Q}$ from $L^2(X)$ onto $L^2(\Xi)^\flat$. Then $\mathcal{Q}$ intertwines the quasi-regular representations of $\mathrm{Aut}(X)$ on $L^2(X)$ and $L^2(\Xi)^\flat$. (Received January 24, 2020)

1156-20-217 **Christopher P. Bendel**, **Daniel K. Nakano**, **Cornelius Pillen** and **Paul Sobaje*** (psobaje@georgiasouthern.edu). *New results on the Tilting Module Conjecture.*

Let G be a simple and simply connected algebraic group over a field of characteristic $p > 0$. It has become clear that answers to various classical problems about G-modules are dependent upon a better understanding of the indecomposable tilting modules for G whose highest weight is in the "p-restricted region above the Steinberg weight." Recently, the authors found an example of such a tilting module that was not indecomposable over the Lie algebra of G, contradicting a longstanding conjecture of Donkin. In this talk, we will discuss continued work on this problem, including new cases in which the authors have verified Donkin's conjecture. (Received January 24, 2020)

1156-20-232 **Peter A Brooksbank*** (pbrooksb@bucknell.edu), Department of Mathematics, 1 Dent Drive, Lewisburg, PA 17837. *Computing with groups generated by involutions.*

Efforts to classify certain types of incidence structures—and to discover new families of them—often rely heavily on computer searches. My own work has led to the creation and accumulation of code that somewhat recently coalesced into a fledgling software package SGGI (String Groups Generated by Involutions) written for the Magma system. In this talk I will discuss features of this package, compare it to existing methods in Magma, and outline some problems I would like to use the package to investigate. Also, since my outlook thus far has been very narrow, I am especially interested to hear what additional functionality would be useful to others. (Received January 24, 2020)

1156-20-327 **Carolyn Abbott*** (abbott@math.columbia.edu), **Nick Miller** and **Priyam Patel**. *Actions of big mapping class groups on the arc complex.*

Given a finite-type surface (i.e., one with finitely generated fundamental group), there are two important objects naturally associated to it: a group, called the mapping class group, and an infinite-diameter hyperbolic graph, called the curve graph. The mapping class group acts by isometries on the curve graph, and this action has been extremely useful in understanding the algebraic and geometric properties of mapping class groups. One particularly important class of elements of the mapping class group are those which act as loxodromic isometries of the curve graph; these are called "pseudo Anosov" elements. Given an infinite-type surface with an isolated puncture, one can associate two analogous objects: the so-called big mapping class group, and the (relative) arc graph. In this talk, we will consider the action of big mapping class groups on the arc graph, and, in particular, we will construct an infinite family of "infinite-type" elements that act as loxodromic isometries of the arc graph, where an infinite-type element is one which is not supported on any finite-type subsurface. This is joint work with Nick Miller and Priyam Patel. (Received January 27, 2020)

1156-20-337 **Corey Bregman*** (cbregman@brandeis.edu), Goldsmith 218 MS050, 415 South Street, Waltham, MA 02453. *Isometries of CAT(0) Cube Complexes and Rank Rigidity.*

We study the isometries of CAT(0) cube complexes, and in particular, those which are not cubical automorphisms. Given a CAT(0) cube complex X, we prove that its isometry group coincides with its automorphism group unless X decomposes as a product with a (topological) R^n, for some n. As an application, if X is 1-ended, cocompact and hyperbolic, then $\mathrm{Aut}(X) = \mathrm{Isom}(X)$ unless X is quasi-isometric to the hyperbolic plane. We also extend the rank rigidity result of Caprace-Sageev to the full isometry group. (Received January 27, 2020)

1156-20-346 **Emily Stark*** (stark@math.utah.edu) and **Daniel J. Woodhouse**. *Action rigidity for free products of hyperbolic manifold groups.*

The relationship between the large-scale geometry of a group and its algebraic structure can be studied via three notions: a group's quasi-isometry class, a group's abstract commensurability class, and geometric actions on proper geodesic metric spaces. A common model geometry for groups G and G' is a proper geodesic metric space on which G and G' act geometrically. A group G is action rigid if every group G' that has a common model geometry with G is abstractly commensurable to G. For example, a closed hyperbolic n-manifold group is not action rigid for all n at least three. In contrast, we show that free products of closed hyperbolic manifold groups are action rigid. Consequently, we obtain the first examples of Gromov hyperbolic groups that are quasi-isometric but do not virtually have a common model geometry. This is joint work with Daniel Woodhouse. (Received January 27, 2020)

1156-20-347 **Brendan Burns Healy*** (healyb@uwm.edu) and **G Chris Hruska**. *Model Spaces for Relatively Hyperbolic Pairs.*

Relatively hyperbolic pairs are groups with preferred peripheral subgroups and are meant to generalize the behavior of non-uniform lattices in rank one symmetric spaces of noncompact type. While in geometric actions the preferred hyperbolic space is well-defined up to quasi-isometry, cusp-uniform actions by relatively hyperbolic pairs require two choices to be made in order to determine a QI-class of hyperbolic space. We examine the symmetric space case for the motivation behind the choice of a preferred type of space and prove these model spaces exist and are uniquely determined. In doing this we examine different kinds of horospherical geometry, prove internal geometry conditions sufficient for uniform perfection of the boundary of a hyperbolic space when acted on cusp-uniformly, note the connection between space quasi-isometries and boundary quasi-symmetries, and demonstrate existence of some classes of cusp-uniform actions on non-model spaces. (Received January 27, 2020)

1156-20-385 **Tim Susse*** (tsusse@simons-rock.edu), Great Barrington, MA 01230, **Jason Behrstock**, New York, NY , and **Victor Falgas-Ravry**, Umea, Sweden. *Sharp thresholds for random right-angled Coxeter groups.*

The presentation for a right-angled Coxeter group is given bya graph: vertices correspond to generating reflections, and two such reflections commute if and only if the corresponding vertices are joined by an edge. This encoding allows us to use random graphs to generate random right-angled Coxeter groups, and study their geometric and algebraic properties.

In this talk, we will focus on the Erdos–Renyi model of random graphs, and ask when the corresponding random right-angled Coxeter group has quadratic divergence. In particular, using work of Dani, Thomas and Levcovitz, we will connect this question to percolation techniques developed by Bollobas and Riordan, and determine a sharp threshold for these phenomena. (Received January 28, 2020)

1156-20-394 **Radhika Gupta**, **Kasia Jankiewicz** and **Thomas Ng*** (thomas.ng@temple.edu). *Quantifying the strong Tits alternative for groups acting on nonpositively curved spaces.* Preliminary report.

Groups that satisfy the strong Tits alternative contain subgroups that are either virtually abelian or have exponential growth. Exponential growth of groups can be controlled by efficiently constructing free semigroups. I will discuss how nonpositive curvature can be leveraged to give new bounds on exponential growth. Specifically, I will describe joint work with Radhika Gupta and Kasia Jankiewicz where we use these ideas to show uniform exponential growth of many groups acting on CAT(0) cube complexes. (Received January 28, 2020)

1156-20-407 **Francesco Lucente Stabile*** (f_lucentestabile@salemstate.edu), **Carey Patrick Atkins** (c_atkins1@salemstate.edu) and **Arthur James Rosenthal** (arosenthal@salemstate.edu). *A new asymmetric encryption algorithm involving both group and number theory: Derivation of the Lucente Stabile Atkins Cryptosystem using Gauss's Generalization of Wilson's Theorem.*

Our research led to the discovery of an asymmetric encryption algorithm that follows Kerckhoff's principle, relies on a specific case of Gauss's Generalization of Wilson's Theorem and involves both group and number theory. Unlike prime factorization based algorithms, the eavesdropping cryptanalyst has no indication that he has successfully decrypted the cyphertext. For this reason, we aim to show that this algorithm is not only more secure than existing asymmetric algorithms, but it has the potential to be significantly computationally faster. (Received January 28, 2020)

1156-20-408 **Carey Patrick Atkins*** (c_atkins1@salemstate.edu), MA, and **Francesco Lucente Stabile** (f_lucentestabile@salemstate.edu) and **Arthur James Rosenthal** (arosenthal@salemstate.edu). *Practical issues involved in implementing the Lucente Stabile Atkins (LSA) Cryptosystem.*

We discuss practical issues involved in implementing the Lucente Stabile Atkins (LSA) cryptosystem, a new asymmetric encryption algorithm that involves both group and number theory. The LSA algorithm requires a key exchange at the beginning of the process to define the group that will be used for information sharing. We show that the LSA cryptosystem can be implemented with an algorithm that generates an arbitrarily large number of keys of various lengths using a computationally efficient method. By using this list of keys to determine the cyclic group to be used, we have developed a cryptosystem that can be more secure than existing encryption algorithms while also being computationally faster. (Received January 28, 2020)

22 ► Topological groups, Lie groups

1156-22-89 **Sigurdur Helgason*** (helgason@mit.edu), 5 Benton Road, Belmont, 02478. *New Spherical Functions on noncompact Symmetris Spaces.*

Let X = G/K be a symmetric space of the noncompact type, d an irreducible representation of K on a space H(d). We call a function F on X a spherical function of type d if it is an eigenfunction of each G-invariant differential operator on X, has values in Hom(H(d), H(d)), and satisfies F(k.x) = d(k) F(x) for all x in X, k in K. If d = 1 these F are the zonal spherical functions. An example for F is the Eisenstein integral of type d. We prove that each F above is related to the Eisenstein integral and can be expressed as a derivative of the zonal spherical function by means of a certain differential operator on X, which is right invariant when lifted to G. We also prove a series expansion for F involving new c functions C(s) for each s in the Weyl group of X. They are meromorphic and have values in Hom(H(d), H(d)). For X of rank 1 these C(s) are explicitly determined and the asymptotic behavior of the F above is established in terms of these C(s). For X of higher rank, we determine the determinant of the C(s) as well as their Hilbert Schmidt norm. These results are in continuation of my book Geometric Analysis on Symmetric Spaces (AMS 2008). They are related to work of Harish Chandra and joint work of K. Johnson and N. Wallach. (Received January 13, 2020)

1156-22-92 **Vignon Oussa*** (voussa@bridgew.edu), Bridgewater, MA 02325, and **Jordy Timo Van Velthoven** (jordy-timo.van-velthoven@univie.ac.at), Vienna, Austria. *Integrability of relative discrete series representations of solvable Lie groups.*

It is known that for simply connected nilpotent Lie groups, unitary irreducible representations, which are square-integrable-modulo the center are also integrable modulo the center. The main aim of this talk is to present an extension of this result to the class exponential solvable Lie groups. (Received January 14, 2020)

1156-22-135 **Raul Gomez*** (raul.gomez.rgm@gmail.com), Av. Universidad s/n. Ciudad Universitaria, 66451 San Nicolas de los G, Nuevo Leon, Mexico, and **Birgit Speh** (bes12@cornell.edu), 10 Malott Hall, Cornell University, Ithaca, NY 14853. *Invariant Trilinear forms on induced representations of $PGL(2,\mathbb{R})$.*

The study of invariant trilinear forms on induced representations of $PGL(2,\mathbb{R})$ started with the classical work of Bernstein and Zelevinsky and has received continuous attention and generalizations during the years following its publication. In this talk we will describe a new technique, developed in joint work with Birgit Speh, that allows the use of homological algebra and the theory of Schwartz functions on Nash manifolds to simplify the computations and provide a complete answer. (Received January 20, 2020)

1156-22-141 **Tian An Wong*** (twong33@smith.edu). *Basic functions and the stable trace formula.* Preliminary report.

I will report on work in progress establishing a stable trace formula whose discrete spectral terms are weighted by automorphic L-functions, as suggested by Langlands in his Beyond Endoscopy proposal. This involves extending the results of Finis, Lapid, and Müller on the continuity of Arthur's noninvariant trace formula to the stable trace formula, while incorporating the use of basic functions. (Received January 20, 2020)

1156-22-150 **Pierre Clare*** (prclare@wm.edu), 200 Ukrop Way, Williamsburg, VA 23185. *Essential representations of real reductive groups.*

The tempered dual of a real reductive group G equipped with the Fell topology is in general a non-Hausdorff space. From the point of view of Noncommutative Geometry, it is natural to consider the associated reduced C*-algebra $C_r^*(G)$ and study its K-theory. The Connes-Kasparov isomorphism relates the K-theory to the ring of representations of a maximal compact subgroup of G and specifies a certain class of representations of G, called *essential*. In this talk, we will discuss these representations in terms of R-groups and Dirac cohomology. This is joint work with Nigel Higson, Yanli Song and Xiang Tang. (Received January 21, 2020)

1156-22-164 **Daniel Orr*** (dorr@vt.edu). *Equivariant K-theory of the semi-infinite flag manifold as a nil-DAHA module.*

The equivariant K-theory of the semi-infinite flag manifold, as developed recently by Kato, Naito, and Sagaki, carries commuting actions of the nil-double affine Hecke algebra (nil-DAHA) and a q-Heisenberg algebra. The action of the latter is free of rank $|W|$, where W is the (finite) Weyl group, which enables one to express the nil-DAHA action in terms of $W \times W$ matrices over the q-Heisenberg algebra. We give an explicit algebraic construction of these matrices as a limit from the full DAHA in simply-laced type. Our construction reveals that multiplication by equivariant scalars, when expressed in terms of the Heisenberg algebra, is given by the nonsymmetric q-Toda system introduced by Cherednik and the author. (Received January 22, 2020)

1156-22-179 **Job Kuit*** (jobkuit@math.upb.de), Institute of Mathematics, Paderborn University, Warburger Strasse 100, 33098 Paderborn, Germany. *Discrete series representations for real spherical spaces.* Preliminary report.

Let $Z = G/H$ be a homogeneous space attached to a real reductive group G and a closed subgroup H. A principal objective in the harmonic analysis of Z is the understanding of the G-equivariant spectral decomposition of the space $L^2(Z)$ of square integrable half-densities. The irreducible components of $L^2(Z)$ are of particular interest, they comprise the discrete series for Z. In this talk I will focus on real spherical homogeneous spaces Z and present some recent results. (Received January 22, 2020)

1156-22-196 **Gestur Olafsson*** (olafsson@math.lsu.edu), Baton Rouge, LA 70803, and **Robert J. Stanton**. *Extension of representations, the real case.*

Assume that G/H is a non-compactly causal symmetric space. Then there exists a maximal parabolic subgroup $P = LAN \subset G$ such that $H/H \cap L \subset G/P$ has a realization as a real symmetric bounded domain. In this talk we discuss the following:

(1) Extension of finite dimensional representations of $H \cap K$ to L and hence extension of homogeneous vector bundles over $H/H \cap K$ to G/P

(2) We introduce the real crown, or Matsuki crown, of $H/H \cap L$ and extend the Krötz-Stanton Theorem of holomorphic extension of matrix coefficients to the real crown.

The talk is based on a joint project with R. Stanton. (Received January 23, 2020)

1156-22-200 **Nolan R. Wallach*** (nwallach@ucsd.edu), Department of Mathematics, University of Califormia, San Diiego, 9500 Gilman Dr., La Jolla, CA 92093. *The inverse functor to the K-finite functor and a remarkable family of universal Harish-Chandra modules.* Preliminary report.

The Casselman-Wallach (CW) theorem asserts that the K-finite functor is an isomorphism of categories between the category of finitely generated, admissible, smooth, Fréchet modules of moderate growth and the category of Harish-Chandra modules of a real reductive group (with a chosen maximal compact subgroup, K). We describe the inverse functor and show that it takes analytic families to continuous families. For the case of one dimensional parameter space this result follows from a theorem of Van der Noort. Our proof uses some of Van der Noort's methods and a class of "universal" Harish-Chandra modules with remarkable analytic and algebraic properties. Special cases of these modules played an important role in our proof of the CW theorem for non-linear groups and in our paper with J-S Huang and T. Oshima. These modules will be studied and a sketch of the proof of the continuity theorem will be given. (Received January 23, 2020)

1156-22-245 **Leonardo C. Mihalcea** (lmihalce@math.vt.edu), Department of Mathematics, Virginia Tech University, Blacksburg, VA 24061, and **Changjian Su*** (changjiansu@gmail.com), Toronto, ON M5S 2E4, Canada. *Whittaker functions from motivic Chern classes.*

We prove a 'motivic' analog of the Weyl character formula, computing the Euler characteristic of a line bundle on a flag manifold G/B multiplied either by a motivic Chern class of a Schubert cell, or a Segre analog of it. The result, given in terms of Demazure-Lusztig (D-L) operators, recovers formulas found by Brubaker, Bump and Licata for the Iwahori-Whittaker functions of the principal series representation of a p-adic group. In particular, we obtain a new proof of the classical Casselman-Shalika formula for the spherical Whittaker function. Joint work with Leonardo C. Mihalcea. (Received January 25, 2020)

1156-22-296 **Henrik Schlichtkrull***, Dept of Math, University of Copenhagen, Universitetsparken 5, DK2100 Copenhagen Ø, Denmark. *Harmonic analysis on real spherical spaces.*

Let G be a real reductive Lie group. A homogeneous space Z of G is called real spherical if a minimal parabolic subgroup of G possesses an open orbit on Z. For example, the Bruhat decomposition of G implies that $Z = G$ is real spherical for the two-sided action of $G \times G$. More generally, all symmetric spaces of G are real spherical. A survey will be given on progress in the direction of a generalization to real spherical spaces of Harish-Chandra's Plancherel theory for G. The main results presented are joint work with Delorme, Knop and Krötz. (Received January 27, 2020)

1156-22-328 **Sean M Taylor*** (smtaylor@shc.edu) and **Matthew Glenn Dawson** (matthew.dawson@cimat.mx). *Frechet sheaves and representation theory.* Preliminary report.

In this talk, we present a beginning of a theory of Frechet sheaves. This combination of functional analysis and sheaf theory is desirable to be able to fully connect the "algebraic" and "analytic" approaches to representation theory of semisimple groups. (Received January 27, 2020)

1156-22-406 **Carl Mautner*** (carlm@ucr.edu). *Nearby cycles to the enhanced nilpotent cone.* Preliminary report.

Let $\mathcal{N}$ be the nilpotent cone of $GL(V)$, where V is an n-dimensional complex vector space. I will discuss nearby cycles and a conjecture of Achar on the structure of the category of $GL(V)$-equivariant perverse sheaves on the enhanced nilpotent cone $\mathcal{N} \times V$. This is based on conversations with Achar and Riche. (Received January 28, 2020)

1156-22-426 **J. Matthew Douglass*** (mdouglas@nsf.gov). *Equivariant coherent sheaves on a point and Kazhdan-Lusztig bases.*

Suppose G is a connected, reductive, complex, algebraic group with Lie algebra $\mathfrak{g}$. In this talk I will discuss (1) a refined version of a question of Ostrik that describes the $G \times \mathbb{C}^*$-equivariant coherent sheaves on the nilpotent cone of $\mathfrak{g}$ that are supported at $\{0\}$ in terms of the Kazhdan-Lusztig basis of the extended, affine Hecke algebra, and (2) how this description is connected with the computation of the $G \times \mathbb{C}^*$-equivariant K-theory of some generalized Steinberg varieties. (Received January 29, 2020)

26 ► Real functions

1156-26-386 **Ryan Gibara*** (ryan.gibara@concordia.ca) and **Riikka Korte**. *Accessible parts of the boundary for domains in metric measure spaces.*

In the setting of Q–Ahlfors regular PI–spaces, we prove that if a domain Ω has uniformly large boundary when measured with respect to the s–dimensional Hausdorff content, then its visible boundary has large t–dimensional Hausdorff content for every $0 < t < s \leq Q - 1$. The visible boundary is the set of points that can be reached by a John curve from a fixed point $z_0 \in \Omega$. This generalizes recent results by Koskela-Nandi-Nicolau (in $\mathbb{R}^2$) and Azzam (in $\mathbb{R}^n$). In particular, our approach shows that the phenomenon is independent of the linear structure of the space. (Received January 28, 2020)

28 ► Measure and integration

1156-28-222 **Jerez Chen*** (yc85@rice.edu). *Optimal Stationary One-dimensional Structures.*

Given an equilibrated force system F of compact support, we investigate whether there exists a lowest cost structure made by given materials that balances the force system. Michell's model considers finitely many bars under extension and bars under compression. Such a bar corresponds to an interval in Euclidean space with a real multiplicity where the absolute value of the multiplicity indicates the stress on the material and the sign of the multiplicity indicates whether it is a bar under extension or compression. Given such a system, however, optimal structures may not be achieved by finite structures. The connection with geometric measure theory is that the notion of extension stress in Michell's model coincides with the first variation of mass, and the compression stress coincides with the negative first variation. We show the general existence of an optimal structure using varifolds, as studied in geometric measure theory. A further goal is to describe the regularity of minimizers. (Received January 24, 2020)

30 ► Functions of a complex variable

1156-30-52 **Eriko Hironaka*** (ehironaka@amsbooks.org). *Augmented deformation space of rational maps.*

We discuss questions surrounding connectivity of deformation space and augmented deformation space as subsets of the Teichmueller and Augmented Teichmueller spaces of rational maps of a fixed topological type. (Received January 06, 2020)

1156-30-279 **Emilio Bujalance**, **Javier Cirre*** (jcirre@mat.uned.es) and **Jesús Rodríguez**. *Abelian group actions on pseudo-real Riemann surfaces.*

A compact Riemann surface is called pseudo-real if it admits anti-conformal (orientation-reversing) automorphisms, but no anti-conformal automorphism of order two. In the talk we will describe necessary and sufficient conditions for a finite abelian group to act as a group of automorphisms containing orientation reversing elements, on some pseudo-real surface of genus bigger than one. Some applications, as the minimum genus and

maximum order problems, will also be considered. The main tool is the combinatorial theory of non-euclidean crystallographic groups. (Received January 26, 2020)

32 ► Several complex variables and analytic spaces

1156-32-142 **Sudeb Mitra*** (sudeb.mitra@qc.cuny.edu), Department of Mathematics, Queens College, CUNY, 65-30 Kissena Blvd, Flushing, NY 11367. *Real-analytic sections for Teichmüller curves.* Preliminary report.

In an important paper, Clifford J. Earle showed that Teichmüller spaces (in general) do not admit holomorphic sections. Later, in his famous thesis, John H. Hubbard showed that Teichmüller curves (in general) do not admit holomorphic sections. In our talk, we will use Douady-Earle sections for Teichmüller spaces to study real-analytic sections for Teichmüller curves. (Received January 20, 2020)

1156-32-154 **Allen Broughton** and **Antonio F. Costa*** (acosta@mat.uned.es), Calle Senda del Rey, 9, Facultad de Ciencias, UNED, 28040 Madrid, Madrid, Spain, and **Milagros Izquierdo**. *One Dimensional Equisymmetric Strata in Moduli Spaces of Riemann Surfaces.*

The Moduli Space admits an equisymmetric stratification by irreducible complex algebraic varieties consisting in surfaces whose automorphism groups have topologically equivalent actions. We describe explicitly those strata of dimension one, as punctured Riemann surfaces in terms of the action data of the automorphism group. (Received January 21, 2020)

33 ► Special functions

1156-33-285 **Vladimir Dragovic*** (vladimir.dragovic@utdallas.edu), 800 W Campbell Road, FO 35, Richardson, TX 75080. *Extremal polynomials, integrable billiards, and Painleve VI equations.*

A comprehensive study of periodic trajectories of the billiards within ellipsoids in the d-dimensional Euclidean space is presented. The novelty of the approach is based on a relationship established between the periodic billiard trajectories and the extremal polynomials of the Chebyshev type on the systems of d intervals on the real line. A new combinatorial object, billiard partitions are introduced and studied. For d=2, the dynamics of extremal polynomials is related to Painleve VI equations. Some results are obtained jointly with George E. Andrews, some with Milena Radnovic, and some with Vasilisa Shramchenko. (Received January 26, 2020)

1156-33-289 **Philip J. Feinsilver*** (phfeins@siu.edu), 2107 Sunset Court, Carbondale, IL 62901. *Canonical Polynomial Sequences: Inverse Pairs.*

For $V(z)$, analytic in a neighborhood of $0 \in \mathbb{C}$, $V(0) = 0$, there is an associated sequence of polynomials, *canonical polynomials*, that is a generalized Appell sequence with lowering operator $V(d/dx)$. Correspondingly, the inverse function $Z(v)$ has an associated canonical polynomial sequence. The coefficients of the two sequences form two mutually inverse infinite matrices. We detail the operator calculus for the pair of systems, illustrating the duality of operators and variables between them. Examples are presented including a connection between Gegenbauer and Bessel polynomials.Touchard polynomials appear as well. Alternative settings would include an umbral calculus approach as well as the Riordan group. Here we use the approach through operator calculus. (Received January 26, 2020)

1156-33-356 **Howard S. Cohl*** (howard.cohl@nist.gov), Mission Viejo, CA 92694, and **Roberto S. Costas-Santos** (rscosa@gmail.com), Alcala de Henares, Spain. *Basic hypergeometric transformations from symmetric and q-inverse sub-families of the Askey-Wilson polynomials in the q-Askey scheme.*

By starting with known basic hypergeometric representations of the Askey-Wilson polynomials, we derive basic hypergeometric representations for the q-inverse Askey-Wilson polynomials and a set of sub-families for both of these these poynomials. These sub-families are obtained by repeatedly setting one of the free parameters (not q) equal to zero until no parameters are left. These subfamilies, are the continuous dual q-Hahn, Al-Salam-Chihara, continuous big q-Hermite, the continuous q-Hermite polynomials, and their q-inverse analogues. From the basic hypergeometric representations of these polynomials, and due to symmetry in their free parameters, and under the mapping $\theta \mapsto -\theta$, we are able to derive terminating basic hypergeometric transformation formulae which these polynomials satisfy. We also derive non-terminating basic hypergeometric transformations from the

Ramanujan and Ismail functions which appear in the Plancherel-Rotach asymptotics of Al-Salam-Chihara and continuous q-Hermite polynomials. (Received January 27, 2020)

34 ► Ordinary differential equations

1156-34-175 **W Riley Casper***, Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803, and **Milen Yakimov**. *The Matrix Bochner Problem.*

We present a solution of the matrix Bochner problem, a long-standing open problem in the theory of orthogonal polynomials, with applications to diverse areas of research including representation theory, random matrices, spectral theory, and integrable systems. Our solution is based on ideas applied by Krichever, Mumford, Wilson and others, wherein the algebraic structure of an algebra of differential operators influences the values of the operators in the algebra. By using a similar idea, we convert the matrix Bochner problem to one about noncommutative algebras of GK dimension 1 which are module finite over their centers. Then the problem is resolved using the representation theory of these algebras. (Received January 22, 2020)

35 ► Partial differential equations

1156-35-98 **Dehua Wang*** (dwang@math.pitt.edu). *Global Solutions for the Active Hydrodynamics.*

The active hydrodynamics arising in biology/biophysics is described by the Q-tensor liquid crystal framework. The global weak solutions with large initial data will be discussed. (Received January 14, 2020)

1156-35-99 **Nathan Glatt-Holtz***, Tulane University, New Orleans, LA 70118. *A Bayesian Approach to Quantifying Uncertainty in Divergence Free Flows.*

We treat the statistical regularization of the ill-posed inverse problem of estimating a divergence free flow field $\mathbf{u}$ from the partial and noisy observation of a passive scalar θ. Our solution is a Bayesian posterior distribution, that is a probability measure μ which precisely quantifies uncertainties in $\mathbf{u}$ once one specifies models for measurement error and a prior knowledge for $\mathbf{u}$. We present some of our recent work which analyzes μ both analytically and numerically. In particular we discuss a posterior contraction (consistency) result as well as some Markov Chain Monte Carlo (MCMC) algorithms which we have developed and refined and rigorously analyzed to effectively sample from μ. This is joint work with Jeff Borggaard, Justin Krometis and Cecilia Mondaini. (Received January 14, 2020)

1156-35-108 **David Isaacson*** (isaacd@rpi.edu), Rensselaer Polytechnic Institute, Mathematics Department, Troy, NY 12180. *Electrical Impedance Tomography.*

Electrical Impedance Tomography (EIT) systems apply patterns of currents to the surface of a body and measure the resulting patterns of voltages. From this data they reconstruct approximations to the electrical conductivity inside the body. This is of interest in monitoring heart and lung function because as blood leaves the heart its conductivity decreases, and as it enters the lungs their conductivity increases. For this reason, impedance images of the inside of the chest show how the lungs are being perfused in real time. Mathematically, reconstructing the interior conductivity is called the Calderon problem and is an inverse boundary value problem for a low frequency approximation to Maxwell's equations, where one approximates the coefficients in the conductivity equation from the Neumann to Dirichlet map on the boundary. This problem, and why the spectral properties of the N to D map played a significant role in the design of our adaptive current tomography systems will be explained. Movies will be shown. (Received January 15, 2020)

1156-35-124 **Brian J Choi*** (choigh@bu.edu), 111 Sciarappa St, APT 6, Cambridge, MA 02141. *Fourth-Order Perturbation of NLS.*

We discuss the adiabatic limit of the Quantum Zakharov System on the real line and the 1-D torus for square-integrable initial data. We show that as the quantum parameter goes to zero, we obtain the NLS limit when the time element is restricted to a compact interval. When the periodic boundary condition is imposed, we prove L^4 and L^6 versions of modified Strichartz estimates. These estimates have applications in showing that certain sub-critical non-linear equations are well-posed at low regularities. (Received January 17, 2020)

1156-35-156 **Yernat Assylbekov** and **Ting Zhou*** (t.zhou@notheastern.edu). *Direct and Inverse Problems for Nonlinear Time-harmonic Maxwell's Equations.*

We consider an inverse boundary value problem of electromagnetism in a nonlinear Kerr medium or with the second harmonic generation nonlinearity. We show the unique determination of the electromagnetic material

parameters and the nonlinear susceptibility parameters of the medium by making electromagnetic measurements on the boundary. We are interested in the case of the time-harmonic Maxwell equations. (Received January 21, 2020)

1156-35-185 **Jongkeun Choi**, **Hongjie Dong** and **Zongyuan Li*** (zongyuan_li@brown.edu), 182 George Street, Division of Applied Mathematics, Brown University, Box F, Providence, RI 02912. *Mixed boundary value problem on irregular domains.*

We consider the mixed Dirichlet-conormal problem on irregular domains in $\mathbb{R}^d$. Two types of regularity results will be discussed: the $W^{1,p}$-regularity and a non-tangential maximal function estimate. The domain is assumed to be Reifenberg-flat, and the interfacial boundary is either Reifenberg-flat of co-dimension 2 or is locally sufficiently close to a Lipschitz function of m variables, where $m = 1, \dots, d-2$. For the non-tangential maximal function estimate, we also require the domain to be Lipschitz. This is based on a joint work with J. Choi and H. Dong. (Received January 22, 2020)

1156-35-187 **Wenjie Lu*** (lu000005@umn.edu), 206 Church St SE, Minneapolis, MN 55455. *Local well-posedness of the De Gregorio modification of the Constantin-Lax-Majda model in a critical space.*

In this paper, we prove the local-in-time well-posedness of the De Gregorio modification of the Constantin-Lax-Majda model in the Besov space $B^{\frac{1}{2}}_{2,1}(\mathbb{S}^1)$, which is critical under the natural scaling of the model. (Received January 23, 2020)

1156-35-263 **Damir Kinzebulatov*** (damir.kinzebulatov@mat.ulaval.ca), **Yuli A Semenov** (semenov.yu.a@gmail.com) and **Karol Szczypkowski** (karol.szczypkowski@pwr.edu.pl). *Heat kernel of fractional Laplacian with Hardy drift via desingularizing weights.*

We establish sharp two-sided bounds on the heat kernel of a semigroup that is not ultracontractive (due to presence of a critical-order singularity in the drift term) by transferring it to an appropriate weighted space where it becomes ultracontractive. (Received January 26, 2020)

1156-35-283 **Jamil Chaker*** (jchaker@math.uchicago.edu), University of Chicago, Mathematics Department, Chicago, IL 60637, and **Luis Silvestre**. *Coercivity estimates for integro-differential operators.*

In this talk, we provide a general condition on the kernel of an integro-differential operator such its associated quadratic form satisfies a coercivity estimate in terms of fractional Sobolev norms:

$$\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} |u(x)-u(y)|^2 K(x,y)\,dx\,dy \geq c\|u\|^2_{H^s}.$$

This talk is based on a joint work with Luis Silvestre. (Received January 26, 2020)

1156-35-304 **Daniela De Silva*** (desilva@math.columbia.edu). *A viscosity approach to the regularity of variational problems.*

In this talk we discuss some extensions of the classical Krylov-Safonov Harnack inequality. After reviewing the standard regularity theory, we will introduce a weaker notion of viscosity solutions. The novelty is that we consider functions that do not necessarily satisfy an infinitesimal equation but rather exhibit a two-scale behavior. Roughly, our viscosity solutions satisfy comparison in a neighborhood of a touching point whose size depends on the properties of the test functions. As an application, we recover the $C^{1,\alpha}$ estimates of Almgren and Tamanini for quasi-minimizers of the perimeter functional. Further applications will be also discussed. This is also the core idea to obtain regularity of the free boundary for almost minimizers of one-phase type problems. (Received January 27, 2020)

1156-35-316 **Matias G. Delgadino**, **Xukai Yan*** (xukai.yan@math.gatech.edu) and **Yao Yao**. *Uniqueness and non-uniqueness of steady states of aggregation-diffusion equations.*

In this talk, I will discuss a nonlocal aggregation equation with degenerate diffusion, which describes the mean-field limit of interacting particles driven by nonlocal interactions and localized repulsion. When the interaction potential is attractive, it is previously known that all stationary solutions must be radially decreasing up to a translation, but uniqueness (for a given mass) within this class was open, except for some special interaction potentials. For general attractive potentials, we show that the uniqueness/non-uniqueness criteria are determined by the power of the degenerate diffusion, with the critical power being $m = 2$. In the case $m \geq 2$, we show the stationary solution for any given mass is unique for any attractive potential, by tracking the associated energy functional along a novel interpolation curve. In the case $1 < m < 2$, we construct examples of smooth

attractive potentials, such that there are infinitely many radially decreasing stationary solutions of the same mass. (Received January 27, 2020)

1156-35-317 **Geng Chen***, Department of Mathematics. *A Lipschitz Finsler type optimal transport metric for nonlinear wave equations.*

In this talk, I will discuss the Lipschitz Finsler type optimal transport metric for some quasilinear wave equations. The solutions of these equations in general include cusp singularity. (Received January 27, 2020)

1156-35-321 **Jiawei Li*** (jiaweil4@andrew.cmu.edu) and **Yu Gu**. *Fluctuations of a non-linear stochastic heat equation in dimensions three and higher.*

We study the solution to the non-linear stochastic heat equation

$$\partial_t u = \Delta u + \beta\sigma(u)\dot{W}_\phi(t,x),$$

in $d \geq 3$ with constant initial date, where σ is a Lipschitz function and $\beta > 0$. The equation is driven by a Gaussian multiplicative noise $\dot{W}_\phi$ that is white in time and smooth in space. In a weak disorder regime, we prove that the solution converges to the stationary distribution in large time and the diffusive scale fluctuations are described by the Edwards-Wilkinson equation. (Received January 27, 2020)

1156-35-349 **P. Robert Kotiuga***, 8 Saint Mary's Street, Boston, MA 02215. *Ramifications, old and new, of the curl Eigenvalue problem.* Preliminary report.

The operator d* acts on coclosed 1-forms on a compact 3-dimensional manifold with boundary, and when subject to suitable boundary conditions, it is self-adjoint (S-A). When the formalism of differential forms on a Riemannian manifold is restricted to 3-d Euclidian space E^3, this operator becomes the curl operator of Gibbsian vector analysis. As a S-A operator, its spectral theory has a surprisingly diverse set of applications. In E^3 the eigenforms are known as Beltrami flows; they characterize force-free magnetic fields. Such eigenforms are also computed on $SU(2)$, the conformal compactification of E^3. The spectral theory also arises on 3-d cocompact double coset spaces such as $\Gamma\backslash SL(2,\mathbb{R})$ or $\Gamma\backslash SL(2,\mathbb{C})/SU(2)$, where Γ is a discrete group arising in analytic number theory. The present work considers the curl EVP on these constant curvature spaces, the relationship to analytic torsion, and a seven dimensional analog of the force-free magnetic field problem involving the exceptional Lie group G_2. (Received January 27, 2020)

1156-35-376 **Punit Gandhi**, **Sarah Iams*** (siams@seas.harvard.edu) and **Mary Silber**. *A fast-slow switching model of banded vegetation pattern formation in drylands.*

From infiltration of water into soil during rainstorms to seasonal plant growth and death, ecohydrological processes thought to be relevant to the formation of banded vegetation patterns in drylands occur across multiple timescales. We propose a fast-slow switching model to capture these processes on appropriate timescales within a modeling framework based on reaction-advection-diffusion equations. The fast system captures hydrological processes that occur on the minute to hour timescales associated with major rainstorms, assuming a fixed vegetation distribution. The slow system acts between rain events, on a timescale of days to months, and evolves vegetation and soil moisture. Modeling processes at the appropriate timescales reduces the number of parameters that are chosen to fit pattern characteristics, and eliminates the artificial slowing of fast processes required to align their timescales with the biomass dynamics. We explore the fast-slow switching model through numerical simulation on a one-dimensional hillslope, finding agreement with observations of band spacing and upslope colonization rates. This fast-slow model framework introduces a tool for investigating the possible impact of changes to frequency and intensity of rain events in dryland ecosystems. (Received January 28, 2020)

1156-35-388 **Jiayi Wang*** (jw9yj@virginia.edu), charlottesville, VA 22903. *Balance of the vorticity direction and the vorticity magnitude in 3D fractional Navier-Stokes equations.*

Fractional Navier-Stokes equations–featuring a fractional Laplacian–provide a 'bridge' between the Euler equations (zero diffusion) and the Navier-Stokes equations (full diffusion). The problem of whether an initially smooth flow can spontaneously develop a singularity is a fundamental problem in mathematical physics, open for the full range of models–from Euler to Navier-Stokes. The purpose of this work is to present a hybrid, geometric-analytic regularity criterion for solutions to the 3D fractional Navier-Stokes equations expressed as a balance–in the average sense–between the vorticity direction and the vorticity magnitude, key geometric and analytic descriptors of the flow, respectively. (Received January 28, 2020)

1156-35-402 **Samuel J Ferguson*** (sjf370@nyu.edu). *Stochastic Hilbert's 19th Problem in Homogenization.*

After providing physical intuition from the random conductance model, we introduce homogenization of PDE arising from the Lagrangians of Hilbert's 19th problem. We then focus on Lagrangians that are sampled by

probability distributions. Although small-scale regularity estimates for solutions fail in general, we prove large-scale regularity estimates for solutions. Moreover, based on the idea that linearization and homogenization should commute, we prove regularity for the homogenization of the stochastic Lagrangian under suitable hypotheses. This gives a result analogous to the solution of Hilbert's 19th problem. (Received January 28, 2020)

37 ► Dynamical systems and ergodic theory

1156-37-15 **Partick Ingram**, Department of Mathematics and Statistics, York University, N520 Ross, 4700 Keele Street, Toronto, Ottowa ON M3J 1P3, Canada, **Rohini Ramadas**, Mathematics Department, Box 1917, Brown University, Providence, RI 02912, and **Joseph H Silverman*** (jhs@math.brown.edu), Mathematics Department, Box 1917, Brown University, 151 Thayer Street, Providence, RI 02912. *Post-Critically Finite Maps on $\mathbb{P}^n$ for $n \geq 2$ are Sparse.*

Let $f : \mathbb{P}^n \to \mathbb{P}^n$ be a morphism of degree $d \geq 2$. The map f is said to be *post-critically finite* (PCF) if there exist integers $k \geq 1$ and $\ell \geq 0$ such that the critical locus Crit_f satisfies $f^{k+\ell}(\mathrm{Crit}_f) \subseteq f^{\ell}(\mathrm{Crit}_f)$. The smallest such ℓ is called the *tail-length*. We discuss the proof that for $d \geq 3$ and $n \geq 2$, the set of PCF maps f with tail-length at most 2 is not Zariski dense in the the parameter space of all such maps. In particular, maps with periodic critical loci, i.e., with $\ell = 0$, are not Zariski dense. (Received December 02, 2019)

1156-37-38 **Enrique Pujals***, 365 5th Ave, New York, NY, NY 10016. *Renormalization of Henon maps.*

For surface diffeomorphisms, a strictly positive entropy is associated with the existence of 'horseshoes'. I will focus on surface diffeomorphisms with zero entropy: can the dynamics of these 'simple' systems be described? how does it bifurcate to positive entropy systems?

These questions will be discussed for a class of volume-contracting surface diffeomorphisms whose dynamics is intermediate between one-dimensional dynamics and general surface dynamics, named mild dissipative. It includes the dynamics of any Henon diffeomorphism with Jacobian smaller than 1/4.

With S. Crovisier and C. Tresser we show that any mild dissipative diffeomorphisms of the disk (that includes the Henon up to Jacobian 1/4) with zero entropy can be renormalized. As a consequence, we obtained a two-dimensional version of Sharkovsky's theorem about the set of periods of interval maps. (Received December 26, 2019)

1156-37-45 **David Aulicino***, Room 1156, Ingersoll Hall, 2900 Bedford Avenue, Brooklyn, NY 11210-2889, and **Artur Avila** and **Vincent Delecroix**. *Weak Mixing for Translation Surfaces with Intermediate Orbit Closures.* Preliminary report.

Work of Avila and Forni established weak mixing for the generic straight line flow on generic translation surfaces, and the work of Avila and Delecroix determined when weak mixing occurs for the straight line flow on a Veech surface. Following the work of Eskin, Mirzakhani, Mohammadi, which proved that the orbit closure of every translation surface has a very nice structure, one can ask how the orbit closure affects the weak mixing of the straight line flow. In this talk all of the necessary background on translation surfaces and weak mixing will be presented followed by the answer to this question. This is joint work with Artur Avila and Vincent Delecroix. (Received January 02, 2020)

1156-37-47 **Benjamin Call**, **Daniel J Thompson*** (thompson@math.osu.edu) and **Tianyu Wang**. *The K-property and a Lindeberg CLT for equilibrium states in non-positive curvature.*

Equilibrium states for geodesic flows over compact rank 1 manifolds and sufficiently regular potential functions were studied by Burns, Climenhaga, Fisher and myself. We showed that if the higher rank set does not carry full topological pressure then the equilibrium state is unique. In this talk, I will describe joint work with Ben Call, which shows that these equilibrium states have the Kolmogorov property. If time permits, I will also describe a new result with Tianyu Wang, which shows that certain sequences of measures supported on collections of regular closed geodesics asymptotically satisfy a type of Central Limit Theorem. This extends ideas of Denker-Senti-Zhang who recently obtained analogous results in a uniform discrete-time setting. (Received January 04, 2020)

1156-37-48 **Victor Donnay*** (vdonnay@brynmawr.edu), PA , and **Daniel Visscher** (dvisscher@ithaca.edu). *A new proof of the existence of embedded surfaces with Anosov geodesic flow.*

We give a new proof of the existence of compact surfaces embedded in R^3 with Anosov geodesic flows. This proof starts with a non-compact model surface whose geodesic flow is shown to be Anosov using a uniformly strictly invariant cone condition. Using a sequence of explicit maps based on the standard torus embedding, we produce compact embedded surfaces that can be seen as small perturbations of the Anosov model system and hence are themselves Anosov. We are now working to quantify this approach, with the aim of giving a non-optimal lower bound on the genus for which a surface supports an isometrically embedded Anosov geodesic flow. Preliminary estimates put the genus at roughly 10^9. (Received January 05, 2020)

1156-37-59 **Diana Davis*** (dianajdavis@gmail.com), Swarthmore, PA 19081, and **Samuel Lelièvre**. *Periodic paths on the pentagon.*

This talk is accessible to graduate students and advanced undergraduates.

Mathematicians have long understood periodic billiard trajectories on the square billiard table, which occur when the slope of the trajectory is rational. In this talk, I'll explain my work on periodic trajectories on the regular pentagon, describing their geometry, symbolic dynamics, and group structure. The periodic trajectories are very beautiful, and some of them exhibit a surprising "dense but not equidistributed" behavior. There will be lots of pictures. This is joint work with Samuel Lelièvre. It is possible that, by the time of the talk, I will also be able to report on periodic paths on the heptagon and nonagon. (Received January 20, 2020)

1156-37-63 **Jing Zhou*** (jingzhou@umd.edu), Department of Mathematics, University of Maryland, 4176 Campus Drive, College Park, MD 20742. *A Rectangular Billiard with Moving Platforms.*

We describe an exponential Fermi accelerator in a two-dimensional billiard with moving platforms. We have found a mechanism of trapping regions which provides the exponential acceleration for almost all initial conditions with sufficiently high initial energy. Under an additional hyperbolicity assumption, we estimate the waiting time after which most high-energy orbits start to gain energy exponentially fast. (Received January 09, 2020)

1156-37-64 **Paul Apisa***, 10 Hillhouse Ave., Yale University, New Haven, CT 06511, and **Alex Wright**. *Large orbit closures in Teichmueller dynamics are trivial!*

The moduli space of translation surfaces admits a GL(2, R) action coming from Teichmueller geodesic flow and scalar multiplication. This action can be described concretely as the action of GL(2, R) on polygons in the plane. By work of McMullen in genus two and Eskin, Mirzakhani, and Mohammadi in general, every GL(2, R) orbit closure is, up to orbifold singularities, a linear manifold. In this talk I will begin by describing the GL(2, R) action on moduli space and the aforementioned orbit closure theorem. In the second half, I will describe joint work in progress with Alex Wright showing that when the dimension of an orbit closure is sufficiently large, i.e. larger than approximately half the genus of the underlying surface, then the orbit closure is a locus of branched covers of lower genus surfaces. In other words, large orbit closures in Teichmueller dynamics only exist for topological reasons. (Received January 09, 2020)

1156-37-65 **Aimee S Johnson*** (aimee@swarthmore.edu), 500 College Ave, Department of Mathematics and Statistics, Swarthmore, PA 19081, and **Van Cyr**, **Bryna Kra** and **Ayse Sahin**. *Low Complexity and Loosely Bernoulli Dynamical Systems.* Preliminary report.

We investigate the connections between an $\mathbb{Z}$-action ergodic dynamical system being loosely Bernoulli and having low complexity. (Received January 09, 2020)

1156-37-75 **Thomas Sharland***, Department of Mathematics, Lippitt Hall, University of Rhode Island, Kingston, RI 02881. *Matings of (cubic) polynomials.*

Informally, the mating of polynomials provides a way of combining the dynamics of two polynomials on the complex plane to construct a rational map on the Riemann sphere, unless the mating is obstructed. We discuss the theory of polynomial matings, with particular focus on the cubic case where both polynomials have a fixed critical point. Along the way, we will draw parallels with the (far better understood) quadratic case, and show that the only possible irreducible obstruction in the cubic case with a pair of fixed critical points is a Levy cycle: a periodic cycle of simple closed curves under the mating. We also describe how these Levy cycles arise from the combinatorics of the two polynomials. (Received January 11, 2020)

1156-37-77 **Mark F Demers***, 1073 North Benson Road, Fairfield, CT 06824. *Projective Cones for Dispersing Billiards.*

We construct a version of projective cones on which the transfer operator corresponding to a finite horizon Sinai billiard map acts as a strict contraction. By proving explicit estimates on the diameter of the cone, we obtain precise control over the rate of convergence to equilibrium for the billiard map. This technique has a wide variety of applications, including to sequential billiards, open billiards with relatively large holes, and some models with random configurations of scatterers. This is joint work with Carlangelo Liverani. (Received January 12, 2020)

1156-37-78 **Mark F Demers***, 1073 North Benson Road, Fairfield, CT 06824. *Unique measure of maximal entropy for the finite horizon periodic Lorentz gas.*

While the existence and properties of the SRB measure for the billiard map associated with a periodic Lorentz gas are well understood, there are few results regarding other types of measures for dispersing billiards. We begin by proposing a naive definition of topological entropy for the billiard map, and show that it is equivalent to several classical definitions. We then prove a variational principle for the topological entropy and proceed to construct a unique probability measure which achieves the maximum. This measure is Bernoulli and positive on open sets. An essential ingredient is a proof of the absolute continuity of the unstable foliation with respect to the measure of maximal entropy. This is joint work with Viviane Baladi. (Received January 12, 2020)

1156-37-87 **Jasmine Powell*** (jtpowell@umich.edu). *Families of rational maps and complex tori.*

Local dynamical properties of certain families of rational maps give rise to a correspondence between a map and a complex torus. The parameter spaces of these rational maps have interesting topology. We discuss how topological questions about these parameter spaces can inform geometric questions about the associated tori, and vice versa. (Received January 13, 2020)

1156-37-88 **Ethan Farber*** (farbere@bc.edu). *From expanders to pseudo-Anosovs and flat surfaces: creating 2-dimensional dynamics from one-dimensional systems.* Preliminary report.

For a dynamicist, the most interesting surface homeomorphisms are pseudo-Anosovs. These can be thought of as linear models for isotopic homeomorphisms, since a given pseudo-Anosov comes with coordinates with respect to which it is locally affine. An analogous phenomenon occurs in the setting of self-maps of a compact interval, wherein suitable endomorphisms admit a semiconjugacy to a so-called "uniform expander," which is piecewise linear with slope of constant magnitude. In this talk we describe a procedure by which one can lift a uniform expander satisfying certain properties to a pseudo-Anosov on some closed surface. This construction immediately provides coordinates with respect to which the pseudo-Anosov may be realized as a piecewise affine map on $\mathbb{R}^2$, and whose limit set is naturally identified with a the hyperelliptic quotient of the surface, allowing us to easily endow the surface with a flat structure. This construction was first given by Thurston. We examine properties of these pseudo-Anosovs and the surfaces they live on, giving a characterization of such pairs that can be constructed in this way. (Received January 13, 2020)

1156-37-91 **Spencer Dowdall** and **Grace Work*** (gwork@mit.edu). *Discretely shrinking targets in moduli space.*

Consider a nested family of target sets shrinking at a specified rate and a given flow, we want to understand the set of points whose orbits hit these targets infinitely often under the flow. One way to examine this set is to determine under what conditions this set has full measure. This question is closely related to the Borel-Cantelli Lemma and also gives rise to logarithm laws. In joint work with Spencer Dowdall we examine this particular question for Teichmuller flow on the moduli space of unit-area quadratic differentials. (Received January 14, 2020)

1156-37-93 **W. Patrick Hooper***, Department of Mathematics, City College of New York, 160 Convent Ave, New York, NY 10031, and **Anna Tao**, Department of Mathematics, Brown University, Box 1917, 151 Thayer Street, Providence, RI 02912. *A family of infinite IETs related to an irrational rotation.* Preliminary report.

We will consider a parameter space of infinite IETs with flips. I will give an example of an infinite IET with flips which is measurably conjugate to an irrational rotation of the circle by a measurable map whose graph is self-similar. We will show that under certain perturbations of this IET, a bifurcation occurs producing a dense full-measure collection of periodic points, but the dynamics on the aperiodic set remains conjugate to the same irrational rotation. We will consider the degree to which systems arising from this bifurcation are stable. (Received January 14, 2020)

1156-37-112 **Terrence Adams** (csilva@williams.edu) and **Cesar E Silva*** (csilva@williams.edu). *Constructions of infinite measure-preserving and nonsingular actions satisfying weak mixing type properties.* Preliminary report.

We will discuss infinite measure-preserving and nonsingular transformations and actions satisfying various properties related to weak mixing. The talk will include a brief survey of such constructions. (Received January 15, 2020)

1156-37-114 **Jonathan Chaika** and **Giovanni Forni***, University of Maryland, College Park, MD 20742. *There exists a weakly mixing polygonal billiard.*

The main result of this talk is that there exists a Gdelta dense set of polygons with weakly mixing billiard flow with respect to the Liouville measure (on the unit tangent bundle to the billiard table). Kerchoff, Masur and Smillie proved in 1986 there exists a Gdelta dense set of polygons with ergodic billiard flow by a fast approximation, Baire category based on the unique ergodicity of arbitrary rational billiards. Our result follows in a similar way from showing that for any translation surface the product of the flows in almost every pair of directions is ergodic with respect to Lebesgue measure. This in turn is proven by showing that for every translation surface the flows in almost every pair of directions do not share non-trivial common eigenvalues. This is joint work with Jon Chaika. (Received January 15, 2020)

1156-37-115 **Vincent R Martinez***, 695 Park Ave, Department of Mathematics & Statistics, New York, NY 10065. *Unique ergodicity and exponential mixing for the damped-driven stochastic KdV equation.*

We discuss a proof for the uniqueness and regularity of the invariant measure for the damped-driven stochastic Korteweg-de Vries equation, where the noise is additive and sufficiently non-degenerate. It is shown that a simple, but versatile control strategy, typically employed to establish exponential mixing for strongly dissipative systems, can nevertheless be applied in this weakly dissipative setting to obtain unique ergodicity, albeit without mixing rates. Under the assumption of large damping, however, exponential mixing can be recovered. Time permitting, application of this approach to other weakly dissipative systems, such as the damped-driven Nonlinear Schrodinger equation will also be discussed. This is joint work with Nathan Glatt-Holtz and Geordie Richards. (Received January 15, 2020)

1156-37-161 **Jason P Bell**, **Jeffrey Diller** and **Mattias Jonsson*** (mattiasj@umich.edu). *Transcendental dynamical degrees.*

Dynamical degrees are fundamental invariants in algebraic dynamics. I will present joint work with Jason Bell and Jeffrey Diller in which we construct rational selfmaps of the projective plane for which the (first) dynamical degree is a transcendental number. (Received January 22, 2020)

1156-37-223 **Enrique R Pujals***, 365 5th Ave, New York, NY, NY 10016. *Fifty years of the stability conjecture.*

For a dynamicist, stability means in loose terms that the relevant dynamical behavior of a system is actually the same, in some appropriate sense, for all near-by systems. Stable systems have a central role in dynamics, and the stability conjecture formulated by Palis and Smale fifty-two years ago proposes a characterization of them.

We will discuss the meaning of that conjecture, its significance and its current status. (Received January 24, 2020)

1156-37-225 **Kurt W Vinhage*** (kwv104@psu.edu), **Daren Wei** and **Adam Kanigowski**. *Kakutani Equivalence of Unipotent Flows.*

Kakutani equivalence is a notion of equivalence which is stronger than orbit equivalence but weaker than measurable conjugacy. Ratner developed an entropy-like invariant of Kakutani equivalence to exhibit two homogeneous flows which were not Kakutani equivalent. We compute this invariant for every unipotent flow on compact homogeneous spaces of semisimple groups. The answer is strikingly different from the case of nilflows. We also compute the slow entropy, introduced by Katok and Thouvenot, for such flows. Joint with Adam Kanigowski and Daren Wei. (Received January 24, 2020)

1156-37-240 **Christopher Bose** (cbose@uvic.ca), **Anthonu Quas** (aquas@uvic.ca) and **Matteo Tanzi*** (mt3986@nyu.edu), 14 Washington Place, Unit 5H, New York, NY 10003. *Limit laws for random LSV maps sampled on any bounded interval in parameter space.*

Liverani-Saussol-Vaienti maps form a family of piecewise differentiable dynamical systems of the unit interval depending on one positive real parameter. The maps are everywhere expanding apart from a neutral fixed point and it is well known that, depending on the amount of expansion close to this point, they have either

an absolutely continuous invariant measure and polynomial decay of correlations, or a unique physical measure that is singular and concentrated at the neutral point. I will present results on the composition of LSV maps whose parameters are randomly sampled so that the two contrasting behaviours are mixed. We will show that if the parameters for which there is an absolutely continuous measure are sampled with positive probability, then suitably rescaled Birkhoff averages converge to limit laws. (Received January 25, 2020)

1156-37-251 **Tanya Firsova*** (tanyaf@math.ksu.edu), **Nikita Selinger** and **Jeremy Kahn**.
Deformation Spaces of Rational Maps.

Building on the work of W.Thurston on postcritically finite maps, A.Epstein introduced deformation spaces into holomorphic dynamics. He suggested them as an approach to study varieties defined by postcritically finite relations. We study topological properties of deformation spaces. We will discuss theorems and conjectures on the boundary behavior of the deformation spaces. In particular, when the behavior can be described in terms of postcrtically finite maps. This is joint work with J.Kahn and N.Selinger. (Received January 25, 2020)

1156-37-276 **Kasun Fernando*** (kasun.akurugodage@utoronto.ca) and **Françoise Pène** (francoise.pene@univ-brest.fr). *Edgeworth Expansions for Hyperbolic Dynamical Systems.*

Given a dynamical system which shows hyperbolicity on a large part of phase space, one would expect it to exhibit good statistical properties like rapid decay of correlations, the Central Limit Theorem, Large Deviation Principle and etc. In this talk, I will discuss sufficient conditions for such mostly hyperbolic dynamical systems to admit Edgeworth expansions in the Central Limit Theorem. Our focus is on systems that admit a Young tower with return times with an exponentially decaying tail. (Received January 26, 2020)

1156-37-280 **Thomas Silverman*** (tjsilver@umich.edu). *Type I repelling density for non-archimedean polynomials.* Preliminary report.

One of the earliest and most fundamental theorems in one-variable complex dynamics is that the repelling periodic points are dense in the Julia set. It is a theorem of Rivera-Letelier that the same result holds for all Berkovich repelling periodic points in the non-archimedean setting. However, it remains an open question whether the classical or type I repelling periodic points are always dense in Julia set of a non-archimedean rational function. I will discuss a proof of the polynomial case and some applications. (Received January 26, 2020)

1156-37-308 **Maxwell Musser*** (maxwell.musser@yu.edu), **Marian Gidea** (marian.gidea@yu.edu) and **Rafael de la Llave** (rll6@math.gatech.edu). *Effect of non-conservative perturbations on homoclinic and heteroclinic orbits.*

The motivation of this work comes from astrodynamics. Consider a spacecraft traveling between the Earth and the Moon. Assume that the spacecraft follows a zero-cost orbit by coasting along the hyperbolic invariant manifolds associated to periodic orbits near the equilibrium points, at some fixed energy level. We would like to make a maneuver – impulsive or low thrust in order to jump to the hyperbolic invariant manifold corresponding to a different energy level. Mathematically, turning on the thrusters amounts to a adding a small, non-conservative, time-dependent perturbation to the original system. Given such an explicit perturbation, we would like to estimate its effect on the orbit of the spacecraft.

We study this question in the context of two simple models: the pendulum-rotator system, and the planar circular restricted three-body problem. Homoclinic/heteroclinic excursions can be described via the scattering map, which gives the future asymptotics of an orbit as a function of the past asymptotics. We add a time-dependent, non-conservative perturbation, and provide explicit formulas, in terms of convergent integrals, for the perturbed scattering map. This is based on joint work with M. Gidea and R. de la Llave. (Received January 27, 2020)

1156-37-418 **Alba M Málaga Sabogal*** (alba_malaga_sabogal@brown.edu) and **Serge Troubetzkoy**. *Generic dynamics of the Ehrenfest wind-tree model.*

The Ehrenfest wind-tree model is a billiard played in the complement of some number of square-shaped scatterers on the Euclidean plane. This talk will present the set of wind-tree configurations and inside it, G_δ dense sets of wind-tree configurations with nice dynamics (minimality, ergodicity, etc). Similar (Received January 28, 2020)

41 ► Approximations and expansions

1156-41-177 **Jens Christensen*** (jchristensen@colgate.edu) and **Gestur Olafsson**. *Atomic decomposition of Bergman spaces on bounded symmetric domains.* Preliminary report.

We present new atomic decompositions of Bergman spaces on bounded symmetric domains $D = G/K$. The construction relies on coorbit/wavelet theory for the holomorphic discrete series representations of the simple group G, and the method provides a wealth of new atoms compared to earlier work by Coifman and Rochberg. (Received January 22, 2020)

42 ► Fourier analysis

1156-42-84 **Dongmian Zou*** (dzou@umn.edu) and **Gilad Lerman**. *Scattering Transform on Graphs.*

Modern data are usually high-dimensional with noise and corruption. A useful representation of data has to be robust and address the data structure. In this talk, I will present a class of robust models called the scattering transform that can be used to generated features from graph data. In graph scattering transforms, the representation is generated in an unsupervised manner based on graph wavelets. It is approximately invariant to permutations and stable to signal or graph manipulations. Numerical results show that it works effectively for classification and community detection problems. (Received January 13, 2020)

1156-42-133 **Michael Perlmutter*** (perlmut6@msu.edu), **Feng Gao**, **Guy Wolf** and **Matthew Hirn**. *Scattering Neural Networks on Graphs and Manifolds.*

The scattering transform is a mathematical model of convolutional neural networks (CNNs) introduced for functions defined on Euclidean space by Stephané Mallat. It differs from traditional CNNs by using predesigned, wavelet filters rather than filters which are learned from training data. This leads to a network which provably has desirable mathematical properties such as translation invariance and diffeomorphism stability. Moreover, in situations where the wavelets can be designed in correspondence to underlying physics, it can produce numerical results which rival state of the art CNNs.

However, many data sets of interest have an intrinsically non-Euclidean structure and are better modeled as graphs or manifolds. This motivates us to construct geometric versions of the scattering transform using the spectral decompostitions of Laplace-Beltrami operator and Graph Laplacian. We will discuss applications of these networks to a variety of geometric deep learning tasks and show that, analogously to their Euclidean predecessor, the geometric scattering transforms possess desirable invariance and stability properties. (Received January 20, 2020)

1156-42-184 **Kasso Okoudjou*** (okoudjou@umd.edu). *Sobolev orthogonal polynomials on fractals.* Preliminary report.

In the first part of the talk, we shall introduce the analogs of the Legendre orthogonal polynomials (OP) on the Sierpinski gasket (SG). After reviewing some of the elementary properties of these OPs, we will present some preliminary work on a theory of Sobolev OPs on SG. The talk is based on a joint work with M. Jiang, T. Lan, S. Sule, R. Strichartz, S. Venkat, and X. Wang. (Received January 22, 2020)

1156-42-241 **Benoit Florent Sehba*** (bfsehba@ug.edu.gh), Department of Mathematics, University of Ghana, Accra, Great Accr LG 62, Ghana. *A generalization of Carleson lemma and weighted norm inequalities for the maximal functions in the Orliz settings.* Preliminary report.

The Carleson's (embedding) lemma asserts the following: Let σ be a weight on $\mathbb{R}^d$, and $\alpha \geq 1$. Assume $\{\lambda_Q\}_{Q\in\mathcal{D}}$ is a sequence of positive numbers indexed over the set of dyadic cubes $\mathcal{D}$ in $\mathbb{R}^d$. Then the following are equivalent.

(i) There exists a constant $A > 0$ such that for any cube $R \in \mathcal{D}$,
$$\sum_{Q\subseteq R, Q\in\mathcal{D}} \lambda_Q \leq A(\sigma(R))^\alpha.$$

(ii) There exists a constant $B > 0$ such that for all $p \in (1, \infty)$,
$$\sum_{Q\in\mathcal{D}} \lambda_Q |m_\sigma(f, Q)|^{p\alpha} \leq B \left(\int_{\mathbb{R}^d} |f(x)|^p \sigma(x) dx \right)^\alpha,$$
where $m_\sigma(f, Q) = \frac{1}{\sigma(Q)} \int_Q f(t)\sigma(t)dt$. Provided the left handside is finite.

In this talk, I will present an extension of the above result when the function $t \to t^p$ and $t \to t^{p\alpha}$ are replaced by Young functions of an appropriate subclass. I will then provide applications to weighted integral inequalities and weighted norm estimates for the fractional maximal function.
This a joint work with J. M. Tanoh Dje (Abidjan) and Justin Feuto (Abidjan). (Received January 25, 2020)

43 ► Abstract harmonic analysis

1156-43-165 **Koichi Kaizuka*** (k-kaizu@nms.ac.jp), 1-7-1 Kyounan-cho, Musashino-shi, Tokyo 180-0023, Japan. *Scattering theory for the Laplacian on symmetric spaces of noncompact type and its application.*

We develop the scattering theory for the Laplacian on symmetric spaces of noncompact type based on detailed analysis for the Helgason Fourier transform and the elementary spherical function. In the scattering theory, asymptotic analysis for the resolvent of the Laplacian plays an essential role and yields a characterization of solutions to the Helmholtz equation. Since a higher rank symmetric space of noncompact type has a certain singular geometric structure at infinity, unlike in the case of the Euclidean and hyperbolic spaces, asymptotic behavior of the resolvent kernel of the Laplacian strongly depends on spherical coordinates in the higher rank case. This causes an essential difficulty in establishing asymptotic analysis for the resolvent. We overcome the difficulty by introducing a certain type of a face decomposition of the Weyl chamber. Furthermore, we give a uniform weighted estimate for the solutions to the Helmholtz equation. As an application of our scattering theory, we prove a conjecture by R. S. Strichartz (J. Funct. Anal. (1989)) concerning a characterization of a family of generalized eigenfunctions of the Laplacian. (Received January 22, 2020)

1156-43-207 **Hiroshi Oda** (hoda@la.takushoku-u.ac.jp) and **Nobukazu Shimeno*** (shimeno@kwansei.ac.jp). *Spherical functions for small K-types.*

Let G be a noncompact connected real semisimple Lie group with finite center and K a maximal compact subgroup. Wallach introduced a class of K-types called small. We classify small K-types for all simple Lie groups and prove except one small K-type for a split real Lie group of type G_2 that each elementary spherical function for each small K-type π can be expressed as a product of hyperbolic cosines and a Heckman-Opdam hypergeometric function. As an application, the inversion formula for the π-spherical transform is obtained from Opdam's theory on hypergeometric Fourier transforms. We also give the inversion formula for the π-spherical transform for each small K-type of a split real Lie group of type G_2 using a different method.

[1] H. Oda and N. Shimeno, *Spherical functions for small K-types* in *Geometric and Harmonic Analysis on Homogeneous Spaces*, A. Baklouti and T. Nomura (Eds.), Springer Proceedings in Mathematics & Statistics (2019), 121–168. arXiv:1710.02975
[2] H. Oda and N. Shimeno, *Spherical inversion for a small K-type on the split real Lie group of type G_2*, arXiv:1812.08963
(Received January 23, 2020)

1156-43-216 **Giovanni Alberti** and **Francesca Bartolucci*** (francesca.bartolucci@sam.math.ethz.ch), Rämistrasse 101, 8092 Zurich, CH, Switzerland, and **Filippo De Mari** and **Ernesto De Vito**. *Unitarization and Inversion Formulae for the Radon Transform between Dual Pairs.*

We consider the Radon transform associated to dual pairs (X, Ξ) in the sense of Helgason, with $X = G/K$ and $\Xi = G/H$, where G is a locally compact group and K and H are closed subgroups of G. Under some technical assumptions, we prove that if the quasi regular representations of G acting on $L^2(X)$ and $L^2(\Xi)$ are irreducible, then the Radon transform admits a unitarization intertwining the two representations. If, in addition, the representations are square integrable, we provide an inversion formula for the Radon transform based on the voice transform associated to these representations. (Received January 24, 2020)

1156-43-257 **Fulton B Gonzalez** and **Jue Wang*** (jue.wang@tufts.edu), Bromfield-Pearson Hall, 503 Boston Avenue, Medford, MA , and **Tomoyuki Kakehi**. *Surjectivity of Convolution Operators on Noncompact Symmetric Spaces.*

A function F on $\mathbb{C}^n$ is called *slowly decreasing* if there exists a constant $A > 0$, such that for every $x \in \mathbb{R}^n$, there exists $z \in \mathbb{C}^n$ satisfying $\|z - x\| \leq A \log(2 + \|x\|)$ and $|F(z)| \geq A(1 + \|x\|)^{-A}$. Let $\mu \in \mathcal{E}'(\mathbb{R}^n)$, we denote $\mu^*(\xi)$ be the Fourier transform, and c_μ be the convolution operator: $c_\mu(T) = T * \mu$. In 1960, Ehrenpreis showed that the following are equivalent:

- $c_\mu : \mathcal{E}(\mathbb{R}^n) \to \mathcal{E}(\mathbb{R}^n)$ is surjective;

- $c_\mu : \mathcal{D}'(\mathbb{R}^n) \to \mathcal{D}'(\mathbb{R}^n)$ is surjective;
- $\mu^*(\xi)$ is slowly decreasing.

In this talk, we shall generalize the result to symmetric spaces. Let $X = G/K$ be a noncompact Riemannian symmetric space and $\mu \in \mathcal{E}'_K(X)$. Our main result is that the convolution operator $c_\mu : \mathcal{E}(X) \to \mathcal{E}(X)$ is surjective if and only if the spherical Fourier transform $\tilde{\mu}(\xi)$ is a slowly decreasing function on $\mathfrak{a}^*_{\mathbb{C}}$. Likewise, we may also substitute the $\mathcal{E}(X)$ above by $\mathcal{D}'_K(X)$, the space of K-invariant distributions on X. (Received January 25, 2020)

1156-43-260 **Yunfeng Zhang*** (yunfeng.zhang@uconn.edu). *Strichartz estimates for the Schrödinger equation on compact symmetric spaces.*

The Schrödinger equation $iu_t = -\Delta u$ and its nonlinear versions have attracted a lot of attention in the analysis community, partly due to the richness of techniques one can apply to establish results. The cases when the underlying Riemannian manifold M is a symmetric space of the compact type seem to be an underexplored territory, and only special cases like spheres and tori have been studied with considerable effort. In this talk, I will make an exploration from the perspective of scaling invariant Strichartz estimates:

$$\|e^{it\Delta} f\|_{L^p(I\times M)} \leq C\|f\|_{H^s(M)},$$

where I is a finite interval of t, and H^s denotes the L^2-based Sobolev space, with $s = \frac{d}{2} - \frac{d+2}{p}$. These estimates involve some Weyl type exponential sums associated with spherical functions. I will demonstrate how to estimate these sums when the underlying space is a compact Lie group, by combining Weyl's differencing technique with the pseudo-polynomial behavior of characters near the origin, and specifying different scenarios regarding the distance of the variable in the maximal torus from the walls of the alcoves/cells. A tentative section of this talk is to discuss how one may extend this result to other high rank compact symmetric spaces. (Received January 27, 2020)

46 ► Functional analysis

1156-46-41 **Vira Babenko*** (vira.babenko@drake.edu) and **Vladyslav Babenko**. *Optimal recovery and best approximation in function L-spaces.*

We consider the problem of approximation of unbounded positively homogeneous operators in L-spaces using Lipschitz operators. In this talk, we discuss its connection to the problem of computing modulus of continuity of the unbounded operator on the class of elements, as well as, to the problem of optimal recovery of an unbounded operator by a Lipschitz one on the class of elements given with an error. As applications, we consider the problem of approximation of unbounded operator, that for functions with values in some L-space puts in a correspondence Hukuhara-type derivatives, by Lipschitz operators. Further, we discuss the solution of the problem of the optimal recovery of this operator on the class of functions that have Hukuhara-type derivative with the given majorant of the modulus of continuity. (Received December 29, 2019)

1156-46-199 **Michael Anshelevich*** (manshel@math.tamu.edu), Department of Mathematics, Mail Stop 3368, Texas A&M University, College Station, TX 77843-3368. *Hermite trace polynomials.* Preliminary report.

Trace polynomials were introduced by Procesi in the early 1970s. We will define the trace version of Hermite polynomials, justify this definition, and prove some of their basic combinatorial properties. (Received January 23, 2020)

1156-46-324 **Nathaniel Eldredge*** (neldredge@unco.edu), **Leonard Gross** and **Laurent Saloff-Coste**. *Strong hypercontractivity on stratified complex Lie groups.*

On Euclidean space $\mathbb{R}^n$ equipped with Gaussian measure μ, the Ornstein–Uhlenbeck semigroup is hypercontractive: it improves $L^p(\mu)$ integrability at a specific rate. This property is closely related to the logarithmic Sobolev inequality, and has many important applications. If one works instead on $\mathbb{C}^n$ and applies the semigroup only to holomorphic functions, it was shown by S. Janson that it actually improves integrability faster, a property called "strong hypercontractivity". We study an extension of this phenomeon to stratified complex Lie groups, which generalize the translation and dilation structure of $\mathbb{C}^n$ in a non-abelian fashion, and where the natural analogue of Gaussian measure is a hypoelliptic heat kernel. An essential tool is an orthogonal polynomial decomposition of holomorphic L^2 space with respect to the heat kernel. (Received January 27, 2020)

47 ► Operator theory

1156-47-122 **Raul Quiroga-Barranco*** (quiroga@cimat.mx), Centro de Investigacion en Matematicas, Jalisco S/N, Colonia Valenciana, 36023 Guanajuato, Guanajuato, Mexico. *Bergman spaces on bounded symmetric domains, representation theory and commuting Toeplitz operators.*

Let D be a circled bounded symmetric domain and K the isotropy subgroup of biholomorphisms at 0. Let $\mathcal{H}^2_\lambda(D)$ be the weighted Bergman space of L^2 holomorphic functions for $d\nu_\lambda(z) = c_\lambda h(z,z)^{\lambda-p}$, where $h(z,z)$ is the Jordan determinant for D, p is its genus and $\lambda > p-1$. This is a Hilbert space with reproducing kernel $K_\lambda(z,w) = h(z,w)^{-\lambda}$. We define on $\mathcal{H}^2_\lambda(D)$ the Toeplitz operator $T^{(\lambda)}_a$ with symbol $a \in L^\infty(D)$ by

$$T^{(\lambda)}_a(f)(z) = \int_D a(w)f(w)K_\lambda(z,w)d\nu_\lambda(w).$$

For a subgroup H of K let $\mathcal{A}^H$ be the space of symbols that are H-invariant, and $\mathcal{T}^{(\lambda)}(\mathcal{A}^H)$ the C^*-algebra generated by Toeplitz operators with such symbols. We show, using representation theory, that $\mathcal{T}^{(\lambda)}(\mathcal{A}^K)$ is commutative. We also show that $\mathcal{T}^{(\lambda)}(\mathcal{A}^T)$ is commutative for T a maximal torus in K if and only if D is biholomorphic to a unit ball. And we exhibit domains D and subgroups $T \subset H \subsetneq K$ so that $\mathcal{T}^{(\lambda)}(\mathcal{A}^H)$ is commutative. (Received January 17, 2020)

49 ► Calculus of variations and optimal control; optimization

1156-49-36 **Charles L. Byrne*** (charles_byrne@uml.edu), 116 Hanks Street, Lowell, MA 01852. *Iterative Algorithms in Inverse Problems.*

Let C be a nonempty subset of an arbitrary set X and $f : X \to \mathbb{R}$. The objective is to minimize $f(x)$ over $x \in C$. We get x^k, for $k = 1, 2, ...$, by minimizing $G_k(x) = f(x) + g_k(x)$ over all $x \in X$. We call this approach an *auxiliary-function* (AF) method if $g_k : X \to [0, +\infty]$, $g_k(x^{k-1}) = 0$, and $g_k(x) < +\infty$ if and only if $x \in C$. Then $\{f(x^k)\} \downarrow \beta^* \geq -\infty$. We consider conditions on the auxiliary functions g_k that guarantee that $\beta^* = \beta \doteq \inf_{x \in C} f(x)$.

An AF algorithm is said to be in the SUMMA class if the SUMMA Inequality, $G_k(x) - G_k(x^k) \geq g_{k+1}(x)$, for all $x \in X$, holds for all k, in which case it follows that $\beta^* = \beta$. We consider a variety of AF algorithms that either are in the SUMMA class or can be reformulated to be such. We also study some AF algorithms that are not in the SUMMA class, but for which $\beta^* = \beta$. This leads to a larger class, the SUMMA2 class of AF algorithms.

An AF algorithm is a proximal minimization algorithm (PMA) if $g_k(x) = d(x, x^{k-1})$, where $d : X \times X \to [0, +\infty]$ is a distance, so that $d(x, y) = 0$ if and only if $x = y$. Optimization transfer (OT) algorithms in statistics can be reformulated as PMA. (Received December 26, 2019)

1156-49-201 **Peter Stechlinski*** (peter.stechlinski@maine.edu), Department of Mathematics and Statistics, University of Maine, Orono, ME. *Sensitivity analysis for nonsmooth dynamical systems.*

Nonsmooth dynamical systems exhibit hybrid-like behavior (i.e., a mixture of continuous and discrete) and are broadly applicable, with motivating problems found in systems biology, atmospheric chemistry, and process systems engineering, among others. Formalisms of this modeling framework include nonsmooth ODEs (i.e., those with nonsmooth right-hand side functions), nonsmooth differential-algebraic equations, optimization-constrained differential equations, and complementarity systems. However, conventional treatments of these models may fail because of the presence of nonsmoothness. Sensitivity analysis is presented for this modeling framework, using recent advancements in generalized derivatives theory. Mirroring classical results, nonsmooth sensitivity systems can be derived whose solutions characterize (local) sensitivity information. Numerical implementations of the theory are practical, with the sensitivity information furnished from this approach suitable for use in dedicated nonsmooth numerical methods (e.g., equation-solving) and large-scale dynamic optimization. (Received January 23, 2020)

1156-49-238 **Brian R Wall***, wallb@wit.edu. *Forward and Backward Neural Networks.* Preliminary report.

This talk explores simple feed forward and update backward neural networks on a fundamental level to investigate how they function and what their capabilities are. It goes through perceptrons, sigmoid neurons, stochastic gradient descent, the backpropagation algorithm, cross-entropy and quadratic cost functions, overfitting and

regularization, and weight and bias initialization. It presents a visual proof that neural networks can abstractly approximate any continuous function. Included are also applications to the classic hand-written digit recognition problem which utilizes the MNIST database. Results of eight different neural networks are compared to see the difference in each network's accuracy as well as the run time of the program needed for learning. The networks vary in cost function, weight initialization, regularization parameter, and hidden layer architecture; while displaying the differences between each network. Using all of these discussed techniques, it is possible to produce a 98.2% accurate network with a single hidden layer of 200 neurons. (Received January 24, 2020)

1156-49-414 **N. Azevedo**, **D. Pinheiro** and **S. Pinheiro*** (scoutopinheiro@qcc.cuny.edu), 222-05, 56th Avenue Bayside, Department of Mathematics & Computer Science, Bayside, NY 11364. *Dynamic programming for semi-Markov modulated SDEs.*

We consider a stochastic optimal control problem with state variable dynamics described by a stochastic differential equation of diffusive type modulated by a semi-Markov process with a finite state space. The time horizon is both deterministic and finite. Within such setup, we obtain a dynamic programming principle and use it to characterize the value function as a viscosity solution of the corresponding Hamilton-Jacobi-Bellman equation. Time allowing, we will illustrate our results with an application to Mathematical Finance: the generalization of Merton's optimal consumption-investment problem to financial markets with semi-Markov switching. (Received January 28, 2020)

51 ► Geometry

1156-51-50 **Sunrose T Shrestha*** (sunrose.shrestha@gmail.com). *Statistics of Random Square-tiled Surfaces.* Preliminary report.

Square-tiled surfaces are finite branched covers of the standard square torus, with branching over exactly one point. These can be also thought of as surfaces obtained by taking finitely many Euclidean unit squares, and gluing the edges in parallel pairs. In this talk we will see topological and geometric statistics of random square-tiled surfaces using a simple combinatorial model. (Received January 05, 2020)

1156-51-70 **Barry Monson*** (bmonson@unb.ca), Department of Mathematics and Statistics, UNB, Fredericton, NB E3B5A3, Canada. *The Richness of Cubes.* Preliminary report.

How rich Eulidean space $\mathbb{E}^n$ is with its unending supply of treasures! Think of the n-cube, perhaps the most familiar regular convex polytope. Today I will survey some beautiful, new objects derived from cubes.

First I discuss a current project with Asia Weiss, where we investigate the realization space of the cubical toroid $\mathcal{T}_b$ (an abstract regular polytope). Think of $\mathcal{T}_b$ as a $b \times \cdots \times b$ n-cube, with customary identifications of opposite facets. We find that the (pure) realizations of $\mathcal{T}_b$ are parametrized in a wonderful way by the orbits of 'integer' points in $\mathcal{T}_b$ under the action of its point group. (This generalizes our work from 1999 in the case $n = 2$.)

Next, as time allows, I look at *Roli's cube* $\mathcal{R}$ (Bracho, Hubard, Pellicer; 2014). Actually, $\mathcal{R}$ isn't a cube at all, although it shares the 1-skeleton of a 4-cube. Instead, $\mathcal{R}$ is a chiral 4-polytope of type $\{8, 3, 3\}$, faithfully realized in $\mathbb{E}^4$ (a situation once thought impossible). I will give some new insights into $\mathcal{R}$, touching on its regular covers and connections to the Möbius-Kantor configuration 8_3. (Received January 10, 2020)

1156-51-113 **Daniel McKenzie*** (mckenzie@math.ucla.edu), Department of Mathematics, UCLA, 520 Portola Plaza, Los Angeles, CA 155505, and **Anna Little** and **James Murphy**. *Power weighted shortest path distances and their applications to data science.* Preliminary report.

A power weighted shortest path distance is a data-driven metric defined on a given point cloud $\mathcal{X} \subset \mathbb{R}^D$. To compute this distance, one first creates a complete weighted graph $G^p_{\mathcal{X}}$ with edge weights $\|\mathbf{x}_i - \mathbf{x}_j\|^p$. The p-weighted distance between $\mathbf{x}_\alpha$ and $\mathbf{x}_\beta$ is then the minimum path length from $\mathbf{x}_\alpha$ to $\mathbf{x}_\beta$ in $G^p_{\mathcal{X}}$.

In this talk we will describe a new result on efficiently computing such distances for large data sets. Time permitting we will also highlight some applications of power weighted distances in contemporary data science. (Received January 15, 2020)

1156-51-208 **Charles Anthony Camacho** and **Dami Lee*** (damilee@uw.edu). *Geometric realization of genus three Fermat quartic as a triply periodic polyhedral surface.*

We study a specific infinite polyhedral surface whose compact quotient is a genus three Riemann surface that is an eightfold cyclic cover over a sphere branched over three points. Its symmetries allow us to construct hyperbolic structures and various translation structures. With this explicit data, one can find its algebraic

description, automorphism group, etc. We will show that the vertices on the polyhedral surface correspond to the Weierstrass points on the quotient Riemann surface and moreover that the surface is a geometric realization of the genus three Fermat quartic. (Received January 23, 2020)

1156-51-209 **Charles Anthony Camacho*** (camachoc@uw.edu) and **Dami Lee**. *Geometric Realizations of Quasiplatonic Cyclic 8-gonal Surfaces.*

We study infinite polyhedral representations of compact Riemann surfaces. In this talk, we discuss the quasiplatonic cyclic 8-gonal surfaces, including Bolza surface of genus two and Fermat quartic of genus three. We show that these surfaces elicit coverings which are realizable as triply periodic polyhedral surfaces. (Received January 23, 2020)

1156-51-211 **Mandy Cheung**, **Yu-Wei Fan*** (ywfan@berkeley.edu) and **Yu-Shen Lin**. *Geometric realization of stable objects in certain quiver Calabi-Yau categories.*

We show that the base of certain Lefschetz fibration admits an embedding into the space of Bridgeland stability conditions on certain quiver 3-Calabi-Yau category, and identify the special Lagrangian Lefschetz thimbles with the stable objects. Joint work with Mandy Cheung and Yu-Shen Lin. (Received January 23, 2020)

1156-51-307 **Siu-Cheong Lau** (lau@math.bu.edu), Mathematics and Statistics Department, Boston University, 111 Cummington Mall, Boston, MA 02215, **Tsung-Ju Lee*** (tjlee@cmsa.fas.harvard.edu), 20 Garden Street, Cambridge, MA 02138, and **Yu-Shen Lin** (yslin@bu.edu), Mathematics and Statistics Department, Boston University, 111 Cummington Mall, Boston, MA 02215. *On the complex affine structures of SYZ fibration of del Pezzo surfaces.*

Given any smooth cubic curve $E \subset \mathbb{P}^2$, we show that the complex affine structure of the special Lagrangian fibration of $\mathbb{P}^2 \setminus E$ constructed by Collins–Jacob–Lin coincides with the affine structure used in Carl–Pomperla–Siebert for constructing the mirror. (Received January 27, 2020)

1156-51-355 **Abigail Ward*** (arward@stanford.edu). *Homological mirror symmetry for the Hopf surface.*

We will present homological mirror symmetry results for the Hopf surface ($S^1 \times S^3$). (Received January 27, 2020)

1156-51-396 **Sara Tukachinsky***, 1 Einstein Drive, Princeton, NJ 08540. *Relative quantum cohomology.*

We define a quantum product on the cohomology of a symplectic manifold relative to a Lagrangian submanifold, with coefficients in a Novikov ring. The associativity of this product is equivalent to an open version of the WDVV equations, for an appropriate disk superpotential we define via Fukaya A_∞ structures and bounding chains. This is joint work with Jake Solomon. No prior knowledge of the subject matter will be assumed. (Received January 28, 2020)

1156-51-410 **Bjoern Muetzel*** (bjorn.mutzel@gmail.com), **Peter Buser** and **Eran Makover**. *Short non-separating loops on hyperbolic hyperelliptic surfaces.* Preliminary report.

For any hyperelliptic hyperbolic surface of genus $g \geq 2$, there exist g homologically non-trivial loops $(\alpha_k)_{k=1,\dots,g}$, such that

$$\ell(\alpha_k) \leq 4 \log \left(2 \cdot \left(\frac{g-1}{g-k+1} \right) + 2 \right), \text{ for all } k \in \{2, \dots, g\}.$$

Especially the median length ($k = \frac{g}{2}$) is bounded by $4\log(6)$ and the $(\alpha_k)_k$ are bounded by $4\log(2g)$.
We conjecture that at least $\frac{g}{2}$ of these loops are homologically independent. (Received January 28, 2020)

1156-51-412 **Jeffrey Carlson**, **Elisheva Adina Gamse*** (eagamse@math.toronto.edu) and **Yael Karshon**. *Realisation of fixed-point data for torus actions.*

Consider a torus T acting on a compact oriented manifold M with a T-equivariant stable complex structure. Suppose that the fixed points of the action are isolated. The isotropy weights at these fixed points satisfy certain identities that are dictated by the ABBV localisation theorem. We ask whether any collection of weights satisfying these identities must be the fixed point data of a torus action on some manifold. This is joint work with Jeffrey Carlson and Yael Karshon. (Received January 28, 2020)

52 ► Convex and discrete geometry

1156-52-51 **Valeriu Soltan*** (vsoltan@gmu.edu), George Mason University, Department of Mathematical Sciences, 4400 University Drive, Fairfax, VA 22030. *On M-decomposable sets.* Preliminary report.

M-*predecomposable set* in n dimensions is a convex set which is the sum of a compact convex set and a closed convex cone. Complementing the existing results on M-decomposable sets, we study their extreme, exposed, and asymptotic properties. Also, we consider M-*polyhedral sets* which are sums of compact convex sets and polyhedral cones and establish some characteristic properties of such sets. (Received January 05, 2020)

1156-52-61 **Gabe Cunningham*** (gabriel.cunningham@gmail.com). *The many faces of abstract polytopes.*

In this talk aimed at graduate students and non-specialists, I will introduce abstract polytopes and the many ways we represent them, including:

- As partially-ordered sets,
- As regular, properly edge-colored graphs,
- As finitely presented groups with a distinguished set of generators, and
- As maps on surfaces.

I will also discuss some recent problems and results that highlight the advantages of each approach. (Received January 08, 2020)

1156-52-62 **Antonio Montero*** (amontero@yorku.ca). *On the Schläfli type of extensions of chiral polytopes.*

Given an abstract n-polytope $\mathcal{K}$, an abstract $(n+1)$-polytope $\mathcal{P}$ is an extension of $\mathcal{K}$ if all the facets of $\mathcal{P}$ are isomorphic to $\mathcal{K}$. All but the last of the entries of the Schläfli symbol of $\mathcal{P}$ (if it exists) are determined by $\mathcal{K}$. In the talk we discuss some possibilities for this last entry when strong symmetry conditions are imposed over $\mathcal{P}$ and $\mathcal{K}$. (Received January 08, 2020)

1156-52-104 **Karoly Bezdek*** (kbezdek@ucalgary.ca), Dept. of Math. and Stats., 2500 University Drive NW, Calgary, Alberta T2N1N4, Canada. *Volumetric bounds for intersections of congruent balls.*

We investigate the intersections of balls of radius r, called r-ball bodies, in Euclidean d-space. An r-lense (resp., r-spindle) is the intersection of two balls of radius r (resp., balls of radius r containing a given pair of points). We prove that among r-ball bodies of given volume, the r-lense (resp., r-spindle) has the smallest inradius (resp., largest circumradius). In general, we upper (resp., lower) bound the intrinsic volumes of r-ball bodies of given inradius (resp., circumradius). This complements and extends some earlier results on volumetric estimates for r-ball bodies. (Received January 14, 2020)

1156-52-105 **Marjorie Senechal***, senechal@smith.edu, MA. *Parallelohedra and the rhombic triacontahedron.* Preliminary report.

Throughout the 115 years since he published it, E. S. Fedorov's short list of parallelohedra in R^3 has galvanized geometers and crystallographers to amend it, clarify it, rigorize it, and extend it. (Is his list complete? Are there necessary and sufficient conditions for belonging to it? Is convexity essential? What about higher dimensions?) I will discuss a family of skeletal parallelohedra hidden in the rhombic triacontahedron that raise such questions and illustrate a conjecture of Grünbaum (2010). This talk is dedicated to Branko's memory. (Received January 15, 2020)

1156-52-138 **Alathea Irene Jensen*** (jensena@susqu.edu), 514 University Ave, Selinsgrove, PA 17870. *The Kobon Triangle Problem.* Preliminary report.

The Kobon triangle problem is a long-standing open problem in discrete geometry that asks for the maximum number of triangular cells in an arrangement of lines in the Euclidean plane. This talk will give an overview of the current state of the problem and discuss some new observations and directions for research. (Received January 20, 2020)

1156-52-151 **Steven Simon*** (ssimon@bard.edu). *Orbit Polytope Partitions of a Tverberg Type.*

Tverberg's Theorem, a foundation of convex and discrete geometry, states that any $T(r,d) := (r-1)(d+1)+1$ points in $\mathbb{R}^d$ can be partitioned into r subsets whose convex hulls have non-empty r-fold intersection. The number $T(r,d)$ is very tight in that any generic collection of fewer points cannot be so divided, and in such

circumstances it is natural to search for weaker symmetry on the convex hulls. Our central result guarantees the following such condition: if G is any finite group and $\rho : G \to O(d)$ is any faithful orthogonal representation not containing the trivial representation, then almost any $N(G, \rho; d) := T(|G|, d) - d$ points in $\mathbb{R}^d$ can be partitioned by $|G|$ subsets so that there are $|G|$ points, one from each of the $|G|$ resulting convex hulls, which form a full G-orbit and the vertex set of of a (necessarily vertex transitive) generic orbit polytope. Again, the number $N(G, \rho; d)$ is optimal in the strong sense above. In low dimensions, these points may be taken to be the vertices of any regular polygon in the plane; of nearly uniform copies of all regular prisms, the icosahedra, and five of the Archimedean solids in three-space; and infinitely many facet transitive four-polytopes, three of which are regular. (Received January 24, 2020)

1156-52-174 **Gabe Cunningham** and **Mark Mixer*** (mixerm@wit.edu). *Vertex-Faithful Regular Polyhedra.* Preliminary report.

A regular polytope is called vertex-faithful if each symmetry of the polytope is determined by its action on the polytope's vertices. Many interesting examples of regular polytopes can be found by considering the vertex-faithful ones. In this talk, I will present results on these examples of regular polyhedra. In particular, utilizing results on vertex-faithful polyhedra, I will give the classification of all regular polyhedra where the number of vertices is prime or twice a prime. (Received January 22, 2020)

1156-52-235 **Florian Frick*** (frick@cmu.edu). *Convex neural codes and local obstructions.*

A place cell is a type of neuron, which fires precisely when an organism is in a certain region, usually convex, called a place field. Characterizing the patterns of neural activity that may arise in this way appears to be an intractable problem. The difficulty lies in the geometric rigidity of the problem: We show that if the convexity requirement is replaced by the more flexible topological notion of contractability, the absence of one local obstruction characterizes the corresponding patterns. More precisely, we characterize the intersection patterns of open sets that form a good cover of a subset of Euclidean space.

This is joint work with Aaron Chen and Anne Shiu. (Received January 24, 2020)

1156-52-262 **Alexey Balitskiy*** (balitski@mit.edu). *Subadditivity of the projection inradii for convex coverings.* Preliminary report.

If a convex body in $\mathbb{R}^n$ is covered by the union of convex bodies, multiple subadditivity questions can be asked. The subadditivity of the width is the subject of the celebrated plank theorem of T. Bang, whereas the subadditivity of the inradius is due to V. Kadets. We adapt the existing proofs of these results to prove a theorem on coverings by certain generalized non-convex "planks". A corollary of this theorem establishes the subadditivity of the following quantity: the minimal inradius over all k-dimensional orthogonal projections onto subspaces of $\mathbb{R}^n$. This corollary interpolates between Bang's theorem ($k = 1$) and Kadets's theorem ($k = n$). (Received January 26, 2020)

1156-52-266 **Jamie Haddock*** (jhaddock@math.ucla.edu), **Jesus A. De Loera** and **Luis Rademacher**. *The Minimum Euclidean-Norm Point on a Convex Polytope: Wolfe's Combinatorial Algorithm is Exponential.*

The complexity of Philip Wolfe's method for the minimum Euclidean-norm point problem over a convex polytope has remained unknown since he proposed the method in 1974. The method is important because it is used as a subroutine for one of the most practical algorithms for submodular function minimization. We present the first example that Wolfe's method takes exponential time. Additionally, we improve previous results to show that linear programming reduces in strongly-polynomial time to the minimum norm point problem over a simplex. (Received January 26, 2020)

1156-52-269 **Alexander Xue** and **Pablo Soberón*** (pablo.soberon-bravo@baruch.cuny.edu), One Bernard Baruch Way, Mathematics Department, New York, NY 10010. *Equitable convex partitions of the plane for two families of lines.*

Several extensions of the ham sandwich theorem prove the existence of certain partitions of Euclidean spaces into convex parts. In the plane, such partitions can always divide fairly two finite sets of points at the same time. We extend these results to deal with families of lines instead of families of points. Given two finite families A and B of n lines in the plane each and a positive integer r, we prove the existence of partitions of the plane into r convex parts such that the following holds. For each part C of the partition, there exists a subset of roughly $n^{1/r}$ lines of A whose pairwise intersections all lie in C, and the same holds for B. (Received January 26, 2020)

1156-52-326 **Alexey Garber*** (alexeygarber@gmail.com). *On the regularity radius for Delone sets in hyperbolic plane.*

Point set X in a metric space $\mathcal{M}$ is called a *Delone set* if its interpoint distance is bounded from below by some positive number and the size of voids in $\mathcal{M}$ is bounded from above by another positive number.

A natural question for Delone sets is whether equal local structure for a set X implies its global symmetry or rich symmetry group. This question originates from an explanation of crystal grow process through local interaction of atoms. For Delone sets in Euclidean spaces this question was in the focus of local theory approach developed by Delone, Dolbilin, Stogrin, and Galiulin in 1970's.

In this talk I review some known results on local theory in Euclidean space as well as give a way to extend them to Delone sets in hyperbolic plane. Particularly, I'll show that there are local conditions on a Delone set X in $\mathbb{H}^2$ that force X to be regular.

This talks is based on a joint work with Nikolay Dolbilin, Undine Leopold, and Egon Schulte. (Received January 27, 2020)

1156-52-338 **Barry Monson**, University of New Brunswick, Department of Mathematics, Fredericton, NB E3B 5A3, Canada, and **Egon Schulte***, Northeastern University, Department of Mathematics, Boston, MA 02115. *The Assembly Problem for Alternating Semiregular Polytopes.* Preliminary report.

In the classical setting, a convex polytope is semiregular if its facets are regular and its symmetry group is transitive on vertices. The lecture is about semiregular abstract polytopes, which have abstract regular facets, still with combinatorial automorphism group transitive on vertices. We focus in particular on alternating semiregular polytopes, with two kinds of regular facets occurring in an alternating fashion. Group amalgamations can be used to prove that given two compatible regular n-polytopes P and Q, there exists a universal alternating semiregular (n+1)-polytope U which is obtained by freely assembling alternate copies of P and Q. We investigate in what circumstances we can have k copies each of P and Q alternate as the facets of a semiregular (n+1)-polytope S. In some fairly broad situations we can construct the polytope S. However, there are instances of "bad behaviour", with obstructions already existing when k=2. (Received January 27, 2020)

1156-52-362 **Abigail Williams*** (ab.williams@northeastern.edu). *Uniform Polyhedra with Planar and Nonplanar Faces.* Preliminary report.

In this talk, we will discuss a method for generating uniform polyhedra with nonplanar faces. The generated polyhedra have regular polygons as faces and are vertex transitive under a symmetry group of a regular polyhedron. We consider polygons as connected edge sets which may be planar or nonplanar. In the nonplanar case, regularity is derived from the polygon's relationship with a prism or antiprism. (Received January 27, 2020)

1156-52-393 **Undine Leopold*** (undine.leopold@gmail.com). *Discrete symmetry and surfaces – from the abstract to the concrete.*

Given a discrete group G acting on a closed surface S, how can we visualize some, or all, of this action? Various authors have explored this question of realizing symmetry. In this talk, I will give a short overview and highlight some recent results, ranging from highly symmetric maps on surfaces, over surfaces immersed symmetrically in Euclidean 3-space, all the way to (finite) polyhedra of higher genus with high symmetry. (Received January 28, 2020)

1156-52-399 **Eric L. Grinberg*** (eric.grinberg@umb.edu), University of Massachusetts Boston, 100 Morrissey Blvd, Boston, MA 02125. *Sets of Admissibility for Radon Transforms over Finite Fields in Higher Dimensions.* Preliminary report.

We consider finite or combinatorial analogs of Radon transforms that occur in analysis and in small models of tomography. In many familiar contexts these transforms are injective and over-determined. We study admissible sets, i.e., minimal restrictions that retain admissibility, by analogy with the continuous versions introduced by I.M. Gelfand in the 1950s. We explore the structures of such *admissible complexes*, both for their geometry and their graph-theoretic and enumerative properties. Both vector and projective contexts are included. The emphasis is on dimensions higher than those previously explored. (Received January 28, 2020)

1156-52-420 **Nicholas A Matteo*** (nicholas.matteo@colorado.edu). *Symmetry of convex polytopes.*

If a convex polytope's symmetries act transitively on certain of its faces (e.g. vertices and edges), what implications does that have for further symmetries or symmetry type? (Received January 29, 2020)

53 ► Differential geometry

1156-53-12 **Christopher T Woodward*** (woodwardc@gmail.com). *Lagrangian Floer theory in the tropics.*

I will first give an introduction to symplectic geometry, Hamiltonian flows, and Lagrangian Floer theory, starting with the example of rigid body motion and a formula of Archimedes. In the second half of the talk I will discuss whether Lagrangian Floer theory can be made combinatorial along the lines of work of Eleny Ionel and Brett Parker. Finally I will describe a recent joint result with Sushmita Venugopalan (IMSc Chennai) which generalizes work of Fulton and Sturmfels and gives an affirmative answer in good cases. Some applications will be described. (Received November 22, 2019)

1156-53-24 **Jonathan Weitsman*** (j.weitsman@northeastern.edu) and **Peter Crooks**. *Enhanced symmetry in the semiclassical category and characters of loop groups.* Preliminary report.

We consider a version of Weinstein's symplectic category, adapted for the case of quasi-Hamiltonian G-spaces. We show that semiclassical quantization in this setting produces the Kac character formula, in analogy with the construction of the Weyl character formula by Guillemin and Sternberg in the symplectic setting. We show that this construction gives a natural action of the modular group on the Kac characters, which we conjecture agrees with Kac's SL(2,Z) action. We conjecture also that a similar construction should give rise to enhanced symmetries—-that is, symmetries of the quantization that do not arise from symmetries of the underlying classical system—also in some other examples. (Received December 13, 2019)

1156-53-25 **Alexander Mednykh*** (smedn@mail.ru), Sobolev Institute of Mathematics, Novosibirsk State University, Novosibirsk, 630090, Russia. *Automorphisms of graphs and Riemann surfaces acting with fixed points.* Preliminary report.

Let X be a finite connected graph, possibly with loops and multiple edges. We provide each edge of the graph including loops by two possible orientations. An automorphism group of a graph acts *harmonically* if it acts freely on the set of directed edges of the graph. Following M. Baker and S. Norine define a genus g of the graph X to be the rank of the first homology group. A finite group acting harmonically on a graph of genus g is a natural discrete analogue of a finite group of automorphisms acting on a Riemann surface of genus g. In the present report, we give a sharp upper bound for the size of cyclic group acting harmonically on a graph of genus $g \geq 2$ with a given number of fixed points. Similar results, for closed orientable surfaces, were obtained by T. Szemberg, I. Farkas and H. M. Kra.

These are joint results with Grzegorz Gromadzki and Ilya Mednykh. (Received December 13, 2019)

1156-53-49 **Oliver Goertsches*** (goertsch@mathematik.uni-marburg.de). *GKM theory in dimension 6.*

In this talk we will relate Tolman's and Woodward's 6-dimensional Hamiltonian non-Kähler T^2-manifolds to Eschenburg's twisted flag manifold $SU(3)//T^2$.

This is joint work with Panagiotis Konstantis and Leopold Zoller. (Received January 05, 2020)

1156-53-57 **Liat Kessler*** (lkessler@math.haifa.ac.il), University of Haifa. *Equivariant cohomology does not distinguish Hamiltonian S^1 manifolds.*

For Hamiltonian circle actions on 4-manifolds, we give an example of an isomorphism of equivariant cohomology modules that cannot be induced by an equivariant diffeomorphism of spaces. This is in contrast to Masuda's result establishing that in the toric case, the equivariant cohomology module determines the manifold. We also give a soft proof that there are finitely many maximal Hamiltonian circle actions on a fixed closed symplectic 4-manifold.

We achieve these results by giving a generators and relations description in terms of the decorated graph for the even part of the equivariant cohomology as a module over the equivariant cohomology of a point. We then give an explicit combinatorial description of all weak algebra isomorphisms. (Received January 07, 2020)

1156-53-72 **Haniya Azam**, **Catherine Cannizzo**, **Heather Lee*** (heathml@uw.edu) and **Chiu-Chu Melissa Liu**. *Global homological mirror symmetry for genus two curves.* Preliminary report.

Cannizzo's thesis proved an HMS result with the B-model being the bounded derived category of coherent sheaves on genus two curves and the A-model being the Fukaya-Seidel category on the Landau-Ginzburg mirror (Y, W), where Y is a locally toric Calabi-Yau 3-fold and $W : Y \to \mathbb{C}$ is a symplectic fibration. Cannizzo's thesis is for a one parameter family of complex parameters on the genus two curve. We upgrade her result to include

all complex parameters and the corresponding Kahler parameters of the mirror, and we identify the mirror map globally which will exhibit wall crossing phenomena. (Received January 10, 2020)

1156-53-82 **Andrés Pedroza*** (andres_pedroza@ucol.mx), Facultad de Ciencias/Universidad de Colima, Bernal Díaz del Castillo no. 340, Col. Villa San Sebastián, 28045 Colima, Colima, Mexico. *Loops of symplectic diffeomorphisms on the one-point blow up of the complex projective space.*

The complex projective space admits a circle action that preserves the standard symplectic structure. It is know that such loop of diffeomorphisms has finite order in the fundamental group of the group of symplectic diffeomorphisms of the complex projective space.

In the talk we will see how this loop of symplectic diffeomorphisms induces a loop of symplectic diffeomorphisms on the one-point blow up of the complex projective space. Moreover such loop will have infinite order in the fundamental group of the group of symplectic diffeomorphisms. (Received January 13, 2020)

1156-53-95 **Catherine Kendall Asaro Cannizzo*** (ccannizzo@scgp.stonybrook.edu), Simons Center for Geometry and Physics, State University of New York, Stony Brook, NY 11794-3636. *Homological mirror symmetry for a genus two curve.*

We prove a homological mirror symmetry result for a one-parameter family of genus 2 curves obtained by scaling a specific complex structure. First we describe the B-model genus 2 curve in a 4-torus and the geometric construction of the generalized SYZ mirror. Then we set up the Fukaya-Seidel category on the mirror. Finally we will see the main algebraic result on homogenous coordinate rings, i.e. the HMS result on the level of cohomology. The method involves first considering mirror symmetry for the 4-torus, then restricting to the hypersurface genus 2 curve on the complex side and extending to a mirror Landau-Ginzburg model with fiber the mirror 4-torus on the symplectic side. This talk is based on the speaker's PhD thesis under Professor Denis Auroux. It also gives the background for the next talk by Heather Lee, extending to all complex structures. (Received January 14, 2020)

1156-53-127 **Francois S. Monard*** (fmonard@ucsc.edu). *Recent progress on sharp mapping properties for the geodesic X-ray transform.*

The geodesic X-ray transform is an operator arising in certain imaging problems in media with variable index of refraction or in problems of seismology. In both settings, given a compact Riemannian manifold with boundary, this transform integrates a function over geodesics passing through that manifold. The corresponding inverse problem is to reconstruct a given function from its X-ray transform, by assessing what can be reconstructed, how to reconstruct it, and how stable the inversion is.

For this purpose, recent interest has been gained in accurately capturing the mapping properties of this operator, in cases where the geometry is *simple* (that is, strictly convex, without conjugate points, nor infinite-length curves). We will review recent results in this direction. In particular, we will describe the recent results of arXiv:1910.13691, where a sharp answer to this problem can be given for a certain class of Riemannian surfaces (geodesic disks of constant curvature). (Received January 18, 2020)

1156-53-181 **Denis Auroux*** (auroux@math.harvard.edu), Harvard University Department of Mathematics, 1 Oxford Street, Cambridge, MA 02138. *Lagrangian Floer theory in trivalent configurations of curves and homological mirror symmetry.* Preliminary report.

The mirror of a genus g Riemann surface can be viewed as a trivalent configuration of $3g-3$ rational curves meeting in $2g-2$ triple points; more precisely, this singular configuration arises as the critical locus of the superpotential in a 3-dimensional Landau-Ginzburg mirror. In joint work in progress with Alexander Efimov and Ludmil Katzarkov, we attempt to introduce a notion of Fukaya category for such a configuration of rational curves, where objects are embedded graphs with trivalent vertices at the triple points, and morphisms are linear combinations of intersection points as in usual Floer theory. We will describe the proposed construction of the structure maps of these Fukaya categories, attempt to provide some motivation, and outline examples of calculations that can be carried out to verify homological mirror symmetry in this setting. (Received January 22, 2020)

1156-53-271 **Allen Knutson*** (allenk@math.cornell.edu). *Duistermaat-Heckman and Chern-Schwartz-MacPherson classes.*

One version of the Duistermaat-Heckman theorem writes the DH measure as an alternating sum of pushforwards of cones, each of which has the appearance of a DH measure of a vector space. This was given one geometric explanation by Ginzburg-Guillemin-Karshon using symplectic cobordism. I will give another (in the algebraic case), realizing each summand as the DH measure of a Lagrangian cycle in the cotangent bundle. This gives an

enrichment of the DH formula, using the extra circle action on the cotangent fibers; the corresponding summands are essentially the Chern-Schwarz-MacPherson classes of the Morse strata. (Received January 26, 2020)

1156-53-282 **Thibaut Grouy*** (thibaut.grouy@uclouvain.be). *Orbital integrals on Lorentzian symmetric spaces.*

On any pseudo-Riemannian symmetric space, the orbital integral operator associates to any compactly supported continuous function its integrals over the orbits of the isotropy subgroup of the transvection group. The problem that will be addressed in this talk is to determine the function in terms of its orbital integrals. In 1959, Sigurdur Helgason solved this inverse problem on any even-dimensional isotropic Lorentzian symmetric space via a limit formula involving the Laplace-Beltrami operator. In 1987, Jeremy Orloff generalized Helgason's work by inverting the orbital integral operator on any rank-one semisimple pseudo-Riemannian symmetric space. His result encompasses the odd-dimensional isotropic Lorentzian case.

In 1970, Michel Cahen and Nolan Wallach showed that any indecomposable Lorentzian symmetric space is either semisimple or solvable. The semisimple ones are isotropic. Hence the inverse problem is already solved. In this talk, I will consider the orbital integrals on model spaces of the solvable ones, called Cahen-Wallach spaces. I will present the limit formula that expresses any compactly supported smooth function in terms of its orbital integrals. As in the isotropic case, it involves an invariant differential operator. (Received January 26, 2020)

1156-53-333 **Daniel Pomerleano*** (daniel.pomerleano@umb.edu). *Intrinisic mirror symmetry and the Bondal-Orlov conjecture.*

I will describe a relationship between the intrinsic mirror symmetry program of Gross-Hacking-Keel-Siebert and a conjecture of Bondal and Orlov on categorical resolutions of singularities. (Received January 27, 2020)

1156-53-342 **Hansol Hong** and **Yoosik Kim*** (yoosik@brandeis.edu), Department of Mathematics, Goldsmith 105, Brandeis University, 415 South Street, Waltham, MA 02453, and **Siu-Cheong Lau** and **Xiao Zheng**. *T-equivariant disc potentials for toric CY manifolds.*

In this talk, we discuss how to formulate and compute the equivariant disc potentials of immersed SYZ fibers for toric Calabi-Yau manifolds. The potentials are closely related to the open Gromov-Witten invariants of Aganagic-Vafa branes. (Received January 27, 2020)

54 ► General topology

1156-54-110 **Henry Adams*** (henry.adams@colostate.edu), **Aurora Clark** and **Hung Le**. *Descriptors of Energy Landscapes using Topological Analysis (DELTA).*

Many of the properties of a chemical system are described by its energy landscape, a real-valued function defined on a high-dimensional domain. I will explain how topology, and in particular persistent homology, can be used in order to describe some of the pertinent features of an energy landscape. As a motivating example, the low-energy conformation space of an aluminate molecular anion interacting with a simple potassium cation can be well-modeled by the permutohedron. In a recently funded NSF Harnessing the Data Revolution project, the DELTA team is learning how to identify and leverage changing topological features of energy landscapes across a range of chemical conditions in order to predict reactivity. (Received January 15, 2020)

55 ► Algebraic topology

1156-55-7 **Matvei Libine*** (mlibine@indiana.edu). *Localization of integrals of equivariant forms for non-compact group actions.*

The Berline-Vergne integral localization formula for equivariantly closed forms is well-known and requires the acting Lie group to be compact. In this talk I will explain my extension of this result to (non-compact) real reductive Lie groups. This extension requires relaxing the notion of convergence of integrals and redefining integrals as distributions on the Lie algebra. It also requires finding the right balance (or duality) between the equivariant forms and the cycles of integration so that integrals make sense and a Berline-Vergne type localization formula holds. An application of this generalized localization formula includes a proof of the integral character formula for real reductive Lie groups originally due to Rossmann, Schmid, Vilonen. This talk is based on my paper "Integrals of Equivariant forms and a Gauss-Bonnet Theorem for Constructible Sheaves" published in Topology, also available from arXiv:math/0306152. (Received September 27, 2019)

1156-55-40 **Sadok Kallel*** (skallel@aus.edu), American University of Sharjah (UAE), and Laboratoire Painleve, U. Lille (France), and **Walid Taamallah**, University of Tunis Al Manar, Tunisia. *Combinatorial Invariants of Stratified Spaces.* Preliminary report.

There is a generalized Euler characteristic for spaces that are stratified by locally compact subpaces. This characteristic can be viewed as an integral ring morphism on a Grothendieck ring that we construct, well suited for this category of spaces. We explain the constructions then provide explicit Euler characteristic computations for a large number of functorial constructions found in the literature, in particular those stratified by configuration spaces. This talk is a mix of topology, combinatorics and geometry, and is accessible to a wide audience. (Received December 29, 2019)

1156-55-66 **Loring W. Tu*** (loring.tu@tufts.edu), Department of Mathematics, Tufts University, Medford, MA 02155. *Computing the Gysin Map Using Fixed Points.*

A continuous map $f\colon X \to Y$ of topological spaces induces a pushforward map in cohomology $f_*\colon H^*(X) \to H^*(Y)$ called the **Gysin map**. In enumerative geometry, one often needs to know the cohomology class of the image of a cycle A in the space X under the map f. Such a formula has been given for many different maps, for example, when $f\colon X \to Y$ is a projective bundle, a Grassmann bundle, or more generally, a homogeneous-space bundle. Using the localization formula of Atiyah–Bott and Berline–Vergne in equivariant cohomology, we generalize these formulas to a bundle with equivariantly formal fibers. (Received January 10, 2020)

1156-55-155 **Matthias Franz*** (mfranz@uwo.ca). *Syzygies in equivariant cohomology.*

I am going to survey the theory of syzgyies in equivariant cohomology. This is partly joint work with Chris Allday, Volker Puppe and Jianing Huang.

Let X be a manifold with an action of a torus T of rank r. The Borel equivariant cohomology $H_T^*(X)$ (with real coefficients) is a module over the polynomial ring $R = H^*(BT)$. Syzygies are a notion from commutative algebra that interpolates between torsion-free R-modules (first syzygies) and free ones (r-th syzygies).

Starting point is the observation that the Chang–Skjelbred sequence is exact if and only $H_T^*(X)$ is a reflexive R-module (a second syzygy). This generalizes the GKM method for computing $H_T^*(X)$. Higher syzgygies are characterized by the exactness of a certain longer sequence. An important role is played by a suitably defined equivariant homology $H_*^T(X)$.

Assume that X is compact orientable. Then the reflexivity of $H_T^*(X)$ is also equivalent to the perfection of the equivariant Poincaré pairing. Moreover, if $H_T^*(X)$ is a syzygy of order $\geq r/2$, then it is already free over R in this case. Big polygon spaces realize all syzygy orders $< r/2$.

Most results extend to actions of arbitrary compact connected Lie groups. (Received January 21, 2020)

1156-55-213 **Chen He*** (hechen123@gmail.com). *Equivariant formality of isotropy action on a homogeneous space with rank difference one.*

Let H be a closed connected subgroup of a compact connected Lie group G. If rank(G) = rank(H), it is well known that the rational cohomology of G/H can be described using the characteristic classes of G and H. In this talk, I will consider the case where rank(G) - rank(H) = 1. The main result is a characterization of homogeneous spaces with rank difference 1 that satisfy the isotropy formal condition. This work is joint with Jeffrey Carlson. (Received January 24, 2020)

1156-55-353 **Facundo Memoli***, Department of Mathematics, The Ohio State University, Columbus, OH 43210. *Vietoris-Rips Persistent Homology and Geometric Invariants.*

The interval decomposition of the persistence module associated to the Vietoris-Rips filtration of a (finite) metric space is a central object in applied algebraic topology. One question of interest is what is the relationship between these intervals and other, more standard, geometric invariants such as diameter, hyperbolicity, curvature, etc. In this talk I will survey some recent results in this direction obtained in collaboration with Sunhyuk Lim and Osman B. Okutan. (Received January 27, 2020)

1156-55-367 **Chi-Kwong Fok*** (ckfok@nyu.edu), New York University Shanghai, 1555 Century Avenue, Pudong New Area, Shanghai, 200122, Peoples Rep of China. *A stroll through equivariant K-theory.*

Equivariant K-theory, introduced by Graeme Segal in his thesis, is an important example of generalized equivariant cohomology theory which interpolates Atiyah-Hirzebruch's topological K-theory and representation theory of Lie groups which act on the topological spaces under study. It provides the natural framework to formulate the equivariant index theorem which has wide applicability. In this talk, we will review equivariant K-theory from the ground up and sample some of its important results and applications in equivariant topology. Time permitting, we will discuss some of its recent developments. (Received January 27, 2020)

1156-55-400 **Jeffrey D Carlson*** (jcarlson@fields.utoronto.ca), **Oliver Goertsches**, **Chen He** and **Augustin-Liviu Mare**. *Equivariant cohomology rings of cohomogeneity-one actions.*
Cohomogeneity-one actions of Lie groups on manifolds, namely those whose orbit space is one-dimensional, form an important and intensely-studied class of examples in differential geometry. We discuss the computation of the Borel equivariant rational cohomology rings of these actions. This work is joint with Oliver Goertsches, Chen He, and Liviu Mare. (Received January 28, 2020)

57 ► Manifolds and cell complexes

1156-57-129 **Rylee Alanza Lyman*** (rylee.lyman@tufts.edu). *Outer automorphisms of free products of finite groups at infinity.* Preliminary report.
In 1995, Vogtmann showed that $\mathrm{Out}(F_n)$, the group of outer automorphisms of a free group of rank n, is *simply-connected at infinity* when $n \geq 5$. That is, if $\mathrm{Out}(F_n)$ acts on a complex X properly and cocompactly, then the inverse limit of $\pi_1(X - K)$ as K varies over compact subspaces of X, is the trivial group. Later, Bestvina–Feighn showed that in fact $\mathrm{Out}(F_n)$ is $(2n-5)$-connected at infinity. We conjecture that the outer automorphism group of a free product of finite groups is $(n-4)$-connected at infinity, where n is the Kurosh rank of the free product. We describe natural spaces and complexes the outer automorphism group acts on, and progress towards this conjecture. (Received January 18, 2020)

1156-57-170 **Neil Hoffman***, 810 S Kings Street, Stillwater, OK 74074. *Cusp types covered by hyperbolic knot complements.*
This paper completes a classification of the types of orientable and non-orientable cusps that can arise in the quotients of hyperbolic knot complements. In particular, $S^2(2,4,4)$ can not be the cusp cross-section of any orbifold quotient of a hyperbolic knot complement. Furthermore, if a knot complement covers an orbifold with a $S^2(2,3,6)$ cusp, it also covers an orbifold with a $S^2(3,3,3)$ cusp. (Received January 22, 2020)

1156-57-171 **Jacob Russell*** (jrussellmadonia@gradcenter.cuny.edu), **Davide Spriano** and **Hung Cong Tran**. *The local-to-global property for Morse quasi-geodesics.*
A hallmark of hyperbolic spaces is the nice behavior of their quasi-geodesic. In particular, both the Morse property (every quasi-geodesic is close to a geodesic) and the local-to- global property (every local quasi-geodesic is a global quasi-geodesic) characterize hyperbolic spaces. We examine a natural extension of these properties by studying the class of spaces in which every local Morse quasi-geodesic is a global Morse quasi-geodesic. We show that this class of spaces in incredibly rich, encompassing CAT(0) spaces, the mapping class group, Teichmuller space, and 3-manifold groups. We also show a number of consequences for this local-to-global property for Morse geodesic, including a generalization of a theorem of Gitik to combinations of stable subgroups of CAT(0) groups, the mapping class group, and 3-manifold groups. In the case of the mapping class group, this produces a combination theorem for the topologically important convex cocompact subgroups. (Received January 22, 2020)

1156-57-335 **Matthew Haulmark***, m.haulmark@vanderbilt.edu. *Bowditch boundaries of relatively hyperbolic groups.*
In this talk, I will discuss boundaries of relatively hyperbolic groups, splittings, and linear connectedness. (Received January 27, 2020)

1156-57-340 **James Farre*** (james.farre@yale.edu). *Dense representations of discrete groups and bounded cohomology.*
Let Γ be a discrete group, and let $\rho : \Gamma \to \mathrm{Isom}^+(\mathbb{H}^3)$ be a homomorphism. We will define bounded cohomology $H_b^\bullet(\Gamma)$ of Γ and describe an invariant of ρ in $H_b^3(\Gamma)$ defined by computing the (signed) volumes of certain geodesic tetrahedra in $\mathbb{H}^3$. The main theorem is that this invariant is non-zero for representations with dense image. Moreover, if a dense representation exists, then the degree 3 bounded cohomology of Γ has uncountable dimension. Surprisingly, the proof relies on some non-trivial topology and geometry of 3-manifolds. The aim of the talk will be to explain how one can apply the ideas of the proof in dimension $n \geq 4$, given a hyperbolic n-manifold with some prescribed topological and geometric properties (if one exists). The discussion is already interesting when Γ is a non-abelian free group, in which case we do not know if bounded cohomology in degree larger than 3 is (non-)zero. (Received January 27, 2020)

1156-57-368 **Bena Tshishiku*** (bena_tshishiku@brown.edu) and **Mauricio Bustamante**.
Symmetries of exotic negatively curved manifolds.

Let N be a smooth manifold that is homeomorphic but not diffeomorphic to a hyperbolic manifold M. How much symmetry does N have? For example, does $\mathrm{Isom}(M)$ act on N by diffeomorphisms? Farrell-Jones showed that in general the answer is "No"; however, they do not rule out that the orientation-preserving subgroup of $\mathrm{Isom}(M)$ acts on N. In this talk, we will discuss this problem and its relation to Nielsen realization. The main result is a construction of examples of N such that $\mathrm{Isom}(M)$ does act on N, as well as examples where the largest subgroup of $\mathrm{Isom}(M)$ that acts on N has arbitrarily large index. This is joint work with Mauricio Bustamante. (Received January 27, 2020)

58 ► Global analysis, analysis on manifolds

1156-58-119 **Isaac Z Pesenson*** (pesenson@temple.edu), Department of Mathematics, Temple University, Philadelphia, PA 19122. *A Weak Weyl's Law on compact homogeneous sub-Riemannian manifolds.*

It is shown that a number of eigenvalues on an interval $[0, \omega]$ of a sub-Laplacian on a compact homogeneous sub-Riemannian manifold is comparable to cardinality of a certain cover of the manifold by balls of radius $\omega^{-1/2}$. (Received January 16, 2020)

1156-58-239 **Tomoyuki Kakehi*** (kakehi@math.tsukuba.ac.jp), Department of Mathematics, University of Tsukuba, 1-1-1 Tennodai, Tsukuba, Ibaraki 305-8571, Japan. *Mean value operators on symmetric spaces of the noncompact type.*

Let $X = G/K$ be a Riemannian symmetric space of non-compact type. For each $y \in X$ the mean value operator $M^y : C^\infty(X) \to C^\infty(X)$ is defined by

$$M^y f(x) = \int_K f(gk \cdot y)\, dk \qquad (x = g \cdot o \in X).$$

We will talk about the surjectivity of the mean value operator M^y and a related problem on the behavior of the zonal spherical function. This is a joint work with Jens Christensen, Fulton Gonzalez, and Jue Wang. (Received January 25, 2020)

60 ► Probability theory and stochastic processes

1156-60-80 **Vasily Zaburdaev*** (vasily.zaburdaev@fau.de), Cauerstr. 11, Chair of Mathematics in Life Sciences, FAU, 91058 Erlangen, Germany. *Lévy walks: overview and challenges.*

The Lévy walk model has two main features distinguishing it from conventional random walks: a finite velocity of moving particles, which linearly couples the duration and length of the particle's displacement, and the length of displacements that is power-law distributed. The model proved to be very successful in describing a large variety of superdiffusive dispersal phenomena across disciplines. We will discuss the development and the main properties of the model with an emphasis on recent theoretical results. We will demonstrate how the basic model can be modified and extended to better match the challenges posed by complex real-life systems and try to overview a few remaining challenges and possible directions in which the model can be further developed in the future. (Received January 13, 2020)

1156-60-81 **Denis D Patterson***, Department of Mathematics, Brandeis University, 415 South Street, Waltham, MA 02453, and **Simon A Levin**, **Carla Staver** and **Jonathan D Touboul**. *Spatial Effects in Savanna Dynamics.*

We introduce two interacting particle systems of vegetation dynamics (one macroscale and one mesoscale) based on the interaction rules from the mean-field Staver-Levin model of forest-savanna-grassland evolution. Using coupling techniques for stochastic jump processes, we show the convergence of these particle systems towards McKean-Vlasov jump processes — processes solving a stochastic differential equation with self-consistent jump rates depending on the statistics of the solution. The generalized Kolmogorov equations of these processes are systems of integro-differential equations (IDEs) which are more amenable to analysis than the original particle systems and constitute nonlocal generalizations of the classical Staver-Levin model. Mathematical and numerical analysis of the resulting IDEs reveals a rich phenomenology in the dynamics, namely phase skipping waves of invasion, front pinning, (spatial) multistability and pattern formation. Furthermore, some of the solutions we

identify predict novel mechanisms for the real-world forest-savanna transition which can be observed in empirical data. (Received January 13, 2020)

1156-60-109 **Marco Lenci***, Dipartomento di Matematica, Università di Bologna, Piazza di Porta San Donato, 5, 40126 Bologna, BO, Italy. *Random walks in Lévy random media.*

We call 'random walk in a Lévy random medium' a random walk occurring in a random environment which causes the jumps, or "ballistic stretches", to have a long-tailed distribution. These processes are used to model the phenomena of anomalous diffusion observed in Physics and in other sciences (Economics, Social Sciences, Biology, etc.). In this talk I will mostly discuss a family of random walks in 1D Lévy random media: the medium is given by a point process on the real line, the "targets", whose nearest-neighbor distances are i.i.d. with long tails, and the dynamics is a random walk on the targets. This can occur in discrete time (instantaneous jumps) or in continuous time (constant-speed "walk" between targets). The latter system generalizes what in the physical literature is called Lévy-Lorentz gas. I will report on some rigorous results, obtained more or less recently with several collaborators, about the asymptotic behavior of these processes. Both normal and anomalous behaviors are observed, depending on the parameters of the models (even when the media is of Lévy type). If time permits, I will also touch upon work in progress concerning a two-dimensional version of the Lévy-Lorentz gas. (Received January 15, 2020)

1156-60-125 **Jeremiah Birrell***, 26 Willow Lane, Lenox, MA 01240. *Small-Mass Limit of Noisy Inertial Particle Dynamics: Higher-Order Approximations.*

Langevin equations model the dynamics of an inertial particle, coupled to forcing, dissipation, and noise. The small-mass regime is a much studied singular limit of these systems. In this talk, I will discuss the derivation of higher-order approximations to the position degrees of freedom in the limit, improving upon the classical overdamped limiting equation. (Received January 18, 2020)

1156-60-139 **James M. Murphy*** (jm.murphy@tufts.edu), 503 Boston Ave, Medford, MA 02155. *Data-driven diffusions, multiscale geometry, and computational biology.*

Approaches to unsupervised clustering and semi-supervised active learning with data-driven diffusion processes are developed. Latent high-dimensional data is considered as a graph on which a diffusion process propagates. We consider this diffusion process across time scales in order to capture multiscale geometries in the data. These structures are distilled into efficient data-driven metrics which exhibit performance guarantees for unsupervised and semisupervised classification problems. We propose efficient algorithms that scale to large datasets, and demonstrate our methods on real protein-protein interaction networks. (Received January 20, 2020)

1156-60-140 **Sergey Denysov*** (sergiyde@oslomet.no). *Planar Lévy walk loops: an approach to home-range search and foraging.*

Lévy walks are random walks in which the walker moves continuously and with a constant velocity between reorientation events. The durations and lengths of these displacements are power-law distributed. Lévy walks were shown to be a very successful model to describe a variety of anomalous diffusion dispersal phenomena in physics, biology and ecology. In the context of search, Lévy walks were identified as an optimal strategy for finding rare renewable targets. That boosted the research in Lévy foraging strategies in living organisms, and Lévy algorithms for robots performing search tasks.

One important aspect intrinsically present in most living systems but also in robotics — the existence of home range — was not considered before in the context of random search. The fact that a bird needs to return to its nest and a robot to its charging station seems obvious, but the implementation of such processes on the model level is a highly non-trivial task. I will introduce the concept of Lévy walk loops — Lévy walk trajectories returning to the origin after a fixed time. I will show how to tackle the challenge of efficient loop generation and present two approaches to planar search problems. Finally, I will discuss related open problems. (Received January 20, 2020)

1156-60-190 **Giampaolo Cristadoro*** (giampaolo.cristadoro@unimib.it), via R. Cozzi, 55, 20125 Milano, Italy. *Time for Sparre-Andersen.*

Knowledge of the first time at which a stochastic process passes a given value finds application in diverse areas of the natural sciences. In the context of one-dimensional random walks, the celebrated Sparre-Andersen theorem proves the universality of the distribution of the number of jumps occurring before entering for the first time into the positive semi-axis, for a walker starting at the origin. On the other hand, there are many physical situations in which the time spent in doing one step correlates with the jump's length performed by the walker (noteworthy examples are Lévy walks and Lévy-Lorentz gases). In these cases, the number of jumps is often not

the accessible or relevant quantity and the interest concentrates on the (continuous) physical time needed to first enter the specified region. In this talk I will show how to to generalise Sparre-Andersen theorem to deal with such situations, providing exact formulas for rather general spatiotemporal correlations. (Received January 23, 2020)

1156-60-197 **Aurel Iulian Stan***, 1465 Mt. Vernon Ave., Marion, OH 43302, and **Florin Catrina**. *On three-dimensional 1-Meixner random vectors.* Preliminary report.

A definition of 1-Meixner random vectors, in the multi-dimensional case, will be presented first. This definition uses the commutators between the joint semi-quantum operators and multiplication operators, generated by the components of a random vector. We use these commutators to derive a system of partial linear differential equations satisfied by the joint Laplace transform of the components of the random vector, on a neighborhood of the origin. Finally, we focus our attention on the three-dimensional case and describe all 1-Meixner random vectors in this case. Joint work with Florin Catrina from St. John's University, New York. (Received January 23, 2020)

1156-60-198 **W. Bryc*** (wlodzimierz.bryc@uc.edu), **R. Fakhfakh** and **W. Mlotkowski**. *Cauchy-Stieltjes families with polynomial variance functions and generalized orthogonality.*

The topic of this talks are sequences of monic polynomials that satisfy a finite-step recursion. I will recall van Iseghem-Maroni's version of Favard's theorem for higher order recursions, and then I will discuss the following recursion of order 3:

$$\begin{aligned} xp_0(x) &= p_1(x), \\ xp_1(x) &= p_2(x) + ap_1(x) + p_0(x), \\ xp_2(x) &= p_3(x) + ap_2(x) + bp_1(x), \\ xp_n(x) &= p_{n+1}(x) + ap_n(x) + bp_{n-1}(x) + cp_{n-2}(x),\ n \geq 3. \end{aligned}$$

We analyzed this recursion by relating it to Cauchy-Stieltjes families and free probability. However, the statement of the main result does not need such concepts. (Received January 23, 2020)

1156-60-219 **Marcus Michelen***, 851 S Morgan Street, Chicago, IL 60607. *Central limit theorems from zero-free regions.*

To each discrete random variable X, we may associate a generating function $f_X(z) = \sum_k \mathbb{P}(X = k)z^k$. We show that if $f_X(z)$ is zero-free in a complex neighborhood of 1, then X is close to a Gaussian in a precise quantitative sense after recentering and rescaling. Time permitting, applications to random discrete structures will be discussed. Based on joint work with Julian Sahasrabudhe. (Received January 24, 2020)

1156-60-224 **Davit Karagulyan*** (dkaragul@umd.edu), Department of Mathematics, University of Maryland, 1301 Mathematics Building, College Park, MD 20740, and **Dmitry Dolgopyat**. *Transient dynamical random walk with unbounded return times.*

We consider a one dimensional model of a dynamical random walk F defined on $\mathbb{T} \times \mathbb{Z}$ and given as follows; for $(x, n) \in \mathbb{T} \times \mathbb{Z}$ let

$$F(x, n) = \Big((ax + h_n(x)) \bmod 1, n - 1_{W_{n,-1}}(x) + 1_{\mathbb{T}\setminus W_{n,-1}}(x)\Big),$$

where $a \in \mathbb{N}$ and for all $n \in \mathbb{Z}$, $h_n \in C^2(\mathbb{T})$, $W_{n,-1} = \cup_{i=1}^m I_i^n$, where $\{I_i^n\}_{i=1}^m$ is a collection of intervals. We also suppose that there is $\delta > 0$ and a set $W_{-1} \subset \mathbb{T}$, so that for all $n \in \mathbb{Z}$ $\|h_n\|_{C^{1+\mathrm{Lip}}} \leq \delta$ and the Hausdorff distance between $W_{n,-1}$ and W_{-1} is smaller than δ. Under certain conditions on a and the set W_{-1}, we show that for δ sufficiently small, F has a drift and $z_n(x) = \pi_{\mathbb{Z}}(F^n(x, 0))$ satisfies the central limit theorem. (Received January 24, 2020)

1156-60-243 **Matthew Kahle** (mkahle@gmail.com), **Elliot Paquette*** (paquette.30@osu.edu) and **Erika Roldan** (e.roldan.roa@gmail.com). *The fundamental group of 2-dimensional random cubical complexes.*

We study the fundamental group of certain random 2-dimensional cubical complexes which contain the complete 1-skeleton of the d-dimensional cube, and where every 2-dimensional square face has probability p. These are cubical analogues of Linial–Meshulam random simplicial complexes, and also simultaneously are 2-dimensional versions of bond percolation on the hypercube. Our main result is that if p is less than or equal to 1/2, then with high probability the fundamental group contains a nontrivial free group, and if p is greater than 1/2 then with high probability it is trivial. As a corollary, we get the same result for homology with any coefficient ring. We also study the structure of the fundamental group below the transition point, especially its free factorization. (Received January 25, 2020)

1156-60-247 **Hoi H Nguyen*** (nguyen.1261@osu.edu), 231 W, 18th Ave., Columbus, OH 43210. *On roots of random trigonometric polynomials.*

We show that for various general ensembles of random trigonometric polynomials with iid coefficients the variance of the number of roots is asymptotically linear in terms of the expectation. We will also show that the number of roots satisfies an exponential concentration of measure.

Based on joint work with Y. Do and O. Nguyen, and with O. Zeitouni (Received January 25, 2020)

1156-60-258 **Guangyu Xi*** (gxi@umd.edu) and **Xuan Liu**. *On a maximal inequality and its application to SDEs with singular drift.*

We present a Doob type maximal inequality for stochastic processes satisfying the conditional increment control condition. If we assume, in addition, that the margins of the process have uniform exponential tail decay, we prove that the supremum of the process decays exponentially in the same manner. Then we apply this result to the construction of the almost everywhere stochastic flow to stochastic differential equations with singular time dependent divergence-free drift. (Received January 25, 2020)

1156-60-265 **Christoph Börgers*** (cborgers@tufts.edu), Department of Mathematics, Tufts University, 503 Boston Avenue, Medford, MA 02155, and **Claude Greengard**. *Sensitive dependence of Lévy walk asymptotics.*

Various types of asymptotic descriptions of the space-time location of a Lévy walker have been given. For instance, the asymptotics of the mean square displacement (MSD) are known in detail. In a central region, the density is governed by a generalized central limit theorem. Outside the central region, the behavior is described by an "infinite density" (e.g., Rebenshtok et al. 2014). All of these are descriptions valid in the limit as $t \to \infty$. It is important to understand how large t must be before they become accurate approximations.

Cristadoro et al. (2014) observed that even when the MSD grows superlinearly, a linear correction term can be significant for a very long time. We have shown that linear and sublinear correction terms can never be deduced just from tail behavior. They depend on the entire step length distribution, as well as on the initialization of the Lévy walk. We have also shown that the convergence in the generalized central limit theorem may be excruciatingly slow, and that the rate of convergence depends on very subtle details of the tail behavior of the distribution of the summands.

Precise knowledge of the details of the Lévy walk may therefore be needed before the accuracy of asymptotic descriptions can be ascertained. (Received January 26, 2020)

1156-60-267 **David Gamarnik*** (gamarnik@mit.edu), E62-563, 100 Main Street, Cambridge, MA 02140, and **Aukosh Jagannath** (a.jagannath@uwaterloo.ca), MA. *The Overlap Gap Property and Approximate Message Passing algorithms for p-spin models.*

We consider the algorithmic problem of finding a near ground state (near optimal solution) of a p-spin model. We show that for a class of algorithms broadly defined as Approximate Message Passing (AMP), the presence of the Overlap Gap Property (OGP), appropriately defined, is a barrier. We conjecture that when $p \geq 4$ the model does indeed exhibits OGP (and prove it for the space of binary solutions). Assuming the validity of this conjecture, as an implication, the AMP fails to find near ground states in these models, per our result. We extend our result to the problem of finding pure states by means of Thouless, Anderson and Palmer (TAP) based iterations, which is yet another example of AMP type algorithms. We show that such iterations fail to find pure states approximately, subject to the conjecture that the space of pure states exhibits the OGP, appropriately stated, when $p \geq 4$. (Received January 26, 2020)

1156-60-287 **Samuel Punshon-Smith***, punshs@brown.edu, Providence, RI. *Chaotic mixing of scalars in stochastic fluid mechanics.*

The long-time behavior of a passive scalar in an incompressible fluid has long been of interest in physics. In this talk I will discuss several recent rigorous results in this area for a passive scalar that is advected by a number of stochastic fluid models, including the stochastic Navier-Stokes equations. We will see how tools from theory of random dynamics and the ergodic and hypoelliptic theory for stochastic PDE can be used to show that the associated Lagrangian flow has a positive Lyapunov exponent. I will then discuss how to use this chaos to show that the passive scalar almost surely mixes exponentially fast, uniformly in the diffusivity. A phenomenon known in physics as chaotic mixing. (Received January 26, 2020)

1156-60-288 **Souvik Dhara*** (sdhara@mit.edu), 77 Massachusetts Ave, Cambridge, MA 02139, and **Debankur Mukherjee** and **Kavita Ramanan**. *Asymptotic normality of the $r \to p$ norm for random matrices with non-negative entries.*

The $r \to p$ operator norm of a matrix A is defined as $\|A\|_{r\to p} = \sup_{\|x\|_r \le 1} \|Ax\|_p$ for $r, p \ge 1$. We consider the $r \to p$ norm of a class of symmetric random matrices with nonnegative entries, which includes the adjacency matrices of the Erdős-Rényi random graphs and matrices with (nonnegative) sub-Gaussian entries. For $1 < p \le r < \infty$, we establish the asymptotic normality of the appropriately centered and scaled $r \to p$ norm. The special case $r = p = 2$, which corresponds to the largest singular value of matrices, was proved in a seminal paper by Füredi and Komlós (1981). Of independent interest, we further obtain a sharp ℓ_∞-approximation for the maximizer vector. The results also hold for sparse matrices and further the ℓ_∞-approximation for the maximizer vector also holds for a broad class of deterministic sequence of matrices with certain asymptotic 'expansion' properties. (Received January 26, 2020)

62 ► Statistics

1156-62-6 **Emily T Winn***, Brown University, Box F, Providence, RI 02906, and **Lorin Crawford**. *Unified Topological and Geometric Characterization of Genomic Networks Underlying Complex Traits in Genome-wide Association Studies.* Preliminary report.

In many biological applications, graph-based methods have become widely used to represent large genomic data sets due to their ability to not only visualize sophisticated relationships between random variables, but also provide interpretable distance measures for comparing various populations. Unfortunately, current distance metrics are not flexible enough to reliably quantify differences between graphs in high-dimensional data settings where notable deviations can occur at either local or global scales. Here, we propose a novel Bayesian framework for estimating an optimal distance via a mixture of functions frequently used in the literature. Furthermore, to ensure both scalability and interpretability for our approach, the distances we compute are based on groups of nodes (features) defined by biologically relevant annotations (e.g. SNPs to genes, genes to signaling pathways). Our main contribution is a unified method which gives detailed insights about the sub-graphical structures driving the variation between classes of observations (such as case-control studies) and offers a new means of biological enrichment analyses with graphs. (Received September 12, 2019)

1156-62-69 **Abiy Tasissa*** (abiy.tasissa@tufts.edu) and **Rongjie Lai** (lair@rpi.edu). *Matrix Completion in a General Non-orthogonal Basis.*

We consider theoretical analysis of recovering a low rank matrix given a few expansion coefficients with respect to any basis. This approach generalizes the existing analysis for the matrix completion problem. Our analysis is based on dual certificates and dual basis approach but does not assume the restricted isometry property (RIP) condition. We propose a novel sufficient condition for the analysis named the correlation condition. For matrix completion problems with respect to a structured deterministic basis, RIP might not hold or is NP hard to verify. However, the correlation condition can be checked in time $O(n^3)$. Under the assumption that the correlation holds and the true low rank matrix obeys certain coherence conditions, the main result shows that the true matrix can be recovered with very high probability from $O(nr \log^2 n)$ uniformly random expansion coefficients. (Received January 10, 2020)

1156-62-74 **Paolo Bertolotti*** (pbertolo@mit.edu), **Ali Jadbabaie** and **Fotini Christia**. *Tests for Network Cascades via Branching Processes.*

We consider a network setting where branches form under the null of independent edge formation and larger branches form under the alternative. Call detail records provide the motivating example, as large call branches form after disruptive events, yet call branches also form during normal periods. Our goal is to distinguish abnormally large branches, which we term cascades, from the common branches formed under the null. We introduce a test statistic that compares observed branch size to expected branch size under the null. Using ideas from branching processes, we derive the expected size and variance of branches. Our test statistic is semiparametric, consistent, and asymptotically distributed standard normal under the null. Using call detail records from Yemen, we find a significant calling cascade occurred after the Presidential Palace was bombed in 2011. Lastly, we employ our statistic for event detection and successfully detect key violent events during the 2011 Yemeni Revolution. (Received January 11, 2020)

1156-62-90 **Jay S Stanley**, **Gal Mishne*** (gmishne@ucsd.edu), **Eric C Chi** and **Ronald R Coifman**. *Co-Manifold Learning on Tensors.* Preliminary report.

In this talk we introduce a new method for performing joint dimension reduction, or manifold learning, along the modes of a data tensor, in order to re-order the fibers along each mode so that the resulting permuted tensor is smooth. Our approach generalizes recent work on a convex formulation of the co-clustering problem. Like convex co-clustering, our co-manifold learning procedure possesses stability guarantees with respect to perturbations in the input data tensor. We illustrate how our method can identify the coupled intrinsic geometries in simulated and real data examples. (Received January 13, 2020)

1156-62-134 **Vasileios Maroulas*** (vmaroula@utk.edu), **Farzana Nasrin** and **Chistopher Oballe**. *A Bayesian Framework for Persistent Homology.*

Persistence diagrams offer a way to summarize topological and geometric properties latent in datasets. While several methods have been developed that utilize persistence diagrams in statistical inference, a full Bayesian treatment remains absent. This talk, relying on the theory of point processes, presents a Bayesian framework for inference with persistence diagrams relying on a substitution likelihood argument. In essence, we model persistence diagrams as Poisson point processes with prior intensities and compute posterior intensities by adopting techniques from the theory of marked point processes. We then propose a family of conjugate prior intensities via Gaussian mixtures to obtain a closed form of the posterior intensity. Finally we demonstrate the utility of this Bayesian framework, packaged in R under BayesTDA, with a classification problem in materials science using Bayes factors. (Received January 20, 2020)

1156-62-147 **Farzana Nasrin*** (fnasrin@utk.edu), Department of Mathematics, University of Tennessee, Knoxville, TN 37996, and **Vasileios Maroulas**. *Bayesian Computation via Importance Sampling for Persistent Homology.* Preliminary report.

Persistent homology is a tool in topological data analysis for learning about the geometrical/topological structures by detecting different dimensional hole and storing their appearance disappearance scales in persistence diagrams. The Bayesian inference for persistence diagrams is recently introduced relying on the theory of Poisson point processes. However, due to the inherent complexity of real datasets, it is not always possible to obtain a closed-form solution to the posterior distribution of persistence diagrams. This requires the development of an approximation method to compute the posterior distribution. In this talk, we propose an importance sampling scheme for estimating the posterior distributions of persistence diagrams. The importance sampling approach proves to be useful since points of a persistence diagram do not appear uniformly in the domain space. We present a detailed example of the posterior estimation to demonstrate analogies to standard Bayesian inference. Finally, based on the posterior distribution, we develop a robust classification algorithm by employing Bayes factor. (Received January 20, 2020)

1156-62-298 **Yash Deshpande*** (yash@mit.edu), 77, Massachusetts Ave., E17-481, E17-481, Cambridge, MA 02139. *Contextual Stochastic Block Models.*

Stochastic block models are a popular model for inference of latent communities in network data. In this work, we extend the stochastic block model with to annotated networks, wherein we have access to per-vertex covariate information. We provide the first information theoretic tight analysis for inference of latent community structure given a sparse graph along with high dimensional node covariates, correlated with the same latent communities. Our work bridges recent theoretical breakthroughs in the detection of latent community structure without nodes covariates and a large body of empirical work using diverse heuristics for combining node covariates with graphs for inference. The tightness of our analysis implies in particular, the information theoretical necessity of combining the different sources of information. Our analysis holds for networks of large degrees as well as for a Gaussian version of the model. (Received January 27, 2020)

1156-62-319 **Xiaochun (William) Meng***, Jubilee Building, Falmer, Brighton, UK, and **James W Taylor**, **Souhaib Ben Taieb** and **Siran Li**. *Scoring Functions for Multivariate Distributions and Level Sets.*

Scoring functions enable the comparison of forecast accuracy, and can potentially be used for estimation. A scoring function for multivariate distributions that has gained some popularity is the energy score. This is a generalization of the continuous ranked probability score (CRPS), which is widely used for univariate distributions. A little-known, alternative generalization is the multivariate CRPS (MCRPS). We propose a theoretical framework for scoring functions for multivariate distributions, which encompasses the energy score and MCRPS, as well as the quadratic score, which has also received little attention. We demonstrate how this framework can be used to generate new scores. For univariate distributions, it is well-established that the CRPS can be

expressed as the integral over a quantile score. We show that, in a similar way, scoring functions for multivariate distributions can be "disintegrated" to obtain scoring functions for level sets. Using this, we present scoring functions for different types of level set, including those for densities and cumulative distributions. To compute the scoring functions, we propose a simple numerical algorithm. (Received January 27, 2020)

1156-62-322 **Daniel Sanz-Alonso*** (sanzalonso@uchicago.edu), IL , and **Ruiyi Yang**. *Graph Representations of Non-Stationary Matérn Gaussian Fields.*

The aim of this talk is to show that graph-theoretical tools can be useful in pushing forward the applicability of kernel methods in spatial statistics and machine learning. I will derive a graph-based representation of non-stationary Matérn fields that stems from the SPDE approach. Graph representations are natural alternatives to finite element and finite difference representations used in spatial statistics and are useful prior models in machine learning applications. I will establish convergence guarantees and highlight that graph representations give a Markov random field approximation that allows to reduce the computational cost by exploiting sparsity. (Received January 27, 2020)

1156-62-377 **Daniel L Sussman*** (sussman@bu.edu), 111 Cummington Mall, Fl 1, Math Dept., Boston, MA 02215-2411. *Finding the Way with Graph Matching.*

The graph matching problem involves estimating a latent correspondence between the vertex sets of two graphs. As traditionally formulated, this involved trying to minimize the number of edge discrepancies between the aligned networks. In this talk we explore methods for solving the graph matching problem and demonstrate how adjusting the objective function can aid algorithmically to improve search performance. We also argue that in many instances the standard objective function will lead to incorrect solutions but appropriate modifications, often based on spectral graph properties, the correct alignment can be recovered even when the graphs have very different global structure. (Received January 28, 2020)

65 ► Numerical analysis

1156-65-336 **Paul E Barbone*** (barbone@bu.edu), 110 Cummington Mall, Dept of Mechanical Engineering, Boston University, Boston, MA 02215, and **Olalekan A Babaniyi**, School of Mathematical Sciences, Rochester Inst of Technology, 84 Lomb Memorial Drive, Rochester, NY 14623. *Comparison of variational formulations for the direct solution of an inverse problem in linear elasticity.*

Given one or more observations of a displacement field within a linear elastic, isotropic, incompressible object, we seek to identify the material property distribution within that object. This is a mildly ill-posed inverse problem in linear elasticity. While most common approaches to solving this inverse problem use forward iteration, several variational formulations have been proposed that allow its direct solution (i.e. solution without iteration). We review five such direct variational formulations for this inverse problem: Least Squares, Adjoint Weighted Equation, Virtual Fields, Inverse Least Squares, and Direct Error in Constitutive Eqn. We briefly review their derivations, their mathematical properties, and their compatibility with Galerkin discretization and numerical solution. We demonstrate these properties through numerical examples. (Received January 27, 2020)

1156-65-354 **Daniel I Gendin*** (digendin@bu.edu), 110 Cummington Mall, Boston, MA 02215, and **Paul E Barbone**. *Practical Computation of Uncertainty in Inverse Elasticity Problems.* Preliminary report.

We consider an inverse elasticity problem from a Bayesian probabilistic point of view, where we wish to approximate the posterior probability distribution of an elastic parameter field using data (likelihood) and regularization (prior). The Bayesian approach directly provides the posterior distribution of the sought elastic parameter field thus giving us the opportunity to estimate the field itself and to quantify its uncertainty. In practice, however, the discretized posterior distribution over random fields is very high dimensional (e.g. 10^4 dimensions). Furthermore, evaluating the posterior at any point requires solving a boundary value problem. Therefore, exploring the posterior is computationally challenging. We combine techniques from optimization theory and computational linear algebra to efficiently characterize the MAP point and the covariance within a Laplace (Gaussian) approximation of the posterior. In particular, while direct evaluation of the Hessian is computationally prohibitive, we show that a Newton method can be practically applied, following the methods presented in (Villa, et al. 2019). We further show that the Hessian and MAP point can be recovered at no additional computational cost, thus efficiently characterizing uncertainty in the MAP estimate. (Received January 27, 2020)

68 ► Computer science

1156-68-44 **Moon Duchin**, **Tom Needham** and **Thomas Weighill*** (thomas.weighill@tufts.edu). *The Topology of Redistricting.*

Redistricting is the process of dividing a state into districts for the purpose of electing representatives. The fairness of this process has received a lot of attention following high-profile court cases and reform efforts. Research into redistricting involves understanding the relationship between many different datasets such as election results, demographic data and geographic boundaries. In this talk we show how persistent homology (a technique from topological data analysis), coupled with large computer-generated ensembles of maps, can be used to understand the essential features of redistricting in a state. In particular, we examine how district scale and election year affect representation, and analyze maps which are biased towards one party. (Received January 02, 2020)

1156-68-67 **Rose Yu***, Khoury College of Computer Sciences, 177 Huntington Ave #2224, 177 Huntington Ave #2224, Boston, MA 02115. *Understanding the Representation Power of Graph Neural Networks in Learning Graph Topology.*

To deepen our understanding of graph neural networks, we investigate the representation power of Graph Convolutional Networks (GCN) through the looking glass of graph moments, a key property of graph topology encoding path of various lengths. We find that GCNs are rather restrictive in learning graph moments. Without careful design, GCNs can fail miserably even with multiple layers and nonlinear activation functions. We analyze theoretically the expressiveness of GCNs, concluding a modular GCN design, using different propagation rules with residual connections could significantly improve the performance of GCN. We demonstrate that such modular designs are capable of distinguishing graphs from different graph generation models for surprisingly small graphs, a notoriously difficult problem in network science. Our investigation suggests that, depth is much more influential than width, with deeper GCNs being more capable of learning higher order graph moments. Additionally, combining GCN modules with different propagation rules is critical to the representation power of GCNs. (Received January 10, 2020)

1156-68-231 **Diego Cifuentes*** (diegcif@mit.edu) and **Shmuel Onn**. *On the Complexity of Toric Ideals.*

We investigate the computational complexity of problems on toric ideals such as normal forms, Gröbner bases, and Graver bases. We show that all these problems are strongly NP-hard in the general case. Nonetheless, we can derive efficient algorithms by taking advantage of the sparsity pattern of the matrix. We describe this sparsity pattern with a graph, and study the parameterized complexity of toric ideals in terms of graph parameters such as treewidth and treedepth. In particular, we show that the normal form problem can be solved in parameter-tractable time in terms of the treedepth. An important application of this result is in multiway ideals arising in algebraic statistics. We also give a parameter-tractable membership test to the reduced Gröbner basis. This test leads to an efficient procedure for computing the reduced Gröbner basis. Similar results hold for Graver bases computation. (Received January 24, 2020)

70 ► Mechanics of particles and systems

1156-70-366 **Jaime Burgos-Gracia**, **Alessandra Celletti**, **Catalin Gales**, **Marian Gidea** and **Wai-Ting Lam***, waiting.lam@yu.edu. *Hill four-body problem with oblate tertiary: an application to the Sun-Jupiter-Hektor-Skamandrios system.*

We consider a restricted four-body problem with a precise hierarchy between the bodies: two point-mass bigger bodies and a smaller one, all three with oblate shape, and an infinitesimal body in the neighborhood of the oblate body. The three heavy bodies are assumed to move in a plane under their mutual gravity, and the fourth body moves under the gravitational influence of the three heavy bodies, but without affecting them. We start by finding the triangular central configurations of the three heavy bodies; since three bodies are oblate, the triangle is scalene, rather than equilateral as in the point mass case. We assume that the three heavy bodies are in such a central configuration and we perform a Hill's approximation of the equations of motion describing the dynamics of the infinitesimal body in a neighborhood of the oblate body. Through the use of Hill's variables and a limiting procedure, this approximation amounts to sending the two other bodies to infinity. Finally, for the Hill approximation, we find the equilibrium points of the infinitesimal body and determine their stability. As a motivating example, we consider the dynamics of the moonlet Skamandrios of Jupiter's Trojan asteroid Hektor. (based on joint work with J. Burgos-Garcia, A. Celletti, C. Gales and M. Gidea) (Received January 27, 2020)

1156-70-417 **Diogo Pinheiro*** (dpinheiro@brooklyn.cuny.edu), Department of Mathematics, CUNY Brooklyn College, 2900 Bedford Avenue, Brooklyn, NY 11210. *On the motion of a charged particle interacting with a lattice of fixed charges under the influence of a uniform magnetic field.* Preliminary report.

We will discuss some qualitative properties of the motion of a charged particle interacting with a planar lattice of fixed charges under the influence of a uniform magnetic field orthogonal to the plane where the motion takes place. This dynamical system is Hamiltonian with two degrees of freedom. If the interaction potential is of Coulomb type, we prove that, for suitable regime of parameters, there are invariant subsets on which the system contains a suspension of a subshift of finite type with positive entropy. As a consequence, we obtain that the Hamiltonian system is not integrable. (Received January 28, 2020)

74 ► Mechanics of deformable solids

1156-74-415 **Olalekan A Babaniyi*** (obsma@rit.edu), 84 Lomb Memorial Drive, Rochester, NY 14623, and **Assad A Oberai** and **Paul E Barbone**. *Direct error in constitutive equations formulation for inverse diffusion and elasticity problems.*

We consider an inverse diffusion problem: Given ϕ, find κ such that $\nabla\cdot(\kappa\nabla\phi) = 0$. Stable variational formulations exist for this problem treated as a first-order hyperbolic system for κ. The smoothness requirements for these approaches are overly restrictive, however, and are ill-suited to the inverse problem. We present a new mixed variational formulation for the inverse diffusion problem that allows the solution and derivatives of the data to be in $L(\Omega)$. The formulation is a quadratic optimization problem based on minimizing the error in constitutive equations, and can be solved directly (ie. without iterations) for the unknown parameters. We prove that this formulation is well-posed for inverse diffusion with mild conditions on the data. We show through computational examples that a stabilized Galerkin discretization converges with mesh refinement. Finally, we extend the formulation to inverse elasticity and show promising computational results in this system. (Received January 28, 2020)

76 ► Fluid mechanics

1156-76-79 **Yuanyuan Feng*** (yzf58@psu.edu), State College, PA 16802, and **Gautam Iyer**, **Jean-Luc Thiffeault** and **Yu Feng**. *Dissipation enhancement by mixing and suppression of phase separation.*

We quantitatively study the interaction between diffusion and mixing in both the continuous, and discrete time setting. In discrete time, we consider a mixing dynamical system interposed with diffusion. In continuous time, we consider the advection diffusion equation where the advecting vector field is assumed to be sufficiently mixing. We then study the addictive Cahn-Hilliard equation. We prove that when the flow is sufficiently mixed, phase separation will be prevented. (Received January 12, 2020)

81 ► Quantum theory

1156-81-86 **Jeffrey H Boyd*** (jeffreyhboyd@gmail.com), 57 Bethany Woods Road, BETHANY, CT 06524. *A new kind of wave that preserves quantum math, explains the double slit experiment and gets rid of the weirdness from quantum experiments.*

Although quantum mechanics is the most successful of all sciences, it has been known for more than a century that something is wrong with it. We present the Theory of Elementary Waves (TEW) that preserves the strengths of the Schrődinger equation while getting rid of the "weirdness" from quantum experiments. A TEW wave involves a zero energy wave moving to the left: Ψ_L = A $exp(-i(k_L x_L - \omega t))$. Meanwhile a Schrődinger wave packet moves to the right. $E_R\Psi_R = -i\hbar\partial\Psi_R/\partial t = -\hbar^2/2m\partial^2\Psi_R/\partial x_R^2 + u\Psi_R$. These are two aspects of the same thing. If we graph a wave packet moving to the right, in what direction is the line (the sine wave of which it is composed) moving? Although it is usually said to be stationary, we picture the line as being a substrate that is moving to the left, like a left-flowing river on the surface of which is a wave packet moving to the right. With this approach the quantum world is not weird, but is similar to the classical world of everyday experience. First the line, representing the Elementary Wave, moves from detector to particle source. A particle then attaches to the Schrődinger wave packet traveling from the source to the detector. (Received January 13, 2020)

1156-81-101 **Nicolas Crampé**, **Rafael I Nepomechie** and **Luc Vinet*** (vinet@umontreal.ca), Centre Recherches Mathématiques (CRM), Université de Montréal, Montreal, Quebec H3C 3J7, Canada. *Free-Fermion entanglement and orthogonal polynomials.*

We present a construction for a tridiagonal matrix T that commutes with the hopping matrix for the entanglement Hamiltonian $\mathcal{H}$ of open finite free-Fermion chains associated with families of discrete orthogonal polynomials. It is based on the notion of algebraic Heun operator attached to bispectral problems, and the parallel between entanglement studies and the theory of time and band limiting. As an example, we consider Fermionic chains related to the finite Chebychev polynomials. (Received January 14, 2020)

1156-81-206 **Nobuaki Obata*** (obata@tohoku.ac.jp), Sendai, 980-8579, Japan. *Supports of Spectral Measures of Multivariate Orthogonal Polynomials.*

The concept of quantum decomposition, one of the significant characteristics of quantum probability, is essential to link classical and quantum probability theories. For one variable case the connection between orthogonal polynomials and one-mode interacting Fock spaces has been well established, where the analytic methods like Stieltjes transform, continued fraction, Nevanlinna functions and so forth are important. On the other hand, the quantum probability approach to multivariate orthogonal polynomials lacks the method of investigating analytic properties of spectral measures. As preliminary consideration we introduce the concept of deficiency rank of Jacobi operator and derive relation to the supports of spectral measures. We also give some motivating examples. This presentation is mostly based on the recent joint work with A. Dhahri and H. J. Yoo [JMAA 485 (2020)]. (Received January 23, 2020)

1156-81-234 **Kristina VanDusen*** (kvanduse@tulane.edu), c/o Mathematics Department, Tulane University, 6823 St. Charles Ave, New Orleans, LA 70118. *A Survey of Negative Dimensional Integration Techniques.* Preliminary report.

In 1987, Halliday and Ricotta introduced the method of negative dimensional integration to evaluate loop integrals. This technique reduced a complicated integral problem to the problem of solving a system of linear equations. However, this technique still required tedious calculations for multiloop integrals. I will give an overview of negative dimensional integration and discuss some of the refinements to the method which have simplified calculations. In particular, I will discuss the method of brackets which has been extended beyond loop integrals to integrals over the half line, such as Wallis' formula. (Received January 24, 2020)

1156-81-294 **Mihai Stoiciu*** (mstoiciu@williams.edu), Williams College - 33 Stetson Ct, Department of Mathematics and Statistics, Williamstown, MA 01267. *Matrix Models for the Circular Beta Ensemble.* Preliminary report.

We consider various families of random orthogonal polynomials on the unit circle and their corresponding CMV and Joye matrices. We investigate the distribution of the eigenvalues of these random unitary matrices and use them to construct matrix models for the circular beta ensemble. We also study spectral properties of these matrix models for beta approaching zero and infinity. (Received January 26, 2020)

1156-81-344 **Radhakrishnan Balu***, 6509 Great drum Circle, Columbia, MD 21044. *Covariant Fermionic Quantum Stochastic Calculus.* Preliminary report.

Abstract: We develop covariant fermionic quantum stochastic calculus and derive the ito rule. As an application of the formalism we discuss the example of space-like particle forming fermionic ensembles that is relativistically covariant. (Received January 27, 2020)

1156-81-351 **Tamar Friedmann*** (tfriedma@colby.edu), **Carl R Hagen** and **Quincy Webb**. *Obtaining Infinite Product Formulas for Irrational Numbers from Quantum Mechanics.*

I will discuss joint work with C. Hagen on the derivation of the Wallis product formula for π from a variational computation of the spectrum of the hydrogen atom. I will also discuss joint work with Q. Webb in which we generalize said derivation to obtain infinite product formulas for other irrational numbers using the same technique. (Received January 27, 2020)

1156-81-365 **Alexander Teplyaev***, Department of Mathematics, University of Connecticut, Storrs, CT 06269-1009. *Spectrum on self-similar and fractal graphs.*

The talk will deal with spectral properties of the Laplacian on self-similar and fractal graphs. It will be discussed when spectrum is singularly continuous or pure point with localized eigenfunctions. As examples one can consider the Sierpinski lattice, one dimensional weighted graphs, and the Schreier graphs of the basilica self-similar group, which is a joint research with Antoni Brzoska, Courtney George, Samantha Jarvis, Luke G. Rogers. (Received January 27, 2020)

1156-81-419 **Adriana Scanteianu*** (adriana.scanteianu@gmail.com), 76 Plum St, New Brunswick, NJ 08901, and **Xiangyue Wang**, Newell Apts, New Brunswick, NJ 08901. *Particle Trajectories for Compton Scattering in One Space Dimension.* Preliminary report.

We review the work of Kiessling, Lienert and Tahvildar-Zadeh on relativistic interacting dynamics of a two-body quantum-mechanical system consisting of one electron and one photon in one space dimension. Our framework treats both the electron and the photon as relativistic quantum objects whose motions are guided by a two body multi-time wave function. We analyzed the dynamics of a single photon in one space dimension using the photon wave function proposed by Kiessling and Tahvildar-Zadeh, and a single electron using Dirac's relativistic wave equation for the electron. We then examined the two-body system interaction through a boundary condition on the coincidence set. We report on the results of numerical experiments we conducted to explore the role of various parameters in the theory and their influence on the trajectories of the two particles, and we comment on the possibility of applying this theory to the analysis of Compton scattering. (Received January 28, 2020)

86 ► Geophysics

1156-86-244 **Geoffrey Gebbie*** (ggebbie@whoi.edu), 266 Woods Hole Rd., MS #29, WHOI, Woods Hole, MA 02543. *Using Machine Learning to Determine Why Water Masses Describe the Ocean So Well.*

The ocean is spatially and dynamically complex, but only a handful of water masses are needed to successfully describe oceanic property distributions. An interesting clue about the organization of the complex system, namely that the sea surface fills the interior ocean according to a simple power law reminiscent of the 80-20 rule in economics. This shallow power law indicates that many thousand surface locations all significantly influence the interior ocean, in contrast to the traditional view of only a few major water masses. How many water masses are necessary to explain oceanic spatial variability: just a handful or many thousand? The optimal water masses will be the combinations of surface waters that recur the most often. In satellite image analysis, the extraction of the mixture of constituent materials in the visible spectrum, or spectral unmixing, is equivalent to this optimal water mass problem. Using an unsupervised machine learning algorithm, the ocean is spectrally unmixed and the most important water masses are identified and ranked by importance. We find the number of water masses needed to explain 95% of ocean property variability is intermediate to previous high and low estimates. (Received January 25, 2020)

91 ► Game theory, economics, social and behavioral sciences

1156-91-5 **Feng Fu*** (feng.fu@dartmouth.edu), 27 N. Main Street, Hanover, NH 03755. *Coevolution of Homophily and Cooperation in Dynamic Social Networks.*

Social interaction networks exist at many scales of biological organization, from the level of multicellular aggregates to human societies. These networks often have an exquisite structure that promotes cooperation. The formation of social ties and cooperative interactions is endogenously driven by homophily, i.e., the phenomenon that "birds of a feather flock together." Individuals tend to bond with and cooperate with someone else who resembles themselves. In this talk, I will present my recent work on studying coevolutionary dynamics of homophily and cooperation in social networks. These results may help us understand how phenotypic similarity leads to social ties ("structure") and cooperation (adaptive "function") of social networks. (Received September 03, 2019)

1156-91-302 **Naghmeh Momeni*** (nmomeni@mit.edu), 100 Main Street, E62-366, Cambridge, MA 02142. *Conjoining uncooperative societies facilitates evolution of cooperation.*

Evolution occurs in populations of reproducing individuals. The structure of a population can affect which traits evolve and the emergence and maintenance of cooperation. Here, we study the evolutionary dynamics of cooperation in fragmented societies, and show that conjoining segregated cooperation-inhibiting groups, if done properly, rescues the fate of collective cooperation. We highlight the essential role of intergroup ties, which sew the patches of the social network together and facilitate cooperation. We point out several examples of this phenomenon in actual settings. We explore random and non-random graphs, as well as empirical networks. In many cases, we find a marked reduction of the critical benefit-to-cost ratio needed for sustaining cooperation.

Our finding gives hope that the increasing worldwide connectivity, if managed properly, can promote global cooperation. (Received January 27, 2020)

1156-91-309 **Philip S. Chodrow*** (pchodrow@mit.edu) and **Peter J. Mucha** (mucha@unc.edu). *Coevolutionary Graphs: Models, Dynamics, and Approximations.*

Coevolutionary graphs are simple, mechanistic models in which node states and graph topology are intertwined. The rich space of possible dynamics includes opinion-updates, epidemic containment, and evolutionary games. Despite their simplicity, this class of models tends to exhibit a rich class of behaviors. These include disconnected homogeneous subgraphs – reminiscent of the formation of "echo-chambers" or geographically isolated speciation – as well as persistent, interacting heterogeneity. These regimes are usually separated by a critical point, whose calculation has been a topic of intense study over the last fifteen years. We will give a brief survey of this class of literature and discuss some new results for several variants of an opinion-based model. (Received January 27, 2020)

1156-91-413 **Olivia J. Chu***, ochu@princeton.edu. *Evolutionary and Behavioral Dynamics in Group-Structured Populations.* Preliminary report.

It is well known that population structure is crucial to a system's dynamics. In human populations, group memberships are critical. Humans often meet and interact with each other due to common group memberships. There exist network-based models to study human dynamics, but they generally do not allow for the multiple group affiliations that we see in real societies. In this talk, I will discuss a model that explores how behaviors influence how membership in multiple groups materializes in the first place, as well as the dynamics that arise out of interactions in an evolutionary game theory setting in which individuals are already distributed across groups. (Received January 28, 2020)

1156-91-416 **Qi Su*** (qisu1991@gmail.com), University of Pennsylvania, Philadelphia, PA 19104, **Alex McAvoy**, 02138, and **Joshua B. Plotkin**, 19104. *Evolutionary dynamics in multilayer structured populations.*

In many real-world systems, individuals are often situated in multiple interaction scenarios, such as online chat and physical contact, interactions between relatives and between colleagues. The interacting results in different scenarios altogether decide one's ability to survive and compete. A natural question is: how does interaction in one scenario influence the dynamics in other situations? We consider a model of evolutionary game dynamics in multilayer structured populations, where the population structure is described by a multilayer network and each layer represents an interaction scenario. Individuals play games within each layer and accumulate payoffs from all layers. Within this model, we study the evolution of cooperation and derive an analytical condition to predict when selection favors cooperation replacing defection. We find that compared with a single-layer structured population, introducing the second layer can promote cooperation replacing defection. Even if selection opposes cooperation replacing defection in each individual layer, coupling them can promote the replacement in both layers. Our results suggest that the correlation of different interaction scenarios can promote the evolution of cooperation. (Received January 28, 2020)

92 ► Biology and other natural sciences

1156-92-76 **Christine Sample*** (samplec@emmanuel.edu), Emmanuel College, 400 Fenway, Boston, MA 02115. *Quantifying the contribution of habitats and pathways to spatially structured populations.*

Environmental disturbance and management differentially affect populations. Such consequences are difficult to quantify for spatially structured populations because changes in one location carry through to other areas due to species movement. Yet the ability to assess the importance of habitats and pathways is an essential part of making management decisions. By modeling spatial structure as a network and population dynamics with a discrete-time matrix model, we develop two metrics that measure the contribution of habitats and pathways. Both approaches allow for any number of classes, unlimited number of seasons within the annual cycle, and for density-dependent parameters. The first is a generalized per-capita contribution metric, the C-metric, for classifying habitats and pathways as sources or sinks. The second, the G-metric, estimates the demographic contribution of a habitat or pathway to network-wide population growth in the face of environmental change. The G-metric quantifies effects of modifying vital rates in perturbation experiments, which may range from small perturbations to complete habitat or pathway removal. We demonstrate the metrics using case studies of eastern monarch butterflies and elk of the Greater Yellowstone Ecosystem. (Received January 12, 2020)

1156-92-153 **Igor V. Erovenko***, igor@uncg.edu, and **Johann Bauer**, **Mark Broom**, **Karan Pattni** and **Jan Rychtar**. *The effect of network topology on optimal exploration strategies and the evolution of cooperation in a mobile population.*

We model a mobile population interacting over an underlying spatial structure using a Markov movement model. Interactions take the form of public goods games, and can feature an arbitrary group size. Individuals choose strategically to remain at their current location or to move to a neighboring location, depending upon their exploration strategy and the current composition of their group. This builds upon previous work where the underlying structure was a complete graph (i.e., there was effectively no structure). Here, we consider alternative network structures and a wider variety of, mainly larger, populations. Previously, we had found when cooperation could evolve, depending upon the values of a range of population parameters. In our current work, we see that the complete graph considered before promotes stability, with populations of cooperators or defectors being relatively hard to replace. By contrast, the star graph promotes instability, and often neither type of population can resist replacement. We discuss potential reasons for this in terms of network topology. (Received January 21, 2020)

1156-92-162 **Alexandru Hening*** (alexandru.hening@tufts.edu), 503 Boston Ave, Medford, MA 02155. *Coexistence, extinction, and optimal harvesting in discrete-time stochastic population models.*

We analyze the long term behavior of interacting populations which can be controlled through harvesting. The dynamics is assumed to be discrete in time and stochastic due to the effect of environmental fluctuations. We present extinction and coexistence criteria when there are one or two interacting species. We then use these tools in order to see when harvesting leads to extinction or persistence of species, as well as what the optimal harvesting strategies, which maximize the expected long term yield, look like. (Received January 22, 2020)

1156-92-183 **Pauline van den Driessche*** (pvdd@math.uvic.ca). *Age structured discrete time infectious disease models.*

A discrete time model that includes age structure is formulated for the spread of a disease. For a Ricker recruitment function, the population demographic dynamics may persist on a periodic k-cycle. When disease is introduced, an extension of the next generation matrix method is used to calculate the basic reproduction number $\mathcal{R}_0$ for disease spread. Numerical simulations for a model of infectious salmon anaemia virus causing significant mortality show that, for $\mathcal{R}_0 > 1$, the demographic dynamics do not in general drive the disease dynamics. (Received January 22, 2020)

1156-92-212 **Daniel B Cooney*** (dcooney@math.princeton.edu), Program in Applied and Computational Math, Fine Hall, Washington Road, Floor 2, Princeton, NJ 08540. *Assortment and Reciprocity Mechanisms for Promotion of Cooperation in a Model of Multilevel Selection.*

In the study of the evolution of cooperation, many mechanisms have been proposed to help overcome the self-interested cheating that is individually optimal in he Prisoners' Dilemma and other social dilemmas. These mechanisms include assortative or networked social interactions, reciprocity rules to establish cooperation as a social norm, or simultaneous competition between individuals favoring cheaters and competition between groups favoring cooperators. Here, we build on recent mathematical tools describing the dynamics of multilevel selection to consider the role that assortment and reciprocity mechanisms play in facilitating cooperation in concert with multilevel selection. We explore a deterministic partial differential equation variant of the replicator equation which the effects of within-group and between-group competition, and we demonstrate the synergistic effects between population structure within groups and the competitive ability of cooperative groups when groups compete according to collective payoff. (Received January 24, 2020)

1156-92-226 **Mauricio Santillana*** (msantill@g.harvard.edu). *Data driven machine learning approaches to monitor and predict events in healthcare. From population-level disease outbreaks to patient-level monitoring.*

I will describe data-driven machine learning methodologies that leverage Internet-based information from search engines, twitter microblogs, crowd-sourced disease surveillance systems, electronic medical records, and weather information to successfully monitor and forecast disease outbreaks in multiple locations around the globe in near real-time. I will also present data-driven machine learning methodologies that leverage continuous-in-time information coming from bedside monitors in Intensive Care Units (ICU) to help improve patients' health outcomes and reduce hospital costs. (Received January 24, 2020)

1156-92-259 **Benjamin Allen*** (allenb@emmanuel.edu), 400 The Fendway, Boston, MA 02115.
Evolution in spatially structured populations: Drift, selection, and games.

The spatial structure of a population has many consequences for its evolution. These consequences can be modeled mathematically by representing population structure as a weighted graph. The graph structure is known to affect the probability for a new mutation to become fixed in the population. Computing this fixation probability, however, is difficult for an arbitrary weighted graph. I will present new methods for computing evolutionary dynamics in spatially structured populations, in three cases ordered by increasing complexity: (i) neutral drift, in which mutations have no effect on fitness; (ii) constant selection, in which mutations have a fixed fitness advantage or disadvantage, and (iii) games, in which the fitness of a mutation (i.e. a novel strategy) depends on the strategies employed by others. A common theme will be the use of coalescing random walks to calculate fixation probabilities under weak selection. (Received January 25, 2020)

1156-92-305 **Jane M Heffernan*** (jmheffer@yorku.ca), 4700 Keele St, Toronto, Ontario, Toronto, ON M3J 1P3, Canada. *Measles Elimination.* Preliminary report.

It is well-known mathematical modelling result that, to achieve herd immunity against measles infection in a population, a $92-95\%$ vaccine coverage needs to be achieved (critical coverage is equal to $1-1/R_0$ where $R_0 = 12$ - 18). It has been generally accepted that the measles vaccine confers life long immunity (with 95% efficacy). However, measles continues to cause disease in countries with $92-95\%$ coverage. Infection has also been documented in some individuals with measles vaccine history. Some recent systematic reviews on waning immunity from measles vaccine (conducted by Public Health Ontario) have shown some evidence of measles vaccine-induced waning immunity. To better understand the effects of these findings we conduct a mathematical modelling study on measles vaccine waning immunity. Our model includes waning and boosting of immunity and allow individuals that are infected with measles to gain lifelong immunity. Models with and without age structured are utilized. (Received January 27, 2020)

1156-92-312 **Carina Curto**, **Elizabeth Gross**, **Jack Jeffries**, **Katherine Morrison**, **Mohamed Omar**, **Zvi Rosen**, **Anne Shiu** and **Nora Youngs***, nora.youngs@colby.edu. *What makes a neural code convex?*

Neural codes allow the brain to represent, process, and store information about the world. Combinatorial codes, comprised of binary patterns of neural activity, encode information via the collective behavior of populations of neurons. A code is called convex if its codewords correspond to regions defined by an arrangement of convex open sets in Euclidean space. Convex codes have been observed experimentally in many brain areas, including sensory cortices and the hippocampus, where neurons exhibit convex receptive fields. What makes a neural code convex? That is, how can we tell from the intrinsic structure of a code if there exists a corresponding arrangement of convex open sets? In this work, we provide a complete characterization of local obstructions to convexity. This motivates us to define max intersection-complete codes, a family guaranteed to have no local obstructions. (Received January 27, 2020)

1156-92-314 **Elizabeth Gross**, **Nida Obatake** and **Nora Youngs***, nora.youngs@colby.edu. *Neural ideals and stimulus space visualization.*

One major problem in neuroscience is to understand how the brain uses neural activity to form representations of the external world. It is known that combinatorial information in the firing patterns of neurons often reflects important features of the space of stimuli which generated them. How can we efficiently extract such information? In this work, we look at the problem of algorithmically drawing Euler diagrams representing the stimulus space of neural codes using two polynomial ideals: the neural ideal, a pseudomonomial ideal; and the neural toric ideal, a binomial ideal. In particular, we study how these objects are related to the theory of piercings in information visualization, and we show how minimal generating sets of the ideals reveal whether or not a code is 0, 1, or 2-inductively pierced. (Received January 27, 2020)

1156-92-323 **Glenn S Young*** (gyoung19@kennesaw.edu) and **Andrew Belmonte**. *Fast Cheater Migration Stabilizes Coexistence in a Public Goods Dilemma on Networks.*

Cooperation is frequently considered an unsustainable strategy: if an entire population is cooperating, each individual can increase its overall fitness by choosing not to cooperate, thereby still receiving all the benefit of its cooperating neighbors while no longer expending its own energy. Observable cooperation in naturally-occurring public goods games is consequently of great interest, as such systems offer insight into both the emergence and sustainability of cooperation. Here we consider a population that obeys a public goods game on a network of discrete regions (that we call nests), between any two of which individuals are free to migrate. We construct a system of piecewise-smooth ordinary differential equations that couple the within-nest population dynamics

and the between nest migratory dynamics. Through a combination of analytical and numerical methods, we show that if the workers within the population migrate sufficiently fast relative to the cheaters, the network loses stability first through a Hopf bifurcation, then a torus bifurcation, after which one or more nests collapse. Our results indicate that fast moving cheaters can act to stabilize worker-cheater coexistence within network that would otherwise collapse. (Received January 27, 2020)

1156-92-350 **Oana Carja***, 5000 Forbes Avenue, GHC 7413, Pittsburgh, PA 15213. *Evolution on networks (The role of spatial structure in shaping evolutionary dynamics).*

There is a large literature of population genetic studies on the role of spatially changing selection in shaping evolutionary dynamics, starting from the classic 1975 paper of Slatkin and Maruyama. Models usually only assume a few demes and symmetrical structures, simple topologies that can be embedded into two-dimensional continuous Euclidean spaces. In many cases, these symmetrical spatial structures do not change fixation probabilities of new mutations relative to well-mixed populations and fail to capture the complex interactions and the variance in local selection pressure present in natural populations. We use the mathematical formalism of the Moran process on graphs to explore how spatial structure shapes the composition of populations and the outcome of the evolutionary process. (Received January 27, 2020)

1156-92-370 **Abba B Gumel*** (agumel@asu.edu), Arizona State University, Tempe, AZ 85226, and **Iboi Enahoro** and **Steffen Eikenberry**. *Insecticide resistance and the quest for malaria eradication.*

Recent dramatic declines in global malaria burden and mortality can be largely attributed to the large-scale deployment of insecticidal-based measures, namely long-lasting insecticidal nets (LLINs) and indoor residual spraying (IRS). However, the sustainability of these gains, and the feasibility of global malaria eradication by 2040, may be affected by increasing insecticide resistance among the Anopheles malaria vector. This talk is based on the design and use of a new mechanistic model, which incorporates the full, weather-dependent mosquito lifecycle, to assess the population-level impact of the large-scale use of LLINs, under different levels of Anopheles pyrethroid insecticide resistance, on malaria transmission dynamics and control in a community. (Received January 27, 2020)

1156-92-374 **Samuel Cho*** (sc43@princeton.edu) and **Simon A. Levin**. *Mean-field game analysis of epidemiological model with adaptive human behavior.*

Managing epidemics with public policy is to motivate people to behave differently for the societal good. We focus on contact between people which drives the spread of many diseases, but also is inherently tied to day-to-day utility.

We pose a mathematical model of this problem as a mean-field game. In addition to the SIR dynamics, each subpopulation chooses a contact rate to maximize its value function, considering the tradeoff between present and future utility. We can compute the mean-field equilibrium, which is the fixed point of the coupled system of Bellman equation and the SIR differential equations.

We compare the mean-field equilibrium with the socially optimal solution, as well as analyze the resulting policy implications to managing an epidemic. (Received January 28, 2020)

1156-92-379 **Zhisheng Shuai*** (zhisheng.shuai@ucf.edu), University of Central Florida, Orlando, FL 32816. *Population Persistence and Extinction in a Heterogeneous Environment.* Preliminary report.

Spatial heterogeneity and spatial movement play an important role in population dynamics. Such dispersal movement in a discrete heterogeneous environment can be represented by a connectivity matrix and the corresponding digraph/network, and the resulting mathematical model becomes a coupled dynamical system on network. Our studies focus on how coupling strength and topological structure of the dispersal network jointly affect the population dynamics. Specifically, our recent results, jointly with Shanshan Chen (Harbin Institute of Technology), Junping Shi (College of William & Mary) and Yixiang Wu (Middle Tennessee State University), highlight the dichotomy of population persistence vs extinction (infectious disease invasion vs eradication) with respect to the dispersal strength. (Received January 28, 2020)

1156-92-384 **Elizabeth Tripp*** (elizabeth.a.tripp.gr@dartmouth.edu), **Feng Fu** and **Scott Pauls**. *Evolutionary Kuramoto Dynamics.*

Understanding the synchronization of systems of coupled oscillators has a rich history in the study of dynamical systems and applications in numerous fields, where decades of work have demonstrated the complex interplay between the properties of these systems and the oscillators' ability to synchronize. We bring the tools of evolutionary game theory (EGT) to the study of coupled oscillatory systems, with the neurons of the suprachiasmatic

nucleus as a motivating example. We cast the oscillatory system as a collection of agents, one for each oscillator, that play a game with one another where the payoffs are based on their phases. We find that even one of the simplest models of these systems yields a rich diversity of outcomes linked to classical games - the prisoner's dilemma, the snowdrift game, etc. - based on the relative costs and benefits of synchronization in the organism. Within this complexity lies a simplicity as well, as we consistently find a simple condition, under various assumptions, between the cost and benefit parameters which leads to the overall synchronization within the population of agents. This simple framework opens the door to a plethora of mathematical and biological questions for future study. (Received January 28, 2020)

1156-92-390 **Chadi M Saad-Roy*** (csaadroy@princeton.edu). *Eco-evolutionary dynamics of pathogens.* Preliminary report.

Pathogens exhibit rich diversity due to a variety of factors. For example, pathogen infections can lead to immunity, which then leads to strong selection. For adequate disease control, it is thus crucial to understand both epidemiological and evolutionary dynamics of pathogens. In this talk, I will present recent work on pathogen eco-evolutionary dynamics, and will highlight the implications that these results have for control strategies. (Received January 28, 2020)

1156-92-401 **Lenore Cowen**, **Xiaozhe Hu** and **Junyuan Lin*** (junyuan.lin@lmu.edu), 1 LMU Drive, Department of Mathematics, Los Angeles, CA 90045, and **Yue Shen** and **Kaiyi Wu**. *Fast Graph-based Algorithms for Analyzing Protein-Protein Interaction Networks.* Preliminary report.

This research aims to predict proteins' functions from protein-protein interaction (PPI) networks. The PPI networks we study include physical and genetic interactions between proteins and the function labels of a subset proteins. This allows us to predict proteins' unknown functions based on the function labels of closely interacted "neighbors". First, we recall the diffusion state distance (DSD) metric, which sets appropriate distance to measure proteins' proximity. Fast algorithms, such as the unsmoothed aggregation algebraic multigrid method with random projections, are adopt to compute DSD efficiently. Based on the aforementioned method to compute DSD, we then introduce several graph-based methods to construct a k-nearest-neighbor (kNN) graph in DSD metric for function prediction. Our methods are based on random walks combining with random projections, and the computational cost is nearly optimal. Lastly, we demonstrate with some numerical results on the robustness and efficiency of the proposed graph-based algorithms. (Received January 28, 2020)

1156-92-405 **Alessandro Vespignani***, Northeastern University, Boston, MA 02115. *Modeling and forecast of epidemics in structured populations.*

The data science revolution is finally enabling the development of infectious disease models offering predictive tools in the area of health threats and emergencies. Analogous to meteorology, large-scale data-driven models of infectious diseases provide real- or near-real-time forecasts of the size of epidemics, their risk of spreading, and the dangers associated with uncontained disease outbreaks. These models are not only valuable because they predict where and how an epidemic might spread in the next few weeks, but also because they provide rationales and quantitative analysis to support public health decisions and intervention plans. At the same time, the non-incremental advance of the field presents a broad range challenges: algorithmic (multiscale constitutive equations, scalability, parallelization), real time integration of novel digital data stream (social networks, participatory platform for disease monitoring, human mobility etc.). I will review and discuss recent results and challenges in the area, ranging from applied analysis for public health practice to foundational computational and theoretical challenges. (Received January 28, 2020)

1156-92-411 **Elizabeth E Crone*** (elizabeth.crone@tufts.edu), Department of Biology, Tufts University, Medford, MA, **Natalie Z Kerr**, Department of Biology, Duke Unviersity, Durham, NC, **Rosemary L Malfi**, Biology Department, University of Massachusetts, Amherst, MA , and **Neal M Williams**, Department of Entomology, University of California, Davis, CA. *Transient dynamics in bumble bee colony growth.*

In structured populations of plants and animals, populations grow at a constant rate once they reach a stable stage distribution. Transient dynamics refer to how changes in stage structure affect short-term population dynamics. Although a number of ecological population models have explored properties of transient dynamics, their importance has rarely been tested in experiments. We propose that bumble bee colony growth is an excellent model system for studying transient dynamics of structured populations. We demonstrate this utility using colonies of the common western bumble bee, Bombus vosnesenskii. We explore empirically based models of transient dynamics and compare model predictions to experimental manipulation of resource pulses that affect

both colony growth rates and stage structure. Models and experiments produce broadly concordant results, suggesting a number of directions for future mathematical and empirical research. (Received January 28, 2020)

93 ► Systems theory; control

1156-93-73 **Allen Tannenbaum*** (arobertan@cs.stonybrook.edu), Department of Computer Science, New Computer Science Building, Stony Brook, NY 11794. *Generalized Curvature for the Modeling and Control of Complex Networks.*

The importance of studying properties of networks is manifest in diverse fields ranging from engineering, biology, physics, chemistry, neuroscience, and medicine. The functionality of networks with regard to performance, throughput, reliability and robustness is strongly linked to the underlying geometric and topological properties, which is the focus of the proposed research program. The fundamental mathematical abstraction of a network as a weighted graph brings to bear the tools of graph theory–a highly developed subject of mathematical research. But more importantly, recently proposed geometric notions of curvature on very general metric measure spaces, allow us to utilize a whole new set of tools and ideas that help quantify functionality and robustness of graphs. We will thus explore curvature, graph Laplacian, and Ricci flows over networks as well as study limits of stochastic networks as the number of nodes tends to infinity and how these relate to ideas from mean field games via the intrinsic network geometry.

This is joint work with Professor Tryphon T. Georgiou of UCI. (Received January 11, 2020)

94 ► Information and communication, circuits

1156-94-157 **Eric W Clarkson*** (clarksoneric@mac.com), 1230 N Cherry Ave, BSRL 245, Tucson, AZ 85721, and **Meredith K Kupinski** (meredith@optics.arizona.edu), 1630 E University Blvd, Meinel 645, Tucson, AZ 85724. *Quantifying the Loss of Information from Binning List-Mode Data.*

List-mode data is increasingly being used in SPECT and PET imaging, among other imaging modalities. However, there are still many imaging designs that effectively bin list-mode data before image reconstruction or other estimation tasks are performed. Intuitively, the binning operation should result in a loss of information. In this work we show that this is true for Fisher information and provide a computational method for quantifying the information loss. In the end we find that the information loss depends on three factors. The first factor is related to the smoothness of the mean data function for the list-mode data. The second factor is the actual object being imaged. Finally, the third factor is the binning scheme in relation to the other two factors. (Received January 21, 2020)

1156-94-275 **Weilin Li***, weilinli@cims.nyu.edu, and **Wenjing Liao** and **Albert Fannjiang**. *Super-resolution, subspace methods, and non-harmonic Fourier matrices.*

This talk is concerned with the super-resolution problem of recovering a discrete measure on the torus consisting of S atoms, given M consecutive noisy Fourier coefficients. Super-resolution recovery is sensitive to noise when the distance between two atoms is less than 1/M and many algorithms fail in this regime. In this talk, we connect this problem to the minimum singular value of non-harmonic Fourier matrices. New results for the latter are presented, and as consequences, we derive results regarding the super-resolution limit of subspace methods (namely, MUSIC and ESPRIT). These results rigorously establish the super-resolution phenomena of these algorithms that were empirically discovered long ago, and numerical results indicate that our bounds are sharp or nearly sharp. Information theoretic lower bounds imply that both algorithms are near optimal for super-resolution. Joint work with Albert Fannjiang and Wenjing Liao. (Received January 26, 2020)

1156-94-320 **Paul Hand*** (p.hand@northeastern.edu). *Generative Neural Networks as Priors for Compressive Sensing and Phase Retrieval.*

Recovering images from very few measurements is an important task in medical, biological, astronomical, and other image problems. Doing so requires assuming a model of what makes some images natural. Such a model is called an image prior. With the recent developments in machine learning, neural networks have been shown to provide efficient and effective priors for inverse problems arising in imaging. They can, for example, achieve comparable recovery quality as classical sparsity priors with 10x less data. In this talk, we will discuss recovery guarantees that have been proven under generative models. We will additionally see that these models provide the first tractable recovery algorithm with optimal sample complexity for the problem of compressive phase

retrieval under generic phaseless measurements. In contrast, under classical sparsity priors, the compressive phase retrieval problem is notoriously difficult and may even be computationally hard. The works discussed are in collaboration with Vladislav Voroninski, Wen Huang, Reinhard Heckel, and Oscar Leong. (Received January 27, 2020)

1156-94-409 **Maxim Derevyagin**, **Gerald Dunne**, **Gamal Mograby*** (gamal.mograby@uconn.edu) and **Alexander Teplyaev**. *Perfect Quantum State Transfer on Graphs and Fractals.*

Perfect quantum state transfer (denoted PQST) is a rare phenomenon, for which the construction of explicit examples remains rather non-trivial. Several one-dimensional cases, where PQST can be achieved, have been found in some XX-chains with inhomogeneous couplings. In this talk, we present our investigation of PQST in geometries beyond one-dimensional graphs. As a first result, we showed that PQST is possible on a large and diverse class of fractal-type diamond graphs. A significant interest in these graphs lies in the fact that their limit spaces constitute a family of fractals, which present different geometrical properties, including a wide range of Hausdorff and spectral dimensions. Moreover, we provide a tool used to construct a Hamiltonian achieving PQST on such graphs. As a second result, we showed that our approach works for an even larger class of graphs. Having constructed Hamiltonians achieving PQST, we also provide a complete description of their spectra. Our work is complementary to the work of M.Kempton, G.Lippner, and S-T. Yau. This talk is a part of joint work with M.Derevyagin, G.V.Dunne, and A.Teplyaev. (Received January 28, 2020)

97 ► Mathematics education

1156-97-3 **Irina Peterburgsky*** (ipeterburgsky@suffolk.edu). *Some classroom-tested strategies helping college students overcome math anxiety.*

Finding strategies and approaches to overcome students' anxiety, prejudices, and misconceptions of mathematics constitutes one of the most critical challenges in contemporary collegiate mathematics education. It is obvious that improvement of students' attitudes toward the subject and therefore success in learning, as a long-term process, directly and deeply depend on their emotional motivation in mathematics classes and also beyond, while preparing assignments, reviewing work accomplished, or exploring new material. In the present paper, we would like to share our classroom-tested strategies helping college students overcome math anxiety and approach mathematics with the sense of adventure and excitement. (Received May 28, 2019)

00 ► General

1157-00-119 **Guillaume Bal** and **Fatma Terzioglu*** (fterzioglu@uchicago.edu). *Uniqueness Criteria in Multi-Energy CT.*

Multi-Energy Computed Tomography (MECT) is a medical imaging modality aiming to reconstruct the spatial density of materials from the attenuation properties of probing x-rays. MECT measurements are modeled by a nonlinear mapping from the line integrals of the unknown densities of several materials to energy-weighted integrals corresponding to different x-ray source energy spectra. The material densities may thus be recovered by first reconstructing line integrals of the material densities from the available energy measurements; and then reconstructing densities from their line integrals. The second step is the standard linear x-ray CT problem whose invertibility is well-known, so this paper focuses on the first step. We show that MECT admits stable, global inversion provided that (a well-chosen linear transform of) the differential of the transform in the first step satisfies appropriate orientation constraints that makes it a P-matrix. We also derive global stability results for MECT in both determined and overdetermined cases. Numerical simulations based on commonly used materials and well accepted models of source energy spectra show that MECT is often either non-globally injective because it is non-injective locally, or globally injective as soon as it is locally injective. (Received January 22, 2020)

1157-00-309 **Sanjay L. Kumar*** (kumarsa4@msu.edu). *Families of links in S^3 with homeomorphic complements to fundamental shadow links.*

In 2015, Chen and Yang provided evidence that the asymptotics of the Turaev-Viro invariant of a hyperbolic 3-manifold evaluated at the root of unity $\exp\left(\frac{2\pi i}{r}\right)$ have growth rates given by the hyperbolic volume. In 2018, Belletti, Detcherry, Kalfagianni, and Yang prove this volume conjecture for the complements of an infinite family of hyperbolic links in connect sums of $S^1 \times S^2$ known as fundamental shadow links. In this talk, I will present examples of links in S^3 satisfying the volume conjecture through homeomorphisms with complements of fundamental shadow links. (Received February 01, 2020)

1157-00-477 **Yaoyu Zhang*** (yaoyu@ias.edu), **Zhiqin John Xu**, **Tao Luo** and **Zheng Ma**. *A Fourier Analysis Perspective of Training Dynamics of Deep Neural Networks.*

This talk focuses on a general phenomenon of "Frequency-Principle" that DNNs often fit target functions from low to high frequencies during the training. I will present empirical evidences on real datasets and deep networks of different settings. Theoretically, I will talk about a "Linear Frequency-Principle" dynamics derived from two-layer neural networks at the NTK regime, which quantifies the priority of different frequencies during the training and gives rise to an apriori generalization error bound based on the frequency composition of the target function. (Received February 03, 2020)

01 ► History and biography

1157-01-1 **Benjamin Joseph Elkins*** (benjaminjosephelkins@gmail.com), 1512 Jersey Court, Buffalo Grove, IL 60089. *Felix Hausdorff's Das Raumproblem in plan English.* Preliminary report.

On July 4th, 1903, Felix Hausdorff was 35 years old; this paper presents the first English translation of Hausdorff's inaugural lecture, Das Raumproblem, upon receiving his full professorship at The University of Leipzig. Approaching its 116th anniversary, I surmise the complexity and abstractness of the vocabulary utilized has caused this seminal lecture to go untranslated. A solid understanding of Hausdorff's body of mathematical and philosophical contributions are necessary to approach this lecture. Hausdorff's metaphors, allegories and allusions utilized makes this little known work beautiful while simultaneously providing insight into Hausdorff's future mathematical work and beyond. With the field of mathematics ever expanding, younger students are being exposed to ideas that have their roots within Hausdorff's research and ensuring a sound foundation of the genesis of these ideas paramount to providing clear vision for the mathematicians that follow in the wake of today's findings. (Received May 13, 2019)

1157-01-261 **Donald A. Sokol*** (3471das@gmail.com), Donald A.Sokol, 9547Palisades Rd., Burr Ridge, IL 60527. *Plimpton 322: Where none has gone before.*

There is an ancient Babylonian clay tablet in the museum at Columbia University in New York City. It is identified as Plimpton 322, and contains information associated with the equation : $a^2 + b^2 = c^2$; known as the Pythagorean TheoremP.T.. The P.,T., since the days of Euclid has been known to be the result of an algorithm with 3 equations; one for each of a, c and b. A basic set of equations was used by the Babylonians to create the tablet;a variation of the basic equations was used by Euclid to create the algorithm in the Elements, as shown below; Babylonian Euclidean $a = 2x + yya = 2xy$, $c = a + x^2c = x^2 + y^2$, $b^2 = c^2 - a^2b^2 = x^2 - y^2$. Subsequently, there was a change in the Babylonian structure and b became $c-2y^2$, and Euclid added $-y$ to the basic equations to create his variation, thus leaving us to search for answers. The possible "who"and "when" of the changes cited above will be discussed. (Received January 30, 2020)

03 ► Mathematical logic and foundations

1157-03-117 **Samuel Braunfeld*** (sbraunf@umd.edu) and **Michael C. Laskowski** (mcl@math.umd.edu). *Counting siblings in universal theories.*

We say two structures are *siblings* if they are (not necessarily elementarily) bi-embeddable. The number of siblings, up to isomorphism, of any countable relational structure M is conjectured to be 1, $\aleph_0$, or $2^{\aleph_0}$. Using mutual algebraicity, we show that if M is not cellular, then it admits an age-preserving extension with $2^{\aleph_0}$ siblings. (Received January 22, 2020)

1157-03-177 **Omer Mermelstein*** (omer@math.wisc.edu). *Forking in the generic flat pregeometry.*

The theory of the generic flat pregeometry — a Fraïssé-Hrushovski limit of the class of all finite flat pregeoemtries — is ω-stable and not $\aleph_1$-categorical. In this talk we explain dependence and rank in this theory, and compare it to dependence in its ω-stable, non-$\aleph_1$-categorical sibling theories of lower rank: the theory of infinitely branching trees and the theory of a non-collapsed ab initio Hrushovski construction. (Received January 27, 2020)

1157-03-201 **Pablo Cubides-Kovacsics** and **Jinhe Ye***, 221 Hayes-Healy, University of Notre Dame, Notre Dame, IN 46556. *A model theoretic adic space.*

Working in the theory of algebraically closed valued fields, Hrushovski and Loeser used the space $\widehat{V}$ of generically stable types concentrating on V to study the topology of Berkovich analytification V^{an} of V. In this talk we will present an analogous construction which provides a model-theoretic counterpart $\widetilde{V}$ of the Huber's analytification of V. We show that the space $\widetilde{V}$ is strict pro-definable. Furthermore, we will discuss canonical liftings of the deformation retraction developed by Hrushovski and Loeser. (Received January 28, 2020)

1157-03-202 **Carl G. Jockusch** and **Paul E. Schupp*** (schupp@math.uiuc.edu). *Coarse computability, the density metric and the Hausdorff distance between Turing degrees.* Preliminary report.

Coarse computability studies how closely arbitrary sets of natural numbers can be approximated by computable sets. Define subsets A and B of $\mathbb{N}$ to be *coarsely similar*, written $Aapprox_cB$, if their symmetric difference has density 0. Let $\mathcal{S} = \mathcal{P}(\mathbb{N})/\approx_c$. There is a natural density metric on $\mathcal{S}$: The distance between two sets is the upper density of their symmetric difference.

The space $\mathcal{S}$ is very interesting. While neither separable nor compact, it is complete and contractible. Indeed, $\mathcal{S}$ is a geodesic metric space such that there are uncountably many different geodesics between any two distinct point of $\mathcal{S}$. Note that $\mathcal{S}$ is hyperbolic in the sense of Gromov.

The *core* $\kappa(\boldsymbol{d})$ of a Turing degree is the family of all subsets of $\mathbb{N}$ computable from $\boldsymbol{d}$. The closure $\overline{\boldsymbol{d}}$ of $\boldsymbol{d}$ is the closure of $\kappa(\boldsymbol{d})$ in the space $\mathcal{S}$. The distance between two Turing degrees is the Hausdorff distance between their closures in the space $\mathcal{S}$. It turns out that the distance between two Turing degrees is either $0, 1/2$ or 1. (Received January 28, 2020)

1157-03-232 **Neer Bhardwaj*** (nbhard4@illinois.edu). *The Pila-Wilkie Theorem: A variant of the proof.*

In joint work with Lou van den Dries, we prove the the Pila-Wilkie theorem, using o-minimal cell decomposition in a more powerful manner to simplify the deduction from the main ingredients of the proof in the original paper. (Received January 29, 2020)

1157-03-268 **Gareth Owen Jones***, Department of Mathematics, University of Manchester, Oxford Road, Manchester, M13 9PL, United Kingdom. *Powers are easy to avoid.*

Suppose that a set is definable in the expansion of the real field by restricted analytic functions, and is also definable in the expansion of the real field by the restricted exponential function together with all real power functions. Then the set is definable using just the restricted exponential function. So additional exponents can be avoided. I'll discuss the general result behind this, and how it can be seen as a polynomially bounded version of an old conjecture of van den Dries and Miller. This is joint work with Olivier Le Gal. (Received January 31, 2020)

1157-03-297 **Will Johnson** (willij6@berkeley.edu), **Chieu-Minh Tran*** (mtran6@nd.edu), **Erik Walsberg** (ewalsber@uci.edu) and **Vincent Yinhe Ye** (jinhe.ye.15@nd.edu). *Canonical topologies over fields.* Preliminary report.

We obtain common characterizations of the Zariski topologies over algebraically closed fields, the Euclidean topologies over real closed fields, and the valuation topologies over p-adically closed fields. We then show that some of these characterizations coincide in more general settings. In particular, this provides us with canonical topologies over pseudo-finite fields. (Received February 01, 2020)

1157-03-330 **Alexander S. Kechris**, **Maciej Malicki**, **Aristotelis Panagiotopoulos** and **Joseph Zielinski*** (zielinski.math@gmail.com). *Orbit equivalence relations of some classes of non-locally compact Polish groups.*

By results of A.S. Kechris, whenever a locally compact Polish group acts continuously on a Polish space, the orbit equivalence relation of the action is essentially countable—that is, Borel reducible to the orbit equivalence relation of an action of a countable group. It is unknown if this characterizes the locally compact Polish groups.

S. Solecki, after proving an analogous characterization for smooth actions of compact Polish groups, showed this to be true in the case where the group, G, is the additive group of a separable Banach space. The characterization also holds for abelian pro-countable groups, by results of M. Malicki. We discuss recent work on this problem, including an extension of this characterization to some important classes of Polish groups.

This is joint work with A.S. Kechris, M. Malicki, and A. Panagiotopoulos. (Received February 01, 2020)

1157-03-368 **Elliot Kaplan*** (eakapla2@illinois.edu). *Logarithmic Hyperseries and Hyperserial Fields.*

In joint work with Lou van den Dries and Joris van der Hoeven, we constructed the field $\mathbb{L}$ of logarithmic hyperseries. This is a proper class-sized ordered differential field which is also equipped with a composition and hyperlogarithm functions (which act like transfinite iterates of a logarithm function). In subsequent joint work with Vincent Bagayoko and Joris van der Hoeven, we introduced the more general notion of a hyperserial field (of type **On**). These proper class-sized fields admit an external composition over the field of logarithmic hyperseries. We showed that each hyperserial field has a minimal hyperserial field extension in which each hyperlogarithm is bijective on the class of positive infinite elements. In this talk, I will discuss both of these works and their connection with transseries, logarithmic transseries, and the surreal numbers. (Received February 02, 2020)

1157-03-369 **Alexi Block Gorman*** (atb2@illinois.edu). *Companionability dichotomy for the expansion of an o-minimal theory by a dense, divisible subgroup.* Preliminary report.

Recent works on companionability have established natural and broad criteria concerning the existence of model companions and the preservation of certain neostability properties when passing to the model companion. In this talk, I restrict my attention to the o-minimal setting because in doing so we can isolate the sort of necessary and sufficient condition that is elusive in the more general settings that have been considered. The central result is a full characterization for when the expansion of a complete o-minimal theory by a unary predicate that picks out a dense, divisble subgroup has a model companion. We provide examples both in which the predicate is an additive subgroup, and where it is a mutliplicative subgroup. The o-minimal setting allows us to provide a full and geometric characterization for companionability, with a very elegant dividing line when the group operation is multiplication. We conclude with a brief discussion of neostability properties, and give examples that illustrate the lack of preservation for properties such as strong, NIP, and NTP_2, though there are also examples for which some or all three of those properties hold. (Received February 02, 2020)

1157-03-374 **Artem Chernikov*** (chernikov@math.ucla.edu) and **Kyle Gannon**. *Idempotent Keisler measures.* Preliminary report.

Recently a considerable work by many authors went into understanding translation-invariant Keisler measures in NIP groups, and a classification was given in Chernikov, Simon, "Definably amenable NIP groups", JAMS 2018. In this talk, we consider a larger class of idempotent Keisler measures. We provide a complete classification in

the stable case (they are precisely the invariant measures on type-definable subgroups) and discuss some partial results outside of stability. Joint work with Kyle Gannon. (Received February 02, 2020)

1157-03-427 **Ya'acov Peterzil** and **Sergei Starchenko***, Department of Mathematics, University of Notre Dame, Notre Dame, IN 46556. *On Hausdorff limits of o-minimal families in real tori.*

Let $\{X_s : x \in S\}$ be a family of subsets of $\mathbb{R}^n$ definable in some o-minimal expansion of the real field.

Let $\Gamma \subseteq \mathbb{R}^n$ be a lattice and $\pi : \mathbb{R}^n/\Gamma \to \mathbb{T}$ be the quotient map.

In this talk we consider Hausdorff limits of the family $\{\pi(X_s) : s \in S\}$ and formulate conditions when a Hausdorff limit is dense in $\mathbb{T}$. (Received February 03, 2020)

1157-03-436 **Rebecca Coulson*** (rebecca.coulson@westpoint.edu), United States Military Academy - West Point, Department of Mathematics, 609 Swift Road, West Point, NY 10996. *The Bipartite Diameter 3 Metrically Homogeneous Graphs of Generic Type: Their Ages and Their Almost Sure Theories.* Preliminary report.

The class of random graphs famously satisfies a zero-one law: every first-order sentence in the language of graphs is such that the proportion of finite graphs on n vertices which satisfy this sentence goes either to zero or to one as n goes to infinity. The "almost-sure" theory of the class of finite graphs matches the generic theory of its Fraisse limit - the Rado graph. Interestingly, the almost-sure theory of the class of finite triangle-free graphs does not match the theory of the generic triangle free graph. In this talk, we will discuss another class of graphs which are Fraisse limits defined by forbidden configurations, and we examine two such graphs in particular. We show that for one of the them, its generic theory does match the corresponding almost-sure theory, and that for the other, the generic theory does not match the corresponding almost-sure theory. (Received February 03, 2020)

1157-03-441 **Sylvia Carlisle*** (carlisle@rose-hulman.edu) and **C. Ward Henson**. *Endpoints in $\mathbb{R}$-trees.* Preliminary report.

An $\mathbb{R}$-tree is a metric space such that between any two points there is a unique geodesic segment. An $\mathbb{R}$-tree is richly branching if the set of points with at least 3 branches of a non-trivial length is dense. We consider bounded, pointed $\mathbb{R}$-trees as metric structures in an appropriate continuous signature. The theory $\mathrm{rb}\mathbb{R}\mathrm{T}_r$ of "richly branching" pointed $\mathbb{R}$-trees with radius r is the model companion of the theory of $\mathbb{R}$-trees of radius at most r. We take a model M of $\mathrm{rb}\mathbb{R}\mathrm{T}_r$ and consider the space of points at distance r from the basepoint. We show that the theory of such spaces and the theory of richly branching $\mathbb{R}$-trees are bi-interpretable, and outline results about the topology of spaces of endpoints in $\mathbb{R}$-trees. This is joint work with C. Ward Henson. (Received February 03, 2020)

1157-03-447 **Anand Pillay*** (apillay@nd.edu), Department of Mathematics, University of Notre Dame, Notre Dame, IN 46556. *Some model theory of proalgebraic groups.*

This is joint work with Michael Wibmer.

In analogy with a certain (co) model theory of profinite groups, developed by Cherlin, van den Dries, and Macintyre, and further developed by Chatzidakis (80's, 90's), we seek a similar theory for pro (affine) algebraic groups, or equivalently affine group schemes. The motivation for this includes the study of the absolute Picard-Vessiot group of a differential field. There are obstacles for more or less directly carrying over the profinite theory to the proalgebraic case, so we end up axiomatizing the category of representations, namely the associated Tannakiam category. The set up is somewhat complicated, and we prove some rather basic results such as that some natural classes of proalgebraic groups are elementary. (Received February 03, 2020)

1157-03-532 **Clifton Ealy*** (cf-ealy@wiu.edu), Department of Mathematics and Philosophy, Western Illinois University, 1 University Circle, Macomb, IL 61455, and **Deirdre Haskell** and **Pierre Simon**. *Residue field domination in some Henselian valued fields.*

The notion of the domination of a type by its stable part was introduced in the work of Haskell, Hrushovski, and Macpherson. In joint work with Haskell and Maříková, we showed that the key ingredient was not so much stability but the residue field, and we obtained similar results for real closed valued fields (where the residue field is unstable). Recently, with Haskell and Simon, we have generalized these results to Henselian valued fields of equicharacteristic zero with bounded Galois group. (Received February 04, 2020)

1157-03-535 **Pablo Andújar Guerrero*** (pandujar@purdue.edu). *Types and intersection properties in o-minimal theories.*

We consider different intersection properties among definable families of sets in o-minimal structures. The weakest of these is the (p, q)-property for some $p \geq q$, which states that, for every p sets in the family, some q

intersect. Our thesis is as follows: among families of sets of low complexity, a weak intersection property implies a strong one. Our main result states that any definable family in an o-minimal structure with the (p, q)-property for p and q large enough can be partitioned into finitely many subfamilies, each of which expands to a complete definable type. We frame this result in terms of VC theory, in particular Matousek's (p, q)-theorem, and known bounds in VC-density among definable families in o-minimal (and other NIP) theories. If time allows, we will address the application of this result in characterizing the topological notion of definable compactness, and in reproving, in the o-minimal setting, the known equivalence between non-forking of formulas and containment in definable types. (Received February 04, 2020)

1157-03-543 **Caroline Terry*** (caterry@uchicago.edu). *Speeds of hereditary properties and mutual algebricity.*

A hereditary graph property is a class of finite graphs closed under isomorphism and induced subgraphs. Given a hereditary graph property H, the speed of H is the function which sends an integer n to the number of distinct elements in H with underlying set 1,...,n. Not just any function can occur as the speed of hereditary graph property. Specifically, there are discrete "jumps" in the possible speeds. Study of these jumps began with work of Scheinerman and Zito in the 90's, and culminated in a series of papers from the 2000's by Balogh, Bollobás, and Weinreich, in which essentially all possible speeds of a hereditary graph property were characterized. In contrast to this, many aspects of this problem in the hypergraph setting remained unknown. In this talk we present new hypergraph analogues of many of the jumps from the graph setting, specifically those involving the polynomial, exponential, and factorial speeds. The jumps in the factorial range turned out to have surprising connections to the model theoretic notion of mutual algebricity, which we also discuss. This is joint work with Chris Laskowski. (Received February 04, 2020)

1157-03-563 **Steven Lazzaro*** (lazzaros@mcmaster.ca). *Generating Theories of II_1-Factors.* Preliminary report.

In the study of II_1-factors, there are two extensively studied properties known as Property Γ and McDuff's Property, which arose from the work of J. von Neumann and D. McDuff, respectively. It is immediate from definition that McDuff's Property implies Property Γ, and so one may consider partitioning the class of II_1-factors into three subclasses: McDuff factors, Γ-non-McDuff factors, and non-Γ factors. Somewhat recent work by R. Boutonnet, I. Chifan, and A. Ioana shows that there exists continuum many distinct continuous theories of McDuff factors, while followup work by I. Goldbring, B. Hart, and H. Towsner constructs explicit sentences distinguishing these theories. In this presentation, I aim to show how to apply the work of the aforementioned authors to obtain identical results for the class of Γ-non-McDuff factors, and briefly discuss how these techniques might be modified to be able to say something about the class of non-Γ-factors. (Received February 04, 2020)

1157-03-565 **Gregory D Cousins*** (gcousins@alumni.nd.edu), 765 King Street W, Hamilton, Ontario L8S 1J9, Canada. *Some Model Theory of Large Fields.*

Large fields tend to make up most of the "nice" fields which model theorists have come to love: algebraically closed, real closed, PAC, and p-adically closed fields are all large. In this talk, we aim to describe some work on the model theory of large fields under the assumption that they have small absolute galois group and enjoy a strong "algebraic" version of model completeness called "almost quantifier elimination". (Received February 04, 2020)

1157-03-616 **Victoria Noquez*** (vnoquez@iu.edu). *Dp-Minimality in Continuous Logic.* Preliminary report.

In this talk we will propose a framework in which to study dp-rank in the continuous setting. We will establish equivalent characterizations of dp-minimality and examine its behavior and relationship to other notions of minimality, providing examples of dp-minimal continuous theories. (Received February 04, 2020)

05 ► Combinatorics

1157-05-9 **Brigitte Servatius*** (bservat@wpi.edu), Mathematical Sciences, WPI, Worcester, MA 01609. *Rigidity matroids for maps on surfaces.*

We define a rigidity matroid for graphs which are cellularly embedded on a compact surface and show that it agrees, in the case of torus maps, with previous work on generic rigidity properties of doubly periodic graphs. (Received September 14, 2019)

1157-05-16 **Chris Cox*** (cocox@andrew.cmu.edu) and **Boris Bukh**. *Nearly orthogonal vectors.*
How can $d + k$ vectors in $\mathbb{R}^d$ be arranged so that they are as close to orthogonal as possible? In particular, define $\theta(d, k) := \min_X \max_{x \neq y \in X} |\langle x, y \rangle|$ where the minimum is taken over all collections of $d + k$ unit vectors $X \subseteq \mathbb{R}^d$. In this talk, we focus on the case where k is fixed and $d \to \infty$. In establishing bounds on $\theta(d, k)$, we find an intimate connection to the existence of systems of $\binom{k+1}{2}$ equiangular lines in $\mathbb{R}^k$. Using this connection, we are able to pin down $\theta(d, k)$ whenever $k \in \{1, 2, 3, 7, 23\}$ and establish asymptotics for general k. The main tool is an upper bound on $\mathbb{E}_{x,y\sim\mu}|\langle x, y \rangle|$ whenever μ is an isotropic probability mass on $\mathbb{R}^k$, which may be of independent interest. (Received November 07, 2019)

1157-05-17 **Zilin Jiang*** (zilinj@mit.edu), 40 Maple Ave, Apt 40, Cambridge, MA 02139. *Equiangular lines with a fixed angle.*
A set of lines through the origin in a Euclidean space is called equiangular if they are pairwise separated by the same angle. It is known that the maximum size of an equiangular set of lines grows quadratically as the dimension of the Euclidean space grows. However, when the angle is held fixed, a very different yet intricate behavior of the maximum size emerges. In this talk, I will explain how the determination of this behavior connects to some central notions in spectral graph theory, namely, the maximum eigenvalue of a graph and the multiplicity of the second largest eigenvalue.

Joint work with Alexandr Polyanskii, Jonathan Tidor, Yuan Yao, Shengtong Zhang and Yufei Zhao. (Received November 15, 2019)

1157-05-120 **Noha Abdelghany***, Western Michigan University, 1903 W. Michigan Ave., Kalamazoo, MI 49008, and **Jay Wood**, Western Michigan University, 1903 W. Michigan Ave., Kalamazoo, MI 49008. *Failure of the MacWilliams Identities for the Lee Weight Enumerator over $\mathbb{Z}_m$, $m \geq 5$.*
The MacWilliams identities give a relation between the Hamming weight enumerator of a linear code and the Hamming weight enumerator of its dual. We are interested in the question of whether there is some version of the MacWilliams identities for the Lee weight over $\mathbb{Z}_m$, the integers modulo m. The Lee weight and the Hamming weight are equal when $m = 2$ or $m = 3$; thus the MacWilliams identities are valid in those cases. It is also known that the MacWilliams identities are valid for the Lee weight over $\mathbb{Z}_4$. In this paper we show the nonexistence of any version of the MacWilliams identities for Lee weight enumerators over $\mathbb{Z}_m$, $m \geq 5$. (Received January 22, 2020)

1157-05-126 **Reuven Hodges*** (rhodges@illinois.edu), **Venkatramani Lakshmibai** and **Alexander Yong**. *Classifying Levi spherical Schubert varieties.*
Over the last several years, in joint work with V. Lakshmibai, I have been studying Levi subgroup actions on Schubert varieties. In particular, we are interested in when a Schubert variety is a spherical variety for the action of a Levi subgroup. Recently, we gave a classification of the Schubert varieties in the Grassmannian that are spherical varieties. I will discuss this classification as well as work in progress with A. Yong on extending the classification to Schubert varieties in the full flag variety. (Received January 22, 2020)

1157-05-195 **Srilakshmi Pattabiraman**, **Ryan Gabrys** and **Olgica Milenkovic*** (milenkov@illinois.edu), 311 Coordinated Science Lab, Urbana, IL. *Composition multiset error-correcting codes.*
We study the problem of reconstructing strings from their substring composition multiset in the presence of composition errors. Our contributions include two new constructions for composition multiset error-correcting codes that use Catalan paths and evaluation polynomials. The latter framework is related to the Turnpike problem and its polynomial characterization and extends the work by Acharya et al. from 2015. (Received January 28, 2020)

1157-05-245 **Eric Katz*** (katz.60@osu.edu), 231 W 18th Ave, Columbus, OH 43210-1101. *Higher Cycle Operations on Graphs and Iterated Integration.*
The classical cycle pairing on graphs takes a pair of cycles to the signed length of their intersection. Inspired by Chen's iterated integrals, we introduce higher cycle operations that produce a number from a loop and an ordered set of cycles. This is mildly non-abelian data. We explain the connection between these higher cycle operations and classical and p-adic iterated integrals. Based on joint work with Raymond Cheng and with Daniel Litt. (Received January 30, 2020)

1157-05-312 **Dane Miyata**, **Nicholas Proudfoot** and **Eric Ramos*** (eramos@uoregon.edu). *The categorified graph minor theorem and graph configuration spaces.*

Perhaps one of the most well-known theorems in graph theory is the celebrated Graph Minor Theorem of Robertson and Seymour. This theorem states that in any infinite collection of finite graphs, there must be a pair of graphs for which one is obtained from the other by a sequence of edge contractions and deletions. In this talk, I will present work of Nick Proudfoot, Dane Miyata, and myself which proves a categorified version of the graph minor theorem. As an application, we show how configuration spaces of graphs must display some strongly uniform properties. We then show how this result can be seen as a vast generalization of a variety of classical theorems in graph configuration spaces (Received February 01, 2020)

1157-05-349 **Anthony Bonato*** (abonato@ryerson.ca), 350 Victoria St, Toronto, ON M5B2K3, Canada. *Modeling Adversaries in Networks.*

Complex network models are often premised on the formation of edges via positive ties. However, structural balance theory in social network analysis cites several mechanisms to complete triads sharing negative ties. Real-world networks ranging from food webs to market graphs highlight the importance of modeling both positive to negative ties.

We present new research on modeling friends and adversaries. The Iterated Local Model provably simulates many emergent properties of complex networks, and gives rise to graph families that are of interest in their own right. We also present a hypothesis, supported by data mined from real-world networks, characterizing the hidden structure of dynamic competition networks. (Received February 02, 2020)

1157-05-360 **Naomi G Boulware*** (boulware@rose-hulman.edu), **Naihuan Jing** and **Kailash C Misra**. *Smith Normal form of q-Varchenko matrices.*

Any real arrangement of hyperplanes partitions the underlying space into open sets called regions. If R_i and R_j are two regions, then $\#sep(R_i, R_j)$ counts the number of hyperplanes separating them. This combinatorial information about the hyperplane arrangement is recorded in a square matrix called the q-Varchenko matrix. It is indexed by its regions and the (i, j)-entry is q^m where $m = \#sep(R_i, R_j)$. If the q-Varchenko matrices for two hyperplane arrangements have the same Smith Normal form, then the hyperplane arrangements have the same combinatorial structure. It is well known that the Smith Normal form of a q-Varchenko matrix exists over a principal ideal domain. However, very little is known about when a q-Varchenko matrix has a Smith Normal form over the ring of polynomials with integer coefficients. In this talk we will introduce q-Varchenko matrices and illustrate some connections between hyperplane arrangements, symmetry groups and the Smith Normal form of a matrix. (Received February 02, 2020)

1157-05-434 **Ben Lund***, lund.ben@gmail.com, and **Manik Dhar** and **Zeev Dvir**. *Simple proofs for finite field Furstenberg sets.*

A (k, m)-Furstenberg set $S \subset \mathbb{F}_q^n$ over a finite field is a set that has at least m points in common with a k-flat in every direction. The question of determining the smallest size of such sets is a natural generalization of the finite field Kakeya problem. The only previously known bound for these sets is due to Ellenberg and Erman, and uses sophisticated machinery from algebraic geometry. In this work we give new, completely elementary and simple, proofs which significantly improve the known bounds. One of our results relies on an equivalent formulation of the problem using the notion of min-entropy, which could be of independent interest. (Received February 03, 2020)

1157-05-489 **Claudiu Raicu*** (craicu@nd.edu). *Feasibility criteria for high-multiplicity partitioning problems.*

For fixed weights $w_1, \cdots, w_n$, and for $d > 0$, let B denote a collection of $d \cdot n$ balls, with d balls of weight w_i for each $i = 1, \cdots, n$. Consider the problem of assigning the balls to n bins with capacities $C_1, \cdots, C_n$, in such a way that each bin is assigned d balls, without exceeding its capacity. When $d \gg 0$, I will explain sufficient criteria for the feasibility of this problem, which coincide up to explicit constants with the natural set of necessary conditions. The combinatorial problem considered here comes up naturally in the study of the asymptotic behavior of the Castelnuovo-Mumford regularity of symmetric monomial ideals. (Received February 03, 2020)

1157-05-521 **Mihai Ciucu*** (mciucu@indiana.edu), Department of Mathematics, Indiana University, 831 East 3rd Street, Bloomington, IN 47405, and **Ilse Fischer** (ilse.fischer@univie.ac.at), Department of Mathematics, University of Vienna, Oskar-Morgenstern-Platz 1, 1090 Vienna, Austria. *Lozenge tilings of hexagons with removed core and satellites.*

We consider regions obtained from 120 degree rotationally invariant hexagons by removing a core and three equal satellites (all equilateral triangles) so that the resulting region is both vertically symmetric and 120 degree rotationally invariant, and give simple product formulas for the number of their lozenge tilings. We then work out consequences for the correlation of holes, which was the original motivation for this study. (Received February 03, 2020)

1157-05-544 **Sara Faridi*** (faridi@dal.ca) and **Mayada Shahada** (mayada.shahada@dal.ca). *Breaking up homological cycles and subadditivity of betti numbers.*

This paper is inspired by various formulations as what is known as "Hochster's formula", which is used to calculate betti numbers of monomial ideals as the rank of the homology modules of topological objects. In fact, a nonvanishing betti number corresponds to the existence of a homological cycle. So many questions about betti numbers of monomial ideals are reduced to the calculation of homology. This talk is about how the "subadditivity" property for syzygies of monomial ideals - which states that the maximum degree of a betti number in homological degree $a+b$ never exceeds the sum of the maximum degrees of a betti number in homological degrees a and b - could be interpreted in terms of breaking up homological cycles into smaller ones. (Received February 04, 2020)

1157-05-550 **Huy Tai Ha*** (tha@tulane.edu) and **Takayuki Hibi** (hibi@math.sci.osaka-u.ac.jp). *Max min vertex cover and the size of Betti tables.*

Let G be a simple graph over n vertices, that contains no isolated vertices, and let $\tau_{\max}(G)$ denote the largest size of a minimal vertex cover of G. We provide a sharp bound for $\tau_{\max}(G)$, namely, $\tau_{\max}(G) \geq 2\sqrt{n}-2$, and use this to study the projective dimension and the regularity of the squarefree monomial ideal associated to G. (Received February 04, 2020)

1157-05-623 **John Jasper***, South Dakota State University, SAME 209A, Box 2225, Brookings, SD 57007. *Equiangular tight frames from group divisible designs.*

Given positive integers N and d, an equiangular tight frame ETF(d,N) is a type of optimal packing of N lines in a real or complex Hilbert space of dimension d. In the complex setting, the existence of an ETF(d,N) remains unresolved for many choices of d and N. In this talk, we observe that the (d,N) parameters of many of the known constructions of ETFs are of one of two types. We further provide a new method for combining a given ETF of one of these two types with an appropriate group divisible design (GDD) in order to produce a larger ETF of the same type. By applying this method to known families of ETFs and GDDs, we obtain several new infinite families of ETFs. Our approach was inspired by a seminal paper of Davis and Jedwab which both unified and generalized McFarland and Spence difference sets. We provide combinatorial analogs of their algebraic results, unifying Steiner ETFs with hyperoval ETFs and Tremain ETFs. (Received February 04, 2020)

1157-05-636 **Hanbaek Lyu**, **Facundo Mémoli** and **David Sivakoff*** (dsivakoff@stat.osu.edu), 1958 Neil Ave, Cockins Hall 440A, Columbus, OH 43210. *Network motif profiles.*

We propose a class of observables for weighted networks, which includes as special cases homomorphism densities and the average clustering coefficient. These observables can be obtained by sampling random homomorphisms from a small motif into a given network, and as such, we propose MCMC algorithms for efficient approximation. Moreover, we prove that these observables are stable with respect to a renormalized cut distance between networks. (Received February 04, 2020)

06 ► Order, lattices, ordered algebraic structures

1157-06-294 **Luke Harmon*** (harmonluke2@gmail.com), 1826 Woodburn Street, Colorado Springs, CO 80905, and **Greg Oman**. *Lower Finite Modules over Commutative Rings.*

A partially ordered set (P,<) is lower finite provided P is infinite and for each x in P, there are but finitely many elements y in P such that y<x. We will call a module M lower finite if the set of proper submodules of M, partially ordered by set-theoretic containment, is lower finite. We will use the (well-studied) class of Jonsson

modules (along with other classical results) to classify the lower finite modules over a commutative ring with identity. (Received January 31, 2020)

11 ► Number theory

1157-11-33 **Shaoming Guo**, **Zane Kun Li*** (zkli@iu.edu), **Po-Lam Yung** and **Pavel Zorin-Kranich**. *A bilinear proof of decoupling for the moment curve.*

We discuss the relationship between Wooley's nested efficient congruencing and Bourgain-Demeter-Guth's decoupling proofs of Vinogradov's Mean Value Theorem. We will sketch a proof of decoupling for the moment curve that is inspired from nested efficient congruencing. This talk is based on joint work with Shaoming Guo, Po-Lam Yung, and Pavel Zorin-Kranich. (Received December 24, 2019)

1157-11-61 **Lucia Mocz*** (lmocz@math.uchicago.edu), 5550 S Dorchester, APT 303, Chicago, IL 60637. *Heights and p-adic Hodge Theory.*

We discuss connections between p-adic Hodge theory and the Faltings height. Most namely, we show how new tools in p-adic Hodge theory can be used to prove new Northcott properties satisfied by the Faltings height, and demonstrate phenomenon which are otherwise predicted by various height conjectures. We will focus primarily on the Faltings height of CM abelian varieties where the theory can be made to be computational and explicit. (Received January 10, 2020)

1157-11-82 **Stefan Patrikis** and **Shiang Tang*** (shiangtang1989@gmail.com), 1409 W Green St., Urbana, IL 61801. *Potential automorphy for* GSpin_{2n+1}*-valued Galois representations.*

We prove a potentially automorphy theorem for suitable Galois representations $\Gamma_{F^+} \to \mathrm{GSpin}_{2n+1}(\overline{\mathbb{F}}_p)$ and $\Gamma_{F^+} \to \mathrm{GSpin}_{2n+1}(\overline{\mathbb{Q}}_p)$, where Γ_{F^+} is the absolute Galois group of a totally real field F^+. We also prove results on solvable descent for $\mathrm{GSp}_{2n}(\mathbb{A}_{F^+})$ and use these to put representations $\Gamma_{F^+} \to \mathrm{GSpin}_{2n+1}(\overline{\mathbb{Q}}_p)$ into compatible systems of $\mathrm{GSpin}_{2n+1}(\overline{\mathbb{Q}}_\ell)$-valued representations. There are three inputs for our potential automorphy results: the potential automorphy theorems of Barnet-Lamb-Gee-Geraghty-Taylor, Arthur's endoscopic classification, and the work of Kret-Shin associating GSpin_{2n+1}-valued Galois representations to suitable automorphic representations of GSp_{2n}. To put representations $\Gamma_{F^+} \to \mathrm{GSpin}_{2n+1}(\overline{\mathbb{Q}}_p)$ into compatible systems, we use Taylor's Brauer induction argument, which relies on solvable descent. But cyclic descent is not available for GSp_{2n}, we establish instances of it using trace formula arguments. (Received January 15, 2020)

1157-11-156 **Solomon Friedberg***, Department of Mathematics, Boston College, Chestnut Hill, MA 02468, and **David Ginzburg**, School of Mathematical Sciences, Tel Aviv University, 6997801 Ramat Aviv, Tel Aviv, Israel. *A new theta correspondence.* Preliminary report.

The classical theta correspondence establishes a relationship between automorphic representations on orthogonal groups and automorphic representations on symplectic groups or their double covers. The correspondence is achieved by using as integral kernel a theta series on the metaplectic double cover of a symplectic group constructed from the Weil representation. In this work we present an extension of this theta correspondence to higher degree covers. The key issue here is that for higher degree covers there is no analogue of the Weil representation, and additional ingredients are needed. Our work reflects a broader paradigm: constructions in automorphic forms that work for algebraic groups or their double covers should often extend to higher degree covers. (Received January 26, 2020)

1157-11-189 **Serin Hong***, 530 Church Street, Ann Arbor, MI 48109. *Classification of quotients and subbundles on the Fargues-Fontaine curve.*

Vector bundles on the Fargues-Fontaine curve play a central role in much of recent developments in p-adic Hodge theory and the local Langlands program. In this talk, we discuss complete and explicit classification results for quotients and subbundles of a given vector bundle on the Fargues-Fontaine curve. We then present some applications of these results to the study of the flag varieties and the stack of vector bundles on the Fargues-Fontaine curve. Our proof is based on a dimension counting argument for moduli spaces of certain bundle maps using Scholze's theory of diamonds. (Received January 28, 2020)

1157-11-197 **Giorgis Petridis*** (petridis@uga.edu) and **Brandon Hanson**. *On a conjecture of Sarkozy.*

Sarkozy has conjectured that for every prime, the set of non-zero quadratic residues modulo that prime can be decomposed as a sum set only if one of the sets is a singleton. We prove this conjecture for almost every prime. (Received January 28, 2020)

1157-11-199 **T. C. Anderson** and **A. V. Kumchev*** (akumchev@towson.edu), Department of Mathematics, Towson University, 8000 York Road, Towson, MD 21252, and **E. A. Palsson**. *Bounds for discrete bilinear spherical maximal functions.* Preliminary report.

We study the bilinear discrete averaging operator $T_\lambda(f,g)(x) = \lambda^{6-2d}\sum_{u,v} f(x-u)g(x-v)$ where f and g are functions in $\ell^p(\mathbb{Z}^d)$ and the summation is over the manifold in $\mathbb{Z}^d \times \mathbb{Z}^d$ defined by the equations $|u| = |v| = |u-v| = \lambda$, where $|u|$ is the standard Euclidean norm on $\mathbb{R}^d$. We prove an approximation formula for the Fourier multiplier of T_λ and establish the boundedness of the respective maximal operator from $\ell^p(\mathbb{Z}^d) \times \ell^p(\mathbb{Z}^d)$ into $\ell^{p'}(\mathbb{Z}^d)$, where p' is the conjugate exponent to p. (Received January 28, 2020)

1157-11-220 **Chantal David** and **Alexandra Florea***, aflorea@math.columbia.edu, and **Matilde Lalin**. *Non-vanishing for cubic L-functions.*

Chowla conjectured that $L(1/2,\chi)$ never vanishes, for χ any Dirichlet character. Soundararajan showed that more than 87.5% of the values $L(1/2,\chi_d)$, for χ_d a quadratic character, do not vanish. Much less is known about cubic characters. Baier and Young showed that more than $X^{6/7-\epsilon}$ of $L(1/2,\chi)$ are non-vanishing, for χ a primitive, cubic character of conductor of size up to X. In joint work with C. David and M. Lalin, we show that a positive proportion of these central L-values are non-vanishing in the function field setting. The same techniques can be used to prove the analogous result in the number field setting, conditional on the Generalized Riemann Hypothesis. (Received January 29, 2020)

1157-11-222 **Akshat Mudgal*** (amudgal@purdue.edu), Purdue University, West Lafayette, IN 47907-2067. *Arithmetic Combinatorics on Vinogradov systems.*

In this talk, we consider the Vinogradov system, that is, the system of equations

$$\sum_{i=1}^{s} x_i^j = \sum_{i=1}^{s} y_i^j \quad (1 \le j \le k),$$

where s and k are natural numbers. The number of solutions of this system, when the variables x_i, y_i are restricted to a set of real numbers A, has been widely studied by researchers in both Analytic Number Theory and Harmonic Analysis. In particular, there has been a significant amount of work regarding upper bounds for the number of solutions to the above system of equations. The objective of our talk will be of a different flavour, wherein we will try to address the following question - given a set A with many solutions to the Vinogradov system, what other arithmetic properties can we infer about A? (Received January 29, 2020)

1157-11-228 **Kwangho Choiy*** (kchoiy@siu.edu), 1245 Lincoln Dr., Carbondale, IL 62901-4408. *On commuting algebra isomorphisms in pseudo-z-embedding.*

Given a connected reductive p-adic group G, its closed subgroup H having the same derived group, it is of interest which group structural conditions on G, H yield the algebra isomorphism between the so-called commuting algebra (the algebra of G-endomorphisms) of a parabolic induction in G and its counterpart in H. We shall give an answer to the question by employing the notion of pseudo-z-embedding defined by T. Kaletha. This study can be applied to a transfer of reducibility on parabolic induction from G to H and vice versa, and is expected to contribute to the relationships between internal structures of tempered L-packets for G and H in the context of the local Langlands correspondence. (Received January 29, 2020)

1157-11-231 **Simon Marshall*** (marshall@math.wisc.edu), Department of Mathematics, UW Madison, 480 Lincoln Dr, Madison, WI 53706, and **Mathilde Gerbelli-Gauthier**. *Strong uniform admissibility for representations of GL_n and Sarnak's density conjecture.*

Let π be an irreducible admissible representation of $GL_n(\mathbb{Q}_p)$, and let $K(n)$ be the usual principal congruence subgroups. We give a strong uniform bound for the dimension of the subspace of π fixed by $K(n)$ that depends only on n (and not on π), in various cases including when π is supercuspidal or a Speh representation. We present an application of these results to Sarnak's density conjecture on limit multiplicities of nontempered forms, which is joint work in progress with Mathilde Gerbelli–Gauthier. (Received January 29, 2020)

1157-11-263 **Daniel Johnstone*** (johnstod@umn.edu). *The Stable Transfer Factor for the Symmetric Power Lifting From GL_2 to GL_{n+1}.*

I will discuss work in progress towards an explicit computation of the stable transfer factor for the symmetric power lifting from $GL_2(F)$ to $GL_3(F)$ for F a local field. For principal series representations there exists an explicit distributional kernel realizing this functorial lifting and there is strong evidence that an analogous kernel exists for discrete series representations. This is joint work with Zhilin Luo. (Received January 30, 2020)

1157-11-276 **Melissa Emory*** (memory@math.toronto.edu). *Multiplicity One Theorems for Clifford Groups and General Spin Groups.* Preliminary report.

Aizenbud, Gourevitch, Rallis and Schiffmann have proved a multiplicity one theorem for restrictions of irreducible representations of certain p-adic classical groups and Waldspurger proved the same theorem for the special orthogonal groups. We will discuss work in progress to use variants of their methods to prove the same theorem for two non-classical groups, the Clifford groups and general spin groups. (Received January 31, 2020)

1157-11-286 **Kevin Hughes** and **Trevor D Wooley*** (twooley@purdue.edu), Department of Mathematics, Purdue University, 150 N. University St., West Lafayette, IN 47906-2067. *Discrete restriction for (x, x^3) and related topics.*

Defining the truncated extension operator E for a sequence $a(n)$ with $n \in \mathbb{Z}$ by putting

$$Ea(\alpha, \beta) := \sum_{|n| \leq N} a(n) e(\alpha n^3 + \beta n),$$

we obtain the conjectured tenth moment estimate

$$\|Ea\|_{L^{10}(\mathbb{T}^2)} \lesssim_\epsilon N^{\frac{1}{10}+\epsilon} \|a\|_{\ell^2(\mathbb{Z})}.$$

We obtain related conclusions when the curve (x, x^3) is replaced by $(\phi_1(x), \phi_2(x))$ for suitably independent polynomials $\phi_1(x), \phi_2(x)$ having integer coefficients. (Received January 31, 2020)

1157-11-321 **Gebhard Böckle**, **Chandrashekhar Khare** and **Jeffrey Manning*** (jmanning@math.ucla.edu). *The Wiles defect for Hecke algebras that are not complete intersections.*

In his work on modularity theorems, Wiles proved a numerical criterion for a map of rings $R \to \mathbb{T}$ to be an isomorphism of complete intersections. He used this to show that certain deformation rings and Hecke algebras associated to a mod p Galois representation at non-minimal level were isomorphic and complete intersections, provided the same was true at minimal level. In addition to proving modularity theorems, this numerical criterion also implies a connection between the order of a certain Selmer group and a special value of an L-function.

In this talk I will consider the case of a Hecke algebra acting on the cohomology a Shimura curve associated to a quaternion algebra. In this case, one has an analogous map of ring $R \to \mathbb{T}$ which is known to be an isomorphism, but in many cases the rings R and $\mathbb{T}$ fail to be complete intersections. This means that Wiles' numerical criterion will fail to hold. I will describe a method for precisely computing the extent to which the numerical criterion fails (i.e. the 'Wiles defect'), which will turn out to be determined entirely by local information at the primes dividing the discriminant of the quaternion algebra.

This is joint work with Gebhard Bockle and Chandrashekhar Khare. (Received February 01, 2020)

1157-11-332 **Dongming She*** (shed@purdue.edu), 150 N University St, West Lafayette, IN 47907. *Local Langlands Correspondence and analytic stability of gamma-factors.*

In this talk we will discuss local Langlands correspondence for GL(n), give a brief introduction to Langlands-Shahidi method, and then talk about the approach of using the stability of the arithmetic and analytic γ-factors to deduce the equality of the local arithmetic and analytic ϵ- and L-factors. The arithmetic stability has been established by P. Deligne in general. But the analytic stability remains open. So far there are only a few cases are known.

We will use the group ($Gspin$) to define the twisted exterior and symmetric square *epsilon*- and L-factors for $GL(n)$, reduce the problem to the stability of local coefficient, which can be represented as the Mellin transform of certain partial Bessel function. We will then apply some analysis to obtain a formula for the asymptotic behavior of the partial Bessel function. This result generalizes the formula in the work of J.W. Cogdell. F. Shahidi, and T.L. Tsai in 2017, and may give some ideas on how to approach the general analytic stability. (Received February 01, 2020)

1157-11-361 **Xuancheng Shao*** (fernandoshao@gmail.com), 757 patterson office tower, lexington, KY 40506, and **Kaisa Matomaki**. *Higher-degree exponential sum estimates over primes in short intervals.*

In studying primes via Fourier analysis (a.k.a the Hardy-Littlewood circle method), one often faces the task of estimating exponential sums over primes. In this talk we discuss generalizations of this problem, when the primes are restricted to a short interval and when the linear exponential phase functions are replaced by polynomial phases. One of the applications is a short interval version of the Waring-Goldbach problem. (Received February 02, 2020)

1157-11-365 **Catherine M Hsu*** (cathy.hsu@bristol.ac.uk), **Holley Friedlander**, **Elena Fuchs**, **Piper H**, **Katherine Sanden**, **Damaris Schindler** and **Katherine Stange**. *Primes in Apollonian circle packings.*

An Apollonian circle packing is a fractal arrangement formed by repeatedly inscribing circles into the interstices in a Descartes configuration of four mutually tangent circles. The curvatures of the circles in such a packing are often integers, and so it is natural to ask questions about their arithmetic properties. For example, it is known by work of Bourgain-Fuchs that a positive fraction of integers appear as curvatures in any integral Apollonian circle packing. In this talk, we investigate the arithmetic properties of the collection of integers appearing in "thickened prime components" of Apollonian circle packings. (Received February 02, 2020)

1157-11-372 **Heng Du*** (du136@purdue.edu), Department of Mathematics, Purdue University, 150 N. University Street, West Lafayette, IN 47906. *Arithmetic Breuil-Kisin-Fargues modules and their moduli.*

The study of modifications of vector bundles over the Fargues-Fontaine curve plays an important role in some recent works about p-adic Hodge theory, and we know they are related to finite free modules over A_{inf} with additional structures called Breuil-Kisin-Fargues modules. For a p-adic field K, Fargues-Fontaine construct a class of G_K-equivariant modifications over the curve from certain linear-algebraic categories developed by Fontaine. We will give a characterization of those G_K-equivariant modifications in terms of the corresponded Breuil-Kisin-Fargues modules. We will then study their moduli following the method of Emerton-Gee, and give some possible applications. (Received February 02, 2020)

1157-11-375 **Jayce Robert Getz*** (jgetz@math.duke.edu). *On triple product L functions.*

Let $\pi = \pi_1 \otimes \pi_2 \otimes \pi_3$ be a unitary cuspidal automorphic representation of $\mathrm{GL}_3^3(\mathbb{A}_F)$ where F is a number field. Assume that π is everywhere tempered. Under suitable local hypotheses, for a sufficiently large finite set of places S of F we prove that the triple product L-function $L^S(s, \pi, \otimes^3)$ admits a meromorphic continuation to $\mathrm{Re}(s) > \frac{1}{2}$. We also give some information about the possible poles. (Received February 02, 2020)

1157-11-376 **Andrea Dotto***, University of Chicago, Department of Mathematics, 5734 S University Ave, Chicago, IL 60637. *Localizing* $\mathrm{GL}_2(\mathbf{Q}_p)$*-representations.*

We let Z denote a certain chain of projective lines over a field k of characteristic p, and we let A denote the category of smooth (but not necessarily admissible) $\mathrm{GL}_2(\mathbf{Q}_p)$-representations on k-vector spaces (with some fixed central character). Our goal in this talk is to explain how the category A may be "localized" over the scheme Z, and to explain various applications of this localization. This is work-in-progress with Matthew Emerton and Toby Gee. (Received February 02, 2020)

1157-11-389 **Lynnelle Ye*** (lynnelle@stanford.edu), Department of Mathematics, Building 380, Stanford, CA 94305. *Geometry of eigenvarieties.*

We will discuss progress on questions about the geometry of eigenvarieties, particularly those for automorphic forms on definite unitary groups, and particularly over the boundary of weight space. (Received February 03, 2020)

1157-11-396 **Yong Suk Moon*** (ysmoon1@math.arizona.edu). *Compatibility of p-adic and classical Riemann-Hilbert functors.* Preliminary report.

Let X be a smooth algebraic variety over a number field. Recently, Diao-Lan-Liu-Zhu conjectured that their construction of p-adic Riemann-Hilbert functor on the category of geometric p-adic étale local systems on X is compatible with Deligne's classical Riemann-Hilbert correspondence, and proved the conjecture for a certain class of étale local systems on Shimura varieties. In this talk, we will discuss the proof of the conjecture for some special cases. (Received February 03, 2020)

1157-11-415 **Kirsti D. Biggs*** (biggs@chalmers.se). *Efficient congruencing in ellipsephic sets.*

An ellipsephic set is a subset of the natural numbers whose elements have digital restrictions in some fixed prime base. Such sets can be viewed as p-adic Cantor sets, and as such have a fractal structure resembling that of certain real Cantor sets studied by harmonic analysts.

In particular, we wish to consider ellipsephic sets with additive properties inherited from their set of permitted digits. For example, there are few representations of an integer as a sum of two squares, and the ellipsephic set of numbers whose digits are squares consequently provides a key motivating example for our work.

We obtain discrete restriction estimates for the moment curve over ellipsephic sets—in number theoretic terms, we bound the number of ellipsephic solutions to a Vinogradov system of equations—using Wooley's

nested efficient congruencing method. These results generalise previous work of the speaker, on the quadratic case of this problem, to the moment curve of arbitrary degree.

In this talk, I will outline the key ideas from the proof and highlight the importance of the additive structure of our ellipsephic sets. (Received February 03, 2020)

1157-11-430 **Ayla Gafni*** (argafni@olemiss.edu). *Partitions into powers of primes.* Preliminary report.

Given a set $\mathcal{A} \subset \mathbb{N}$, the restricted partition function $p_{\mathcal{A}}(n)$ counts the number of integer partitions of n with all parts in $\mathcal{A}$. In this talk, we will explore the features of the restricted partitions function $p_{\mathcal{P}_k}(n)$ where $\mathcal{P}_k$ is the set of k-th powers of primes. Powers of primes are both sparse and irregular, which makes $p_{\mathcal{P}_k}(n)$ quite an elusive function to understand. We will discuss some of the challenges involved in studying restricted partition functions and what is known in the case of primes, k-th powers, and k-th powers of primes. (Received February 03, 2020)

1157-11-455 **Giacomo Micheli***, gmicheli@usf.edu. *Locally recoverable codes from Galois theory.*

In this talk we present an effective method to construct locally recoverable codes using Galois Theory. The method is based on estimating the number of totally split places of an extension of global function fields. (Received February 03, 2020)

1157-11-466 **John Enns*** (john.enns@northwestern.edu). *Mod p local-global compatibility for* $\mathrm{GSp}_4(\mathbb{Q}_p)$ *and the Jantzen filtration.* Preliminary report.

Given a Galois representation $\bar{\rho} : G_{\mathbb{Q}_p} \to \mathrm{GL}_n(\overline{\mathbb{F}}_p)$, a global construction of Caraiani-Emerton-Gee-Geraghty-Paškūnas-Shin furnishes a smooth $\overline{\mathbb{F}}_p$-representation Π of $\mathrm{GL}_n(\mathbb{Q}_p)$ that one hopes realizes a mod p local Langlands correspondence. It is not obvious that this construction doesn't forget any information about $\bar{\rho}$. I will explain how the data of $\bar{\rho}$ gets encoded in this smooth representation in certain cases by connecting the deformation theory of $\bar{\rho}$ with data arising from Π via Jantzen's filtration of principal series representations for finite Chevalley groups. This allows me to prove new results in an analogous case for $\mathrm{GSp}_4(\mathbb{Q}_p)$. (Received February 03, 2020)

1157-11-508 **Olivia Beckwith*** (obeck@uiuc.edu), Illini Hall, Room 337, 725 S Wright St, Champaign, IL 61820, and **Gene Kopp** (gene.s.kopp@gmail.com). *Polyharmonic Maass forms and real quadratic fields.*

We study polyharmonic Maass forms for $\Gamma(N)$ and show that they are related to ray class extensions of real quadratic fields. In particular, we construct a basis for the space of polyharmonic Maass forms for $\Gamma(N)$ of moderate growth and bounded polyharmonic depth, generalizing a result of Lagarias and Rhoades for $N = 1$. We also show that twisted traces of cycle integrals of certain polyharmonic Maass forms are central values of Hecke L-series for real quadratic fields. (Received February 03, 2020)

1157-11-523 **Bir Kafle*** (bkafle@pnw.edu), 1401 S. U.S. 421, Westville, IN 46391, and **Alain Togbé** and **Benedict Normenyo**. *Repdigits as sums of members of another sequences.*

Let $b > 1$ be an integer. A *base b-repdigit* is a natural number of the form

$$c\left(\frac{b^m - 1}{b - 1}\right), \quad \text{for some} \quad m \geq 1, \quad \text{and} \quad c \in \{1, 2, \ldots, b - 1\}.$$

We consider the particular case of repdigits when $b = 10$.

In this talk, we will discuss about the solutions of the Diophantine equations

$$U_{m_1} + U_{m_2} + \cdots = c\left(\frac{10^m - 1}{9}\right),$$

where U_m is a Fibonacci number, a Lucas number, or a Pell number.

This is a joint work with Alain Togbé and Benedict Normenyo. (Received February 03, 2020)

1157-11-529 **Muthu Krishnamurthy*** (muthu-krishnamurthy@uiowa.edu), 14 MLH, Iowa City, IA 52241, and **Yeongseong Jo**. *Types, covers and the Langlands-Shahidi method.* Preliminary report.

I will explain a method for computing the local coefficient defined by Shahidi through the theory of types and covers in the representation theory of p-adic groups. I will do this through the example of the self-associate maximal parabolic subgroup of a general linear group, but the method is applicable to other similar situations within the context of the Langlands-Shahidi method. This also provides a way to compare the associated local factors with those defined through the Rankin-Selberg theory. (Received February 04, 2020)

1157-11-546 **Yu-Ru Liu*** (yrliu@uwaterloo.ca), 200 University Avenue, West, Waterloo, Ontario N2L 3G1, Canada, and **Wentang Kuo** and **Xiaomei Zhao**. *The asymptotic estimates and Hasse principle for multidimensional Waring's problem.*

Motivated by the asymptotic estimates and Hasse principle for multidimensional Waring's problem via the circle method, we prove for the first time that the corresponding singular series is bounded below by an absolute positive constant without any nonsingular local solubility assumption. The number of variables we need is near-optimal. By proving a more general uniform density result over certain complete discrete valuation rings with finite residue fields, we also establish uniform lower bounds for both singular series and singular integral in $\mathbb{F}_q[t]$. We thus obtain asymptotic formulas and the Hasse principle for multidimensional Waring's problem in $\mathbb{F}_q[t]$ via a variant of the circle method. This is a joint work with Wentang Kuo and Xiaomei Zhao. (Received February 04, 2020)

1157-11-552 **Dihua Jiang***, School of Mathematics, 127 Vincent Hall, Minneapolis, MN 55455. *Non-linear Fourier transforms and standard L-function for symplectic groups: local theory.* Preliminary report.

I will discuss the local theory of the Braverman-Kazhdan proposal, which is to establish the basic properties of the L-functions (Langlands conjecture) via via non-linear Fourier transform and Poisson summation formula, for the standard L-function of $Sp(2n)$. This is a report on a joint work in progress with Zhilin Luo and Lei Zhang. (Received February 04, 2020)

1157-11-561 **Daniel Le***, 40 St. George St, Toronto, Ontario M5S 2E4, Canada. *Serre weights and Breuil–Mézard cycles.*

Given an arithmetic locally symmetric space Y and a mod p system m of Hecke eigenvalues, the weight part of Serre's conjecture seeks to classify the coefficient systems on Y whose cohomology contains m. Through Taylor-Wiles patching, this problem is known to be closely linked to the Breuil-Mézard conjecture, which predicts a representation theoretic recipe for the special fibers of local Galois deformation rings (and more generally, the Emerton-Gee moduli stack of local Galois representations) out of atomic pieces corresponding to Irreducible coefficient systems. We will describe recent progress on both questions in many cases. The main tool is a theory of local models, which relates (portions of) the Emerton-Gee stack to mixed characteristics versions of (deformed) affine Springer fibers. This is joint work with B.V. Le Hung, B. Levin, and S. Morra. (Received February 04, 2020)

1157-11-585 **Rongqing Ye*** (ye271@purdue.edu) and **Elad Zelingher**. *Exterior square gamma factors for cuspidal representations of GL_n: finite field analogs and level zero representations.*

We follow Jacquet-Shalika, Matringe and Cogdell-Matringe to define exterior square gamma factors for irreducible cuspidal representations of $GL_n(\mathbb{F}_q)$. These exterior square gamma factors are expressed in terms of Bessel functions, or in terms of the regular characters associated with the cuspidal representations. We also relate our exterior square gamma factors over finite fields to those over local fields through level zero representations. (Received February 04, 2020)

1157-11-620 **Amita Malik*** (amita.malik@rutgers.edu). *Zeros of derivatives of the completed Riemann zeta function.*

Riemann Hypothesis implies the Riemann Hypothesis for higher order derivatives of the completed Riemann zeta function. The gap distribution of the zeros of these derivatives would provide insight into the existential question of the Landau-Siegel zero. In this talk, we discuss the vertical distribution of these zeros and establish their uniform distribution modulo one. Moreover, we prove that 100% of the zeros of combinations of these derivatives lie on the critical line even though there is no Euler product in this case. (Received February 04, 2020)

1157-11-634 **Brad Rodgers*** (brad.rodgers@queensu.ca) and **Jeffrey C. Lagarias**. *Mimicking zeta-zero statistics with lattices.*

In this talk I will review what information is known about the local spacing of zeros of the Riemann zeta-function and discuss why this information alone is insufficient to rule out a hypothesis that zeros of the zeta function never have gaps smaller than half the average size, a result obtained independently by T. Tao. I hope to discuss the relation of this problem to some extremal problems in harmonic analysis and the theory of point processes. (Received February 04, 2020)

1157-11-645 **Yeongseong Jo*** (yeongseong-jo@uiowa.edu), 14 MacLean Hall, Iowa City, IA 52242.
Rankin-Selberg L-functions via Good Section.

In the pioneering work of Jacquet, Piatetski-Shapiro, and Shalika, they defined and computed the local Rankin-Selberg L-functions for $GL(n) \times GL(m)$ in terms of L-functions for inducing datum at finite places. Cogdell and Piatetski-Shapiro revisited Rankin-Selberg L-functions to develop the method of exceptional poles attached to derivatives due to Bernstein and Zelevinsky. In this talk, we present the modified integrals to incorporate intertwining operator opposed to Fourier transform in the functional equation. In order to define analogous L-functions, we adopt the notion of "Good Sections" introduced by Piatetski-Shapiro and Rallis. We propose a concept of exceptional poles in the context of good section which can be carried out on the analysis of symmetric square or cubic Asai L-functions. (Received February 05, 2020)

1157-11-646 **Aaron Pollack*** (apollack@math.duke.edu). *A quaternionic Saito-Kurokawa lift.*

The Saito-Kurokawa lift is a very special instance of a theta correspondence, which takes holomorphic modular forms on metaplectic SL_2 to classical holomorphic Siegel modular forms on $\mathrm{SO}(2,3) = \mathrm{PGSp}_4$. Moreover, the Fourier coefficients of the lift are neatly related to those of the input. Following work of Gan, Gross, Savin, Wallach, Weissman, and the speaker, there is a good notion of "modular forms" on the groups $\mathrm{SO}(4,n)$ for $n \geq 3$, based on the so-called quaternionic discrete series. I will discuss a very special theta lift from classical holomorphic Siegel modular forms on Sp_4 to these quaternionic modular forms on $\mathrm{SO}(4,4)$, which is a close analogue of the Saito-Kurokawa lift. As an application, one obtains cusp forms on the exceptional group G_2 with all algebraic Fourier coefficients. (Received February 05, 2020)

12 ► Field theory and polynomials

1157-12-503 **Daniel C Smith-Tone*** (daniel.smith@nist.gov), 100 Bureau Drive, Stop 8960, Gaithersburg, MD 20899. *On Superdetermined Instances of the MinRank Problem.*

The MinRank Problem is a natural complexity theoretic problem closely related to cryptanalytic attacks on code- and multivariate-based schemes. Various methods of modelling the computational version of this problem exist, including two interesting algebraic techniques. While one of these two methods, minors modelling, is straightforward to analyze in certain parameter domains, the other, Kipnis-Shamir (KS) modelling, has been much harder to analyze and widely considered inferior in application to MinRank instances arising in crptography.

We present a new Grobner basis calculation technique and provide an analysis showing that for cryptographically significant MinRank instances, the KS method performs much better than minors. We conduct our complexity analysis by exposing non-trivial syzygies through the analysis of the Jacobian of the resulting system, taken with respect to a subset of variables. Our focus is on a particular set of instances that yield a very overdetermined system which we refer to as "superdetermined." We provide a tight complexity estimate for such instances and discuss its implications for key recovery on some cryptographic schemes. (Received February 03, 2020)

1157-12-539 **Francisco Revson Fernandes Pereira*** (revson.ee@gmail.com), School of Science and Technology, Physics Division, Via Madonna delle Carceri 9, 62032 Camerino, Macerata, Italy. *Entanglement-assisted Quantum Codes from Euclidean and Hermitian Cyclic Codes.* Preliminary report.

Entanglement-assisted quantum (QUENTA) codes are a subclass of quantum error-correcting codes that use entanglement as a resource. These codes can provide error correction capability higher than the codes derived from the traditional stabilizer formalism. In the first part of the talk, it is shown a general method to construct QUENTA codes from cyclic codes. Afterward, the method is applied to Reed-Solomon codes, BCH codes, and general cyclic codes. We use the Euclidean and Hermitian construction of QUENTA codes. Two families of QUENTA codes are maximal distance separable (MDS), and one is almost MDS or almost near MDS. The comparison of the codes in this paper is mostly based on the quantum Singleton bound. In the second part of the talk, an algorithm for computing intersection of vector spaces derived from cyclic codes is presented. Using this algorithm, we are able to construct new QUENTA codes from Euclidian and Hermitian cyclic codes. (Received February 04, 2020)

13 ► Commutative rings and algebras

1157-13-14 **Reeve Garrett*** (garrett.425@osu.edu), 415 W 8th Ave, Columbus, OH 43201. *On the inductive construction of rank 1, 2, and 3 valuations on $K(x, y)$ and their associated structures.* Preliminary report.

In a 1935 paper, Saunders Maclane constructed all rank one and some rank two valuations on the field of rational functions $K(x)$ extending a discrete valuation v_0 on a field K via inductively defined sequences of key polynomials in $K[x]$, and in 2009, K.A. Loper and F. Tartarone introduced the notion of an "upside down" valuation to construct the remaining rank two valuations on $K(x)$ extending v_0 on K and used both Maclane's construction and their own to represent integrally closed rings between $\mathbb{Z}_p[x]$ and $\mathbb{Q}[x]$ as a minimal intersection of valuation overrings and classify such rings (when they are Noetherian, Prüfer, Mori, PvMD, etc.) by the form of the valuations corresponding to the valuation overrings in that minimal intersection. In a series of 3 papers published in 1969, 1971, and 1973, H. Inoue attempted to generalize Maclane's approach to $K(x, y)$ but was not able to fully accomplish this goal. This talk will present fixes to the problems in Inoue's approach and some results on the construction of rank 2 and rank 3 valuations on $K(x, y)$ and integrally closed rings between $\mathbb{Z}_p[x, y]$ and $\mathbb{Q}[x, y]$. (Received November 01, 2019)

1157-13-25 **Simplice Tchamna*** (simplice.tchamna@gcsu.edu), 231 W. Hancock Street, Milledgeville, GA 31061. *On ring extension satisfying the star-hash property.*

Let $R \subseteq S$ be a ring extension, and let $\star : \mathcal{J}(\mathcal{R}, \mathcal{S}) \longrightarrow \mathcal{J}(\mathcal{R}, \mathcal{S})$ be a strict star operation on the extension $R \subseteq S$. The extension $R \subseteq S$ is said to satisfy the $\star$-hash property if $\bigcap_{M \in \mathcal{M}_\infty} R_{[M]} \neq \bigcap_{M \in \mathcal{M}_\in} R_{[M]}$ for any two distinct subsets $\mathcal{M}_\infty$ and $\mathcal{M}_\in$ of $\mathcal{M}(\star; \langle \mathcal{R}, \mathcal{S} \rangle)$, where $\mathcal{M}(\star; \langle \mathcal{R}, \mathcal{S} \rangle)$ is the set of all S-regular $\star$-maximal ideals of R. We study ring extensions satisfying the star-hash property. Special attention is given to the case of Prüfer extensions. (Received December 02, 2019)

1157-13-37 **Kuei-Nuan Lin*** (kul20@psu.edu) and **Yi-Huang Shen**. *On the Conjecture of Vasconcelos for Artinian Almost Complete Intersection Monomial Ideals.*

We state a conjecture of Vasconcelos which states that the Rees algebra of any Artinian almost complete intersection monomial ideal is almost Cohen–Macaulay. We explain the motivation and history of the conjecture, and we confirm this conjecture. This is joint work with Yi-Huang Shen. (Received January 03, 2020)

1157-13-106 **Craig Huneke*** (huneke@virginia.edu), VA 22904. *The Containment Problem for Symbolic Powers.* Preliminary report.

This talk will be a report on work in progress, joint with E. Grifo and V. Mukundan, that proves the following theorem: if I is a reduced ideal in a regular ring containing a field and defining a Gorenstein quotient, then the resurgence of I is strictly less than its codimension. As a consequence, the ideal satisfies the stable Harbourne conjecture. (Received January 21, 2020)

1157-13-140 **Thomas G Lucas*** (tglucas@uncc.edu). *Basic A+B construction and beyond.* Preliminary report.

We discuss the basic $A + B$ construction for creating examples and some recent variations. The first example (using the basic $A+B$ technique) will be of a reduced integrally closed ring R where the corresponding polynomial ring R[x] is not integrally closed. The second will be of a strongly Prüfer ring that is not a McCoy ring. Using a variation of the construction we provide examples of Prüfer rings with various local properties. Also we give an example to show that a theorem that appeared in my thesis (in 1983) actually did provide a new (at that time) sufficient condition for R[x] to be integrally closed. Since the session is in honor of the career of Bill Heinzer, each of the domains used to create the examples will be a subring of a power series domain in three or fewer indeterminates over a (sometimes specific) field. (Received January 24, 2020)

1157-13-143 **Selvi Kara Beyarslan*** (selvi@southalabama.edu), University of South Alabama, Department of Mathematics and Statistics, 411 N University Blvd, Mobile, AL 36606. *Weighted Stanley-Reisner Correspondence.* Preliminary report.

In this talk, we will discuss relations between algebraic invariants of Stanley-Reisner ideals of weighted and unweighted simplicial complexes. (Received January 24, 2020)

1157-13-154 **D. D. Anderson***, Department of Mathematics, The University of iowa, Iowa City, IA 52242, **Shiqi Xing** (sqxing@yeah.net), Department of Mathematics, Chengdu University of Technology, Chengdu, Peoples Rep of China, and **Muhammad Zafrullah** (mzafrullah@usa.net), Department of Mathematics, Idaho State University, Pocatello, ID 83209. *Almost Discrete Valuation Domains.*

We consider various generations of discrete valuation domains such as Noetherian almost valuation domains, almost valuation domains whose integral closure is a discrete valuation domain, and quasilocal and local almost principal ideal domains. (Received January 26, 2020)

1157-13-157 **Monica A. Lewis*** (malewi@umich.edu). *The local cohomology of a parameter ideal with respect to an arbitrary ideal.*

Let S be a complete intersection presented as R/J for R a regular ring and J a parameter ideal in R. Let I be an ideal containing J. It is well known that the set of associated primes of $H^i_I(S)$ can be infinite, but far less is known about the set of minimal primes. In 2017, Hochster and Núñez-Betancourt showed that if R has prime characteristic $p > 0$, then the finiteness of Ass $H^i_I(J)$ implies the finiteness of Min $H^{i-1}_I(S)$, raising the following question: is Ass $H^i_I(J)$ always finite? We give a positive answer when $i = 2$ but provide a counterexample when $i = 3$. The counterexample crucially requires Ass $H^2_I(S)$ to be infinite. The following question, to the best of our knowledge, is open: (under suitable hypotheses on R) does the finiteness of Ass $H^{i-1}_I(S)$ imply the finiteness of Ass $H^i_I(J)$? When S is a domain, we give a positive answer when $i = 3$. When S is locally factorial, we extend this to $i = 4$. Finally, if R has prime characteristic $p > 0$ and S is regular, we give a complete answer by showing that Ass $H^i_I(J)$ is finite for all $i \geq 0$. (Received January 26, 2020)

1157-13-164 **Farrah Yhee***, 426 2nd St., Apt. 5, Ann Arbor, MI 48103. *Ulrich modules do not always exist.* Preliminary report.

Ulrich modules were introduced by Bernd Ulrich in 1984 and has since been a very active area of research. The existence of Ulrich modules for complete local domains have powerful applications. For example, existence implies Lech's conjecture: given a flat local map of local rings from R to S, the Hilbert-Samuel multiplicity of S is at least the Hilbert-Samuel multiplicity of R. Until recently, it was unknown if there were any counterexamples to the existence of Ulrich modules for local domains. In this talk, I will introduce the notion of Ulrich modules and discuss implications of their existence. Then I will give a counterexample to the existence of Ulrich modules for complete local domains. (Received January 26, 2020)

1157-13-175 **Sonja Mapes***, 255 Hurley Hall, Notre Dame, IN 46556, and **Kuei-Nuan Lin**. *Computing projective dimension of monomial ideals via associated hypergraphs and lcm-lattices.*

Given a square-free monomial ideal I in a polynomial ring R over a field k, we would like to know the projective dimension of R/I. We recall the definition of the lcm-lattice of a monomial ideal introduced by Gasharov, Peeva and Welker, and the definition of the dual hypergraph of a square-free monomial ideal introduced by Kimura, Terai and Yoshida. Our work focuses on the relationship between the lcm-lattice and the dual hypergraph of a given square-free monomial ideal. We use the properties of lcm-lattice to find whether two different dual hypergraphs have the same projective dimension, and thus are able to extend some of the results by Lin and Mantero which compute the projective dimensions for ideals with certain hypergraphs. (Received January 27, 2020)

1157-13-180 **Jason G Boynton**, Fargo, ND 58102, **Jim Coykendall*** (jcoyken@clemson.edu), School of Mathematical & Statistical Sciences, Clemson University, Clemson, SC 29634, and **Chelsey Morrow**, Fargo, ND 58102. *Half-factoriality and the boundary function.*

An atomic integral domain, R, is said a half-factorial domain (HFD) if given the irreducible factorizations

$$\alpha_1\alpha_2\cdots\alpha_n = \beta_1\beta_2\cdots\beta_m$$

we have that $n = m$.

If R is an HFD with quotient field K then the boundary map is a function $\partial_R : K^* \longrightarrow \mathbb{Z}$ such that $\partial_R(xy) = \partial_R(x) + \partial_R(y)$ for all $x, y \in K^*$ and $\partial_R(\pi) = 1$ for all $\pi \in \text{Irr}(R)$ (∂_R is well-defined precisely when R is an HFD).

The boundary map has been used with some success to determine stability of the half-factorial property in some special ring extensions, and in this talk we will describe how the boundary map can play a utilitarian role in investigating (complete) integral closures of HFDs. We will also explore the open question as to whether nonunits are always contained in the support of the boundary function in the case of atomic integral extensions. (Received January 27, 2020)

1157-13-183 **Matthew Mastroeni*** (mmastro@okstate.edu), **Hal Schenck** and **Mike Stillman**.
Quadratic Gorenstein Rings and the Koszul Property.

Many quadratic Gorenstein rings arising in algebraic geometry, such as the coordinate rings of canonical curves, Grassmannians, and certain sets of points in projective space, are also always Koszul. In this talk, we aim to answer the question: For which integers $c, r \geq 0$ is every quadratic Gorenstein algebra over a field with codimension c and regularity r Koszul? We give an affirmative answer when $c = r + 1$, and time permitting, we will discuss how to produce numerous examples of non-Koszul quadratic Gorenstein rings in almost all other cases, negatively answering a question of Conca, Rossi, and Valla concerning the $r = 3$ case in the process. (Received January 27, 2020)

1157-13-186 **Richard Erwin Hasenauer*** (hasenaue@nsuok.edu), **Kailey Earlywolf**, **Bethany Kubik** and **Chris Sherman**.
τ-elasticity in the integers. Preliminary report.

A factorization of an integer $b = \pm a_1 a_2 \cdots a_k$ is said to be τ_n-factorization if $a_1 \equiv a_2 \equiv ... \equiv a_k \pmod{n}$. An element is a τ_n-atom if all of its τ_n-factorizations are of length one. The ratio of the longest factorization to the shortest factorization is called elasticity. For a given n, we explore the elasticity of all integers that factor into a product of τ_n-atoms. (Received January 27, 2020)

1157-13-188 **Naoki Endo*** (t.nacmath@gmail.com), West Lafayette, IN 47907. *On weakly Arf rings.*

In this talk, we introduce weakly Arf rings and explore the relation between the rings and Arf rings, defined by J. Lipman in 1971. The present purpose is to propose a generalization of Arf rings and develop the basic theory. We provide characterizations for rings to be weakly Arf in contrast with Arf rings. Examples are also explored as well. This talk is based on the recent research jointly with E. Celikbas, O. Celikbas, C. Ciuperca, S. Goto, R. Isobe, and N. Matsuoka. (Received January 28, 2020)

1157-13-210 **Michael Gekhtman*** (mgekhtma@nd.edu), Department of Mathematics, University of Notre Dame, Notre Dame, IN 46556. *Periodic staircase matrices and generalized cluster algebras.*

As is well-known, cluster transformations in cluster algebras of geometric type are often modeled on determinant identities, such short Plücker relations, Desnanot-Jacobi identities and their generalizations. I will present a construction that plays a similar role in a description of generalized cluster transformations and discuss its applications to generalized cluster structures in GL(n) compatible with a certain subclass of Belavin-Drinfeld Poisson-Lie backers, in the Drinfeld double of GL(n) and in spaces of periodic difference operators. (Based on joint work with M. Shapiro and A. Vainshtein). (Received January 29, 2020)

1157-13-225 **Yihui Liang*** (liang226@purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907, and **Giulio Caviglia** (gcavigli@math.purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907. *Explicit Stillman bounds for all degrees.* Preliminary report.

In 2016 Ananyan and Hochster proved Stillman's conjecture by showing the existence of a uniform upper bound for the projective dimension of all homogeneous ideals generated by n forms of degree at most d. Explicit bounds for forms of degrees 5 and higher are not yet known. We give explicit Stillman-type bounds for all degrees through a recurrence relation, which behave like power towers of height d^3. This is done by proving a bound $D(k, d)$, which controls the number of generators of a minimal prime over an ideal of a regular sequence of k or fewer polynomials with degree d, and supplementing it into Ananyan and Hochster's proof. This is joint work with Giulio Caviglia. (Received February 03, 2020)

1157-13-239 **Michael C Loper*** (loper012@umn.edu). *What Makes a Complex Virtual.*

Let S be the Cox ring of a smooth projective toric variety and B be the irrelevant ideal. In 2017, Berkesch, Erman, and Smith introduced virtual resolutions for toric varieties as an analogue of minimal free resolutions for projective space. Virtual resolutions are graded complexes of free S-modules that allow B-torsion homology. I will name two algebraic conditions that determine whether a bounded chain complex of free S-modules is a virtual resolution. This theorem is similar to the depth criterion of exactness that Buchsbaum and Eisenbud published in 1973. (Received January 30, 2020)

1157-13-241 **Benjamin Drabkin*** (benjamin.drabkin@huskers.unl.edu) and **Lorenzo Guerrieri**.
Quasi-equigenerated and Freiman cover ideals of graphs.

A quasi-equigenerate monomial ideal I is a Freiman ideal if the inequality for the number of generators of I^2 arising from Freiman's theorem is an equality. Such ideals are particularly interesting because there exists an

explicit formula for computing the number of generators of I^n for all n. This talk will address the problem of characterizing when cover ideals of simple graphs are Freiman. (Received January 30, 2020)

1157-13-244 **Alessandra Costantini*** (alessanc@ucr.edu). *Cohen-Macaulay property of special fiber rings.* Preliminary report.

Let R be a Noetherian local ring. The special fiber ring, or fiber cone, $\mathcal{F}(E)$ of a module E is defined as the tensor product of the Rees algebra $\mathcal{R}(E)$ with the residue field of R. In this talk, I will discuss a general technique to study the Cohen-Macaulay property of $\mathcal{F}(E)$, and provide sufficient conditions for $\mathcal{F}(E)$ to be Cohen-Macaulay. The latter generalize previous work of Corso, Ghezzi, Polini and Ulrich and of Montaño for the special fiber ring of an ideal. The material presented in this talk is part of my Ph.D. thesis. (Received January 30, 2020)

1157-13-271 **John Eagon**, **Ezra Miller** and **Erika Ordog***, erika.ordog@duke.edu. *Minimal resolutions of monomial ideals.*

It has been an open problem since the 1960s to construct closed-form, canonical, combinatorial minimal free resolutions of arbitrary monomial ideals in polynomial rings. This talk explains how to solve the problem, in characteristic 0 and almost all positive characteristics, using sums over lattice paths of combinatorial data from simplicial complexes, one simplicial complex for each lattice point. Any minimal free resolution of any monomial ideal must – either implicitly or explicitly – produce homomorphisms between various homology groups of these simplicial complexes. Therefore an important aspect of the solution is an explicit way to write down canonical homomorphisms between these homology groups without choosing bases. (Received January 31, 2020)

1157-13-272 **Nicholas J Werner*** (wernern@oldwestbury.edu) and **Eric Swartz**. *Fuchs' Problem for Unit Groups.*

Given a group G, we say that G is *realizable* if there exists a ring R such that the group of units of R is isomorphic to G. Determining whether a group or family of groups is realizable has come to be called *Fuchs' problem* after László Fuchs, who in 1960 posed the problem of determining which groups can be realized as the group of units of a commutative ring. To date, no complete answer has been given to Fuchs' problem, although many partial answers or modifications have been produced. In 1963, Gilmer classified all cyclic groups that can realized by a finite commutative ring; by 1971, Fuchs' problem had been completely solved for all cyclic groups, and for any finite group of odd order. In this talk, we will survey these historical results and discuss significant recent progress on solving Fuchs' problem for families of groups such as symmetric groups; alternating groups; finite simple groups; p-groups; nilpotent groups; and various classes of Abelian groups. (Received January 31, 2020)

1157-13-278 **Susan Morey***, Department of Mathematics, Texas State University, 601 University Dr., San Marcos, TX 78666, and **Huy Tai Ha**. *Rees Algebras, Jacobian Duals, and Graphs.*

Blowup algebras, in particular Rees algebras, form an important link between commutative algebra and algebraic geometry. A natural path to study these algebras is through the ideal of defining equations. A Jacobian dual of a presentation matrix of an ideal I can be a fruitful source of generators of the defining ideal of the Rees algebra of I. In particular, minors of a Jacobian dual have served a key role in determining the defining equations for several classes of ideals. In 1995 Rafael Villarreal determined the generators of the ideal of defining equations of the Rees algebra of the edge ideal of a graph in terms of the structure of the graph. In this talk, an alternate description of the equations of the Rees algebra of an edge ideal will be given in terms of minors of a Jacobian dual matrix. Comparing the two descriptions leads to algebraic approaches to problems in graph theory, including digraphs. (Received January 31, 2020)

1157-13-288 **Nathan Fieldsteel** and **Patricia Klein*** (klein847@umn.edu). *Gröbner bases and the Cohen–Macaulay property of Li's double determinantal varieties.*

We consider double determinantal varieties, a special case of Nakajima quiver varieties. Li Li conjectured that double determinantal varieties are normal irreducible Cohen-Macaulay varieties whose defining ideals have a Gröbner basis given by their natural generators. We use liaison theory to prove this conjecture. (Received January 31, 2020)

1157-13-289 **Uwe Nagel*** (uwe.nagel@uky.edu), University of Kentucky, Lexington. *Segre's Regularity Bound for Fat Point Schemes.*

A 1962 result of Segre inspired Trung and, independently, Fatabbi and Lorenzini in 2001 to conjecture an optimal bound for the Castelnuovo-Mumford regularity of a projective scheme of fat points. In joint work with Bill Trok, we recently established this conjecture. We discuss some of the arguments and, in particular, a decomposition result for matroids. (Received January 31, 2020)

1157-13-290 **Uwe Nagel*** (uwe.nagel@uky.edu), University of Kentucky, Lexington. *Interpolation and Castelnuovo-Mumford Regularity.*

A basic problem in Hermite interpolation theory is to determine the least degree of a homogeneous polynomial that vanishes to some specified order at every point of a given finite set. This is related to the problem of finding the Castelnuovo-Mumford regularity of an ideal generated by powers of linear forms. We discuss results in the case when the number of points or linear forms is small compared to the dimension of their linear span.

This is based on joint work with Bill Trok. (Received January 31, 2020)

1157-13-291 **Uwe Nagel*** (uwe.nagel@uky.edu), University of Kentucky, Lexington. *Stabilization in Sequences of Ideals and Modules.*

Consider an ascending chains of symmetric ideals in more and more variables. We discuss a finiteness results in a suitable category that leads, for a fixed integer j, to a uniform description of j-syzygy modules of all but finitely many ideals in such a chain.

This is based on joint work with Tim Römer. (Received January 31, 2020)

1157-13-302 **Lee Klingler*** (klingler@fau.edu), Department of Mathematical Sciences, Florida Atlantic University, 777 Glades Road, Boca Raton, FL 33431, and **Alan Loper**, **Warren McGovern** and **Matthew Toeniskoetter**. *Semiclean Group Rings.* Preliminary report.

We call the commutative ring R a *clean ring* if every element of R can be written as the sum of a unit and an idempotent. The notion of a clean ring was defined by Nicholson [1977] in a study of exchange rings and lifting idempotents. Ye [2003] introduced the notion of semiclean rings: R is called a *semiclean ring* if every element of R can be written as the sum of a unit and a periodic element, where $r \in R$ is called *periodic* if there are natural numbers $k < n$ such that $r^k = r^n$. In joint work with Alan Loper, Warren McGovern, and Matthew Toeniskoetter, we explore conditions on the commutative ring R and the abelian group G that make the group ring $R[G]$ semiclean. (Received February 01, 2020)

1157-13-315 **Hans Schoutens*** (hschoutens@citytech.cuny.edu). *Closure operations in characteristic zero.*

I will give a brief survey of closure operations on ideals based on the Frobenius endomorphism (in positive characteristic therefore), such as tight closure, and how they can be transferred using ultraproducts to characteristic zero. (Received February 01, 2020)

1157-13-317 **William Heinzer** (heinzer@purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907-1395, **K. Alan Loper** (lopera@math.ohio-state.edu), Department of Mathematics, Ohio State University-Newark, Newark, OH 43055, and **Bruce Olberding*** (bruce@nmsu.edu), Department of Mathematical Sciences, New Mexico State University, Las Cruces, NM 88011. *The tree of quadratic transforms of a regular local ring of dimension two.*

Let D be a 2-dimensional regular local ring and let $Q(D)$ denote the quadratic tree of 2-dimensional regular local overrings of D. We explore the topology of the tree $Q(D)$ and the family $\mathcal{R}(D)$ of rings obtained as intersections of rings in $Q(D)$. If A is a finite intersection of rings in $Q(D)$, then A is Noetherian and the structure of A is well understood. However, other rings in $\mathcal{R}(D)$ need not be Noetherian. We examine topological properties of the quadratic tree $Q(D)$ and the structure of rings in the set $\mathcal{R}(D)$. (Received February 01, 2020)

1157-13-323 **Michael DiPasquale*** (michael.dipasquale@colostate.edu), Colorado State University, Department of Mathematics, 1874 Campus Delivery, Fort Collins, CO 80523, and **Nelly Villamizar**. *A linear bound on the regularity of uniform power ideals.* Preliminary report.

A power ideal is an ideal generated by powers of linear forms; a uniform power ideal is one which is generated by the same power of every linear form. Power ideals appear in a variety of places, perhaps most notably in the Waring problem for homogeneous forms. Our motivation for studying these ideals is drawn from applications to multivariate splines, although we will not elaborate on this connection.

In this talk we present a linear bound on the regularity of a uniform power ideal in which the 'slope' of the linear bound is given in terms of the Waldschmidt constant of the dual set of points. Time permitting, we will specialize to certain power ideals in three variables whose combinatorics are encoded by planar graphs. For such ideals we present a simple regularity bound whose derivation relies on a combinatorial estimation of the Waldschmidt constant due to Cooper, Harbourne, and Teitler. This is joint work with Nelly Villamizar. (Received February 01, 2020)

1157-13-333 **Yevgeniya Tarasova*** (ytarasov@purdue.edu). *Generators of Residual Intersections.* Preliminary report.

Under suitable technical assumptions, a description is given for the generators of s-residual intersections in terms of lower residual intersections, if $s \geq \mu(I) - 2$. This implies that s-residual intersections can be expressed in terms of links, if $\mu(I)$ is less than or equal to the height of I plus 3 and some other hypotheses are satisfied. (Received February 01, 2020)

1157-13-340 **Thiego H Freitas**, **Victor H Jorge Pérez**, **Roger A Wiegand** and **Sylvia M Wiegand*** (swiegand1@unl.edu), 201 Avery Hall, UNL, Lincoln, NE 68588-0130. *Vanishing of Tor over fiber products.*

Let $(S, \mathfrak{m}, k)$ and $(T, \mathfrak{n}, k)$ be local rings, and let R denote their fiber product over their common residue field k. Inspired by work of Nasseh and Sather-Wagstaff, we explore consequences of vanishing of $\operatorname{Tor}^R_{\mathfrak{m}}(M, N)$ for various values of $\mathfrak{m}$, where M and N are finitely generated R-modules. (Received February 02, 2020)

1157-13-347 **Rankeya Datta**, **Nicholas Switala** and **Wenliang Zhang***, 851 S Morgan St, MSCS Department, Chicago, IL 60607. *Annihilators of D-modules in mixed characteristic.*

Let R be either a polynomial ring or formal power series ring over a DVR V of mixed characteristic, and let D be the ring of V-linear differential operators on R. I will discuss some recent results concerning annihilators of D-modules in R. This is a joint work with Rankeya Datta and Nicholas Switala. (Received February 02, 2020)

1157-13-351 **Youngsu Kim*** (yk009@uark.edu), University of Arkansas, 850 W. Dickson ST #225, Fayetteville, AR 72701, **Lance E Miller**, University of Arkansas, Fayetteville, AR 72701, and **Wenbo Niu**, University of Arkansas, Fayetteville, AR 72701. *The log canonical threshold of a generic link.*

Linkage is a classical topic in algebraic geometry and commutative algebra. Fix an affine space A. We say two subschemes X, Y of A are linked if their union is a complete intersection in A, and X and Y do not have a common component. Two linked subschemes share several properties in common. Linkage has been studied by various people, Artin-Nagata, Peskine-Szprio, Huneke-Ulrich, to name a few.

In 2014, Niu showed that if Y is a generic link of a variety X, then $\operatorname{lct}(A, X) \leq \operatorname{lct}(A', Y)$, where lct stands for log canonical threshold and A' is an extension of A. In this talk, we show that if Y is a generic link of a determinantal variety X, then X and Y have the same log canonical threshold. This is joint work with Lance E. Miller and Wenbo Niu. If time permits, we show an extension of this result. (Received February 02, 2020)

1157-13-352 **Claudia Miller*** (clamille@syr.edu) and **Hamid Rahmati**. *Transferring algebra structures on complexes.*

We discuss a homological method for transferring algebra structures on complexes along suitably nice homotopy equivalences. We also show how one can get such homotopy equivalencies, from old ones, using a homological tool called the perturbation lemma. Lastly, we discuss how to use the homotopy on the tautological Koszul complex given by the weighted de Rham map to build a concrete permutation invariant dg-algebra structure on a well-known resolution. This is joint work with Hamid Rahmati. (Received February 02, 2020)

1157-13-353 **Claudia Miller*** (clamille@syr.edu), **Hamid Rahmati** and **Rebecca R.G.**. *Betti numbers of the Frobenius powers of the maximal ideal over general hypersurfaces in 3 variables.*

We discuss the conjectured curious behavior of the Betti numbers of the Frobenius powers of the maximal ideal in hypersurfaces $R = k[x, y, z]/(f)$, where k is a field of positive characteristic p, as described by Kustin and Ulrich. We prove that if f is chosen to be very general (with k algebraically closed) and of degree the opposite parity to p then high enough Frobenius powers of the maximal ideal have identical high graded Betti numbers up to explicit shifts, and we find some structural features of the resolution and its associated matrix factorization. We also find the Hilbert-Kunz function of R. The proof is based on our constructions of free resolutions of relatively compressed Artinian graded algebras. (Received February 02, 2020)

1157-13-358 **Susan M. Cooper*** (susan.cooper@umanitoba.ca), Department of Mathematics, University of Manitoba, 476 Machray Hall, Winnipeg, MB R3T 2N2, Canada. *Symbolic Powers of Edge Ideals.* Preliminary report.

Understanding resolutions of symbolic powers of ideals is challenging, even for square-free monomial ideals. In this talk we investigate a splitting for symbolic powers of edge ideals of graphs. The splitting provides a technique for exploring the graded Betti numbers of edge ideals. We close with a discussion of bounds for graded Betti

numbers when one considers the parallelization of the graph. This is joint work with S. Da Silva, M. Gutkin, and T. Reimer. (Received February 02, 2020)

1157-13-362 **Dolors Herbera**, **Pavel Příhoda** and **Roger A Wiegand*** (rwiegand@unl.edu), Department of Mathematics, University of Nebraska, Lincoln, NE 68588-0130.
Pure-projective modules over Noetherian local domains.

Let $(R, \mathfrak{m})$ be a commutative, local, Noetherian integral domain. An R-module M is *pure-projective* provided the map $\mathrm{Hom}_R(M, B) \to \mathrm{Hom}_R(M, C)$ is surjective whenever $0 \to A \to B \to C \to 0$ is a pure-exact sequence. As with projective modules, there is a structure theorem: M is pure-projective if and only if M is isomorphic to a direct summand of a direct sum of finitely generated modules. It follows from Kaplansky's theorem that every pure-projective is a direct sum of countably generated modules. For a given local domain R, it is interesting to ask whether or not every pure-projective module is actually a direct sum of finitely generated R-modules, and also to study the structure of the monoid $\mathrm{V}^*(X)$ consisting of isomorphism classes of direct summands of $X^{(\omega)}$, where X is a fixed finitely generated module. We will study these questions by making use of descent properties of the inclusion $R \to \widehat{R}$, where $\widehat{R}$ is the $\mathfrak{m}$-adic completion of R. (Received February 02, 2020)

1157-13-371 **Patricia Klein*** (klein847@umn.edu) and **Jenna Rajchgot**. *Geometric vertex decomposition and liaison.*

Geometric vertex decomposition and liaison are two frameworks that can be used to study algebraic varieties. These approaches have been used historically by two distinct communities of mathematicians. In this talk, we will describe a connection between the two. In particular, we will see how each geometrically vertex decomposable ideal is linked by a sequence of ascending elementary G-biliaisons of height 1 to an ideal generated by indeterminates and, conversely, how every G-biliaison of a certain type gives rise to geometric vertex decomposition. As a consequence, we establish that several families of ideals are glicci, including Schubert determinantal ideals, defining ideals of varieties of complexes, and homogeneous defining ideals of lower bound cluster algebras. This talk is based on joint work with Jenna Rajchgot. (Received February 02, 2020)

1157-13-382 **Christine Berkesch*** (cberkesc@umn.edu), Minneapolis, MN. *The geometry of toric syzygies.*

The homological algebra of minimal free resolutions, or syzygies, capture intricate geometric properties of a subvariety in projective space, as first shown in work of Hilbert more than a century ago. However, when the ambient space is a product of projective spaces or a more general smooth projective toric variety, this relationship fails. To remedy this, we consider another homological object, called a virtual resolution, which encodes several geometric invariants in close parallel to the more classical case of syzygies over projective space. (Received February 02, 2020)

1157-13-394 **Andrew W. Snowden***, asnowden@umich.edu. *Two tales from equivariant commutative algebra.*

Suppose that a group G acts on a commutative ring R. One can then formulate equivariant analogs of many basic concepts from commutative algebra: for example, one has a notion of G-prime ideal, or the G-spectrum of R. I will explain a little about the general set-up, and then discuss two results: the first describes $\mathbf{GL}_\infty$-prime ideals using supergeometry; the second (joint with Rohit Nagpal) classifies the $\mathfrak{S}_\infty$-prime ideals in the infinite variable polynomial ring. (Received February 03, 2020)

1157-13-407 **Dario Spirito***, Dipartimento di Matematica, Università di Padova, Via Trieste 63, 35121 Padova, Italy. *Pseudo-monotone sequences and extensions of valuations.*

The concept of pseudo-convergent sequences was introduced by Ostrowski to classify one-dimensional extensions of a valuation domain to the field $K(X)$ of rational functions, and later used by Kaplansky in his study of valuation rings with no immediate extensions; furthermore, they were generalized to pseudo-monotone sequences by Chabert to study the polynomial closure on valuation domains of rank one. Pseudo-convergent sequences were also used by Loper and Werner to construct a class of extensions of V to $K(X)$.

In this talk, I will show how the latter construction can be generalized to arbitrary pseudo-monotone sequences E, obtaining a very wide class of extensions of V to $K(X)$; in particular, if the quotient field of V is algebraically closed then all extensions of V to $K(X)$ can be obtained in this way. I will also discuss the topological properties of the set of extensions induced by pseudo-convergent extensions and how these sequences can be used to obtain counterexamples: in particular, they can be used to construct essential domains that are not PvMDs and to show that the polynomial closure is not a topological closure for valuation domains of rank > 1.

This is a joint work with Giulio Peruginelli. (Received February 03, 2020)

1157-13-409 **Hailong Dao***, 405 Snow Hall, 1460 Jayhawk Blvd, Lawrence, KS 66045, and **Joseph Doolittle** and **Justin Lyle**. *Minimal Cohen-Macaulay complexes.*

We define and study the notion of a minimal Cohen-Macaulay simplicial complex. We prove that any Cohen-Macaulay complex is shelled over a minimal one in our sense, and we give sufficient conditions for a complex to be minimal Cohen-Macaulay. We show that many interesting examples of Cohen-Macaulay complexes in combinatorics are minimal, including Rudin's ball, Ziegler's ball, the dunce hat, and recently discovered non-partitionable Cohen-Macaulay complexes. We further provide various ways to construct such complexes (Received February 03, 2020)

1157-13-411 **Hailong Dao***, 405 Snow Hall, 1460 Jayhawk Blvd, Lawrence, KS 66045. *Ideals invariants and additive combinatorics.*

In this talk I will focus on some old and new ideal-theoretic invariants of and how they connect to additive combinatorics. (Received February 03, 2020)

1157-13-412 **Lindsey Hill*** (hill178@purdue.edu) and **Rachel Lynn**. *Specialization of Integral Closures of Ideals by General Elements.* Preliminary report.

J. Hong and B. Ulrich proved that the integral closure of an ideal of height at least 2 is compatible with specialization by *generic elements*, allowing one to use induction on the height of the ideal to prove results about integrally closed ideals. This technique's applications are limited by the fact that one must extend the base ring to a polynomial ring or localization thereof in order to define generic elements. We have shown that the integral closure of an ideal of height at least 2 is compatible with specialization by *general elements*, allowing one to induct on the height of the ideal without extending the ring. This is based on joint work with Rachel Lynn. (Received February 03, 2020)

1157-13-413 **Mircea Mustaţă**, **Michael Perlman*** (mperlman@nd.edu) and **Claudiu Raicu**. *Hodge ideals for the determinant hypersurface.*

Hodge ideals are invariants of hypersurface singularities that arise from birational geometry and Morihiko Saito's theory of mixed Hodge modules. We compute explicitly these ideals for hypersurfaces defined by the generic determinant. To do so, we take advantage of the rich symmetry of their log resolutions, the spaces of complete collineations. Joint work with Mircea Mustaţă and Claudiu Raicu. (Received February 03, 2020)

1157-13-429 **Eamon Quinlan-Gallego*** (equinlan@umich.edu), East Hall, 530 Church Street, Ann Arbor, MI 48109. *Bernstein-Sato theory in positive characteristic.*

The Bernstein-Sato polynomial of an ideal $\mathfrak{a} \subseteq \mathbb{C}[x_1, \ldots, x_n]$ is an invariant that originated in complex analysis and with now strong applications in birational geometry and singularity theory over $\mathbb{C}$. In this talk we present an analogue of this invariant in positive characteristic and discuss some of its basic properties. (Received February 03, 2020)

1157-13-433 **Caitlyn Booms**, **Daniel Erman*** (derman@math.wisc.edu), **Kevin Kristensen** and **Jay Yang**. *Random Flag Complexes and Torsion for Veronese Syzygies.* Preliminary report.

We use a random simplicial complex model to make predictions about p-torsion for Veronese syzygies. (Received February 03, 2020)

1157-13-456 **Cheng Meng*** (cheng319@purdue.edu), 150 N. University Street, West Lafayette, IN 47907-2067. *Hilbert coefficients of prime ideals and restriction on depth.*

In this paper, we prove that for a prime ideal P of dimension r inside a polynomial ring, if there exist a positive integer $s \leq r$ such that for s general linear forms $l_1, ..., l_s$, the $r - s$-th Hilbert coefficient of P and $(P, l_1, ..., l_s)$ differ by 1, then the depth of S/P is $s - 1$. We prove this by studying the restrictions on the generic initial ideal of a prime ideal inside a polynomial ring. (Received February 04, 2020)

1157-13-457 **Jesse Elliott*** (jesse.elliott@csuci.edu). *Multiplicative ideal theory of rings.*

We outline a strategy for generalizing multiplicative ideal theory of domains to commutative rings, and we also generalize parts of it to modules over noncommutative rings. This talk is a summary of my new book, *Rings, Modules, and Closure Operations.* (Received February 03, 2020)

1157-13-460 **Seth Gannon*** (dsgann01@louisville.edu), 1124 Reutlinger Ave, Apt 204, Louisville, KY 40204, and **Hamid Kulosman**. *The condition for cyclic codes over $\mathbb{Z}_4$ to be LCD.*

A linear code with complementary dual (LCD code) is a linear code C whose intersection with its dual code $C^{\perp}$ is only the zero codeword (i.e. $C \cap C^{\perp} = \{0\}$). These codes are of importance for not only theory, but for application as well. LCD codes give an optimum linear coding solution for a two user binary adder channel,

they have applications to data storage, they simplify the maximum-likelihood decoding problem, and these codes have recently been shown to have applications to cryptography. It was first shown by James Massey a condition of linear codes to have complementary dual. Later, Yang & Massey produced a condition for a cyclic code over a Galios Field to have a complementary dual. It will be shown that a necessary and sufficient condition for a cyclic code C over $\mathbb{Z}_4$ of odd length N to be an LCD code is that $C = (f(x))$, where f is a self-reciporocal monic divisor of $X^N - 1 \in \mathbb{Z}_4[X]$. (Received February 03, 2020)

1157-13-463 **Monte Cooper** and **Edward F. Price III*** (price79@purdue.edu). *Degree Bounds on Defining Equations of Rees Algebras of Certain Determinantal Ideals.* Preliminary report.

Let R be a polynomial ring over a field, and suppose that A is an $m \times n$ matrix over R where all entries are homogeneous of the same degree. Let X be a generic $m \times n$ matrix over R. Then there is a specialization of $R[X]$ which maps X to A. Suppose that I is the ideal generated by the $t \times t$ minors of A and J is the ideal generated by the $t \times t$ minors of X. If I satisfies certain conditions, then one can prove x-degree bounds on the defining equations of the Rees ring $\mathcal{R}(I)$ given sufficient information about powers of J and about $\mathcal{R}(J)$. This talk will explore these conditions and give explicit bounds for classes of examples. (Received February 03, 2020)

1157-13-471 **Louiza Fouli***, New Mexico State University, Las Cruces, NM 88003, and **Ela Celikbas**, **Emilie Dufresne**, **Elisa Gorla**, **Kuei-Nuan Lin**, **Claudia Polini** and **Irena Swanson**. *On the defining equations of the Rees algebra of determinantal ideals.* Preliminary report.

Let $X = (x_{ij})$ be an $m \times n$ sparse matrix of indeterminates, that is a matrix whose entries are either zero or indeterminates, and let I be the ideal of maximal minors of X in a polynomial ring $R = k[x_{ij}]$ over a field k. We present some recent progress in obtaining the defining equations of the Rees algebra of such an ideal I when X is any $2 \times n$ sparse matrix. Moreover, we will discuss various properties of the Rees algebra and the special fiber cone of such an ideal. (Received February 03, 2020)

1157-13-476 **Benjamin Briggs**, **Eloísa Grifo*** (eloisa.grifo@ucr.edu) and **Josh Pollitz**. *Towards a new characterization of complete intersections.* Preliminary report.

Regular local rings R can be characterized homologically by the property that every finitely generated R-module has finite projective dimension. This can be phrased in the language of derived categories: R is regular if and only if every complex with finitely generated homology is a perfect complex.

Recently, Pollitz established an analogous derived category characterization for complete intersections: R is a complete intersection if and only if each complex with finitely generated homology can build some perfect complex in the derived category using finitely many cones and retracts. We investigate whether there is a version of this result using only finitely generated modules, and whether there is a natural finitely generated module that witnesses the failure of R being a complete intersection, analogously to the role that the residue field plays in the characterization of regular rings. (Received February 03, 2020)

1157-13-491 **Hang Huang**, **Michael Perlman**, **Claudia Polini**, **Claudiu Raicu*** (craicu@nd.edu) and **Alessio Sammartano**. *Relations between the* 2×2 *minors of a generic matrix.*

I will explain the proof of a conjecture of Bruns-Conca-Varbaro, describing the minimal relations between the 2×2 minors of a generic matrix. Interpreting these relations as polynomial functors, and applying transpose duality as in the work of Sam-Snowden, this problem is equivalent to understanding the relations satisfied by 2×2 generalized permanents. The proof then combines Koszul homology calculations on the minors side, with a study of subspace varieties on the permanents side, and with the Kempf-Weyman technique (on both sides). Joint work with H. Huang, M. Perlman, C. Polini, and A. Sammartano. (Received February 03, 2020)

1157-13-498 **Josua C Illian*** (jillian@oakland.edu) and **Li Li** (li2345@oakland.edu). *Gröbner basis for the double determinantal ideal.* Preliminary report.

Determinantal ideals provide important examples in commutative algebra and algebraic geometry. We consider a generalization closely related to Nakajima quiver varieties, called double determinantal ideals, and show that the natural generators form a Gröbner basis, using a method of S-pair construction. As a result, we generalize the combinatorial objects called pipe dreams, introduced by Allen Knutson, to explicitly describe the Stanley-Reisner simplicial complex of the double determinantal ideal and prove shellability, thus implying Cohen-Macaulayness. This provides an algebraic-combinatorial proof to a conjecture of the second author which was recently proved by Fieldsteel and Klein using liaison theory. (Received February 03, 2020)

1157-13-502 **Paolo Mantero***, pmantero@uark.edu. *The interpolation problem for star configurations.*
The (homogeneous) interpolation problem for a set X of points in projective space asks to determine (for any d and m) the number of linearly independent equations in degree d passing through X at least m times. The problem is in general extremely challenging, even for general points in $\mathbb{P}^2$.

In the present talk we determine the structure of all symbolic powers of all star configurations (of hypersurfaces) and compute all their graded Betti numbers. In particular, this solves the interpolation problem for this class of sets of points.

Along the way, we prove that star configurations of hypersurfaces have c.i. quotients, and use combinatorial computations to prove an explicit formula for part of their graded Betti numbers. (Received February 03, 2020)

1157-13-520 **Jay Yang*** (jkyang@umn.edu). *Virtual Resolutions of Monomial Ideals.*
Virtual resolutions as introduced by Berkesch-Erman-Smith are free complexes for ideals in the Cox ring of a toric variety which can be simpler and more geometrically meaningful than the minimal free resolution. I discuss recent work using combinatorial techniques from monomial ideals to provide bounds on the length of virtual resolutions, and future directions on this front. (Received February 03, 2020)

1157-13-522 **Claudia Polini**, **Ngo Viet Trung**, **Bernd Ulrich** and **Javid Validashti*** (jvalidas@depual.edu). *Multiplicity Sequence and Integral Dependence.*
Let $I \subset J$ be a pair of arbitrary ideals in an equidimensional and universally catenary Noetherian local ring. We prove that I and J have the same integral closure if and only if they have the same multiplicity sequence introduced by Achilles and Manaresi. We also obtain a Principle of Specialization of Integral Dependence, which gives a condition for integral dependence in terms of the constancy of the multiplicity sequence in families. (Received February 03, 2020)

1157-13-526 **Janet Page*** (jrpage@umich.edu). *Measuring the Gorenstein property using the trace of the canonical module for some combinatorial rings.*
Trace ideals can be defined for any module over a ring and have been studied in various contexts. Recently, Herzog, Hibi, and Stamate studied the trace of the canonical module for local Cohen Macaulay rings and showed it can be viewed as a measure of how close the ring is to being Gorenstein. In this talk, I will introduce trace ideals and discuss recent results on the trace of the canonical module for rings in which we can exploit some combinatorial data, including Hibi rings which are defined in terms of finite posets and some other toric rings. I will discuss some joint work with Jürgen Herzog and Fatemeh Mohammadi, as well as more recent results with Mitsuhiro Miyazaki. (Received February 04, 2020)

1157-13-528 **C-Y. Jean Chan***, Department of Mathematics, PE 214, Central Michigan University, Mt. Pleasant, MI 48859, **I-Chiau Huang**, Institute of Mathematics, Academia Sinica, Taiwan, and **Jung-Chen Liu**, National Taiwan Normal University. *Affine semigroup rings – combinatorial v.s. algebraic properties.* Preliminary report.
Let S be an affine semigroup ring. Let R be an affine semigroup ring contained in S as a subgring. We consider S as an algebra over R (not necessarily a finite algebra). In this talk, we will define Apéry monomials of S with respect to R. This generalizes the notion of the Apéry set in the numerical semigroup rings. Interesting algebraic properties are encoded in the set of Apéry monomials. This is a joint work with I-C. Huang and J.-C. Liu. (Received February 04, 2020)

1157-13-536 **Jason R. Juett*** (juettj@uwstout.edu), **Christoper Park Mooney**, **Lois W. Ndungu** and **Alejandra M. Roman**. *Unique factorization of ideals in rings with zero divisors.* Preliminary report.
Dedekind domains are the ideal-theoretic analog of unique factorization domains in the sense that they are precisely the domains where every nonzero proper ideal is a unique up to order product of nonfactorable ideals, where a proper ideal is *(weakly) nonfactorable* if it is not a(n) (irredundant) product of two nonzero proper ideals. Juett and Mooney recently generalized this classic result by proving that every proper ideal of a commutative ring R is a unique up to order irredundant product of weakly nonfactorable ideals if and only if R is a finite direct product of Dedekind domains and Artinian local principal ideal rings, arriving at a new characterization of general Z.P.I.-rings. In other words, general Z.P.I.-rings are the ideal-theoretic analog of *Fletcher unique factorization rings*, which are rings where every nonunit is a unique up to order and associates product of irreducible elements. In this talk we will discuss this and several other extensions of the above classic "unique ideal factorization domain" result to rings with zero divisors, each analogous to a different kind of "unique factorization ring" that has been studied in the literature. (Received February 04, 2020)

1157-13-540 **Yu Xie***, 21 Central Avenue, Berwyn, PA 19312. *On the containment problem for fat points ideals and Harbourne's conjecture.*

In this talk we show that Harbourne's conjecture is true for symbolic powers of fat points ideals, and we check that the stable version of this conjecture is valid for ideals of very general points (resp. generic points) in $\mathbb{P}^N_{\mathbb{K}}$ (resp. $P^N_{\mathbb{K}(\underline{z})}$). We also show that this conjecture and the Harbourne-Huneke conjecture are true for a class of ideals I defining fat points obtained from line arrangements in $\mathbb{P}^2_{\mathbb{K}}$. (Received February 04, 2020)

1157-13-551 **Selvi Kara Beyarslan**, **Huy Tai Ha** and **Augustine O'Keefe*** (aokeefe@conncoll.edu). *Algebraic properties of toric edge rings.*

Given a graph G, one can define an associated toric ring $k[G]$ via a monomial map generated by the edges of the graph. Doing so tethers the combinatorial structure of the graph to the algebraic structure of the ring. In this talk we will discuss what graph theoretic invariants can tell us about the ring being Cohen-Macaulay in addition to giving us bounds for invariants arising from the minimal free resolution of the ring. (Received February 04, 2020)

1157-13-568 **Selvi K Beyarslan**, **Jennifer Biermann*** (biermann@hws.edu), **Kuei-Nuan Lin** and **Augustine O'Keefe**. *Algebraic invariants of weighted oriented graphs.*

Let $\mathcal{D}$ be a weighted oriented graph and let $I(\mathcal{D})$ be its edge ideal in a polynomial ring R. We give a formula for the Castelnuovo-Mumford regularity of $R/I(\mathcal{D})$ when $\mathcal{D}$ is a weighted oriented path or cycle such that edges of $\mathcal{D}$ are oriented in one direction. Additionally, we compute the projective dimension for this class of graphs. (Received February 04, 2020)

1157-13-579 **Christopher A Francisco**, **Matt Mastroeni**, **Jeff Mermin*** (mermin@math.okstate.edu) and **Jay Schweig**. *Critical exponents for monomial ideals.*

Let I be a monomial ideal over a field of characteristic p. Given a monomial m, we determine which fractional exponents $t = \frac{k}{p^e}$ satisfy $m \in I^{[t]}$, the corresponding fractional Frobenius power. In particular, we determine the critical exponents at which this Frobenius power jumps. (Received February 04, 2020)

1157-13-581 **Jesus A. De Loera** and **Sonja Petrovic***, sonja.petrovic@iit.edu, Illinois Institute of Technology, and **Despina Stasi**. *Random Sampling in Computational Algebra: Violator Spaces.*

We present a randomized algorithm for computing minimal generating sets and certificate sub-systems for large polynomial systems whose expected runtime is linear in the number of input polynomials. This is an adaptation of a sampling technique [GMRS] which has been used in geometric optimization. The fundamental object is called a violator space and allows for probabilistic algorithms for computing bases of linear programming type problems; we show how to consider ideal computations within the LP-type framework. We will also provide some computational evidence and discuss generalizations. (Received February 04, 2020)

1157-13-583 **Nicholas James Bruno*** (bruno.74@osu.edu), 1244 Bunker Hill Blvd, Apt 205, Columbus, OH 43220. *Ideals in the p-adic Ring of Analytic Functions.* Preliminary report.

O. Helmer and M. Henriksen developed the theory of ideals for rings of analytic functions in the complex numbers. We develop an analogous theory for analytic functions where the convergence is determined by the p-adic norm rather than the traditional absolute value norm. Our main result is that this ring is not Bezout; however, finitely generated ideals are 2-generated. We also study the Kruul dimension by creating a correspondence between chains of prime ideals and growth rates on sequences of natural numbers. (Received February 04, 2020)

1157-13-584 **Hailong Dao** and **Jonathan Montaño*** (jmon@nmsu.edu). *Symbolic Analytic Spread.*

The symbolic analytic spread of an ideal is defined in terms of the rate of growth of the minimal number of generators of its symbolic powers. In this talk we show some upper bounds for the symbolic analytic spread under certain assumptions. As applications we provide bounds for the arithmetic rank and the Frobenius complexity. We also show sufficient conditions for an ideal to be a set-theoretic complete intersection. (Received February 04, 2020)

1157-13-589 **Neil Epstein*** (nepstei2@gmu.edu), **Rebecca R.G.** and **Janet C. Vassilev**. *Basically empty ideals and submodules.*

The notion of a *basically full* ideal or submodule, in the context of a Noetherian local ring, was pioneered by Heinzer, Ratliff, and Rush in 2002 [HRR]. It comes equipped with a closure operation on $\mathfrak{m}$-primary submodules of a finitely generated module, called the *basically full closure*. Rebecca R.G. and I, in our earlier paper, developed a means for associating many closure operations with counterpart interior operations, and vice versa.

In the current work, we extend this correspondence to associate closure-related reductions with interior-related expansions, and closure-related cores with interior-related hulls. This applies particularly in the context of a *Nakayama closure* (in which case the associated interior is called a *Nakayama interior* operation), such as basically full closure. As the study of reductions is closely related to generating sets, it is natural to apply this last part to the Heinzer *et al.* notion of basic fullness, to arrive at its counterpart, *basic emptiness*, which can be defined in terms of *co*generating sets. As such, we obtain results dual to those in [HRR], and in particular we obtain a new interior operation on finite length submodules of an artinian module. (Received February 04, 2020)

1157-13-602 **Kevin Tucker*** (kftucker@uic.edu) and **Rankeya Datta**. *On some permanence properties of (derived) splinters.*

Perhaps owing to their simple definition, basic questions about splinters are often devilishly difficult to answer. Following André's celebrated proof of Hochster's direct summand conjecture, it is natural to ask whether splinters satisfy some basic permanence properties enjoyed by other classes of singularities. We show that Noetherian splinters ascend under essentially étale homomorphisms. Along the way, we also prove that the henselization of a Noetherian local splinter is always a splinter and that the completion of a local splinter with geometrically regular formal fibers is a splinter. Finally, we give an example of a (non-excellent) Gorenstein local splinter with mild singularities whose completion is not a splinter. (Received February 04, 2020)

1157-13-612 **Daniel Smolkin*** (smolkin@ou.edu), **Marcus Robinson** and **Dylan Johnson**. *Conic modules of Hibi rings.* Preliminary report.

Given a ring R and a divisor D on $SpecR$, one associates an R-module $R(D)$. If R is a toric ring, then one associates to a R a distinguished subset of the modules called the conic modules of R. These conic modules played an important role in Faber-Muller-Smith's work on non-commutative resolutions. They are also useful in the study if F-singularities since these conic modules are exactly the irreducible R-module summands of $R^{1/q}$, up to isomorphism.

In this talk, we will describe one can use the theory of chip-firing games to achieve a novel classification of the conic modules of Hibi rings. (Received February 04, 2020)

1157-13-613 **Haydee Lindo*** (08hml@williams.edu), Williamstown, MA 01267. *Endomorphism Invariance and Ring Classifications.*

I will speak on modules which are invariant under endomorphisms of their envelopes. This will include connections to the general theory of trace modules with some preliminary applications to ring classifications and conjectures involving modules with no self-extensions. (Received January 20, 2020) (Received February 04, 2020)

1157-13-614 **Jason G. Boynton*** (jason.boynton@ndsu.edu), Department of Mathematics, North Dakota State University, Minard Hall 408, Fargo, ND 58102, and **Jim Coykendall** (jcoyken@clemson.edu), School of Math. and Stat. Sciences, Clemson University, O-110 Martin Hall, Clemson, SC 29634. *An example of an atomic integral domain without the ACCP.*

In this talk, we give another example of an atomic integral domain that without the ascending chain condition on principal ideals. Our example is a ring between D[x] and K[x] and is a generalization of Int(E,D) with E finite. (Received February 04, 2020)

1157-13-621 **Matthew Toeniskoetter*** (mtoeniskoetter@fau.edu), **Papiya Bhattacharjee** and **Warren McGovern**. *h-Local Rings of Continuous Functions.* Preliminary report.

The concept of an h-local domain was originally defined by Matlis, who used it to classify D-domains, and has since had a variety of applications in commutative algebra. As one of the main goals of his recent dissertation, A. Omairi showed that the notion of an h-local domain extends in a natural way to rings with zero divisors, and that most of the equivalent conditions for h-local domains carry over to this new definition. In this talk, we will give background information on rings of continuous functions and on h-local rings, then give a topological characterization for a ring of continuous functions to be an h-local ring. (Received February 04, 2020)

1157-13-627 **Antoni Rangachev*** (rangachev@math.uchicago.edu), Department of Mathematics, University of Chicago, 5734 S University Avenue, Chicago, IL 60637. *Relative integral closure of Noetherian rings.*

Let A be a Noetherian domain and B be a finitely generated A-algebra that is a domain and that contains A. Suppose that A is generically integrally closed in B. Then the integral closure of A in B is the intersection of B and finitely many valuation rings that sit between A and its field of fractions. This result recovers Rees'

valuation theorem for ideals. I will discuss applications to the study of the local volume of a relatively ample invertible sheaf. (Received February 04, 2020)

14 ► Algebraic geometry

1157-14-56 **Dávid Papp*** (dpapp@ncsu.edu) and **Sercan Yildiz**. *Computing rational sum-of-squares decompositions from numerical dual certificates.*

We revisit the problem of computing rational certificates for lower bounds of polynomials. Polynomial and sum-of-squares optimization problems are typically solved numerically using conic optimization algorithms. Turning the numerical solutions into exact (verifiable in rational arithmetic) certificates is a challenging problem that has been studied by Peyrl and Parrilo, Kaltofen et al., and others. The focus of the existing results is to quantify the required precision in the numerical methods (usually high) and the deterioration of the bound as the numerical certificates are "rounded" to rational ones. We present a new, dual approach to rounding numerical certificates to exact ones. The method is motivated by interior-point theory and does not require high-precision numerical solutions, but allows for the exactly certified bound to be more precise than what is achievable using purely numerical methods. Computational results demonstrate that in the extreme case the exactly certified bound can be within machine epsilon of the true minimum of the polynomial. (Received January 09, 2020)

1157-14-64 **Nadia Marie Ott***, ottxx141@umn.edu, and **Alexander Voronov**. *The supermoduli spaces of genus zero super Riemann surfaces with Ramond punctures.*

We give a construction of the moduli space $\mathfrak{M}_{0,\mathfrak{n}_R}$ of genus zero super Riemann surfaces with $\mathfrak{n}_R \geq 4$ Ramond punctures as a Deligne-Mumford stack of dimension $\mathfrak{n}_R - 3|\mathfrak{n}_R/2 - 2$. (Received January 11, 2020)

1157-14-78 **Giuseppe Favacchio**, **Elena Guardo**, **Brian Harbourne** and **Juan Migliore*** (migliore.1@nd.edu). *Unexpected hypersurfaces and when to expect them.*

This talk will primarily describe joint with with Giuseppe Favacchio, Elena Guardo and Brian Harbourne. If $X \subset \mathbb{P}^3$ is a reduced subscheme (actually many results work in $\mathbb{P}^n$), we say that X *admits an unexpected hypersurface of degree* t *and multiplicity* m if the imposition of passing through a general point P with multiplicity m fails to impose the expected number $\binom{m+2}{3}$ of conditions on the linear system of hypersurfaces of degree t containing X. We exhibit geometric and algebraic properties of X that can guarantee that X admits certain kinds of unexpectedness, or guarantee that X admits *no* unexpectedness. For example, if X is a general complete intersection in $\mathbb{P}^3$ of type (a, b) with $2 < a \leq b$ then it must admit an unexpected hypersurface of degree $t = (a-1)(b-1)+1$ and multiplicity $m = (a-1)(b-1)$. On the other hand, if X is degenerate then it admits no unexpected hypersurfaces of any kind. Finally, we begin the study of what effect, if any, the Hilbert function of X has on whether or not X admits unexpected hypersurfaces. (Received January 14, 2020)

1157-14-79 **Juan Migliore*** (migliore.1@nd.edu), **Uwe Nagel** and **Henry Schenck**. *The singular locus of a hyperplane arrangement in* $\mathbb{P}^3$*, and liaison.*

Let $\mathcal{A}$ be a hyperplane arrangement in $\mathbb{P}^3$. (Much of the work actually holds in $\mathbb{P}^n$.) Let J be the Jacobian ideal of $\mathcal{A}$. This ideal is usually not unmixed, or even saturated. We consider two associated ideals, J^{top} and $\sqrt{J}$, both of which *are* unmixed and define equidimensional curves X^{top} and X^{red} in $\mathbb{P}^3$. These curves both have a claim to being called the (unmixed) singular locus of $\mathcal{A}$. If a certain fairly mild combinatorial property $(*)$ of the incidence lattice of $\mathcal{A}$ is satisfied, both X^{top} and X^{red} are arithmetically Cohen-Macaulay (ACM). If $(*)$ does not hold, either or both of these curves may fail to be ACM. In fact, they both can be made to fail to be ACM by as much as you like (as measured by the dimension of the Hartshorne-Rao module). The proofs use tricks from liaison, which in turn prompt liaison-related questions. We will briefly review facts from liaison along the way. (Received January 14, 2020)

1157-14-105 **Sean P Howe*** (sean.howe@utah.edu). *Motivic random variables.*

The average smooth hypersurface in $\mathbb{P}^n$ is $\mathbb{P}^{n-1}$, the cohomology of the universal genus g curve is asymptotically a Poisson random variable, and constructible conditions on Taylor expansions give rise to algebraic families of asymptotically independent Bernoulli random variables on sections of coherent sheaves. In this talk, we introduce the language of motivic random variables that gives mathematical meaning to these statements, and discuss some open problems relating the asymptotic distributions of motivic random variables to other forms of stabilization in topology, arithmetic, and representation theory. (Received January 21, 2020)

1157-14-170 **Aaron Landesman*** (aaronlandesman@gmail.com). *The geometric average size of Selmer groups over function fields.*

We show that the average size of n-Selmer groups of elliptic curves over $\mathbb{F}_q(t)$, in a suitable large q limit, is the sum of divisors of n. Loosely speaking, the n-Selmer group of an elliptic curve measures certain objects which are torsors for the n-torsion of the elliptic curve. We relate the question of computing the average size of the n-Selmer group to demonstrating homological stability for a sequence of moduli spaces of these n-Selmer elements, which we then approach using monodromy arguments. (Received January 26, 2020)

1157-14-213 **Isabelle Shankar*** (isabelle_shankar@berkeley.edu), Berkeley, CA, and **Alexander Heaton** and **Serkan Hosten**. *Symmetry Adapted Gram Spectrahedra.*

Sum of squares (SOS) relaxations are often used to certify nonnegativity of polynomials and are equivalent to solving a semidefinite program (SDP). The feasible region of the SDP for a given polynomial is the Gram spectrahedron. For symmetric polynomials, there are reductions to the problem size that can be done using tools from representation theory. In joint work with Alex Heaton, we used this machinery to disprove a conjecture about symmetric function inequalities. I will give a brief introduction to the theory used. Moreover, I will describe recent work with Serkan Hosten on understanding the geometric structure of the spectrahedra that arise in the study of symmetric SOS polynomials. (Received January 29, 2020)

1157-14-251 **Daniel Lloyd Bath*** (dbath@purdue.edu). *Bernstein–Sato Varieties and the Topological Multivariable Strong Monodromy Conjecture.*

We consider a multivariate generalization of the Bernstein–Sato polynomial due to Budur as well as the associated $D_X[s_1, \ldots, s_r]$-module $D_X[s_1, \ldots, s_r]f_1^{s_1} \cdots f_r^{s_r}$. Generalizing techniques of Walther, we show that for a large class of divisors $f = f_1 \cdots f_r$, the annihilator of $f_1^{s_1} \cdots f_r^{s_r}$ is generated by the simplest possible elements. As an application, we verify Budur's Topological Multivariable Strong Monodromy Conjecture for tame hyperplane arrangements. Time permitting, we show that for tame and free arrangements, the generalized Bernstein–Sato variety contains several combinatorially determined hyperplanes. (Received January 30, 2020)

1157-14-267 **Greg Blekherman*** (greg@math.gatech.edu). *Tropicalization for comparing sets in polynomial optimization.* Preliminary report.

Tropicalization is a widely used operation in algebraic geometry which replaces a variety with a combinatorial object. One can also tropicalize semi-algebraic sets, and I will explain how tropicalization of the moment cone (which is dual to the cone of nonnegative polynomials) and the cone dual to sums of squares can explain the difference between the cones. Tropicalization also highlights the fundamental role of the arithmetic mean-geometric mean inequality as an inequality that cannot be explained via sums of squares. (Received January 31, 2020)

1157-14-281 **Eduardo Esteves*** (esteves@impa.br), IMPA, Estrada Dona Castorina 110, Rio de Janeiro, RJ 22460, Brazil. *Compactified Jacobians of stable curves.*

The compactified Jacobian is a compactification of the (generalized) Jacobian of a singular curve, the variety parameterizing line bundles over the curve with (multi)degree 0. It was first described for irreducible curves by D'Souza, Rego, Altman and Kleiman in the 1970's. The interest arose from work on fixing Severi's proof of the Brill-Noether Theorem, eventually carried out by Griffiths and Harris.

In the early 1980's Eisenbud and Harris gave another proof of the theorem, introducing the notion of limit linear series for curves of compact type. Their theory was used in several applications to explain the geometry of the moduli space of stable curves. Unfortunately though, curves of compact type give only divisorial information about the moduli, thus the search for compactified Jacobians for all stable curves.

In the 90's Caporaso described such a compactification. Unfortunately though, it parameterizes only stable sheaves, which seldom carry enough information about degenerating divisors.

I will describe work in progress, joint with Omid Amini (Ecole Polytechnique, Paris) and Margarida Melo and Filippo Viviani (Roma Tre), to produce another natural compactification of the Jacobian, one that does not rely on stability and accounts for all degenerating divisors. (Received January 31, 2020)

1157-14-356 **Victoria Powers** (reznick@illinois.edu) and **Bruce Reznick*** (reznick@illinois.edu), 1409 W Green St, Urbana, IL 61801. *Mediated simplices and Hilbert's 17th Problem.*

This talk will be taken from the paper "A note on mediated simplices", which can be found at http://front.math.ucdavis.edu/1909.11008. The talk will describe the connection between the following two sets of topics: (1) Hilbert's 17th Problem, the Arithmetic-Geometric Inequality, Motzkin's polynomial and writing polynomials as a sum of squares of polynomials and (2) the configuration of lattice points within lattice point

simplices and how the configuration changes when the simplices are dilated. Godzilla will make a metaphorical appearance. (Received February 02, 2020)

1157-14-384 **Ritvik Ramkumar** and **Alessio Sammartano*** (asammart@nd.edu). *On the tangent space to the Hilbert scheme of points in* $\mathbb{P}^3$.

In this work we study the tangent space to the Hilbert scheme $\mathrm{Hilb}^d\mathbb{P}^3$, motivated by conjectures of Briançon-Iarrobino stated in 1978 and by Haiman's work on $\mathrm{Hilb}^d\mathbb{P}^2$. (Received February 03, 2020)

1157-14-385 **Ritvik Ramkumar** (ritvik@berkeley.edu) and **Alessio Sammartano*** (asammart@nd.edu). *Singular lexicographic points in Hilbert schemes.*

In this work we study the smoothness of lexicographic points on Hilbert schemes, motivated by a famous theorem of Reeves and Stillman for Grothendieck's classical Hilbert scheme. (Received February 03, 2020)

1157-14-449 **Katherine Harris*** (keharri4@ncsu.edu), North Carolina State University, Campus Box 8205, Raleigh, NC 27695, **Johnathan D. Hauenstein** (hauenstein@nd.edu), University of Notre Dame, 152C Hurley Hall, Notre Dame, IN 46556, and **Agnes Szanto** (aszanto@ncsu.edu), North Carolina State University, Campus Box 8205, Raleigh, NC 27695. *Finding real smooth points on semialgebraic sets.*

Many algorithms for determining properties of real algebraic or semialgebraic sets rely upon the ability to compute smooth points. Existing methods to compute smooth points on semialgebraic sets use symbolic quantifier elimination tools. We present a simple algorithm based on computing the critical points of some well-chosen function that guarantees the computation of smooth points in each connected compact component of a real (semi)-algebraic set. Our technique is intuitive in principal, performs well on previously difficult examples, and is straightforward to implement using existing numerical algebraic geometry software. It complements other approaches that compute sample points on real semialgebraic sets, such as computing critical points of the distance function, but our method also guarantees the smoothness of the sample points. The practical efficiency of our approach is demonstrated by solving a conjecture on the number of equilibria of the Kuramoto model for the $n = 4$ case. We also apply our method to design an efficient algorithm to compute the real dimension of (semi)algebraic sets, the original motivation for this research. (Received February 03, 2020)

1157-14-567 **Boris L Tsygan***, B4 Lunt Hall, 2033 North Sheridan Avenue, Evanston, IL 60208. *Noncommutative crystalline complexes.*

For an algebra over a finite field we construct a complex of modules over the Witt vectors. When the algebra admits a lifting to an algebra of Witt vectors, this complex is the completed periodic cyclic complex of the lifting (in particular, the complex does not depend on a lifting, in a strong sense that we will specify). This work is closely related to recent works of Petrov, Layntrob, and Vologodsky. (Received February 04, 2020)

1157-14-586 **Danielle A. Brake**, **Noah S. Daleo**, **Jonathan D. Hauenstein** and **Samantha N. Sherman*** (ssherma1@nd.edu). *Solving critical point conditions for the Hamming and taxicab distances to solution sets of polynomial equations.*

Minimizing the Euclidean distance (ℓ_2-norm) from a given point to the solution set of a given system of polynomial equations can be accomplished via critical point techniques. This article extends critical point techniques to minimization with respect to Hamming distance (ℓ_0-"norm") and taxicab distance (ℓ_1-norm). Numerical algebraic geometric techniques are derived for computing a finite set of real points satisfying the polynomial equations which contains a global minimizer. Several examples are used to demonstrate the new techniques. (Received February 04, 2020)

1157-14-642 **Saugata Basu**, **Antonio Lerario**, **Erik Lundberg** and **Chris Peterson*** (peterson@math.colostate.edu). *Lines on real cubic surfaces and related combinatorial problems in real algebraic geometry.*

A well known result in algebraic geometry asserts that 27 lines lie on a general cubic surface in $\mathbb{C}P^3$. From a homogeneous cubic polynomial in $\mathbb{C}[w, x, y, z]$ one can use elimination theory to determine a degree 27 homogeneous polynomial in two variables whose roots correspond to the 27 lines. If one starts with a homogeneous cubic polynomial in $\mathbb{R}[w, x, y, z]$ and again use elimination theory one obtains a degree 27 homogeneous polynomial in two variables with real coefficients. With probability one, it is known that this polynomial always has 3, 7, 15, or 27 real roots and that each of these cases occurs with positive probability. It is natural to ask for the expected number of real roots (i.e. the expected number of real lines on a general real cubic surface). Of course the answer depends on the probability distribution used to choose the cubic. We show that with respect to the Kostlan distribution (a particular O(4) invariant Gaussian distribution on the space of real cubic polynomials in

four variables), one expects there to be $6\sqrt{2}-3$ real lines. The goal of this talk is to present some of the features that lead to this result. (Received February 05, 2020)

15 ► Linear and multilinear algebra; matrix theory

1157-15-101 **Christina Pospisil*** (pospisil.christina@gmx.de). *Generalization Theory of Linear Algebra II.*

The talk continues the first part of the project Generalization Theory of Linear Algebra I (first part was presented at JMM 2019) and presents inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be applications to physics and other natural sciences be explored. (Received January 20, 2020)

1157-15-211 **David F Gleich*** (dgleich@purdue.edu), 305 N University St, West Lafayette, IN 47906. *Erasure Coding and Fault Tolerance in Numerical Linear Algebra.*

Erasure codes were developed in storage systems to ensure reliability with faulty storage substrates. An erasure coded computation is a new paradigm we have been developing where given a computational problem P and an input x, we create a new input x', such that computing P(x') in the presence of errors or faulty computation still allows us to extract the solution to P(x) through an inexpensive post-processing procedure. This scenario is particularly well suited to numerical linear algebra procedures including linear systems. The erasure coded problems correspond to adding redundant blocks of rows to the equations and we will demonstrate a variety of ways these can be used in existing solvers. (Received January 29, 2020)

1157-15-316 **Xuefeng Xu*** (xu1412@purdue.edu), Department of Mathematics, Purdue University, 150 North University Street, West Lafayette, IN 47907-2067. *New Theory on the Ideal Interpolation Operator in Algebraic Multigrid.*

Various algebraic multigrid algorithms have been developed for solving problems in scientific and engineering computation over the past decades. They have been shown to be well-suited for solving discretized partial differential equations on unstructured grids in practice. One key ingredient of algebraic multigrid algorithms is a strategy for constructing an effective prolongation operator. Among many questions on constructing a prolongation, an important question is how to evaluate its quality. In this talk, we introduce a new theory on the ideal interpolation operator in algebraic multigrid methods. Our theory suggests that, compared with common wisdom, one has more freedom to construct an ideal interpolation operator, which provides some new insights for designing algebraic multigrid algorithms. Moreover, we present an explicit expression for a class of ideal interpolation operators. (Received February 01, 2020)

1157-15-446 **Jianlin Xia*** (xiaj@math.purdue.edu), **Xin Ye** and **Lexing Ying**. *Proxy point method for analytical compression of kernel matrices.*

It has been known in potential theory that, for some kernels matrices corresponding to well-separated point sets, fast analytical low-rank approximation can be achieved via the use of proxy points. The proxy point method gives a surprisingly convenient way to explicitly write out approximate basis matrices for a kernel matrix and avoid expensive algebraic compression. However, the method is lack of clear algebraic understanding of the theoretical background. Moreover, rigorous quantifications of the approximation errors and reliable criteria for the selection of the proxy points were missing. In this work, we rigorously justify the idea in terms of a class of important kernels. We further provide comprehensive accuracy analysis for the analytical compression and show how to choose nearly optimal proxy points. The analytical compression is then combined with fast rank-revealing factorizations to get compact low-rank approximations and also to select certain representative points. This gives a fast and reliable strategy for compressing those kernel matrices. The work provides an intuitive way of understanding the proxy point method and bridges the gap between this useful analytical strategy and practical low-rank approximations. (Received February 03, 2020)

1157-15-469 **David Samuel Bindel*** (bindel@cs.cornell.edu), 425 Gates Hall, Department of Computer Science, Cornell University, Ithaca, NY 14853. *The Structure and Interpretation of Graph Spectral Densities.*

In this talk, we report on the analysis of graphs via global summaries of the eigenvalue distributions and eigenvector behavior. Our approach is drawn from the condensed matter physics literature, where the idea of local and global densities of states is often used to understand the electronic structure of systems, and we describe how these densities play a common role in such seemingly disparate topics as spectral geometry, condensed matter

physics, and the study of centrality measures in graphs. We then discuss how structural motifs manifest in the spectrum, give fast algorithms to estimate spectral densities, and conclude with a discussion of some of our current research directions in applying these tools to the analysis of large-scale graphs. (Received February 03, 2020)

1157-15-470 **Moe Z. Win*** (moewin@mit.edu), Laboratory for Information & Decision Systems, 77 Massachusetts Avenue, Room 32-D658, Cambridge, MA 02139, and **Andrea Conti** (a.conti@ieee.org). *Localization-of-Things: A New Opportunity in Applied Mathematics.*

The availability of positional information is essential for numerous wireless applications including crowdsensing, autonomous driving, and Internet-of-Things. The coming years will see the emergence of location-aware networks with sub-meter localization accuracy even in GNSS-challenged environments. This calls for Localization-of-Things (LoT), a new paradigm that refers to locating, tracking, and navigating collaborative or non-collaborative nodes (sensors, vehicles, and objects). Our work relies on statistics, optimization, and communication theory, and it approaches LoT from different perspectives: theoretical framework, cooperative algorithms, network operations, and network experimentation. This talk will give an overview of LoT and provide our recent research results in this exciting field. (Received February 03, 2020)

1157-15-506 **Angela Robinson***, angela.robinson@nist.gov. *Attacking product structure of LEDAcrypt.*

The 2nd Round submission to NIST's Post Quantum Cryptography standardization process, LEDAcrypt, is based on quasi-cyclic low-density parity check codes. The public parity check matrix of the code in systematic form is generated by taking a product of two sparser matrices with quasi-cyclic structure. We present an attack that exploits the product structure of LEDAcrypt's key in a non-trivial manner by using information set decoding on structured information sets. We believe that in all parameter sets proposed in the LEDAcrypt submission, there are substantial classes of weak keys that can be recovered significantly faster than would be expected from the usual analysis, and in some cases, ithe security level of the average key is also weaker than expected. (Received February 03, 2020)

1157-15-518 **Hsin-Po Wang*** (hpwang2@illinois.edu), **Xiao Li** and **Iwan Duursma**. *Multilinear Algebra for Distributed Storage.*

We use tensor and wedge powers to define linear encoding maps for use in distributed storage. Different nodes store different restrictions of a chosen linear map such that the restrictions meet the requirements for data download (reconstructing the linear map from its restrictions) and node repair (reconstructing any single restriction from partial information about the remaining restrictions). New codes are presented for the important MSR trade off point. (Received February 03, 2020)

1157-15-619 **Xiaozhe Hu***, 503 Boston Ave. Bromfield-Pearson, Medford, MA 02155. *Algebraic Multigrid Methods for Computing Diffusion State Distance on Graphs.*

Recently, diffusion state distance (DSD) was introduced for protein-protein interaction networks and used for protein function prediction. In this talk, we focus on the challenges in computing the DSD metric, especially for large-scale networks. By exploring the algebraic properties of the diffusion state distance, we reformulate the computation of the distance into solving a series of graph Laplacians. Randomized dimensionality reduction techniques and algebraic multigrid methods are developed to compute DSD efficiently. Applications to real-world networks will be presented and possible generalizations will be discussed. (Received February 04, 2020)

16 ► Associative rings and algebras

1157-16-145 **Steve Szabo*** (steve.szabo@eku.edu), 544 Dover Road, Lexington, KY 40505. *More On Codes over Noncommutative Rings.* Preliminary report.

In this talk we explore the considerations on the structure of the ring alphabet and the properties of the bilinear form used in defining duality needed to have the traditional tools necessary for coding theoretic study. (Received January 24, 2020)

1157-16-325 **Yang Mo*** (mo12@purdue.edu). *From Bialgebra to Hopf algebra.*

Bi- and Hopf algebra, which are well-studied algebraic objects, arise in many contexts as higher structures. This gives new impetus to the question whether a given bialgebra admits antipode. Takeushi (1971) showed a bi-algebra over a field admits an antipode if it is pointed and the group-like elements form a group. This generalizes a well-known result that a connected graded bialgebra admits antipode. However, the bi-algebras and Hopf algebras from higher strucutres often have ground ring to be commutative and unital, which is not

necessarily a field. Takeuchi's beatiful result does not apply because in this case definition of pointedness becomes problematic. We will generalize Takeuchi's result to bi-/ Hopf algebras over arbitrary commutative unital rings by relaxing the condition of being pointed. This is expected to have applications to recent construction of Hopf algebras from bi-algebras following Galvez-Kaufmann-Tonks. (Received February 01, 2020)

1157-16-348 **Avi Steiner*** (asteine9@uwo.ca). *Computing D-annihilators of algebraic functions.*
Let u be an algebraic function in the variables $x_1, \ldots, x_n$, and let D be the Weyl algebra in the x's. I will discuss how to use the notions of integration ideal and Weyl closure to easily compute the annihilator $\mathrm{ann}_D(u)$ in Macaulay2. (Received February 02, 2020)

17 ► Nonassociative rings and algebras

1157-17-364 **Curtis J Wendlandt***, Wendlandt.4@osu.edu. *The Yangian double revisited.*
The Yangian $Y_\hbar(\mathfrak{g})$ of a complex semisimple Lie algebra $\mathfrak{g}$ is a remarkable Hopf algebra which arises as a quantization of the Lie bialgebra $\mathfrak{g}[t]$. In the 1990's, S. Khoroshkin and V. Tolstoy introduced a related algebra $\mathrm{D}Y_\hbar(\mathfrak{g})$, called the Yangian double, which deforms the enveloping algebra of the loop algebra $\mathfrak{g}[t^{\pm 1}]$ and conjecturally coincides with the graded Hopf algebra double of $Y_\hbar(\mathfrak{g})$.

In this talk, I will explain how the action of the additive group $\mathbb{C}$ on the Yangian by shift automorphisms induces a family of injections from $\mathrm{D}Y_\hbar(\mathfrak{g})$ into a completion of $Y_\hbar(\mathfrak{g})$, each of which naturally extends to an isomorphism between completions of both algebras. This result holds more generally for $\mathfrak{g}$ taken to be any symmetrizable Kac-Moody algebra. Time permitting, I will discuss some of its consequences and postulate how it can be applied further. (Received February 02, 2020)

18 ► Category theory; homological algebra

1157-18-90 **Alexei Davydov*** (davydov@ohio.edu), Department of Mathematics, Ohio University, Athens, OH 45701. *Schur-Weyl categories.*
Schur-Weyl categories are tensor categories generated by one object, without morphisms between different tensor powers. These categories are particularly useful for classifying quantum groups of type A. (Received January 16, 2020)

1157-18-125 **Pinches Dirnfeld*** (dirnfeld@math.utah.edu), 155 S 1400 E, Room 233, Salt Lake City, UT 84112. *Base Change Along the Frobenius Endomorphism and the Gorenstein Property.*
Let R be a local ring of positive characteristic and X a complex with nonzero finitely generated homology and finite injective dimension. We prove that if derived base change of X via the Frobenius (or more generally, via a contracting) endomorphism has finite injective dimension then R is Gorenstein. (Received January 31, 2020)

1157-18-146 **Ben C Ward*** (benward@bgsu.edu), Department of Mathematics and Statistics, Bowling Green State University, Bowling Green, OH 43403. *Wheel graph homology classes via weak modular operads.*
I will give a new proof of the non-triviality of a family of graph homology classes which, after the work of Willwacher, correspond to conjectural generators of the Grothendieck-Teichmuller Lie algebra. The proof uses a derived version of Koszul duality and an operadic version of Massey products for modular operads. (Received January 24, 2020)

1157-18-301 **Michael Monaco*** (monacom@purdue.edu). *Plus constructions and plethysm products.*
In operad theory, there are several versions of the plus construction. A fundamentally important application of these plus constructions to connect operads to their algebras. For Feynman categories, operads and algebras are functors and the plus construction takes a Feynman category as input and outputs a Feynman category one level deeper. We will give new details on this construction. We will also show that the input of the plus construction can be extended to monoidal categories which are unique factorization categories (UFCs) so that the output is still a Feynman category.

Closely related to these results is the existence of a plethysm product. A well-known example is the one for operads which exhibits an operad as a monoid in the category of Sigma-modules. We will show that a plethysm product exists if the Feynman category is an output of the plus construction. This reproduces the result for operads. Moreover, by extending the plus construction to UFCs, we get the PROP version of this result.

These results are also important for the comparison between Feynman categories and the operadic categories of Batanin and Markl. We will present the general construction and the examples above in detail. (Received February 01, 2020)

1157-18-319 **Radmila Sazdanovic*** (rsazdan@ncsu.edu), Department of Mathematics NC State University, PoBox 8205, Raleigh, NC 27695, and **Mikhail Khovanov**. *Diagrammatic categorification of the polynomial ring $\mathbb{Z}[x]$.*

We develop a diagrammatic categorification of the polynomial ring $\mathbb{Z}[x]$ that leads to categorification of some families of orthogonal polynomials including Chebyshev and Hermite polynomials. Our categorifications satisfy a version of the Bernstein-Gelfand-Gelfand reciprocity property with the indecomposable modules corresponding to the monomials and standard modules to the orthogonal polynomials in the Grothendieck ring. (Received February 01, 2020)

1157-18-497 **Benjamin Briggs**, **Srikanth B Iyengar**, **Janina C Letz*** (letz@math.utah.edu) and **Josh Pollitz**. *A homotopical characterization of l.c.i. maps.* Preliminary report.

This talk will be about locally complete intersection maps of commutative noetherian rings. Results of Dwyer, Greenlees, and Iyengar, and Pollitz lead to a characterization of the complete intersection property for a noetherian ring in terms of the structure, as a triangulated category, of the bounded derived category of the ring. In my talk I will present a similar characterization for locally complete intersection maps. This is part of ongoing joint work with Briggs, Iyengar, and Pollitz. (Received February 03, 2020)

20 ► Group theory and generalizations

1157-20-237 **Michael C. Steward*** (michael.steward@westpoint.edu), **Elie Alhajjar** and **Travis Russell**. *Numerical Semigroups and Kunz Polytopes.*

A numerical semigroup S is an additive submonoid of $\mathbb{N}$ whose complement is finite. The cardinality of $\mathbb{N} \setminus S$ is called the genus of S and is denoted by $g(S)$. The first nonzero element of S is called the multiplicity of S and is denoted by $m(S)$. In this paper, we focus on the number $N(m, g)$ of numerical semigroups with parameters m and g. It is known that $N(m, g)$ can be formulated as the number of integer points in a certain family of rational polytopes and hence coincides with a quasi-polynomial in g of degree $m-2$. We show that the leading coefficient is constant and provide an interpretation for it. Moreover, we relate $N(m, g)$ to the number $MED(m, g)$ of maximally embedded numerical semigroups with the same parameters, hence proving a conjecture posed by Kaplan (J Pure Appl Algebra 216(5):1016–1032, 2012. https://doi.org/10.1016/j.jpaa.2011.10.038). We close the paper by discussing the special case $m = 4$. (Received January 30, 2020)

1157-20-277 **You Qi*** (yq2dw@virginia.edu), 141 Cabell Drive, Kerchof Hall, P.O. Box 400137, Charlottesville, VA 22904. *On the center of small quantum groups.*

Since its introduction by Lusztig, the center of quantum groups at roots of unity remains an interesting open problem. We will report some recent progress on the problem of determining the centers of small quantum groups at a root of unity or restricted enveloping algebras, which points to surprising connections towards Haiman's diagonal coinvariant algebra. This is based on joint work with A. Lachowska. (Received January 31, 2020)

1157-20-343 **Sema Gunturkun*** (sgunturkun@amherst.edu) and **Andrew Snowden**. *The Representation theory of the increasing monoid.*

The increasing monoid is the monoid $\mathcal{I}$ of the order-preserving injections from $\mathbb{Z}_{>0}$ to $\mathbb{Z}_{>0}$, and its crucial role in the study of $\mathfrak{S}_\infty$-Noetherianity of $K[x_1, x_2, \ldots]$ is known. Connections of this theory with other areas such as commutative algebra and representation stability have motivated us to investigate it in more details. In this talk, we discuss some results in the representation theory of $\mathcal{I}$. (Received February 02, 2020)

1157-20-377 **Mark Sapir*** (mark@gmail.com), 8025, Claytie Circle, Nashville, TN 37221, and **Alexander Olshanskii**, TN. *Conjugacy and power conjugacy problems in finitely presented groups.* Preliminary report.

This is a joint work with A.Yu. Olshanskii. The power conjugacy problem asks if for a pair of elements in the group some powers these elements are conjugate. We construct a finitely presented group which has solvabe word and power conjugacy problems and unsolvabl conjugacy problem. (Received February 02, 2020)

1157-20-416 **Simon André*** (simon.andre@vanderbilt.edu), Vanderbilt University, Department of Mathematics, 1326 Stevenson Center, Nashville, TN 37240. *Acylindrically hyperbolic groups and elementary equivalence.*

Zlil Sela proved that, given a non-cyclic torsion-free hyperbolic group G, the groups G and $G*\mathbb{Z}$ are elementarily equivalent. In my talk, I will present the following partial generalization of this result: if G is acylindrically hyperbolic, say with no non-trivial normal finite subgroup, then G and $G*\mathbb{Z}$ have the same $\forall\exists$-theory. It is an open question whether or not G and $G*\mathbb{Z}$ have the same elementary theory. Joint work with Jonathan Fruchter. (Received February 03, 2020)

1157-20-445 **Michael Hull*** (mbhull@uncg.edu). *Equationally noetherian groups and families.*

A group is equationally noetherian if every algebraic set over that group can be defined by finitely many equations. This property can be viewed as a group-theoretic version of the Hilbert Basis Theorem. It has applications to the algebraic properties and the elementary theory of a group, as well as applications to questions in topology. We will give a brief survey of examples and properties of equationally noetherian groups and equationally noetherian families of groups. This work is joint with D. Groves and H. Liang. (Received February 03, 2020)

1157-20-483 **Jeremy Miller**, **Peter Patzt**, **Jennifer C. H. Wilson*** (jchw@umich.edu) and **Dan Yasaki**. *The high-degree cohomology of the special linear group.*

In this partly-expository talk, I will describe some current efforts to understand the high-degree rational cohomology of $\mathrm{SL}_n(\mathbb{Z})$, or more generally the cohomology of $\mathrm{SL}_n(\mathcal{O})$ when $\mathcal{O}$ is a number ring. Although the groups $\mathrm{SL}_n(\mathcal{O})$ do not satisfy Poincare duality, they do satisfy a twisted form of duality, called Bieri–Eckmann duality. Consequently, their high-degree rational cohomology groups are governed by an $\mathrm{SL}_n(\mathcal{O})$-representation called the Steinberg module. I will overview some results and open problems about these representations. (Received February 03, 2020)

1157-20-533 **Mark Pullins*** (mark.pullins@colorado.edu), 3152 Redstone Lane, Boulder, CO 80305. *Generic 2-Transitive Actions with Solvable Point Stabilizers.* Preliminary report.

In 2005, Popov calculated the maximal degree of generic transitivity for reductive linear algebraic groups. These results were extended by Devyatov examining actions on multiple flag varieties. Generic transitivity also turned out to be a natural area of study for permutation groups of finite Morley rank and generated research centered around questions posed by Borovik and Cherlin in 2008. In this talk, I will begin by discussing a class of generic 2-transitive actions with solvable point stabilizers. The focus will be on the primarily group theoretic and combinatorial techniques used to identify them. Then, I will conclude with a brief foray into avenues forward and possible applications in the context of permutation groups of finite Morley rank. (Received February 04, 2020)

1157-20-545 **Jason Green***, jag1032@wildcats.unh.edu, and **Dmitri Nikshych**. *On the braid group representations coming from weakly group-theoretical fusion categories.*

Braided fusion categories give rise to representations of the braid groups B_n. A natural problem is to investigate the images of these representations. In this talk, we extend previously known results by proving that representations of the braid groups coming from weakly group-theoretical braided fusion categories have finite image. (Received February 04, 2020)

1157-20-605 **Mark A Pengitore***, 100 Math Tower, 231 W. 18th. Ave., Columbus, OH 43210. *Residual Finiteness of solvable groups and distortion of cyclic subgroups.*

We provide polynomial lower bounds for residual finiteness of residually finite, finitely generated solvable groups that admit infinite order elements in the Fitting subgroup of strict distortion at least exponential. For this class of solvable groups which include polycyclic groups with a nontrivial exponential radical and the metabelian Baumslag-Solitar groups, we improve the lower bounds found in the literature. Additionally, for the class of residually finite, finitely generated solvable groups of infinite Prüfer rank that satisfy the conditions of our theorem, we provide the first nontrivial lower bounds. (Received February 04, 2020)

22 ► Topological groups, Lie groups

1157-22-48 **Tian An Wong*** (twong33@smith.edu). *Basic functions and the stable trace formula.* Preliminary report.

I will report on work in progress establishing a stable trace formula whose discrete spectral terms are weighted by automorphic L-functions. This involves extending the results of Finis, Lapid, and Müller on the continuity of

Arthur's noninvariant trace formula to the stable trace formula, while incorporating the use of basic functions. (Received January 07, 2020)

1157-22-112 **Omer Offen*** (offen@brandeis.edu). *An explicit Plancherel formula for the space of p-adic alternating matrices.*

In this talk we explicate the Plancherel formula and consequently the spectral decomposition of the space of non-degenerate alternating matrices in 2n variables over a p-adic field in terms of that of the general linear group in n variables. This is joint work with Erez Lapid. It is based on some recent new ideas of Raphael Beuzart-Plessis. (Received January 22, 2020)

1157-22-163 **Dubravka Ban*** (dban@siu.edu), Department of Mathematics, Suthern Illinois University, Carbondale, IL 62901, and **Joseph Hundley**. *Intertwining operators on p-adic principal series.* Preliminary report.

Let G be a split p-adic group and K a maximal compact subgroup of G. We study continuous principal series representations on p-adic Banach spaces. We prove that the intertwining algebra of any principal series of G is one-dimensional. Surprisingly, the intertwining algebra of any principal series of K is also one-dimensional. For the proof, we use the duality theory of Schneider and Teitelbaum, and work with the corresponding Iwasawa modules. (Received January 26, 2020)

1157-22-208 **david ginzburg*** (ginzburg@tauex.tau.ac.il) and **david soudry**. *Derivation of Global Integrals.*

It is well known that for certain L functions there is more than one Rankin-Selberg integral which represents this L function. Typically, when unfolding these integrals we obtain an integral which involves a certain functional. This functional may be unique, for example like the Whittaker coefficient, or it may be non-unique. These latter cases are referred to as integrals of The New Way type. In this work we explain why so many such integrals exist and introduce a recipe how to derive these integrals. To do that we use a certain identity derived from a doubling integral and also use certain types of identities between Eisenstein series. (Received January 29, 2020)

1157-22-275 **Birgit Speh*** (bes12@cornell.edu), NY. *Period integrals for some representations of SO(p,q).* Preliminary report.

In this talk I discuss branching laws for certain unitary representations of SO(p,q). This is done on the smooth vectors via a version of the *period integrals*, studied in number theory, It is also closely connected to the *symmetry-breaking operators* introduced by T. Kobayashi. We exhibit non-vanishing symmetry breaking operators for the restriction of a representation Π in the discrete spectrum for real hyperboloids to representations of smaller orthogonal groups. I will also discuss that these results suggest a generalization of the Gross Prasad conjectures to non tempered representations of orthogonal groups. The first part of lecture is joint work with B. Orsted and and the second part is joint work with T. Kobayashi. (Received January 31, 2020)

1157-22-346 **Chris Jantzen*** (jantzenc@ecu.edu), Department of Mathematics, East Carolina University, Greenville, NC 28590, and **Baiying Liu**. *Generic representations for quasi-split similitude groups.*

In this talk, we discuss the classification of irreducible generic representations for quasi-split p-adic similitude groups, as well as some of the background needed to determine the classification. (Received February 02, 2020)

28 ► Measure and integration

1157-28-27 **Alan Chang***, 5734 S University Avenue, Chicago, IL 60637, and **Marianna Csörnyei**. *The Kakeya needle problem for rectifiable sets.*

We show that the classical results about rotating a line segment in arbitrarily small area, and the existence of a Besicovitch and a Nikodym set hold if we replace the line segment by an arbitrary rectifiable set. This is joint work with Marianna Csörnyei. (Received December 09, 2019)

1157-28-266 **Valentino Magnani***, Largo Bruno Pontecorvo 5, 56127 Pisa, Italy. *Area formulas for intrinsic regular submanifolds in the Heisenberg.*

We present an explicit integral formula for the spherical measure of intrinsic regular submanifolds in any Heisenberg group. Since these submanifolds are not smooth in the classical sense, our approach is based on intrinsic differentiability of the parametrizing mapping and a chain rule for intrinsic differentiable mappings. A delicate

lemma to compute the Jacobian of projections among vertical subgroups also plays an important role. These results have been obtained in collaboration with Francesca Corni. (Received January 31, 2020)

1157-28-531 **Fernando Y. Roman-Garcia*** (romanga2@illinois.edu). *A Fourier coefficient approach to Hausdorff dimension in the Heisenberg group.*

In this talk I discuss a, recently developed, alternative formulation of the Fourier transform on the Heisenberg group and used it to show that energies of measures can be computed via integrals on an appropriate frequency space. This in turns opens the possibility of using Fourier methods in the computation of Hausdorff dimension of sets. (Received February 04, 2020)

30 ► Functions of a complex variable

1157-30-109 **Mohamed Nasser*** (mms.nasser@qu.edu.qa), Department of Math, Stat and Phy, Qatar University, Doha, Qatar. *Computation of conformal invariants.*

We study numerical computation of conformal invariants of domains in the complex plane. In particular, we provide an algorithm for computing the conformal capacity of a condenser. The algorithm can be applied for a wide kind of geometries where the domains are assumed to have smooth or piecewise smooth boundaries. Our proposed method is based on the boundary integral equation with the generalized Neumann kernel. A characteristic feature of this method is that, with small changes in the code, a wide spectrum of problems can be treated. We compare the performance and accuracy to previous results in the literature when numerical data are available and also for several examples with known exact solutions. (Received January 22, 2020)

31 ► Potential theory

1157-31-320 **Xuemei Chen*** (xchen@nmsu.edu), **Douglas P Hardin** and **Edward B Saff**. *On the Search for Tight Frames of Low Coherence.*

We introduce a projective Riesz s-kernel for the unit sphere $\mathbb{S}^{d-1}$ and investigate properties of N-point energy minimizing configurations for such a kernel. We show that these configurations, for s and N sufficiently large, form frames that are well-separated (have low coherence) and are nearly tight. Our results suggest an algorithm for computing well-separated tight frames which is illustrated with numerical examples. (Received February 01, 2020)

1157-31-557 **Dmitriy Bilyk**, **Feng Dai** and **Ryan W Matzke*** (matzk053@umn.edu), 206 Church St SE, Office 557, Minneapolis, MN 55455. *Generalized Stolarsky Principle.*

The classical Stolarsky Invariance Principle connects the spherical cap L^2 discrepancy of a finite point set on the sphere to the pairwise sum of Euclidean distances between the points. We extend these results to arbitrary measures on the sphere and arbitrary notions of discrepancy, and use this generalization to provide new proofs for some conjectures of Fejes Tóth. (Received February 04, 2020)

32 ► Several complex variables and analytic spaces

1157-32-158 **Damin Wu***, 341 Mansfield Road U1009, Storrs, CT 06269. *Some old and new results on the invariant metrics.*

In this talk I shall recall some classical results on the Kobayashi-Royden metric, the Bergman metric, and the complete Kahler-Einstein metric with negative Ricci curvature. I will then report some recent development on these metrics. (Received January 26, 2020)

1157-32-303 **Julius Ross***, juliusro@uic.edu. *The Minimum Principle.*

Although the minimum of two convex functions is not in general convex, it is true that if $f(x, y)$ is a convex function in two variables then the so-called marginal function

$$g(x) := inf_y f(x, y)$$

is a convex function in x. Although rather basic, and easy to prove, this minimum principle is fundamental in a large part of the theory of convex functions. There exists a complex version of this minimum principle for plurisubharmonic functions due to Kiselman, and a version connected to Lagrangian graphs due to Darvas-Rubinstein.

In this talk we will discuss a broad generalization of this minimum principle in the context of F-subharmonic functions in the sense of Harvey-Lawson. The long-term goal is to apply this to planar elliptic flows in the sense of Khavinson–Mineev-Winstein–Putinar therby generalizing previous work of the speaker with Witt-Nyström that connects the Hele-Shaw flow to the Dirichlet problem of the complex Monge-Ampere equation. (Received February 01, 2020)

33 ► Special functions

1157-33-230 **Alexander Its** and **Andrei Prokhorov*** (andreip@umich.edu), 530 Church street, EH 1847, Ann Arbor, MI 48109. *Asymptotic of solution of three-component Calogero-Painlevé II system.* Preliminary report.

Consider Calogero-Painlevé system

$$\frac{dy_k}{dt} = 2x_k^3 + tx_k + \theta + 2g^2 \sum_{j\neq k} \frac{1}{(x_k - x_j)^3}, \quad \frac{dx_k}{dt} = y_k, \quad k = 1, ..., n$$

Here g and θ are parameters. It comes from matrix Painlevé-II equation and has Lax pair structure. Its solution appears in the random matrix theory in the description of β Tracy-Widom distribution for $\beta = 2n$.

Our goal is to find monodromy data corresponding to the solution mentioned above. It would let us to obtain extra information about β Tracy-Widom distribution. For values $\beta = 2, 4$ Tracy-Widom distribution is well studied. We consider case $\beta = 6$.

The boundary condition at infinity for x_k consists of two parts: power terms and exponentially small terms. We managed to find the monodromy data which allows us to match the power asymptotic and we are still working on matching of the exponentially small asymptotic. The method we are using is nonlinear steepest descent analysis for Riemann-Hilbert problem. (Received February 01, 2020)

34 ► Ordinary differential equations

1157-34-107 **Mingji Zhang*** (mingji.zhang@nmt.edu) and **Zhenshu Wen**. *Dynamics of ionic flows via steady-state Poisson-Nernst-Planck systems with multiple cations. Part I: Permanent charge (small) effects on current-voltage relations.*

We analyze a quasi-one-dimensional steady-state Poisson-Nernst-Planck system for ionic flow through a membrane channel with nonzero but small permanent charge. The system involves *three* ion species, two positively charged with the same valences and one negatively charged. We treat the model as a boundary value problem of a singularly perturbed differential system. Under the framework of geometric singular perturbation theory, together with specific structures of this concrete model, the existence and local uniqueness of solutions to the boundary value problem is established. In particular, treating the permanent charge as a small parameter, under electroneutrality conditions, we are able to derive an approximation of the I-V (current-voltage) relation, from which the effects on ionic flows from the small permanent charge are analyzed in great detail. Critical potentials are identified and their roles in characterizing permanent charge effects are carefully studied. (Received January 22, 2020)

35 ► Partial differential equations

1157-35-31 **Michael Sever*** (sever@math.huji.ac.il), 76 Shenkin St., apt. 4, 65223 Tel-Aviv, Israel. *Slightly inadmissible weak solutions.*

Admissibility failure of discontinuous weak solutions of nonlinear systems of conservation laws is quantified. Such allows experimentally obtained solutions to be classified generally as "slightly inadmissible", by reason of underspecified boundary data, likely unavoidable. It is determined that the traditional definition of well-posedness typically does not hold in the generality suggested by the results of computations. However, a weakened statement of well-posedness suffices to reconcile experimental results with existing theory. (Received December 18, 2019)

1157-35-38 **xianpeng HU*** (xianpehu@cityu.edu.hk). *Wellposedness of compressible viscoelasticity with zero shear viscosity.*

In this talk, the multi-dimensional compressible viscoelastic system with zero shear viscosity is discussed. With the help of potentials of solutions and the Green's function, we construct the global existence of solutions for

small perturbations around the equilibrium by combining the energy and decay estimates based on the vector fields. (Received January 03, 2020)

1157-35-39 **Huanyao Wen*** (mahywen@scut.edu.cn), School of Mathematics, South China University, GUANGZHOU, Guangdong 510641, Peoples Rep of China. *Some recent progress on mathematical analysis of a two-fluid type model.*

In this talk, I will show how to derive a two-fluid type model in some different ways. Then some recent results with associated challenges will be mentioned. As a by-product, we obtain the existence of global weak solutions to the compressible MHD system without resistivity in two dimensions under the action of vertical magnetic field only. (Received January 04, 2020)

1157-35-40 **Yuanzhen Shao*** (yshao8@ua.edu), Tuscaloosa, AL 35487, and **Nikolaos Roidos**. *The Fractional Porous Medium Equation on Manifolds with Conic Singularities.*

Due to the need to model long range diffusive interaction, during the last decade there has been a growing interest in considering diffusion equations involving non-local operators, e.g. the fractional powers of differential operators. In this talk, I will report some recent work with Nikolaos Roidos on the fractional porous medium equation on manifolds with cone-like singularities. I will show that most of the properties of the usual (local) porous medium equation, like existence, uniqueness of weak solution, comparison principle, conservation of mass, are inherited by the non-local version. (Received January 04, 2020)

1157-35-42 **Mimi Dai*** (mdai@uic.edu), IL , and **Han Liu** (hliu94@uic.edu). *Wild solutions for the magnetohydrodynamics with Hall effect.*

We will discuss some wild behaviors exhibited by weak solutions of the magnetohydrodynamics with Hall effect. It includes lack of uniqueness of weak solutions in the Leray-Hopf class and construction of finite energy weak solutions that do not conserve magnetic helicity and magnetic energy. (Received January 05, 2020)

1157-35-53 **Brian Street*** (street@math.wisc.edu). *Atlases in Analysis.*

A manifold is a topological space together with an atlas of coordinate charts. Many different atlases give rise to the same manifold structure, and the first thing that a student learns about manifolds is that if a problem is intrinsic to the manifold structure, it does not matter which atlas you use to study it. Contrary to this abstract principle, many techniques from analysis can be very dependent on the choice of atlas. In fact, many problems use infinitely many different choices of equivalent atlases simultaneously. In this talk, we discuss this phenomenon first in the familiar context of elliptic equations where the atlases are easy to write down explicitly, and then in the context of maximally hypoelliptic equations where one merely proves appropriate atlases exist.

This talk is aimed at people who have seen the definition of a manifold and a vector field before and are familiar with what a partial differential operator is (e.g., the Laplacian on Euclidean space). (Received January 08, 2020)

1157-35-57 **Xiang Xu*** (x2xu@odu.edu), Department of Mathematics and Statistics, ECSB 2300, Norfolk, VA 23529. *Global wellposedness of a gradient flow arising from a triblock copolymer system.*

We study the global well-posedness of the Allen-Cahn Ohta-Nakazawa model with fixed volume constraint. Utilizing the gradient flow structure of its free energy, we give an existence and uniqueness proof of solutions. (Received January 09, 2020)

1157-35-63 **John Lewis*** (johnl@uky.edu), Math Department Room 971, Lexington, KY 40506, and **Andrew Vogel** and **Murat Akman**. *On a Theorem of Wolff Revisited.* Preliminary report.

In this talk we outline a proof which shows for fixed $p, 1 < p < \infty, p \neq 2$, and N a large positive integer ($N \geq N_0$) that there exists $V = V(re^{i\theta}), p$ harmonic in $B(0,1) = \{z : |z| < 1\}$ which is $2\pi/N$ periodic in the θ variable, and Lipschitz continuous on $\partial B(0,1)$, with $c^{-1} \leq |\int_0^{2\pi} V(e^{i\theta})d\theta| \leq c, V(0) = 0$, and Lipschitz norm on $\partial B(0,1) \leq cN$, where $c \geq 1$ depends only on p. In case $p \geq 2$ we give a more or less explicit example of V. Time permitting we discuss how this result can be used to extend Wolff's nonexistence results on Fatou type theorems for $\mathbf{R^2}_+$ to $B(0,1)$ as well as work of Llorente, Manfredi, and Wu, concerning non additivity of p harmonic measure on $\partial\mathbf{R^2}_+$ to $\partial B(0,1)$. (Received January 11, 2020)

1157-35-87 **Yang Yang*** (yangy5@msu.edu), **Wei Li** and **Yimin Zhong**. *Inverse Problems in Fluorescence Ultrasound Modulated Optical Tomography.*

Fluorescence optical tomography (FOT) is an imaging technology that localizes fluorescent targets in tissues. FOT is unstable and of poor resolution in strongly scattering media where the propagation of multiply-scattered light is highly diffusive. We study a hybrid imaging modality called fluorescent ultrasound-modulated optical tomography (fUMOT). It combines FOT with acoustic modulation to produce high-resolution images of optical properties. The principle of fUMOT is to perform multiple measurements of photon currents at the boundary as the optical properties undergo a series of perturbations by acoustic radiation, then the internal information of the optical field can be extracted from the measurement. We set up a mathematical model for fUMOT, prove well-posedness for certain choices of parameters, and present reconstruction algorithms and numerical experiments for the well-posed cases. This is joint work with Wei Li and Yimin Zhong. (Received January 15, 2020)

1157-35-147 **M. Paul Laiu*** (laiump@ornl.gov), **Zheng Chen** and **Cory Hauck**. *A fast implicit solver for semiconductor models in one space dimension.*

Several different approaches are proposed for solving fully implicit discretizations of a simplified Boltzmann-Poisson system with a linear relaxation-type collision kernel. This system models the evolution of free electrons in semiconductor devices under a low-density assumption. At each implicit time step, the discretized system is formulated as a fixed-point problem, which can then be solved with a variety of methods. A key algorithmic component in all the approaches considered here is a recently developed sweeping algorithm for Vlasov-Poisson systems. A synthetic acceleration scheme has been implemented to accelerate the convergence of iterative solvers by using the solution to a drift-diffusion equation as a preconditioner. The performance of four iterative solvers and their accelerated variants has been compared on problems modeling semiconductor devices with various electron mean-free-path. (Received January 24, 2020)

1157-35-148 **M. Paul Laiu*** (laiump@ornl.gov), **J. Austin Harris**, **Ran Chu** and **Eirik Endeve**. *Nonlinear Solvers for Two-species Neutrino-matter Equations in Core-Collapse Supernovae Simulations.*

Efficient numerical schemes are needed for high-fidelity simulations of neutrino transport in nuclear astrophysics applications, where the dominating computational cost is in modeling neutrino-matter interactions. With implicit/implicit-explicit time discretization, simulating neutrino-matter interactions requires solutions to nonlinear systems that couple the neutrino transport equation and the matter equation. We compare various iterative solvers for these nonlinear systems arising from four-momentum and lepton exchange due to emission and absorption, scattering, and pair processes. The numerical tests are performed on a multi-group two-moment model employing discontinuous Galerkin phase space discretization and implicit-explicit time integration. We analyze the performance of these iterative solvers on CPUs and discuss preliminary results from porting these solvers to GPUs. (Received January 24, 2020)

1157-35-153 **Richard S Laugesen*** (laugesen@illinois.edu). *Well-posedness of Weinberger's center of mass by energy minimization.*

Weinberger's result that the ball maximizes the first nonzero Neumann eigenvalue of the Laplacian, among bounded domains Ω of given volume, relies on orthogonality of the trial function to a constant. That is, the trial function must be translated suitably to achieve a "center of mass" normalization

$$\int_\Omega g(|x+y|)\,\frac{x+y}{|x+y|}\,dy = 0,$$

for some $x \in \mathbb{R}^n$. Here g is the radial part of the ball eigenfunction, extended as a constant outside the ball.

I will show by energy minimization that the point x exists, is unique, and depends continuously on the domain Ω. The method extends to measures $d\mu(y)$, and can handle unbounded supports by renormalizing the energy and extending to the sphere at infinity. (Received January 26, 2020)

1157-35-165 **Gieri Simonett*** (gieri.simonett@vanderbilt.edu) and **Giusy Mazzone** (giusy.mazzone@queensu.ca). *On the motion of a rigid body with a cavity filled with a viscous liquid.*

I will consider the motion of a rigid body with an interior cavity that is completely filled with a viscous fluid. The equilibria of the system will be characterized and their stability properties are analyzed. It will be shown that the fluid exerts a stabilizing effect, driving the system towards a state where it is moving as a rigid body with constant angular velocity. In addition, I will characterize the critical spaces for the governing evolution equation, and I will show how parabolic regularization in time-weighted spaces affords great flexibility in establishing

regularity and stability properties for the system. The approach is based on the theory of Lp-Lq maximal regularity. (Received January 26, 2020)

1157-35-178 **Geng Chen***, Department of Mathematics, University of Kansas, and **Tao Huang** and **Weishi Liu**. *Poiseuille flow of nematic liquid crystals via Ericksen-Leslie model.*

In this talk, we will discuss a recent global existence result on the Poiseuille flow of nematic liquid crystals via full Ericksen-Leslie model. The existing results on the Ericksen-Leslie model for the liquid crystals mainly focused on the parabolic and elliptic type models by omitting the kinetic energy term. In this recent progress, we established a new method to study the full model. A singularity formation result will also be discussed, together with the global existence result showing that the solution will in general live in the Holder continuous space. The talk is based on a joint work with Tao Huang & Weishi Liu. (Received January 27, 2020)

1157-35-192 **Daniel Phillips*** (phillips@purdue.edu) and **Patricia Bauman** (baumanp@purdue.edu). *Nematic Liquid Crystal Structures in Slabs.* Preliminary report.

We examine minimizers for the Landau - de Gennes energy in a slab $\Omega \times (-\delta, \delta)$ where the cross section Ω is a bounded simply connected domain in $\mathbb{R}^2$. The minimizers are to satisfy tangential boundary conditions on the top and bottom faces and prescribed boundary conditions on the lateral surface. We determine the defect structures, textures and estimates on the energies for minimizers. In particular it is shown how these features depend on the relative sizes of the Landau - de Gennes order parameter ϵ and the slab thickness 2δ. (Received January 28, 2020)

1157-35-194 **Jiuyi Zhu***, Department of Mathematics, Lousiana State University, Baton Rouge, LA 70803, and **Carlos Kenig**, Department of Mathematics, University of Chicago, Chicago, IL 60637. *Propagation of smallness in elliptic periodic homogenization.* Preliminary report.

We consider a family of second order operators $\mathcal{L}_\epsilon$ in divergence form with rapidly oscillating and periodic coefficients. The paper is mainly concerned with an approximate three-ball inequality for solutions in elliptic periodic homogenization. It is the first time such an approximate three-ball inequality for homogenization theory is obtained. It implies an approximate quantitative propagation of smallness. The proof relies on a representation of the solution by the Poisson kernel and the Lagrange interpolation technique. (Received January 28, 2020)

1157-35-206 **Brian Seguin*** (bseguin@luc.edu), 1261 W Granville #2, Chicago, IL 60660. *Multi-component multiphase flow through a poroelastic medium.*

I will present an axiomatic development for a continuum description of a multi-component multiphase porous flow in an elastic medium. Rather that using mixture theory, the presented development is motived by results in homogenization theory and uses Darcy's law to simplify the resulting system of PDEs. The principle of local thermodynamic equilibrium is used to determine the distribution of the components on the microscale, and the Coleman—Noll procedure is used to derive constitutive restrictions which guarantee that the resulting model satisfies an appropriate statement of the second law of thermodynamics and a corresponding dissipation inequality, which can be used to obtain useful bounds. This is joint work with Noel Walkington. (Received January 28, 2020)

1157-35-216 **Samuel Cogar*** (samuel.cogar@rutgers.edu). *A trace class Stekloff eigenvalue problem for nondestructive testing of materials.*

The Stekloff eigenvalue problem has been a recent problem of interest due to its potential use in nondestructive testing of materials, but it suffers from two shortcomings that are common among this class of eigenvalue methods: it is not known in general if Stekloff eigenvalues exist in the case of complex coefficients, and there exists no mechanism in which to increase their sensitivity to changes in the material properties of the medium under consideration. We present a variation of the Stekloff eigenvalue problem which overcomes both of these issues by the introduction of a trace class operator in the boundary condition, and we provide numerical examples to examine the practical consequences of this variation. We discuss this idea in the broader context of what might be called tailored eigenvalue methods, in which the eigenvalue problem is designed to ensure favorable theoretical or practical results. (Received January 29, 2020)

1157-35-218 **Roman Shvydkoy*** (shvydkoy@uic.edu), MC 249, 851 S Morgan St, Chicago, IL 60607. *Hierarchy of entropies in pressured Euler Alignment systems.*

We discuss a new method of obtaining global existence results for 1D compressible models of fluid and alignment dynamics. (Received January 29, 2020)

1157-35-234 **Mimi Dai*** (mdai@uic.edu). *Ill-posedness of magnetohydrodynamics.*
We will talk about certain ill-posedness behavior of the three dimensional magneto-hydrodynamic models. In particular, we will address the norm inflation phenomena and existence of discontinuous weak solutions. (Received January 29, 2020)

1157-35-236 **Alexandre Girouard***, Alexandre.Girouard@mat.ulaval.ca, and **Antoine Henrot** and **Jean Lagacé**. *From Steklov to Neumann eigenvalues via homogenization.*
We study a new link between the Steklov and Neumann eigenvalues of Euclidean domains. This is obtained through an homogenization limit of the Steklov problem on a periodically perforated domain, converging to a family of eigenvalue problems with dynamical boundary conditions. For this problem, the spectral parameter appears both in the interior of the domain and on its boundary. This intermediary problem interpolates between Steklov and Neumann eigenvalues of the domain. As a corollary, we recover some isoperimetric type bounds for Neumann eigenvalues from known isoperimetric bounds for Steklov eigenvalues. The interpolation also leads to the construction of planar domains with first perimeter-normalized Stekov eigenvalue that is larger than any previously known example. (Received January 30, 2020)

1157-35-238 **Sarah Biesenbach*** (biesenbach@eddy.rwth-aachen.de). *Optimal convergence rates for the Cahn-Hilliard equation on the real line.*
We explain how to derive optimal algebraic-in-time relaxation rates to a "bump" for the Cahn-Hilliard equation on the line under the condition that the initial condition has a finite L^1-distance to the bump. The result extends the relaxation method developed previously by the second and the third author for a single transition layer (the "kink") to the case of two transition layers (the "bump"); as in the previous work, the method exploits the gradient flow structure, Nash-type inequalities, duality arguments, and Schauder estimates. For both the kink and the bump, the energy gap is translation invariant and its decay alone cannot specify to which member of the family of minimizers the solution converges. Whereas in the case of the kink, the conserved quantity singles out the longtime limit, in the case of a bump, a new argument is needed. This is joint work in progress with Felix Otto and Maria G. Westdickenberg. (Received January 30, 2020)

1157-35-243 **M. Burak Erdogan*** (berdogan@illinois.edu), 1409 W. Green Street (MC-382), Urbana, IL 61801. *Dispersive estimates for Dirac operators.*
We will review recent results on dispersive decay estimates for the Dirac operators in $\mathbb{R}^n$, $n \geq 2$, with a decaying self-adjoint potential. These include $L^1 \to L^\infty$, Strichartz, and Kato smoothing estimates. We also discuss the classification of threshold resonances and their effect on dispersive estimates, and the limiting absorption principle for Dirac operators. The talk is based on joint works with M. Goldberg (U. Cincinnati), W. Green (Rose-Hulman), and E. Toprak (Rutgers). (Received January 30, 2020)

1157-35-250 **Thomas Beck*** (tdbeck@email.unc.edu). *Eigenvalue estimates and localization of the first Dirichlet eigenfunction.*
In this talk, we will discuss the first Dirichlet eigenvalue and eigenfunction of a convex domain of high eccentricity. Our aim is to obtain quantitative estimates on the shape of the level sets of this eigenfunction, which are uniform in this high eccentricity setting. We will describe a sharp estimate showing that the eigenfunction cannot localize to too small a subset of the domain. A key step in the proof is to obtain sufficiently sharp eigenvalue estimates using the variational formulation of the eigenvalue, in order to estimate the first derivatives of the eigenfunction. (Received January 30, 2020)

1157-35-292 **Anup Lamichhane*** (a-lamichhane@onu.edu), Department of Mathematics and Statistics, Ohio Northern Univerisy, 525 South Main Street, Ada, OH 45810. *Three-dimensional implicit surface reconstruction using method of approximate particular solutions.*
We reconstruct the three-dimensional surface from a set of scattered cloud data. Recently, a partial differential equation (PDE)-based algorithm is proposed for obtaining the reconstruction of the surface. In this method, we need to obtain the numerical solution of a PDE that will help us to reconstruct the implicit surface. We use the method of approximate particular solutions, a meshless method, to obtain the solutions of the PDE. Several reconstructed surfaces will be presented. (Received January 31, 2020)

1157-35-318 **Evans M Harrell*** (harrell@math.gatech.edu), Skiles Building, 686 Cherry St., Atlanta, GA 30332-0160, and **Luigi Provenzano** and **Joachim Stubbe**. *Complementary Estimates for Eigenvalue Means.*

We use the averaged variational principle to obtain asymptotically sharp inequalities for the Dirichlet and Neumann eigenvalues of the Laplacian on a domain, which are complementary to the familiar Berezin–Li–Yau and Kröger inequalities. (Received February 01, 2020)

1157-35-327 **Shixu Meng***, 530 Church Street, 3060 East Hall, Ann Arbor, MI 48109. *A convergent low-wavenumber, high-frequency homogenization of the wave equation in periodic media with a source term.*

We pursue a low-wavenumber, second-order homogenized solution of the time-harmonic wave equation in periodic media with a source term whose frequency resides inside a band gap. Considering the wave motion in an unbounded medium $\mathbb{R}^d$ ($d \geqslant 1$), we first use the Bloch transform to formulate an equivalent variational problem in a bounded domain. By investigating the source term's projection onto certain periodic functions, the second-order model can then be derived via asymptotic expansion of the Bloch eigenfunction and the germane dispersion relationship. We establish the convergence of the second-order homogenized solution, and we include numerical examples to illustrate the convergence result. (Received February 01, 2020)

1157-35-350 **Ru-Yu Lai*** (rylai@umn.edu), 206 Church St SE, Minneapolis, MN 55455, and **Qin Li** and **Gunther Uhlmann**. *Inverse problems for the stationary transport equation in the diffusion scaling.*

We consider the inverse problem of reconstructing the optical parameters of the radiative transfer equation (RTE) from boundary measurements in the diffusion limit. In the diffusive regime, the forward problem for the stationary RTE is well approximated by an elliptic equation. However, the connection between the inverse problem for the RTE and the inverse problem for the elliptic equation has not been fully developed. This problem is particularly interesting because the former one is mildly ill-posed, with a Holder type stability estimate, while the latter is well known to be severely ill-posed with a logarithmic type stability estimate. In this talk, we will illustrate stability estimates for the inverse problem for the RTE and its dependence on Kn. In particular, the stability is Holder in all regimes, but the coefficient deteriorates exponentially, making the inverse problem of the RTE severely ill-posed when Kn is small. In this way we connect the two inverse problems. Numerical results agree with the analysis of worsening stability as the Knudsen number gets smaller. (Received February 02, 2020)

1157-35-355 **Yulong Lu***, Department of Mathematics, Duke University, Durham, NC 27705. *Accelerating Langevin sampling with birth-death processes.*

A fundamental problem in Bayesian inference and statistical machine learning is to efficiently sample from multimodal distributions. Due to metastability, multimodal distributions are difficult to sample using standard Markov chain Monte Carlo methods, such as the algorithms based on Langevin diffusion. We propose a new particle algorithm based on a birth-death mechanism to accelerate the mixing of Langevin diffusion. Our algorithm is motivated by its mean field partial differential equation (PDE), which is a Fokker-Planck equation supplemented by a nonlocal birth-death term. This PDE can be viewed as a gradient flow of the Kullback-Leibler divergence with respect to the Wasserstein-Fisher-Rao metric. We prove that under some assumptions the asymptotic convergence rate of the nonlocal PDE is independent of the potential barrier, in contrast to the exponential dependence in the case of the Langevin diffusion. Some analytical and numerical examples demonstrate that the proposed particle algorithm far outperforms the Langevin diffusion (Received February 02, 2020)

1157-35-363 **chunmei wang*** (chunmei.wang@ttu.edu), 2500 Broadway, Lubbock, TX 79409. *Primal-Dual Weak Galerkin Finite Element Methods for First-Order Transport Problems.*

The speaker will introduce a new numerical method for first-order transport problems by using the primal-dual weak Galerkin (PD-WG) finite element method recently developed in scientific computing. The PD-WG method is based on a variational formulation of the modeling equation for which the differential operator is applied to the test function so that low regularity for the exact solution of the original equation is sufficient for computation. The PD-WG finite element method indeed yields a symmetric system involving both the original equation for the primal variable and its dual for the dual variable (also known as Lagrangian multiplier). For the linear transport problem, it is shown that the PD-WG method offers numerical solutions that conserve mass locally on each element. Optimal order error estimates in various norms are derived for the numerical solutions arising from the PD-WG method with weak regularity assumptions on the modelling equations. A variety of numerical

results are presented to demonstrate the accuracy and stability of the new method. (Received February 02, 2020)

1157-35-373 **Fernando Charro**, **Alaa Haj Ali**, **Monica Torres** and **Peiyong Wang*** (pywang@wayne.edu), 656 W. Kirby, 1150 FAB, Detroit, MI 48202. *A bifurcation phenomenon in a singularly perturbed two-phase free boundary problem of phase transition.*

We prove a bifurcation phenomenon exists in a two-phase singularly perturbed free boundary problem of phase transition. This is a continuation of the study of a bifurcation in the one-phase problem. More precisely, the uniqueness of a solution of the two-phase problem breaks down as the boundary data decreases through a threshold value. Besides the minimizer of the associated functional and the harmonic solution without a free boundary, a third solution is proved to exist as a critical point of the functional being minimized through the application of the Mountain Pass Lemma. We further prove convergence of the evolution with initial data near a minimizer or the harmonic solution, while the evolution deviated away from a saddle point solution of the free boundary problem as time goes by. (Received February 02, 2020)

1157-35-386 **Lotfi Hermi*** (lhermi@fiu.edu), Department of Mathematics and Statistics, Florida International University, 11200 S.W. 8th Street, Miami, FL 33199. *Isoperimetric Inequalities for Wedge-Like Membranes and Convex Cones.*

By introducing geometric factors and physical parameters that lend themselves to the Payne interpretation in Weinstein fractional space, we prove new isoperimetric inequalities for the fundamental eigenvalue of the Dirichlet problem and relate it to "relative torsional rigidity" which will be introduced. We motivate, conjecture, and prove these isoperimetric inequalities. They generalize and improve classical sharp inequalities dating back to the works of Payne-Weinberger (1960), Payne-Rayner (1973), Kohler-Jobin (1972), and Chiti (1982). They are forms of the classical Faber-Krahn and Kohler-Jobin inequalities in manifolds with density. Numerical evidence will be provided for certain shapes. (Received February 03, 2020)

1157-35-408 **Isaac Harris*** (harri814@purdue.edu), 150 N. University Street, office: MATH 612, West Lafayette, IN 47907. *Reconstructing a Generalized Impedance Inclusion with ElectroStatic Sampling.* Preliminary report.

We derive a Sampling Method to solve the inverse shape problem of recovering an inclusion with a generalized impedance condition. The generalized impedance condition is a second-order differential operator applied to the boundary of the inclusion. We assume that the Dirichlet-to-Neumann mapping is given on the outer boundary from an imposed voltage. A simple numerical example is given to show the effectiveness of the proposed inversion method. We also consider the inverse impedance problem of determining the material parameters from the Dirichlet-to-Neumann mapping and give a uniqueness result. (Received February 03, 2020)

1157-35-410 **Denis Bonheure**, **Jean Dolbeault**, **Maria Esteban**, **Ari Laptev** and **Michael Loss*** (loss@math.gatech.edu). *An application of the sharp Caffarelli-Kohn-Nirenberg inequalities.*

I will present an application of the sharp Caffarelli-Kohn-Nirenberg inequalities to non-linear functionals of Aharonov-Bohm type, i.e., to problems that include a magnetic flux concentrated at one point. These functionals are rotationally invariant and, as I will discuss, depending on the magnitude of the flux, the optimizers are radially symmetric or not. (Received February 03, 2020)

1157-35-417 **Ali Pakzad*** (apakzad@iu.edu), 831 E. 3rd St., Bloomington, IN 47405. *The challenge of finding ε in turbulence.*

Turbulence models, which model the universal effect of small scales on large scales, are introduced to account for sub-mesh scale effects on a coarse mesh. The key to getting a good approximation for a turbulence model is to correctly calibrate the energy dissipation in the model on a coarse mesh. In this talk, I present a calculation of the energy dissipation in a turbulence model discretized on a coarse mesh. (Received February 03, 2020)

1157-35-422 **Michael S Jolly** and **Anuj Kumar*** (kumar22@iu.edu), 831 E 3rd St, Bloomington, IN 47405, and **Vincent R Martinez**. *Gevrey regularity for the supercritical generalized SQG equations.*

We study two families of active scalar equations of the form $\partial_t\theta + \nu\Lambda^\kappa\theta + u\cdot\nabla\theta = 0$ where the velocity field u is given in terms of the scalar temperature θ by an integral operator which is more singular than the (usual) Reisz transform. In the above equation, Λ denotes the Zygmund operator $(-\Delta)^{\frac{1}{2}}$. The first family has velocity fields given by $u = -\nabla^\perp\Lambda^{\beta-2}\theta$ for $1 < \beta \leq 2$. With initial data in the critical space $H^{1+\beta-\kappa}$, we establish local well-posedness of Gevrey class solutions for arbitrary initial data and global well-posedness for small initial

data. The second family has velocity fields given by $u = -\nabla^{\perp}(\ln(I-\Delta))^{\mu}\theta$ for $\mu > 0$. In that case, we prove local well-posedness of Gevrey class solutions with initial data in H^3.

We also consider a logarithmically regularized inviscid counterpart of the first family above and establish local well-posedness in the borderline Sobolev space $H^{1+\beta}$. This was done by D. Chae and J. Wu for the range $0 \leq \beta \leq 1$. We extend their result to the range $1 < \beta \leq 2$. (Received February 03, 2020)

1157-35-437 **Zoran Grujic** (zg7c@virginia.edu), Department of Mathematics, Kerchof Hall, Charlottesville, VA 22904, and **Liaosha Xu*** (lx5af@virginia.edu), Department of Mathematics, Kerchof Hall, Charlottesville, VA 22904. *Asymptotic criticality of the Navier-Stokes regularity problem.*

In this talk I will present a general framework for the study of the regularity problems for some PDE models with super-criticality (e.g. the Navier-Stokes equations). The spatial intermittency of all the derivatives of the solution can be realized if the chain of all the higher order derivatives are placed in certain order. The key idea is: the scaling gap is closing (i.e. the equation becomes critical) as we include more higher order derivatives in the Z_{α}-framework as introduced in a recent article by Bradshaw, Farhat and Grujić, and the solution is regular if 'the infinite chain' is dynamically restricted under an asymptotic curve. This is a joint work with Zoran Grujić. (Received February 03, 2020)

1157-35-442 **Rafael D. Benguria*** (rbenguri@uc.cl), Instituto de Fisica, Casilla 306, Santiago 22, Santiago, RM, Chile, and **Cristobal Vallejos** and **Hanne Van Den Bosch**. *Existence and non-existence of minimizers for Poincaré–Sobolev inequalities.*

In this talk I discuss the existence and non–existence of minimizers for a type of (critical) Poincaré–Sobolev inequalities. We show that minimizers do exist for smooth domains in $\mathbb{R}^d$, and also for some polyhedral domains. On the other hand, we show the non-existence of minimizers in the rectangular isosceles triangle in $\mathbb{R}^d$. (Received February 03, 2020)

1157-35-465 **Patricia Bauman*** (baumanp@purdue.edu), Dept. of Mathematics, Purdue University, 150 No. University Street, West Lafayette, IN 47907, and **Daniel Phillips** (phillips@purdue.edu), Dept. of Math, Purdue University, 150 No. University Street, West Lafayette, IN 47907. *Regularity of Minimizers for Constrained Energies with Applications to Landau-de Gennes Models.*

We prove regularity of minimizers for constrained energies in two-dimensional domains. Our results have applications to Landau-de Gennes Q-tensor energies for liquid crystals, including all elasticity terms and a Maier-Saupe bulk term. In particular, we show that minimizers of this energy are Q-tensor valued, Holder continuous functions whose eigenvalues are bounded strictly between -1/3 and 2/3. (Received February 03, 2020)

1157-35-472 **Derek Kielty*** (dkielty2@illinois.edu) and **Mark Ashbaugh**. *Spectral gaps of 1-D Robin Schrödinger operators.*

Lavine proved in 1994 that the spectral gap of a Schrödinger operator on an interval with convex potential is minimized when the potential is constant. His result holds for both Neumann and Dirichlet boundary conditions. Viewing the Robin boundary condition as interpolating between the Neumann and Dirichlet cases, we ask: What convex potential minimizes the spectral gap of a Robin Schrödinger operator?

After showing the minimizing potential is a constant, we go further to explore negative values of the Robin parameter (lying outside the interpolating regime). This minimization result implies a sharp lower bound on the spectral gap for each real value of the Robin parameter. (Joint work with Mark Ashbaugh.) (Received February 03, 2020)

1157-35-478 **Giovanna Citti*** (giovanna.citti@unibo.it), Dipartimento di Matematica, Università di Bologna, Piazza di Porta S. Donato 5, 40126 Bologna, Italy. *Schauder estimates at non characteristic boundary points in Carnot groups.*

We present a joint work with A. Baldi and G.Cupini on Schauder estimates at the boundary for sub-Laplacian type operators in Carnot groups. Up to now sub-Riemannian estimates at the boundary are known only in the Heisenberg groups. The proof of these estimates in the Heisenberg setting, due to Jerison, is based on the Fourier transform and cannot be repeated in general Lie groups. We consider a non characteristic point, and assume that the boundary is a sub-Riemannian surface in a neighbourhood of the selected point. Under this assumption we build a Poisson kernel using the fundamental solution, and its restriction to the boundary. From here we deduce the estimates. (Received February 03, 2020)

1157-35-494 **L Mercredi Chasman***, chasmanm@morris.umn.edu, and **Jeffrey Langford**. *Low Eigenvalues of the Robin Plate.*

Inspired by recent results by Freitas and Laguesen for the Robin Laplacian, we consider the Robin bi-Laplacian (plate) under tension ($\Delta^2 - \tau\Delta$). Using similar methodology and a careful treatment of special functions, we are able to obtain a similar result for the second eigenvalue of the Robin plate for certain parameter ranges. (Received February 03, 2020)

1157-35-495 **Don Colladay**, **Emily Dryden** and **Jeffrey Langford*** (jeffrey.langford@bucknell.edu), Department of Mathematics, Bucknell University, 1 Dent Drive, Lewisburg, PA 17837, and **Patrick McDonald**. *Exit Time Moments and Eigenvalue Estimates.*

Motivated by the relationship between torsional rigidity and the principal Dirichlet eigenvalue, we consider expectations of powers of exit times of Brownian motion paths, called exit time moments. We show how to sharply estimate eigenvalues of the Dirichlet Laplacian using quantities involving exit time moments. (Received February 03, 2020)

1157-35-534 **Selma Yildirim*** (selma@math.uchicago.edu), 5734 S University Ave, Chicago, IL 60637. *Extensions of some results about the eigenvalues for the Laplacian.*

After giving an overview of some results about eigenvalues of the Laplacian, I will present some old and new related work extending these ideas and methods to prove analogues results for the Laplacian. (Received February 04, 2020)

1157-35-537 **HaiGang LI*** (hgli@bnu.edu.cn), 19 Xinjiekou Waidajie, Beijing, 100875, Peoples Rep of China. *Babuska Problem in Composite Materials and its Applications to Stokes Equations.*

In high-contrast composite materials, the stress concentration is a common phenomenon when inclusions are densely distributed and may possibly touch. It always causes damage initiation and even ultimate failure. This problem was proposed mathematically by Ivo Babuska, concerning the system of linear elasticity, modeled by a class of second order elliptic system of divergence form with discontinuous coefficients.

I will first review some of our results on upper and lower bounds of the gradients by developing an iteration technique with respect to the energy integral to overcome the essential difficulty from the lack of maximal principle for elliptic systems, then present our very recent results on the gradients' asymptotics to close this problem and show the key role of the geometry of the inclusions played in such blow-up analysis. Finally, we present its application to suspension problem in Stokes equation setting. (Received February 04, 2020)

1157-35-538 **Michael Levitin**, **Leonid Parnovski**, **Iosif Polterovich** and **David A. Sher*** (dsher@depaul.edu), 2320 N Kenmore Ave, Chicago, IL 60614. *Steklov eigenvalue asymptotics for curvilinear polygons.*

We consider the Steklov eigenvalue problem on curvilinear polygons in the plane, with all interior angles measuring less than π. In this setting, we formulate and prove precise spectral asymptotics, with error converging to zero as the spectral parameter increases. The problem turns out to have an interesting relationship to a scattering-type eigenvalue problem on the one-dimensional boundary of the polygon, viewed as a quantum graph. (Received February 04, 2020)

1157-35-582 **Chase O Mathison*** (cmathiso@purdue.edu), 1736 N 9TH ST, APT 15, Lafayette, IN 47904. *Sampling in thermoacoustic tomography.*

Thermoacoustic tomography is a medical imaging technique which uses electromagnetic waves to create thermal expansion in an object that we wish to image. This in turn creates ultrasound waves, which we can measure by a number of methods. In this talk, we explore the effect of sampling rates of the measurements (which are Fourier integral operators) on reconstructed images in thermoacoustic tomography. We start with sampling requirements on Mf given f satisfying certain conditions. After this we discuss the resolution limit on f posed by the sampling rate of Mf without assuming any conditions on these sampling rates. Next we discuss aliasing artifacts when Mf is known to be under sampled in one or more of its variables, and finally, we discuss averaging of measurement data and resulting aliasing and artifacts. (Received February 04, 2020)

1157-35-587 **G. Chen*** (gchen@lanl.gov), MS B284, Los Alamos National Laboratory, P.O. Box 1663, Los Alamos, NM 87545, and **Guangye Chen** (chacon@lanl.gov). *Energy- and Charge-conserving semi-implicit particle-in-cell algorithms for the relativistic Vlasov-Maxwell system.*

Particle-in-cell (PIC) methods for the Vlasov-Maxwell equations have enabled efficient simulations of many problems of interest in collisionless plasmas. It has been widely used for many first-principle simulations in fusion studies, laser-plasma interactions, space plasmas, etc. However, standard PIC implementations employ explicit time integration schemes, and are therefore unsuitable for long-term multiscale simulations owing to the lack of conservation properties and the presence of secular stochastic heating. Recently, a semi-implicit PIC scheme has been developed for the relativistic Vlasov-Maxwell system [1] that overcome some of these limitations by enforcing important discrete conservation properties such as energy and charge conservation. I will discuss the details of this algorithm, and demonstrate it with numerical examples of varying complexity. [1] G. Chen, L. Chacon, L. Yin, B.J. Albright, D.J. Stark, R.F. Bird, a semi-implicit, energy- and charge-conserving particle-in-cell algorithm for the relativistic Vlasov-Maxwell equations. J. Comput. Phys. 407, 109228, 2020 (Received February 04, 2020)

1157-35-606 **Mark Ashbaugh*** (ashbaughm@missouri.edu), Department of Mathematics, University of Missouri, Columbia, MO 65211-4100, and **Richard Laugesen**. *Open Problems Session.* Preliminary report.

We will lead a session on "Open Problems" on sharp eigenvalue estimates and inequalities for differential operators. We welcome all participants to offer new problems or to discuss possible strategies for solving open problems in the area. (Received February 04, 2020)

1157-35-632 **John Goldman*** (tyolcu@bradley.edu), 1501 W Bradley Ave, Bradley University, Mathematics, Peoria, IL 61625. *Sharper estimates for the eigenvalues of the fractional Laplacian.*

In recent years there has been considerable interest in the eigenvalue inequalities estimating how big or small they can be. It is a fairly active research area in which generalizations and sharper (and elegant) proofs of such inequalities have been abundantly explored. Like the Drichlet Laplacian, the eigenvalues estimates of the fractional Laplacian have also been of interest. In this talk, we will focus on the mechanics of recent developments that will help further develop research experience in this field. (Received February 04, 2020)

37 ► Dynamical systems and ergodic theory

1157-37-84 **Vincent R Martinez***, 695 Park Ave, Department of Mathematics & Statistics, New York, NY 10065. *Unique ergodicity and exponential mixing for the damped-driven stochastic KdV equation.*

We discuss a proof for the uniqueness and regularity of the invariant measure for the damped-driven stochastic Korteweg-de Vries equation, where the noise is additive and sufficiently non-degenerate. It is shown that a simple, but versatile control strategy, typically employed to establish exponential mixing for strongly dissipative systems, can nevertheless be applied in this weakly dissipative setting to obtain unique ergodicity, albeit without mixing rates. Under the assumption of large damping, however, exponential mixing can be recovered. Time permitting, application of this approach to other weakly dissipative systems, such as the damped-driven Nonlinear Schrodinger equation will also be discussed. This is joint work with Nathan Glatt-Holtz and Geordie Richards. (Received January 15, 2020)

1157-37-198 **Jesse Capecelatro*** (jcaps@umich.edu), 1231 Beal Ave, Ann Arbor, MI 14809, and **Sarah Beetham** (snverner@umich.edu), 1231 Beal Ave, Ann Arbor, MI 14809. *Turbulence modeling of strongly-coupled gas-particle flows using machine learning.*

Many natural and industrial processes involve the flow of solid particles or liquid droplets whose dynamical evolution and morphology are intimately coupled with a carrier gas. A peculiar behavior of disperse multiphase flows are their ability to give rise to large-scale structures (hundreds to thousands of times the size of individual particles), from dense clusters to nearly-particle-free voids. Such coupling can effectively 'demix' the underlying flow, which has enormous consequences in engineering systems. In this talk, high-resolution simulations particle-laden flows will be presented to reveal how multiphase interactions at the particle scale augment or restrict large-scale flow processes, and provide unique insight into the budget of turbulent kinetic energy. A rigorous derivation of the Reynolds-averaged Navier-Stokes equations for coupled fluid-particle flows will be presented,

with specific emphasis made on the unclosed terms unique to multiphase systems. In an effort to transform the simulation data into a robust and predictive turbulence model, a new data-driven approach based on sparse regression will be presented. The method will be demonstrated on both single-phase and multiphase flows, with the aim of generating compact, algebraic turbulence closure. (Received January 28, 2020)

1157-37-559 **Michael Shapiro*** (mshapiro@msu.edu), Department of Mathematics, Michigan State University, 619 Red Cedar Rd C212 Wells Hall, East Lansing, MI 48824. *Noncommutative pentagram map.*

A pentagram map is a discrete integrable transformation on the space of projective classes of (twisted) n-gons in projective plane. We will discuss a classical and non-commutative version of pentagram map and its integrability properties. (Received February 04, 2020)

41 ► Approximations and expansions

1157-41-29 **Haizhao Yang*** (haizhao@purdue.edu), 150 N University St, West Lafayette, IN 47907, and **Qiang Du**, **Hadrien Montanelli**, **Zuowei Shen** and **Shijun Zhang**. *Deep Network Approximation.*

Deep learning has been successfully applied to many high-dimensional problems including computer vision, speech recognition, and numerical PDEs. In this talk, we first introduce explicit error characterization of deep network approximation. Second, we present connections between deep network approximation, Monte Carlo sampling, random orthogonal projection, and Kolmogorov–Arnold representation theorem on the curse of dimensionality. These connections lead to new error estimates for the approximation of multivariate functions by deep networks, for which the curse of the dimensionality is lessened. (Received December 16, 2019)

1157-41-638 **Miaojung Yvonne Ou***, 408 Ewing Hall, University of Delaware, Newark, DE 19716, and **Chuan Bi** and **Richard Spencer**. *A multi-regularization method for NMR relaxometry.*

The mathematical core of NMR relaxametry is the inverse Laplace transform of real-valued measurement data subjected to noise. In this talk, we will present a new method inspired by the learning theory and Reproducing Kernel Hilbert space. Numerical results will also be presented. (Received February 04, 2020)

42 ► Fourier analysis

1157-42-2 **Bingyang Hu*** (bhu32@wisc.edu), Department of Mathematics, Room 101-4a, Madison, WI 53706, **Theresa Anderson** (tcanderson@purdue.edu), 150 N. University St., West Lafayette, IN, West Lafayette, IN 47907, and **Joris Roos** (jroos@math.wisc.edu), 480 Lincoln Drive, Madison, WI 53706. *Sparse bounds of singular Radon transform and its discrete analog.*

In this talk, we will first talk about the sparse bound of singular Radon transforms, which can be viewed as a finer quantification of the L^p boundedness of such operators due to Christ, Nagel, Stein and Wainger in 1999. As an application, we also establish the sparse bound of discrete singular Radon transforms along a general class of polynomials, which generalizes the result due to Culiuc, Kesler and Lacey in 2018. The second part of this work is a joint work with Theresa Anderson and Joris Roos. (Received September 03, 2019)

1157-42-10 **Benjamin Baker Bruce*** (bbruce4@wisc.edu), Van Vleck Hall, 480 Lincoln Drive, Madison, WI 53706. *Fourier Restriction to a Hyperbolic Cone.*

We resolve the Fourier restriction problem for a conical hypersurface in $\mathbb{R}^4$ whose cross sections are hyperbolic paraboloids. Earlier work of S. Lee established an optimal L^2-based bilinear restriction theorem, as well as the full range of off-scaling linear restriction estimates, for this "hyperbolic cone." We obtain the remaining scale-invariant estimates by adapting a bilinear-to-linear argument introduced by B. Stovall in the context of restriction to the hyperbolic paraboloid in $\mathbb{R}^3$. (Received September 30, 2019)

1157-42-34 **Michael Greenblatt***, 851 S Morgan St, SEO 322, Mail Code 249, Chicago, IL 60607-7045. *Fourier transforms of indicator functions, lattice point discrepancy problems, and related matters.*

We describe some sharp estimates for Fourier transforms of indicator functions of bounded open sets in $\mathbb{R}^n$ with real analytic boundary. These estimates are closely connected to corresponding sharp estimates on Fourier transforms of hypersurface measures. The estimates have immediate number theoretical applications, providing

nontrivial lattice point discrepancy results for a large class of domains. Unlike most previous results in this subject, no convexity condition is required on the domains. The estimates also have applications to maximal averages and local stability theorems for integrals of negative powers of real-analytic functions, which will be described if time permits. (Received December 30, 2019)

1157-42-35 **Xiaochun Li*** (xcli@illinois.edu), 904 Green Street, Champaign, IL 61801. *Roth's type theorems on (polynomial) progressions.*

The arithmetic progression problems were posed by Erd́os-Turan, answered affirmatively by Szemerédi. However, there are still many questions remained on precise quantitative description on how large a subset shall be in oredr to guarantee a progression in it. Involving with Fourier analysis, considerable work had been accomplished recently. We will give a survey on those progress, and report our recent progress on quantitative version of Roth's type theorem on (polynomial) progressions of short length. (Received January 02, 2020)

1157-42-45 **Matthew Fickus*** (matthew.fickus@afit.edu) and **Benjamin R. Mayo**. *Mutually unbiased equiangular tight frames.*

An equiangular tight frame (ETF) yields a type of optimal packing of lines in a Euclidean space. ETFs seem to be rare, and all known infinite families of them arise from some type of combinatorial design. In this paper, we introduce a new method for constructing ETFs. We begin by showing that it is sometimes possible to construct multiple ETFs for the same space that are "mutually unbiased" in a way that is analogous to the quantum-information-theoretic concept of mutually unbiased bases. We then show that taking certain tensor products of these mutually unbiased ETFs with other ETFs sometimes yields infinite families of new complex ETFs. (Received January 07, 2020)

1157-42-49 **Xiumin Du**, Department of Mathematics, 4176 Campus Drive - William E. Kirwan Hall, College Park, MD 20742, and **Ruixiang Zhang***, UW-Madison Department of Mathematics, Van Vleck Hall, 480 Lincoln Drive, Madison, WI 53703. *The pointwise convergence problem for the free Schrodinger equation.*

Carleson proposed a problem on a.e. convergence for free Schrodinger solutions as time goes to zero. Recently it got a sharp answer (up to the endpoint) in all dimensions. I will talk about the new result in dimensions $n+1$ for all $n>2$ and ideas behind it (joint work with Xiumin Du). (Received January 07, 2020)

1157-42-55 **Gennady Uraltsev*** (guraltsev@math.cornell.edu), Malott Hall - Dept of Mathematics, Cornell University, Ithaca, NY 14850. *Uniform Bounds for the Bilinear Hilbert Transform.*

The Bilinear Hilbert Transform is a prototypical modulation invariant multi-linear singular operator.

$$\mathrm{BHT}_\alpha(f_1,f_2)(x) := \int_{\mathbb{R}}\int_{\mathbb{R}} \widehat{f}_1(\xi_1)\widehat{f}_2(\xi_2)1_{[0,+\infty)}(\xi_1-\alpha\xi_2)e^{2\pi i(\xi_1+\xi_2)x}d\xi_1\xi_2$$

It arises in many contexts including Cauchy integrals along Lipschitz curves and in the study of Calderón commutators relevant for semi-linear non-constant-coefficient t ellyptic PDEs.. The bounds

$$\|\mathrm{BHT}_\alpha(f_1,f_2)(x)\|_{L^{p'_3}(\mathbb{R})} \leq C_{p_1,p_2,\alpha}\|f_1\|_{L^{p_1}(\mathbb{R})}\|f_2\|_{L^{p_2}(\mathbb{R})}$$

hold for any $p_{1,2,3}\in(1,+\infty)$ with $\sum_{i=1}^3 p_i^{-1}=1$.

A longstanding open problem is weather the above bounds holds with a uniform constant in the parameter α. The dyadic analog of this problem has been solved in Oberlin, Thiele '10.

We will give a general overview of the topic. We will then sketch how the framework of outer measure L^p spaces in time-scale-frequency space plays a crucial role in proving the result and how the problem gives useful insights into embeddings of functions into times-scale-frequency space. (Received January 09, 2020)

1157-42-58 **Alex Barron*** (aabarron@illinois.edu), **Burak Erdogan** and **Terence L.J. Harris**. *Decay of hyperbolic averages of the Fourier transform.*

We discuss upper and lower bounds for the decay of hyperbolic averages of the Fourier transform of measures. Techniques developed recently by Du and Zhang for parabolic averages play an important role in our argument, though some new ideas are needed to handle the negative curvature setting. (Received January 09, 2020)

1157-42-60 **D Maldague***, dmal@mit.edu. *Proof sketch of a generalized multilinear Kakeya inequality.* Preliminary report.

A generalized multilinear Kakeya inequality follows from a certain generalized Brascamp-Lieb inequality. We sketch the inductive proof. (Received January 10, 2020)

1157-42-65 **Larry Guth** and **Hong Wang***, hong.wang1991@gmail.com, and **Ruixiang Zhang**. *Incidence estimates with application to harmonic analysis.*

We are going to discuss an incidence estimate between points and tubes, with some applications to local smoothing for the wave equation. (Received January 11, 2020)

1157-42-76 **Joris Roos** (jroos@math.wisc.edu) and **Andreas Seeger*** (seeger@math.wisc.edu). *L^p improving bounds for circular maximal operators.*

For suitable functions $f \in L^p(\mathbb{R}^2)$ let $A_r f(x)$ be the mean of f over the circle of radius r centered at x. Given a set $E \subset [1,2]$ one aims for sharp $L^p \to L^q$ estimates for the maximal function $\sup_{r\in E} |A_r f(x)|$. We review results from a previous paper with Theresa Anderson, Kevin Hughes and Joris Roos regarding the corresponding problem for spherical maximal functions in higher dimensions and discuss the new features in two dimensions. The outcomes relate to various dimensional assumptions on E or subsets of E. We also report on some more recent results concerning the shape of the possible type set. (Received January 14, 2020)

1157-42-96 **Thomas Hill*** (hill2ts@mail.uc.edu), 2216 Ohio Ave Apt. 1, Cincinnati, OH 45219. *New Method for Dispersive Estimates of One-Dimensional Schrödinger Equations.*

We prove dispersive estimates for the Schrödinger equation with Hamiltonians $H = -\Delta + V$ on $\mathbb{R}$ using a new approach inspired by work in the three-dimensional setting. Instead of constructing the resolvent of H out of scattering data for the potential, we identify its properties by working inside a Wiener algebra of operator-valued functions. The approach allows us to handle potentials satisfying $(1+|x|)V \in L^1$ in all cases.

This is the largest known class of potentials that supports an $L^1 \to L^\infty$ dispersive estimate in both the resonant and non-resonant case. We replicate both of those results and prove an additional weighted dispersive estimate with $|t|^{-\frac{3}{2}}$ decay (in the non-resonant case) that previously required stronger decay conditions on V. (Received January 18, 2020)

1157-42-480 **Matthew Hirn*** (mhirn@msu.edu), Michigan State University, Dept. of Comp. Math, Science & Engineering, Dept. of Mathematics, East Lansing, MI 48824. *Invariant multiscale graph representations.*

Modeling data as a graph is ubiquitous, e.g., molecules, gene-gene interactions networks, and social networks. Even for data that does not have a network structure, it is often useful to model relationships between data points through a graph that encodes similarities between them. Data-driven graphs often have rich, multiscale patterns that must be extracted for downstream tasks, be it clustering, graph classification, community detection, or others. Graph neural networks (GNNs) have revolutionized the way in which we extract information from graphs, but numerous mathematical mysteries remain. In this talk, we introduce a model for GNNs called a geometric wavelet scattering transform (GWST), which computes a cascade of graph wavelet transforms and nonlinearities. We prove the GWST encodes a (locally) permutation invariant representation of a graph that is stable to graph perturbations. When the graph is a mesh sampled from a manifold, we prove the manifold counterpart of the GWST is (locally) invariant to isometric transformations of the manifold and stable to diffeomorphisms. We will also discuss nonlinearities that deform the graph, leading to a multiscale, iterative condensation process that gradually collapses the graph and results in a continuously hierarchical clustering. (Received February 03, 2020)

1157-42-496 **Shujie Kang** (kangsj@umd.edu), Department of Mathematics, College Park, MD 20742, and **Kasso Okoudjou*** (kasso@mit.edu), Department of Mathematics, MIT, 77 Massachusetts Ave, Cambridge, MA 02139. *Minimizing the p-frame potentials of finite Gabor systems.* Preliminary report.

Given $p \in (0,\infty]$, and integers N, d with $N \geq d \geq 2$, we consider the function

$$Z_{p,N,d}(g) = \sum_{k,\ell=1}^{N} |\langle g, M^k T^\ell g\rangle|^p$$

defined for $g \in \mathbb{C}^d$ and $\|g\| = 1$. In this talk, we report on preliminary results dealing with minimizing the function $Z_{p,N,d}$ over the unit sphere in $\mathbb{C}^d$. In particular, we focus on the case $N = d^2$ which is related to the Zauner conjecture. (Received February 03, 2020)

1157-42-549 **Brandon Hanson*** (brandon.w.hanson@gmail.com), 320 Southview Drive, Athens, GA 30605. *The Littlewood Problem for Multidimensional Sets.*

The Littlewood Problem is about lower bounds for the L^1 norm of a trigonometric polynomial with frequencies from a finite set A. While nearly best-possible estimates have been known since the resolution of the problem

in the early eighties, it remains open which sets A are extremal. We discuss results that show such an A must be appropriately low-dimensional. (Received February 04, 2020)

1157-42-615 **Shaoming Guo**, **Changkeun Oh**, **Joris Roos*** (jroos@math.wisc.edu), **Po-Lam Yung** and **Pavel Zorin-Kranich**. *Decoupling for two quadratic forms in three variables: a complete characterization.*

The talk will be about a recent work, where we prove decoupling inequalities for all degenerate surfaces of codimension two in $\mathbb{R}^5$ given by two quadratic forms in three variables. Together with previous work by Demeter, Guo, and Shi in the non-degenerate case, this provides a classification of decoupling inequalities for pairs of quadratic forms in three variables. (Received February 04, 2020)

1157-42-622 **Jameson Cahill***, jamesonc@nmsu.edu, and **Andres Contreras** and **Andres Contreras Hip**. *Group invariant representations.*

In image and audio signal classification, a major problem is to build stable representations that are invariant under rigid motions and, more generally, to small diffeomorphisms. Translation invariant representations of signals in C^n are of particular importance. The existence of such representations is intimately related to classical invariant theory, inverse problems in compressed sensing and deep learning. We construct low dimensional representations of signals that are invariant under finite unitary group actions, as a special case we establish the existence of low-dimensional set of measurements which separates the orbits of any cyclic group action, of which translation is one example. Furthermore our map is Lipschitz with respect to the natural metric on the space of orbits. Our construction is closely related to methods use in phase retrieval, so we will give an overview of these methods. (Received February 04, 2020)

1157-42-641 **Alejandra Gaitan***, 150 N. University Street, West Lafayette, IN 47907-2067. *Boundedness of the bilinear Hilbert transform associated to general polynomials.*

Let P be a generalized polynomial, i.e., an expression of the form $P(t) := \sum_{i=1}^{d} a_j t^{\alpha_j}$ with $\{a_j\}_{j=1}^{d}, \{\alpha_j\}_{j=1}^{d} \subset \mathbb{R}$ and $d \in \mathbb{N}$. Define the bilinear Hilbert transform along the curve $(t, -P(t))$ as

$$H_P(f,g)(x) := pv \int_{\mathbb{R}} f(x-t)g(x+P(t))\frac{dt}{t}.$$

In this talk, we address the $L^2 \times L^2 \to L^1$ boundedness of H_P.

V. Lie showed the boundedness of H_P for a class of curves that includes, in particular, any generalized (Laurent) real polynomial with no linear term. For general polynomials (containing linear term), the problem is harder, since it involves the results of the estimates for the classical bilinear Hilbert transform. (Received February 05, 2020)

43 ► Abstract harmonic analysis

1157-43-640 **Amineh Farzannia*** (afarzannia@dom.edu), Dominican University, River Forest, IL. *Harmonic Frames and Graphs.*

In this talk, we discuss the relationship between biangular and two-distance frames. We have already used partial difference sets and Divisible difference sets to construct such frames. In this the new paper, we use Graphs to characterize and classify two-distance Harmonic frames. We show the connection of two-distance Frames and their angles with a specific family of graphs called Cayley Graphs. Later in the work, we show the connection of Biangular frames and another family of graphs called divisible design graphs. Also, we show how such frames can be linked to another combinatorial object called Divisible designs. (Received February 05, 2020)

44 ► Integral transforms, operational calculus

1157-44-519 **Yang Zhang***, Department of Mathematics, 150 N. University Street, West Lafayette, IN 47907-2067. *Recovery of singularities for the weighted cone transform appearing in the Compton camera imaging.*

We study the weighted cone transform I_κ of distributions with compact support in a domain M of $\mathbb{R}^3$, over cone surfaces whose vertexes are located on a smooth surface away from M and opening angles are limited to an open interval of $(0, \pi/2)$. This transform models data are obtained by a Compton camera with attenuation and a realistic angle of view. We show that (Received February 03, 2020)

1157-44-609 **Siamak Rabieniaharatbar*** (srabieni@purdue.edu), 102 N Salisaboury St, West Lafayette, IN 47906. *Inverse Problems and Dynamical X-ray Tomography.*

Mathematical Tomography is an Inverse Problem which aims to find the internal structure of a non-transparent object by sending Electromagnetic waves of various frequencies or Acoustic waves.

To motivate our work, let A be a 2-dimensional dynamic operator integrating over a family of level curves in X. We investigate the operator A when the object changes between the measurement over time. We use analytic microlocal analysis to determine which singularities can be recovered by the data-set. Our results show that not all singularities can be recovered as the object moves with a speed lower than the X-ray source. We establish stability estimates and prove that the injectivity and stability are of a generic set if the dynamic operator satisfies the visibility, no conjugate points, and local Bolker conditions. We also illustrate the implementation of this result fan-beam geometry. (Received February 04, 2020)

49 ► Calculus of variations and optimal control; optimization

1157-49-77 **Gabor Pataki*** (gabor@unc.edu), NC. *Combinatorial characterizations in semidefinite programming duality: how elementary row operations help.*

To solve semidefinite programs (SDPs), we rely on a natural dual problem. However, SDP duality is quite a messy affair: for example, SDPs may not attain their optimal value, and their optimal value may differ from that of their dual.

I will show that many of these pathologies can be understood using a surprisingly simple tool: we can transform SDPs to normal forms that makes some of the pathologies trivial to recognize. The transformations mostly use elementary row operations from Gaussian elimination. The normal forms have computational uses, for example, in several cases they help us recognize infeasibility.

The talk assumes only knowledge of elementary linear algebra. (Received January 14, 2020)

1157-49-114 **Chun Liu*** (cliu124@iit.edu) and **Yiwei Wang** (ywang487@iit.edu). *Energetic Variational Approaches in Dynamics arising from Biology.* Preliminary report.

(with Yiwei Wang and Pei Liu) In this talk, I will discuss several dynamical systems arising from the modeling of molecular biology systems. In particular, these are the systems involving general diffusion and chemical reactions. We will employ the general energetic variational approaches to capture the coupling and competition of various mechanisms. Focus will be on several analytical and numerical issues and challenges. (Received January 22, 2020)

1157-49-151 **Nicolas Garcia Trillos*** (garciatrillo@wisc.edu), **Ryan Murray** and **Matthew Thorpe**. *From clustering with graph cuts to isoperimetric inequalities: quantitative convergence rates of Cheeger cuts on data clouds.*

Graph cuts have been studied for decades in the mathematics and computer science communities, and in modern applications in machine learning have been used to formulate optimization problems for data clustering. A canonical example with historical motivation is the so called Cheeger cut problem. This problem is on the one hand intuitively motivated, but on the other, is highly non-convex with a pessimistic NP hard label stamped on it (at least in a worst case scenario setting). Despite this, in the past decade or so, several algorithmic improvements made the minimization of Cheeger cuts more feasible, and at the same time there was a renewed interest in studying statistical properties of Cheeger cuts.

The purpose of this talk is to present new theoretical results where, for the first time, high probability converge rates of Cheeger cuts of proximity graphs over data clouds are deduced. These quantitative convergence rates are obtained by building bridges between the original clustering problem and another field within the mathematical analysis community that has seen enormous advancements in the past few years: quantitative isoperimetric inequalities. (Received January 26, 2020)

1157-49-196 **Qinfeng Li*** (qinfeng.li@utsa.edu), 6515 W Hausman Road, San antonio, TX 78249, and **Guiqiang Chen**, **Changfeng Gui**, **Yeyao Hu** and **Monica Torres**. *Interior approximation of sets of finite perimeter and applications to traces and extensions of divergence measure fields and BV functions.*

In this talk, I will present new smooth interior approximation results for sets of finite perimeter. As an application, we can generalize previous results on extensions and traces of BV functions and divergence measure fields. This

talk combines several joint works with Guiqiang Chen, Changfeng Gui, Yeyao Hu and Monica Torres. (Received January 28, 2020)

1157-49-203 **Ryan Murray*** (rwmurray@ncsu.edu). *Stein variational descent as a gradient flow: towards a rigorous theory for kernel selection.*

Stein variational gradient descent is a recently developed algorithm for efficiently generating posterior samples in Bayesian statistics. It has been formally derived as a gradient flow, in a modified Wasserstein space, which replaces the L^2 type tangent spaces with reproducing kernel Hilbert spaces. This talk will discuss several directions for analyzing these problems, with a focus on the problem of kernel design. Rigorous analytical results will be presented for different classes of kernels. Some numerical results based upon linearized analysis will also be presented. (Received January 28, 2020)

1157-49-331 **Michael M Dotzel*** (mackerel@mit.edu), 143 Albany Street, Apt 013, Cambridge, MA 02139, and **Carlo Morpurgo** (morpurgoc@missouri.edu). *Minimization of trace functionals on graphs with symmetries.*

Given an undirected graph whose edges are weighted by a symmetric transition matrix P, we discuss the problem of minimizing Tr[F (P)] over all such P, for a given function F. Does a minimizer exist? If so, can it be characterized? We provide an answer to these questions in terms of certain symmetries of the graph. In particular we give a sufficient condition guaranteeing that the entries of the minimizer are equal. This optimization question lends itself to applications in chemistry and information theory. We will give such an example pertaining to the computation of bond order in molecules minimizing a graph energy function, as well as an application to finding Laplacians representing information networks that maximize the von Neumann entropy. (Received February 01, 2020)

1157-49-392 **Stephan Sturm** and **Jiaxuan Ye***, 100 Institute Road, WPI - Stratton Hall, Worcester, MA 01609. *Intertemporal Consumption Optimization for Fund Managers under Taxation.* Preliminary report.

We study an intertemporal consumption optimization under taxation for fund managers who are forced to invest their earnings back to the fund. Optimal strategy is characterize with calculus of variation. If managers' personal discounting factor is smaller than the management fee rate, the best strategy is to use the unique solution to Euler-Lagrange equation in infinite horizon. Otherwise, managers consume according to this solution until their accounts returns zero, and then maintain their accounts zero after. (Received February 03, 2020)

1157-49-514 **Wing Fung Chong*** (wfchong@illinois.edu), Department of Mathematics, 235 Illini Hall, MC-382, 1409 W. Green Street, Urbana, IL 61801. *Pricing and Hedging Equity-Linked Life Insurance Contracts Beyond the Classical Paradigm: The Principle of Equivalent Forward Preferences.*

By applying the principle of equivalent forward preferences, this talk revisits the pricing and hedging problems for equity-linked life insurance contracts. The equity-linked contingent claim depends on, not only the future lifetime of the policyholder, but also the performance of the reference portfolio in the financial market for the segregated account of the policyholder. For both zero volatility and non-zero volatility forward utility preferences, prices and hedging strategies of the contract are represented by solutions of random horizon backward stochastic differential equations. Numerical illustration is provided for the zero volatility case. The derived prices and hedging strategies are also compared with classical results in the literature. (Received February 03, 2020)

1157-49-530 **Ian Tobasco*** (itobasco@uic.edu), University of Illinois at Chicago, 851 S. Morgan Street (M/C 249), Chicago, IL 60607. *Convex computation of extremal trajectories in dynamical systems.*

The problem of finding optimal trajectories in dynamical systems will be considered. A trajectory is considered optimal if it maximizes the long-time average of a given quantity defined on phase space, the selection of which picks out one or another trajectory in a non-trivial way. Examples include Rayleigh–Bénard convection where the average heat transport is thought of as being optimized or, as a much simpler model to get started, the Lorenz equations where a given polynomial in the state is to be optimized on average in time. We discuss a method for locating optimal trajectories without solving the underlying dynamics, which makes use of a duality between invariant measures and certain "auxiliary functionals" playing the role of Lyapunov functionals here. Recent progress will be highlighted showing the relation between the well-known "background method" to prove bounds on transport and the general auxiliary functional method. We close with questions regarding the sharpness of this method for infinite-dimensional systems whose well-posedness is yet unknown. (Received February 04, 2020)

51 ► Geometry

1157-51-21 **Meera Sitharam*** (sitharam@cise.ufl.edu), University of Florida PO box 11-6120, CSE Bldg, Gainesville, FL 32605. *Convex Cayley Parameterization and Applications.*

Configuration spaces of *Linkages* (graphs whose edges represent distance constraints) show up in a diverse scenarios and are notoriously difficult to explore, analyze or compute with, due to their topological and geometric complexity.

Our recent work has given explicit forbidden-minor characterizations of graphs for which - using non-edge distances or *Cayley* parameters – this configuration space (up to 3 dimensions) can be nearly bijectively mapped to a convex set of Cayley configurations, with an efficient pre-image computation to recover the actual configuration. The ideas generalize when distances are non-Euclidean L_p, and polyhedral norms. The existence of a convex, Cayley configuration space is directly related to *graph d-flattenability* (old name: *d*-realizability) for any dimension *d*.

Cayley convexification theory has extended the notion of DR-plans (recursive rigid decompositions) to give fast realization algorithms for a much larger class of distance constraint graphs; algorithms for a complete analysis including path finding in configuration spaces of many mechanical linkages and for atlasing and analysis of energetics, entropy and kinetics of molecular configuration spaces. (Received November 27, 2019)

1157-51-23 **Alexander Heaton***, Inselstraße 22, 04103 Leipzig, Saxony, Germany. *Deciding local rigidity by numerical algebraic geometry.*

A well-known combinatorial algorithm can decide generic rigidity of a graph embedded in the plane by determining if the graph is of Laman type. Methods from matroid theory have been used to prove other interesting results, again under the assumption of generic configurations. There are some exceptional results which apply also to non-generic configurations, for example Connelly showed that any convex triangulated polytope in dimension 3 is second-order rigid, and hence locally rigid. If we allow edges to be bars, cables, or struts, the problem becomes more difficult still. Given a not-necessarily-generic embedded graph, can we decide its local rigidity? If the rank of the rigidity matrix is $nd - \binom{d+1}{2}$, then the graph is infinitesimally rigid, and therefore locally rigid as well. But what if there are additional infinitesimal mechanisms? We will present an approach to solving this problem using numerical algebraic geometry, dealing directly with the semi-algebraic set of configurations satisfying the member constraints, which take the form of a system of polynomial equations and inequalities. (Received November 29, 2019)

1157-51-181 **Jun Zhang*** (jun.zhang.3@umontreal.ca), Pavillon André-Aisenstadt, 2920 Chemin de la tour, bureau 6151, Montreal, Quebec H3T 1J8, Canada. *Relative growth rate and contact Banach-Mazur distance.*

Recent years have seen a significant improvement in the quantitative methods of comparing domains in symplectic geometry, for instance, the invention of symplectic Banach-Mazur distance between star-shaped domains of a Liouville manifold. In this talk, I will present an analogue study in the contact geometry set-ups. Explicitly, I will consider contact domains in two types of contact manifolds: the contactization of a Liouville manifold $W \times S^1$ and a closed Liouville-fillable contact manifold M. To quantitatively compare the contact domains, I will introduce a non-linear version of Banach-Mazur distance called contact Banach-Mazur distance. This is partially motivated by the relative growth rate introduced and studied by Eliashberg and Polterovich on the contactomorphism groups. Moreover, I will present some large-scale geometric results in terms of this distance, which are implied by the contact shape invariant. This talk is based on joint work with Daniel Rosen. (Received January 27, 2020)

1157-51-214 **Mireille Boutin** (mboutin@purdue.edu) and **Gregor Kemper*** (kemper@ma.tum.de). *A Drone Can Hear the Shape of a Room.*

In previous work we addressed the question of whether a point configuration can be reconstructed from the unlabeled set of mutual distances between the points. Our approach was based on algebraic relations between the distances.

Recently, we used such relations again to work on the following question: Can a drone detect the walls of a room from hearing their echoes with various microphones mounted on the drone? The problem is how to work out which of the echoes heard by the microphones come from the same wall. We propose a matching algorithm based on relations. Our proof that this works with probability one leads to problems about ideals in polynomial rings and requires massive computer algebra computations. (Received January 29, 2020)

1157-51-435 **Mutlu Akar*** (makar@yildiz.edu.tr), Yildiz Technical University, College of Arts, and Sciences, Department of Mathematics, Esenler, 34210 Istanbul, Turkey, and **Nikolay Metodiev Sirakov** (nikolay.sirakov@tamuc.edu), Texas A&M University-Commerce, Department of Mathematics, P.O. Box 3011, Commerce, TX 75429. *Multiplication of 2-vectors and 3-vectors in Clifford Algebra $C\ell_{5,0}$.*

The classification with Clifford Support Vector Machines (CSVM) is gaining popularity, in recent years, among the Computer Vision society. The main component of a CSVM is the Clifford kernel. In the paper [1] we developed a classification approach in the subspaces of the Clifford Algebra $C\ell_{5,0}$. In the study [2] the authors provided general formulas for multiplication of multivectors, but no guidelines are given on their detailed implementation. In the present study, we provide detailed formulas for Clifford multiplication between 2-vectors and multiplication between 3-vectors in $C\ell_{5,0}$. The new formulas lie on the basis of novel CSVM kernels we do develop for skin lesion classification. The 2-vectors, 3- vectors Clifford multiplication formulas are validated with a Matlab code, which uses 5D vectors describing skin lesions.

[1] M. Akar, N. M. Sirakov, "Support Vector Machine Skin Lesion Classification in Clifford Algebra Subspaces", Applications of Mathematics, Vol. 64, Issue 5, (2019), 581-598.

[2] E. J. Bayro-Corrochano, N. Arana-Daniel, "Clifford Support Vector Machines for Classification, Regression, and Recurrence", IEEE Trans. Neural Networks 21 (2010), 1731–1746. (Received February 03, 2020)

1157-51-461 **Ling Xu*** (xu511@purdue.edu). *Superrigidity of mapping class groups with finite volume quotient.*

Teichmüller space $\mathcal{T}(S)$ is the space of structures on a Riemann surface S up to isotopy, and the mapping class group $MCG(S)$ is the group of isotopy classes of orientation-preserving homeomorphisms of S. Teichmüller space is nonpositively curved when equipped with the Weil-Petersson metric. In this paper, we first prove the existence of equivariant harmonic maps from a symmetric space into the Weil-Petersson completion of a Teichmüller. Then give a uniform proof of the superrigidity theorem of the mapping class group by using harmonic map approach and the Bochner formula. (Received February 03, 2020)

1157-51-475 **Chris Gartland*** (cgartla2@illinois.edu). *Markov Convexity of Carnot Groups.* Preliminary report.

The Ribe program is the research program concerned with generalizing local properties of Banach spaces to biLipschitz invariant properties of metric spaces. Among such generalizations that have been found is the notion of Markov p-convexity, proven by Mendel-Naor to generalize uniform p-convexity. One of the first important spaces for which this invariant has been calculated is the Heisenberg group, proven by Li to be Markov p-convex for every p at least 4 and not Markov p-convex for any p less than 4. In this talk, we'll explain how to extend this result to filiform groups - a class of Carnot groups generalizing the Heisenberg group - and discuss some applications. (Received February 03, 2020)

1157-51-527 **Bahar Acu*** (baharacu@northwestern.edu), Evanston, IL, and **Orsola Capovilla-Searle**, **Agnes Gadbled**, **Aleksandra Marinkovic**, **Emmy Murphy**, **Laura Starkston** and **Angela Wu**. *Weinstein Handlebodies for Complements of Smoothed Toric Divisors.*

Toric manifolds together with their divisors are of great importance in four-dimensional symplectic geometry. These divisors initially have "normal crossing singularities" and are important classes of symplectic divisors. It is, moreover, known that the complement of the normal crossing divisor is symplectically T^*T^2. Any of the normal crossing singularities can be smoothed so the divisor has fewer singularities. This smoothing operation changes the topology of the complement. In this work, the fruit of the meeting at the 2019 Research Collaboration Conference for Women in Symplectic and Contact Geometry and Topology, we develop a method to draw a Weinstein handlebody decomposition for the complement of any smoothed toric divisor. (Received February 04, 2020)

1157-51-556 **R Abhishek Shankar*** (shanka18@purdue.edu), **Mohit Singh** (mohit.mnnit@gmail.com) and **Byunghoo Jung** (jungb@purdue.edu). *Personal Area Position Tracking and its Applications.*

With the advent of fast computation devices, there has been an increase in the demand for a fast, portable and precise personal area position tracking system. The applications for such tracking systems include body tracking for sports training and medical purposes, gesture recognition, gaming, home automation and robotics. The commonly used technologies are camera or infrared based. They have line-of-sight and occlusion problems, and tend to be power hungry, compute intensive and expensive. The system proposed here achieves 3D position and orientation tracking, using AC magnetic fields and inertial measurement units (IMU). The system consists

of a single-axis transmitter coil with respect to which a tri-axis receiver coil is tracked, the IMUs are used for finding the orientation. The usage of a single-axis transmitter coil has advantages in terms of computation complexity, latency, form factor and power consumption as compared to traditional magnetic tracking systems. Further improvements in latency and accuracy are achieved by modifying and optimizing the existing Madgwick and Kalman filters used for sensor fusion. The system offers a 400Hz update rate with 5ms of latency. The system has an accuracy of 1cm/1deg and a precision of 0.1cm/0.1deg at 1m from the transmitter coil. (Received February 04, 2020)

1157-51-573 **Hang Huang*** (hhuang@math.tamu.edu), **Mateusz Michalek** and **Emanuele Ventura**. *Vanishing Hessian and Wild Polynomials.*

Notions of ranks and border rank abounds in the literature. Polynomials with vanishing hessian and their classification is also a classical problem. Motivated by an observation of Ottaviani, we show that polynomials with vanishing Hessian and of minimal border rank are wild, i.e. their smoothable rank is strictly larger than their border rank. The main tool we are using is the rencent work of Buczyńska and Buzyński relating the border rank of polynomials and tensors to multigraded Hilbert scheme. From here, we found two infinite series of wild polynomials and we will try to describe their border varieties of sums of powers, which is an analogue of the variety of sums of powers. (Received February 04, 2020)

1157-51-624 **Mireille Boutin*** (mboutin@purdue.edu), 465 Northwestern Avenue, West Lafayette, IN 47907, and **Gregor Kemper**. *On reconstructing a point configuration from noisy unlabeled pairwise distances.* Preliminary report.

Given is a generic point configuration in $p_1, p_2, \ldots, p_n \in \mathbb{R}^n$. It has been shown in previous work that the point configuration can be uniquely reconstructed, up to a rigid motion, from the multiset of its unlabeled pairwise distances. In this talk, we will discuss the challenge of extending this result to the case where the pairwise distance measurements are noisy, and include some results for the case $n = 2$. (Received February 04, 2020)

1157-51-648 **R Abhishek Shankar*** (shanka18@purdue.edu), **Mohit Singh** (mohit.mnnit@gmail.com) and **Byunghoo Jung** (jungb@purdue.edu). *Personal Area Position Tracking and its Applications.*

With the advent of fast computation devices, there has been an increase in the demand for a fast, portable and precise personal area position tracking system. The applications for such tracking systems include body tracking for sports training and medical purposes, gesture recognition, gaming, home automation and robotics. The commonly used technologies are camera or infrared based. They have line-of-sight and occlusion problems, and tend to be power hungry, compute intensive and expensive. The system proposed here achieves 3D position and orientation tracking, using AC magnetic fields and inertial measurement units (IMU). The system consists of a single-axis transmitter coil with respect to which a tri-axis receiver coil is tracked, the IMUs are used for finding the orientation. The usage of a single-axis transmitter coil has advantages in terms of computation complexity, latency, form factor and power consumption as compared to traditional magnetic tracking systems. Further improvements in latency and accuracy are achieved by modifying and optimizing the existing Madgwick and Kalman filters used for sensor fusion. The system offers a 400Hz update rate with 5ms of latency. The system has an accuracy of 1cm/1deg and a precision of 0.1cm/0.1deg at 1m from the transmitter coil. (Received February 05, 2020)

52 ► Convex and discrete geometry

1157-52-22 **Daniel Irving Bernstein*** (dibernst@mit.edu), 77 Massachusetts Avenue, E17-481D, Cambridge, MA 02139, and **Robert Krone**. *The tropical Cayley-Menger variety.*

Varieties that arise in the algebraic study of matrix completion and rigidity theory come embedded in a vector space whose coordinates are indexed by some graph. A recurring problem is to find combinatorial descriptions, in terms of these graphs, of the algebraic matroids underlying these varieties. Tropical geometry can be used to solve such problems. In this talk, I will show that the tropicalization of the Cayley-Menger variety of points in the plane has a simplicial complex structure that can be described in terms of rooted trees. Then, I will show how one can use this to obtain a new proof of Laman's theorem. (Received November 27, 2019)

1157-52-205 **P Boyvalenkov** and **P Dragnev*** (dragnevp@pfw.edu), Department of Mathematical Sciences, Purdue University Fort Wayne, 2101 E Coliseum Blvd., Fort Wayne, IN 46825, and **D Hardin**, **E Saff** and **M Stoyanova**. *Universal upper bounds for energies of spherical codes of given cardinality and separation.*

We introduce a linear programming framework for obtaining upper bounds for the potential energy of spherical codes of fi xed cardinality and minimum distance. Using Hermite interpolation we construct polynomials to derive corresponding bounds. These bounds are universal in the sense that they are valid for all absolutely monotone potential functions and the required interpolation nodes do not depend on the potentials. (Received January 28, 2020)

1157-52-338 **Oleg R Musin*** (oleg.musin@utrgv.edu), 1 W University blvd, SMSS UTRGV, Brownsville, TX 78520. *Optimal spherical configurations in low dimensions.*

In this talk we will discuss Tammes', kissing number and several other sphere packings problems in dimensions three and four. In dimension 4 the uniqueness of the maximal kissing arrangement is conjectured to be the 24-cell but not yet proven. Recently, we obtain new SDP bounds for the distance distribution and distance graphs of spherical codes. We consider extensions of these bounds for optimal spherical configurations. (Received February 02, 2020)

1157-52-467 **Joseph W. Iverson*** (jwi@iastate.edu), **Gary Greaves**, **John Jasper** and **Dustin G. Mixon**. *Equiangular tight frames over finite fields.* Preliminary report.

A basic problem in projective packings is to arrange a given number of lines through the origin of $\mathbb{C}^d$ without creating sharp angles. Using the coherence as a metric, optimal line packings that achieve equality in the Welch bound enjoy a simpler characterization as equiangular tight frames (ETFs). General existence of complex ETFs is a major open problem with implications in quantum information theory and algebraic number theory. Here, part of what makes the complex case so difficult is that the signature matrix can take entries anywhere in the torus, corresponding to an infinite search space. In the hope of creating a more tractable version of this problem, we introduce the notion of equiangular tight frames over finite fields. We explain how finite fields capture the complexity of general ETFs, and give some surprising results concerning their existence. (Received February 03, 2020)

1157-52-488 **Josiah Park*** (j.park@gatech.edu), 686 Cherry St NW, Atlanta, GA 30332. *Optimal measures for three-point energies and semidefinite programming.* Preliminary report.

Given a potential function of three vector arguments, $f(x, y, z)$, which is $O(n)$-invariant, $f(Qx, Qy, Qz) = f(x, y, z)$ for all Q orthogonal, we use semidefinite programming bounds to determine optimizing probability measures for interaction energies of the form $\int\int\int f(x, y, z)d\mu(x)d\mu(y)d\mu(z)$ over the sphere. This approach builds on previous use of such bounds in the discrete setting by Bachoc-Vallentin, Cohn-Woo, and Musin, and is successful for kernels which can be shown to have expansions in a particular basis, for instance certain symmetric polynomials in inner products $u = \langle x, y\rangle$, $v = \langle y, z\rangle$, and $t = \langle z, x\rangle$. For other kernels we pose conjectures on the behavior of optimizers, partially inferred through numerical studies. (Received February 03, 2020)

1157-52-509 **Douglas Hardin** (doug.hardin@vanderbilt.edu), **Edward Saff** (edward.b.saff@vanderbilt.edu) and **Oleksandr Vlasiuk*** (ovlasiuk@fsu.edu). *Asymptotic properties of the minimizers of short-range scale-invariant interaction energies.* Preliminary report.

For a collection of N distinct points ω_N in a compact set $\Omega \subset \mathbb{R}^p$, define the *truncated Riesz energy* as

$$E^k(\omega_N) = \sum_{i=1}^{N} \sum_{j \in I_{i,k}} \|x_i - x_j\|^{-s}, \qquad s > 0,$$

where $I_{i,k}$ denotes the set of indices of the k nearest neighbors to the point x_i in ω_N. We discuss the asymptotic properties of the configurations ω_N^*, minimizing E^k, and obtain the asymptotics of the minimal energy and the limiting weak* distribution. We then show how to augment the above energy functional with a weight and an external field term, allowing to recover a distribution with the given density with respect to the Hausdorff measure on Ω. (Received February 03, 2020)

1157-52-515 **Ciprian S. Borcea** (borcea@rider.edu) and **Ileana Streinu*** (istreinu@smith.edu), Computer Science Department, Northampton, MA 01063. *Periodic frameworks and auxetic bracing.*

The notion of auxetic behavior emerged in materials science and refers to lateral widening upon stretching. A natural context for a rigorous mathematical model is provided by the deformation theory of periodic bar-and-joint frameworks. Adopting this geometrical model, we present a general solution for the following problem. If a periodic framework can deform with several degrees of freedom and allows auxetic deformations, how can one constrain the structure (by additional bars and joints) to follow (locally) a one-parameter auxetic deformation? (Received February 03, 2020)

1157-52-541 **Dennis G Collins*** (d_collins_pr@hotmail.com), 1519 S. State Rd, 119, Apt. 2, Winamac, IN 46996. *Convexity of Symmetry Measures for Discrete n-point Patterns.* Preliminary report.

Following up on the Author's 2011 Symmetry patent, it is possible to measure the maximum symmetry MAXSYM of an n-point pattern as (n+1) n (n-1) (n-2)/8 and by the second derivative test for convex functions, to show this function is convex for n >=2. On the other hand, 2-dimensional grid growth is generally not convex. For example the SYM of a 5x5 =25 point symmetric grid is 4242 versus a 24-point grid with corner point removed is 3736 and a 26-point grid with specific extra point added is 4698. The midpoint of the straight line connecting (24,3736) and (26,4698) is (25,4217) where $4217 < 4242$ rather than greater than 4242 as would be required for convex growth. Thus to obtain maximum symmetry growth other methods of organization beyond 2-dim grid growth are required, as possibly related to punctuated evolution. Other examples are discussed (Received February 04, 2020)

53 ► Differential geometry

1157-53-81 **Tamas Darvas*** (tdarvas@umd.edu), **Eleonora Di Nezza** and **Chinh H Lu**. *The metric geometry of singularity types.*

Let X be a compact Kähler manifold. Given a big cohomology class, there is a natural equivalence relation on the space of quasi-psh functions giving rise to the space of singularity types. We introduce a natural pseudometric on this space that is non-degenerate on the space of model singularity types and whose atoms are exactly the relative full mass classes. In the presence of positive mass we show that this metric space is complete. As applications, we show that solutions to a family of complex Monge-Ampère equations with varying singularity type converge as governed by the resulting topology. (Received January 15, 2020)

1157-53-88 **Ben Weinkove***, 2033 Sheridan Road, Evanston, IL 60208. *PDEs on compact complex surfaces.*

I will give a survey of some recent work on elliptic and parabolic nonlinear PDEs on compact (non-Kahler) complex surfaces, and indicate possible directions for future research. (Received January 16, 2020)

1157-53-89 **Ivan Contreras*** (icontreraspalacios@amherst.edu), 31 Quadrangle Drive, Amherst, MA 01002, and **Nicolas Martinez Alba**. *Poly-Symplectic Structures and the AKSZ Formalism.*

We extend the Alexandrov-Kontsevich-Schwarz-Zaboronsky (AKSZ) formulation of the Poisson sigma model to more general target spaces: graded manifolds equipped with a poly-Poisson structure. We introduce the general theory of graded geometry for poly-symplectic and poly-Poisson structures, that allows us to prove a Schwarz-type theorem and a transgression result for graded poly-symplectic structures. (Received January 16, 2020)

1157-53-123 **Richard W Montgomery*** (rmont@ucsc.edu), Mathematics, UC Santa Cruz, 1156 High Street, Santa Cruz, CA 95064-1058. *Bicycling, elastica, and metric lines for SE(2).* Preliminary report.

$SE(2)$, the group of rigid motions of the plane, admits a canonical left-invariant subRiemannian metric, up to scale. It also comes with two equivariant projections onto the plane, which we call the back and front projections. The subRiemannian geodesics for $SE(2)$ had been previously investigated in the context of vision, taking the perspective of the back projection. But the front projection is a submetry and the back projection is not. (The back projection's fibers are tangent to the underlying contact distribution.) We use 'bicycling mathematics' as developed in arxiv: to show that the front projection of any geodesic is an Euler elastica, and tha, among all these elastica, only the straight lines and Euler kinks correspond to metric lines, where a metric line is a globally

minimizing geodesic whose domain is the entire real line. The length of the bicycle frame turns out to be the scale. This is joint work with Andrei Ardentov, Gil Bor, Enrico LeDonne and Yuri Sachkov. (Received January 22, 2020)

1157-53-169 **Ailana Fraser*** (afraser@math.ubc.ca), University of British Columbia, Vancouver, BC V6T 1Z2, Canada, and **Pam Sargent** and **Richard Schoen**. *Some results on higher eigenvalue optimization.*

We will discuss some recent results concerning the optimization higher eigenvalues of manifolds with boundary, both in two and higher dimensional cases. This talk is based on joint work with P. Sargent and R. Schoen. (Received January 26, 2020)

1157-53-176 **Gabor Szekelyhidi*** (gszekely@nd.edu), 255 Hurley Hall, Department of Mathematics, Notre Dame, IN 46556. *Calabi-Yau metrics on $\mathbb{C}^n$.*

I will discuss the existence and uniqueness of certain complete Calabi-Yau metrics on $\mathbb{C}^n$, with maximal volume growth. (Received January 27, 2020)

1157-53-187 **Kuang-Ru Wu*** (wu739@purdue.edu), Department of Mathematics, Purdue University, 150 N. University Street, West Lafayette, IN 47907. *A Wess–Zumino–Witten type equation in the space of Kähler potentials in terms of Hermitian–Yang–Mills metrics.*

We prove that the solution of a Wess–Zumino–Witten type equation from a domain D in $\mathbb{C}^m$ to the space of Kähler potentials can be approximated uniformly by Hermitian–Yang–Mills metrics on certain vector bundles. The key is a new version of Berndtsson's theorem on the positivity of direct image bundles. (Received January 27, 2020)

1157-53-240 **Tian-Jun Li**, **Cheuk Yu Mak** and **Jie Min*** (minxx127@umn.edu), Minneapolis, MN 55414. *Symplectic log Calabi-Yau pairs - contact aspects.*

We study the rigidity of log Calabi-Yau surfaces and its implications on symplectic fillings. To be precise, we classify which cycle of sphere can be realized by log Calabi-Yau pairs. As an application, we determine the (Stein) symplectic fillability of contact torus bundles with contact structure induced as the boundary of symplectic caps. (Received January 30, 2020)

1157-53-260 **Daniel Alvarez-Gavela*** (da21@princeton.edu). *Hamiltonian egg-beater maps.*

Following Polterovich and Shelukhin, we consider Hamiltonian egg beater maps on a symplectic surface of genus at least four. These maps can be approximated by compositions of piecewise shears on a union of cylinders, i.e. products of matrices. This makes some aspects of their Floer theory explicitly computable. By computing the asymptotic growth of a certain boundary depth, we show that the free group on two generators embeds into any asymptotic cone of the group of Hamiltonian diffeomorphisms. The result stabilizes to products with symplectically aspherical manifolds. Joint with Victoria Kaminker, Asaf Kislev, Konstantin Kliakhandler, Andrei Pavlichenko, Lorenzo Rigolli, Daniel Rosen, Ood Shabtai, Bret Stevenson and Jun Zhang. (Received January 30, 2020)

1157-53-339 **Nicholas McCleerey*** (njm2@math.northwestern.edu) and **Jianchun Chu**. *$C^{1,1}$ Regularity of Geodesics of Singular Kähler Metrics.*

We study the regularity of weak geodesics of plurisubharmonic functions; in particular, we focus on the case when the functions are singular. This requires us to develop new estimates to control the behavior of the geodesic at singular points on the boundary. We then show how these can be applied to prove $C^{1,1}$-regularity of geodesics of Kähler metrics on a possibly singular Kähler variety. This is joint work with Jianchun Chu. (Received February 02, 2020)

1157-53-370 **Richard K Hind*** (hind.1@nd.edu), Department of Mathematics, University of Notre Dame, Notre Dame, IN 46556. *Packing and squeezing Lagrangian tori.*

Starting with a symplectic manifold and an arrangement of Lagrangian submanifolds, like the integral product tori in Euclidean space, one can ask about the symplectic size of the complement. The talk will describe some constructions of balls and Lagrangian tori which show the size is larger than expected. This is based on joint work with Ely Kerman. (Received February 02, 2020)

1157-53-381 **Daniel R. Irvine*** (dirvine@umich.edu), 5080 East Hall, 530 Church St, Ann Arbor, MI 48109. *Low-dimensional versus high-dimensional symplectic geometry.*

It is well known that low-dimensional differential topology happens in dimensions up to and including dimension 4. This low-versus-high dimensional dichotomy is usually attributed to a delicate computation in the proof of the

h-cobordism theorem, because high-dimensional manifolds exhibit a degree of flexibility. Symplectic manifolds, on the other hand, exhibit an interesting mixture of rigidity and flexibility, and they are strictly even-dimensional. Symplectic geometry also has a low-versus-high dimensional dichotomy; the low dimensions are 2 and 4. This talk will discuss the computational problems that make high-dimensional symplectic geometry difficult. (Received February 02, 2020)

1157-53-443 **Kyler Bryce Siegel*** (kyler@math.columbia.edu), 435 West 119th Street, Apt 7B, New York, NY 10027. *Computing higher symplectic capacities.*

I will describe a purely algebraic framework which computes "higher symplectic capacities" for ellipsoids and more generally convex toric domains. With the aid of a computer, these give many new obstructions to stabilized symplectic embeddings. I will also mention some open questions about the behavior of these capacities. (Received February 03, 2020)

1157-53-490 **Facundo Memoli***, Department of Mathematics, The Ohio State University, Columbus, OH 43210. *Some rigidity questions related to the local and global distance distributions.*

We will describe some recent work on the characterization of curves, surfaces, and metric graphs via distance distributions. This includes (1) some examples of pairs of smooth planar curves and pairs of smooth surfaces which are not discriminated by the global distribution of distances, and (2) some positive results for metric graphs under local distance distribution. (Received February 03, 2020)

1157-53-493 **Olguta Buse**, School of Mathematics, IUPUI, Indianapolis, IN 46202, and **Jun Li*** (lijungeo@umich.edu), 530 Church street, Ann Arbor, MI 48109. *On $Symp(X, w)$ of non-minimal ruled surfaces.*

We prove the stability of $Symp(X, w)$ for one point blowup of ruled surfaces. This allows us to describe the topological limit of $Symp(X, w)$ as the area ratio of the base and fiber is large. (Received February 03, 2020)

1157-53-512 **Dan Cristofaro-Gardiner*** (dcristof@ucsc.edu), Santa Cruz, CA 95060, and **Vincent Humiliere** and **Sobhan Seyfaddini**. *Proof of the simplicity conjecture.*

I will explain recent joint work proving that the group of compactly supported area preserving homeomorphisms of the two-disc is not a simple group; this answers the "Simplicity Conjecture" in the affirmative. Our proof uses new spectral invariants, defined via periodic Floer homology, that I will introduce: these recover the Calabi invariant of monotone twists. (Received February 03, 2020)

1157-53-524 **Bahar Acu*** (baharacu@northwestern.edu), Evanston, IL. *Planarity in higher-dimensional contact manifolds.*

Planar contact manifolds, those that correspond to an open book decomposition with genus zero pages, have been intensively studied to understand several aspects of 3-dimensional contact topology. In this talk, we present a higher-dimensional notion of planarity, iterated planarity, and provide several generalizations of results for planar contact manifolds, to higher dimensions. This is partly joint work with A. Moreno. (Received February 03, 2020)

1157-53-588 **Daniel R. Irvine*** (dirvine@umich.edu), 5080 East Hall, 530 Church St, Ann Arbor, MI 48105. *Symplectic embedding functions.*

The problem of embedding one symplectic four manifold, say the ellipsoid $E(1, x)$, into another symplectic four manifold Y can be encoded into a single function

$$f(x) = \inf_{\lambda}\{E(1, x) \text{ symplectically embeds into } \lambda Y\}.$$

This function has been computed for several examples, including $Y = B^4(1)$, the symplectic ball, and $Y = C^4(1)$, the symplectic cube. These examples are surprisingly rich, and their computation gave an indication that this problem is quite delicate. Higher dimensional embedding problems are even harder, but a useful simplification is the stabilized embedding problem. A stabilized symplectic manifold is a symplectic four manifold crossed with N copies of the complex plane. This talk will discuss the aforementioned embedding functions, as well as new results concerning the embedding function of stabilized polydiscs

$$f(x, a) = \inf_{b}\{P(1, x) \times \mathbb{C}^N \text{ symplectically embeds into } P(a, b) \times \mathbb{C}^N, \text{ with } b \geq a\}.$$

(Received February 04, 2020)

1157-53-610 **James Dibble*** (james-dibble@uiowa.edu). *Splitting theorems for manifolds with no conjugate points.*

An unresolved question in Riemannian geometry is the extent to which rigidity results for nonpositively curved manifolds extend to the case of no conjugate points. For example, it is not known whether a compact manifold with no conjugate points must split isometrically as a product with a flat factor when the center of its fundamental group is nontrivial. A partial result in this direction is the following: Each compact manifold with no conjugate points admits a family of functions whose integrals vanish exactly when central Busemann functions split linearly. These functions vanish when central Busemann functions are sub- or superharmonic. As a consequence, one obtains generalizations of splitting theorems of O'Sullivan for manifolds with no focal points and Eberlein for those with nonpositive curvature. (Received February 04, 2020)

1157-53-631 **Julian Chaidez*** (jchaidez@berkeley.edu). *Combinatorial Reeb Dynamics and Rotation Numbers.*

Many open problems in contact geometry center on quantitative properties of Reeb flows. For instance, the Viterbo conjecture proposes an isoperimetric inequality relating the minimal period of a closed Reeb orbit on the boundary of a convex domain in $\mathbb{C}^n$ to that domain's volume. One challenge to our understanding of these questions is a general lack of computationally tractable examples. In this talk, I will discuss an ongoing series of projects developing a theory of Reeb flows on certain triangulated spaces, and as a specific starting point, convex polytopes in $\mathbb{C}^2$. One can use some of this theory to write computer programs and probe quantitative questions about Reeb flows on convex domains by doing numerical experiments. I will present some of the interesting outcomes of these experiments. (Received February 04, 2020)

1157-53-637 **Weiwei Wu*** (mathwww@gmail.com) and **Jun Li**. *Topology of symplectomorphism groups and ball-swappings.*

In this talk, we give a survey on some recent progress on the topology of symplectomorphism groups, and discuss a problem that connects it to the topology of configuration spaces. (Received February 04, 2020)

1157-53-639 **Nikolaos Eptaminitakis*** (neptamin@uw.edu), Department of Mathematics, University of Washington, Box 354350, C-138 Padelford, Seattle, WA 98195-4350, and **C. Robin Graham** (robin@math.washington.edu), Department of Mathematics, University of Washington, Box 354350, C-138 Padelford, Seattle, WA 98195-4350. *Geodesic X-Ray Transform on Asymptotically Hyperbolic Manifolds.* Preliminary report.

Asymptotically hyperbolic manifolds are a class of non-compact, complete manifolds whose behavior near infinity resembles in many ways that of hyperbolic space. In part due to the behavior of geodesics and their infinite length, and the fact that those spaces have infinite volume, the study of the geodesic X-ray transform becomes particularly interesting and challenging. In this talk we will discuss the problem of injectivity of the geodesic X-ray transform on asymptotically hyperbolic manifolds, identify the issues induced by the geometry, and present some progress in extending to this setting the celebrated local injectivity result proved by Uhlmann and Vasy on compact manifolds with boundary. (Received February 04, 2020)

54 ► General topology

1157-54-499 **Deniz Kutluay*** (dkutluay@indiana.edu), Rawles Hall, 831 E 3rd St, Bloomington, IN 47405. *Winding homology of knotoids.*

Knotoids were introduced by V. Turaev as open-ended knot-type diagrams that generalize knots. Turaev defined a two-variable polynomial invariant of knotoids which encompasses a generalization of the Jones polynomial to knotoids. We define a triply-graded homological invariant of knotoids categorifying the Turaev polynomial. Forgetting one of the three gradings gives a generalization of the Khovanov homology to knotoids. (Received February 03, 2020)

55 ► Algebraic topology

1157-55-28 **Javier Zúñiga*** (zuniga_jj@up.edu.pe), Av Salaverry 2020, Jesús María, Lima, Lima , Peru, and **Ralph Kaufmann**. *Bordered Stable Ribbon Graphs.* Preliminary report.

Deformations of Riemann surfaces with marked points are encoded by the combinatorics of stable ribbon graphs. This fact is reflected by the existence of an orbicell decomposition of a real-blowup of the Deligne-Mumford compactification of the moduli space of decorated Riemann Surfaces. Bordered Riemann surfaces can be doubled

along the boundary to obtain a compact surface without border with an induced involution. By tracking the combinatorics of stable ribbon graphs along this construction one can define bordered ribbon graphs encoding the deformations of bordered Riemann surfaces. (Received December 09, 2019)

1157-55-83 **Zachary Himes*** (himesz@purdue.edu), 150 N. University Street, West Lafayette, IN 47906. *Secondary stability and periodicity for unordered configuration spaces.*

Secondary stability is a stability pattern in a range that homological stability does not hold. The first example of secondary stability is Galatius–Kupers–Randal-Williams results on mapping class groups. We prove secondary stability for unordered configuration spaces of manifolds. The main difficulty is the closed case (the open case was previously known by some experts). In the closed case, there are no obvious stabilization maps and the homology does not stabilize but is periodic. We resolve this issue by constructing a chain-level stabilization map for configurations of closed manifolds. The main technical lemma involves describing obstructions for multiplication by an element of an E_n-algebra to be homotopic to a map of modules over the E_{n-1}-Hochschild chains of the E_n-algebra. (Received January 15, 2020)

1157-55-242 **Alexander Kupers*** (kupers@math.harvard.edu). *The Drinfel'd-Kohno Lie algebras and their applications to high-dimensional manifolds.*

We will discuss the Drinfel'd-Kohno Lie algebras. These arise from the rational homotopy groups of configuration spaces of Euclidean spaces, and thus come with actions of symmetric groups. I will explain some of their representation-theoretic properties, and explain how they appear in the study of diffeomorphisms of high-dimensional manifolds. This includes joint work with Jeremy Miller, and with Oscar Randal-Williams. (Received January 30, 2020)

1157-55-247 **Manuel Rivera*** (manuelr@purdue.edu) and **Zhengfang Wang**. *The Goresky-Hingston algebra and singular Hochschild cohomology.*

The Goresky-Hingston algebra on the reduced Hochschild homology of a differential graded (dg) Frobenius algebra is an algebraic model for a geometric string topology operation constructed by Goresky and Hingston on the homology of the free loop space of a manifold relative its constant loops. In this talk I will explain a surprising relationship between this operation and a construction in homological algebra. The main statement is that the Goresky-Hingston algebra on the reduced Hochschild homology of a simply connected dg Frobienus algebra A can be identified with a sub algebra of the "singular Hochschild cohomology algebra of A", i.e. the endomorphism algebra of A in the Verdier quotient of the derived category of A-bimodules modulo perfect objects. I will also discuss some consequences of this fact, for example, the invariance of the Goresky-Hingston algebra structure with respect to quasi-isomorphisms of dg associative algebras. This is joint work with Z. Wang. (Received January 30, 2020)

1157-55-265 **Calista Bernard*** (calista@stanford.edu). *Twisted Dyer-Lashof operations.* Preliminary report.

In the 70s, Fred Cohen and Peter May gave a description of the mod p homology of a free E_n algebra in terms of certain homology operations, known as Dyer-Lashof operations, and the Browder bracket. I will discuss a framework to generalize these operations to homology with certain twisted coefficient systems and explain computational results that show the existence of additional operations in the twisted case. (Received January 31, 2020)

1157-55-274 **Ronno Das*** (ronnodas@gmail.com) and **Benjamin O'Connor**. *Noncollinear points on the projective plane.*

The space F_n of n points in the projective plane such that no 3 are collinear is an open subspace of $(\mathbb{P}^2)^n$. The cohomology of F_n with the S_n action by permuting coordinates is representation stable, but explicit computations are hard. We compute the cohomology of F_6 by counting such 6-tuples over the finite field $\mathbb{F}_q$ (with a 'twist') and using the Grothendieck–Leftschetz trace formula. Unfortunately $n > 6$ is still hard. (Received January 31, 2020)

1157-55-293 **Daniel A Ramras*** (dramras@iu.edu) and **Mentor Stafa** (mstafa@tulane.edu). *Homological stability for spaces of representations.*

I will discuss recent work with Mentor Stafa on homological stability for various spaces built from commuting elements in Lie groups. These results depend on rational models for these spaces due to T. Baird, and the theory of FI_W–modules developed by Church, Farb, and Wilson. A key aspect of this work is the relationship between stability in ordinary and in equivariant (co)homology. Time permitting, I'll also discuss what is (and is not) known about stability for character varieties of free groups and surface groups. (Received January 31, 2020)

1157-55-296 **Daniel A Ramras*** (dramras@iu.edu). *Flat connections and the topological Atiyah-Segal map.*

Given a closed, aspherical manifold M, the topological Atiyah–Segal is a map of ring spectra relating the topological K-theory of M to a spectrum built from unitary representations of $\pi_1 M$. The homotopy fiber of this map can be described in terms of spaces of flat connections on bundles over M. I'll explain this set-up, and describe some applications. (Received January 31, 2020)

1157-55-311 **Bena Tshishiku*** (bena_tshishiku@brown.edu). *Geometric cycles and characteristic classes.*

We give a construction of nontrivial cohomology classes of arithmetic subgroups of $\mathrm{SO}(p,q)$. These classes lie just outside the stable range and are dual to geometric cycles in the locally symmetric space. The classes we construct have a nice interpretation as characteristic classes, and as an application we obtain nontrivial characteristic classes for certain manifold bundles that are not in the algebra generated by the stable classes. (Received February 01, 2020)

1157-55-326 **Philip Hackney**, **Marcy Robertson*** (marcy.robertson@unimelb.edu.au) and **Donald Yau**. *Higher Modular Operads.*

We define a weak version of modular operads whose compositions and contractions are only defined up to homotopy. This homotopy theory takes the form of a Quillen model structure on the collection of simplicial presheaves for a certain category of undirected graphs. This new category of undirected graphs, denoted **U**, plays a similar role for modular operads that the dendroidal category Ω plays for operads. (Received February 01, 2020)

1157-55-329 **Philip Hackney*** (hackney@louisiana.edu). *Configuration spaces form an ∞-operad.*

We exhibit a Segal semi-dendroidal space whose value on a one-vertex tree with n-leaves is the ordered configuration space of n points in $\mathbb{R}^k$. This semi-dendroidal space is homotopy equivalent to the nerve of the little k-disks operad D_k, extending the well-known equivalences $D_k(n) \simeq \mathrm{Conf}(\mathbb{R}^k, n)$. (Received February 01, 2020)

1157-55-388 **Alice Patania*** (apatania@iu.edu), 1001 E SR 45/46 Bypass, Bloomington, IN 47408, and **Jean-Gabriel Young** and **Elizabeth E Bruch**. *The shape of social hierarchies.* Preliminary report.

Many complex networks encode hierarchies, where ties between nodes denote a local ordering of individuals—say, a pecking order or social signal of dominance. A series of recent advances have focused on finding rankings from such data. But a lot needs to be done beyond simply ranking nodes, because these hierarchies interact with many other properties of social systems (e.g., community structure).

We propose to tackle this problem by adapting the highly effective "Mapper" algorithm to work with networks and hierarchies data. Mapper is a state-of-the-art algorithm that takes large datasets as input (N¿¿1 vectors of D features), and that outputs a coarse topological summary of how these data points interact with one another (a graph where nodes are clusters of input vectors and ties represent shared vectors). The network version of Mapper allows us to mix and mash pre-existing embedding and clustering algorithms in a principled way, and obtain various topological summary of real hierarchies. By doing so, we find accurate summaries of the input network that differs from the current standards such as community detection, because embedding the nodes allows Mapper to reveal large scale structures not accounted for by typical detection algorithms. (Received February 03, 2020)

1157-55-423 **Alexander Berglund**, **Gregory Michel**, **TriThang Tran** and **Craig Westerland*** (cwesterl@umn.edu). *Second order terms in arithmetic statistics.* Preliminary report.

It's an old conjecture that the the cardinality of the set of number fields of degree n and discriminant bounded by X grows asymptotically linearly in X. This is classical when $n = 2$, due to Davenport-Heilbronn for $n = 3$, and Bhargava for $n = 4, 5$. Work of Taniguchi-Thorne and Bhargava-Shankar-Tsimerman establishes a second-order term for cubic extensions.

In this talk, we will report on work in progress aiming to establish a second order term in the function field setting, as well as giving a general heuristic for what we believe the degree of the second-order term to be. Our tools are homological algebraic, coming from the Koszul complex of the Nichols algebra associated to the Galois data; this gives refined information on the cohomology of the Hurwitz moduli spaces of branched covers with this Galois data. (Received February 03, 2020)

1157-55-428 **Irving Dai**, **Matthew Hedden** and **Abhishek Mallick***, mallicka@math.msu.edu. *Cork Involutions and Heegaard Floer homology.*

Building on the algebraic framework developed by Hendricks, Manolescu, and Zemke, we introduce and study a set of Floer-theoretic invariants aimed at detecting corks. Our invariants obstruct the extension of a given involution of a 3-manifold over any homology ball (rather than a particular contractible manifold). We call such a 3-manifold with involution a strong cork. The first example of a strong cork was given by Lin-Ruberman-Saveliev. Unlike previous approaches, our method does not explicitly reference closed 4-manifolds or contact structures. Instead, we utilize the formalism of local equivalence coming from involutive Heegaard Floer homology. As an application, we define a modification of the homology cobordism group $\Theta^3_{\mathbb{Z}}$ which takes into account involutions acting on homology spheres, and prove that this admits a $\mathbb{Z}^\infty$-subgroup generated by strong corks. Our group $\Theta^3_{\mathbb{Z}}$ can be viewed as a refinement of the classical bordism group of diffeomorphisms. Using our invariants, we establish several new families of strong corks and prove that various known examples are also strong. (Received February 03, 2020)

57 ► Manifolds and cell complexes

1157-57-36 **David Auckly*** (dav@ksu.edu) and **Daniel Ruberman**. *Exotic Homotopy Classes in Diffeomorphism Groups.*

There is a jungle full of exotic phenomena associated with four-dimensional manifolds. In this talk, we will describe the large difference between the homotopy of the diffeomorphism group and the homotopy of the homeomorphism group that appears for many four-manifolds. Analogous behavior is found in spaces of embeddings. Such results may lead to a better understanding of configurations of surfaces in four-manifolds. (Received January 02, 2020)

1157-57-41 **Colin Adams***, Bascom Hall, 33 Stetson Court, Williamstown, MA 01267, and **Or Eisenberg**, **Jonah Greenberg**, **Kabir Kapoor**, **Zhen Liang**, **Kate O'Connor**, **Natalia Pacheco-Tallej** and **Yi Wang**. *TG-Hyperbolicity and Turaev Hyperbolicity of Classical and Virtual Knots.*

We extend the theory of hyperbolicity of links in the 3-sphere to tg-hyperbolicity of virtual links, using the fact that the theory of virtual links can be translated into the theory of links living in closed orientable thickened surfaces. When the boundary surfaces are taken to be totally geodesic, we obtain a tg-hyperbolic structure with a unique associated volume. We prove that all virtual alternating links are tg-hyperbolic. We then consider bounds on volumes of virtual links and include a table for volumes of the 116 nontrivial virtual knots of four or fewer crossings, all of which, with the exception of the trefoil knot, turn out to be tg-hyperbolic. We further employ a construction of Turaev to associate a family of hyperbolic 3-manifolds of finite volume to any classical or virtual link, even if non-hyperbolic. These are in turn used to define the *Turaev volume* of a link, which is the minimal volume among all the hyperbolic 3-manifolds associated via this Turaev construction. In the case of a classical link, we can also define the *classical Turaev volume*, which is the minimal volume among all the hyperbolic 3-manifolds associated via this Turaev construction for the classical projections only. We investigate these new invariants. (Received January 05, 2020)

1157-57-43 **William Rushworth*** (will.rushworth@math.mcmaster.ca). *Ascent concordance.*

Let Σ_g be a closed orientable surface of genus g, and $L_1 \hookrightarrow \Sigma_{g_1} \times I$ and $L_2 \hookrightarrow \Sigma_{g_2} \times I$ links in thickenings of Σ_{g_1} and Σ_{g_2}. A concordance between L_1 and L_2 is a pair (S, M), where M is a compact orientable 3-manifold with $\partial M = \Sigma_{g_1} \sqcup \Sigma_{g_2}$, and S a disjoint union of annuli properly embedded in $M \times I$ such that each annulus has a boundary component in both L_1 and L_2.

Given a concordance between L_1 and L_2, how complex need the 3-manifold M be? We show that there exist representatives of the same concordance class that are not concordant if one restricts to 3-manifolds that are Morse-theoretically simple. We use an augmented version of Khovanov homology to do so. Such links provide counterexamples to an analogue of the Slice-Ribbon Conjecture. (Received January 06, 2020)

1157-57-59 **Ciprian Manolescu**, **Marco Marengon*** (marengon@math.ucla.edu), **Sucharit Sarkar** and **Michael Willis**. *A generalization of Rasmussen's invariant, with applications to surfaces in some four-manifolds.*

Building on previous work of Rozansky and Willis, we generalise Rasmussen's s-invariant to $\#^r(S^1 \times S^2)$. Such an s-invariant can be computed by approximating the Khovanov-Lee complex of a link in $\#^r(S^1 \times S^2)$ with that of appropriate links in S^3. We use the approximation result to compute the s-invariant of a family of

links in S^3 which seems otherwise inaccessible, and use this computation to deduce an adjunction inequality for null-homologous surfaces in a (punctured) connected sum of $\overline{CP}^2$. This inequality has several consequences: first, the s-invariant of a link in S^3 does not increase under the operation of adding a null-homologous full twist. Second, the s-invariant cannot be used to distinguish S^4 from homotopy 4-spheres obtained by Gluck twist on S^4. We also prove a connected sum formula for the s-invariant of links in S^3, improving a previous result of Beliakova and Wehrli. We define two s-invariants for links in $S^1 \times S^2$. One of them gives a lower bound to the slice genus in $S^1 \times B^3$ and the other one to the slice genus in $D^2 \times S^2$. Lastly, we give a combinatorial proof of the slice Bennequin inequality in $S^1 \times S^2$. (Received January 09, 2020)

1157-57-62 **Artem Kotelskiy*** (artofkot@iu.edu). *Immersed curves in Khovanov homology.*

Consider a 2-sphere S intersecting a knot K in 4 points. This defines decomposition of the knot into two 4-ended tangles. In a recent work we showed that Khovanov homology Kh(K), and its deformation due to Bar-Natan, are isomorphic to wrapped Lagrangian Floer homology of a pair of specifically constructed immersed curves on the dividing 4-punctured sphere S. We will explain how to modify this construction to recover annular sutured Khovanov homology as Lagrangian Floer homology of two curves. This is joint work with Liam Watson and Claudius Zibrowius. (Received January 11, 2020)

1157-57-70 **Francis Bonahon*** (fbonahon@usc.edu). *Asymptotics of quantum invariants of surface diffeomorphisms.* Preliminary report.

The Kashaev-Murakami-Murakami Volume Conjecture asserts that, for a knot K in the 3-sphere S^3, the evaluation $J_n^K(\mathrm{e}^{\frac{2\pi\mathrm{i}}{n}})$ of the n–th colored Jones polynomial of K at the root of unity $\mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}$ grows exponentially as $n \to \infty$, with exponential growth rate related to the hyperbolic volume $\mathrm{vol}(S^3 - K)$ of the knot complement. We will discuss a closely related conjecture for diffeomorphisms of surfaces, based on the representation theory of the quantum Teichmüller space and/or the Kauffman bracket skein algebra of the surface. We will present experimental evidence for this conjecture, and describe partial results in work in progress with Helen Wong and Tian Yang. (Received January 13, 2020)

1157-57-71 **Qayum Khan*** (qkhan@indiana.edu) and **Gerrit N Smith**. *Stable existence of incompressible 3-manifolds in 4-manifolds.*

Given a separating embedded connected 3-manifold in a closed 4-manifold, the Seifert–vanKampen theorem implies that the fundamental group of the 4-manifold is an amalgamated product along the fundamental group of the 3-manifold.

In the other direction, given a closed 4-manifold whose fundamental group admits an injective amalgamated product structure along the fundamental group of a 3-manifold, is there a corresponding geometric-topological decomposition of the 4-manifold in a stable sense?

We find an algebraic-topological splitting criterion in terms of the orientation classes and universal covers. Also, we equivariantly generalize the Lickorish–Wallace theorem to regular covers. (Received January 13, 2020)

1157-57-74 **Rob Schneiderman*** (robert.schneiderman@lehman.cuny.edu) and **Peter Teichner**. *Homotopy versus isotopy for spheres with duals in 4-manifolds.*

David Gabai's smooth 4-dimensional Light Bulb Theorem states roughly that 'for embedded 2-spheres in an orientable 4-manifold, homotopy implies isotopy in the presence of a geometric dual sphere and the absence of 2-torsion in the ambient fundamental group'. In joint work with Peter Teichner this result is extended to arbitrary orientable 4-manifolds by showing that an invariant of Mike Freedman and Frank Quinn which takes values in a quotient of the Z/2Z-vector space on the 2-torsion fundamental group elements gives the complete obstruction to the existence of an isotopy between homotopic embedded 2-spheres having a common geometric dual. We also give an alternative approach to Gabai's proof. (Received January 14, 2020)

1157-57-80 **Charles Frohman*** (charles-frohman@uiowa.edu), Professor, and **Adam Sikora**. *Finite generation of the SL_3-skein algebra of surfaces.*

Let F be a finite type surface with at least one puncture and negative Euler characteristic. A non-elliptic web is an embedding of a trivalent graph W into F whose edges have been oriented so that every vertex is a source or sink, so that all the faces of the web W have a least 6 sides. We prove that you can minimize the number of points of intersection of a web up to isotopy with the ideal triangulation by using four local moves, in such a way that cardinality of the intersection of the web with the edges of the ideal triangulation is monotone decreasing. This leads to coordinates for isotopy classes webs in F. The coordinates form a finite union of integral cones. The addition in the integral cones corresponds to multiplication in the SL_3-skein algebra of F. We use this machinery to prove that the SL_3-skein algebra of F is finitely generated. (Received January 15, 2020)

1157-57-85 **Zhenghan Wang***, 2237 CNSI Bldg, UC Santa Barbara, Santa Barbara, CA 93106-6105. *Representations of motion groups of links in the three sphere from (3+1)-TQFTs.*

Motion groups of links in the three sphere are generalizations of braids. Just as (2+1)-TQFTs lead to representation of braid groups, more general mapping class groups, (3+1)-TQFTs should give rise to representations of motion groups of links. But such representations are hard to describe explicitly even from the Dijkgraaf-Witten TQFTs. We will discuss progress for the motion groups of n-copies of a torus knot. This is a joint work with my student Yang Qiu. (Received January 15, 2020)

1157-57-92 **Carmen Caprau**, **Nicolle Gonzalez**, **Christine Ruey Shan Lee*** (crslee@southalabama.edu), **Radmila Sazdanovic** and **Melissa Zhang**. *Plamenevskaya's invariant and the stable Khovanov homology of twisted torus knots.*

A transverse link is a link in the 3-sphere that is everywhere transverse to the standard contact structure. Transverse links are considered up to transverse isotopy, with classical invariants such as the self-linking number and regular isotopy class. One of the first connections between transverse links and quantum invariants was made by Plamenevskaya in 2006, when she defined a invariant of transverse links from Khovanov homology. Since then, the relationship between this fascinating invariant and contact geometry remains of intense interest. In this talk, I will discuss open questions relating Plamenevskaya's invariant to the contact-geometric properties of braids such as the Fractional Dehn Twist Coefficient, and how my recent works with Hubbard, and with Caprau, Gonzalez, Sazdanovic, and Zhang apply to study those questions. (Received January 17, 2020)

1157-57-103 **Hans U. Boden** and **Homayun Karimi*** (homayun.karimi@gmail.com), 1280 Main Street West, Hamilton, Ontario L8S 4L8, Canada. *The Jones-Krushkal polynomial and minimal diagrams of surface links.*

We prove a Kauffman-Murasugi-Thistlethwaite theorem for alternating links in thickened surfaces. It states that any reduced alternating diagram of a link in a thickened surface has minimal crossing number, and any two reduced alternating diagrams of the same link have the same writhe. This result is proved more generally for link diagrams that are adequate, and the proof involves a two-variable generalization of the Jones polynomial for surface links defined by Krushkal. The main result is used to establish the first and second Tait conjectures for links in thickened surfaces and for virtual links. This is joint work with Hans U. Boden. (Received January 21, 2020)

1157-57-113 **Christopher William Davis***, 105 Garfield Avenue, P.O. Box 4004, Eau Claire, WI 54703. *Concordance and homotopy of knots in homology spheres.*

Any knot in S^3 may be reduced to a slice knot by crossing changes. Indeed, this slice knot can be taken to be the unknot. In this paper we study the question of when the same holds for knots in homology spheres. We show that a knot in a homology sphere is nullhomotopic in a smooth homology ball if and only if that knot is smoothly concordant to a knot which is homotopic to a smoothly slice knot. As a consequence, we prove that the equivalence relation on knots in homology spheres given by cobounding immersed annuli in a homology cobordism is generated by concordance in homology cobordisms together with homotopy in a homology sphere. (Received January 22, 2020)

1157-57-115 **Christopher William Davis*** (daviscw@uwec.edu), 105 Garfield Avenue, P.O. Box 4004, Eau Claire, WI 54703, and **Jung Hwan Park**. *Concordance to links with unknotted components.*

Dating back to 1991 various results of Cha, Cochran, Orr, and Rubermann demonstrate the existence of links $L_1 \cup L_2$ with L_2 unknotted and L_1 slice which are not concordant to any link with first component unknotted. By a study of Alexander modules we demonstrate the existence of link with every component slice but which is not concordant to any link with even one unknotted component. (Received January 22, 2020)

1157-57-121 **P Feller**, **A M Miller**, **M Nagel**, **P Orson*** (orson@bc.edu), **M Powell** and **A Ray**. *Embedding spheres in knot traces.*

Given a compact topological 4-manifold X, when can a homotopy class in $\pi_2(X)$ be represented by a locally flatly embedded 2-sphere? This basic question has been studied by many authors in the case that X has no boundary. We consider the question for knot traces. These are manifolds with boundary, homotopic to the 2-sphere, and obtained by attaching a 2-handle to the 4-ball along a framed knot in the 3-sphere. We characterise when this homotopy 2-sphere can be represented by a locally flat embedding with abelian exterior fundamental group. The answer is in terms of classical and computable invariants of the knot. (Received February 02, 2020)

1157-57-124 **Eric Samperton*** (smprtn@illinois.edu). *Towards hyperbolic quantum computing and such.* Preliminary report.

As Freedman, Kitaev, Larsen, and Wang famously showed, there exist 3-dimensional unitary topological quantum field theories where the ability to make certain kinds of approximations of the resulting 3-manifold invariants is equivalent to the usual BQP circuit model. Their result theoretically justifies the idea of topological quantum computing. Can we "geometrize" their result by implementing a kind of "hyperbolic quantum computing? More precisely, can we efficiently encode quantum circuits not just with arbitrary 3-manifolds, but with *hyperbolic* 3-manifolds? I'll report on progress I've made on this question. (Received January 22, 2020)

1157-57-129 **Lei Chen*** (chenlei1991919@gmail.com), 250 S Oak Knoll Ave 207, Pasadena, CA 91101. *Some algebraic properties of the power subgroup of mapping class groups.*

In this talk, I will discuss some algebraic properties of the power subgroup of mapping class group; i.e., the normal closure of a big power of a Dehn twist. For example, I will discuss the restriction of the Euler class on it, its orbit on curve complexes and possibly other properties. I will also talk about some open problems. (Received January 22, 2020)

1157-57-134 **Yu Pan** and **Dan Rutherford*** (rutherford@bsu.edu). *Augmentations and immersed Lagrangian fillings.*

Let $\Lambda \subset \mathbb{R}^3$ be a Legendrian knot with respect to the standard contact structure. The Legendrian contact homology (LCH) DG-algebra, $\mathcal{A}(\Lambda)$, of Λ is functorial for exact Lagrangian cobordisms in the symplectization of $\mathbb{R}^3$, i.e. a cobordism $L \subset Symp(\mathbb{R}^3)$ from Λ_- to Λ_+ induces a DG-algebra map, $f_L : \mathcal{A}(\Lambda_+) \to \mathcal{A}(\Lambda_-)$. In particular, if L is an exact Lagrangian filling ($\Lambda_- = \emptyset$) the induced map is an augmentation $\epsilon_L : \mathcal{A}(\Lambda_+) \to \mathbb{Z}/2$. In this talk, I will discuss an extension of this construction to the case of immersed, exact Lagrangian cobordisms based on considering the Legendrian lift Σ of L. When L is an immersed, exact Lagrangian filling a choice of augmentation α for Σ produces an induced augmentation $\epsilon_{(L,\alpha)}$ for Λ_+. Using the cellular formulation of LCH (developed with M. Sullivan), we are able to show that any augmentation of Λ may be induced by such a filling. (Received January 23, 2020)

1157-57-142 **Rhea Palak Bakshi*** (rhea_palak@gwu.edu), **Dionne Ibarra**, **Gabriel Montoya-Vega**, **Józef H. Przytycki** and **Deborah Weeks**. *Framing Changes of Knots and Links in 3-Manifolds.*

We show that the only way of changing the framing of a knot or a link by ambient isotopy in an oriented 3-manifold is when the manifold admits a properly embedded non-separating S^2. This change of framing is given by the Dirac trick, also known as the light bulb trick or belt trick. The main tools we use are based on McCullough's work on the mapping class groups of 3-manifolds and Dehn homeomorphisms. (Received January 24, 2020)

1157-57-160 **Greg Friedman*** (g.friedman@tcu.edu), TCU Dept. of Mathematics Box 298900, Fort Worth, TX 76129, and **Efton Park**. *Topological Linear Algebra.*

We will discuss ongoing work with Efton Park (TCU) to classify the unitary equivalence classes of normal matrices with coefficients in the ring $C(X)$ of continuous complex-valued functions on the space X. Such matrices can be thought of as bundles of complex-valued matrices over X, and their properties depend in interesting ways on the algebraic topology of X. (Received January 26, 2020)

1157-57-161 **Honghao Gao*** (gaohongh@msu.edu), 619 Red Cedar Road, Wells Hall C212, East Lansing, MI 48824. *Fillings and clusters on augmentation varieties.* Preliminary report.

Cluster variety is a type of algebraic variety with very strong algebraic constriants. A cluster variety has a collection of algebraic toric charts (or clusters), and the transition maps between charts are governed by a quiver and its quiver mutations. We will use this algebraic structure to study Legendrian knots. For a positive braid closure, we prove the coordinate ring of its augmentation variety is canonically isomorphic to an upper cluster algebra, and decomposable exact Lagrangian fillings induce K_2 clusters. This is a joint work with Linhui Shen and Daping Weng. (Received January 26, 2020)

1157-57-162 **Honghao Gao*** (gaohongh@msu.edu), 619 Red Cedar Road, Wells Hall C212, East Lansing, MI 48824. *Infinitely many Lagrangian fillings.*

A filling is an oriented surface bounding a link. Lagrangian fillings can be constructed via local moves in finite steps, but it was unknown whether a Legendrian link could admit infinitely many Lagrangian fillings. In this talk, I will show that Legendrian torus links other than (2,m), (3,3), (3,4), (3,5) indeed have infinitely many fillings. These fillings are constructed using Legendrian loops, and proven to be distinct using the microlocal

theory of sheaves and the theory of cluster algebras. This is a joint work with Roger Casals. (Received January 26, 2020)

1157-57-173 **Anthony Conway***, anthonyyconway@gmail.com, and **Mark Powell**, mark.a.powell@durham.ac.uk. *Enumerating homotopy ribbon discs.*

Given a knot K and a group G, we explore classifications of surfaces $F \subset B^4$ with boundary K and $\pi_1(B^4 \setminus F) = G$. Much of the talk will focus on the case where G is infinite cyclic, and is based on joint work in progress with Mark Powell. (Received January 27, 2020)

1157-57-174 **Kyle Hayden*** (hayden@math.columbia.edu) and **Lisa Piccirillo**. *Exotic 4-manifolds and piecewise linear surfaces.*

The "trace embedding lemma" is a classical observation about certain 4-manifolds that embed in 4-space. I'll describe new applications of this lemma to the study of exotic phenomena in dimension four. In particular, we'll use this to help distinguish 4-manifolds that are homeomorphic but not diffeomorphic, build simple "spineless" 4-manifolds, and find (locally flat) topological embeddings of surfaces into 4-manifolds that are not homotopic to smooth or even piecewise linear embeddings. This is joint work with Lisa Piccirillo. (Received January 27, 2020)

1157-57-179 **Rhea Palak Bakshi** and **Dionne Ibarra***, 801 22nd Street NW, Washington DC, DC 20052, and **Gabriel Montoya-Vega**, **Jozef H. Przytycki** and **Deborah Weeks**. *Skein modules on framed links in 3-manifolds.*

In this talk we apply the framing skein module, q-homology skein module, and the Kauffman bracket skein module to orientable 3-manifolds focusing on the fact that the only way of changing the framing of a link by ambient isotopy in an oriented 3-manifold is when the manifold has a properly embedded non-separating S^2. We also define the spin twisted homology from the Kauffman bracket skein module. (Received January 27, 2020)

1157-57-184 **Geunho Lim*** (limg@indiana.edu), Department of Mathematics, Indiana University, 831 East 3rd St., Bloomington, IN 47405. *Enhanced universal bounds for Cheeger-Gromov L^2 ρ-invariants of 3-manifolds.* Preliminary report.

Cheeger and Gromov defined L^2 ρ-invariants to Riemannian manifolds and showed L^2 ρ-invariants have universal bounds using deep analytic arguments. Chang and Weinberger extended the definition to topological manifolds. Cha proved the existence of universal bounds for L^2 ρ-invariants of topological manifolds and found explicit bounds in terms of the complexity of given 3-manifolds. Cha's construction is motivated by the Baumslag-Dyer-Heller proof of the acyclicity of the acyclic container. In this talk, I introduce a more direct approach, and construct a new and more economic bounds guided primarily by the relations in the Baumslag-Dyer-Heller's acyclic container. (Received January 27, 2020)

1157-57-212 **Vladimir Turaev***, Dep of Mathematics, Indiana University, 831 E. Third st, Bloomington, IN 47405. *Quasi-surfaces and related algebraic structures.*

Quasi-surfaces are topological spaces generalizing surfaces with boundary by allowing singular (non-surface) parts. We discuss several homotopy-invariant operations on loops in quasi-surfaces which generalize standard operations on loops in surfaces (the homological intersection form, the Goldman bracket, etc.). This leads us to notions of a quasi-Lie algebra, a quasi-Lie bialgebra, and a quasi-Poisson algebra. (Received January 29, 2020)

1157-57-217 **Keiko Kawamuro***, 14 MacLean Hall, Iowa City, IA 52242. *Twist left-veering open books.*

I will define a twist left-veering open book and discuss its properties and application to knot theory and contact geometry. (Received January 29, 2020)

1157-57-219 **Radmila Sazdanovic*** (rsazdan@ncsu.edu), Department of Mathematics NC State University, Raleigh, NC 27695, and **Adam Lowrance**. *Torsion of Khovanov homology.*

In the integral Khovanov homology of links, the presence of odd torsion is rare. Khovanov homology of homologically thin links only contains $\mathbb{Z}_2$ torsion. We prove a local version of this result and apply it to an infinite family of 3-braids, strictly containing all 3-strand torus links. This provides a partial answer to Sazdanovic and Przytycki's conjecture that 3-braids have only $\mathbb{Z}_2$ torsion in Khovanov homology. We provide explicit computations of integral Khovanov homology for all links in this family. Additionally, we prove an upper bound on the order of the torsion part of Khovanov homology in terms of the crossing number of the link. This is joint work with A. Chandler, A. Lowrance and V. Summers (Received January 29, 2020)

1157-57-253 **Scott Baldridge*** (sbaldrid@lsu.edu), Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803, and **Lou H Kaufman** and **Ben McCarty**. *Unoriented Khovanov Homology.*

The Jones polynomial and Khovanov homology of a classical link are invariants of oriented links—they depend upon an initial choice of orientation for the link. In this talk, we describe a Khovanov homology theory and Jones polynomial for unoriented links, and discuss some of the difficulties that needed to be overcome to define them for virtual links. (Received January 30, 2020)

1157-57-258 **Matthew Harper*** (harper.946@buckeyemail.osu.edu). *Link Invariants from Unrolled Restricted Quantum* $\mathfrak{sl}_3$.

In this talk, we discuss the R-matrix invariant of links coming from unrolled restricted quantum $\mathfrak{sl}_3$ at a fourth root of unity. This symmetric two-variable Laurent polynomial is obtained by coloring links with a Borel induced representation $V(\underline{t})$, with $\underline{t} = (t_1, t_2) \in (\mathbb{C}^\times)^2$, of quantum dimension zero. We prove that this invariant coincides with the Alexander polynomial of knots precisely when substituting $\underline{t}$ corresponding to reducible $V(\underline{t})$. We also give the value of this invariant for various knots and observe that it distinguishes the mutant pair 11_{34} and 11_{42}. (Received January 30, 2020)

1157-57-270 **Sujoy Mukherjee** and **Dirk Schuetz***, Department of Mathematical Sciences, Durham University, Durham, United Kingdom. *Torsion in the Khovanov cohomology of composite links.*

The Khovanov cohomology of a composite link can be calculated by using basepoints to give the Khovanov chain complexes the structure of $R = \mathbb{Z}[X]/X^2$-modules, and then forming the tensor product over R of these complexes. We show that this can lead to surprising behaviour for the torsion in the resulting cohomology groups. (Received January 31, 2020)

1157-57-279 **Paolo Aceto**, **Corey Bregman**, **Christopher Davis** and **JungHwan Park***, junghwan.park@math.gatech.edu, and **Arunima Ray**. *Isotopy and equivalence of knots in 3-manifolds.*

It is a well-known and often used fact that the notions of (ambient) isotopy and equivalence coincide for knots in S^3, since any orientation-preserving homeomorphism of S^3 is isotopic to the identity. We compare the notions of equivalence and isotopy for knots in more general 3-manifolds. We show that any orientation-preserving homeomorphism of a prime, oriented 3-manifold which preserves free homotopy classes is isotopic to the identity, except in the single case of the Gluck twist acting on $S^1 \times S^2$. We give infinitely many examples of knots in $S^1 \times S^2$ whose isotopy classes are changed by a Gluck twist. This is joint work with Paolo Aceto, Corey Bregman, Christopher Davis, and Arunima Ray. (Received January 31, 2020)

1157-57-280 **Paolo Aceto**, **Jeffrey Meier**, **Allison Miller**, **Maggie Miller** and **JungHwan Park***, junghwan.park@math.gatech.edu, and **Andras Stipsicz**. *Branched covers bounding rational homology balls.*

Prime-power-fold cyclic branched covers along smoothly slice knots all bound rational homology balls. This phenomenon, however, does not characterize slice knots: In this talk, we give examples of non-slice knots that have the above property. This is joint work with Aceto, Meier, A. Miller, M. Miller, and Stipsicz. (Received January 31, 2020)

1157-57-282 **Irving Dai**, **Jennifer Hom**, **Matthew Stoffregen** and **Linh Truong***, School of Mathematics, Institute for Advanced Study, 1 Einstein Drive, Princeton, NJ 08540. *An infinite rank summand of the homology cobordism group.*

We prove that the homology cobordism group of integer homology three-spheres contains an infinite rank summand. The proof uses an algebraic modification of the involutive Heegaard Floer package of Hendricks-Manolescu and Hendricks-Manolescu-Zemke. This is joint work with Irving Dai, Jen Hom, and Matt Stroffregen. (Received January 31, 2020)

1157-57-287 **Nicole Vorderobermeier*** (nicole.vorderobermeier@sbg.ac.at). *Real analyticity of knots.*

Is it possible to determine an "optimal" representative of every knot class? And how "optimal" are such embeddings? These questions can be tackled by introducing geometric knot energies. These energies are designed to produce particularly "optimal" knots as local minimizers.

In this talk, we give a short introduction to geometric knot energies with focus on the so-called Möbius energy. We already know that minimizers of the Möbius energy exist in every prime knot class and that merely bounded

energy of a critical knot implies smoothness. We conclude the talk by presenting an essential technique to show the real analyticity of smooth critical knots of the Möbius energy and discuss the consequences of this result. (Received January 31, 2020)

1157-57-295 **Langte Ma*** (ltmafixer@brandeis.edu), 66 Baker Circle, Chestnut Hill, MA 02467.
Surgery for Gauge Theoretical Invariants of Integral Homology $S^1 \times S^3$.

Given an integral homology $S^1 \times S^3$, one can define the Furuta-Ohta invariant λ_{FO} using the Yang-Mills theory and Casson-Seiberg-Witten invariant λ_{SW} using the Seiberg-Witten theory. In this talk, I will discuss a torus-surgery formula for those two invariants. I will also discuss some computations resulting from the surgery formula, which provides us with more evidence on the equivalence of those invariants. (Received January 31, 2020)

1157-57-300 **Samantha Allen** and **Charles Livingston*** (livingst@indiana.edu), Department of Mathematics, Indiana University, Bloomington, IN 47405. *Unknotting with a single twist.* Preliminary report.

The torus knot $T(2,3)$ can be unknotted by performing a full negative twist on either two or three coherently oriented parallel strands. It can also be unknotted by performing a full positive twist on a pair of oppositely oriented strands. Thus, we say the unknotting linking set for $T(2,3)$, denoted $U(T(2,3))$, includes $\{-3,-2,+0\}$. In this talk I will describe how classical knot invariants, Casson-Gordon theory, and Heegaard Floer theory, can be applied to study $U(K)$ for arbitrary knots K. (Received February 01, 2020)

1157-57-308 **Bena Tshishiku*** (bena_tshishiku@brown.edu) and **Mauricio Bustamante**.
Symmetries of exotic negatively curved manifolds.

Let N be a smooth manifold that is homeomorphic but not diffeomorphic to a hyperbolic manifold M. How much symmetry does N have? For example, does $\mathrm{Isom}(M)$ act on N by diffeomorphisms? Farrell-Jones showed that in general the answer is "No"; however, they do not rule out that the orientation-preserving subgroup of $\mathrm{Isom}(M)$ acts on N. In this talk, we will discuss this problem and its relation to Nielsen realization. The main result is a construction of examples of N where the largest subgroup of $\mathrm{Isom}(M)$ that acts on N has arbitrarily large index. (Received February 01, 2020)

1157-57-335 **Ka Ho Wong*** (daydreamkaho@tamu.edu). *Volume conjecture, geometric decomposition and deformation of hyperbolic structure.*

The volume conjecture of the Turaev-Viro invariant is a new topic in quantum topology. It has been shown that the $(2N+1)$-th Turaev-Viro invariant for a link complement can be expressed as a sum of norm squared of the colored Jones polynomial of the link evaluated at $t = \exp\left(\frac{2\pi i}{N+\frac{1}{2}}\right)$. This leads to the study of the asymptotics for the M-th colored Jones polynomials of links evaluated at $(N+\frac{1}{2})$-th root of unity, with a fixed limiting ratio of M to $N+\frac{1}{2}$. In this talk, I will first discuss how the asymptotics of the colored Jones polynomials of the Whitehead link is related to the (not necessarily complete) hyperbolic structures on its complement. Then, we will focus on some satellite links whose complements have more than one hyperbolic piece in the geometric decomposition, and relate the asymptotics of their colored Jones polynomials to the geometric structures on the geometric pieces. (Received February 02, 2020)

1157-57-337 **Clayton McDonald*** (mcdonafi@bc.edu). *Doubly slice odd pretzel knots.*

A knot K in S^3 is slice if it is the cross section of an embedded sphere in S^4, and it is doubly slice if the sphere is unknotted. Although slice knots are very well studied, doubly slice knots have been given comparatively less attention. We prove that an odd pretzel knot is doubly slice if it has 2n+1 twist parameters consisting of n+1 copies of a and n copies of -a for some odd integer a. Combined with the work of Issa and McCoy, it follows that these are the only doubly slice odd pretzel knots. (Received February 02, 2020)

1157-57-345 **Miriam Kuzbary*** (kuzbary@gatech.edu), 686 Cherry St NW, Atlanta, GA 30332. *Pure Braids and Link Concordance.* Preliminary report.

The set of knots in the 3-sphere with the connected sum operation forms a group when its quotient is taken by a 4-dimensional equivalence relation called concordance. This knot concordance group, defined by Fox and Milnor in 1966, can be contextualized as organizing problems about 3- and 4-dimensional spaces and the relationships between them. However, one of its limitations is that every 3-manifold is surgery on a link, not necessarily a knot, and thus it may be useful to understand such a group for links. Unfortunately, the connected sum of links is not well defined; in 1988, Le Dimet constructed the string link concordance group for links with n components to get around this problem. Notably, the knot concordance group is abelian while, for each n, the string link

concordance group on n strands is non-abelian as it contains the pure braid group on n strands as a subgroup. In this talk, I will discuss my recent result that even the quotient of each string link concordance group by its pure braid subgroup is non-abelian. (Received February 02, 2020)

1157-57-354 **Peter Feller**, **Allison N. Miller*** (allison.miller@rice.edu) and **Juanita Pinzon-Caicedo**. *Satellite operations and knot genera.*

The satellite construction, which associates to a pattern knot P in a solid torus and a companion knot K in the 3-sphere the so-called satellite knot $P(K)$, features prominently in knot theory and low-dimensional topology. Besides the intuition that $P(K)$ is "more complicated" than either P or K, one can attempt to quantify how the complexity of a knot changes under the satellite operation. In this talk, I'll discuss how several notions of complexity based on the minimal genus of an embedded surface whose boundary is the given knot and which satisfies various additional constraints change under satelliting. In the case of the classical Seifert genus of a knot, Schubert gives an exact formula. In the 4-dimensional context the situation is more complicated, and depends on whether we work in the smooth or topological category: the smooth category is asymptotically similar to the classical setting, but our main results show that the topological category is much weirder. This talk is based on joint work with Peter Feller and Juanita Pinzón-Caicedo. (Received February 02, 2020)

1157-57-367 **Paul Drube*** (paul.drube@valpo.edu). *Divisibility of the Jones-Kauffman Polynomial for Virtual Links.*

For any virtual link $L = S \cup T$ that may be decomposed into a pair of oriented n-tangles S and T, an oriented local move of type $T \mapsto \phi(T)$ is a replacement of T by another n-tangle $\phi(T)$ in a way that preserves the overall orientation of L. In this talk, we analyze the Jones-Kauffman polynomials of virtual links L_1, L_2 that differ via a local move of type $T \mapsto \phi(T)$. Divisibility conditions on the difference of polynomials $V(L_1) - V(L_2)$ are derived for broad classes of local moves that include the Δ-move and the double-Δ-move as special cases. One consequence of these divisibility results is a new necessary condition for any pair of classical knots to be S-equivalent. (Received February 02, 2020)

1157-57-379 **Jozef H. Przytycki*** (przytyck@gwu.edu), George Washington Universuty, Phillips Hall, Washington, DC 20052-0001. *Knot theory problems in Mathathon: past and future.*

I will describe the ideas behind the annual research projects undertaken in each December at GWU, which we call MATHATHON.

I will summarize the first five MATHATHONs:

1.(December 2015) Strict unimodality of a Plucking polynomial.

2.(December 2016) Search for torsion in Khovanov homology.

3.(December 2017) On multiplying curves in the Kauffman bracket skein algebra of the thickened four-holed sphere.

4.(December 2018) A Generalization of the Gram determinant of type A.

5.(December 2019) On Framing of knots in 3-manifolds;

(research at Mathathon V will be described at this special session by Dionne Ibarra and by Rhea Palak Bakshi).

I will also discuss a few problems considered for the future Mathathons (e.g. the problem of the shortest skein relation for the Fox n-colorings). (Received February 02, 2020)

1157-57-380 **Maggie Miller*** (maggiem@math.princeton.edu). *Dehn surgery on links vs. the Thurston norm.*

Gabai proved the Property R conjecture (surgery on a nontrivial knot cannot yield $S^1 \times S^2$) by studying the effect of Dehn surgery on Thurston norm. Specifically, he showed that if F is a Thurston norm-minimizing surface in a knot complement $S^3 \setminus \nu(K)$, then $\widehat{F}$ is norm-minimizing in the zero-surgery $S^3_0(K)$. We extend these techniques to study multi-component links in rational homology spheres. If $L \subset Y$ is such an n-component link ($n > 1$) with nondegenerate Thurston norm on its complement, then outside of an $(n-2)$-dimensional set of rays in $H_2(Y \setminus \nu(L), \partial; \mathbb{R})$, a norm-minimizing surface F remains norm-minimizing when capped off into $Y_{\partial F}(L)$. In particular, if $n = 2$, then $\widehat{F}$ is norm-minimizing for all but finitely many primitive $[F]$.

In this talk, I will sketch the motivation and main techniques used in the proof and then discuss some interesting further questions. (Received February 02, 2020)

1157-57-390 **Carmen Caprau*** (ccaprau@csufresno.edu), **Nicolle Gonzalez**, **Christine Ruey Shan Lee**, **Radmila Sazdanovic** and **Melissa Zhang**. *Link homologies and ribbon concordance.* Preliminary report.

In this talk, we explain how a ribbon concordance between two links induces injective maps on corresponding $sl(n)$ homologies, for each $n \geq 2$. (Received February 03, 2020)

1157-57-403 **Dongtai He**, **Diana Hubbard*** (diana.hubbard@brooklyn.cuny.edu) and **Linh Truong**. *Right-veering open books and the Upsilon invariant.*

Fibered knots in a three-manifold Y can be thought of as the binding of an open book decomposition for Y. One can ask how properties of the open book decomposition relate to properties of the corresponding knot. In this talk I will describe work with Dongtai He and Linh Truong that explores this question: we give a sufficient condition for the monodromy of an open book decomposition of a fibered knot to be right-veering from the concordance invariant Upsilon. I will discuss an application of this work to the Slice-Ribbon conjecture. (Received February 03, 2020)

1157-57-418 **Inanc Baykur*** (baykur@math.umass.edu) and **Mustafa Korkmaz** (korkmaz@metu.edu.tr). *Geography of surface bundles over surfaces.* Preliminary report.

An outstanding problem for surface bundles over surfaces, closely related to the symplectic geography problem in dimension four, is to determine for which fiber and base genera there are examples with non-zero signature. I will report on our recent progress (joint with Korkmaz), which resolves the question for all fiber and base genera except for about 20 pairs at the time of writing. (Received February 03, 2020)

1157-57-421 **Kursat Sozer*** (ksozer@indiana.edu). *Two-dimensional extended HQFTs.*

Topological quantum field theories (TQFTs), inspired by theoretical physics, produce manifold invariants behaving well under gluing. For every discrete group G, homotopy quantum field theories (HQFTs) are G-equivariant versions of TQFTs. In this talk, we review 1- and 2-dimensional HQFTs and their connections with representations of G and certain G-algebras respectively. Later generalizing both, we define 2-dimensional extended HQFTs and state a theorem about their classification. As an application, we discuss the connection of this result with the (G x SO(2))-structured cobordism hypothesis due to J. Lurie. If time permits we discuss the ongoing project about a generalization of the result in a different direction. (Received February 03, 2020)

1157-57-448 **Miriam Kuzbary*** (kuzbary@gatech.edu), 686 Cherry St NW, Atlanta, GA 30332. *An Analogue of Milnor's Invariants for Knots in 3-Manifolds.*

Milnor's invariants are some of the more fundamental oriented link concordance invariants; they behave as higher order linking numbers and can be computed using combinatorial group theory (due to Milnor), Massey products (due to Turaev and Porter), and higher order intersections (due to Cochran). In this paper, we generalize the first non-vanishing Milnor's invariants to oriented knots inside a closed, oriented 3-manifold M. We call this the Dwyer number of a knot and show methods to compute it for null-homologous knots inside connected sums of $S^2 \times S^1$. We further show in this case the Dwyer number provides the weight of the first non-vanishing Massey product in the knot complement in the ambient manifold. Additionally, we prove the Dwyer number detects a family of knots K in $\#^\ell S^2 \times S^1$ bounding smoothly embedded disks in $\natural^\ell S^2 \times D^2$ which are not concordant to the unknot. (Received February 03, 2020)

1157-57-464 **Nickolas A Castro**, **Mark C Hughes**, **Gabriel Islambouli** and **Maggie Miller*** (maggiem@math.princeton.edu). *Trisections of non-compact 4-manifolds.*

I will define a trisection of a non-compact 4-manifold. I will discuss their existence and uniqueness (after first discussing the situation for Heegaard splittings of non-compact 3-manifolds, studied by Taylor and then Kirker) and give some interesting examples. (Received February 03, 2020)

1157-57-492 **Siddhi Krishna*** (siddhi.krishna@bc.edu). *Taut Foliations and Positive Braids.* Preliminary report.

I will describe a construction for building taut foliations in manifolds obtained by Dehn surgery along positive braid knots (i.e. knots obtained by taking the closure of a positive braid). Namely, we'll discuss how we can reduce building foliations to a combinatorial problem. This has implications towards the L-space conjecture, which predicts that every non-L-space admits a taut foliation. No background in Heegaard Floer homology or foliation theory will be assumed. (Received February 03, 2020)

1157-57-501 **McKee Krumpak*** (mkrumpak@brandeis.edu). *Cobordism maps in monopole Floer and twisted h-invariants.* Preliminary report.

I will discuss ongoing work constructing a generalization of Frøyshov's h-invariant and a conjectural framework for computing cobordism maps in monopole Floer homology. (Received February 03, 2020)

1157-57-542 **Micah Chrisman** (chrisman.76@osu.edu) and **Puttipong Pongtanapaisan*** (puttipong-pongtanapaisan@uiowa.edu). *Double null concordance of knots in thickened surfaces.* Preliminary report.

In 2008, Turaev introduced a notion of concordance for knots in thickened surfaces. This generalizes the classical idea of concordance of knots in the 3-sphere by allowing the concordance annulus to be embedded in a more general thickened 3-manifold that is not necessarily the thickened 3-sphere. In this talk, I will discuss joint work with Micah Chrisman where we generalize the notion of double null concordance to the setting of knots in thickened surfaces. More specifically, I will present examples and non-examples of double null concordant knots in thickened surfaces. (Received February 04, 2020)

1157-57-548 **Thomas E. Mark***, PO Box 400137 Kerchof Hall, University of Virginia, Charlottesville, VA 22904, and **Bulent Tosun**. *Rationally and holomorphically convex domains with specified boundary.*

A fundamental problem in low-dimensional topology is the determination of those homology 3-spheres Y that bound smooth, contractible 4-manifolds; or, relatedly, those that embed smoothly in 4-dimensional Euclidean space. More tools can be brought to bear on more refined versions of the question, for example a holomorphic version that asks whether Y is homeomorphic to the boundary of a Stein domain in complex 2-space, or a symplectic version that requires Y to be the boundary of a rationally convex domain. As pointed out by Weimin Chen, examples satisfying some of these conditions but not others both highlight differences between Stein topology in complex dimension 2 compared to higher dimensions, and also may have bearing on the existence question for smooth 4-manifolds homeomorphic but not diffeomorphic to the projective plane. We will discuss ideas, constructions, and obstructions related to these questions, some of which are joint work with Bülent Tosun. (Received February 04, 2020)

1157-57-570 **Wade Bloomquist***, wbloomquist3@gatech.edu, and **Thang Le**. *Stated Skein Algebras at Roots of Unity.* Preliminary report.

The stated skein algebra of a surface is a refinement of the usual Kauffman bracket skein algebra of a surface which introduces diagrams containing stated arcs meeting the boundary of the surface. The power of this refinement follows from a splitting homomorphism between algebras induced by cutting a surface along an ideal arc. Additionally, there is an isomorphism between the stated skein algebra of the bigon and $\mathcal{O}_{q^2}(\mathrm{SL}(2))$. We discuss some algebraic properties of these algebras, explore the extension to stated skein modules of 3-manifolds, and keep an eye toward applications in low dimensional topology. (Received February 04, 2020)

1157-57-572 **Kelvin Guilbault*** (kguilbau@iu.edu). *Biperiodic Knots.*

Define a knot $K \subset S^3$ to be (m,n)-biperiodic if there is a group action $C_m \times C_n \curvearrowright S^3$ leaving K invariant such that $C_m \times 1$ and $1 \times C_n$ have disjoint circular fixed sets which combine to form a Hopf link. In this talk we review some work by Murasugi in periodic knots and explain why this work suggests the above definition as a natural generalization. We go on to see to what extent Murasugi's congruence relation on the Alexander polynomial of periodic knots generalizes to this context, deriving a factorization for any biperiodic knot polynomial. We conclude by discussing a few consequences. (Received February 04, 2020)

1157-57-574 **Alexandra Kjuchukova** (sashka@mpim-bonn.mpg.de) and **Kent E. Orr*** (korr@indiana.edu), 831 3rd St, Rawles Hall, Department of Mathematics, Bloomington, IN 47405. *Extending dihedral representations of knot groups over groups of surfaces complements in the 4-disk.* Preliminary report.

Remarkably, I have not found this classical sounding problem solved in the literature. This work is motivated by a ribbon obstruction derived from Kjuchukova's Ξ invariant. Given a dihedral representation of a knot group, I will define a new invariant which vanishes if and only if the representation extends over the group of a surface exterior in the 4-disk. Additionally, I'll give computable and equivalent (although not obviously so) statements of the theorem, with briefly described examples. (Received February 04, 2020)

1157-57-590 **Dylan P Thurston*** (dpthurst@indiana.edu), 831 E Third St', Bloomington, IN 47401, and **Roland van der Veen**. *Sutured manifolds and Hopf algebras.* Preliminary report.

Extending on the work of Greg Kuperberg and Rohit Thomas, we sketch how to construct invariants of (framed, marked) sutured 3-manifolds from (open) Hopf algebra expressions in a finite-dimensional Hopf algebra, without any assumption that the manifold is involutory or balanced. In particular, this should give a direct, geometric interpretation of the Drinfel'd double construction. (Received February 04, 2020)

1157-57-608 **Charles D Frohman** and **Joanna Kania-Bartoszynska***, jkaniaba@nsf.gov. *Geometric Kauffman Bracket in Closed Hyperbolic 3-Manifolds.*

Given a closed hyperbolic 3-manifold M and a "good" representation of its fundamental group into SL(2,C) we obtain an invariant of links in M that generalizes the Kauffman bracket. (Received February 04, 2020)

1157-57-626 **Ziva Myer***, zmyer@math.duke.edu. *Comparing Legendrian Invariants.* Preliminary report.

In contact geometry there are a few different tools used to construct invariants for Legendrian submanifolds: pseudoholomorphic curves, generating families, and constructible sheaves. While the underlying theory of these techniques employ vastly distinct flavors of mathematics, there has been some evidence that certain Legendrian invariants defined through these methods contain the same information. I will elaborate more on this phenomenon and highlight where some of my work appears in this story. (Received February 04, 2020)

58 ► Global analysis, analysis on manifolds

1157-58-46 **Jingzhi Tie*** (jtie@uga.edu), Department of Mathematics, Athens, GA 306027403, and **Der-Chen Chang**, **Shu-Cheng Chang** and **Yingbo Han**. *A CR Analogue of Yau's Conjecture On Pseudoharmonic Functions of Polynomial Growth.*

Cheng and Yau derived the well-known gradient estimate for positive harmonic functions and obtained the classical Liouville theorem, which states that any bounded harmonic function is constant in the complete noncompact Riemannian manifolds with nonnegative Ricci curvature. I will talk about the joint work with Der-Chen Chang, Shu-Cheng Chang, and Yingbo Han. We study the CR analog of Yau's conjecture on the space $\mathcal{H}^d(M)$ consisting of all pseudoharmonic functions of polynomial growth of degree at most d in a complete noncompact pseudohermitian $(2n+1)$-manifold (see Definition **??**). We will show that the first part of CR Yau's conjecture holds for pseudoharmonic functions of polynomial growth via Peter Li's method, which is more adaptable to the case of pseudohermitian geometry. (Received January 07, 2020)

1157-58-47 **Fabrice Baudoin***, 341 Mansfield Road, Storrs, CT 06269-1009, and **Erlend Grong**, **Gianmarco Molino** and **Luca Rizzi**. *Sub-Laplacian comparison theorems in H-type sub-Riemannian manifolds.*

On H-type sub-Riemannian manifolds we establish sub-Hessian and sub-Laplacian comparison theorems which are uniform for a family of approximating Riemannian metrics converging to the sub-Riemannian one. We also prove a sharp sub-Riemannian Bonnet-Myers theorem that extends to this general setting results previously proved on contact and quaternionic contact manifolds. (Received January 07, 2020)

1157-58-159 **Silvia Anjos**, **Jun Li** and **Tian-Jun Li*** (lixxx248@umn.edu), 206 Church Street SE, Minneapolis, MN 55455, and **Martin Pinsonnault**. *Stability of the symplectomorphism group of rational surfaces.*

We apply Zhang's almost Kähler Nakai-Moishezon theorem to establish a complete stability result for the symplectomorphism group of a rational surface with Euler number up to 12. (Received January 26, 2020)

1157-58-167 **Mikhail Karpukhin*** (mkarpukh@uci.edu), 340 Rowland Hall (Bldg.# 400), University of California, Irvine, Irvine, CA 92617. *Isoperimetric inequalities for Laplacian eigenvalues on surfaces.*

The study of upper bounds for Laplacian eigenvalues on Riemannian manifolds under the volume constraint is a classical problem of spectral geometry going back to J. Hersch, P. Li and S.-T. Yau. The particular interest to this problem stems from a surprising connection to the theory of minimal surfaces in spheres. In the present talk we will survey some recent results in the area, including an optimal isoperimetric inequality for all Laplacian eigenvalues on the sphere and the projective plane. (Received January 26, 2020)

1157-58-207 **Xiaolong Han*** (xhan25@illinois.edu). *Eigenvalue estimate on Geometrically Finite Hyperbolic Manifolds with Infinite Volume.* Preliminary report.

In this talk, we compare the eigenvalues of geometrically finite hyperbolic manifolds with infinite volumes with the eigenvalues of compact nice core part of the manifold. We decompose the manifold into some nice metric structures, like tubes, cusps, ends with infinite volumes. We then use the estimate on those parts to derive information about the eigenvalues on the core part. It can have potential use in estimating the number of small eigenvalues in the manifold with infinite volume. (Received January 29, 2020)

1157-58-248 **Nate Eldredge**, **Maria Gordina** and **Laurent Saloff-Coste*** (lps2@cornell.edu). *Uniform doubling on SU(2).*

We consider the collection of all left-invariant Riemannian (or sub-Riemannian) metrics on the compact group SU(2) (the 3-sphere). We show that these metrics satisfy the doubling volume property, uniformly. (Received January 30, 2020)

1157-58-328 **Ely Kerman*** (ekerman@illinois.edu) and **Yuanpu Liang**. *On capacities and their blind spots.*

In this talk I will describe some refinements of the formulae for the Gutt-Hutchings capacities of convex and concave toric domains, and the properties of capacities that these refinements reveal. This is joint work with Yuanpu Liang. (Received February 01, 2020)

1157-58-414 **Nathan Fisher*** (nathan.fisher@tufts.edu), 503 Boston Ave, Medford, MA 02155. *Sub-Finsler metrics and the Heisenberg group at infinity.* Preliminary report.

In this talk, we will discuss a class of sub-Finsler metrics on the real Heisenberg group $H(\mathbb{R})$, arising as asymptotic cones of word metrics on the integer Heisenberg group $H(\mathbb{Z})$ for various generating sets. We will describe new results on boundaries for these Carnot-Carthéodory metrics, with applications to the study of random walks on $H(\mathbb{Z})$. This is joint work with Sebastiano Nicolussi Golo. (Received February 03, 2020)

1157-58-511 **Robert Neel*** (robert.neel@lehigh.edu) and **Ludovic Sacchelli**. *Uniform, localized asymptotics for sub-Riemannian heat kernels and diffusions.*

We show how Molchanov's method provides a systematic approach to studying the small-time asymptotics of the heat kernel on a sub-Riemannian manifold away from any abnormal minimizers, including local, uniform bounds for the heat kernel and its derivatives. Moreover, the method extends naturally to log-derivatives of the heat kernel and the closely related law of large numbers for the corresponding bridge process. (Received February 03, 2020)

1157-58-553 **Luís Diogo**, Universidade Federal Fluminense, Niterói, Brazil, and **Samuel Lisi*** (stlisi@olemiss.edu), University of Mississippi, Department of Mathematics, PO Box 1848, University, MS 38677-1848. *Computations of Symplectic Homology for affine algebraic varieties.*

Symplectic homology is an important Floer-theoretic invariant of Liouville domains (with extensions to more general domains). In work with Luis Diogo, we have developed a method for computing the symplectic homology of a class of examples: affine algebraic varieties, obtained as the complement of a smooth hyperplane section. Our computation describes symplectic homology in terms of Gromov-Witten invariants and of relative Gromov-Witten invariants. I will explain our existing work, and will discuss some extensions to understanding higher operations in symplectic homology by means of our construction. (Received February 04, 2020)

60 ► Probability theory and stochastic processes

1157-60-12 **Shuyang Bai*** (bsy9142@gmail.com), 331 Turtle Creek Dr, Winder, GA 30680. *New representations of Hermite processes.* Preliminary report.

Hermite processes are a class of self-similar processes with stationary increments. They often arise in limit theorems under long-range dependence. We introduce new representations of Hermite processes involving the local time of intersecting stationary stable regenerative sets. (Received October 16, 2019)

1157-60-30 **Cristina Tone*** (cristina.tone@louisville.edu). *A Central Limit Theorem for Quadruple-Wise Independent Arrays of Random Variables.*

A CLT is proved to hold for a triangular array of strongly mixing random variables satisfying quadruple-wise independence and the Lindeberg condition. (Received December 17, 2019)

1157-60-66 **Timo O Seppäläinen*** (seppalai@math.wisc.edu), Mathematics Department, UW-Madison, 480 Lincoln Drive, Madison, WI 53706. *Busemann measures and the geometry of the corner growth model.*

The corner growth model is a last-passage percolation model of random growth on the square lattice. It lies at the nexus of several branches of mathematics: probability, statistical physics, queueing theory, combinatorics, and integrable systems. It has been studied intensely for 40 years. We introduce an analytical tool, the Busemann measure, that relates naturally to certain geometric properties of the geodesics of the corner growth model. The Busemann measure is the Lebesgue-Stieljes measure of the Busemann function on the space of spatial directions. In the exactly solvable exponential case properties of the Busemann measures are explicitly available and can be used to elucidate a global picture of uniqueness and coalescence of semi-infinite geodesics. Based on collaborations with Louis Fan (Indiana), Firas Rassoul-Agha and Chris Janjigian (Utah). (Received January 12, 2020)

1157-60-68 **Michael Damron*** (mdamron6@gatech.edu), 686 Cherry St., Atlanta, GA 30324, **David M Harper** (dharper80@gatech.edu), 686 Cherry St., Atlanta, GA 30324, **Jack Hanson** (jhanson@ccny.cuny.edu), 160 Convent Ave., New York, NY 10031, and **Wai-Kit Lam** (wlam@umn.edu), 206 Church St. SE, Minneapolis, MN 55255. *Critical first-passage percolation in two dimensions.* Preliminary report.

In first-passage percolation, one places i.i.d. nonnegative weights (t_e) on the edges of a graph and studies the induced pseudometric T. In the case of $\mathbb{Z}^2$, it is known that when $p := \mathbb{P}(t_e = 0)$ is $< 1/2$, $T(0, x)$ scales linearly in x, whereas when $p > 1/2$, it is stochastically bounded in x. I will discuss the so-called critical case, where $p = 1/2$. After reviewing work with X. Wang and W.-K. Lam, in which we solved for the exact scaling of $T(0, x)$ as a function of the distribution of t_e, I will talk about ongoing work with J. Hanson, D. Harper, and W.-K. Lam on a dynamical version of this process in which edge-weights update according to i.i.d. exponential clocks. I will present preliminary results on the dimension of sets of exceptional times during which there are (or are not) infinite paths of finite total passage time. (Received January 13, 2020)

1157-60-91 **Sayan Banerjee**, **Fabrice Baudoin**, **Maria Gordina** and **Phanuel Mariano*** (pmariano@newhaven.edu). *The Cheng-Yau gradient estimate for Carnot groups and sub-Riemannian manifolds.*

We recall the Cheng-Yau gradient inequality and its implications in the Riemannian setting. We discuss how the Cheng-Yau inequality can be proven on two classes of sub-Riemannian manifolds: Carnot groups and sub-Riemannian manifolds satisfying a generalized curvature-dimension inequality. Two approaches will be presented. A coupling approach and an analytic approach using a curvature-dimension condition. (Received January 17, 2020)

1157-60-94 **Dan Pirjol*** (dpirjol@gmail.com), Stevens Institute of Technology, School of Business, Babbio Center room 303A, Hoboken, NJ 07030, and **Lingjiong Zhu**. *Asymptotics of the log-normal SABR model.*

The log-normal SABR model is one of the simplest stochastic volatility models used in financial practice. We propose a new time discretization scheme for this model, and study analytically its asymptotic properties in the limit of a very large number of time steps. We derive almost sure, fluctuations and large deviations for the log-asset price, and obtain an exact representation for the implied volatility surface for arbitrary maturity and strike in a certain small vol-of-vol limit. (Received January 18, 2020)

1157-60-95 **Wei-Kuo Chen***, 127 Vincent Hall 206 Church St. SE, Minneapolis, MN 55455, and **Wai-Kit Lam**. *Order of Fluctuations of the SK Model at Critical Temperature.*

I will discuss the order of fluctuations in the Sherrington-Kirkpatrick mean field spin glass model. In particular, I will focus on the predictions concerning the free energy and present an elementary approach for obtaining a logarithmic bound on its variance at the critical temperature. (Received January 18, 2020)

1157-60-98 **Xuefeng Gao**, **Mert Gurbuzbalaban** and **Lingjiong Zhu*** (zhu@math.fsu.edu), 1017 Academic Way, Room 208, Tallahassee, FL 32306. *Global Convergence of Stochastic Gradient Hamiltonian Monte Carlo for Non-Convex Stochastic Optimization.*

Stochastic gradient Hamiltonian Monte Carlo (SGHMC) is a variant of stochastic gradient with momentum where a controlled and properly scaled Gaussian noise is added to the stochastic gradients to steer the iterates towards a global minimum. Many works reported its empirical success in practice for solving stochastic non-convex optimization problems, in particular it has been observed to outperform overdamped Langevin Monte Carlo-based methods such as stochastic gradient Langevin dynamics (SGLD) in many applications. In this

work, we study two variants of SGHMC based on two alternative discretizations of the underdamped Langevin diffusion. We provide finite-time performance bounds for the global convergence of both SGHMC variants for solving stochastic non-convex optimization problems with explicit constants. Our results lead to non-asymptotic guarantees for both population and empirical risk minimization problems. For a fixed target accuracy level, on a class of non-convex problems, we obtain complexity bounds for SGHMC that can be tighter than those for SGLD. These results show that acceleration with momentum is possible in the context of global non-convex optimization. (Received January 19, 2020)

1157-60-100 **Carl Mueller*** (carl.e.mueller@rochester.edu), Department of Mathematics, University of Rochester, Rochester, NY 14627, and **Robert Dalang** and **Yimin Xiao**. *Polarity of Almost All Points for Systems of Non-Linear Stochastic Heat Equations in the Critical Dimension.*

We study vector-valued solutions $u(t,x) \in \mathbb{R}^d$ to systems of nonlinear stochastic heat equations with multiplicative noise:

$$\frac{\partial}{\partial t}u(t,x) = \frac{\partial^2}{\partial x^2}u(t,x) + \sigma(u(t,x))\dot{W}(t,x).$$

Here $t \geq 0$, $x \in \mathbb{R}$ and $\dot{W}(t,x)$ is an $\mathbb{R}^d$-valued space-time white noise. We say that a point $z \in \mathbb{R}^d$ is polar if

$$P\{u(t,x) = z \text{ for some } t > 0 \text{ and } x \in \mathbb{R}\} = 0.$$

We show that in the critical dimension $d = 6$, almost all points in $\mathbb{R}^d$ are polar. (Received January 20, 2020)

1157-60-102 **Yen Q Do*** (yendo@virginia.edu), Charlottesville, VA 22904. *Real roots of random polynomials.*

Finding the real roots for a polynomial is often a challenging talk, even numerically, especially when the polynomial has high degree. In this talk I will describe some recent work (by myself and also with coauthors, H. Nguyen, O. Nguyen, V. Vu) about the distribution of the real roots of random polynomials. I will also give a brief survey of the history of the subject. (Received January 20, 2020)

1157-60-132 **Alfredo Garbuno-Inigo** and **Franca Hoffmann***, 1200 E California Blvd, Pasadena, CA 91125, and **Wuchen Li** and **Andrew M. Stuart**. *Kalman-Wasserstein gradient flows for solving inverse problems.*

We study a class of interacting particle systems that may be used for optimization. By considering the mean-field limit one obtains a nonlinear Fokker-Planck equation. This equation exhibits a novel gradient structure in probability space, based on a modified Wasserstein distance which reflects particle correlations: the Kalman-Wasserstein metric. This setting gives rise to a methodology for calibrating and quantifying uncertainty for parameters appearing in complex computer models which are expensive to run, and cannot readily be differentiated. This is achieved by connecting the interacting particle system to ensemble Kalman methods for inverse problems.

This is joint work with Alfredo Garbuno-Inigo (Caltech), Wuchen Li (UCLA) and Andrew Stuart (Caltech). (Received January 23, 2020)

1157-60-133 **Elena Kosygina*** (elena.kosygina@baruch.cuny.edu), **Thomas Mountford** and **Jonathon Peterson**. *Convergence of random walks on Markovian cookie stacks to Brownian motion perturbed at extrema.* Preliminary report.

We consider one-dimensional excited random walks (ERWs) with i.i.d. Markovian cookie stacks in the non-boundary recurrent regime. We prove that under the diffusive scaling these walks converge in the standard Skorokhod topology to a constant multiple of a Brownian motion perturbed at extrema (BMPE). More precisely, let X_n denote the position of the ERW at time $n \geq 0$ by X_n and θ^+, θ^- be its parameters. These parameters always satisfy $v := 1 - \theta^+ - \theta^- > 0$ and have explicit formulas in terms of the transition probabilities matrix of the cookie Markov chain, the distribution of the first cookie in the stack, and the drifts assigned to each cookie type. We show that in the non-boundary recurrent regime, i.e. when $\theta^+, \theta^- \in (-1, 1)$, $\left((vn)^{-1/2}X_{\lfloor nt \rfloor}\right)_{t \geq 0}$ converge weakly as $n \to \infty$ in the standard Skorokhod topology to (θ^+, θ^-)-perturbed BMPE $(W(t))_{t \geq 0}$. The process $(W(t))_{t \geq 0}$ is the unique path-wise solution of the equation

$$W(t) = B(t) + \theta^+ \sup_{s \in [0,t]} W(s) + \theta^- \inf_{\in [0,t]} W(s), \quad t \geq 0,$$

where $(B(t))_{t \geq 0}$ is the standard Brownian motion. The techniques are based on generalized Ray-Knight theorems and couplings with discretized versions of the limiting BMPE. (Received January 23, 2020)

1157-60-136 **Yingxin Mu** and **Eviatar B Procaccia*** (procaccia@tamu.edu), Mailstop 3368, Department of Mathematics, Texas A&M University, College Station, TX 77843, and **Yuan Zhang**. *Scaling limit of DLA on a long line segment.*

We prove that the bulk of DLA (Diffusion Limited Aggregation) starting from a long line segment on the x-axis has a scaling limit to the stationary DLA process (SDLA). The main phenomenological difficulty is the multi-scale, non-monotone interaction of the DLA arms. We overcome this via a coupling scheme between the two processes and an intermediate DLA process with absorbing mesoscopic boundary segments. (Received January 23, 2020)

1157-60-155 **PANKI KIM** (pkim@snu.ac.kr) and **Renming Song*** (rsong@illinois.edu), Department of Mathematics, University of Illinois, 1409 W Green St, Urbana, IL 61801, and **Zoran Vondracek** (pkim@snu.ac.kr). *On potential theory of Markov processes with jump kernels decaying at the boundary.*

We study processes in an open set $D \subset \mathbb{R}^d$ defined via Dirichlet forms with jump kernels of the form $J^D(x,y) = j(|x-y|)B(x,y)$ and critical killing functions. Here $j(|x-y|)$ is the Lévy density of an α-stable process in $\mathbb{R}^d$. The main novelty is that the term $B(x,y)$, depending on three parameters $\beta_1, \beta_2, \beta_3$, tends to 0 when x or y approach the boundary of D. Under some general assumptions on $B(x,y)$, we prove that the Harnack inequality and Carleson's estimate.

In the second part, we specialize to the case of the half-space $D = \mathbb{R}^d_+$ and the killing function $\kappa(x) = cx_d^{-\alpha}$. Our main result is a boundary Harnack principle which says that, for any $p > (\alpha-1)_+$, there are values of $\beta_1, \beta_2, \beta_3$ and the constant c such that non-negative harmonic functions of the process must decay at the rate x_d^p if they vanish near a portion of the boundary for any $p > (\alpha-1)_+$, there are values of the parameters $\beta_1, \beta_2, \beta_3$ and the constant c such that We further show that there are values of $\beta_1, \beta_2, \beta_3$ for which the boundary Harnack principle fails despite the fact that Carleson's estimate is valid. (Received January 26, 2020)

1157-60-166 **Tomoyuki Ichiba*** (ichiba@pstat.ucsb.edu), South Hall 5607 A, Department of Statistics, University of California, Santa Barbara, Santa Barbara, CA 93106. *Stochastic games in infinite directed chain.*

We shall consider stochastic game problems of infinitely many players on an infinite directed chain graph, where each player associated with a vertex of the graph controls the drift of a diffusion process driven by a Brownian motion and minimizes an expected cost functional which depends on the states of players in the neighborhood in the directed graph. We derive a Nash equilibrium of such game and compare with that of the mean-field game. We also compare it with the stochastic game of finitely many players in a finite directed chain graph. Then we discuss connection to systemic risk problem in financial network. Part of research is joint work with Jean-Pierre Fouque and Yichen Feng. (Received January 26, 2020)

1157-60-171 **Christopher Janjigian*** (janjigia@math.utah.edu), 155 S 1400 E, LCB 207, Salt Lake Cty, UT 84112-0090, and **Elnur Emrah** and **Timo Seppäläinen**. *Geodesics in inhomogeneous exponential last passage percolation.* Preliminary report.

We study geodesics in the exactly solvable model of directed last passage percolation with exponential weights, where the weights are independent but not identically distributed. The impact of inhomogeneity can be seen at many different levels in the model. In this talk, we investigate the structure of geodesics and how they differ from the classical, homogeneous model with i.i.d. weights. (Received January 26, 2020)

1157-60-172 **Dohyun Ahn**, **Nan Chen** and **Kyoung-Kuk Kim*** (catenoid@kaist.ac.kr), 291 Daehak-ro, Yuseong-gu, Daejeon, 34141, South Korea. *Systemic Risk Quantification via Shock Amplification in Financial Networks.*

We consider the Eisenberg-Noe model for financial networks, focusing on random shocks to financial institutions. Using duality, we characterize shock amplification caused by the network structure and the condition when a specific group of banks (e.g., SIFI) fails. This finding enables us to improve our understanding of shock propagation in financial networks. To be specific, we obtain robust bounds of default probabilities when only partial network information is available, and we observe that the link structure of the network contains crucial information. This is also confirmed by looking at asymptotic default probabilities in a small shock regime. With such analytical tools, systemic risk capital which prevents the default of target banks is discussed using chance-constrained optimization. All the claims are numerically illustrated by an actual European banking network. (Received January 27, 2020)

1157-60-185 **José E. Figueroa-López**, **Ruoting Gong*** (rgong2@iit.edu) and **Yuchen Han**.
Optimum Thresholding Using Mean Square Error for Tempered-Stable-Like Lévy Models.

We consider a semimartingale model X with jumps for the log-return of an asset price. Separately identifying the contribution of the Brownian part and the one of the jumps to the price variations is crucial for model assessing and for improving volatility forecasting. With discrete observations, one of the most efficient estimators for non-parametrically disentangling the jumps from integrated variance is the so-called Truncated Realized Variance (TRV), especially in the presence of infinite activity jump component. When X is a Levy process, it was shown by Figueroa-López and Mancini 2019 (FLM19) that there exists a unique threshold ε_h^* that minimizes the mean-square error of the TRV estimator of the continuous variance. The asymptotic behavior of ε_h^*, as the time step $h \to 0$, was derived in FLM19 for the cases when X has finite activity jumps or X is symmetric and strictly stable. In this talk, we generalize the above work to the so-called tempered-stable-like (TSL) Lévy models. We show that the asymptotic behavior of the optimal threshold, obtained in FLM19 holds true for TSL models. Our numerical study demonstrates that the finite-sample performance of the estimator is superior to any other method proposed in the literature. (Received January 27, 2020)

1157-60-190 **Otavio Menezes*** (omenezes@purdue.edu) and **Milton Jara**. *Scaling limits for a slowed random walk driven by symmetric exclusion.*

We look at a simple symmetric exclusion process in the 1d lattice and at a random walk on the same space. When on top of particles, the walker has a drift to the left, when on top of holes it has a drift to the right. Under weakly asymmetric scaling, we prove a law of large numbers and a functional central limit theorem for the position of this random walk. The proof uses techniques from the field of hydrodynamic limits to study the fluctuations of the number of particles in large boxes around the walker. (Received January 28, 2020)

1157-60-193 **Zaoli Chen** and **Gennady Samorodnitsky*** (gs18@cornell.edu), School of Operations Research, and Information Engineering, Ithaca, NY 14850. *Extremal clustering under moderate long range dependence and moderately heavy tails.*

We study clustering of the extremes in a stationary sequence with subexponential tails in the maximum domain of attraction of the Gumbel distribution. We obtain functional limit theorems in the space $D[0,\infty)$ and in the space of random sup-measures. The limits have the Gumbel distribution if the memory is only moderately long. However, as our results demonstrate rather strikingly, the "heuristic of a single big jump" could fail even in a moderately long range dependence setting. As the tails become lighter, the extremal behavior of a stationary process may depend on multiple large values of the driving noise. (Received January 28, 2020)

1157-60-221 **Erkan Nane*** (ezn0001@auburn.edu), 221 Parker Hall, Auburn, AL 36849. *Blow-up results for space–time fractional Dynamics.*

Consider non-linear time-fractional stochastic reaction-diffusion equations of the following type,

$$\partial_t^\beta u_t(x) = -\phi(-\Delta)u_t(x) + I_t^{1-\beta}[b(u) + \sigma(u)\ \dot{F}\ (t,x)]$$

in $(d+1)$ dimensions, where $\beta \in (0,1)$, and the operator ∂_t^β is the Caputo fractional derivative while $\phi : (0,\infty) \to (0,\infty)$ is a Bernstein function, and $I_t^{1-\beta}$ is the Riesz fractional integral operator. The forcing noise denoted by $\dot{F}\ (t,x)$ is a Gaussian noise. These equations might be used as a model for materials with random thermal memory. We derive non-existence (blow-up) of global random field solutions under some additional conditions, most notably on b, σ and the initial condition.

These results are our recent joint work with Sunday Asogwa, Chansong Deng, Mohammud Foondun, Wei Liu, and Jebessa Mijena. (Received January 29, 2020)

1157-60-249 **Kiseop Lee** (kiseop@purdue.edu), 250 N. University St, Purdue Department of Statistics, West Lafayette, IN 47907, and **Hyoeun Lee*** (hyoeun@illinois.edu), 725 South Wright Street, Illini Hall, Room 101, Champaign, IL 61820. *Optimal execution with liquidity risk in a diffusive order book market.*

We study an optimal order placement strategy with the presence of a liquidity cost. In this problem, a stock trader wishes to clear her large inventory by a predetermined time horizon (T). A trade can be placed using both limit and market orders, and a large market order faces an adverse price movement caused by the liquidity risk. First, we study a single period model where the trader places limit orders and/or market orders at the beginning. We show the behavior of optimal amount of market order (m^*) and optimal placement of limit order (y^*) under different market conditions. Next, we extend to a multi-period model, where the trader can cancel the non-executed limit orders and resubmit market or limit orders to clear the remaining inventory at multiple time steps.

This is joint work with Professor Kiseop Lee. (Received January 30, 2020)

1157-60-252 **Cheuk-Yin Lee** (leecheu1@msu.edu), 619 Red Cedar Road, Department of Statistics and Probability, Michigan State University, East Lansing, MI 48824, and **Yimin Xiao*** (xiao@stt.msu.edu), 619 Red Cedar Road, Department of Statistics and Probability, Michigan State University, East Lansing, MI 48824. *Regularity Properties and Singularity Propagation of Stochastic Wave Equation.* Preliminary report.

Consider the linear stochastic wave equation (SWE) driven by a Gaussian noise which is white in time and colored in space. We show that the solution satisfies a certain form of strong local nondeterminism and we use this property to derive regularity properties such as the exact uniform modulus of continuity for the solution.

On the other hand, by applying general Gaussian methods, we show the existence and the propagation of singularities of the solution of SWE. The combined results show interesting fine structures of the sample function of the solution. (Received January 30, 2020)

1157-60-256 **Rodrigo Banuelos*** (banuelos@purdue.edu), **Phanuel Mariano** (pmariano@newhaven.edu) and **Jing Wang** (jingwang@purdue.edu). *Extremal problems for eigenvalues and torsion functions.* Preliminary report.

We will discuss several extremal problems that have received a lot of attention in recent years concerning comparison inequalities for the lowest eigenvalue of the Dirichlet Laplacian and maximal expected life times for Brownian motion (torsion function) in domains of Euclidean space. We list improvements to various inequalities in the literature and discuss some conjectures on the shape of the domains which are expected to be extremal for these problems. (Received January 30, 2020)

1157-60-298 **Hyungbin Park*** (hyungbin@snu.ac.kr), Seoul, 08826, South Korea, and **Jonghwa Park** (tmfmfkal01@snu.ac.kr). *Pricing and hedging short-maturity Asian options in local volatility models.*

This talk discusses the short-maturity behavior of Asian option prices and hedging portfolios. We consider the risk-neutral valuation and the delta value of the Asian option having a Hölder continuous payoff function in a local volatility model. The main idea of this analysis is that the local volatility model can be approximated by a Gaussian process at short maturity T. By combining this approximation argument with Malliavin calculus, we conclude that the short-maturity behaviors of Asian option prices and the delta values are approximately expressed as those of their European counterparts with the Asian volatility

$$\sigma_A(T) := \sqrt{\frac{1}{T^3}\int_0^T \sigma^2(t, S_0)(T-t)^2\, dt}\,,$$

where $\sigma(\cdot,\cdot)$ is the local volatility function and S_0 is the initial value of the stock. In addition, we show that the convergence rate of the approximation is determined by the Hölder exponent of the payoff function. (Received February 01, 2020)

1157-60-304 **Yongjia Xie*** (xie287@purdue.edu). *Functional Weak Limit of Random Walks in Cooling Random Environment.*

In the last few decades, many results of random walks in dynamic random environment have been made. One of the dynamics of the environment is called the cooling random environment which lets the underlying environment be refreshed on a deterministic time sequence. This model was first introduced by Luca Avena and Frank den Hollander in 2017. Later, they worked with Yuki Chino and Conrado da Costa to prove several properties such as LLN, CLT, and recurrent condition in different cooling regimes. In this talk, I will first introduce the basic setup and the above existing results of random walks in cooling random environment(RWCRE). Then I'll show my recent work about the functional weak limit of RWCRE under two cooling cases, namely, polynomial and exponential. (Received February 01, 2020)

1157-60-305 **Charles K Smart*** (smart@math.uchicago.edu). *Localization and unique continuation on the integer lattice.*

I will discuss recent results on localization for the Anderson–Bernoulli model. This will include my work with Ding as well as work by Li–Zhang. Both develop new unique continuation results for the Laplacian on the integer lattice. (Received February 01, 2020)

1157-60-310 **Maria Gordina*** (maria.gordina@uconn.edu). *Hypoelliptic diffusions in infinite dimensions.*

We will review a number of recent developments concerning what can be considered as infinite-dimensional hypoelliptic diffusions. One of the settings is Wiener or heat kernel measures on infinite-dimensional sub-Riemannian manifolds such as Heisenberg group. Another is path spaces over sub-Riemannian manifolds. Smoothness of measures (understood as quasi-invariance) will be discussed. (Received February 01, 2020)

1157-60-314 **Alec N. Kercheval***, 1017 Academic Way, Rm. 208, Department of Mathematics, Florida State University, Tallahassee, FL 32306-4510, and **Navid Salehy** and **Nima Salehy**. *Self-excited Black-Scholes models.*

The Black-Scholes option pricing model is well known to be a limit of binomial tree models. What happens if the branching times of the binomial tree are given by a random point process, such as the self-exciting Hawkes process commonly used to model order arrivals in the electronic limit order book that determines the price of the stock? In this case, the limit is a version of the Black-Scholes model based on a time-changed Brownian motion calibrated to the market clock of the limit order book. In certain cases we can explicitly compute the price of a European call option as a function of the accumulated intensity of price changes over the life of the option. (Received February 01, 2020)

1157-60-359 **Le Chen*** (le.chen@emory.edu), Department of Mathematics, Emory University, 400 Dorman Drive, Atlanta, GA 30322, and **Kunwoo Kim** (kunwoo@postech.ac.kr), 77 Cheongam-Ro, Nam-Gu, Pohang, Gyeongbuk 37673, South Korea. *Stochastic comparison principles for stochastic heat equations.*

In this paper, we establish the stochastic comparison principles, including moment comparison principle as a special case, for solutions to the following nonlinear stochastic heat equation on $\mathbb{R}^d$

$$\left(\frac{\partial}{\partial t} - \frac{1}{2}\Delta\right) u(t,x) = \rho(u(t,x))\, \dot{M}(t,x),$$

where $\dot{M}$ is a spatially homogeneous Gaussian noise that is white in time and colored in space, and ρ is a Lipschitz continuous function that vanishes at zero. These results are obtained for rough initial data and under Dalang's condition, namely, $\int_{\mathbb{R}^d}(1+|\xi|^2)^{-1}\hat{f}(\mathrm{d}\xi) < \infty$, where $\hat{f}$ is the spectral measure of the noise. We establish the comparison principles by comparing either the diffusion coefficient ρ or the correlation function of the noise f. As corollaries, we obtain Slepian's inequality for SPDEs and SDEs. (Received February 02, 2020)

1157-60-393 **Jian Ding**, **Ryoki Fukushima**, **Rongfeng Sun** and **Changji Xu*** (changjixu@galton.uchicago.edu). *Random walk among Bernoulli obstacles.*

Place an obstacle with probability $1-p$ independently at each vertex of $\mathbb{Z}^d$ and consider a simple symmetric random walk that is killed upon hitting one of the obstacles. This is called *random walk among Bernoulli obstacles*. We will discuss some recent results about the behaviors of the random walk conditional on the event that it survives for a long time, in quenched, annealed, and biased setting. This talk is based on joint works with Jian Ding, Ryoki Fukushima, and Rongfeng Sun. (Received February 03, 2020)

1157-60-398 **Kiseop Lee***, Department of Statistics, Purdue University, West Lafayette, IN 47907, and **Hyunjin Jang** and **Kyungsub Lee**. *Systemic Risk in Market Microstructure: A Hawkes Flocking Model Approach.*

We propose a novel class of Hawkes-based model that assesses two types of systemic risk in high-frequency price processes: the endogenous systemic risk within a single process and the interactive systemic risk between a couple of processes. We examine the existence of systemic risk at a microscopic level via an empirical analysis of the futures markets of the West Texas Intermediate (WTI) crude oil and gasoline and perform a comparative analysis with the conditional value-at-risk as a benchmark measure of the proposed model. Throughout the analysis, we uncover remarkable empirical findings in terms of the high-frequency structure of the two markets: for the past decade, the level of endogenous systemic risk in the WTI market was significantly higher than that in the gasoline market. Moreover, the level at which the gasoline price affects the WTI price was constantly higher than in the opposite case. Although the two prices interact with each other at the transaction-unit level, the degree of relative influences on the two markets, that is, from the WTI to the gasoline and vice versa, was very asymmetric, but that difference has reduced gradually over time. (Received February 03, 2020)

1157-60-399 **Prakash Chakraborty*** (chakra15@purdue.edu), 250 N University St, MATH 535, Purdue Department of Statistics, West Lafayette, IN 47906, and **Kiseop Lee**. *Bond Prices with Insufficient Information.* Preliminary report.

We study an information asymmetry problem in a bond market. Especially we derive bond price dynamics of traders with different levels of information. A fully informed trader observes all information which affect the bond price while a partially informed trader observes only a part of it. In addition, the exclusive information of the informed trader has jumps, which represent more dramatic movements. We first obtain the bond price dynamic under the full information, and also derive the bond price of the uninformed trader using stochastic filtering method. The main idea is to change the measure so that the dynamic under the changed measure becomes computationally efficient. (Received February 03, 2020)

1157-60-402 **Leo Neufcourt*** (neufcour@msu.edu), Wells Hall, Office C442, Michigan State University, East Lansing, MI 48824. *Continuous expansion of a filtration with a stochastic process: the information drift.*

In this joint work with Philip Protter we consider a general market with a semimartingale asset price and study the situation where an "insider" agent has access to a continuous flow of additional information generated by a stochastic process. Assuming no arbitrage opportunity is created, the price process remains a semimartingale for the expanded filtration; it is then characterized by an (additional) information drift. The information drift is a key proxy to the statistical advantage provided by the additional information. The core of our results consists in a series of convergence theorems for semimartingale decompositions based on L^p norms, which provides a representation of the information drift for continuous expansions. These tools are employed to study a new class of models for the information accessible to high-frequency traders. (Received February 03, 2020)

1157-60-406 **Antonio Auffinger*** (tuca@northwestern.edu). *Counting saddles in high-dimensions.*

How many critical values does a typical random Morse function have on a high dimensional manifold? How many of given index, or below a given level? What is the topology of level sets?

In this talk, I will answer some of these classic questions and explain recent work with Julian Gold that provide almost sure asymptotics for the number of saddles of finite index in the pure p-spin model. (Received February 03, 2020)

1157-60-419 **Fabrice Baudoin*** (fabrice.baudoin@uconn.edu), 341 Mansfield Road, Mansfield, CT 06250, and **Nizar Demni** and **Jing Wang**. *Brownian windings in dimension 4.*

We define and study the windings of Brownian paths in the 4 dimensional Euclidean, spheres and hyperbolic spaces. In particular, the asymptotic laws of these windings are shown to be Gaussian for the flat and spherical geometries while the hyperbolic winding exhibits a different long time-behavior. The corresponding asymptotic law seems to be new and is related to the Cauchy relativistic distribution. (Received February 03, 2020)

1157-60-450 **Yi Shen*** (yi.shen@uwaterloo.ca) and **Zhenyuan Zhang**. *On discrete-time self-similar processes with stationary increments.*

In this talk we study the self-similar processes with stationary increments in a discrete-time setting. Different from the continuous-time case, it is shown that the scaling function of such a process may not take the form of a power function. More precisely, its scaling function can belong to one of three types, among which one type is degenerate, one type has a continuous-time counterpart, while the other type is new and unique for the discrete-time setting. We then focus on this last type of processes, construct two classes of examples, and prove a special spectral representation result for the processes of this type. We also derive basic properties of discrete-time self-similar processes with stationary increments of different types. (Received February 03, 2020)

1157-60-452 **David Nualart*** (nualart@ku.edu), Kansas University, Department of Mathematics, Lawrence, KS 66045. *Rate of convergence in the Breuer-Major theorem.*

We will survey the Malliavin-Stein method to derive estimates for the total variation distance in Gaussian approximations. As an application, we present an estimate for the total variation distance, in the framework of the Breuer-Major theorem, assuming that the underlying function is only once weakly differentiable and, together with its derivative, has finite fourth moment with respect to the standard Gaussian density. This result is proved using a combination of Gebelein's inequality and some new estimates involving Malliavin operators. (Received February 03, 2020)

1157-60-482 **Xia Chen***, 12235 Patagonia LN, Knoxville, TN 37922. *Parabolic Anderson models: large scale asymptotics.*

The parabolic Anderson equations are relevant to some problems arising from physics models such as particle movement, population dynamics and KPZ equations.

The speaker will talk about the large scale asymtotics of this system formulated in terms of long term and high moment limiting behaviors (Received February 03, 2020)

1157-60-507 **Benjamin Cooper Boniece** (bcboniece@wustl.edu), One Brookings Drive, St. Louis, MO 63130-4899, and **Jose E. Figueroa-Lopez*** (figueroa-lopez@wustl.edu), One Brookings Drive, St. Louis, MO 63130-4899. *Optimal Jump Detection and Volatility Estimation Using Wavelet And Multiscale Analysis.* Preliminary report.

The power of wavelet and multiscale analysis to study the fine path properties of functions has long been recognized and applied in many fields, including financial econometrics. So, it is not a surprise that wavelets have been applied for jump detection and integrated variance estimation of semimartingale models. These

methods exploit the sharp difference in the rate of convergence of the wavelet transformation at small scales near points of discontinuity compared to the rate near smooth points. By using a thresholding or shrinkage technique of the wavelet coefficients, researchers have been able to develop jump detection methods and applied them to integrated variance estimation. In this literature, the universal threshold of Donoho and Johnstone has become the gold standard. In this talk, we present new optimality characterizations of the universal thresholding for a general class of finite jump activity semimartingales. Comparisons with other detections methods are also presented as well as extensions to certain infinite jump activities models. (Received February 04, 2020)

1157-60-554 **Elton P Hsu*** (ehsu@math.northwestern.edu), Department of Mathematics, Northwestern University, 2033 Sheridan Road, Evanston, IL 60208. *The Parisi Formula for Spin Glass Models via Stochastic Analysis.*

The Parisi formula is a fundamental result in spin glass theory. In this talk I will present a new approach to the proof of Guerra's identity, from which the upper bound in the Parisi formula follows immediately. Among the techniques from stochastic analysis we will use include path space integration parts for the Wiener measure (Brownian motion), the Girsanov transform exponential martingales). We hope that this approach will shed some lights on the much more difficult lower bound in the Parisi formula. The basic ideas of this new approach will be outlined the first, and then some details of the proof will be discussed if time permits. (Received February 04, 2020)

1157-60-555 **Agostino Capponi**, **Jose E. Figueroa-Lopez** and **Chuyi Yu*** (chuyi@wustl.edu). *Optimal Market Making Strategies with Random Linear Demand And Overnight Inventory Costs.*

A market-maker aims to optimize his/her cumulative reward by recurrently updating his/her bid and ask quotes in a Limit Order Book (LOB) through the day. In this work, we study an optimal discrete-time market making decision problem with Poissonian order flow and overnight inventory costs. The number of filled orders during each period is assumed to depend linearly on the distance between the midprice and the market maker's limit order quotes with random coefficients. We explicitly characterize the optimal market making strategy and find conditions for its admissibility. We then proceed to calibrate our model to real LOB data and compare the model-based optimal strategy to a model free reinforcement learning solution. (Received February 04, 2020)

1157-60-560 **Rafal Kulik*** (rkulik@uottawa.ca), 150 Louis Pasteur, Ottawa, ON K1N6N5, Canada. *Tail empirical process for infinite variance functionals of long memory Gaussian processes.*

In this talk I will present some results on asymptotics for tail empirical processes of infinite variance long memory processes. I will start with stochastic volatility models for which dichotomous behaviour of the tail empirical process is proven. Then, I will consider infinite variance functionals of long memory Gaussian sequences. I will present the existing results and open problems.

The talk is based on my joint work with Clemonell Bilayi. (Received February 04, 2020)

1157-60-562 **Fabrice Baudoin**, **Nathaniel Eldredge*** (neldredge@unco.edu) and **Ximo Zhang**. *Transportation cost and functional inequalities for sub-Riemannian heat semigroups.* Preliminary report.

Regularizing properties of a heat semigroup P_t can be studied in terms of optimal transport: for instance, under appropriate conditions, the action of P_t on probability measures may be a contraction with respect to the Wasserstein distance. Recent work of Luise and Savaré obtained an interesting contraction result relating Wasserstein-type distances to the Hellinger distance, which is equivalent to total variation. However, their results are in the setting of $RCD(K,\infty)$ spaces, which involve curvature constraints that rule out sub-Riemannian manifolds. We discuss how these results can be generalized to include many sub-Riemannian examples, as well as an adaptation of their techniques that gives information about entropic distances. The key ingredients turn out to be reverse Poincaré and reverse logarithmic Sobolev inequalities. (Received February 04, 2020)

1157-60-564 **Fabrice Baudoin**, **Maria Gordina** and **Tai Melcher***, 141 Cabell Drive, Kerchof Hall, Charlottesville, VA 22904. *Quasi-invariance for generalized Kolmogorov diffusions in infinite dimensions.*

The Kolmogorov diffusion stands as one of the simplest and most natural examples of hypoelliptic diffusions, that is diffusions whose end point distribution is smooth with respect to Lebesgue measure. We consider different infinite-dimensional versions of these diffusions and show that they retain smoothness properties. These results are attained by establishing uniform control over coefficients appearing in certain functional analytic inequalities for approximating finite-dimensional diffusions.

This is joint work with Fabrice Baudoin and Maria Gordina. (Received February 04, 2020)

1157-60-566 **Elizabeth Meckes** and **Tai Melcher***, 141 Cabell Drive, Kerchof Hall, Charlottesville, VA 22904, and **Jing Wang**. *Improved log Sobolev coefficients for compact Lie groups.*

It is a now classical result in stochastic analysis that a lower bound on the Ricci curvature of a Riemannian manifold M implies that a log Sobolev inequality holds for the heat kernel measure on M. In particular, if the Ricci curvature is non-negative, the log Sobolev coefficient is linear. Using basic properties of Lie groups and Brownian motion, we improve this standard estimate for log Sobolev coefficients of the heat kernel measure on compact Lie groups equipped with a bi-invariant metric (and thus of non-negative curvature).

This is joint work with Elizabeth Meckes and Jing Wang. (Received February 04, 2020)

1157-60-569 **Christopher E. O'Lone**, **Harpreet S. Dhillon*** (hdhillon@vt.edu) and **R. Michael Buehrer**. *Performance Analysis of Wireless Localization using Stochastic Geometry.*

Measurement-based wireless localization techniques involve the exchange of wireless signals between nodes termed *anchors*, whose positions are known, and a *target*, whose position is unknown, in order to estimate the target's position. The performance of these localization techniques fundamentally depends upon: (i) the number of anchors participating in the localization procedure, (ii) the locations of the participating anchors, and (iii) the quality of the positioning observations gathered from the participating anchors. These factors in turn strongly depend upon the underlying geometry of the anchor nodes. The most common benchmark used to assess the performance of these techniques is the Cramér-Rao lower bound, which is characterized for the fixed positions of the anchors and the target. In this work, we develop a new approach using tools from stochastic geometry that focuses on the network-wide performance characterization of these techniques. More specifically, we model the locations of both the anchors and the targets as point processes and characterize the localization performance with respect to the *typical* target. In this talk, we will discuss several concrete results that demonstrate the relevance and mathematical tractability of this approach. (Received February 04, 2020)

1157-60-576 **Bei Wu*** (bei.wu@wustl.edu), 6605 Clayton Ave, Apt 111, St. Louis, MO 63139, and **Jose Figueroa-Lopez**. *Optimal Kernel Estimation of Spot Volatility with Leverage and Microstructure Noise Using Pre-Averaging.*

We present a new type of pre-averaging/kernel estimator for spot volatility in a general continuous Itô semi-martingale model under the presence of leverage and microstructure noise. We prove central limit theorems for the estimation error with an optimal rate and study the problem of optimal bandwidth and kernel selection. A feasible implementation of the proposed estimators is also developed and the superior performance of the proposed estimators are illustrated by Monte Carlo experiments. (Received February 04, 2020)

1157-60-577 **Xiaoqin Guo*** (xguo234@wisc.edu) and **Hung Vinh Tran**. *Heat kernel estimates for random walks in an i.i.d. balanced random environment.* Preliminary report.

We consider a random walk in an i.i.d. random environment in the integer lattice Z^d with no drift. It is known that the quenched heat kernel of this RWRE does not have deterministic Gaussian bounds. In this talk, we will present lower and upper bounds for the heat kernel which are Gaussian up to a random multiplicative factor which has exponential moment. Joint work with H.V.Tran (Madison). (Received February 04, 2020)

1157-60-578 **Fabrice Baudoin** and **Li Chen*** (li.4.chen@uconn.edu), 341 Mansfield Road, Department of Mathematics, University of Connecticut, Storrs, CT 06269. *Second Riesz transforms on some Lie groups.*

In this talk, we study explicit L^p bounds for second order Riesz transforms on Heisenberg groups $\mathbb{H}^{2n+1}$ and $\mathbb{SU}(2)$ via martingale transform techniques. In particular, Beurling-Ahlfors operators on $\mathbb{H}^3$ and $\mathbb{SU}(2)$ will be discussed. This is a joint work with Fabrice Baudoin. (Received February 04, 2020)

1157-60-591 **Aaron M Lattanzi*** (alattanz@umich.edu), **Shankar Subramaniam** (shankar@iastate.edu) and **Jesse Capecelatro** (jcaps@umich.edu). *Stochastic formulations for particle dispersion.*

Particle-laden flows are often characterized by the formation of large-scale coherent structures due to strong coupling with a carrier gas. Volume-filtered Euler-Lagrange (EL) frameworks are capable of capturing meso-scale structuring in strongly coupled gas-solids flows. However, drag force closures developed for EL frameworks only capture the mean interphase force; when in reality, flow past a collection of particles will result in a distribution of drag forces. Here, we consider stochastic formulations to account for the effect of drag force variance on particle dispersion. Introducing random increments at the particle position, velocity, and force level lead to different Fokker-Planck (FP) equations. Moments of the FP equations are utilized to derive evolution equations for dispersion and velocity variance. Dispersion of a particle cloud leads us to consider the effect of a time varying velocity variance on the dispersion of particles following an Ornstein-Uhlenbeck process. It is

shown that additional time scales are introduced into the 'ballistic' regime, but the classic stationary solution of Taylor (1922 *Proceedings of the London Mathematical Society*, s2-20: 196-212) is recovered when time is large with respect to the Stokes time scale. (Received February 04, 2020)

1157-60-597 **Wai-Tong (Louis) Fan*** (waifan@iu.edu), 2409 E Oakmont Dr, Bloomington, IN 47401-8185, and **Timo Seppäläinen**. *Joint distribution of Busemann functions in the corner growth model.*

We present the joint distribution of the Busemann functions, in all directions of growth, of the exactly solvable corner growth model (CGM). This gives a natural coupling of all stationary CGMs and leads to new results about geodesics. Properties of this joint distribution are accessed by identifying it as the unique invariant distribution of a multiclass last passage percolation model. This is joint work with Timo Seppäläinen. (Received February 04, 2020)

1157-60-611 **Yuchen Han*** (y.han@wustl.edu) and **Jose Figueroa-Lopez**. *Infill Asymptotic Properties Of Optimal Iterative Estimation Methods Of Volatility Using Truncated Realized Variations.*

We consider a stochastic process driven by a diffusion and jump component, which are common models for high frequency data in finance. We are interested in estimating the integrated variance of the process, which is a global measure of riskiness of the asset during a fixed time interval. To filter out or mitigate the effect of the jump component from the realized quadratic variation, we use truncated realized quadratic variations with optimum thresholding, followed by a "fixed point" iterative method for its implementation. The proposed iterative method starts with a data-driven initial guess of the true parameter of interest and aims to iteratively improve this guess by using a formula for the optimal threshold that depends on the parameter of interest itself. The iterative estimator eventually settles down to some fixed value. We have characterized the asymptotic infill or high-frequency behavior of this final estimate, and proved the consistency and central limit results in the case of finite activity jumps under relatively mild assumptions. (Received February 04, 2020)

1157-60-649 **Oana Mocioalca*** (oana@math.kent.edu), Department of Mathematical Sciences, PO Box 5190, Kent, OH 44242. *Ito formula for additive summable processes.*

We consider Banach-valued processes $X : \mathbb{R}_+ \times \Omega \to E$, with E continuously embedded into $L(F, G)$ for some Banach spaces F and G. We work with the class of *additive summable* processes, which are a generalization of *summable* processes, introduced by Dinculeanu and Brooks, which in turn generalize the notion of Banach-valued semimartingales. We develop a stochastic calculus, including an Ito formula, for these processes. (Received February 05, 2020)

62 ► Statistics

1157-62-108 **Saddam Akber Abbasi*** (sabbasi@qu.edu.qa), 101-C03, Qatar University, Doha, 00000, Qatar. *Efficient Nonparametric EWMA Chart for Monitoring Process Variability.*

Control chart is the most important tool in the Statistical Process Control tool-kit. The monitoring of process variability and process location is mostly of interest. Most control chart structures are based on the assumption that the process observations are normally distributed. When the assumption of normality is violated, the use of non-parametric charts is usually recommended. In this study, a new control chart structure is proposed for efficient monitoring of process variability. The new proposal is based on using the sign statistic under the progressive mean setup. The performance of the newly proposed chart will be compared with the existing nonparametric variability charts using different run length characteristics such as average run length, standard deviation of run length and extra quadratic loss measures. It will help quality practitioners to choose an efficient non-parametric chart for monitoring of process variability. (Received January 22, 2020)

1157-62-139 **Chanwoo Lee** and **Miaoyan Wang*** (miaoyan.wang@wisc.edu), Department of Statistics, University of Wisconsin-Madison, Madison, WI 53503. *Tensor denoising and completion based on ordinal observations.*

Higher-order tensors arise frequently in applications such as neuroimaging, recommendation system, social network analysis, and psychological studies. We consider the problem of low-rank tensor estimation from possibly incomplete, ordinal-valued observations. Two related problems are studied, one on tensor denoising and another on tensor completion. We propose a multi-linear cumulative link model, develop a rank-constrained M-estimator, and obtain theoretical accuracy guarantees. Our mean squared error bound enjoys a faster convergence rate than

previous results, and we show that the proposed estimator is minimax optimal under the class of low-rank models. Furthermore, the procedure developed serves as an efficient completion method which guarantees consistent recovery of an order-K $(d, \dots, d)$-dimensional low-rank tensor using only $O(Kd)$ noisy, quantized observations. We demonstrate the outperformance of our approach over previous methods on the tasks of clustering and collaborative filtering. (Received February 03, 2020)

1157-62-152 **Rida Benhaddou*** (benhaddo@ohio.edu), Department of Mathematics, Morton Hall 321, Ohio University, Athens, OH 45701, and **Qing Liu**. *Anisotropic functional deconvolution with long-memory noise: the case of a multi-parameter fractional Wiener sheet.*

We look into the minimax results for the anisotropic two-dimensional functional deconvolution model with the two-parameter fractional Gaussian noise. We derive the lower bounds for the L^p-risk, $1 \leq p < \infty$, and taking advantage of the Riesz poly-potential, we apply a wavelet-vaguelette expansion to de-correlate the anisotropic fractional Gaussian noise. We construct an adaptive wavelet hard-thresholding estimator that attains asymptotically optimal or quasi-optimal convergence rates in a wide range of Besov balls. Such convergence rates depend on a delicate balance between the parameters of the Besov balls, the degree of ill-posedness of the convolution operator and the parameters of the fractional Gaussian noise under regular-smooth convolution, whereas the rates are not affected by long-memory under super-smooth convolution. A limited simulations study confirms the theoretical claims of the paper. The proposed approach is extended to the general r-dimensional case, with $r > 2$, and the corresponding convergence rates do not suffer from the curse of dimensionality. (Received January 26, 2020)

1157-62-200 **Xiaowu Dai*** (xwdai.stat@gmail.com), 410 Evans Hall, University Drive, Berkeley, CA 94720. *Multi-Layer Kernel Machines: A Fast and Accurate Approach for Large-Scale Supervised Learning.*

This article proposes an approximation of kernel ridge regression (KRR) based on random features and a multi-layer structure. KRR is popular in statistics and machine learning for nonparametric regressions over reproducing kernel Hilbert spaces. We study the minimum number of random features and the size of layers can be chosen for preserving minimax optimality of the approximate KRR estimate. We show that the multi-layer kernel machines only require $O(n^{3/2} \log n)$ time and $O(n \log n)$ memory, which is significantly better compared to $O(n^3)$ time and $O(n^2)$ memory in computing KRR. For various classes of random features, we prove that the multilayer structure is more effective in reducing the computational complexity than the single-layer while keeping statistical minimax optimality. The analysis is supported by simulation and real data examples. (Received January 28, 2020)

1157-62-395 **Ziwei Ji** and **Matus Telgarsky*** (mjt@illinois.edu). *Polylogarithmic width suffices for gradient descent to achieve arbitrarily small test error with shallow ReLU networks.*

Recent theoretical work has guaranteed that overparameterized networks trained by gradient descent achieve arbitrarily low training error, and sometimes even low test error. The required width, however, is always polynomial in at least one of the sample size n, the (inverse) target error $\frac{1}{\epsilon}$, and the (inverse) failure probability $\frac{1}{\delta}$. This work shows that $\mathcal{O}(\frac{1}{\epsilon})$ iterations of gradient descent with $\Omega(\frac{1}{\epsilon^2})$ training examples on two-layer ReLU networks of any width exceeding $\operatorname{poly}\log(n, \frac{1}{\epsilon}, \frac{1}{\delta})$ suffice to achieve a test misclassification error of ϵ. The analysis further relies upon a margin property of the limiting kernel, which is guaranteed positive, and can distinguish between true labels and random labels. (Received February 03, 2020)

1157-62-425 **Alexandra Chronopoulou*** (achronop@illinois.edu). *Optimal sampling schemes in fractional volatility models.*

The goal of this talk is to propose efficient sampling schemes for sampling fractional (both rough and long memory) stochastic volatility models, when the number and type of observations is constrained due to limited resources. We find that the sampling mechanism depends on the value of the Hurst parameter, H, and that the equidistant sampling scheme is optimal only when H=0.5. (Received February 03, 2020)

1157-62-487 **Facundo Memoli***, Department of Mathematics, The Ohio State University, Columbus, OH 43210. *The Wasserstein Transform.*

We introduce the Wasserstein transform, a method for enhancing and denoising datasets defined on general metric spaces. The construction draws inspiration from Optimal Transportation ideas. We establish the stability of our method under data perturbation and, when the dataset is assumed to be Euclidean, we also exhibit a precise connection between the Wasserstein transform and the mean shift family of algorithms. We then use this connection to prove that mean shift also inherits stability under perturbations. We study the performance of the

Wasserstein transform method on different datasets as a preprocessing step prior to clustering and classification tasks. (Received February 03, 2020)

1157-62-618 **B Cooper Boniece*** (bcboniece@wustl.edu), **Gustavo Didier**, **Patrice Abry** and **Herwig Wendt**. *On wavelet-based estimation of Hurst exponents in high dimensions.*

In multi-dimensional contexts, the scaling behavior of a self-similar processes is governed by matrix parameter H; the eigenvalues of H – the so-called Hurst exponents – determine the scaling laws that contribute to its dynamics.

In this talk, we consider the problem of estimation of Hurst exponents in a high-dimensional setting via a wavelet-based method. (Received February 04, 2020)

1157-62-651 **Omar De la Cruz Cabrera*** (odelacru@kent.edu), Department of Mathematical Sciences, PO Box 5190, Kent, OH 44242. *Issues in Kernel Methods for Exploratory Data Analysis.*

Multivariate methods like Principal Components Analysis (PCA) are extremely useful in the exploratory analysis (EDA) of complex data; however, the interpretation of the results is often fraught with difficulties. Kernel methods add a great amount of power to a large variety of multivariate techniques, by easily allowing the introduction of non-linearities, but in the exploratory case, unfortunately, the interpretation of the results becomes even more delicate.

We will show how some peculiarities that occur in Kernel PCA (KPCA) help explain phenomena that occur in classical PCA, especially in high-dimensional settings; we argue that conclusions derived from cursory examination of the result of PCA (especially in genomics) need to be reexamined critically, in the light of the facts presented here. We also comment on the critical issue of bandwidth choice, and some the the computational challenges involved. (Received February 05, 2020)

65 ► Numerical analysis

1157-65-3 **Xiaofeng Cai** (xfcai@udel.edu), Math department, university of Delaware, newark, DE 19716, and **jingmei qiu*** (jingqiu@udel.edu), math department, university of Delaware, newark, DE 19716. *Eulerian-Lagrangian discontinuous Galerkin method for linear advection problems and its application to nonlinear Vlasov dynamics.*

We propose a new Eulerian-Lagrangian (EL) discontinuous Galerkin (DG) method. The method is designed as a generalization of the semi-Lagrangian (SL) DG method for linear advection problems proposed in [**?**], which is formulated based on an adjoint problem and tracing upstream cells by tracking characteristics curves exactly (or highly accurately). Depending on the velocity field, the shape of upstream cells could be of arbitrary shape, for which a more sophisticated approximation is required to get high order approximation.For linear advection problems, we propose a more general formulation, named the EL DG method, for which the scheme is based on a modified adjoint problem for which upstream cells are always quadrilaterals. This leads to a new formulation of ELDG method, which avoids the need to use QC quadrilaterals to better approximate upstream cells in the original SL DG algorithm. The newly proposed EL DG method can be viewed as a new general framework, in which both the classical RK DG formulation and the SL DG formulation can fit in. Numerical results on linear advection problems, as well as the nonlinear Vlasov dynamics using the exponential RK framework, will be presented to demonstrate the effectiveness of the proposed approach. (Received August 31, 2019)

1157-65-4 **Guosheng Fu*** (gfu@nd.edu), 201D Crowley Hall, Notre Dame, South Bend, IN 46556. *"Explicit" divergence-free discontinuous Galerkin methods for incompressible flow.*

We present an explicit divergence-free DG method for incompressible Euler and Navier-Stokes equations based on velocity formulation only. A globally divergence-free finite element space is used for the velocity field, and the pressure field is eliminated from the equations by design. The resulting ODE system is discretized by the explicit third order strong-stability preserving Runge-Kutta method of Shu and Osher. Our spatial discretization produces the identical velocity field as the divergence-conforming DG method of [Cockburn et al., JSC 31(2007), pp 61-73] based on a velocity-pressure formulation, when the same DG operators are used for the convective and viscous parts.

We present a key result on the efficient implementation of the scheme by identifying the equivalence of the mass matrix inversion of the globally divergence-free finite element space to a standard (hybrid-)mixed Poisson solver. Hence, in each time step, a Poisson solver is used, which reflects the global nature of the incompressibility

condition. Since we treat viscosity explicitly, our method shall be best suited for unsteady high-Reynolds number flows so that the viscous CFL constraint is not too restrictive.

The extension to incompressible MHD will also be discussed. (Received September 01, 2019)

1157-65-44 **Ren-Cang Li*** (rcli@uta.edu), **Li Wang**, **Lei-hong Zhang** and **Zhaojun Bai**. *SCF Iteration for Orthogonal Canonical Correlation Analysis.*

Canonical Correlation analysis (CCA) is a standard statistical technique and widely-used feature extraction paradigm for two sets of multidimensional variables. Mathematically, CCA is an optimization problem that can be turned into a singular value problem. Orthogonal CCA (OCCA) is a term that was coined broadly as a collection of variants of CCA imposing orthogonality among basis vectors. The most simple variant is the plain CCA followed by performing the Gram-Schmidt orthgonalization process on the two sets of basis vectors. A more straightforward way is to impose orthogonality while optimizing the same objective function as CCA. It has been observed that orthonormal bases by the latter are more effective than the most simple way in data science applications. However, directly optimizing the objective function with orthogonality constraints on the basis vectors is nontrivial. In this talk, we will present an alternating numerical scheme whose core is a sub-maximization problem in the trace-fractional form with an orthogonal constraint. Experimental results show that our methods not only perform competitively to or better than baselines but also are more efficient. (Received January 06, 2020)

1157-65-50 **Jingwei Hu** and **Ruiwen Shu*** (rshu@cscamm.umd.edu). *On the uniform accuracy of implicit-explicit backward differentiation formulas (IMEX-BDF) for stiff hyperbolic relaxation systems and kinetic equations.*

Many hyperbolic and kinetic equations contain a non-stiff convection/transport part and a stiff relaxation/collision part (characterized by the relaxation or mean free time ε). To solve this type of problems, implicit-explicit (IMEX) Runge-Kutta or multistep methods have been widely used and their performance is understood well in the non-stiff regime ($\varepsilon = O(1)$) and limiting regime ($\varepsilon \to 0$). However, in the intermediate regime (say, $\varepsilon = O(\Delta t)$), some uniform accuracy or order reduction phenomena have been reported numerically without much theoretical justification. In this work, we prove the uniform accuracy (an optimal *a priori* error bound) of a class of IMEX multistep methods – IMEX backward differentiation formulas (IMEX-BDF) for linear hyperbolic systems with stiff relaxation. The proof is based on the energy estimate with a new multiplier technique. For nonlinear hyperbolic and kinetic equations, we numerically verify the same property using a series of examples. (Received January 08, 2020)

1157-65-51 **xiaofeng cai**, **jingmei qiu*** (jingqiu@udel.edu) and **hong xue**. *An Eulerian-Lagrangian discontinuous Galerkin method for transport problems and its application to nonlinear Vlasov dynamics.*

We propose a new Eulerian-Lagrangian (EL) discontinuous Galerkin (DG) method. The method is designed as a generalization of the semi-Lagrangian (SL) DG method, which is formulated based on an adjoint problem and tracing upstream cells by tracking characteristics curves highly accurately. In the SLDG method, depending on the velocity field, upstream cells could be of arbitrary shape. Thus, a more sophisticated approximation to sides of upstream cells is required to get high order approximation. In this paper, for linear advection problems, we propose a more general formulation, named the ELDG method. The scheme is formulated base on a modified adjoint problem for which upstream cells are always quadrilaterals, which avoids the need to use QC quadrilaterals in the SLDG algorithm. The newly proposed ELDG method can be viewed as a new general framework, in which both the classical Eulerian Runge-Kutta DG formulation and the SL DG formulation can fit in. Numerical results on linear transport problems, as well as the nonlinear Vlasov dynamics using the exponential RK time integrators, are presented to demonstrate the effectiveness of the ELDG method. (Received January 08, 2020)

1157-65-52 **Dinh-Liem Nguyen*** (dlnguyen@ksu.edu), Department of Mathematics, Kansas State University, Manhattan, KS 66506. *Orthogonality Sampling Method for Solving the Electromagnetic Inverse Scattering Problem.*

This talk is about solving the electromagnetic inverse scattering problem that aims to determine the location and shape of anisotropic scattering objects from far field data (at a fixed frequency). This is an important inverse problem that has been extensively studied thanks its mathematical interests and applications in non-destructive testing, medical imaging, radar etc. We will talk about our recent results on the orthogonality sampling method (OSM) for solving the inverse problem. Theoretical analysis and numerical simulations of the OSM for both cases of one-wave data and multi-static data will be discussed. The advantages of the OSM are that it is very

robust with noisy data, simple, fast, and does not require a priori information of the solution. This is joint work in part in Isaac Harris. (Received January 08, 2020)

1157-65-67 **Zhonggang Zeng*** (zzeng@neiu.edu), Department of Mathematics, Northeastern Illinois University, Chicago, IL 60625. *Sensitivities of singular problems.* Preliminary report.

Sensitivity measures in terms of condition numbers have been well established for nonsingular problems. For singular algebraic problems, however, the notion of sensitivity is still questionable while the condition numbers are generally defined as infinity as a convention. As a result, singular problems are often avoided for being highly sensitive ill-posed problems. This characterization, as Kahan pointed out, may be a "misconception" since their solutions can be stable with respect to constrained perturbations maintaining singularities. From a different perspective, singular problems can have inherently stable numerical solutions with bounded sensitivities. We shall elaborate the sensitivities of singular linear systems, singular nonlinear systems and defective eigenvalue problems. For those problems, finite sensitivity measures exist so that reformulated numerical solutions can be accurately computed using floating point arithmetic even if the problem is given with empirical data. The resulting error bounds are computationally verifiable and have many applications in practical computation. (Received January 12, 2020)

1157-65-69 **Yi Zhang*** (y_zhang7@uncg.edu). *Numerical Algebraic Geometry and Semidefinite Programming.*

Standard interior point methods in semidefinite programming can be viewed as tracking a solution path for a homotopy defined by a system of bilinear equations. By considering this in the context of numerical algebraic geometry, we employ numerical algebraic geometric techniques such as adaptive precision path tracking, endgames, and projective space to accurately solve semidefinite programs. We develop feasibility tests for both primal and dual problems which can distinguish between the four feasibility types of semidefinite programs. Finally, we couple our feasibility tests with facial reduction to develop a solving approach that can handle every scenario, including problems with nonzero duality gap. This is joint work with Jonathan D. Hauenstein, Alan C. Liddell, Jr and Sanesha McPherson. (Received January 13, 2020)

1157-65-86 **Yong Liu*** (yong_liu@brown.edu), 182 George Street, Providence, RI 02912, and **Chi-Wang Shu** and **Mengping Zhang**. *Optimal error estimates of the semidiscrete discontinuous Galerkin methods for two dimensional hyperbolic equations on Cartesian meshes using P^k elements.*

In this talk, we present the optimal error estimates of the classical discontinuous Galerkin method for time-dependent 2-D hyperbolic equations using P^k elements on uniform Cartesian meshes, and prove that the error in the L^2 norm achieves optimal $(k+1)$-th order convergence when upwind fluxes are used. For the linear constant coefficient case, the results hold true for arbitrary piecewise polynomials of degree $k \geq 0$. For variable coefficient and nonlinear cases, we give the proof for piecewise polynomials of degree $k = 0, 1, 2, 3$ and $k = 2, 3$, respectively, under the condition that the wind direction does not change. The theoretical results are verified by numerical examples. (Received January 15, 2020)

1157-65-93 **Li Wang***, 645 N1st St., Apt. 303, Minneapolis, MN 55401. *Fisher information regularization schemes for Wasserstein gradient flows.*

We propose a variational scheme for computing Wasserstein gradient flows. The scheme builds upon the Jordan–Kinderlehrer–Otto framework with the Benamou-Brenier's dynamic formulation of the quadratic Wasserstein metric and adds a regularization by the Fisher information. This regularization can be derived in terms of energy splitting and is closely related to the Schrodinger bridge problem. It improves the convexity of the variational problem and automatically preserves the non-negativity of the solution. As a result, it allows us to apply sequential quadratic programming to solve the sub-optimization problem. We further save the computational cost by showing that no additional time interpolation is needed in the underlying dynamic formulation of the Wasserstein-2 metric, and therefore, the dimension of the problem is vastly reduced. (Received January 17, 2020)

1157-65-97 **Zhiyan Ding***, 480 Lincoln Dr., Madison, WI 53706. *Validity of Ensemble Kalman Inverse.*

Ensemble Kalman Inverse (EKI), as an algorithm, has been successfully applied on many inverse problems. Why it works, however, is far from being thoroughly understood. In this talk, we will translate the EKI method to a coupled stochastic differential equation, and then in the linear setting use mean-field limit argument to equate the coupled SDE system with a Fokker-Planck type PDE. This leads to the first result that reveals the convergence rate of EKI. (Received January 18, 2020)

1157-65-116 **lei wang*** (wang256@uwm.edu), 3200 N Cramer St, milwaukee, WI 53211. *A Fast Kernel-independent treecode for General Rotne-Prager-Yamakawa tensor.*

A particle-cluster treecode based on barycentric Lagrange interpolation is presented for fast summation of hydrodynamic interactions through general Rotne-Prager-Yamakawa(RPY) tensor in 3D. The interpolation nodes are taken to be Chebyshev points of the 2nd kind in each cluster. The barycentric Lagrange interpolation is scale-invariant that promotes the treecode's efficiency. Numerical results show that the treecode CPU time scales as $O(N \log N)$, where N is the number of beads in the system. The treecode is applied to speed-up the Lanczos iteration in computing the squre-root of the RPY tensor matrix. The kernel-independent treecode is a relatively simple algorithm with low memory consumption, and this enables a straightforward OpenMP parallelization. (Received January 22, 2020)

1157-65-122 **Jichun Li*** (jichun.li@unlv.edu), 4505 Maryland PKWY; Box 454020, Las Vegas, NV 89154-4020. *Mathematical analysis and finite element simulation of surface plasmon polaritons in graphene.*

Graphene is a 2D nanomaterial, which was rediscovered in 2004. Due to its outstanding electrical, mechanical, magnetic, and thermal properties, graphene has gained significant interest among scientists as evidenced by the award of Nobel Prize in Physics 2010 to two graphene experts. In this talk, I will focus on the development and analysis of a mathematical model for simulating the surface plasmon polaritons. Finite element method for solving this model will be presented. Finally, I will show some numerical simulations. This is a joint work with Yunqing Huang and Wei Yang at Xiangtan University. (Received January 22, 2020)

1157-65-127 **Xiaozhi Zhu** and **Yong-Tao Zhang***, 102G Crowley Hall, Dept. of Applied and Comp. Math. and Stat., University of Notre Dame, Notre Dame, IN 46556. *Fast sparse grid simulations of fifth order WENO scheme for high dimensional hyperbolic PDEs.*

WENO schemes are a popular class of high order numerical methods for solving hyperbolic PDEs. However when the spatial dimensions are high, the number of grid points increases significantly. It leads to large amount of costs in the simulations. How to achieve fast simulations by WENO methods is a challenging and important question. In the literature, sparse-grid technique has been developed as a very efficient approximation tool for high dimensional problems. Recently in [Lu, Chen and Zhang, Pure and Applied Mathematics Quarterly, 2018], a third order WENO method with sparse-grid combination technique was designed to solve multidimensional hyperbolic PDEs. In applications, higher than third order WENO schemes are often preferred in order to efficiently resolve the complex solution structures. In this paper, we extend the approach to higher order WENO simulations specifically the fifth order WENO scheme. Benchmark problems are first solved to show that significant CPU times are saved while both fifth order accuracy and stability of the WENO scheme are preserved for simulations on sparse grids. The fifth order sparse grid WENO method is then applied to kinetic problems modeled by high dimensional Vlasov based PDEs to further demonstrate large savings of computational costs. (Received January 22, 2020)

1157-65-128 **Hugo Casquero**, **Carles Bona-Casas**, **Deepesh Toshniwal**, **Thomas J.R. Hughes**, **Hector Gomez*** (hectorgomez@purdue.edu) and **Jessica Zhang**. *The divergence-conforming immersed boundary method.*

The divergence-conforming immersed boundary (DCIB) method is presented to tackle a long-standing issue of immersed boundary (IB) numerical methods for fluid-structure interaction (FSI), namely, the challenge of accurately imposing the incompressibility constraint at the discrete level. In the DCIB method, the Eulerian velocity-pressure pair is discretized using divergence-conforming B-splines, which leads to inf-sup stable, H^1-conforming, and pointwise divergence-free Eulerian solutions. In order to discretize the higher-order derivatives that appear in vesicle and capsule formulations, we use C^2-continuous cubic B-splines with periodic knot vectors and C^1-continuous bi-cubic analysis-suitable T-splines in 2D and 3D settings, respectively. Non-negligible spurious changes of the fluid volume inside of closed co-dimension one solids is a well-known deficiency of IB methods. The DCIB method results in volume changes various orders of magnitude lower than conventional IB methods. Benchmark and application problems of vesicle and capsules are solved, including mesh-independence studies and comparisons with other numerical methods. (Received January 22, 2020)

1157-65-131 **Guang Lin*** (guanglin@purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907, and **Sheng Zhang** (zhan2694@purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907. *Data-driven discovery of differential equations with random initial and boundary conditions using subsampling-based robust Bayesian method.*

Data-driven discovery of physical laws is a grand challenge in many science and engineering research areas, and many fundamental laws are formulated in the form of differential equations. We propose a new method capable of discovering the governing differential equations from data. Our method is based on Bayesian inference and uses a subsampling technique so that it is robust to noisy data. Numerical examples show that our new method is significantly better than the other model-discovering methods and traditional regression methods. The examples include predator-prey model, shallow water equations, heat diffusion, and fish-harvesting problem. (Received January 22, 2020)

1157-65-135 **Wenrui Hao***, 208 Mcallister Bldg, University Pk, PA 16802-6401. *A homotopy training algorithm for fully connected neural networks and applications in solving PDEs.*

In this talk, I will introduce a homotopy training algorithm (HTA) to solve optimization problems arising from fully connected neural networks with complicated structures. The HTA dynamically builds the neural network starting from a simplified version and ending with the fully connected network via adding layers and nodes adaptively. Therefore, the corresponding optimization problem is easy to solve at the beginning and connects to the original model via a continuous path guided by the HTA, which provides a high probability of obtaining a global minimum. Then I will apply the HTA to compute the multiple solutions of nonlinear PDEs with neural network discretization. (Received January 23, 2020)

1157-65-149 **Yang Yang*** (yyang7@mtu.edu), 1400 Townsend Drive, Houghton, MI 49931, and **Xiaofeng Cai** and **Jingmei Qiu**. *Optimal convergence and superconvergence of semi-Lagrangian discontinuous Galerkin methods for linear convection equations in one space dimension.*

In this talk, we apply semi-Lagrangian discontinuous Galerkin (SLDG) methods for linear hyperbolic equations in one space dimension and analyze the error between the numerical and exact solutions under the L^2-norm. In all the previous works, the theoretical analysis of the SLDG method would suggest a suboptimal convergence rate due to the error accumulation over time steps. The analysis in this talk contains three main difficulties: 1. The error analysis on overlapping meshes. Due to the nature of the semi-Lagrangian time discretization, we need to introduce the background Eulerian mesh and the shifted mesh. The two meshes are staggered. 2. The superconvergence of time dependent terms under the L^2-norm. We need to construct a sequence of projections and improve the superconvergence rates gradually. 3. The stopping criterion of the sequence of projections. The sequence of projections are basically of the same form. We will seek a "hidden" condition for the existence of the projections. In this talk, we will solve all the three difficulties and construct several local projections to prove the optimal convergence and superconvergence rates. Numerical experiments verify the theoretical findings. (Received January 24, 2020)

1157-65-150 **LIU LIU*** (lliu@ices.utexas.edu), E 24th St, Austin, TX 78712. *A Bi-fidelity method for multiscale kinetic equations with uncertainties.*

In this talk, we introduce a bi-fidelity numerical method for solving high-dimensional parametric kinetic equations. We first briefly discuss about the Boltzmann equation and its fluid dynamic limit, then introduce a bi-fidelity stochastic collocation method for its uncertainty quantification problem. By combining computational efficiency of the low-fidelity model–chosen as the compressible Euler system–with high accuracy of the high-fidelity (Boltzmann) model, our bi-fidelity approximation can successfully capture well the macroscopic quantities of solution to the Boltzmann equation in the random space. A uniform error estimate of the bi-fidelity method, based on a series of our theoretical work on hypocoercivity for the uncertain kinetic equations, will be shown. Lastly we present numerical results to validate the efficiency and accuracy of our proposed method. (Received January 25, 2020)

1157-65-168 **Juntao Huang*** (huangj75@msu.edu) and **Yingda Cheng**. *An adaptive multiresolution discontinuous Galerkin method with artificial viscosity for scalar hyperbolic conservation laws in multidimensions.*

In this paper, we develop an adaptive multiresolution discontinuous Galerkin (DG) scheme for scalar hyperbolic conservation laws in multidimensions. Compared with previous work for linear hyperbolic equations, a class of interpolatory multiwavelets are applied to efficiently compute the nonlinear integrals over elements and edges in DG schemes. The resulting algorithm therefore can achieve similar computational complexity as the sparse grid

DG method for smooth solutions. Theoretical and numerical studies are performed taking into consideration of accuracy and stability with regard to the choice of the interpolatory multiwavelets. Artificial viscosity is added to capture the shock and only acts on the leaf elements taking advantage of the multiresolution representation. Adaptivity is realized by auto error thresholding based on hierarchical surplus. Accuracy and robustness are demonstrated by several numerical tests. (Received January 26, 2020)

1157-65-182 **Yuan Gao*** (yg86@duke.edu). *Gradient flow formulation and numerical simulation for motion by mean curvature and contact line dynamics on rough surface.*

We study the dynamics and equilibrium of a droplet placing on an inclined rough surface. The capillary effect, which contributes the leading behaviors of the geometric motion of the droplet, is characterized by the surface tension on interfaces and greatly affects the dynamics of the contact angle and the contact line, where three phases (gas, liquid and solid) meet. Based on explicit moving boundaries and semi-Lagrangian method, we propose unconditionally stable first/second order numeric schemes to simulate the geometric motion of the droplet described using motion by mean curvature with moving contact lines. Challenging examples are demonstrated and the accuracy and long-time validation is compared with quasi-static solution obtained by desingularized differential-algebraic system of equations. (Received January 27, 2020)

1157-65-209 **Shuwang Li***, 10 West 32nd Street, Room 220, Chicago, IL 60616, **Yue Cao**, 10 West 32nd Street, Room 220, Chicago, IL 60616, and **Wenjun Ying**. *Kernel-free boundary integral method for boundary value and interface problem in doubly-connected domain.*

We present a kernel-free boundary integral method (KFBIM) for variable coefficients partial differential equations (PDEs) in doubly-connected domain. We are interested in boundary value problems (BVP) and interface problems. The KFBIM computes boundary or volume integrals by equivalently solving an interface problem without requiring the the explicit form of the Green function. We first decompose the problem defined in a doubly-connected into two separate boundary value problems or interface problems. Then we evaluate integrals in a finite difference framework and solve solutions using a Krylov subspace iterative method. The method is second order accurate in space, and its complexity is linearly proportional to mesh node number. Numerical examples show the method is robust for variable coefficients PDEs with large diffusion coefficients ratio and complex geometries where two interfaces are close. (Received January 29, 2020)

1157-65-223 **Zhaoxing Li**, **Zhiliang Deng** and **Jiguang Sun*** (jiguangs@mtu.edu), Fisher 313, 1400 Townsend Dr., Houghton, MI 49931-1295. *Extended-Sampling-Bayesian Method for Limited Aperture Inverse Scattering Problems.*

Limited aperture inverse scattering problems arise from many important applications. In this talk, we discuss a new method combining the extended sampling method (ESM) and Bayesian approach for the inverse acoustic scattering problem to reconstruct the shape of a sound-soft obstacle using the limited aperture data. The problem is formulated as a statistical model using the Baye's formula. The well-posedness is proved in the sense of the Hellinger metric. A modified ESM is proposed to obtain the obstacle location, which is critical to the convergence of the MCMC algorithm. An extensive numerical study is presented to illustrate the performance of the method. (Received January 29, 2020)

1157-65-224 **Mirjeta Pasha*** (mpasha1@kent.edu), Current 1300 Lefton Esplanade, OH, and **Lothar Reichel** and **Alessandro Buccini**. *Linearized Krylov subspace Bregman iteration with nonnegativity constraint.*

Bregman-type iterative methods have attracted considerable attention in recent years due to their ease of implementation and the high quality of the computed solutions they deliver. However, these iterative methods may require a large number of iterations and this reduces their attractiveness. This talk discusses a computationally attractive linearized Bregman algorithm by projecting the problem to be solved into an appropriately chosen low-dimensional Krylov subspace. The projection reduces both the number of iterations and the computational effort required for each iteration. A variant of this solution method, in which nonnegativity of each computed iterate is imposed, also is described. Convergence of the latter method is shown. Extensive numerical examples illustrate the performance of the proposed methods. (Received January 29, 2020)

1157-65-227 **Mirjeta Pasha*** (mpasha1@kent.edu), Current 1300 Lefton Esplanade, Kent, OH 44116, and **Lothar Reichel** and **Alessandro Buccini**. *Modulus-based iterative methods for constrained ℓ_p-ℓ_q minimization.*

The need to solve discrete ill-posed problems arises in many areas of science and engineering. The solutions of these problems, if they exist, are very sensitive to perturbations in available data. Regularization replaces the original problem by a nearby regularized problem, whose solution is less sensitive to the error in the data.

The regularized problem contains a fidelity term and a regularization term. Recently, the use of a p-norm to measure the fidelity term and a q-norm to measure the regularization term has received considerable attention. The balance between these terms is determined by a regularization parameter. In many applications, such as in image restoration, the desired solution is known to live in a convex set, such as the nonnegative orthant. It is natural to require the computed solution of the regularized problem to satisfy the same constraint(s). This talk will describe modulus-based iterative methods for computing a constrained approximate solution of the regularized problem. Convergence of the iterative method is shown, and numerical examples that illustrate the performance of the proposed method are presented. (Received January 29, 2020)

1157-65-233 **Zhiguo Yang*** (yang1508@purdue.edu), 2550 Yeager road, Apt 6-4, West Lafayette, IN 47906, and **Suchuan Dong**. *A roadmap for discretely energy-stable schemes for general dissipative systems based on gPAV method.*

In this talk, I will present a framework, termed as generalized Positive Auxiliary Variable (gPAV) method for devising discretely energy-stable schemes for general dissipative systems based on a generalized auxiliary variable. A unified procedure for treating the dissipative governing equations and the generalized auxiliary variable on the discrete level has been presented. The discrete energy stability and the positivity of the computed auxiliary variable have been proved for general dissipative systems. The gPAV method requires only the solution of linear algebraic equations within a time step. With appropriate choice of the operator in the algorithm, the resultant linear algebraic systems upon discretization involve only constant and time-independent coefficient matrices, which is highly efficient. (Received January 29, 2020)

1157-65-235 **Will Pazner*** (pazner1@llnl.gov), Center for Applied Scientific Computing, Lawrence Livermore National Laboratory, Livermore, CA 94550. *Robust and efficient matrix-free solvers for high-order finite element discretizations.*

High-order methods feature high accuracy per degree of freedom and high arithmetic intensity, both of which are attractive properties for GPU-based supercomputers. However, the cost of assembling or even storing the matrix associated with these discretizations can be prohibitively expensive, thus necessitating the use of matrix-free methods and solvers. Common preconditioners such as those arising from domain decomposition methods typically require dense blockwise factorizations, which are not possible without an explicit representation of the matrix.

In this talk, I will describe recent work on the development of matrix-free preconditioners and solvers for very high-order finite element discretizations. One approach is based on a sparse, low-order refined discretization which is spectrally equivalent to the high-order discretization. The resulting preconditioners are robust with respect to the mesh spacing and polynomial degree. These preconditioners can be extended to discontinuous Galerkin methods, for which they are also robust with respect to the penalty parameter. Applications of these methods to radiative transfer and incompressible fluid flow will be discussed. (Received January 29, 2020)

1157-65-254 **Abner J. Salgado*** (asalgad1@utk.edu), Department of Mathematics, University of Tennessee, Knoxville, TN 37996, and **Steven Wise** and **Jea-Hyun Park** (jpark79@utk.edu). *An ODE model for Nesterov's accelerated gradient descent method.* Preliminary report.

The so-called Nesterov accelerated descent method has become very popular since it is first order and converges substantially faster than classical steepest descent. This has motivated many researchers to find an explanation to why such an acceleration does in fact occur. In this talk, we will provide some partial answers to this question by showing that Nesterov can be seen as a discretization of a second-order ODE. (Received January 30, 2020)

1157-65-255 **Abner J. Salgado*** (asalgad1@utk.edu), Deparment of Mathematics, University of Tennessee, Knoxville, TN, and **Enrique Otarola** and **Alejandro Allendes**. *A posteriori error estimation for fluid models under singular forcing.*

We develop and analyze a posteriori error estimators for the stationary Stokes and Navier Stokes problems in the case that the force is a combination of deltas supported at points inside the domain of interest. We show that our estimators are reliable and locally efficient, and illustrate their performance within an adaptive method. We briefly comment on ongoing work regarding the Bousinessq system. (Received January 30, 2020)

1157-65-259 **xiaofeng yang***, 1523 Greene St, Columbia, SC 29208. *Efficient schemes with unconditionally energy stabilities for anisotropic phase field models.*

We consider numerical approximations for anisotropic phase field models, by taking the anisotropic Cahn-Hilliard/Allen-Cahn equations with their applications to the faceted pyramids on nanoscale crystal surfaces and the dendritic crystal growth problems, as special examples. The main challenge of constructing numerical

schemes with unconditional energy stabilities for these type of models is how to design proper temporal discretizations for the nonlinear terms with the strong anisotropy. We combine the recently developed IEQ/SAV approach with the stabilization technique, where some linear stabilization terms are added, which are shown to be crucial to remove the oscillations caused by the anisotropic coefficients, numerically. The novelty of the proposed schemes is that all nonlinear terms can be treated semi-explicitly, and one only needs to solve some coupled/decoupled, but linear equations at each time step. We further prove the unconditional energy stabilities rigorously, and present various 2D and 3D numerical simulations to demonstrate the stability and accuracy. (Received January 30, 2020)

1157-65-262 **Tong Qin*** (qin.428@osu.edu), **Zhen Chen**, **John Jakeman** and **Dongbin Xiu**.
Data-driven governing equations approximation using deep neural networks.

In this talk, I will introduce our recent work on using neural networks to learn dynamics from data. In particular, basing on the one-step integral form of the ODEs, we propose to use the residual network (ResNet) as the basic building block for dynamics recovery. The ResNet block can be considered as an exact one-step integration for autonomous ODE systems. Autonomous dynamical systems will first be used to illustrate the general framework. And then extensions to dynamical systems with random or time-dependent parameters will be discussed. (Received January 30, 2020)

1157-65-264 **Zhen Chen*** (chen.7168@osu.edu), 231 W. 18th Ave, MA 438, Columbus, OH 43210, and **Dongbin Xiu**. *On generalized residue network for deep learning of unknown dynamical systems.*

We present a general numerical approach for learning unknown dynamical systems using deep neural networks (DNNs). Our method is built upon recent studies that identified residue network (ResNet) as an effective neural network structure. In this paper, we present a generalized ResNet framework and broadly define "residue" as the discrepancy between observation data and prediction made by another model, which can be an existing coarse model or reduced order model. In this case, the generalized ResNet serves as a model correction to the existing model and recovers the unresolved dynamics. When an existing coarse model is not available, we present numerical strategies for fast creation of coarse models, to be used in conjunction with the generalized ResNet. These coarse models are constructed using the same data set and thus do not require additional resource. The generalized ResNet is capable of learning the underlying unknown equations and producing predictions with accuracy higher than the standard ResNet structure. This is demonstrated via several numerical examples, including long-term prediction of a chaotic system. (Received January 30, 2020)

1157-65-273 **James A Rossmanith*** (rossmani@iastate.edu), 411 Morrill Road, Ames, IA 50011, and **Christine Vaughan**. *Asymptotic-preserving Lax-Wendroff schemes for quasi-exponential moment-closure approximations of kinetic models.*

The dynamics of gas and plasma can be accurately modeled using kinetic Boltzmann equations. These equations are posed in a high-dimensional phase space, which is typically comprised of the spatial coordinates and the velocity coordinates. If the system is sufficiently collisional the kinetic equations may be replaced by a fluid approximation that is posed in physical space (i.e., a lower dimensional space than the full phase space). The precise form of the fluid approximation depends on the choice of the moment-closure. In general, finding a suitable robust moment-closure is still an open scientific problem.

In this work we consider a specific moment-closure based on a nonextensive entropy formulation. In particular, the true distribution is replaced by a Maxwellian distribution multiplied by a quasi-exponential function. We develop a high-order, locally-implicit, discontinuous Galerkin scheme to numerically solve resulting fluid equations. The numerical update is broken into two parts: (1) an update for the background Maxwellian distribution, and (2) an update for the non-Maxwellian corrections. We also develop limiters that guarantee that the inversion problem between moments of the distribution function and the parameters in the quasi-exponential function is well-posed. (Received January 31, 2020)

1157-65-284 **Qing Cheng*** (qcheng4@iit.edu), 3256 Wells street, Chicago, IL 60616, **Chun Liu** (cliu124@iit.edu), 10 West 35th Street, Chicago, IL 60616, and **Jie Shen** (shen7@purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907. *A new Lagrange multiplier approach for gradient flows.*

In this talk, we introduce a new Lagrange Multiplier approach to design unconditional energy stable schemes for gradient flows. The new approach leads to unconditionally energy stable schemes that are as accurate and efficient as the recently proposed SAV approach, but enjoys two additional advantages: (i) schemes based on the new approach dissipate the original energy, as opposed to a modified energy in the recently proposed SAV

approach; and (ii) they do not require the nonlinear part of the free energy to be bounded from below as is required in the SAV approach. The price we pay for these advantages is that a nonlinear algebraic equation has to be solved to determine the Lagrange multiplier. (Received January 31, 2020)

1157-65-285 **Qing Cheng*** (qcheng4@iit.edu), 3256 Wells street, Chicago, IL 60616, and **Jie Shen** (shen7@purdue.edu), Department of Mathematics, West Lafayette, IN 47907. *Global constraints preserving SAV schemes for gradient flows.*

In this talk, we introduce several efficient numerical schemes which preserve exactly the global constraints for constrained gradient flows. Our schemes are based on the SAV approach combined with the Lagrangian multiplier approach. They are as efficient as the SAV schemes for unconstrained gradient flows, i.e., only require solving linear equations with constant coefficients at each time step plus an additional nonlinear algebraic system which can be solved at negligible cost, can be unconditionally energy stable, and preserve exactly the global constraints for constrained gradient flows. Ample numerical results for phase-field vesicle membrane and optimal partition models are presented to validate the effectiveness and accuracy of the proposed numerical schemes. (Received January 31, 2020)

1157-65-313 **Qiang Ye*** (qye3@uky.edu), Department of Mathematics, University of Kentucky, Lexington, KY 40513. *Cayley Transform and Orthogonal Recurrent Neural Networks.*

Deep neural networks have emerged as one of the most powerful machine learning methods. Recurrent neural networks (RNNs) are special architectures designed to efficiently model sequential data by exploiting temporal connections within a sequence and handling variable sequence lengths in a dataset. However, they suffer from so-called vanishing or exploding gradient problems. Recent works address this issue by using a unitary/orthogonal recurrent matrix. In this talk. we will present a simpler and novel RNN that maintains orthogonal or unitary recurrent matrix using a scaled Cayley transform. The resulting method, called scaled Cayley orthogonal recurrent neural network, achieves superior results with fewer trainable parameters than other variants of RNNs in a variety experiments. A complex version with a unitary recurrent matrix is also developed that allows direct training of the scaling matrix in the Cayley transform. This is a joint work with K. Helfrich, D. Willmott, and K.D. Maduranga. (Received February 01, 2020)

1157-65-334 **Junshan Lin*** (jzl0097@auburn.edu), Department of Mathematics, Auburn University, Auburn, AL 36830. *A Super-resolution Imaging Approach by using Subwavelength Hole Resonances.*

Based on our recent studies on subwavelength hole resonances, we present a new imaging modality with illumination patterns generated from a collection of coupled resonant holes. When the incident frequencies are close to the resonant frequencies, the corresponding patterned illuminations encompass both low frequency and highly oscillatory waves, which allow for probing both the low and high spatial frequencies components of the imaging sample to achieve super-resolution. Under the weak scattering scenario, the linear imaging problem essentially boils down to a deconvolution problem that can be solved efficiently. The imaging setup, the underlying mathematical framework and the computational results will be exemplified in two dimensions. (Received February 02, 2020)

1157-65-341 **Mustafa Danis*** (danis@iastate.edu), 485 Carver 411 Morrill Rd., Ames, IA 50011-2104. *A High Order Positivity-Preserving Discontinuous Galerkin Method for Compressible Navier-Stokes Equations.*

We present a new high order positivity-preserving limiter with interior penalty discontinuous Galerkin (IPDG) method for compressible Navier-Stokes equations. By only applying a CFL condition and satisfying a weak-positivity criterion at Gauss-Lobatto quadrature points, we prove that the density and pressure approximations are nonnegative at all time levels with uniform high order of accuracy. The main advantage of this approach is that it is compatible with the standard definitions of the numerical flux definitions for convective and viscous terms. Therefore, the numerical penalty variables in Lax-Friedrichs and IPDG fluxes are not required to be modified. In addition, since limiter applies a CFL condition on the first-order forward Euler time discretization, this algorithm is very easy to implement and accessible to strong stability preserving (SSP) high order time discretizations. The simplicity of the formulation allows extension of this limiter to other DG schemes such as Direct DG (DDG). (Received February 02, 2020)

1157-65-357 **Zheng Sun*** (sun.2516@osu.edu), 231 W 18th Ave, 100 Math Tower, Columbus, OH 43210, and **Chi-Wang Shu**. *Error analysis of Runge-Kutta discontinuous Galerkin methods for linear time-dependent partial differential equations.* Preliminary report.

In this talk, we present error estimates of fully discrete Runge–Kutta discontinuous Galerkin (DG) schemes for linear time-dependent partial differential equations. The analysis applies to explicit Runge–Kutta time discretizations of any order. For spatial discretization, a general discrete operator is considered, which covers various DG methods, such as the upwind-biased DG method, the central DG method, the local DG method and the ultra-weak DG method. We obtain error estimates for stable and consistent fully discrete schemes, if the solution is sufficiently smooth and a spatial operator with certain properties exists. Applications to schemes for hyperbolic conservation laws, the heat equation, the dispersive equation and the wave equation are discussed. In particular, we provide an alternative proof of optimal error estimates of local DG methods for equations with high order derivatives in one dimension, which does not rely on energy inequalities of auxiliary unknowns. (Received February 02, 2020)

1157-65-378 **Yulong Xing*** (xing.205@osu.edu), Ohio State University, Columbus, OH 43210. *Energy conserving local discontinuous Galerkin methods for the improved Boussinesq equation.*

The Boussinesq-type equations describe the propagation of weakly non-linear long waves in shallow waters and are widely applied to model water waves in shallow seas and harbors. In this presentation, we propose a high-order local discontinuous Galerkin method to solve the improved Boussinesq equation, coupled with both explicit leap-frog and implicit midpoint energy-conserving time discretization. The proposed full-discrete method can be shown to conserve the discrete versions of both mass and energy of the continuous solution. The error estimate with optimal order of convergence is provided for the semi- discrete method. Our numerical experiments confirm optimal rates of convergence as well as the mass and energy conserving property, and show that the errors of the numerical solutions do not grow significantly in time due to the energy conserving property. A series of numerical experiments are provided to show that the proposed method has the capability to simulate the interaction between two solitary waves, single wave break-up and blow-up behavior well. (Received February 02, 2020)

1157-65-438 **Jianlin Xia*** (xiaj@math.purdue.edu). *Preconditioners based on enhanced structured incomplete factorization (eSIF) for general SPD matrices.* Preliminary report.

In this talk, we show a type of effective and robust preconditioners for general symmetric positive definte (SPD) matrices based on structured incomplete factorization (SIF), called enhanced SIF (eSIF) preconditioners. The original SIF preconditioning strategy by Xia, et al. applies compression to scaled off-diagonal blocks to yield structured preconditioners. As compared with the original SIF preconditioners, the eSIF preconditioners have three significant advantages:

1. With a similar truncation tolerance in the approximation of certain subblocks, the preconditioner can approximate the original matrix to a much higher accuracy.

2. The preconditioner is much more effective in the sense that the reduction of condition numbers is much more significant, and the eigenvalues of the preconditioned matrix are much better clustered.

3. The multilevel eSIF precondiitoners are guaranteed to be positive definite, while the original multilevel SIF preconditioner has an impractical strict requirement in order to preserve positive definiteness. (Received February 03, 2020)

1157-65-440 **Shuhao Cao***, Roland Hall, Irvine, CA 92697-3875. *A posteriori error estimates for a conforming virtual element method.*

An posteriori error analysis for the virtual element method (VEM) applied to a model Poisson problem is presented. The local error indicator is constructed based on a recovered equilibrated flux in the $H(\mathrm{div})$-conforming VEM space, which can be computed relying only on quantities available, namely the degrees of freedom and element-wise polynomial projections. An h-refinement strategy for polytopal meshes and hanging node automation are also discussed. This is a joint work with Long Chen. (Received February 03, 2020)

1157-65-458 **Carl F Jagels*** (jagels@hanover.edu), 517 Ball Drive, Hanover, IN 47243, and **Lothar Reichel** (reichel@math.kent.edu), Math dept, Kent State University, Kent, OH 44240. *Generalized averaged rational Gauss quadrature.*

Kyrlov subspace methods may be used to approximate matrix functionals of the form $v^T Av$ where A is defined by a large, sparse or structured, symmetric definite matrix. The methods are polynomial based and are associated with symmetric tridiagonal matrices and Gaussian quadrature rules. Modifications of these rules have been extensively studied. One recent modification is the generalized averaged Gauss quadrature introduced by Spalević

for estimating the quadrature error obtained when applying Gauss quadrature over an interval on the real axis. We examine the modification when applied to rational Gauss quadrature rules; rules that are associated with Laurent polynomials. In particular, the modification produces an extended set of Laurent polynomials. We discuss their properties and present some numerical examples. (Received February 03, 2020)

1157-65-459 **Alex Pothen*** (apothen@purdue.edu), Computer Science Department, Purdue University, West Lafayette, IN 47907, and **Yu-Hong Yeung** (yu-hong.yeung@pnnl.gov), Pacific Northwest National Laboratory, P O Box 999, Richland, WA 99352. *Real-time Simulations of Surgery with Augmented Matrices.*

We present the Augmented Matrix for Principal Submatrix update (AMPS) algorithm, a method that combines principal submatrix updates and Schur complement techniques, for interactive simulations of deformation and cutting of finite element meshes. Our approach features real-time solutions to the updated stiffness matrix systems to account for interactive changes in mesh connectivity and boundary conditions. Updates are accomplished by an augmented matrix formulation of the stiffness equations to maintain its consistency with changes to the underlying model without re-factorization at each time-step. As changes accumulate over multiple simulation time-steps, the augmented solution algorithm enables tens or hundreds of updates per second. Acceleration schemes that exploit sparsity, memoization and parallelization lead to the updates being computed in real-time. The complexity analysis and experimental results for this method demonstrate that it scales linearly with the number of nonzeros of the factors of the stiffness matrix. (Received February 03, 2020)

1157-65-462 **Xueyu Zhu*** (xueyu-zhu@uiowa.edu), Department of Mathematics, University of Iowa, Iowa City, IA 52242, and **Chuan Lu** (chuan-lu@uiowa.edu), Department of Mathematics, University of Iowa, Iowa city, IA 52242. *Bifidelity data-assisted Neural Network Approximation in Reduced Order Modeling for time-dependent PDEs.* Preliminary report.

In this talk, we will discuss a neural network-based reduced basis method with bifidelity models to accurately approximate the reduced solutions, particularly for time-dependent PDEs. We show that augmenting the bifidelity feature can help improve the accuracy of the neural networks as the approximator, and this method demonstrates its ability to produce accurate results with a limited number of high-fidelity simulations with an affordable computational cost. We also provide several numerical examples to illustrate the effectiveness of this method. (Received February 03, 2020)

1157-65-468 **Mykhailo Kuian**, **Lothar Reichel*** (reichel@math.kent.edu) and **Sergij Shiyanovskii**. *Optimally Conditioned Vandermonde-Like Matrices.*

Vandermonde matrices arise frequently in computational mathematics in problems that require polynomial approximation, differentiation, or integration. These matrices are defined by a set of n distinct nodes $x_1, x_2, \ldots, x_n$ and a monomial basis. A difficulty with Vandermonde matrices is that they typically are quite ill-conditioned when the nodes are real and n is not very small. The ill-conditioning often can be reduced significantly by using a basis of orthonormal polynomials $p_0, p_1, \ldots, p_{n-1}$, with $\deg(p_j) = j$. This was first observed by Gautschi. The matrices so obtained are commonly referred to as Vandermonde-like. Gautschi analyzed optimally conditioned and optimally scaled square Vandermonde and Vandermonde-like matrices with real nodes. We extend Gautschi's analysis to rectangular Vandermonde-like matrices with real nodes, as well as to Vandermonde-like matrices with nodes on the unit circle in the complex plane. (Received February 03, 2020)

1157-65-473 **Thi-Phong Nguyen*** (nguy4105@umn.edu), 500 Pillsbury Drive SE, Minneapolis, MN 55455, and **Bojan Guzina**. *Sampling methods for identifying cracks in rock from surface seismic measurements.*

I will present in this talk the theoretical analysis and numerical implementation of a so-called Generalized Linear Sampling Method (GLSM) to reconstruct a partially-closed crack in rock from surface seismic measurements. For the mathematical viewpoint, the crack is characterized by a second order tensor synthesizing its special stiffness - relation the jump in seismic field cross the crack to the contact traction. Since this tensorial quantity intrinsically dependent on the cracks' aperture and the surface roughness, that makes the elastic wave equation partially heterogeneous. Therefore, solving this problem is difficult. The advantage of GLSM is to build an indicator function, which allows to detect if a trial curve belongs to the crack. Some related numerical results will be presented at the end. (Received February 03, 2020)

1157-65-479 **Javier A. Almonacid*** (javiera@sfu.ca), 8888 University Drive, Burnaby, BC V5A 1S6, Canada. *Attractors and the Spectrum of a Zeroth-Order Pseudo-Differential Operator.* Preliminary report.

Physicists have observed that in the presence of bathymetry, the forced propagation of internal waves can lead to the development of singular patterns (called "attractors") which accumulate most of the wave energy. On the other hand, recent results have pointed out a connection between this phenomenon and the spectral theory of zeroth-order pseudo-differential operators. In this talk, we explore this connection from a Numerical Analysis perspective. First, we will review the origins of the model, as well as the main characteristics that the solution is expected to have. Then, we focus on the eigenvalue problem, how the eigenvalues of the viscous problem approach those of the original problem (even when both are completely different operators), and how some of the eigenfunctions seem to provide information about the long-time behaviour of the solution to the evolution problem. (Received February 03, 2020)

1157-65-525 **Juntao Huang*** (huangj75@msu.edu), **Weifeng Zhao** and **Chi-Wang Shu**. *Positivity-preserving time discretizations for production-destruction equations with applications to non-equilibrium flows.*

In this talk, we will present a family of second-order and third-order modified Patankar Runge-Kutta (MPRK) methods, which are conservative and unconditionally positivity-preserving, for production-destruction equations. We derive necessary and sufficient conditions to obtain the designed order of accuracy. This ordinary differential equation solver is then extended to solve a class of semi-discrete schemes for PDEs. Combining this time integration method with the positivity-preserving finite difference weighted essentially non-oscillatory (WENO) schemes, we successfully obtain a positivity-preserving WENO scheme for non-equilibrium flows. Various numerical tests are reported to demonstrate the effectiveness of the methods. (Received February 03, 2020)

1157-65-571 **Jiafeng Jin*** (jjin3@kent.edu), **Omar De la Cruz Cabrera** and **Lothar Reichel**. *New Models for Multi-Class Networks.*

Networks model complex systems and consist of a set of nodes and a set of edges. Any two nodes may be connected by an edge. Multi-class networks arise when the nodes belong to different classes, with the nodes in each class modeling a different type of entity. This paper introduces new models for this kind of network. These models differ from available ones in how the adjacency matrix for the network is formed. This matrix describes the relationship between the nodes and classes. The importance of a node depends both on the importance it receives from nodes in the same class, and how it relates to nodes in other classes. This importance is captured by computing the left Perron vector of the adjacency matrix. Perturbation results shed light on how the relative importance of a node changes by adding an extra edge between nodes. Experiments with both synthetic and real data sets illustrate features of the models discussed. (Received February 04, 2020)

1157-65-575 **Yuwei Luo**, **Tingran Gao** and **Lek-Heng Lim*** (lekheng@galton.uchicago.edu). *Differential geometry of Nesterov's accelerated gradient method.*

Nesterov's accelerated gradient method (NAG) and its derivatives have grown to paramount importance in machine learning, in part due to their ability to achieve optimal rates of convergence among all first-order methods for convex smooth functions. We proffer the following insight: NAG in the limiting case as step size goes to zero defines a spray on a manifold. This insight has two immediate implications: (i) it gives a geometric generalization of NAG to any Riemannian manifold in a completely natural way; (ii) it leads to a new spray accelerated gradient method (SAG) that improves NAG. In the limiting case, both SAG and NAG converge to the same spray. However, for finite time steps, SAG remains a faithful discretization of the spray; is numerically stabler; and preserves the continuous trajectories of the spray — none of which holds for NAG. Our spray approach is a natural next step in the modeling of Nesterov's scheme using second-order ordinary differential equations (ODEs). The trajectories of these ODEs depend on time, making it a challenge to find a discretization that respects the ODEs' dynamical behavior. By resorting to sprays, we circumvent the time dependence and obtain an entire family of geometry-preserve algorithms that generalize easily to manifolds. (Received February 04, 2020)

1157-65-592 **Yuwen Li*** (yuwli@psu.edu). *Adaptivity in finite element exterior calculus.*

Using the language of differential forms, Arnold, Falk, Winther developed a unified framework named "finite element exterior calculus (FEEC)" for analyzing numerical problems posed on de Rham complexes, e.g., Poisson, Hodge-Laplace, and Maxwell equations. A priori error estimates based on FEEC are well-established in the literature. On the other hand, a posteriori error estimation and analysis of adaptive algorithms are limited in FEEC. In this talk, I will start with basic concepts in adaptive finite element methods and review recent progress

on a posteriori error estimates, convergence, and complexity results of adaptive methods in FEEC. (Received February 04, 2020)

1157-65-593 **Kevin Williamson*** (kwillia1@umbc.edu) and **Bedřich Sousedík** (sousedik@umbc.edu). *A posteriori error estimates and adaptive mesh refinement for the Stokes-Brinkman problem.*

The Stokes–Brinkman equations model flow in heterogeneous porous media by combining the Stokes and Darcy models of flow into a single system of equations. With suitable parameters, the equations can model either flow without detailed knowledge of the interface between the two regions. Thus, the Stokes–Brinkman equations provide an alternative to coupled Darcy–Stokes models. After a brief review of the Stokes–Brinkman problem and its discretization using Taylor-Hood finite elements, we present a residual-based a posteriori error estimate and use it to drive an adaptive mesh refinement process. We compare several strategies for the mesh refinement, and demonstrate its effectiveness by numerical experiments in both 2D and 3D. (Received February 04, 2020)

1157-65-594 **Weizhang Huang*** (whuang@ku.edu). *Phase-field modeling of brittle fracture and its moving mesh finite element computation.*

Brittle fracture is the cracking of a metallic object or other elastic material under stress where the material exhibits little or no evidence of plastic deformation before the fracture occurs. It usually occurs very rapidly and can be catastrophic in engineering practice. Understanding the initiation and propagation of brittle fracture is vital to the prevention of fracture failure in engineering design, however its mathematical modeling has been extremely challenging due to the discontinuity nature of cracking. In recent years the phase-field mathematical modeling of brittle fracture has gained considerable attention, where the so-called phase-field function is introduced and used to characterize the damage level of the material, cracks are represented as sharp but continuous layers, and the total free energy is modified to involve cracking effects through the phase-field function. In this talk, we will discuss some of the mathematical and computational challenges with the phase-field modeling of brittle fracture. We will also present an adaptive moving mesh finite element method for the numerical solution of the model and numerical results obtained with the method. (Received February 04, 2020)

1157-65-595 **Lin Mu*** (linmu@uga.edu). *A Posteriori Error Estimates for Weak Galerkin Methods for Stokes Equations on Polygonal Meshes.*

In this talk, we shall introduce a simple posteriori error estimate of the weak Galerkin (WG) finite element method. This residual type a posteriori error estimate can be applied to general polygonal meshes or meshes with hanging nodes. The reliability and efficiency of the designed error estimator have been proved and validated. These results demonstrate the effectiveness of the adaptive mesh refinement guided by the proposed error estimator. (Received February 04, 2020)

1157-65-596 **Shun Zhang*** (shun.zhang@cityu.edu.hk). *Battling Gibbs Phenomenon: Approximation of Discontinuous Solutions of PDEs by Adaptive Finite Elements.*

For both linear and nonlinear hyperbolic partial differential equations, the solution can be discontinuous. For singularly perturbed problems, when the transient layer is very sharp, the solution can also be viewed as discontinuous. When piecewise polynomials are used to approximate these discontinuous solutions, the numerical solutions often overshoot near a discontinuity. Can this be resolved by adaptive mesh refinements? Shall we use continuous or discontinuous approximation spaces? In this talk, we explain by simple examples that the piecewise discontinuous constant approximation with adaptive mesh refinement is the best choice to achieve accuracy without overshootings. For discontinuous piecewise linear approximations, non-trivial overshootings will be observed unless the mesh is matched with discontinuity. For continuous piecewise linear approximations, non-trivial overshootings will always be observed under regular meshes. We calculate the explicit overshooting values for several typical cases. We also will discuss the overshoot for numerical methods based on L1 minimization. As an example, for a transport problems with discontinuous solutions and boundary conditions, we propose a flux-based least-squares finite element method. (Received February 04, 2020)

1157-65-598 **Cuiyu He*** (cuiyu.he@uga.edu). *A Posteriori Error Estimates with Boundary Correction for a Cut Finite Element Method.*

Unfitted finite element methods, in particular, Cut Finite Element Method (CutFEM), have been intensively studied in the last decade. The topic of this talk focuses on the residual based a posteriori error estimation for the CutFEM method applied to an elliptic model problem. We consider the problem with non-polygonal boundary and the error analysis takes into account both the geometry and data approximations on the boundary.

The reliability and efficiency are theoretically proved. Moreover, constants are robust with respect to the mesh-domain intersection. (Received February 04, 2020)

1157-65-599 **Guosheng Fu*** (gfu@nd.edu) and **Mark Ainsworth** (mark_ainsworth@brown.edu). *Fully computable a posteriori error bounds for HDG methods for diffusion problems.*

We derive a posteriori error estimates for the HDG methods, including both the primal and mixed formulations, for the approximation of a linear second-order elliptic problem on conforming simplicial meshes in two and three dimensions. We obtain fully computable, constant free, a posteriori error bounds on the broken energy seminorm and the HDG energy (semi)norm of the error. The estimators are also shown to provide local lower bounds for the HDG energy (semi)norm of the error up to a constant and a higher-order data oscillation term. Numerical examples aregiven illustrating the theoretical results. (Received February 04, 2020)

1157-65-601 **Lina Ma*** (lina.ma@trincoll.edu), **Lina Ma** and **Chun Liu**. *Model Reduction on Langevin Dynamics.*

Langevin dynamics models arise from a wide variety of problems, especially where a mechanical system is subject to random forces that can be modeled by white noise. In this talk, I will present a derivation of a coarse-grained description, in the form of a generalized Langevin equation, from the Langevin dynamics model that describes the dynamics of bio-molecules. To facilitate the construction and implementation of the reduced models, the problem is then formulated as a reduced-order modeling problem. The reduced models can then be directly obtained from a Galerkin projection to appropriately defined Krylov subspaces. (Received February 04, 2020)

1157-65-604 **Yiwei Wang*** (ywang487@iit.edu) and **Chun Liu** (cliu124@iit.edu). *Lagrangian schemes for gradient flows and diffusions: a discrete energetic variational approach.*

In talk, we present a systematic framework of deriving variational-structure-preserving numerical schemes by a discrete energetic variational approach, which performs energetic variational approach at a semi-discrete level. In particular, we apply such an approach to construct Lagrangian schemes for some problems modeled by gradient flows and diffusions, such as porous medium equation, equations for phase-field and liquid crystals. Numerical examples show the advantages of our Lagrangian schemes in capturing singularities, thin diffuse interfaces, and free boundaries. (Received February 04, 2020)

1157-65-607 **Jing Yang*** (yang648@purdue.edu). *A posteriori error estimator for the dual finite element approximation to a singularly perturbed reaction-diffusion problem.*

We derive a residual-type a posteriori error estimate for the dual finite element approximation to a singularly perturbed reaction-diffusion problem. It is robust with respect to the singularly perturbed limit. The robust upper and lower bound are derived based on a quasi- Helmholtz decomposition of H(div) space. Finally, the application of the estimator for the dual finite element approximation is presented. (Received February 04, 2020)

1157-65-625 **Yue Yu*** (yuy214@lehigh.edu), Math Department, Lehigh University, Bethlehem, PA 18015. *A Multiscale/Multiphysics Coupling Framework for Bioprosthetic Heart Valves (BHVs) Damage.*

Bioprosthetic heart valves (BHVs) are the most popular artificial replacements for diseased valves that mimic the structure of native valves. However, the life span of BHVs remains limited to 10-15 years, and the mechanisms that underlie BHVs failure remain poorly understood. Therefore, developing a unifying mathematical framework which captures material damage phenomena in the fluid-structure interaction environment would be extremely valuable for studying BHVs failure. In this talk, I will focus on developing a fluid—peridynamics coupling framework to capture the fluid-induced material damage in BHVs failure. In the peridynamic and other nonlocal models the loading boundary conditions should be defined in a nonlocal way, while in fluid—structure interfaces the hydrodynamic loadings from the fluid side are typically provided on a sharp co-dimension one surface. To overcome this challenge, we have proposed a new nonlocal Neumann-type boundary condition which provides an approximation of physical boundary conditions on a sharp surface. Based on this nonlocal boundary condition, we have developed a stable and asymptotically compatible fluid—peridynamics coupling framework without overlapping regions. (Received February 04, 2020)

1157-65-635 **Ding LU*** (ding.lu@uky.edu), Department of Mathematics, LEXINGTON, KY 40506. *On the local convergence of SCF iteration for eigenvector-dependent nonlinear eigenvalue problems.*

Eigenvector-dependent nonlinear eigenvalue problems (NEPv) have long played a key role in electronic structure calculations, and have more recently become increasingly popular in many data science applications. The Self-Consistent Field (SCF) iteration is one of the most general and widely-used methods for solving NEPv's. Despite simple in the formulation, SCF is also prone to slow convergence and may not even converge at all. This has triggered considerable research interests in the past years for the convergence analysis. In this talk, we will revisit those existing studies and present our most recent result on the local convergence of SCF. A key insight made in our analysis is that SCF can be simplified locally as a linear operator but in the "tangent matrix" associated with the iteration vectors. This allows a closer look at the iteration error and a better insight into the local convergence factor, using the spectral radius of the linear operator. Furthermore, numerical experiments show our new convergence factor can precisely capture the local convergence behavior of SCF. (Received February 04, 2020)

1157-65-643 **Edgar Solomonik*** (solomon2@illinois.edu), Thomas M. Siebel Center for Computer Science, 201 North Goodwin Avenue, Urbana, IL 61801. *Efficient Algorithms and Scalable Software for Tensor Decompositions.*

Tensor decompositions have numerous applications within scientific computing and data analysis. The canonical polyadic (CP) decomposition is the perhaps the most prominent of these. This decomposition can be used for discovery of fast bilinear algorithms, obtaining reduced scaling quantum chemistry algorithms, as well as compressing and understanding datasets from various domains. We describe a new algorithm for CP decomposition, pairwise perturbation, which improves state of the art approaches in theory and practice. Further, we provide a comparative study of the accuracy and performance (in sequential and distributed settings) of various CP decomposition methods. We identify a new robust regularization approach and provide the first effective parallelizations of some of these methods for dense and sparse tensors. We also describe a new approach for tensor completion with CP decomposition, which yields the first high-level distributed algorithm implementations for this problem. Finally, we provide a software library with high-level Python implementations of many tensor decomposition algorithms, which can run sequentially or leverage the Cyclops library for parallel execution. (Received February 05, 2020)

1157-65-644 **Ignacio Tomas*** (itomas@sandia.gov), Sandia National Laboratories, PO Box 5800, Org 1442, Albuquerque, NM, and **Jean-Luc Guermond** and **Bojan Popov**. *Invariant domain preserving methods and convex limiting.*

The first step in the development of a high-order scheme is the construction of a robust first-order method (ideally) endowed with a sound theoretical basis. With that goal in mind, we develop a general framework of first-order fully-discrete numerical schemes that are guaranteed to preserve every convex invariant of the hyperbolic system and satisfy every entropy inequality. This framework is not tied to any particular space discretization technique (e.g. finite volumes, continuous or discontinuous finite elements) and it is of general applicability: it works in any mesh, in any space dimension, and uses neither exact nor approximate Riemann solvers. We then proceed to present a new flux-limiting (hybridization) technique. This technique does not preserve or enforce bounds on conserved variables, but rather bounds on quasiconvex/concave functionals of the conserved variables. This flux-limiting technique, is suitable to enforce/preserve convex constraints of the numerical solution which are natural of hyperbolic systems. Examples of such bounds are positivity of the internal energy and minimum principle of the specific entropy in the context of Euler's equations. We have coined this technique "convex limiting". (Received February 05, 2020)

1157-65-650 **Xiaofeng Ou*** (ou17@purdue.edu), Room 441, Math Science Building, 150 N University St, West Lafayette, IN 47906, and **Jianlin Xia** (xiaj@purdue.edu), Room 442, Math Science Building, 150 N University St, West Lafayette, IN 47906. *A Matlab Package for Superfast Divide-and-Conquer Hermitian Eigenvalue Decompositions.*

We discuss a Matlab package for the quick eigenvalue decompositions of some useful Hermitian matrices (tridiagonal, banded, diagonal plus low rank, Toeplitz, hierarchically semiseparable, etc.). The package is based on a superfast divide-and-conquer algorithm that uses structured matrices and the fast multiple method to accelerate all the intermediate operations. The algorithm has nearly O(n) complexity for Hermitian matrices with small off-diagonal (numerical) ranks. The package tackles a series of challenges in the accuracy and efficiency of the structured accelerations of the divide-and-conquer algorithm. For dense rank structured matrices, it does not

an extra tridiagonal reduction. The Matlab routine is significantly faster than the Matlab eig function when the matrix size increases. (Received February 05, 2020)

68 ► Computer science

1157-68-99 **Michael Perlmutter*** (perlmut6@msu.edu), **Feng Gao**, **Guy Wolf** and **Matthew Hirn**. *Scattering Neural Networks for Graph Structured Data.*

The scattering transform is a mathematical model of convolutional neural networks (CNNs) introduced for functions defined on Euclidean space by Stephané Mallat. It differs from traditional CNNs by using predesigned, wavelet filters rather than filters which are learned from training data. This leads to a network which provably has desirable mathematical properties such as translation invariance and diffeomorphism stability. Moreover, in situations where the wavelets can be designed in correspondence to underlying physics, it can produce numerical results which rival state of the art CNNs. However, many data sets of interest have an intrinsically non-Euclidean structure and are better modeled as graphs. This motivates us to construct geometric versions of the scattering transform using the spectral decomposition of the Graph Laplacian. We will discuss applications of these networks to a variety of geometric deep learning tasks such as graph classification and show that, analogously to its Euclidean predecessor, the graph scattering transforms possess desirable invariance and stability properties. (Received January 20, 2020)

1157-68-111 **Petros Drineas***, Lawson Computer Science Building (LWSN), 305 N University Street, West Lafayette, IN 47907. *Randomization in Numerical Linear Algebra (RandNLA): fast regression solvers.*

The introduction of randomization in the design and analysis of algorithms for matrix computations (such as matrix multiplication, regression, the Singular Value Decomposition (SVD), etc.) over the past 20 years provided a new paradigm and a complementary perspective to traditional numerical linear algebra approaches. These novel approaches were motivated by technological developments in many areas of scientific research that permit the automatic generation of Big Data, which are often modeled as matrices. In this talk, we will outline how such approaches can be used to design fast solvers for least-squares problems, ridge-regression problems, and even linear programs. (Received January 22, 2020)

1157-68-138 **Jiefeng Chen**, **Xi Wu**, **Vaibhav Rastogi**, **Yingyu Liang*** (yliang@cs.wisc.edu) and **Somesh Jha**. *Robust Attribution Regularization.*

An emerging problem in trustworthy machine learning is to train models that produce robust interpretations for their predictions. We take a step towards solving this problem through the lens of axiomatic attribution of neural networks. Our theory is grounded in the recent work, Integrated Gradients (IG), in axiomatically attributing a neural network's output change to its input change. We propose training objectives in classic robust optimization models to achieve robust IG attributions. Our objectives give principled generalizations of previous objectives designed for robust predictions, and they naturally degenerate to classic soft-margin training for one-layer neural networks. We also generalize previous theory and prove that the objectives for different robust optimization models are closely related. Experiments demonstrate the effectiveness of our method, and also point to intriguing problems which hint at the need for better optimization techniques or better neural network architectures for robust attribution training. (Received January 23, 2020)

1157-68-246 **Ivan Dokmanić***, ivan.dokmanic@unibas.ch, and **Shuai Huang**, **Mona Zehni** and **Zhizhen Zhao**. *Reconstructing Point Sets from Distance Distributions: Turnpikes, Beltways, and a bit More.*

The unassigned distance geometry problem (uDGP) asks to reconstruct the geometry of a point set from a subset of interpoint distances without knowing which distances belong to which pairs of points. It models room geometry reconstruction from echoes, positioning by multipath, and nanostructure determination by powder diffraction. It can be seen as assembling a pile of sticks into a structure without labels telling us which sticks go together.

I will describe our approach to 1D unassigned DGP, known as the turnpike on a line or as the beltway on a loop. The usual backtracking method is slow and performs poorly in noise. We approximate the problem by discretizing the domain and representing the points via a density supported on the discrete grid. We then show that the distance distribution is quadratic in the density and solve the resulting quadratic inverse problem by projected gradient descent. Combined with an adaptation of the spectral initialization technique from phase retrieval, this leads to state-of-the-art performance across a number of numerical experiments. I will conclude

by explaining how our ideas can be generalized to higher dimensions and describe some curious connections to unknown-view tomography. (Received January 30, 2020)

1157-68-299 **Hunter Chase*** (hchase2@uic.edu). *Query learning with random counterexamples.* Preliminary report.

Several notions of complexity of set systems correspond both with model-theoretic dividing lines and notions of machine learning. Set systems of finite VC-dimension, which correspond to NIP formulas, are exactly those which are PAC-learnable, while set systems of finite Littlestone dimension, which correspond to stable formulas, are exactly those which are online-learnable. Recently, a connection between query learning and stable NFCP formulas has been described.

In equivalence query learning, a learner attempts to identify a set by submitting hypotheses and learning from counterexamples to those hypotheses. We consider query learning where counterexamples are chosen randomly. We give bounds in this setting and describe connections with model theory. No background in model theory or machine learning will be required.

This is joint work with James Freitag and Kyle Gannon. (Received February 01, 2020)

1157-68-307 **Karl RB Schmitt*** (karl.schmitt@valpo.edu), **James P Canning**, **Emma E Ingram**, **Sammantha Nowak-Wolff**, **Adriana M Ortiz**, **Cody Packer**, **Arezu Mansuri**, **Charles Morris** and **Casey Primozic**. *Developments and Advances in Network Classification using Machine Learning.*

Networks are often categorized according to the underlying phenomena that they represent, such as re-tweets, protein interactions, or web page links. It is generally believed that networks from different categories have inherently unique network characteristics. In this work, we find strong evidence supporting this hypothesis by learning a classification model $f : \mathbf{x} \to y$ that is able to *accurately* predict the category of a new arbitrary unknown network G' described only by a D-dimensional feature vector x' where $y \in \{1, 2, \ldots, K\}$ is the class label (category). These results will be compared to previous work on network classification using community decomposition and an alternative classification algorithm.

Efforts to make the work more robust and deployable by increasing the underlying data, classification mechanism and more will be discussed along with some open challenges for the Network Science community. (Received February 01, 2020)

1157-68-444 **Fengguang Song*** (fgsong@cs.iupui.edu), Department of Computer Science, 723 W. Michigan St., SL 280, Indianapolis, IN 46202. *From textbook algorithms to scalable parallel linear algebra algorithms.*

Implementing parallel linear algebra (LA) algorithms to achieve scalable performance on massively parallel many-core systems requires a comprehensive design and demand taking advantage of various architectural strengths. This talk first presents the state of the art in modern computer architectures. Motivated by emerging computer architectures' strengths, weaknesses and challenges, the talk then introduces a class of new parallel linear algebra algorithms, which is able to maximize data locality, degree of parallelism, data reuse, and minimize load imbalance, synchronization and communication costs. The parallel library builds on a decentralized data-flow driven dynamic scheduling runtime system, named Task BLAS (TBLAS), to automatically execute parallel algorithms efficiently on large-scale high performance computing (HPC) systems. The talk is summarized with a list of general-purpose principles for designing scalable parallel algorithms, and successfully applying the principles to a new problem domain of lattice Boltzmann methods (LBM). (Received February 03, 2020)

1157-68-617 **Tianshi Xu*** (xuxx1180@umn.edu), **Vasilis Kalantzis**, **Geoffrey Dillon**, **Yuanzhe Xi**, **Ruipeng Li** and **Yousef Saad**. *Recent Development of Generalized Multilevel Low-Rank Preconditioning and Solution Package.*

This talk summarizes the recent development and implementation of GeMSLR (Generalized Multilevel Schur complement Low-Rank), a distributed-memory preconditioner for the solution of large and sparse (non)Hermitian linear systems of equations. The GeMSLR preconditioner is purely algebraic and is based on a multilevel reordering of the original set of equations/variables. At each given level, GeMSLR decouples the solution of the current linear system into one associated with the interior variables and another associated with the interface ones. The first subproblem is block-diagonal and can be solved in parallel by applying some form of ILU preconditioning. The recursive nature of the preconditioner appears on the second subproblem where the Schur complement system is preconditioned by the interface coupling matrix. Low-rank correction terms can be added at each level to further enhance robustness, and these are applied using the Woodbury formula. The user can increase the accuracy of the low-rank correction terms if the convergence is not satisfactory. GeMSLR is

implemented in MPI and has been tested on E5-2680v3 processors. We demonstrate the potential of GeMSLR by presenting numerical tests performed on several 2D and 3D problems, and both strong and weak scaling is discussed. (Received February 04, 2020)

70 ► Mechanics of particles and systems

1157-70-481 **Cody Reeves*** (codyreeves2015@u.northwestern.edu) and **Petia Vlahovska**.
Hydrodynamics for Active Suspensions of Self-Rotating Particles.

There has been a recent surge in experiments being conducted on active self-rotating particles such as Quincke rotors and ferromagnetic colloids. These systems have demonstrated a wide array of complex collective behaviors: including phase separation, structure formation, and turbulent-like fluid flows. We introduce a continuum model for a suspension of active spinners and the corresponding induced hydrodynamics. This model derivation utilizes the physical dynamics of discrete spinners and, using the Smoluchowski equation, coarsens the dynamics into a continuous distribution. This model has been successful in reproducing and understanding results observed in recent experiments and discrete simulations: including turbulent-like flows and collective formations. We can then study the dynamics of these spinners while under both rigid and deformable confinement. The coupling between a boundary and the spinner-driven hydrodynamics allows for the structure formation. (Received February 03, 2020)

1157-70-629 **Marcial Gonzalez*** (marcial-gonzalez@purdue.edu), School of Mechanical Engineering, Purdue University, West Lafayette, IN 47907. *Contact mechanics as a reversible upscaling method: from continuum contact mechanics to mechanistic nonlocal contact laws and back.*

The complex properties of granular systems depend primarily on the nature of the interaction between particles and between particles and their environment. These interactions often involve different length and time scales, requiring quantifying mechanical behaviors across the different scales and identifying connections between them. In this presentation, I will demonstrate that if inter-particle contact laws are derived from an analytical upscaling of continuum contact mechanics of deformable bodies, then the upscaling procedure is reversible, i.e., the particle deformed configuration and its stress-strain fields are recoverable. In addition, this analytical procedure provides a framework for deriving nonlocal contact formulations that account for the interplay of deformations due to multiple contact forces acting on a single particle—characteristic of confined granular systems. This is in sharp contrast to traditional contact mechanics theories, which are strictly local and assume that contacts are independent regardless the confinement of the particles. For definiteness, I will restrict attention to elastic and elasto-plastic spheres and demonstrate how to further bridge particle scale mechanics to the continuum scales of interest for engineering applications. (Received February 04, 2020)

76 ► Fluid mechanics

1157-76-110 **El Mostafa Kalmoun*** (ekalmoun@qu.edu.qa), Department of Math, Stat and Physics, College of Arts and Sciences, Qatar Univeristy, Doha, Qatar. *2D Point Vortex Dynamics in Doubly-Connected Polygonal Domains.*

2D point vortex motion has sparked intense research due to its important applications in fluid dynamics and geophysics. A point vortex is a model of potential flow in which vorticity of the surrounding flow is present only at a single point. The dynamics of point vortices is governed by a Hamiltonian system, which can be derived from the Euler equations describing the velocity field and pressure of a 2D incompressible and inviscid flow. In this work, we consider the case of a single point vortex in doubly-connected polygonal domains. By the help of the MATLAB toolbox PlgCirMap, our method utilize first conformal mappings to transfer the polygonal domains onto circular ones where the Crowdy and Marshall analytic formula can be applied. Then, we employ the Schottky-Klein prime function to compute the Hamiltonian governing the point vortex motion in circular domains. This prime function is numerically computed by either a spectral method or a boundary integral equation with the generalized Neumann kernel. Our objective is to compare between the topological structures of the contour lines of the Hamiltonian in symmetric and asymmetric domains. We discuss the effect of symmetry breaking, and obstacle location and shape on the behavior of vortex motion. (Received January 22, 2020)

1157-76-118 **Arezoo M Ardekani***, 585 Purdue Mall, West Lafayette, IN 47907, and **Nikhil Desai** (desai63@purdue.edu), Many marine bacteria, particularly those in r, West Lafayette, IN 47907. *Bio-remediation and motility: how hydrodynamics and chemotaxis affect bacterial foraging.*

Many marine bacteria, particularly those in regions experiencing natural oil spills, can break down the hydrocarbons in oil and use them for nutrition. The fluid flow induced by a bacterium, via its appendages, depends acutely on its proximity to surfaces/interfaces. This has important consequences on a bacterium's ability to populate any nutrient source. In this talk, we theoretically demonstrate the phenomenon of 'hydrodynamic trapping' of bacteria around spherical obstacles (rigid or fluid) and discuss its implications on nutrient-source colonization and consumption. We begin by showing how surfactant-laden drops act as more effective bacterial traps than clean drops. Next, we explain the important differences between hydrodynamic trapping around stationary versus translating obstacles. Finally, we show how hydrodynamic interactions are ably complemented by chemotaxis. In fact, hydrodynamic and chemotactic motility combine to non-trivially alter the distribution of marine bacteria around both stationary oil drops and sinking marine snow particles effusing hydrocarbon/nutrient plumes. Thus, we present a theoretical framework which allows us to study a broad range of scenarios that any foraging marine microbe is likely to experience. (Received January 22, 2020)

1157-76-130 **Vishal Anand** and **Ivan C. Christov*** (christov@purdue.edu), School of Mechanical Engineering, 585 Purdue Mall, WEST LAFAYETTE, IN 47907-2088, and **Tanmay C. Shidhore** and **Xiaojia Wang**. *Mathematical modeling of viscous flows in deformable conduits.*

The pressure-driven flow profile through a channel of arbitrary cross-section can be calculated from exact unidirectional solutions of the steady Stokes equations, via classical methods of mathematical physics. Recently, however, experiments on internal flows in rectangular channels with a soft boundary have shown that coupling between the wall deformation and the pressure gradient driving the flow lead to a flow profile that is not described by the classical solutions. To address this knowledge gap, we take a perturbative approach to solving the coupled problem of viscous flow and elastic deformation. Specifically, for a long and slender geometry, the flow problem is reduced to unidirectional Stokes flow in a slowly-varying geometry. In parallel, the deformation problem is reduced to a two-dimensional linear elasticity boundary value problem in each flow-wise cross-section. Importantly, this mathematical approach yields closed-form solutions for the flow and deformation profiles, which, under the assumption of steady flow, predict the hydraulic resistance (i.e., pressure drop required to maintain a given volumetric flow rate). Our mathematical model compares favorably to microfludic experiments.

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1157-76-137 **Abhinandan Chowdhury***, chowdhury@savannahstate.edu, Savannah, GA 31404. *Stochastic Functional Expansion for Identifying the Effective Heat Conductivity Coefficient of Polydisperse Suspension.*

We consider a random two-phase medium which represents a matrix containing an array of non-overlapping spherical inclusions with random radii. A statistical theory of transport phenomena in the medium is constructed by means of the functional (Volterra-Wiener) series approach for identifying the effective heat conductivity of a polydisperse spherical suspension. An approximate model based on power-series expansion of the kernels with respect to the volume fraction is developed. The functional series for the temperature is rendered virial in the sense that its truncation after the p-tuple term asymptotically correct to the order γ^p where γ is the mean number of spheres per unit volume – also proportional to the volume fraction. The case $p = 2$ is considered in detail and the needed kernels of the functional series are found to the order γ^2. The truncated Volterra-Wiener expansion is applied consistently to derive the equations for the kernels and their contributions to the overall (effective) modulus are identified. In this way, not only the effective conductivity, but also all needed correlation functions can be expressed in closed form, correct to the said order. (Received January 23, 2020)

1157-76-141 **Chun Liu** (cliu124@iit.edu) and **Jan-Eric Sulzbach*** (jsulzbach@hawk.iit.edu). *Thermal Effects in Transport of Ideal Gas.* Preliminary report.

We present a unified energetic variational approach in the derivation of the thermodynamics of compressible fluids. Starting from the free energy of the materials and the entropy production, we obtained the coupled dynamical system. We show the existence of local-in-time weak solutions for these systems. More difficulties and open questions will be discussed in the talk (Received January 24, 2020)

1157-76-144 **Rishabh V More*** (morer@purdue.edu) and **Arezoo M Ardekani** (ardekani@purdue.edu). *Effect of particle surface roughness on the rheology of dense non-Brownian suspensions.*

The presence of irregularities on the particle surface leads to the breaking of the lubrication film and particles come into contact in dense suspensions resulting in contact stresses and friction. This suggests roughness has a significant impact on the rheological behavior of suspensions. To this end, we numerically investigate the effects of systematically increasing the particle surface roughness on the rheology of dense non-Brownian suspensions. First, we show that increasing roughness leads to an increase in the viscosity of suspension in agreement with recent experiments. Further, increasing the asperity size reduces the critical shear rate for shear thickening (ST) transition and results in an early onset of discontinuous shear thickening in terms of solid loading. In addition, roughness enhances the strength of the shear thickening effect. Denser contact networks leading to high contact stresses and reduction in the jamming fraction with increasing roughness size provide an explanation for this behavior. We summarize the role of roughness on the ST behavior of dense suspensions with the help of the viscosity phase diagram in the shear rate - volume fraction plane. Finally, we provide a constitutive model to quantify the role of roughness on ST behavior. (Received January 24, 2020)

1157-76-257 **Luoding Zhu*** (luozhu@iupui.edu), LD270G, Math Dept, 402 N Blackford Street, Indianapolis, IN 46202, and **Kaoru Sakai**. *Modeling and simulation of blood flow past the distal arteriovenous graft with intimal hyperplasia.* Preliminary report.

Late-stage-kidney-disease patients rely on hemodialysis for the maintenance of their regular lives. Arteriovenous graft (AVG) is one of the commonly used devices for dialysis. However, this artificial created shunt may get clotted and eventually causes the dialysis to fail. The culprit behind the AVG clotting and failure is the intimal hyperplasia (IH), the gradual thickening of vein-wall in the vicinity of the vein-graft conjunction.

In this work we investigate the effects of the IH development, including its location and severity on the flow and force fields in the AVG anastomosis. The stenosis is modelled in the shape of a Gaussian function with different height, spread, and location. The blood is modelled as a viscous incompressible fluid, and the blood flow (pulsatile) is modelled by the lattice Boltzmann equations (D3Q19). The fluid-structure-interaction is modelled by the immersed boundary (IB) framework. Our computational results show that the IH severity has the most influence on the WSS, WNS, and the axial OSI. The stenosis location and flow pulsatility have little effect on flow and force fields. Our results indicate that the progression of the IH tends to accelerate the closure of the vein lumen. (Received January 30, 2020)

1157-76-306 **Cheng-Wei Tai*** (tai11@purdue.edu), Forney Hall of Chemical Engineering, 480 W. Stadium Ave., West Lafayette, IN 47906, **Shiyan Wang** (wang2502@purdue.edu), Forney Hall of Chemical Engineering, 480 W. Stadium Ave, West Lafayette, IN 47906, and **Vivek Narsimhan** (vnarsim@purdue.edu), Forney Hall of Chemical Engineering, 480 W. Stadium Ave, West Lafayette, IN 47906. *Cross-stream migration of non-spherical particles in arbitrary quadratic flows.*

Cross-stream particle migration in viscoelastic suspensions plays a crucial role in many applications in the biomedical community and the chemical/oil industries. Previous studies in viscoelastic media provided insight on the dynamics of spherical particles in simple shear viscoelastic flows, yet the effect of more complex flow profiles and particle shapes is underexplored. In this study, we develop approximate, analytical expressions for the polymeric force and torque on an arbitrary-shaped particle in a second-order fluid subject to a general quadratic flow field. The expressions will be exact for the case when the first and second normal stress coefficients satisfy $\psi_1 = -2\psi_2$. We first investigate how particle shape alters cross-stream migration and particle orientation in pressure-driven flows. For prolate particles, the lift is initially larger than an equal-volume sphere due to its tumbling, but then becomes smaller as it transitions to a log-rolling state (aligned along vorticity direction). This observation leads to a smaller average migration speed for larger aspect ratio particles in slit-like microchannels, but different migration behaviors in other cross-sectional channel geometries. We conclude on how to extend our theories to more general situations. (Received February 01, 2020)

1157-76-405 **Vera Mikyoung Hur***, 1409 W Green Street, Urbana, IL 61801. *Stokes Waves in a Constant Vorticity Flow: Theory and Numerics.*

Stokes in the 1800s made formal but far-reaching considerations about periodic waves at the surface of water, under the influence of gravity, propagating a long distance at a practically constant velocity without change of form. In an irrotational flow, for instance, he observed that crests become sharper and troughs flatter as the amplitude increases, and that the so-called wave of greatest height, or extreme wave, possesses a 120 degree's angle at the crest. The irrotational flow assumption is justified in many situations, and facilitates rigorous

analysis and numerical computation. On the other hand, rotational effects are significant in many others. I will review recent progress in a constant vorticity flow. Numerical findings include folds and gaps in the wave speed vs. amplitude plane, and a profile enclosing multiple bubbles of fluids. I will discuss analytical and numerical applications if time permits. (Received February 03, 2020)

1157-76-454 **Arkadz Kirshtein*** (akirshtein@umass.edu), Department of Mathematics and Statistics, Lederle Graduate Research Tower, 1623D, 710 N. Pleasant Street, Amherst, MA 01003, and **Chun Liu** and **James J Brannick**. *Modeling and simulation of multi-component fluid flows.*

In this talk I will introduce the systematic energetic variational approach for dissipative systems applied to multi-component fluid flows. These variational approaches are motivated by the seminal works of Rayleigh and Onsager. The advantage of this approach is that we have to postulate only energy law and some kinematic relations based on fundamental physical principles. The method gives a clear, quick and consistent way to derive the PDE system. We employ diffusive interface method to capture the dynamics of the interface between different components. The diffusive interface method is an approach for modeling interactions among complex substances. The main idea behind this method is to introduce phase field labeling functions in order to model the contact line by smooth change from one type of material to another. (Received February 03, 2020)

1157-76-504 **Wenxiao Pan*** (wpan9@wisc.edu), 1513 University Ave, Madison, WI 53706, and **Wei Hu**, **Zisheng Ye** and **Nathaniel Trask**. *A spatially adaptive high-order meshless method for fluid-solid interactions.*

In this talk, we present a scheme implementing an a posteriori refinement strategy in the generalized moving least squares (GMLS) method for problems involving point singularities and fluid-solid interfaces. The GMLS method provides high-order compatible discretization of the Stokes problems and offers a high-fidelity simulation tool for problems with moving boundaries. The meshless nature of the discretization is particularly attractive for adaptive h-refinement, especially when resolving the near-field aspects of variables and point singularities governing lubrication effects in fluid-solid interactions. We demonstrate that the resulting spatially adaptive GMLS method is able to achieve optimal convergence in the presence of singularities for both the div-grad and Stokes problems. Further, we present a series of simulations for flows of colloid suspensions, in which the refinement strategy efficiently achieved highly accurate solutions, particularly for colloids with complex shapes. In particular, the lubrication effects can be accurately resolved without any subgrid-scale lubrication model or correction. (Received February 03, 2020)

1157-76-513 **Pei Liu***, 234 Vincent Hall, Minneapolis, MN 55455. *Non-Isothermal Electrokinetics: Energetic Variational Approach.*

A number of ion channels are observed to be sensitive to the temperature changes. These temperature-gated ion channels can detect the temperature thus regulate the internal homoeostasis and disease-related processes such as the thermal adaptation and the fever response. In order to understand how the temperature affects the ion channel mechanics, we develop a Poisson–Netnst–Planck–Fourier (PNPF) system through the energetic variational approach. With given form of the free energy functional and the entropy production, we achieve the mechanical equations and a temperature equation, which satisfy the laws of thermodynamics automatically. From the energy point of view, we also develop the numeric scheme which satisfy the discrete energy dissipation. (Received February 03, 2020)

1157-76-600 **Charlie Lin*** (lin891@purdue.edu) and **Vivek Narsimhan** (vnarsim@purdue.edu). *Shape stability of osmotically deflated vesicles in general linear flows.*

Vesicles have become paradigmatic for studying cell-like dynamics. Vesicle dynamics in simple shear or extensional flows have been thoroughly studied, but flow types in devices or biology are not always described by those flows alone. We use the boundary element method to investigate the shape stability of osmotically deflated vesicles in a general linear flow (i.e., linear combinations of extensional and rotational flows). We model a vesicle as a droplet with an incompressible and bending resistant interface. We simulate flow types from purely shear to extensional at viscosity ratios from 0.01 to 5.0 and reduced volumes from 0.60 to 0.70. The vesicle's viscosity ratio plays a minimal role in describing its shape and stability for many mixed flows, even in cases when significant flows are present in the vesicle interior. We find in these cases that the bending critical capillary number for shape instabilities collapse onto similar values if the capillary number is scaled by an effective extensional rate. Our simulations suggest one can accurately quantify the shape and stability of vesicles from the purely extensional flow results for flow types not close to pure shear flow. We only start to see deviations in the observations discussed above for flow types very nearly shear flow. (Received February 04, 2020)

1157-76-603 **Meet Patel*** (meetm@umich.edu), G029, AL (W.E. Lay Auto Lab), 1231 Beal, Ann Arbor, MI 48109, **Gregory Shallcross** (grshall@umich.edu), 2016, AL (W.E. Lay Auto Lab), 1231 Beal, Ann Arbor, 48109, and **Jesse Capecelatro** (jcaps@umich.edu), 2011, AL (W.E. Lay Auto Lab), 1231 Beal, Ann Arbor, MI 48109. *Modeling and simulation of compressible particle-laden flows.*

Despite decades of progress towards understanding and modeling the dynamics of inertial (heavy) particles suspended in turbulent flows, much less attention has been made to flows with significant gas-phase compressibility. Strong gas-phase acceleration and finite Mach number effects present in many compressible flows challenge the validity of existing models. In this work, we present a rigorous derivation of the volume-filtered Navier–Stokes equations for compressible particle-laden flows. Compared to incompressible flows, many new unclosed terms appear. Particular attention is paid to the pseudo-turbulent kinetic energy, which is shown to systematically reduce the local gas-phase pressure and consequently increase the local Mach number. The resulting set of volume-filtered equations are implemented within a high-order Eulerian–Lagrangian framework. Three-dimensional simulations of particle-laden under-expanded jets will be presented. The simulation configuration is guided by a companion experimental setup at Johns Hopkins University. Velocity and acceleration statistics are compared to high-resolution experimental data to assess the accuracy of existing models. (Received February 04, 2020)

1157-76-628 **Ashley Earls*** (earls@stolaf.edu) and **Maria-Carme Calderer**. *Flexoelectricity and three-dimensional solitons in nematic liquid crystals.*

We model a novel type of three-dimensional, chevron soliton found in nematic electrolytes with negative dielectric and conducting anisotropies, subject to an alternating electric (AC) field. (These solitons were observed by the Lavrentovich group.) Flexoelectricity couples electric field to splay and bend distortions in the nematic, and was proposed as the driving mechanism of the solitons. In this talk, we first discuss the dynamic equations for a flexoelectric liquid crystal with ionic impurities. We then linearize about the initial uniform state, find that it is unstable to soliton-like perturbations, and recover estimates for the size of the traveling wave. (Received February 04, 2020)

78 ► Optics, electromagnetic theory

1157-78-485 **Peijun Li** and **Xu Wang*** (wang4191@purdue.edu). *Inverse random source scattering for the Helmholtz equation with attenuation.*

In this talk, a new model is introduced for the inverse random source scattering problem of the Helmholtz equation with attenuation. The source is assumed to be driven by a fractional Gaussian field whose covariance is represented by a classical pseudo-differential operator. By establishing the connection between fractional Gaussian fields and rough sources characterized by their principal symbols, we demonstrate that the micro-correlation strength of the random source can be uniquely determined by the passive measurements of the wave field in a set which is disjoint with the support of the strength function. (Received February 03, 2020)

81 ► Quantum theory

1157-81-204 **Philippe Mathieu*** (pmathieu@nd.edu), 255 Hurley Bldg, Notre Dame, IN 46556-4618, and **Alexander Schenkel**, **Nicholas Joshua Teh** and **Laura Wells**. *Homological perspective on edge modes in linear Yang-Mills theory.*

I will present a homological construction of the extended phase space for linear Yang-Mills theory on an oriented and time-oriented Lorentzian manifold M with a time-like boundary ∂M. This formalizes many of the rather ad hoc constructions for edge modes appearing in the theoretical physics literature. (Received January 28, 2020)

1157-81-229 **Luan Bezerra*** (luanpere@iupui.edu) and **Evgeny Mukhin**. *Braid actions on quantum toroidal superalgebras.*

We discuss braid actions on quantum toroidal superalgebras associated with $\mathfrak{gl}_{m|n}$. For each choice of root system $\mathbf{s}$ of $\mathfrak{gl}_{m|n}$, a corresponding quantum toroidal superalgebra $\mathcal{E}_{\mathbf{s}}$ is defined. We show that all such superalgebras are isomorphic.

The superalgebra $\mathcal{E}_{\mathbf{s}}$ contains two distinguished subalgebras, both isomorphic to the quantum affine superalgebra $U_q\widehat{\mathfrak{gl}}_{m|n}$, called vertical and horizontal subalgebras. We show the existence of Miki automorphism of $\mathcal{E}_{\mathbf{s}}$, which exchanges the vertical and horizontal subalgebras. (Received January 29, 2020)

1157-81-324 **Ralph M Kaufmann*** (rkaufman@purdue.edu), 150 N University St, West Lafayette IN, IN 47907, and **Birgit Wehefritz-Kaufmann**, 150 N University St, West Lafayette IN, IN 47907. *Combinatorial Algebras and Topological Realizations.*

We will discuss several combinatorial algebras of the type of the Temperley-Lieb algebra and show relations between them. This leads to a surprising connection to well studied topological objects, such as the little discs operad and surfaces. (Received February 01, 2020)

1157-81-336 **Shuang Ming*** (sming@math.tamu.edu) and **Greg Kuperberg**. *On images of SU(2) and SO(3) TQFT representations of mapping class groups with boundary.*

(2+1)-dimensional TQFT provide interesting examples of finite dimensional representations of mapping class groups. In this talk, I will discuss the recent progress on studying the images of these representations. This is joint work with Greg Kuperberg. (Received February 02, 2020)

1157-81-397 **Sachin Gautam*** (gautam.42@osu.edu), Department of Mathematics, 231 W 18th Ave, The Ohio State University, Columbus, OH 43210, and **Valerio Toledano Laredo** and **Curtis Wendlandt**. *The meromorphic R-matrix of the Yangian.*

The Yangian associated to a simple Lie algebra is a Hopf algebra which comes equipped with a universal R-matrix. This R-matrix, as introduced by Drinfeld in 1985, is a formal series in 1/u, with values in a two-fold tensor product of the Yangian.

In this talk, I will present a new, explicit construction of this R-matrix, on a pair of finite-dimensional representations. Our construction yields two meromorphic, matrix-valued functions of a complex variable, which are related by a unitary relation. Moreover, Drinfeld's formal R-matrix, once evaluated on such a pair, is the asymptotic expansion of these two meromorphic functions, as the complex parameter goes to infinity in two distinct sectors.

This talk is based on a joint work with V. Toledano Laredo and C. Wendlandt. (Received February 03, 2020)

1157-81-404 **Pavel Mnev*** (pmnev@nd.edu). *Two-dimensional perturbative scalar field theory with polynomial potential and cutting-gluing.*

We study perturbative path integral quantization of the two-dimensional scalar field theory with a polynomial (or power series) interaction potential on manifolds with boundary. The perturbative partition function defined in terms of configuration space integrals on the surface satisfies an Atiyah-Segal type gluing formula. Moreover, partition functions (modified by an interesting nonlocal boundary term) do organize into a functor (in the sense of Segal's axiomatics), from Riemannian cobordism category to the category of Hilbert spaces. A crucial role in the result is played by the tadpoles (short loops). We will discuss functorial assignments of tadpoles and the relation to RG flow in the space of potentials. This is a report on a joint work with Santosh Kandel and Konstantin Wernli, arXiv:1912.11202 [math-ph]. (Received February 03, 2020)

1157-81-431 **Anton Zeitlin*** (zeitlin@lsu.edu), Deapartment of Mathematics, Louisiana State University, Baton Rouge, LA 70803. *A geometric view of the Bethe ansatz equations: q-opers.*

The geometric realization of integrable systems related to quantum groups recently drew a lot of attention, primarily in the framework of enumerative geometry. In particular, the celebrated XXZ Bethe ansatz equations were realized as relations for the quantum K-theory ring of a certain variety. In this talk, I will give an overview of the different approach to obtain the Bethe ansatz equation via a group-valued q-difference version of connections, called q-opers. This is an appropriate q-deformation of relation between oper connections and Bethe ansatz equations for the Gaudin model, which is a nontrivial manifestation of the geometric Langlands correspondence. If time allows, I will relate this q-oper approach with the quantum q-Langlands correspondence and constructions of quantum K-theory. (Received February 03, 2020)

1157-81-439 **Vitaly Tarasov** and **Filipp Uvarov***, filuvaro@iu.edu. *Duality for Bethe algebras acting on polynomials in anticommutative variables.*

We consider the actions of the current Lie algebras gl(n)[t] and gl(k)[t] on the space of polynomials in kn anticommuting variables. We show that the images of the Bethe algebras under these actions coincide. To prove the statement we use the Bethe ansatz description of eigenvalues of the actions of the Bethe algebras via spaces of quasi-exponentials and establish explicit correspondence between these spaces. (Received February 03, 2020)

1157-81-451 **Yilong Wang*** (yilongwang@lsu.edu). *Higher central charges and their applications.*
In this talk, we will introduce the notion of higher central charges of premodular categories. We will discuss their arithmetic properties and their applications to the study of the Witt group of modular categories. (Received February 03, 2020)

1157-81-547 **Joshua Jeishing Wen*** (jeiwenno@gmail.com). *Formal eigenfunctions for affine-elliptic integrable systems.* Preliminary report.
I will report on ongoing work on solving the bispectral problem for the elliptic Ruijsenaars-Schneider integrable system and its less well-known dual. (Received February 04, 2020)

82 ► Statistical mechanics, structure of matter

1157-82-322 **Kang Lu***, 402 N. Blackford, LD 270, INDIANAPOLIS, IN 46202. *Frobenius integrability of quantum spin chains.* Preliminary report.
We introduce the Frobenius integrability of quantum spin chains and conjecture that quantum spin chains with suitable conditions are Frobenius integrable. The notion of Frobenius intergability is motivated by the Bethe ansatz conjecture, geometric Langlands correspondence, and general facts about Frobenius algebra which could deduce "simple spectrum" of Bethe algebra. We also discuss several questions and known results related to Frobenius integrability.

This talk is partially based on a joint work with E. Mukhin and V. Tarasov. (Received February 01, 2020)

90 ► Operations research, mathematical programming

1157-90-24 **Yuehaw Khoo*** (yuehaw.khoo@gmail.com), Chicago, IL 60637, **Amit Singer** (amits@math.princeton.edu), Princeton, NJ 08540, and **David Cowburn** (david.cowburn@einstein.yu.edu), The Bronx, NY 10461. *Convex optimization approach to distance geometry and related problems in NMR spectroscopy.*
Nuclear magnetic resonance (NMR) spectroscopy is the most-used technique for protein structure determination besides X-ray crystallography. In this talk, the computational problem of protein structuring from residual dipolar coupling (RDC) will be discussed.

Typically the 3D structure of a protein is obtained through finding the coordinates of atoms subject to pairwise distance constraints. RDC measurements provide additional geometric information on the angles between bond directions and the principal-axis-frame. The optimization problem involving RDC is non-convex and we present a novel convex programming relaxation to it by incorporating quaternion parameterization. In simulations we attain the Cramer-Rao lower bound with relatively efficient running time. From real data, we obtain the protein backbone structure for ubiquitin with 1 Angstrom resolution. (Received November 30, 2019)

1157-90-54 **Jeffrey Zhang*** (jeffz@princeton.edu) and **Amir Ali Ahmadi**. *On Local Minima of Cubic Polynomials.* Preliminary report.
We study local minima of cubic polynomials. We first give a characterization of local minima, and show that this characterization can be checked in polynomial time. We then give an SDP approach for finding local minima of cubic polynomials. (Received January 08, 2020)

1157-90-75 **Henry Wolkowicz*** (hwolkowicz@uwaterloo.ca), Department of Combinatorics and Optimization, University of Waterloo, Waterloo, Ontario N2L3G1, Canada. *Facial reduction in cone optimization with applications to hard combinatorial problems and low rank matrix completions.*
Strict feasibility is at the heart of convex optimization. This is needed for optimality conditions and stability. New optimization modelling techniques and convex relaxations for hard nonconvex problems have shown that the loss of strict feasibility is a much more pronounced phenomenon than has previously been realized. Such new developments suggest a reappraisal. In this talk we describe the various reasons for the loss of strict feasibility, whether due to poor modelling choices or (more interestingly) rich underlying structure, and describe ways to cope with it. In particular, we look at applications to the quadratic assignment, max-cut, problems, and to robust low rank matrix completions. (Received January 14, 2020)

1157-90-387 **Mareike Dressler***, mdressler@ucsd.edu. *Global Optimization via the Dual SONC Cone and Linear Programming.*

In this talk I will report on recent developments in (polynomial) optimization: Using the dual cone of sums of nonnegative circuits (SONC), we provide a relaxation of the global optimization problem to minimize an exponential sum and, as a special case, a multivariate real polynomial. This approach builds on two key observations. First, that the dual SONC cone is contained in the primal one. Hence, containment in this cone is a certificate of nonnegativity. Second, we show that membership in the dual cone can be verified by a linear program. Finally, we present initial experimental results comparing our method to existing approaches. Joint work with Janin Heuer, Helen Naumann, and Timo de Wolff. (Received February 03, 2020)

1157-90-517 **Ruixin Wang**, **Prateek Jaiswal** and **Harsha Honnappa*** (honnappa@purdue.edu), 315 N. Grant St, West Lafayette, IN 47906. *Variational Inference Using Coupled Neural Differential Equations.*

Neural ordinary and stochastic differential equations (henceforth, "Neural DE") have emerged as important methodological innovations for predicting time-series using deep neural networks. Motivated by applications in operations research, in this talk we present two extensions of this emerging methodology for learning stochastic process models from trace data. First, we introduce an extension of the Neural DE framework to learn a system of coupled differential equations. The latter are crucial for modeling doubly-stochastic and non-stationary stochastic process models. Second, we use this framework to learn the (latent) stochastic intensity, modeled as a stochastic differential equation, of a doubly-stochastic point process, using a variational autoencoder. We illustrate our methodology by presenting examples for learning Cox or doubly stochastic Poisson processes, and infinite server queues driven by Cox and Hawkes processes. These models are highly relevant to performance prediction and decision support in operations research. Time permitting, we will also discuss our on work on de-biasing the variational autoencoder. (Received February 03, 2020)

1157-90-558 **Dmitriy Kunisky*** (kunisky@cims.nyu.edu). *Sum-of-squares lower bounds from highly symmetric examples (or, discrete geometry for optimization).*

I will describe the process, carried out in a sequence of recent works, of devising a new construction to show that the sum-of-squares hierarchy of convex relaxations cannot effectively bound certain natural random quadratic programs. At a key step, this process is guided by thoroughly studying a class of highly structured deterministic examples arising from (real-valued) *equiangular tight frames*. These examples, once properly understood, suggest the right general construction to apply to random problem instances. Finally, that general construction sheds light back onto the deterministic examples, clarifying the intricate combinatorial structure present in the sum-of-squares extensions of Gram matrices of equiangular tight frames. (Based on joint work with Afonso Bandeira.) (Received February 04, 2020)

1157-90-630 **Raghu Pasupathy*** (pasupath@purdue.edu), Dept. of Statistics, HAAS Hall, West Lafayette, IN 47907, and **Aaron Yip** and **David Newton**. *Adaptive Sequential SAA as a Universal Paradigm for Stochastic Optimization.*

The Stochastic Approximation (SA) recursion, also known as Stochastic Gradient Descent (SGD), is the "workhorse" recursion for solving stochastic optimization problems arising in machine learning and elsewhere. For smooth, strongly convex objectives, it is well-known that SA iterates achieve the information theoretic Cramer-Rao (CR) lower bound when the step size is chosen correctly. Such correct choice, however, depends on unknown curvature constants; moreover, the CR bound as a measure of performance ignores transient evolution and is thus not fully reflective of finite-time performance. This explains why SGD is rarely implemented in its native form. Instead, heuristic variations, e.g., ADAM, have come to be widely preferred (see, for example, TensorFlow). In this talk, I will argue for an alternative to SGD and its variants. Specifically, I will re-introduce a decades-old idea and rigorize it in a way that allows well-established deterministic optimization methods. The power of the framework is that it naturally retains the structure that is often inherent to sample-paths, and is expressly equipped to exploit deterministic optimization methods. I will demonstrate that the proposed framework retains the CR lower bound to within a factor in the convex smooth case. (Received February 04, 2020)

91 ► Game theory, economics, social and behavioral sciences

1157-91-19 **D. C. Arangno*** (darangno@hcc-nd.edu), South Bend, IN 46556. *AN APPLICATION OF MATHEMATICAL STRUCTURES TO A PHILOSOPHICAL DEBATE.*

In contemporary metaethics, realist-antirealist debates centering on the ontology of moral properties continue to generate significant and exciting research. We have explored the use of methods from Graph Theory to clarify a debated position of Meta-Ethics, previously encumbered by intrinsic vagueness and ambiguity. We employ rigorous mathematical formalism to symbolize, parse, and thus disambiguate, particular philosophical questions regarding ethical ontological materialism of the reductivist variety. In our research, we seek to revisit the once vexed question regarding the multiple-realizability of moral properties by employing the mathematical machinery of hypergraphs. The utilization of hypergraphs offers explanatory flexibility to the hypothesis that there are, potentially, an infinite number of unrelated subvening facts (viz., the subvening sets of natural non-moral properties) which have the possibility of coming together as subsets (hyperedges) to form supervening moral properties. We hope our research will be of interest to both the Analytic Philosophy and the applied Mathematics communities, and will help facilitate discussions of metaethics, particularly pertaining to ethical naturalism. We believe there is much research yet to be done (Received November 21, 2019)

1157-91-269 **Scott Philip Robertson*** (scottrob@bu.edu), 595 Commonwealth Ave, Boston, MA 02215. *Dynamic Noisy Rational Expectations Equilibrium with Heterogeneous Information.*

In this talk, we consider equilibria in the presence of asymmetric information and misinformed agents (noise traders). We establish existence of two equilibria. First, a full communication one where the informed agents' signal is disclosed to the market, and static policies are optimal. Second, a partial communication one where the signal disclosed is affine in the informed and noise traders' signals. Here, information asymmetry creates a demand for a dark pool with endogenous participation where private information trades can be implemented. Markets are endogenously complete and equilibrium prices have a three factor structure. Results are valid for multiple dimensions; constant absolute risk averse investors; fundamental processes following a general diffusion; and non-linear terminal payoffs. Asset price dynamics and public information flows are endogenous, and are established using multiple filtration enlargements, in conjunction with predictable representation theorems for random analytic maps. Rational expectations equilibria are special cases of the general results. This is joint work with J. Detemple and M. Rindisbacher of Boston University. (Received January 31, 2020)

1157-91-400 **Gu Wang***, Department of Mathematical Sciences, 100 Institute Road, Worcester, MA 01609, and **Paolo Guasoni**. *Sharing Profits in the Sharing Economy.*

A monopolist platform (the principal) shares profits with a population of affiliates (the agents), heterogeneous in skill, by offering them a common nonlinear contract contingent on individual revenue. The principal cannot discriminate across individual skill, but knows its distribution and aims at maximizing profits. This paper identifies the optimal contract, its implied profits, and agents' effort as the unique solution to an equation depending on skill distribution and agents' costs of effort. If skill is Pareto-distributed and agents' costs include linear and power components, closed-form solutions highlight two regimes: If linear costs are low, the principal's share of revenues is insensitive to skill distribution, and decreases as agents' costs increase. If linear costs are high, the principal's share is insensitive to the agents' costs and increases as inequality in skill increases. (Received February 03, 2020)

1157-91-426 **Sergey Nadtochiy*** (snadtochiy@iit.edu), 10 W. 32nd St, Rettaliata Engineering Center, Dept of Applied Mathematics, Chicago, IL 60616. *A simple microstructural explanation of concave price impact.*

I will present a simple model of market microstructure which explains the concavity of price impact. In the proposed model, the local relationship between the order flow and the fundamental price (i.e. the local price impact) is linear, which makes the model dynamically consistent. Nevertheless, the expected impact on midprice from a large sequence of co-directional trades is nonlinear and asymptotically concave. The main practical conclusion of the model is that, throughout a meta-order, the volumes at the best bid and ask prices change (on average) in favor of the executor. This conclusion, in turn, relies on two more concrete predictions of the model, one of which is tested using publicly available market data without the information about meta-orders. I will present the theoretical results and will confirm them with empirical analysis. (Received February 03, 2020)

92 ► Biology and other natural sciences

1157-92-13 **Carlile Lavor*** (clavor@unicamp.br), IMECC - UNICAMP, Campinas, SP 13083-859, Brazil. *Rigidity and Protein Graphs.*

Nuclear Magnetic Resonance (NMR) experiments provide distances between nearby atoms of a protein and the problem is how to calculate its 3D structure using such distances. We present an atomic order, based on information from the chemistry of proteins and NMR experiments, which allows us to formulate the problem as a combinatorial search. This order also tells us what kind of NMR distance information is crucial to understand the cardinality of the solution set of the problem and its computational complexity. (Received October 18, 2019)

1157-92-424 **Weishi Liu*** (wsliu@ku.edu), Department of Mathematics, 1460 Jayhawk Blvd., Snow Hall 405, Lawrence, KS 66045. *Flux ratios for permanent charge effects on ionic flow.*

Permanent charge is the crucial structure of a protein (ion channel) on a cell membrane for ionic flow properties of the protein.

In this talk, we will report analytical and numerical investigations on the flux ratio for effects of permanent charges on ionic flow via a quasi-one-dimensional Poisson-Nernst-Planck model.

For flow of two ion species, the flux ratios are examined analytically for the two extremal cases of small and large permanent charges. A universal property and all specifics of flux ratios demonstrate a quite rich interplay between permanent charges and boundary conditions as well as other physical parameters.

Guiding by the analytical results, our numerical studies are able to (i) verify analytical predictions on flux ratios for both small and large permanent charges, (ii) provide a detailed behavior of flux ratios for intermediate values of permanent charges, and (iii) reveal bifurcations of flux ratios as one varies the permanent charge values. (Received February 03, 2020)

1157-92-474 **Qian Zhang*** (qian.zhang_jennifer@northwestern.edu), **Boem-Kyeong Park**, **Qinyuan Liu**, **Scott Barnett** and **Peter Voorhees**. *Modeling and Analysis on Distribution of Oxygen Partial Pressure in Electrolytes and Morphological Evolution in the Ni/YSZ Electrode : Explaining Degradation of Solid Oxide Electrolyzer Cells.*

A number of degradation mechanisms have been observed during the long-term operation of solid oxide electrolysis cells (SOEC). Since many of these can be attributed to the extreme oxygen potentials and difference of gradient of electric field between the grain boundaries and grains in the electrolyte during electrolysis and Nicole phase migration in the electrode, it is important to accurately model these potentials and phase migration.

Using charge carrier transport model, we quantify the oxygen potentials and electric field distribution across the electrolyte and phase migration in the electrode and thereby provide insights into these degradation mechanisms.

In particular, we find when a GDC barrier layer is present in the electrolyte, a maximum in the oxygen pressure is found at the interface of the YSZ and GDC that can result in degradation or fracture at this location. To guide experiments and minimize degradation of the two-layer electrolyte, an analytical estimate of the oxygen partial pressure at the YSZ/GDC interface is obtained. (Received February 03, 2020)

1157-92-516 **Pei Liu***, 234 Vincent Hall, Minneapolis, MN 55401. *Liquid Crystal Structure of Bacteriophage DNA in a Capsid.* Preliminary report.

DNA of bacteriophage is known to form hexagonal chromonic liquid crystal phases in the capsid. The ionic condition in the enviroment is crucial to the stucture and pressure of the DNA. We combine the liquid crystal theory with electrostatic interaction to model the DNA structure in both the disordered core region and the ordered liquid crystal phase. (Received February 03, 2020)

94 ► Information and communication, circuits

1157-94-15 **Jintai Ding***, 8770 Wellerstation Drive, Cincinnati, OH 45249. *Post-Quantum Key Exchange from the LWE.*

In this lecture, we present practical and provably secure (authenticated) key exchange protocol and password authenticated key exchange protocol, which are based on the learning with errors problems. These protocols are conceptually simple and have strong provable security properties. This type of new constructions were started in 2011-2012. These protocols are shown indeed practical. We will explain that all the existing LWE based key

exchanges are variants of this fundamental design. In addition, we will explain some issues with key reuse and how to use the signal function invented for KE for authentication schemes. (Received November 05, 2019)

1157-94-191 **Heide Gluesing-Luerssen*** (heide.gl@uky.edu). *MRD codes and their proportion.*
Rank-metric codes are linear spaces of matrices over a finite field such that the rank of each nonzero matrix is lower bounded by a given number d, the rank distance. MRD codes are rank-metric codes with maximum possible size for the given rank distance. They exist for all choices of parameters. In this talk, I will discuss the proportion of MRD codes within the space of all rank-metric codes of the same dimension and, more specifically, will focus on the asymptotic behavior of this proportion as the field size tends to infinity. As it turns out, $[3 \times 2; 2]$-MRD codes have an asymptotic proportion of $1/3$, whereas for $[3 \times 3; 3]$-MRD codes this asymptotic proportion is zero. The proofs for these two facts make use of two very different classical results. (Received January 28, 2020)

1157-94-215 **Jintai Ding*** (jintai.ding@gmail.com), 8770 wellerstation dr, cincinnati, OH 45249. *Cryptanalysis of LUOV a NIST second round post-quantum candidate.*
In this talk, we will show a new method to break the NIST second round post-quantum candidate, Lifted Unbalance Oil-Vinegar scheme (LUOV). LUOV is a multivariate signature scheme submitted a research group led by Professor Bart Preneel in Belgium. (Received January 29, 2020)

1157-94-342 **Alexander Barg*** (abarg@umd.edu). *Cyclic codes with hierarchical locality.*
We extend the framework of constructing cyclic codes with local erasure recovery proposed a few years ago to the case of locally recoverable codes with several levels of hierarchy (H-LRC codes). We obtain a new range of parameters for optimal H-LRC codes and present other related results. This talk is based on a joint work with Zitan Chen (University of Maryland). (Received February 02, 2020)

1157-94-344 **Jon-Lark Kim*** (jlkim@sogang.ac.kr), Baekbeom-ro 35, Sogang University, Ricci Hall 1401, Department of Mathematics, Seoul, 04107, South Korea. *Perfect Code Based Steganography.*
Steganography is the science of communicating a secret message by hiding it in a cover object. Steganography has become more valuable and popular in the network transmission and DNA cryptography. The well known algorithm F5 is based on Hamming codes, which are perfect. Unfortunately there are a few classes of perfect codes in the Hamming graph. It is natural to consider perfect codes in a general graph. In this talk we consider perfect codes in various graphs including circulant graphs, Abelian Cayley graphs, a Gaussian graph based on Gaussian integers and describe their application to steganography. (Received February 02, 2020)

1157-94-391 **Hiram H. Lopez*** (h.lopezvaldez@csuohio.edu). *Rank-metric codes and q-polymatroids.*
In this talk we give an introduction to rank-metric codes. We introduce q-polymatroids, the q-analogue of polymatroids, and study their basic properties. We associate a pair of q-polymatroids to a rank-metric code and show that several invariants and structural properties of the code, such as generalized weights, the property of being MRD or an optimal anticode, and duality, are captured by the associated combinatorial object. (Received February 03, 2020)

1157-94-432 **Carolyn Mayer*** (cdmayer@wpi.edu) and **William J. Martin**. *Domination Mappings for Low-Power Cooling Codes.* Preliminary report.
Low-power cooling (LPC) codes (introduced by Chee, Etzion, Kiah, Vardy, and Wei '18) are designed to control the power consumption and temperature of interconnects of wires by limiting the number of state transitions allowed per transmission and avoiding transitions on the hottest wires. LPC codes can be constructed using *domination mappings*, injective maps subject to certain constraints given by a bipartite graph. We will present a family of domination mappings and discuss a conjecture of Chee et al. about the graphs for which domination mappings exist. (Received February 03, 2020)

1157-94-453 **Hunter Lehmann*** (hunter.lehmann@uky.edu) and **Heide Gluesing-Luerssen**. *Distance Distributions of Nearly-Optimal Cyclic Orbit Codes.*
Subspace codes are collections of subspaces of the finite vector space $\mathbb{F}_q^n$ under the subspace metric. The distance distribution of such a code is the vector whose i^{th} entry counts the number of pairs of codewords with subspace distance i. Constant dimension cyclic orbit codes, which are contained in a Grassmannian $\mathcal{G}_q(n,k)$ and are the orbit of a subspace under an action of $\mathbb{F}_{q^n}^*$ on $\mathcal{G}_q(n,k)$, are of particular interest. Optimal cyclic orbit codes have size $\frac{q^n-1}{q-1}$ and minimum distance $2k-2$, the best possible for a code of this size. Nearly-optimal cyclic

orbit codes, then, have the same size and the next best possible minimum distance $2k-4$. We will discuss the distance distributions of these codes. (Received February 03, 2020)

1157-94-484 **Jenya Soprunova*** (soprunova@math.kent.edu), **Kyle Meyer** and **Ivan Soprunov**.
Evaluation codes from 3-d lattice polytopes. Preliminary report.

Fix a convex lattice polytope P in $\mathbb{R}^m$, and define $\mathcal{L}(P)$ to be the $\mathbb{F}_q$-vector space spanned by the monomials whose exponent vectors lie in P. The codewords of the corresponding toric code are obtained by evaluating polynomials in $\mathcal{L}(P)$ at the points of the torus $(\mathbb{F}_q^*)^m$, taken in some fixed order. For large values of q, minimum weight codewords come from polynomials in $\mathcal{L}(P)$ with the largest number of irreducible factors. The number, as well as the monomial structure, of such factors is governed by the geometry of the lattice polytope P. In this talk, I will explain how this connection leads to new bounds on the minimum distance for toric codes with $m = 3$. This is joint work with Kyle Meyer and Ivan Soprunov. (Received February 03, 2020)

1157-94-500 **Felice Manganiello*** (manganm@clemson.edu), **Jessalyn Bolkema** and **Harrison Eggers**. *Constructions of CSS-T codes for universal quantum computation.*

CSS codes take care of errors a special type of stabilizer codes that are able to correct Clifford gates. Universal quantum computation though requires the implementation of a logical non-Clifford gate. The easiest fault-tolerant realization of such a gate is a transversal operation which splits into gates on individual qubits such as the T and the $T^\dagger$ gates. In a recent work, Rengaswamy et al. show that among non-degenerate stabilizer codes, CSS are optimal when supporting the transversal T gate. In this seminar we focus on understanding the results of Rengaswamy et al. and give some initial constructions and bounds on CSS-T codes, meaning CSS codes that support the transversal T gate. (Received February 03, 2020)

1157-94-580 **Kathryn Haymaker*** (kathryn.haymaker@villanova.edu) and **Justin O'Pella**. *Locally recoverable codes from planar graphs.*

In this talk we will apply Kadhe and Calderbank's definition of locally recoverable codes from convex polyhedra and planar graphs to analyze the codes resulting from 3-connected regular and almost regular planar graphs. The resulting edge codes are locally recoverable with availability two. We show that the minimum distance of an LRC from a planar graph is equal to the girth of the graph, and we also establish a new bound on the rate of planar graph LRCs. Constructions of regular and almost regular planar graphs will be presented with their associated code parameters. In certain cases the code families meet the rate bound. (Received February 04, 2020)

1157-94-633 **Christine A Kelley*** (ckelley2@unl.edu). *Designing Graph-based Codes for Window Decoding.* Preliminary report.

Low-density parity-check (LDPC) codes are a class of linear codes defined by sparse parity-check matrices, and have corresponding sparse bipartite graph representations. They have been shown to be capacity-achieving over many channels using low-complexity graph-based iterative decoders. Spatially-coupled LDPC (SC-LDPC) codes are a special class that are amenable to window decoding, in which the nodes are decoded in groups from one end to the other, making them applicable for data streaming and other applications. In this talk we explore how the graph structure of the overall code compares to that seen by the window and derive results on the corresponding distance and other properties relevant for effective decoding. (Received February 04, 2020)

00 ► General

1158-00-3 **Joseph Teran***. *Elastoplasticity Simulation with the Material Point Method*

Hyperelastic constitutive models describe a wide range of materials. Examples include biomechanical soft tissues like muscle, tendon, skin etc. Elastoplastic materials consisting of a hyperelastic constitutive model combined with a notion of stress constraint (or feasible stress region) describe an even wider range of materials. A very interesting class of these models arises from frictional contact considerations. I will discuss some recent results and examples in computer graphics and virtual surgery applications. Examples include simulation of granular materials like snow in Walt Disney's "Frozen" as well as frictional contact between thin elastic membranes and shells for virtual clothing simulation. I will also discuss practical simulation of these materials with some recent algorithmic modifications to the Particle-In-Cell (PIC) technique, the Material Point Method (MPM). (Received June 11, 2019)

1158-00-73 **Cristina Tortora*** (cristina.tortora@sjsu.edu), One Washington square, San Jose, CA 95112. *Adding flexibility in directional outlier detection and model-based learning.*

Model-based clustering assumes that the data were generated from a convex combination of densities. The choice of the density function is crucial; the multivariate contaminated normal distribution (MCN) was proposed to model datasets characterized by the presence of outliers. The MCN is a two-component Gaussian mixture; one of the components, with a large prior probability, represents the good observations, and the other, with a small prior probability, the same mean, and an inflated covariance matrix, represents the outliers. Mixtures of MCN distributions can detect outliers and perform cluster analysis improving the clustering performance when compared to normal mixtures and representing an alternative to t mixtures. However, the mixture of MCN distributions uses univariate parameters to model the proportion of outliers and their impact on the inflation parameter, i.e., they are the same for all the variables. This is a limit because the outliers may be different in each dimension. To overcome this issue, we propose a multiple scaled contaminated normal distribution with a p-dimensional proportion of outliers and degrees of contamination, where p is the number of variables. (Received February 20, 2020)

1158-00-98 **Jacqueline Alvarez*** (jalvarez94@ucmerced.edu), 5200 N Lake Rd, Merced, CA 95343, and **Omar DeGuchy** and **Roummel F Marcia**. *Image Classification In Synthetic Aperture Radar Using Reconstruction From Learned Inverse Scattering.*

Synthetic aperture radar (SAR) is a remote sensing technique used to obtain high-resolution images, where image classification is a primary application. However, reconstructing SAR data is difficult which makes the classification of these images even more challenging. We present a study that investigates techniques for classifying SAR images using machine learning. In particular, we compare the classification accuracy using three different training data: raw SAR observation data, reconstructions using Kirchoff migration, and reconstructions by learning an approximation to the inverse of the SAR sensing operator. We consider two different architectures, namely a multi-layer perceptron and a convolutional neural network. The training set is composed of 50,000 images from the CIFAR-10 dataset. We find that the images reconstructed using the learned approximate inverse have a higher classification accuracy than that from the SAR measurement and the Kirchoff migration approach. (Received February 24, 2020)

1158-00-228 **Asma Azizi*** (aazizibo@asu.edu) and **A Mubayi**. *Impact of Ecological Risk Factors on the Dynamics of Alcohol Drinking Amongst ASU Students: A simulation based model Implemented in NetLogo.*

College students drinking became a public health issue needed to be addressed. Approximately \$62—72 at least once per month, with an average number of drinking days ranged in \$4-10\$. Drinking, indirectly, causes more than \$60\$ types of diseases and injuries and results in approximately \$88,000\$ deaths each year, Global Status Report on Alcohol and Health. University environment can affect drinking among students. We developed and analyzed an spatial agent based model to examine students interactions with environments that support alcohol drinking at the population level. We examine how contextual factors and movement patterns affect alcohol spread. The data collected from the survey on health behaviors of students conducted on ASU

Tempe campus are used to estimate most parameters of model. We used global sensitivity analysis the to quantify variability in drinking prevalence resulting from model unknown parameters. The primary result of model identified critical environmental mechanisms driving patterns of drinking behaviors. Alcohol drinking depends on certain social contexts and duration of time in those social venues more than any other factors in the model. (Received March 02, 2020)

1158-00-242 **Andrea Arauza Rivera***, 25800 Carlos Bee Blvd, Hayward, 94542. *Fractals, Binary counting, and Towers of Hanoi.*

We discuss an unexpected connection between the Towers of Hanoi puzzle, binary counting, and fractal geometry. The concepts we explore are suitable for middle school students and can be explored further by mathematicians. Come with pen and paper in hand! (Received March 02, 2020)

01 ► History and biography

1158-01-203 **Michael Reynolds*** (jreynold@irsc.edu). *Some Key Points in the History of Exponential Notation.*

In this talk, we trace the history of exponential notation, starting as early as Hippocrates of Chios in the fifth century BC. We will explore the evolution of both the mathematical notation, as well as the vocabulary used, for the operation of exponentiation, through the Middle Ages, the Renaissance, and up to the modern notation with which we are all familiar. (Received March 01, 2020)

1158-01-364 **Agnes Tuska*** (agnest@csufresno.edu). *George Polya's influence on mathematics competitions in the USA.* Preliminary report.

Soon after George Polya settled in Palo Alto, he published his most widely read book How to Solve It: A New Aspect of Mathematical Method, and started to collaborate on running a high school mathematics competition at Stanford University. Polya also served in constructing the first post-war contest for the William Lowell Putnam Mathematical Competition, and was a member of the Putnam Prize Committee from 1948 until 1950. Polya's international perspective to learning and doing mathematics made a palpable impact on the justifications for conducting mathematics competitions, the types of problems posed, and the ways to effectively prepare for the competitions. I plan to highlight samples of this impact in the USA during my presentation. (Received March 03, 2020)

03 ► Mathematical logic and foundations

1158-03-34 **Lavinia Ciungu***, 1575 Foster Rd, Iowa City, IA 52245. *Monadic operators on quantum B-algebras.*

In this presentation we define the monadic quantum B-algebras and investigate their properties. Given a monadic quantum B-algebra $(X, \exists, \forall)$ where the quantifiers are isotone, we show that $(\exists, \forall)$ is a residuated pair. Moreover, $\exists$ is a closure operator, while $\forall$ is an interior operator on the poset $(X \leq)$. If one of the two quantifiers is surjective, then we prove that it is the identity map. Special properties are studied for the particular case of monadic quantum B-algebras with pseudo-product. The monadic filters of monadic quantum B-algebras are defined and their properties are studied. We prove that there is an isomorphism between the lattice of all filters of a monadic quantum B-algebra and the lattice of all filters of its subalgebra of fixed elements. Another main result consists of proving that there is a one-to-one correspondence between the set of all monadic operators on a monadic quantum B(pP)-algebra X and the set of all relative complete subalgebras of X, and also, there is a one-to-one correspondence between the set of all monadic operators on a strict monadic quantum B(pP)-algebra X and the set of all m-relative complete subalgebras of X. (Received February 11, 2020)

1158-03-99 **Rasulkhozha Sultonkhozhaevich Sharafiddinov*** (rasulkhozha@yahoo.com), Theoretical Physics Division, Institute of Nuclear Physics, Uzbekistan Academy of Sciences, Tashkent, Ulugbek 100214, Uzbekistan. *Unification Theorems of Elements of a Set.*

One feature of each fully regular set is what unites all its elements in a unified whole. As a consequence, the difference, for example, in sizes with a_{1B} and b_{1B} constitutes their linearly ordered set $A = \{a_{1B}, \ b_{1B}\}$ of a definite B symmetry. Another definite L symmetry characterizes the class $C = \{a_{2L}, \ b_{2L}\}$ so that it has a linear order. We formulate and prove here the unification theorems of elements of a set $D = \{H, \ J\}$ with subsets $H = \{a_{1B}, \ b_{2L}\}$ and $J = \{b_{1B}, \ a_{2L}\}$. **Theorem 1.** If the two functions with some individual variables

correspond in a set D to each of its B and L symmetries, they are in it the solutions of the same united equation in these two forms of unknowns. **Theorem 2.** If the two equalities with some own variables correspond in a set D to each of its B and L symmetries, they are in it the solutions of the same united inequality in these two forms of unknowns. **Theorem 3.** If an intrasystem feature of B and L symmetries is their simultaneous violation, their coexistence or both, an object of latent system unification within a set D becomes functional series. (Received February 24, 2020)

1158-03-102 **Valentina Harizanov*** (harizanv@gwu.edu), Department of Mathematics, Washington, DC 20052. *Orderable quandles.*

We investigate algebraic and computability-theoretic properties of orderable quandles and, more generally, magmas. A magma is an algebraic structure with a single binary operation. Important examples of magmas include groups and semigroups, but also those that come from knot theory and do not necessarily have an associative operation, such as quandles. A magma M is right-orderable (left-orderable) if there is a linear ordering of the domain of M, which is right-invariant (left-invariant) with respect to the magma operation. If M has a right order that is also a left order, then M is bi-orderable. A computable orderable magma does not necessarily have a computable order. There is a natural topology on the set of all (left, right, bi-) orders of M. These spaces are compact, and in some cases homeomorphic to the Cantor set. (Received February 25, 2020)

1158-03-118 **Can Başkent***, Department of Computer Science, Middlesex University, London, NW4 4BT, United Kingdom. *Learning from and Working with Contradictions in Heuristics.*

Contradictions (or *monsters*) are essential part of Lakatosian heuristics. Depending on the characteristics of monster, the theory is revised. Similarly, Hintikka "brackets" abnormalities in order to obtain a classical, contradiction-free system for epistemic inquiries.

But, if contradictions are such central elements of dynamic heuristics, why are they excluded? If we choose to work with them, how can we achieve it?

In this talk, I will argue that the underlying logic of reasoning for Lakatosian and even Hintikkan heuristics is non-classical. I will also maintain that it is a kind of logic that is inconsistency-tolerant.

Particularly, I will give an example of how Lakatosian heuristics works and how it is sympathetic to contradictions. Next, I will do the same for Hintikka's interrogative models of inquiry. I will then conclude that meta-mathematical heuristic reasoning must be inconsistency- and incompleteness-tolerant based on the aforementioned cases. This will also substantiate the thesis that inquiry and heuristics are indeed "games" between rational players who may have inconsistent knowledge or incomplete strategies. (Received February 26, 2020)

1158-03-214 **Yuri Movsisyan***, Alex Manoogian 1, Yerevan State University, 0025 Yerevan, Armenia. *On Boole-De Morgan Bilattices.*

It is commonly known that the free Boolean algebra on n free generators is isomorphic to the Boolean algebra of Boolean functions of n variables. The free bounded distributive lattice on n free generators is isomorphic to the bounded distributive lattice of monotone Boolean functions of n variables (R. Dedekind, 1897). In this talk we introduce the concept of Boole-De Morgan bilattice and prove a functional representation theorem for finitely generated free Boole-De Morgan bilattices. As a consequence we characterize finitely generated free distributive bilattices via functional representation. New functions alternative to Boolean functions are presented here which are applicable in Cryptography, discrete mathematics and the theory of quantum computers, too. (Received March 02, 2020)

1158-03-220 **Dario Verta***, Phillips Hall, Room 739, Washington, DC 20052. *Computability theoretic complexity of properties of structures.*

We investigate computability-theoretic complexity of determining whether an algebraic structure has a certain property, relative to the algorithmic description that presents a structure. We study computable structures, but also, taking examples from algebraic categories, consider structures admitting a recursive presentation in terms of generators and relators. We establish that certain properties (such as Markov properties) are hard in a given class of structures. We can apply general results to various specific structures including various magmas such as semigroups, racks and quandles. (Received March 02, 2020)

05 ► Combinatorics

1158-05-50 **Hayan Nam*** (nckaplan@math.uci.edu), 4225 Lincoln Swing St, unit 34, Ames, IA 50014, and **nathan kaplan**. *Counting numerical semigroups using polytopes.*

A numerical semigroup is an additive monoid that has a finite complement in the set of non-negative integers. For a numerical semigroup S, the genus of S is the number of elements in $\mathbb{N} \setminus S$ and the multiplicity is the smallest nonzero element in S. In 2008, Bras-Amorós conjectured that the number of numerical semigroups with genus g is increasing as g increases. Later, Kaplan posed a conjecture that implies Bras-Amorós conjecture. In the second half, we prove Kaplan's conjecture when the multiplicity is 4 or 6 by counting the number of integer points in a polytope. Moreover, we find a formula for the number of numerical semigroups with multiplicity 4 and genus g. (Received February 16, 2020)

1158-05-57 **Jennifer Morse**, **Jianping Pan**, **Wencin Poh** and **Anne Schilling***, Department of Mathematics, University of California, One Shields Avenue, Davis, CA 95616. *Crystal for stable Grothendieck polynomials.*

We introduce a new crystal on decreasing factorizations on 321-avoiding elements in the 0-Hecke monoid which we call $\star$-crystal. This crystal is a K-theoretic generalization of the crystal on decreasing factorizations in the symmetric group of the first and last author. We prove that under the residue map the $\star$-crystal intertwines with the crystal on set-valued tableaux recently introduced by Monical, Pechenik and Scrimshaw. We also define a new insertion from decreasing factorization in the 0-Hecke monoid to pairs of (transposes of) semistandard Young tableaux and prove several properties about this new insertion, in particular its relation to the Hecke insertion and the uncrowding algorithm. The new insertion also intertwines with the crystal operators.

This is joint work with Jennifer Morse, Jianping Pan and Wencin Poh (see https://arxiv.org/abs/1911.08732). (Received February 18, 2020)

1158-05-59 **Joshua P Swanson*** (jswanson@ucsd.edu) and **Nolan R Wallach**. *Harmonic differential forms for pseudo-reflection groups.*

A great deal of research activity in algebraic combinatorics concerns diagonal actions of the symmetric group $\mathfrak{S}_n$ on k sets of n commuting indeterminants and ℓ sets of n anti-commuting indeterminants. The *coinvariant ideal* is the ideal generated by non-constant homogeneous $\mathfrak{S}_n$-invariants, and the *coinvariant algebra* is the corresponding quotient. When $k = 1, \ell = 0$, Chevalley famously determined the Hilbert series of the coinvariant algebra. When $k = 2, \ell = 0$, Haiman determined the bigraded Hilbert series using deep results of algebraic geometry. A recent conjecture of Zabrocki relates the $k = 2, \ell = 1$ coinvariant algebra to the Delta Conjecture of Haglund–Remmel–Wilson. We present progress towards Zabrocki's conjecture focusing on the $k = 1, \ell = 1$ specialization and generalizations to pseudo-reflection groups. In particular, we construct bases of semi-invariants and derive certain vanishing bounds in agreement with Zabrocki's conjecture. (Received February 18, 2020)

1158-05-68 **Oliver Pechenik*** (pechenik@umich.edu), **David E Speyer** and **Anna Weigandt**. *Castelnuovo–Mumford polynomials.* Preliminary report.

We study the highest-degree homogeneous parts of Grothendieck polynomials, which will call the Castelnuovo–Mumford polynomials $\mathfrak{CM}_w$. In particular, we give a formula for the degrees of these polynomials, determining the Castelnuovo–Mumford regularity of matrix Schubert varieties and answering a question of Jenna Rajchgot. Indeed, we compute the leading terms of all Castelnuovo–Mumford polynomials and give a complete description of when two Castelnuovo–Mumford polynomials agree up to scalar multiple. Our formulas rely on some new facts that we establish about major index of permutations. (Received February 20, 2020)

1158-05-69 **Cara Monical**, **Oliver Pechenik*** (pechenik@umich.edu) and **Travis Scrimshaw**. *Crystal structures for symmetric Grothendieck polynomials.*

The symmetric Grothendieck polynomials representing Schubert classes in the K-theory of Grassmannians are generating functions for semistandard set-valued tableaux. We construct a type A crystal structure on these tableaux. Applications include a new combinatorial formula for decomposing symmetric Grothendieck polynomials into Schur polynomials. For rectangular shapes, we give a new interpretation of Lascoux polynomials (K-analogues of Demazure characters) by constructing a K-theoretic analogue of crystals with an appropriate analogue of a Demazure subcrystal. (Received February 20, 2020)

1158-05-80 **Sami Assaf*** (shassaf@usc.edu). *The Coxeter graph of a permutation as a ranked poset.*

Tits considered the graph whose vertices are labeled by reduced words with an edge between two words when they are related by a single Coxeter relation. We consider an inversion statistic on reduced words, analogous to Coxeter length for permutations, such that this graph becomes a ranked poset with unique maximal element.

As an application, we recover old and obtain new bounds for the diameter of the graph. (Received February 21, 2020)

1158-05-83 **Robert Page** and **Ruriko Yoshida*** (ryoshida@nps.edu), 1411 Cunningham Road, Naval Postgraduate School, Monterey, CA 93943, and **Leon Zhang**. *Tropical Principal Component Analysis.*

In 2004, Speyer and Sturmfels showed that a space of phylogenetic trees with fixed set of leaves is a tropical Grassmannian under the max-plus tropical semiring, i.e., a tropical linear space defined by the tropicalization of linear equations. Therefore it is natural to apply tropical arithmetics to conduct statistical analyses over a tree space under the tropical metric, such as principal component analysis (PCA). PCA is a widely-used method for the dimensionality reduction of a given data set in a high-dimensional Euclidean space. Here we define and analyze two analogues of PCA in the setting of tropical geometry. In one approach, we study the Stiefel tropical linear space of fixed dimension closest to the data points in the tropical projective torus; in the other approach, we consider the tropical polytope with a fixed number of vertices closest to the data points. In this talk we focus on tropical PCA as a tropical polytope. We end this talk with applications to empirical datasets: African coelacanth, Apicomplexa genomes, and Hemagglutinin (HA) sequences for influenza in the state of New York from 1993 to 2017. This is joint work with Mr. Leon Zhang, Ms. Xu Zhang, and Maj. Robert (Rory) Page. (Received February 21, 2020)

1158-05-105 **Sami Assaf*** (shassaf@usc.edu). *Combinatorial models for Schubert calculus.*

Schubert calculus began in the 1880s when Hermann Schubert began asking enumerative questions in geometry, such as how many lines in space are incident to four given lines. Efforts to build a rigorous foundation for these questions led to the development of cohomology rings and modern intersection theory. In the 1980s, Lascoux and Schutzenberger defined an explicit basis for polynomials, called Schubert polynomials, whose structure constants precisely compute these intersection numbers. Despite this concrete model, we still lack a combinatorial understanding of Schubert structure constants. In this talk, I present several generating polynomial models for Schubert polynomials that compute these intersection numbers for various cases. (Received February 25, 2020)

1158-05-108 **Colleen Robichaux*** (cer2@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801, **Harshit Yadav** (yadav7@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801, and **Alexander Yong** (ayong@illinois.edu), 273 Altgeld Hall, 1409 W. Green St., Urbana, IL 61801. *The ABCDs of Schubert calculus.*

We collect *Atiyah-Bott Combinatorial Dreams* (A·B·C·Ds) in Schubert calculus. One result relates equivariant structure coefficients for two isotropic flag manifolds, with consequences to the thesis of C. Monical. We contextualize using work of N. Bergeron-F. Sottile, S. Billey-M. Haiman, P. Pragacz, and T. Ikeda-L. Mihalcea-I. Naruse. The relation complements a theorem of A. Kresch-H. Tamvakis in quantum cohomology. Results of A. Buch-V. Ravikumar rule out a similar correspondence in K-theory. (Received February 25, 2020)

1158-05-133 **Jane Ivy Coons*** (jicoons@ncsu.edu), NC , and **Seth Sullivant**. *Maximum Likelihood Estimation for Independence Models with Structural Zeros.*

A two-way independence model with structural zeros is obtained from a two-way independence model by forcing the probability of some outcomes to be zero. We use algebra and combinatorics to approach the problem of maximum likelihood estimation in these models. In cases where the structural zeros of the model avoid certain patterns, we give an explicit formula for the maximum likelihood estimate as a rational function of the data. (Received February 27, 2020)

1158-05-143 **Maria M Gillespie*** (maria.gillespie@colostate.edu), Department of Mathematics, Colorado State University, Fort Collins, CO 80523, and **Brendon P Rhoades**. *Higher Specht Bases for Generalizations of the Coinvariant Ring.* Preliminary report.

The classical coinvariant ring R_n is the quotient of a polynomial ring in n variables by the positive-degree S_n-invariants, under the natural action of the symmetric group S_n by permuting the variables. Ariki, Terasoma, and Yamada recently discovered a basis for R_n that respects its decomposition into irreducible S_n-modules, consisting of the *higher specht polynomials*.

We provide an extension of the higher Specht basis to the generalized coinvariant rings $R_{n,k}$ introduced by Haglund, Rhoades, and Shimozono. We also give a conjectured higher Specht basis for the Garsia-Procesi modules R_μ, and provide a proof of the conjecture in the case of two-row partition shapes μ. (Received February 27, 2020)

1158-05-147 **Zachary Hamaker*** (zhamaker@ufl.edu). *(Almost) symmetries of sorting networks.*
Sorting networks are reduced words for the reverse permutation. In 1984, Stanley showed they are equinumerous with standard Young tableaux of staircase shape. By visualizing sorting networks as wiring diagrams in a square, one can ask the natural question: how many sorting networks are there that are invariant under various symmetries? Naively, the answer is none. However, by imposing mild constraints, we identify many classes of sorting networks with simple enumeration formulas that almost satisfy these symmetries. Our approach offers a unified perspective for results of Haiman as well as the author with Marberg and Pawlowski, while also identifying new conjectures for enumeration. We conclude with comments on conjectural properties of random sorting networks satisfying (almost) symmetries. (Received February 28, 2020)

1158-05-148 **Zachary Hamaker*** (zhamaker@ufl.edu), **Oliver Pechenik** (pechenik@umich.edu) and **Anna Weigandt** (weigandt@umich.edu). *Grobner degeneration through ice.*
Matrix Schubert varieties are a vast generalization of determinantal varieties, which have been studied since the 19th century in the context of degeneracy loci. We will discuss how Grobner degeneration is used to compute properties of these varieties. Grobner degenerations are determined by term orders. Historically, people used diagonal term orders, but a major breakthrough by Knutson and Miller showed that antidiagonal terms orders are more natural, and used them to describe the Hilbert series of these varieties combinatorially. Deficiencies of diagonal term orders were discussed in work by Knutson, Miller and Yong. I will outline this history and discuss joint work with Oliver Pechenik and Anna Weigandt where we make new progress with diagonal term orders, including a conjectural way forward in this setting using alternating sign matrices. (Received February 28, 2020)

1158-05-150 **Susanna Fishel*** (sfishel1@asu.edu), **Jessica Gatica**, **Luc Lapointe** and **Maria-Elena Pinto**. *Quasisymmetric functions in superspace.*
Abstract: Symmetric functions in superspace were developed to study the supersymmetric version of the quantum Calogero-Sutherland model of identical particles on a circle. They are a generalization of symmetric functions: we still have the variables $x_1, x_2, \ldots$ and additionally we have anticommuting variables $\theta_1, \theta_2, \ldots$. Superspace analogues of Macdonald, Jack, and Schur polynomials have been defined in a series of papers by Desrosiers and others. I will discuss quasisymmetric functions in superspace. This is work in progress, joint with Jessica Gatica, Luc Lapointe, and Maria-Elena Pinto. (Received February 28, 2020)

1158-05-151 **Carlos E Arreche** and **Nathan Williams*** (nathan.williams1@utdallas.edu). *Normal Subgroups of Reflection Groups.*
We study normal reflection subgroups of complex reflection groups. Our approach leads to a refinement of the theorem of Orlik and Solomon that the generating function for fixed-space dimension over a reflection group is a product of linear factors involving generalized exponents. (Received February 28, 2020)

1158-05-153 **Richard M. Green*** (rmg@colorado.edu), Department of Mathematics, University of Colorado Boulder, Campus Box 395, Boulder, CO 80309-0395. *Reduced words arising from convex subheaps.*
A heap is a certain type of partially ordered set whose elements are labeled by the vertices of a graph. A finite heap over a Coxeter graph gives rise to an expression for an element of the associated Coxeter group, by refining the partial order to a total order and reading the labels in order. This associates a heap to every reduced expression, giving a bijection between reduced expressions (up to commutation) and their heaps (up to isomorphism). This talk will study the set of Coxeter group elements that can be obtained from the finite convex subheaps of a fixed infinite heap. For certain choices of infinite heaps, the reduced words arising from convex subheaps have interesting points of contact with representation theory. (Received February 28, 2020)

1158-05-161 **Eric Marberg***, Department of Mathematics, HKUST, Clear Water Bay, Kowloon, Hong Kong. *Bumping operators and insertion algorithms for supercrystals.*
Results of Morse and Schilling show that the set of increasing factorizations of the reduced words for a permutation is naturally a crystal for the general linear Lie algebra. Hiroshima has recently constructed two analogues of these crystals for queer Lie superalgebras. This talk will explore how Hiroshima's crystal structures interact with shifted analogues of the Edelman-Greene correspondence and the Little bumping algorithm. As applications, we will explain how to prove that Hiroshima's crystals are normal and identify their connected components. (Received February 29, 2020)

1158-05-164 **Zachary Hamaker**, **Oliver Pechenik**, **David E Speyer** and **Anna Weigandt*** (weigandt@umich.edu). *Derivatives of Schubert polynomials.*

We describe the action of a differential operator when applied to the Schubert basis. We use this to give a short proof of the Macdonald identity. We also discuss an application to the study of the (strong) Sperner property of the weak order on the symmetric group. This property was proven by Gaetz and Gao (2018). Our description of the differential operator leads to a proof of a determinant conjecture of Stanley (2017), which also implies the Sperner property. (Received February 29, 2020)

1158-05-174 **Kelly Isham***, ishamk@uci.edu. *A formula for the number of n-arcs in projective 3-space.* Preliminary report.

An n-arc in $\mathbb{P}^k$ is a set of n points so that no $k+1$ lie on a hyperplane. Glynn's Theorem gives a formula for the number of n-arcs in a projective plane of order q in terms of a polynomial in q and realizations of special point-line incidence structures called superfigurations. In this talk, we generalize this framework to count n-arcs in projective 3-space of order q. (Received February 29, 2020)

1158-05-181 **Jiaxi Nie*** (jin019@ucsd.edu) and **Jacques Verstraete** (jacques@ucsd.edu). *Randomized greedy algorithm for independent sets in regular uniform hypergraphs with large girth.*

We consider a randomized greedy algorithm for independent sets in r-uniform d-regular hypergraphs G on n vertices with girth g. By analyzing the expected size of the independent sets generated by this algorithm, we show that the independence number $\alpha(G) \geq (f(d,r) - \epsilon(g,d,r))n$, where $\epsilon(g,d,r)$ converges to 0 as $g \to \infty$ for fixed d and r, and $f(d,r)$ is determined by a differential equation. This extends earlier results of Gamarnik and Goldberg for graphs. (Received March 01, 2020)

1158-05-207 **Eduardo Torres Davila*** (etdavila10@gmail.com), **Tara Gomes** and **Christopher O'Neill**. *Minimal Presentation of Kunz Posets.*

Several recent papers have examined a rational polyhedron P_m whose integer points are in bijection with the numerical semigroups (cofinite, additively closed subsets of the non-negative integers) containing m. A combinatorial description of the faces of P_m was recently introduced, one that can be obtained from the divisibility posets of the numerical semigroups a given face contains. We extend the notion of a minimal presentation of a numerical semigroup to posets and use it to determine the dimension of the corresponding face. (Received March 01, 2020)

1158-05-211 **Sheila Sundaram*** (shsund@comcast.net). *The homology of subposets of subword order.* Preliminary report.

We investigate the homology representation of the symmetric group on rank-selected subposets of subword order. We show that the homology module for the subposet consisting of words of bounded length decomposes into a sum of tensor powers of the S_n-irreducible indexed by the partition $(n-1,1)$, recovering, as a special case, a theorem of Björner and Stanley for words of length at most k. For arbitrary ranks we show that the homology is an integer combination of the jth tensor powers of the reflection representation, $j = 0, \ldots, k$, where k is the maximum length of a word; we conjecture that this combination is nonnegative. Bell and Stirling numbers feature prominently in our analysis. (Received March 02, 2020)

1158-05-223 **Jennifer E Elder***, Jennifer.E.Elder@asu.edu. *Graphs of Reduced Words.* Preliminary report.

We are working on understanding the structure of the graph whose vertices are reduced words for a permutation w in $\mathfrak{S}_n$, and whose edges are related to braid or commutation moves from one word to another. We are working towards a lower bound on the number of braid classes for a permutation in relation to the size of the full set of reduced words. We will discuss the results that we have already completed, and the steps we still need to take to complete the problem. (Received March 02, 2020)

1158-05-236 **Martha Yip*** (martha.yip@uky.edu). *Newton polytopes arising from Schur polynomials and symmetric Grothendieck polynomials.* Preliminary report.

A Newton polytope associated to a polynomial is a lattice polytope associated to a multivariable polynomial. Recently, Monical, Tokcan and Yong initiated the study of the Newton polytope of many classes of polynomials from algebraic combinatorics, but open questions remain.

We study some properties of two families of Newton polytopes arising from symmetric functions: the Schur polynomials s_λ and the symmetric Grothendieck polynomials G_λ. Specifically, we show that $\mathrm{Newt}(s_\lambda)$ and $\mathrm{Newt}(G_\lambda)$ have the integer decomposition property, and discuss when these polytopes are reflexive. We also

compute the h^*-vector for reflexive Newt(s_λ) and show that it is unimodal, thereby providing a family of lattice polytopes as an example which supports a conjecture of Stanley.

This is joint work with Bayer, Goeckner, Hong, McAllister, Olsen, Pinckney and Vega. (Received March 02, 2020)

1158-05-247 **Farid Aliniaeifard** and **Nathaniel Thiem*** (thiemn@colorado.edu). *Representation theoretic faces of noncommutative symmetric functions.* Preliminary report.

Supercharacter theories have given us new ways to think about categorifying combinatorial Hopf algebras via towers of groups. In fact, the approach is sufficiently flexible that we typically get more than one choice for how to realize a given Hopf algebra. This talk explores different ways to categorify the Hopf algebra of noncommutative symmetric functions using permutation modules. These approaches not only imply generalized "labelled" versions of NSym but also naturally lift to Hopf monoids. (Received March 02, 2020)

1158-05-257 **Marino Romero*** (mar007@sas.upenn.edu), mar007@sas.upenn.edu. *Delta eigenoperators and e-positivities.*

In this talk, we will give a wide variety of e-positivity results. They arise from the study of Delta eigenoperators of the modified Macdonald basis. One of the open questions regarding Delta operators conjectures that $\Delta_{s_\lambda} e_n$ is Schur positive for any partition λ. Mark Haiman has proved this to be true for $\Delta_{s_\lambda e_n} e_n$ by showing that it is the Frobenius image of a certain S_n-module. In particular $\Delta_{e_n} e_n$ is the Frobenius characteristic for the space of Diagonal Harmonics. Zabrocki has recently conjectured a module for $\Delta_{e_k} e_n$. However, there are no explicit Schur function expansions for any of these expressions. We will solve the problem of expanding a general expression of the form $\Delta_F G$ whenever one of the parameters q or t is set to 1. In turn, we get a surprising connection between Delta operators and parallelogram polyominoes. (Received March 02, 2020)

1158-05-276 **Helene Barcelo*** (hbarcelo@msri.org), MSRI, Berkeley, CA 94720, and **Curtis Greene**, **Abdul Jarrah** and **Volkmar Welker**. *On the vanishing of the discrete singular cubical homology for graphs.*

In the early 2000, a discrete notion of homotopy for graphs and simplicial complexes was developed. The motivation came from the desire to find invariants for dynamic processes that could be encoded via (combinatorial) simplicial complexes. The invariants are topological in nature, while also sensitive to the combinatorics encoded in the complex, in particular the level of connectivity of simplices. The ensuing discrete homotopy groups differ from the classical homotopy groups in significant ways that will be reviewed in this talk. In the early 2010, the associated notion of discrete (singular cubical) homology was developed.

In this talk we show that if a graph G has no 3-cycles and 4-cycles, then the discrete cubical homology group of G is trivial in dimension d, for all $d \geq 2$. We also provide a sequence $\{G_d\}$ of graphs such that the homology is non-trivial in dimension d for $d \geq 1$. (Received March 02, 2020)

1158-05-299 **Eugene Gorksy**, **Mazin Mikhail** and **Vazirani Monica***, Department of Mathematics, One Shields Ave, Davis, CA 95616. *Recursions for rational q,t-Catalan numbers.*

Catalan numbers enumerate Dyck paths and q,t-Catalan numbers refine that count with respect to the `area` and `dinv` statistics. This has a rational analogue moving from Dyck paths in a square to those in an $n \times m$ rectangle—or equivalently *normalized* n, m-invariant subsets of $\mathbb{Z}$. The combinatorics as well as related algebraic, geometric, and topological structures get far more complicated when n, m are not coprime. Some aspects are easier to control if one passes to the q,t-Catalan power series and one relaxes the normalization condition on invariant subsets. In particular as one reads a "window" of the subset, interpreted as a binary sequence of length $n+m$, and monitors the changes in `area` and `dinv`, one gets a recursion that computes the rational q,t-Catalan power series. This recursion is inspired by the recursion of Hogancamp-Mellit and is similar to that of Elias-Hogancamp-Mellit, obtained in their studies of the homology of n, m-torus links. (Received March 03, 2020)

1158-05-303 **Susanna Fishel*** (sfishel1@asu.edu), **Elizabeth Milicevic**, **Rebecca Patrias** and **Bridget Tenner**. *Enumerations Relating Braid and Commutation Classes.*

The set of reduced words of a permutation possesses a rich combinatorial structure that has been studied from many differrent perspectives. Quotients of this set under the Coxeter relations hold particular interest, but most work has focused on the commutation classes. I will discuss recent work with Mile 'cević, Patrias, and Tenner. We obtain an upper and lower bound for the number of reduced words for a permutation in terms of the number of braid classes and the number of commutation classes of the permutation. We classify the permutations that achieve each of these bounds, and enumerate both cases. (Received March 03, 2020)

1158-05-311 **Jessica C De Silva*** (jdesilva1@csustan.edu), One University Circle, Turlock, CA 95382. *Graphs with few spanning pseudoforests.*

Graphs with extremal properties are often found within the class of threshold graphs. Threshold graphs can be obtained from any graph by repeated application of the Kelmans transformation. This transformation has been shown to have a monotonic effect on the number of spanning substructures of a graph, including the number of spanning trees. In this talk, we will show that the Kelmans transformation decreases the number of spanning pseudoforests of a graph. This result implies that there exists a threshold graph on n vertices and e edges with the minimum number of spanning pseudoforests. (Received March 03, 2020)

1158-05-316 **Theo McKenzie*** (mckenzie@math.berkeley.edu). *Eigenvalue Multiplicity Bounds on the Adjacency Matrix.*

The maximum multiplicity of the second eigenvalue of the adjacency matrix of a graph has recently been shown to solve the equiangular lines problem. Moreover, the multiplicity of this eigenvalue gives information on the speed of convergence of the power method. Here we give bounds on the multiplicity of the second eigenvalue using netting arguments and the theory of nodal domains. (Received March 03, 2020)

1158-05-318 **Theo McKenzie*** (mckenzie@math.berkeley.edu), **Hermish Mehta** and **Luca Trevisan**. *A New Algorithm for the Robust Semi-Random Independent Set Problem.*

We study the independent set problem in a semi-random model proposed by Feige and Kilian. This model selects a graph with a planted independent set of size k and then allows an adversary to modify a large fraction of edges: the subgraph induced by the complement of the independent set can be modified arbitrarily, and the adversary may add edges from the independent set to its complement. In particular, the adversary can create a graph in which the initial planted independent set is not the largest independent set. Feige and Kilian presented a randomized algorithm, which with high probability recovers an independent set of size k when $k = \alpha n$ where α is a constant, and the probability of a random edge $p > (1+\epsilon)\ln n/\alpha n$. We give a new deterministic algorithm in the Feige-Kilian model that finds an independent set of size at least $.99k$ provided that the planted set has size $k = \Omega(n^{2/3}/p^{1/3})$, and finds a list of independent sets, one of which is the planted one provided that $k = \Omega(n^{2/3}/p)$. This improves on the algorithm of Feige and Kilian by working for smaller k if $p = \Omega(1/n^{1/3})$. The ability to find a good approximation of the largest independent set is new when $p < \ln n/k$. (Received March 03, 2020)

1158-05-319 **Nathan Williams***, nathan.williams1@utdallas.edu. *Braid Groups and Shards.*

An element w of a finite Coxeter group W naturally lifts to an element $\widetilde{w}$ of the corresponding braid group $B(W)$ by interpreting any reduced word in simple reflections for w as a word in the standard generators of $B(W)$. N. Reading defined a method for cutting hyperplanes into pieces called shards; for Coxeter arrangements, this gives geometric meaning to certain lattice-theoretic properties of the weak order. We prove that the set $\{\widetilde{w} \cdot \widetilde{s} \cdot \widetilde{w}^{-1} : s \text{ an ascent of } w\}$ is naturally indexed by shards by showing that two of Salvetti's generators for the pure braid group $P(W)$ are homotopic if and only if they go around the same shard. (Received March 03, 2020)

1158-05-336 **Oscar Vega*** (ovega@csufresno.edu), **Elizabeth Compton** (lizzy4brown@mail.fresnostate.edu), **Maria Diaz** (mdiaz1824@mail.fresnostate.edu) and **Yuliana Segura** (segura_yuli@mail.fresnostate.edu). *The commuting and the power graphs of split metacyclic groups.* Preliminary report.

The commuting graph of a group G has vertex set equal to $G \setminus Z(G)$, and edges existing only when two elements commute. The (directed or undirected) power graph of a group G has the set of elements of G as vertex set, and an edge connecting g_1 with g_2 if and only if $< g_1 > \subseteq < g_2 >$ or $< g_2 > \subseteq < g_1 >$.

In this talk, we will discuss the structure of these graphs for large families of split metacyclic groups. (Received March 03, 2020)

1158-05-340 **Sami Assaf** and **Danjoseph Quijada*** (dquijada@usc.edu), 6178 Fremont Way, Buena Park, CA 90620. *A Pieri rule for key polynomials.*

We give a signed, cancellation-free Pieri formula for key polynomials, a nonsymmetric generalization of Schur polynomials. (Received March 03, 2020)

1158-05-349 **Brendan Pawlowski*** (br.pawlowski@gmail.com) and **Eric Marberg**. *Principal specializations of Schubert polynomials in classical types.*

Macdonald found a surprising formula for the principal specialization of a Schubert polynomial as a generating function over reduced words. We give similar formulas for the principal specializations of the Billey-Haiman

Schubert polynomials (actually power series) of types B, C, and D, and of the Grothendieck polynomials of types A, B, C, D. (Received March 03, 2020)

1158-05-350 **Samantha Dahlberg*** (sdahlber@asu.edu) and **Younghwan Kim** (ykim117@asu.edu). *Diameters of graphs of reduced words for some permutation avoidance classes.* Preliminary report.

In this talk we consider the diameter of the graph created from reduced words of a single permutation in the symmetric group connected by short and long braid moves. Reiner and Roichman conjectured an upper and lower bound for this diameter related to the enumeration of certain co-dimension two hyperplanes. We resolve the conjecture for permutations that avoid 321, 231 or 312. Also, we find a recursive formula for the diameter of the longest word, which matches the diameter found by Reiner and Roichman. (Received March 03, 2020)

1158-05-352 **Matthew Hogancamp*** (m.hogancamp@northeastern.edu) and **Anton Mellit**. *Superpolynomials of torus links.*

In this talk I will discuss joint work with Anton Mellit, in which we find simple recursions which calculate the "super-polynomial" of torus links $T(m,n)$. By combining various results of mine and Elias, Gorsky, or Mellit, these super-polynomials can often be recognized as graded dimensions of some (more-or-less) familiar objects in combinatorics, Hilbert schemes, and representation theory. Most famously, the super-polynomial for $T(n,n+1)$ is the graded dimension of the space of diagonal harmonics, also known as the q,t Catalan (this was first conjectured by Gorsky), so for general m,n (not necessarily coprime) these superpolynomials may be regarded as generalized q,t Catalans.

The focus of the talk will be on the recursions for superpolynomials with Mellit, but I will also survey what is known (or not known) about their representation-theoretic interpretation. (Received March 03, 2020)

1158-05-379 **Benjamin J. Young*** (bjy@uoregon.edu). *Single and the double dimer models in enumerative geometry.*

I'll explain how Jenne's recent work on the double-dimer model can be applied to prove a combinatorial conjecture arising in the enumerative geometry of toric Calabi-Yau Threefolds. (Received March 04, 2020)

06 ► Order, lattices, ordered algebraic structures

1158-06-18 **Harold Polo***, Department of Mathematics, University of Florida, P.O. Box 118105, Gainesville, FL 32611. *On the sets of lengths of Puiseux monoids generated by multiple geometric sequences.*

It is well known that a Puiseux monoid M is atomic if and only if M contains a minimal set of generators. This characterization, whose verification involves a great deal of effort given that Puiseux monoids are not, in general, finitely generated, motivated a series of papers studying the atomicity of Puiseux monoids generated by structured sets such as monotone and geometric sequences. In this paper, we focus on the study of rational multicyclic monoids, that is, Puiseux monoids generated by multiple geometric sequences. In particular, we provide a complete description of the rational multicyclic monoids that are hereditarily atomic. Additionally, we show that the sets of lengths of certain family of atomic rational multicyclic monoids are finite unions of infinite multidimensional arithmetical progressions, a result we use to realize infinite arithmetic progressions as delta sets of some Puiseux monoids. (Received February 01, 2020)

08 ► General algebraic systems

1158-08-205 **Roberto Pelayo*** (rcpelayo@uci.edu). *Visualization Methods for Numerical Semigroups.*

The study of numerical semigroups has benefited from a careful analysis of factorization quantities (e.g., delta set, ω-primality, elasticity) over a single semigroup and other invariants (e.g., Frobenius number, Apery set) over a family of semigroups. Many of these results were motivated from a graphical analysis of these quantities and illustrate a general trend of eventual quasi-polynomial behavior. In this talk, we illustrate some of these graphical techniques and highlight general trends that have emerged from these investigations. (Received March 01, 2020)

11 ► Number theory

1158-11-8 **quan yong han*** (hyq131501@163.com), huairou bingjing, beijing, beijing 101400. *Research on Prime Formula and Goldbach Conjecture.* Preliminary report.

There are also formulas for prime numbers. The formula for prime numbers can be written as: prime number = n + a, where n is a natural number greater than 1, and a is an integer. Let a, n1, and n2 be natural numbers, n1> 1, n2> 1, and n1> a. Then two prime numbers are taken from the deformation of the prime formula: n2 + a = prime number, n1-a = prime number. The sum of two odd numbers must be even, because prime numbers greater than 2 must exist in odd numbers, and because 2 + 2 = 4 is even. (N1-a) + (n2 + a) = n1 + n2 = even, it must be true. n1 + n2 indicates that the condition of all even numbers is that n1 and n2 are even or odd at the same time, because n1 and n2 are natural numbers and can be satisfied, so n1 + n2 indicates that all even numbers are shown; it can also be proved as: n1> 1 n2> 1 and n1> a, so (n1-a) + (n2 + a) = n1 + n2> a + 1, a is a natural number, a can be sufficiently large, and a + 1 can be sufficiently large, so either is sufficient Large even numbers can be expressed as the sum of two prime numbers; Goldbach's conjecture is correct. (Received December 03, 2019)

1158-11-19 **Lenny Fukshansky*** (lenny@cmc.edu) and **Yingqi Shi**. *Positive semigroups and generalized Frobenius numbers over totally real number fields.*

Frobenius problem and its many generalizations have been extensively studied in several areas of mathematics. We study semigroups of totally positive algebraic integers in totally real number fields, defining analogues of the Frobenius numbers in this context. We use a geometric framework recently introduced by Aliev, De Loera and Louveaux to produce upper bounds on these Frobenius numbers in terms of a certain height function. We discuss some properties of this function, relating it to absolute Weil height and obtaining a lower bound in the spirit of Lehmer's conjecture for algebraic vectors satisfying some special conditions. We also obtain bounds on the size of representations and number of elements of bounded height in such positive semigroups of totally real algebraic integers. (Received February 01, 2020)

1158-11-30 **Derek Orr*** (djo15@pitt.edu), 301 Thackeray Avenue, Pittsburgh, PA 15213. *Rational zeta series for $\zeta(2n)$ and $\zeta(2n+1)$.*

I will begin by using the cotangent function to find rational zeta series with $\zeta(2n)$ in terms of $\zeta(2k+1)$ and $\beta(2k)$, the Dirichlet beta function. I then develop a certain family of generalized rational zeta series using the generalized Clausen function $\mathrm{Cl}_m(x)$ and use those results to discover a second family of generalized rational zeta series. As a special case of my results from Theorem 3.1, I prove a conjecture given in 2012 by F.M.S. Lima. Later, I use the same analysis but for the digamma function $\psi(x)$ and negapolygammas $\psi^{(-m)}(x)$. With these, I extract the same two families of generalized rational zeta series with $\zeta(2n+1)$ on the numerator rather than $\zeta(2n)$. Afterwards, I look into the applications of these rational zeta series and how they are related to other special functions such as the multiple zeta function. (Received February 08, 2020)

1158-11-46 **Madeline Locus Dawsey*** (mdawsey@uttyler.edu) and **Ken Ono**. *Recent developments in moonshine.*

The most famous example of a connection between representation theory and number theory is the Monstrous Moonshine Conjecture, proven by 1998 Fields Medalist Richard Borcherds, which asserts that the character dimensions of the Monster group, the largest of the sporadic finite simple groups, are dictated by the Fourier expansions of a distinguished set of modular functions called McKay-Thompson series. Recently DeHority, Gonzalez, Vafa, and Van Peski established that a weak form of moonshine holds for every finite group. Weak moonshine only relies on group character tables, which are not isomorphism class invariants, and so non-isomorphic groups can have the same McKay-Thompson series. In joint work with Ken Ono, we address this issue by defining the notion of "higher width moonshine." We prove that complete width 3 moonshine determines groups of fixed order up to isomorphism. (Received February 14, 2020)

1158-11-87 **Ayomikun Adeniran***, ayoijeng@math.tamu.edu, and **S. Butler**, **C Defant**, **Y Gao**, **P. E. Harris**, **C Hettle**, **Q Liang**, **H Nam** and **A Volk**. *Invariants of quotients of a numerical semigroup.*

There are several fundamental invariants of a numerical semigroup S among which are the *Frobenius number* and *genus* of S, denoted $F(S)$ and $g(S)$, respectively. The *quotient* of a numerical semigroup S by a positive integer d is the set $S/d = \{x \in \mathbb{N}_0 : dx \in S\}$ which is also a numerical semigroup. In this talk, we present a recent result showing the relation between the genus of S/d and the genus of S. Also, we will show certain

identities relating the Frobenius numbers and the genus of quotients of numerical semigroups that are generated by certain types of arithmetic progressions. (Received February 22, 2020)

1158-11-233 **Kathrin Bringmann**, **Chris Jennings-Shaffer*** (christopher.jennings-shaffer@du.edu) and **Karl Mahlburg**. *Asymptotics for Statistics on Unimodal Sequences.*

A unimodal sequence of a positive integer n is a sequence of positive integers that sum to n, while also being unimodal in the sense that the sequence is weakly increasing to a peak and is then weakly decreasing past this peak. One statistic on such sequences measures how far the peak is from the center of the sequence. A method for deriving the asymptotic behavior of such a statistic is based on identities for basic hypergeometric series, asymptotic expansions of holomorphic functions, and the method of moments from probability. We give a summary of this method, some recent results, and mention some variations. (Received March 02, 2020)

1158-11-296 **Cezar Lupu*** (cezar.lupu@ttu.edu), 1108 Memorial Circle, Lubbock, TX 79409. *A Zagier-type formula for special multiple Hurwitz zeta values.*

In this talk, we provide a Zagier-type formula for the multiple t-values (special Hurwitz zeta values),

$$t(k_1,k_2,\dots,k_r)=2^{-(k_1+k_2+\ldots+k_r)}\zeta\left(k_1,k_2,\dots,k_r;-\frac{1}{2},-\frac{1}{2},\dots,-\frac{1}{2}\right)=$$

$$=\sum_{1\leq n_1<n_2<\ldots<n_r}\frac{1}{(2n_1-1)^{k_1}(2n_2-1)^{k_2}\dots(2n_r-1)^{k_r}}.$$

Our formula is similar with Zagier's formulas for MZVs $\zeta(2,\dots,2,3)$ and will involve $\mathbb{Q}$-linear combinations of powers of π and odd zeta values. The derivation of the formula for $t(2,\dots,2,3)$ relies on a rational zeta series approach via a Gauss hypergeometric argument. (Received March 03, 2020)

1158-11-300 **Jesse Elliott*** (jesse.elliott@csuci.edu). *The prime counting function, the exponential integral, and the digamma function.*

We provide several strengthenings of the prime number theorem, as well as various equivalents of the Riemann hypothesis, using the exponential integral function, the digamma function, and the harmonic numbers. (Received March 03, 2020)

1158-11-307 **Zhi-an Luan*** (zhianluanwl@gmail.com), Vancouver, BC 14350, Canada. *Number structures of quantum gravity fields.*

Here we establish the Number Theory framework of Generalized Newton's Gravity Laws (GNGL).-Zhi-an Luan, CAP-Congress (Theoretical Physics), 2019 4 June Burnaby, #2448.

1. We found that the Fibonacci sequence $\sqrt{5}/2$, $(7-\sqrt{5})/2$ are representation of quantum gravity on 2-torus geometric group momentum equals to 0.618012897, inverse is 1.61809209, which near above irrational algebraic numbers of the Fibonacci sequence.

2. We exactly determine Boltzmann constant is $8\sqrt{3}$; Planck constant is $2\pi\sqrt{3}$, reduced is $2\sqrt{3}$, then $(2\sqrt{3})**2$ is 12, $(24\sqrt{3})**2=12^{***}=1728$!

3. We obtain that maximum gravity growth rate is $27=(3\sqrt{3})**2$, coherent radius is $\sqrt{3}$ in normal transition, then light velocity is 3 million km/s lower than real photon velocity $3.89953\dots$.

4. Eight quantum gravity spectrums totally include key number $\pm 1/2\ \pm\sqrt{3}i/2$, $\pm\sqrt{3}/2\ \pm i/2$. Newton's constant is 2/3.

(Received March 03, 2020)

1158-11-313 **Amita Malik*** (amita.malik@rutgers.edu). *Zeros of derivatives of the completed Riemann zeta function.*

Riemann Hypothesis implies the Riemann Hypothesis for higher order derivatives of the completed Riemann zeta function. The gap distribution of the zeros of these derivatives would provide insight into the existential question of the Landau-Siegel zero. In this talk, we discuss the vertical distribution of these zeros and establish their uniform distribution modulo one. Moreover, we prove that 100% of the zeros of combinations of these derivatives lie on the critical line even though there is no Euler product in this case. (Received March 03, 2020)

1158-11-315 **Santanu Chakraborty*** (santanu.chakraborty@utrgv.edu), School of Math and Stat Sc., UTRGV, 1201 West University Drive, Edinburg, TX 78539. *Some Interesting Observations with respect to Collatz Conjecture.* Preliminary report.

Collatz Conjecture concerns the following problem of generating a sequence. Start with a positive integer. If it is even, divide it by 2. If it is odd, multiply it by 3 and then add 1. Each subsequent term is obtained from

the previous term by following the same philosophy. The conjecture is that, whatever be the first term in the sequence, it always leads to 1 which means this process generates a terminating sequence with the last term in the sequence being 1.

This conjecture is more than 80 years old. There have been several unsuccessful/partially successful attempts to prove this conjecture. While we do not claim to prove this conjecture, we make some interesting observations with respect to this conjecture from Markov Chain point of view and from binary number representation point of view. (Received March 03, 2020)

1158-11-327 **Trevor Ferguson*** (tferggie@mail.fresnostate.edu). *q-Series and Ramanujan's Congruences.* Preliminary report.

A q-series is a series whose coefficients contain $\prod_{k=0}^{n-1}(1-aq^k)$ where $a, q \in \mathbb{C}$ and $n \in \mathbb{Z}$. In this talk, we will use famous q-series identities to prove two of Ramanujan's Congruences and their generalizations. (Received March 03, 2020)

1158-11-356 **Vlad Matei***, Department of Mathematics, University of California Irvine, Irvine, CA 92697. *A geometric approach to counting norms in cyclic extensions of function fields.*

We present an explicit version of a function field analogue of a classical result of Odoni about norms in number fields in the case of a cyclic Galois extensions. In the particular case of a quadratic extension, we recover the result of Bary-Soroker, Smilanski, and Wolf which deals with finding asymptotics for a function field version on sums of two squares, improved upon by Gorodetsky , and reproved by the author in his Ph.D thesis using the method of this paper. The main tool is a twisted Grothendieck Lefschetz trace formula, inspired by the work of Church, Farb and Ellenberg on representation stability and asymptotic for point counts on varieties. Using a combinatorial description of the cohomology we obtain a precise quantitative result which works in the $q^n \to \infty$ regime, and a new type of homological stability phenomena, which arises from the computation of certain inner products of representations. (Received March 03, 2020)

1158-11-370 **Daniel J. Katz*** (daniel.katz@csun.edu), Department of Mathematics, California State University, Northridge, 18111 Nordhoff Street, Northridge, CA 91330-8313. *A new generating function for calculating the Igusa local zeta function.*

Consider a polynomial $f(x_1, \ldots, x_n)$ with integer coefficients. Let p be a prime and k a positive integer let $N_k(f)$ be the number of zeroes in $(\mathbb{Z}/p^k\mathbb{Z})^n$ of f modulo p^k. The Igusa local zeta function Z_f is a generating function that organizes these zero counts for all k. Its poles tell us about the p-divisibility of the counts. More generally, f can have coefficients in a ring R of integers of a p-adic field, and the Igusa local zeta function for f organizes the counts $N_k(f)$ of zeroes of f modulo π^k, where π is a uniformizing parameter for R.

We devise a new method for calculating the Igusa zeta function via a new kind of generating function G_f, which is a projective limit of a family of generating functions, and contains more data than the local zeta function. Our G_f resides in an algebra whose structure is naturally compatible with operations on the underlying polynomials, thus facilitating calculation of local zeta functions. This new method enables us to calculate Igusa local zeta functions for a much wider range of quadratic polynomials over 2-adic fields than have been determined previously. (This is joint work with Raemeon A. Cowan and Lauren M. White.) (Received March 03, 2020)

12 ► Field theory and polynomials

1158-12-163 **Paul Baginski*** (pbaginski@fairfield.edu), Dept. of Mathematics, Fairfield University, 1073 North Benson Rd., Fairfield, CT 06824. *Elasticity and Factorization in Numerical Semigroup Rings.*

Given a numerical semigroup S and a field K, we can construct the numerical semigroup ring $K[S] \subseteq K[x]$, consisting of polynomials with coefficients in K where the only powers of x permitted are those integers that appear in S. These integral domains have nonunique factorization and their factorization properties have been studied extensively. We will give an update on what is known about factorization in these rings, focusing in particular on the elasticity. Given a nonzero nonunit a, the elasticity $\rho(a)$ is the length of a longest factorization of a, divided by the shortest factorization length of a. The elasticity of the ring R is $\rho(R) = \sup_{a \in R} \rho(a)$. We will discuss the value of $\rho(K[S])$, the distribution of elasticities of elements in $K[S]$, and how these questions relate to the analogous answers for S itself. We will also discuss how the class group of $K[S]$ can be described in terms of the numerical monoid S. (Received February 29, 2020)

13 ► Commutative rings and algebras

1158-13-51 **Saeed Nasseh*** (snasseh@georgiasouthern.edu). *DG homological algebra and vanishing of Ext over local rings.* Preliminary report.

The use of techniques from differential graded (DG) homological algebra was established by Avramov, Buchsbaum, Eisenbud, Foxby, Halperin, Kustin, Miller, and Weyman in commutative algebra. It has been shown recently that these techniques can be applied to solve non-trivial problems in commutative algebra. These applications include a solution to a conjecture posed by Vasconcelos in 1974 about semidualizing modules given by Nasseh and Sather-Wagstaff and introducing several new classes of commutative local rings by Avramov, Iyengar, Nasseh, and Sather-Wagstaff that satisfy the Auslander-Reiten Conjecture.

In this talk, we will give a brief survey of these results along with discussing some progress on the (weak) lifting problem of DG modules and its relation to the Auslander-Reiten Conjecture. This is based on a joint work with Maiko Ono and Yuji Yoshino. (Received February 17, 2020)

1158-13-94 **Joseph Gubeladze*** (soso@sfsu.edu), Department of Mathematics, San Francisco State University, San Francisco, CA 94132. *From numerical to affine and back.*

Although the numerical monoids are the special rank one case of general affine monoids, most of the works on these classes of monoids are organized around different ideas and challenges. In the talk we will focus on instances when a same approach leads to interesting yet different observations on both types of monoids. These include the structures of associate graded rings and arithmetic properties, such as factorizations and normality. (Received February 24, 2020)

1158-13-114 **K. R. Goodearl*** (goodearl@math.ucsb.edu) and **M. T. Yakimov** (yakimov@math.lsu.edu). *Poisson Cluster Algebras and Unique Factorization.*

Many cluster algebras are known to be unique factorization domains in which the cluster variables are irreducible elements. Moreover, large classes of algebras that are known or conjectured to be cluster algebras have compatible Poisson structures. (This is, in fact, a natural expectation for semiclassical limits of quantum algebras.) We will discuss the reverse direction – how combinations of unique factorization and Poisson structures lead to cluster algebra structures. This applies to algebras from a large class of Poisson UFDs. (Received February 25, 2020)

1158-13-120 **Hugh Roberts Geller*** (hgeller@clemson.edu). *DG-Structures for Fiber Products.* Preliminary report.

A construction of Tate shows that every algebra over a ring R possess a DG-algebra resolution over R. These resolutions are not always minimal and Avramov even shows that certain algebras cannot have a minimal resolution with a DG-algebra structure. This talk gives an explicit construction of a DG-structure for certain fiber products and criteria for determining when the structure is a DG-module, DG-algebra, or minimal DG-algebra. (Received February 26, 2020)

1158-13-146 **Hannah Altmann** and **Sean Sather-Wagstaff*** (ssather@clemson.edu). *Strongly Tor-independent Modules over Local Rings.* Preliminary report.

Let $(R, \mathfrak{m})$ be a local ring, and let $M_1, \dots, M_n$ be finitely generated R-modules that are strongly Tor-independent, i.e., such that $\mathrm{H}_{\geq 1}(M_{i_1} \otimes^{\mathbf{L}}_R \cdots \otimes^{\mathbf{L}}_R M_{i_t}) = 0$ for all $1 \leq i_1 < \cdots < i_t \leq n$. Gerko proved that if R is artinian, then the existence of such a sequence implies that $\mathfrak{m}^n \neq 0$. (A version when R is Cohen-Macaulay follows naturally.) We use DG algebra techniques to investigate the case where R is not Cohen-Macaulay. (Received February 28, 2020)

1158-13-149 **Francesca Gandini***, 1200 Academy St, Kalamazoo, MI 49006. *Noether's degree bound in the exterior algebra.*

When we consider the action of a finite group on a polynomial ring, a polynomial unchanged by the action is called an invariant polynomial. A famous result of Noether states that in characteristic zero the maximal degree of a minimal invariant polynomial is bounded above by the order of the group. Our work establishes that the same bound holds for invariant skew polynomials in the exterior algebra. Our approach to the problem relies on a theorem of Derksen that connects invariant theory to the study of ideals of subspace arrangements. We adapt his proof over the polynomial ring to the exterior algebra, reducing the question to establishing a bound on the Castelnuovo-Mumford regularity of intersections of linear ideals in the exterior algebra. We prove the required regularity bound using tools from representation theory. In particular, the proof relies on the existence of a functor on the category of polynomial functors that translates resolutions of ideals of subspace arrangements over the polynomial ring to resolutions of ideals of subspace arrangements over the exterior algebra. (Received February 28, 2020)

1158-13-172 **Carmelo Cisto**, **Michael DiPasquale*** (michael.dipasquale@colostate.edu), **Gioia Failla**, **Zachary Flores**, **Chris Peterson** and **Rosanna Utano**. *Generalizing Wilf's conjecture to higher dimensions.*

Wilf's conjecture is a long standing open problem about the density of holes in a numerical semigroup, which is a submonoid of the natural numbers ($\mathbb{N}$) with finite complement. We propose a generalization of Wilf's conjecture for submonoids of $\mathbb{N}^d$ with finite complement (called generalized numerical semigroups). We prove this conjecture for several large classes of generalized numerical semigroups, including irreducible, symmetric, and monomial classes. We also discuss the relationship of our conjecture to a different generalization of Wilf's conjecture proposed by García-García, Marín-Aragón, and Vigneron-Tenorio. Time permitting we'll discuss some extensions of our results which are in progress. This is joint work with C. Cisto, G. Failla, Z. Flores, C. Peterson, and R. Utano. (Received February 29, 2020)

1158-13-176 **Claudia Miller*** (clamille@syr.edu) and **Hamid Rahmati** (hrahmati2@unl.edu). *Transferring algebra structures on complexes, part 2.*

We discuss how to use the homotopy on the tautological Koszul complex given by the weighted de Rham map to build a concrete permutation invariant dg-algebra structure on a well-known resolution. This is joint work with Hamid Rahmati. (Received February 29, 2020)

1158-13-178 **Claudia Miller** and **Hamid Rahmati*** (hrahmati2@unl.edu). *Transferring algebra structures on complexes, part 1.*

We discuss a homological method for transferring algebra structures on complexes along suitably nice homotopy equivalences. We also show how one can get such homotopy equivalencies, from old ones, using a homological tool called the perturbation lemma. This is joint work with Claudia Miller. (Received March 01, 2020)

1158-13-183 **Rachel N. Diethorn*** (rngettin@syr.edu). *Generators of Koszul homology.*

One approach to understanding Koszul homology is to find the generators. In the first part of this talk, I will provide explicit formulas for the generators of Koszul homology on the minimal generators of an ideal J with coefficients in what we call a J-closed module. This generalizes work of Herzog and of Corso, Goto, Huneke, Polini, and Ulrich. In the second part of the talk I will demonstrate the utility of such formulas. In particular, I will discuss an application to the Koszul homology algebra of quotients by certain edge ideals, and I will give an answer, for such rings, to a question of Avramov about the Koszul homology algebra of a Koszul algebra. (Received March 01, 2020)

1158-13-184 **Mark E Walker*** (mark.walker@unl.edu), Department of Mathematics, University of Nebraska, 203 Avery Hall, Lincoln, NE 68588-0130. *Lower bounds on the homology of dg-modules over exterior algebras.* Preliminary report.

Let E be an exterior algebra with n generators over a field, and let P be a perfect dg-E-module that is not acyclic. In this talk I will discuss what is known about the smallest possible value of the dimension of the homology of P. (Received March 01, 2020)

1158-13-190 **Lars Winther Christensen*** (lars.w.christensen@ttu.edu), **Sergio Estrada** and **Peder Thompson**. *Singularity Categories of Sheaves.* Preliminary report.

In local algebra, the quotient of the bounded derived category by the subcategory of perfect complexes is often referred to as the *singularity category.* The quotient is trivial for a regular local ring, and for a Gorenstein ring it is triangulated equivalent the stable category of Gorenstein projective modules or, if one prefers that point of view, to the homotopy category of totally acyclic complexes of projective modules. I will discuss an ongoing project in which we extend these ideas to sheaves and reconcile them with other categorical measures of the Gorenstein property. (Received March 03, 2020)

1158-13-202 **Lars Winther Christensen*** (lars.w.christensen@ttu.edu) and **Srikanth B Iyengar**. *Dimension and codimension of homologically finite complexes.*

Let R be a commutative noetherian ring. In the 1970s Foxby and Iversen extended the notions of dimension, depth, and codimension from R-modules to R-complexes. While their notions agree up to a normalization, the notion of codimension used in Bruns and Herzog's treatment of the homological conjectures (not mostly theorems) does not quite fit in; in fact, it is not even "obviously" homological. I will discuss how to align these invariants. (Received March 01, 2020)

1158-13-229 **Luchezar L. Avramov** and **Nicholas R. Packauskas*** (nicholas.packauskas@cortland.edu), 81 Maple Avenue, Cortland, NY 13045, and **Nicholas R Packauskas**. *Quasi-Polynomial Growth of Betti Sequences.*

Let Q be a regular local ring and I an ideal generated by a regular sequence of c elements in the square of the maximal ideal. It is known that over the complete intersection $R = Q/I$ that any finitely generated module M has Betti numbers eventually given by quasi-polynomial of degree less than c. That is, there are integer-valued polynomial functions p_+^M and p_-^M with the same leading term such that $\beta_{2i}^R(M) = p_+^M(2i)$ and $\beta_{2i+1}^R(M) = p_-^M(2i+1)$ for i sufficiently large. We will show that if q is the height of the ideal generated by the quadratic initial forms of I in the associated graded ring of Q, then the degree of $p_+^M - p_-^M$ is less than $c-q-1$. (Received March 02, 2020)

1158-13-237 **John Myers***, 7060 State Route 104, Oswego, NY 13126. *Koszul-Tate resolutions as twisted tensor products.* Preliminary report.

Let R be a standard graded commutative algebra over a field k and let K be the Koszul complex on a minimal set of generators of the augmentation ideal of R. Then $\mathrm{Tor}^R(k,k)$ is well-known to be a Hopf algebra (e.g., as the homology of the bar construction), but such a structure can also be realized through a Koszul-Tate resolution of k over R and a coproduct originally defined by Assmus. Furthermore, by factoring out a certain coideal (with respect to the Assmus coproduct) one passes from $\mathrm{Tor}^R(k,k)$ to $\mathrm{Tor}^K(k,k)$, the latter Tor being a "derived" version computed via semifree resolutions.

In this talk, for certain classes of algebras we will show how the Assmus coproducts on $\mathrm{Tor}^R(k,k)$ and $\mathrm{Tor}^K(k,k)$ conspire with R and K to factor the Koszul-Tate resolution as twisted tensor products $R\otimes_\tau \mathrm{Tor}^R(k,k)$ and $K\otimes_{\tau'}\mathrm{Tor}^K(k,k)$, where τ and τ' are certain acyclic twisting cochains. Applications will be included. (Received March 02, 2020)

1158-13-259 **David A. Jorgensen*** (djorgens@uta.edu), **Liana Sega** and **Peder Thompson**. *Asymptotic Behavior of Ext over graded complete intersections.*

It is well known that for a pair of modules over a complete intersection, when all higher Ext modules have finite length, then the lengths of these higher Ext modules grow polynomially. In fact, the lengths are determined by two polynomials, one giving the lengths of the even Ext, and one giving the lengths of the odd Ext. In this talk we will discuss conditions when these two polynomials have the same leading coefficient. This is equivalent to the vanishing of the Herbrand difference, or Hochster's Theta invariant, a phenomenon studied by several authors in recent papers. This is Joint work with Lian Sega and Peder Thompson. (Received March 02, 2020)

1158-13-264 **Scott T. Chapman**, **Jim Coykendall**, **Felix Gotti*** (felixgotti@ufl.edu) and **William W. Smith**. *Other-half-factoriality on monoids and domains.*

An atomic monoid M is called *other-half-factorial* or an *OHFM* provided that for all $x \in M$ and any two factorizations z and z' of x the equality $|z| = |z'|$ implies that $z = z'$, where $|w|$ denotes the number of atoms in a given factorization w (counting repetitions). Note that other-half-factoriality is a notion that complements that of half-factoriality. Other-half-factoriality was first considered by Jim Coykendall and William W. Smith in 2011, where the authors established some properties of OHFMs and then proved that any other-half-factorial domain is a UFD. We will provide further insight of OHFMs and discuss, in the context of both monoids and domains, some properties weaker than being other-half-factorial. (Received March 02, 2020)

1158-13-273 **Fairfield University*** (jstriuli@fairfield.edu), North Benson Rd, Fairfield, 06824, and **Claudia Miller** and **Hamidreza Rahmati**. *Annihilator of Koszul Homology.*

Let R be a local ring and I and ideal of R. In this talk we present results on the annihilator of the first Koszul Homology of I which, in several examples, can be computed from the ideal generated by the entries of the matrix representing the first differential in a free resolution of I. This is a preliminary report. (Received March 02, 2020)

1158-13-279 **Rebekah J Aduddell*** (rebekah.aduddell@uta.edu). *The Critical and Cocritical Degrees of a Totally Acyclic Complex.* Preliminary report.

Let $(Q, \mathfrak{m}, k)$ be a commutative local Noetherian ring, $f_1, \dots, f_c$ a Q-regular sequence in $\mathfrak{m}$, and $R = Q/(f_1, \dots, f_c)$. We present a generalization of critical degree for R-modules to the category of totally acyclic R-complexes, $\mathbf{K_{tac}}(R)$. In particular, this talk will discuss how the critical and cocritical degrees of an R-complex change under certain operations, such as the mapping cone and the pinched tensor product. (Received March 02, 2020)

1158-13-282 **Tyler D Anway*** (tyler.anway@uta.edu). *Classifying Totally Acyclic Complexes over Rings of Finite CM Type.* Preliminary report.

Let $(Q, \mathfrak{m}, k)$ be a Henselian CM local ring of finite CM type. This implies there are only finitely many indecomposable totally acyclic complexes over Q. Using these complexes, we define a new classification scheme and explore how to use this scheme, in conjunction with a technique of approximation, to compare the properties of complexes over quotients of Q. (Received March 02, 2020)

1158-13-301 **Jay Daigle*** (gdaigle@oxy.edu), **Dan Huth**, **Andrea Stine** and **Vena Zhang**. *Delta sets of arithmetic and geometric semigroups with respect to non-minimal generating sets.*

The delta set is an invariant that describes the factorization structure in a numerical semigroup, and recent research has revealed a great deal about the structure of delta sets, especially in semigroups with relatively simple minimal presentations.

We generalize delta sets to consider factorizations with respect to a fixed non-minimal generating set, such as $\{4, 6, 9, 20\}$. We specifically investigate semigroups generated by arithmetic or geometric sequences with one additional generator, and find complex patterns in their delta sets related to the Euclidean algorithm. (Received March 03, 2020)

1158-13-321 **Rasoul Ahangari Maleki** and **Liana M Şega*** (segal@umkc.edu). *Free resolutions over compressed artinian level algebras of socle degree three.*

We will discuss rationality of Poincaré series and linearity of free resolutions in the case of compressed artinian level algebras of socle degree three. (Received March 03, 2020)

1158-13-331 **Haydee M Lindo*** (08hml@williams.edu), MA 01267. *Endomorphism invariant modules and ring classifications.* Preliminary report.

I will speak on modules which are invariant under endomorphisms of their envelopes. This will include connections to the general theory of trace modules with some preliminary applications to ring classifications and conjectures involving modules with no self-extensions. (Received March 03, 2020)

1158-13-337 **Jesus A. De Loera**, **Robert Krone**, **Lily Silverstein*** (lsilverstein@cpp.edu) and **Zekai Zhao**. *Algorithm selection for Hilbert series computations.*

We look at the problem of selecting the best among several algorithms for computing the Hilbert series of a monomial ideal. Which algorithm is fastest varies with the problem instance, so a neural network trained to automate this choice for each input can outperform any of the individual algorithms. (Received March 03, 2020)

1158-13-375 **Tara N Gomes*** (tngomes@sdsu.edu). *Numerical semigroups, Polyhedra, and Posets: Exploring the Kunz Polyhedra.*

A numerical semigroup is a subset of the non-negative integers that contains zero and is closed under addition. There are several longstanding open conjectures related to these objects. Recently, the connection between numerical semigroups and a family of polyhedral cones has become a trending area of research. In this presentation we seek to develop the bijection between semigroups and polyhedral integer points, and motivate the use of geometry to approach these conjectures.

We explored these ideas during the 2019 San Diego State REU program, as well as through continued undergraduate research

Submitted paper: https://arxiv.org/abs/1912.04460 (Received March 04, 2020)

14 ► Algebraic geometry

1158-14-5 **Shanna Nicole Dobson*** (sdobson@calstatela.edu), 5151 State University Drive, Los Angeles, CA 90039. *Crystalline Topos and the Profinite Completion of the Adic Fargues-Fontaine Curve.* Preliminary report.

We interpret Gaitsgory and Arinkin's notions of crystals and cohomological support of coherent sheaves on a locally complete intersection scheme as a profinite completion of the adic Fargues-Fontaine Curve, where the perfectoid field Cb is replaced by a perfectoid space S of characteristic p. Using the result that the category of crystals is equivalent to that of D-modules, we formulate a pro-crystalline topos, where crystals on X are quasi-coherent sheaves on the de Rham prestack X_{dR}. (Received October 09, 2019)

1158-14-56 **David Kahle**, **Christopher ONeill*** (cdoneill@sdsu.edu) and **Jeff Sommars**. *Using Macaulay2 from within R: the M2R Package.*

The R package m2r provides a persistent interface to the algebra system Macaulay2, making it an ideal utility for algebraic statistics computations. In this talk, we will explore a variety of m2r functionality, and discuss several design decisions that make m2r easy to use for Macaulay2 novices but flexible enough to accommodate advanced usage by Macaulay2 experts. We will also preview some exciting new features currently under development. (Received February 17, 2020)

1158-14-122 **Isabelle Shankar***, 739 Evans Hall, Berkeley, CA , and **Serkan Hosten**. *Symmetry Adapted Gram Spectrahedra.*

Sum of squares (SOS) relaxations are often used to certify nonnegativity of polynomials and are equivalent to solving a semidefinite program (SDP). The feasible region of the SDP for a given polynomial is the Gram Spectrahedron. For symmetric polynomials, there are reductions to the problem size that can be done using tools from representation theory. In recent work with Serkan Hosten, we investigate the geometric structure of the spectrahedra that arise in the study of symmetric SOS polynomials, the Symmetry Adapted PSD cone and the Symmetry Adapted Gram Spectrahedron. (Received February 26, 2020)

1158-14-263 **Margaret H Regan*** (mregan9@nd.edu), **Jonathan D Hauenstein**, **Tingting Tang**, **Edgar A Bernal** and **Dhagash Mehta**. *Machine learning the discriminant locus.*

Parameterized polynomial systems arise in many problems in science and engineering. The number of real solutions to these systems change as the parameters are varied with the boundaries forming the discriminant locus. Using techniques in numerical algebraic geometry, sample points perturbed from the discriminant locus are collected and labeled with the number of real solutions. These data sets are used as testing and training data for a machine learning algorithm that determines classification boundaries, where the classes are the different number of real solutions. We explore nearest neighbor and deep learning algorithms applied to various examples. In addition, we use the learned boundaries to perform parameter homotopy continuation to find the real solutions by tracking only real solution paths. Examples will be discussed to show the speed-up this technique provides. (Received March 02, 2020)

1158-14-268 **Dhagash Metha**, **Tianran Chen**, **Tingting Tang*** (ttang2@sdsu.edu) and **Jonathan Hauenstein**. *Understanding the impact of regularization on loss surface of deep linear network using algebraic geometry.*

Deep linear networks are neural networks with multiple hidden layers but no nonlinearity between layers. They are often used as toy models to used as toy models to understand the loss surface of deep neural networks. We explore properties of the optimization landscapes (loss surface) of the deep linear neural network models with a general L2 regularization term using numerical algebraic geometry. In particular, we establish upper bounds on the number of isolated stationary points of these networks with the help of algebraic geometry. Using these upper bounds and utilizing a numerical algebraic geometry method, we find all stationary points for modest depth and matrix size. We show that in the presence of the non-zero regularization, deep linear networks indeed possess local minima which are not the global minimum. (Received March 02, 2020)

1158-14-343 **Robert Krone*** (rckrone@ucdavis.edu). *Phylogenetic cycle network model dimensions.*

Group-based phylogenetic networks are a family of statistical models describing the evolutionary history of a set of related species. Correctly distinguishing the network structure requires understanding the geometry of these models. We describe the dimensions for most cycle networks and group structures, using tropical geometry. This dimension information allows us to classify which of these network features are identifiable and which are not, building off of the previous work of Gross and Long. (Received March 03, 2020)

15 ► Linear and multilinear algebra; matrix theory

1158-15-14 **Christina Pospisil*** (pospisil.christina@gmx.de). *Generalization Theory of Linear Algebra II.*

The talk continues the first part of the project Generalization Theory of Linear Algebra I (first part was presented at JMM 2019) and presents inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be applications to physics and other natural sciences be explored. (Received January 20, 2020)

1158-15-189 **Linda J Patton*** (lpatton@calpoly.edu), Mathematics Department, Cal Poly, San Luis Obispo, CA 93407, and **Mackenzie Cox**, **Weston Grewe**, **Grace Hochrein** and **Ilya Spitkovsky**. *Flat portions on the boundary of numerical ranges of 4-by-4 nilpotent matrices.* Preliminary report.

A 4-by-4 matrix has at most four flat portions on the boundary of its numerical range; three, if it is unitarily irreducible. It is easy to see that a unitarily reducible 4-by-4 nilpotent matrix has either one or no such portions, and it was conjectured by Gau and Wu in 2008 that there are at most two flat portions in the case of 4-by-4 unitarily irreducible nilpotent matrices. In our talk we will discuss the validity of this conjecture, which was proved by Militzer, Patton, Spitkovsky and Tsai [*Operator Theory: Advances and Applications*, **259**, 2017], and will also present further details on possible configurations of these portions. Construction of matrices having the respective shapes of these numerical ranges will be given as well. (Received March 01, 2020)

1158-15-381 **Youngseo Yoon***, Oklahoma City, OK , and **Amanda Kyung**, Demarest, NJ. *Study on a New Scheme to Enhance the Quality of Bio-image Using Inverse Fourier Transformation(IFT).*

The process of transformation from a frequency domain to an image domain requires lots of operation and time because Inverse Fourier Transformation(IFT) takes every frequency point to generate the final output image. However, if a relevant function is manipulated onto the frequency space, it will save a substantial amount of work time.

In this research, to remove ringing artifacts and improve the quality of a bio-image, mathematical and computational simulations were performed with several modified functions.

In the digital imaging process, when the domain of the proposed function over the frequency domain is narrow, it showed that the resolution of the final image was low due to the insufficient amount of frequency data from K-space. In this research, new filters as a new low pass filters were applied to sample bio-images and proved to save operation count and time.

By performing simulations with modified filters with a goal of determining an efficient filter that enhances bio-image resolution and reduces Ringing Artifact, this research shows how the proposed low pass filters improve the quality of the resolution of an image using mathematical, statistical and computational analysis. (Received March 04, 2020)

16 ► Associative rings and algebras

1158-16-38 **Jason Gaddis** and **Xingting Wang***, Department of Mathematics, Howard University, Washington, DC. *Zariski cancellation for Poisson algebras.*

We study the Zariski cancellation problem for Poisson algebras asking whether $A[x]$ is isomorphic to $B[y]$ implies A is isomorphic to B when A and B are Poisson algebras. We resolve this affirmatively in the cases when A and B are both connected graded Poisson algebras finitely generated in degree 1 without degree one Poisson central elements and when A is a Poisson integral domain of Krull dimension two with nontrivial Poisson bracket. We further introduce the Poisson analogues of Makar-Limanov invariant and discriminant to deal with the Zariski cancellation problem for some families of Poisson algebras. This is joint work with Jason Gaddis. (Received February 12, 2020)

1158-16-44 **Pablo Ocal*** (pso@math.tamu.edu). *On the Gerstenhaber bracket in relative Hochschild cohomology of associative algebras.*

We will begin by briefly presenting the inception of the Gerstenhaber bracket in Hochschild cohomology, which helped in capturing in an algebraic way the infinitesimal information stored in the cohomology of the algebra. On the first uses of the bracket by Gerstenhaber and Schack, they essentially claimed that everything that can be done on Hochschild cohomology can also be done in relative Hochschild cohomology. However, they required a separability condition to obtain relative projective resolutions when working with diagrams of algebras. This additional requirement motivates contextualizing our work to relative homological algebra. This is a less general context but it has multiple advantages: we can remove the separability condition, proofs are approachable, computations can be carried out, and an there is an interpretation of the bracket as a dg Lie algebra structure on a complex. We will also comment on recent results by Kaygun, who constructed a Jacobi-Zariski long exact sequence, and by Cibils, Lanzilotta, Marcos, Schroll, and Solotar, who described aspects of the Hochschild cohomology of bounded quiver algebras using relative cohomological tools. This strongly suggests that this context may be adequate for a better understanding of the cohomology of associative algebras. (Received February 13, 2020)

1158-16-52 **Xin Tang*** (xtang@uncfsu.edu), Math & Computer Science, Fayetteville State University, 1200 Murchison Road, Fayetteville, NC 28301, **James J. Zhang** (zhang@math.washington.edu), Dept. of Mathematics, Box 354350, University of Washington, Seattle, WA 98195, and **Xiangui Zhao** (zhaoxg@hzu.edu.cn), Dept. of Mathematics, Huizhou University, Huizhou, Guandong 516007, Peoples Rep of China. *Cancellation of Morita and Skew Types.*

There has been some interest in the study of various cancellation problems. In this talk, we will present a couple of results on the Morita cancellative and skew cancellative properties for noncommutative algebras; and we will explore what classes of noncommutative algebras are Morita cancellative (respectively, skew cancellative). Several new results concerning these two types of cancellations, as well as the classical cancellation, will be proved. This is joint work with James J. Zhang and Xiangui Zhao. (Received February 17, 2020)

1158-16-75 **Jon F Carlson** and **Srikanth B. Iyengar***, Department of Mathematics, University of Utah, 155 South 1400 East, Room 233, Salt Lake City, UT 84112-0090. *Cup products in the cohomology of local Hopf algebras.* Preliminary report.

Given a Hopf algebra A over a field k, the coproduct on A induces an action of the cohomology ring, $\mathrm{Ext}^*_A(k,k)$, of A on $\mathrm{Ext}^*_A(M,M)$, for any A-module M. In my talk, I will present some results about the (in)dependence of this action on the coproduct. This is based on join work with Jon Carlson. (Received February 20, 2020)

1158-16-93 **Alexandru Chirvasitu*** (achirvas@buffalo.edu), **Ryo Kanda** and **Paul S. Smith**. *Elliptic algebras.*

Feigin and Odesskii's generalizations $Q_{n,k}(E,\tau)$ of Sklyanin algebras are parametrized by an elliptic curve E, a point τ on it, and a pair of coprime integers $1 \leq k \leq n$. They have the same Hilbert series as the polynomial ring in n variables and hence are non-commutative analogues of projective spaces. Their "classical points" are parametrized by certain products of symmetric powers of E which can then be regarded as subschemes of $(n-1)$-dimensional projective space that "survive the deformation". The main results are that the algebras $Q_{n,k}(E,\tau)$ have the expected Hilbert series and global dimension and are Koszul for all choices of parameters. They are also AS-regular for generic choices of $\tau \in E$.

(joint with R. Kanda and S.P. Smith) (Received February 23, 2020)

1158-16-103 **Chelsea Walton**, **Elizabeth Wicks** and **Robert Won*** (robwon@uw.edu). *Algebraic structures in comodule categories over weak bialgebras.*

A weak bialgebra is both an algebra and a coalgebra satisfying compatibility conditions which are weaker than those defining a bialgebra. These weakened compatibility conditions still ensure that the categories of modules and comodules over a weak bialgebra admit a monoidal structure. We will discuss algebra objects and coalgebra objects in comodule categories over weak bialgebras, and their relationships to comodule algebras and comodule coalgebras. This work is joint with Chelsea Walton and Elizabeth Wicks. (Received February 25, 2020)

1158-16-104 **Zachary Cline** and **Jason Gaddis*** (gaddisj@miamioh.edu). *Actions of quantum linear spaces on quantum algebras.*

Due to the principle of quantum rigidity, noncommutative algebras exhibit few classical symmetries. That is, their graded automorphism group is small compared to their commutative counterparts. In the search for more symmetries, one is naturally led to the study of Hopf algebra actions. In this talk, I will focus on Taft algebras, their generalizations and higher-dimensional analogues. I will discuss a classification of linear of actions of quantum linear spaces on quantum affine spaces and quantum matrix algebras. (Received February 25, 2020)

1158-16-112 **Birge Huisgen-Zimmermann*** (birge@math.ucsb.edu) and **Ken Goodearl** (goodearl@math.ucsb.edu). *Families of representations determined by hypergraphs.* Preliminary report.

We consider directed hypergraphs in which each arrow has a single starting vertex, but may have multiple terminal vertices. These graphs pin down families of representations of path algebras modulo relations, and all representations of fixed dimension belong to one of a finite collection of such graph-induced families. We outline how structural properties of the modules which are thus tied together may be effectively analyzed with a mix of combinatorial and geometric tools. (Received February 25, 2020)

1158-16-113 **Birge Huisgen-Zimmermann*** (birge@math.ucsb.edu) and **Manuel Saorin** (msaorinc@um.es). *Dualities induced by tilting modules.*

Arbitrary tilting bimodules over finite dimensional algebras give rise to partial equivalences of the corresponding module categories. The main point of the talk is to supplement these thoroughly explored covariant functors

by dualities. In fact, we show that all dualities defined on certain types of representation-theoretically relevant subcategories of module categories are induced by tilting modules in this way. We discuss the impact of these dualities and outline applications. (Received February 25, 2020)

1158-16-115 **K. R. Goodearl*** (goodearl@math.ucsb.edu) and **S. Launois** (s.launois@kent.ac.uk). *Catenarity in Quantum Nilpotent Algebras.* Preliminary report.

A guiding principle in the study of quantized coordinate rings is the expectation that these algebras display properties which parallel those of their classical counterparts. We will discuss the important property of *catenarity*, namely that all saturated chains of prime ideals between any two fixed primes have the same length. It holds in the classical coordinate rings of affine varieties and has been extended to many quantized coordinate rings. Recently, catenarity has been established for the large class of quantum nilpotent algebras. (Received February 25, 2020)

1158-16-160 **Cris Negron*** (cnegron@email.unc.edu). *Support data for Hopf algebras via noncommutative hypersurfaces.* Preliminary report.

In recent work with J. Pevtsova, we develop an approach to support theory for Hopf algebras via noncommutative hypersurfaces. As a starting point, one considers a Hopf algebra u which admits a smooth deformation $U \to u$ by a Noetherian Hopf algebra U of finite global dimension. One uses this deformation to produce a rank variety for u which takes values in an associated projective space. Our work is inspired by earlier contributions of Avramov and Buchweitz, which concerned support for (commutative) local complete intersections. I will discuss some modular examples, functions on finite group schemes and Drinfeld doubles of infinitesimal group schemes, and also quantum groups over the complexes. I will discuss how one can use this hypersurface approach to address the tensor product property in certain "solvable" examples. (Received February 28, 2020)

1158-16-196 **Luigi Ferraro*** (ferrarl@wfu.edu), **Frank Moore** and **Desiree Martin**. *The Taylor Resolution over a Skew Polynomial Ring.*

Let I be a monomial ideal in the polynomial ring $R = k[x_1, \ldots, x_n]$ over a field k. In her thesis, Taylor introduced a complex which provides a multi-graded free resolution for R/I as an R-module. Later, Gemeda provided a differential graded structure on this complex while Avramov showed that this DG algebra admits a divided power structure. We generalize these results to monomial ideals J in a skew polynomial ring S. As an application we show that if one fixes the number of generators of the ideal J, then there are finitely many isomorphism classes for $\pi^{\geq 2}(S/J)$, where $\pi(S/J)$ is the homotopy color Lie algebra of S/J, an invariant which was introduced and studied by the first and last author in a different work. As a result it follows that there are finitely many possibilities for the Poincaré series of k over S/J, if the number of generators of J is fixed. (Received March 01, 2020)

1158-16-198 **Stephen Doty*** (doty@math.luc.edu), Mathematics & Statistics, Loyola University Chicago, Chicago, IL 60660. *Integral second fundamental theorem of invariant theory for partition algebras.*

(Joint work with Chris Bowman and Stuart Martin.) There is a Schur-Weyl duality for a tensor power of a vector space, regarded as a bimodule for a symmetric group and partition algebra. We prove that the kernel of the action of the group algebra of the symmetric group is a cell ideal with respect to the alternating Murphy basis. This provides an analogue of the second fundamental theory of invariant theory for the partition algebra. The results are valid over an arbitrary integral domain. In consequence, the centralizer algebras of the partition algebra are cellular, and we provide an explicit cellular basis. We also prove similar results for the "half" partition algebras. (Received March 01, 2020)

1158-16-215 **Susan J Sierra***, s.sierra@ed.ac.uk. *Chain conditions and growth in enveloping algebras of infinite-dimensional Lie algebras.*

Let W be the Witt algebra, the Lie algebra of derivations of the complex affine line. We show that, although the universal enveloping algebra $U(W)$ of W, has infinite GK-dimension, any proper quotient of it has polynomial growth: that is, $U(W)$ has *just infinite GK-dimension*. We prove similar statements for the enveloping algebras of several related Lie algebras. This is joint work with Natalia Iyudu.

In 2013 we proved with Walton that $U(W)$ is neither left or right noetherian by constructing a non-noetherian factor ring. This factor, however, satisfies the ascending chain condition on two-sided ideals. Our work with Iyudu provides supporting evidence for the conjecture that $U(W)$ itself has ACC on two-sided ideals, and if time permits we will discuss other evidence in favour of this conjecture. (Received March 02, 2020)

1158-16-243 **Peter Goetz*** (peter.goetz@humboldt.edu). *1-generation of section rings associated to certain inverse systems of projective schemes.* Preliminary report.

In seminal work Artin, Tate and Van den Bergh introduced the following fundamental construction. Let $\mathbb{K}$ be a field. Suppose that $\{Z_d \subset (\mathbb{P}^n)^{\times d}\}_{d\geq 1}$ is a family of closed subschemes such that

$$\pi_{1,d-1}(Z_d) \subset Z_{d-1} \quad \text{and} \quad \pi_{2,d}(Z_d) \subset Z_{d-1} \quad \text{for all } d,$$

where the $\pi_{i,j}$ are the obvious projection maps. Let $\mathcal{L}_d$ be the restriction of the invertible sheaf $\mathcal{O}(1,1\ldots,1)$ on $(\mathbb{P}^n)^{\times d}$ to Z_d. Define $B_0 = \mathbb{K}$ and $B_d = H^0(Z_d, \mathcal{L}_d)$ for $d \geq 1$. Then one can define a canonical multiplication map $B_i \otimes B_j \to B_{i+j}$ so that $B = \oplus B_d$ becomes a connected $\mathbb{N}$-graded $\mathbb{K}$-algebra.

We call B the *section ring associated to the family* $\{Z_d\}$. In this talk I will discuss a new theorem that gives several conditions that ensure that B is generated as a $\mathbb{K}$-algebra by B_1. If Z_d is the scheme of truncated point modules of length $d+1$ associated to an Artin-Schelter regular algebra of dimension 3, then B is isomorphic to a twisted homogeneous coordinate ring and we recover the theorem that B is 1-generated. (Received March 02, 2020)

1158-16-244 **Ian Christopher Lim*** (ian.lim@mavs.uta.edu) and **Michaela Vancliff** (mvancliff@gmail.com). *Quadratic Quantum $\mathbb{P}^3$s with a Linear Line Scheme.*

It is believed that quadratic Artin-Shelter regular (AS-regular) algebras of global dimension four (sometimes called quadratic quantum $\mathbb{P}^3$s) can be classified using a geometry similar to that developed in the 1980's by Artin, Tate, and Van den Bergh. Their geometry involved studying a scheme (later called the point scheme) that parametrizes the point modules over a graded algebra. The notion of line scheme (which parametrizes line modules) was introduced later by Shelton and Vancliff. It is known that "generic" quadratic AS-regular algebras of global dimension four have a finite point scheme and one-dimensional line scheme. We will discuss a family of quadratic AS-regular algebras of global dimension four where each member has a line scheme that is a union of lines. Moreover, we prove that if a quadratic AS-regular algebra A of global dimension four is an Ore extension of a quadratic AS-regular algebra B of global dimension three, then the point scheme of B is embedded in the line scheme of A. (Received March 02, 2020)

1158-16-266 **Daniel Rogalski*** (drogalski@ucsd.edu), **Robert Won** and **James Zhang**. *The Brown-Goodearl conjecture for weak Hopf algebras.*

Brown and Goodearl conjectured that any Noetherian Hopf algebra should have finite injective dimension. The conjecture is known to be true in some cases, in particular for PI Hopf algebras. Weak Hopf algebras are an important generalization of Hopf algebras in which the axioms on the unit and counit are weakened. Just as for Hopf algebras, the category of modules over a weak Hopf algebras has a monoidal structure, and this has important consequences for the homological properties of the algebra. We study the extension of the Brown-Goodearl conjecture to the case of weak Hopf algebras, and show that a weak Hopf algebra which is finite over an affine center has finite injective dimension, and is a direct sum of AS Gorenstein algebras. (Received March 02, 2020)

1158-16-270 **David J Benson**, **Srikanth B Iyengar**, **Henning Krause** and **Julia Pevtsova***, julia@math.washington.edu. *Support theory for Elementary supergroups.* Preliminary report.

Elementary supergroup schemes arise as a detecting family in the theory of supports for finite supergroup schemes. As such, they play a similar role to finite supergroup schemes as elementary abelian p-groups play for finite groups, as known from the classical work of Quillen and Chouinard. In this talk I'll describe the theory of varieties, the calculation of the Balmer spectrum and the Benson-Iyengar-Krause stratification for the singularity category of an elementary supergroup scheme. An interesting and novel feature of the theory is that it combines the π-point approach of Friedlander-Pevtsova with the hypersurface approach of Avramov-Iyengar as an attempt to construct Carlson's rank varieties in the super context. (Received March 02, 2020)

1158-16-272 **Andrew B. Conner*** (abc12@stmarys-ca.edu) and **Peter D. Goetz**. *Noncommutative Projective Geometry of Certain Twisted Tensor Products.* Preliminary report.

Let $T = k\langle x_0, \ldots, x_n\rangle$ denote the free associative k-algebra on generators of degree 1. Let $A = T/I$ be the quotient by a finitely-generated, homogeneous ideal I. For $d \geq 1$, let $Z_d \subset (\mathbb{P}^n)^{\times d}$ be the scheme of common zeros of elements of I_d, viewed as functions $(T_1^*)^{\otimes d} \to k$. From the geometric data of the schemes Z_d, one can define a ring structure on $B = \bigoplus_d H^0(Z_d, i^*\mathcal{O}_{(\mathbb{P}^n)^{\times d}}(1))$ where $i : Z_d \to (\mathbb{P}^n)^{\times d}$ is the inclusion.

If A is an Artin-Schelter (AS) regular algebra on three generators, then $Z_d \cong Z_2$ for all $d \geq 2$, and Z_2 is the is the graph of an automorphism σ on a scheme X. In this case, B is isomorphic to a twisted homogeneous

coordinate ring on the data $(X, \sigma, \mathcal{L})$ where $\mathcal{L}$ is an invertible sheaf. If A is not AS-regular, the sequence $\{Z_d\}$ need not stabilize.

In this talk we describe the schemes Z_d and the associated ring structure of B in the case where A is a quadratic twisted tensor product of $k[x, y]$ and $k[z]$. Such twisted tensor products were recently classified by the authors, and the classification includes both AS-regular and non-noetherian examples. (Received March 02, 2020)

1158-16-308 **Naomi L Krawzik*** (naomikrawzik@my.unt.edu). *Graded Hecke algebras arising from the symmetric group acting in positive characteristic.*

We discuss skew group algebras and their deformations. Deformations of skew group algebras have been well studied when the characteristic of the underlying field is zero. In that case, every Lusztig graded Hecke algebra is isomorphic to a Drinfeld Hecke algebra as filtered algebras. However, new algebras arise when the characteristic of the field divides the order of the group acting, specifically when the symmetric group acts on a polynomial ring in positive characteristic. We present a classification of the resulting graded Hecke algebras using PBW conditions. (Received March 03, 2020)

17 ► Nonassociative rings and algebras

1158-17-97 **Gregory P. Wene*** (gpwene2011@hotmail.com), 7342 Oak Manor Dr, APT 7109, San Antonio, TX 78229. *The Hypercube Semigroup.*

An m-dimensional hypercube A, $m \geq 1$, as an array of n^m elements belonging to a field; the elements are denoted by $A_{ij\ldots u}$ where there are m subscripts, and each varies from 1 to n. Knuth (J. Algebra 2 (1965), 182-217) introduced hypercubes as a generalization of the display of the structure constants of a finite semifield. We examine the semigroup formed by these under a binary composition suggested by Knuth in that article. (Received February 24, 2020)

1158-17-152 **Cody Tipton*** (cat1184@uw.edu). *Hopf actions on A_∞-algebras.*

We will talk about Hopf actions on A_∞ algebras, and some extensions of classical results to A_∞ algebras. (Received February 28, 2020)

1158-17-227 **Raphael Rouquier*** (rouquier@math.ucla.edu), -, -. *Representation categories as 2-representations.*

I will explain how 2-representation theory provides a description of some non-semisimple representation categories from more elementary algebraic data. (Received March 02, 2020)

18 ► Category theory; homological algebra

1158-18-4 **Shanna Nicole Dobson*** (sdobson@calstatela.edu), Shanna Dobson, 5151 State University Drive, Los Angeles, CA 90039. *Higher Categorical Traces on $Crys^r(X)^\heartsuit$.* Preliminary report.

Let X be a DG scheme which is locally almost of finite type. Following Gaitsgory and Rozenblyum, to each prestack (locally almost of finite type) Y, we assign a stable ∞-category Y Crys(Y) that has a left realization functor to QCoh(Y) and a right realization functor to IndCoh(Y). Using their results that $Crys^r(X)^\heartsuit$ is a Grothendieck abelian category that is locally Noetherian and that $Crys^r(X)$ has finite cohomological dimension with respect to its t-structure,we investigate higher categorical traces on $Crys^r(X)^\heartsuit$ using the trace functor terminology of Hayois. (Received February 12, 2020)

1158-18-25 **Yiby Morales**, **Monique Muller**, **Ana Ros Camacho**, **Julia Plavnik**, **Angela Tabiri** and **Chelsea Walton*** (notlaw@illinois.edu). *Algebras in group-theoretical fusion categories.*

Introduced by Pavel Etingof, Dmitri Nikshych, and Victor Ostrik (2005), "group-theoretical fusion categories" (GTFCs) are a certain kind of monoidal category whose construction depends on group-theoretic data. They are a vital part of the classification program of general fusion categories, and due to their explicit construction, they also serve as a go-to testing ground for results about fusion categories. The goal of this talk is to present results on the representation theory of GTFCs, which by work of Ostrik (2003), boils down to understanding algebraic structures in GTFCs. The main result is a construction of explicit Morita equivalence class representatives of indecomposable, separable algebras in GTFCs. This is joint work with Yiby Morales, Monique Müller,

Julia Plavnik, Ana Ros Camacho, and Angela Tabiri, and is available here: https://arxiv.org/abs/2001.03837. (Received February 07, 2020)

1158-18-31 **Nicolás Andruskiewitsch***, andrus@famaf.unc.edu.ar, and **Iván Angiono**, **Julia Pevtsova** and **Sarah Witherspoon**. *Cohomology rings of finite-dimensional pointed Hopf algebras over abelian groups.* Preliminary report.

It was asked by Friedlander and Suslin whether the cohomology ring of a finite-dimensional Hopf algebra is finitely generated. It will be explained our approach to answer affirmatively this question for finite-dimensional pointed Hopf algebras over abelian groups (in characteristic 0), extending work of several authors. (Received February 09, 2020)

1158-18-39 **Victor Ostrik*** (vostrik@uoregon.edu), Department of Mathematics, University of Oregon, Eugene, OR 97403. *Examples of symmetric tensor categories.*

I will report on joint work with D.Benson and P.Etingof. We construct new examples of non-semisimple abelian tensor categories over fields of positive characteristic and describe their properties. The construction is based on the theory of tilting modules for the group $SL(2)$. (Received February 12, 2020)

1158-18-81 **Daniel K. Nakano*** (nakano@math.uga.edu), **Kent B. Vashaw** and **Milen T. Yakimov**. *Noncommutative Tensor Triangular Geometry.*

In this talk, I will show how to develop a general noncommutative version of Balmer's tensor triangular geometry that is applicable to arbitrary monoidal triangulated categories (MΔCs). Insight from noncommutative ring theory is used to obtain a framework for prime, semiprime, and completely prime (thick) ideals of an MΔC, **K**, and then to associate to **K** a topological space–the Balmer spectrum Spc **K**. The novel feature of our approach is to define the noncommutative Balmer spectrum Spc **K** and support data for **K** in terms of tensoring of thick ideals of **K**, and not object-wise tensoring. (Received February 21, 2020)

1158-18-88 **Jade Master*** (jmast003@ucr.edu), 1181 Minerva Ct, Riverside, CA 92507. *Generalized Petri Nets.*

We give a definition of Q-Net; a generalization of Petri nets based on a Lawvere theory Q for which many existing variants of Petri nets are a special case. This definition is functorial with respect to change in Lawvere theory and we exploit this to explore the relationships between different kinds of Q-nets. To justify our definition of Q-net, we construct a family of adjunctions for each Lawvere theory explicating the way in which Q-nets present free models of Q in Cat. This gives a functorial description of the operational semantics for an arbitrary category of Q-nets. We show how this can be used to construct the semantics for Petri nets, pre-nets, integer nets, and elementary net systems. (Received February 22, 2020)

1158-18-92 **Christian Williams*** (williams@math.ucr.edu), 720 El Cerrito Drive, Unit 10, Riverside, CA 92507. *Predicate Calculus for Algebraic Type Theory.* Preliminary report.

An algebraic theory T is a category which encapsulates a form of algebraic structure. We present a notion of predicate for algebraic theories, which admits a calculus of both logical and algebraic operations. We thereby extend equational logic by first-order logic, and provide a natural type theory for algebraic structures. The process can be understood as providing the theory T with a polymorphic type system.

We demonstrate the idea with the theory of monoids, and derive the example predicate "prime". This and many algebraic examples are useful only when they are mapped from a theory into models. However our motivation is programming languages: we focus on applying the idea to a more general notion of theory with variable binding.

The motivation of this work is a logic for concurrency, known as Namespace Logic. This applies to the reflective higher-order π calculus, which is the language of the distributed computing platform RChain. We demonstrate the predicate calculus by constructing Namespace Logic. This gives a general framework for logic computation, and provides a glimpse into a large field of potential application. (Received February 23, 2020)

1158-18-96 **Mohsen Gheibi*** (mohsen.gheibi@uta.edu), Department of Mathematics, 411 S. Nedderman Drive, Pickard Hall 445, Arlington, TX 76019, **David A. Jorgensen** (djorgens@uta.edu), Department of Mathematics, 411 S. Nedderman Drive, Pickard Hall 429, Arlington, TX 76019, and **Ryo Takahashi** (takahashi@math.nagoya-u.ac.jp), Graduate School of Mathematics, Nagoya University, Nagoya, 464-8602, Japan. *Quasi-Projective Dimension.*

In this talk, I will introduce a homological invariant namely quasi-projective dimension, which is a generalization of the projective dimension. I will discuss the basic properties of the quasi-projective dimension and compare

it with other homological dimensions. In particular, I will show that the modules with finite quasi-projective dimension, in many cases, behave similarly as modules of finite complete intersection dimension. I also will address some open problems about the quasi-projective dimension. (Received February 24, 2020)

1158-18-175 **Zbigniew Oziewicz*** (oziewicz.zbigniew@gmail.com), Universidad Nacional Autonoma de Mexico, Facultad de Estudios Superiores, C.P. 54715 Cuautitlan Izcalli, Mexico, **Hilda Maria Colin Garcia** (hildacolin@gmail.com), Facultad Estudios Superiores Cuautitlan, Universidad Nacional Autonoma de Mexico, C.P. 54715 CUAUTITLAN IZCALLI, Mexico, and **William Stewart Page** (bill.page@newsynthesis.org), 6148 Temperance Lake Road, Ontario K0E 1B0, Athens, Canada. *Groupoid category how is distinct from a group category.*

Heinrich Brandt's groupoid is distinct from the group. Groupoid category is distinct from the group as a category. Within the group as a category the composition of the inverse morphism from the left and from the right, with the given morphism, are always equal. Whereas these compositions are never equal in a groupoid category. Moreover every morphism in a group-category possesses all powers, element of a group can be composed with themself many times. Within the groupoid category this is impossible, does not exists the power of a groupoid morphism. These most important facts are ignored in 22-pages 'Introduction to groupoids' by Piotr Stachura in Journal of Knot Theory June 2018. Iso-morphism of groupoid category can never be endo-morphism - must not be presented as a matrix. Metric tensor is an iso-morphism in a groupoid category, thus neither Minkowski metric, nor any other metric tensor must be understand as a matrix, possessing diagonal elements, therefore the condition (3.6) in 1841005-paper in Journal of Knot Theory need to be re-discussed. The same remarks concern symplectic tensor for which some Author use matrix notation in Journal of Knot Theory June 2018. Symplectic tensor is an iso-morphism in a groupoid category, and do not possess minimal polynomial. (Received March 04, 2020)

1158-18-186 **Daniel Nakano**, **Kent Vashaw*** (kvasha1@lsu.edu) and **Milen Yakimov**. *Noncommutative tensor triangular geometry.*

We describe a general theory of the prime spectrum of non-braided monoidal triangulated categories. These notions are a noncommutative analogue to Paul Balmer's prime spectra of symmetric tensor-triangulated categories. Noncommutative monoidal triangulated categories appear naturally as stable module categories for non-quasitriangular Hopf algebras and as derived categories of bimodules of noncommutative algebras. In stable module categories of Hopf algebras, the support theory of the category, as described by Benson-Iyengar-Krause, is linked to the Balmer spectrum, which is shown to be the final support datum. We will describe how this connection can be used to compute Balmer spectra in general, and we will compute the Balmer spectra for stable module categories of the small quantum group of a Borel subalgebra at a root of unity, and the stable module categories for smash coproduct Hopf algebras of group algebras with coordinate rings of groups. This is joint work with Daniel Nakano and Milen Yakimov. (Received March 01, 2020)

1158-18-188 **Beren Sanders*** (beren@ucsc.edu). *Finite étale algebras belong in the heart.*

In the context of tensor triangular geometry, understanding the finite étale morphisms out of a tensor triangular category amounts to classifying the finite étale algebras (a.k.a. compact tt-rings) in the category. For an equivariant tensor triangulated category, restricting to a finite-index subgroup is an example of such a finite étale morphism. The question of whether these restriction functors are the only finite étale morphisms, say on the stable module category of a finite group, remains an open question in general. In this talk I will discuss recent work, joint with Paul Balmer, which addresses these questions in the case when your category is equipped with a weight structure. Our theorem reduces the problem to classifying the finite étale algebras in the heart of the weight structure. It provides a new perspective on the fact (due to John Rognes) that the sphere spectrum is separably closed, and also has applications to understanding the étale topology of the equivariant stable homotopy category. The latter was our original motivation. (Received March 01, 2020)

1158-18-224 **Kent Vashaw*** (kvasha1@lsu.edu) and **Milen Yakimov**. *Prime spectra of abelian 2-categories and categorification of Richardson varieties.*

Balmer's tensor triangular geometry provides a powerful tool for studying symmetric tensor-triangulated categories. We will describe a generalization of this theory to the context of abelian 2-categories, and discuss the application of this theory to algebraic categorification. The Serre prime spectrum of a 2-category is linked to the set of Serre primes of its Grothendieck ring. We construct a categorification of the quantized coordinate rings of Richardson varieties for symmetric Kac-Moody groups, by constructing Serre completely prime ideals of monoidal categories of modules of KLR algebras and taking Serre quotients with respect to them. These

Serre completely prime ideals are constructed by analysis of the canonical bases of quantum groups and the compatibility of these bases with the quantized Richardson varieties. This is joint work with Milen Yakimov. (Received March 02, 2020)

1158-18-241 **Pinches Dirnfeld*** (dirnfeld@math.utah.edu). *Complexes of Injectives Over Commutative Noetherian Rings.*

M. Hopkins and A. Neeman have classified the lattice of thick subcategories of the category of perfect complexes over a commutative Noetherian ring R, in terms of specialization closed subsets of $\mathrm{Spec}(R)$. This talk will be about the corresponding problem for bounded complexes of injective modules with finitely generated cohomology. (Received March 02, 2020)

1158-18-297 **James Cunningham Cameron*** (jcameron@math.ucla.edu). *Homological residue fields and tensor triangulated fields.* Preliminary report.

There has been recent work on using abelian avatars of tensor triangulated fields to study tensor triangulated categories in situations where tensor triangulated residue fields lack good properties or are not known to exist. These abelian categories are the homological residue fields. In the stable module category of a finite group group scheme tensor triangulated residue fields do exist and are given by π-points, but π-points lack a few of the properties that one might want from a residue field functor.

In this talk I will discuss the relationship between π-points and the abelian homological residue fields for the stable module category of a finite group scheme. This is joint work in progress with Paul Balmer. (Received March 03, 2020)

1158-18-347 **Nicolle Gonzalez*** (nicolle@math.ucla.edu) and **Matt Hogancamp**. *A skein-theoretic $A_{q,t}$ algebra.* Preliminary report.

The $A_{q,t}$ algebra was originally introduced by Carlsson and Mellit in their proof to the shuffle conjecture. This algebra arises as an extension of the affine Hecke algebra by certain creation and annihilation operators and has a natural action on a family of polynomial rings. In this talk I will discuss ongoing work entailing a skein-theoretic formulation of the $A_{q,t}$ algebra specialized at $t = q^{-1}$ that recovers the original algebraic formulation and lays the groundwork for a categorical analogue. (Received March 03, 2020)

20 ► Group theory and generalizations

1158-20-16 **Alfred Geroldinger*** (alfred.geroldinger@uni-graz.at), Institute for Mathematics and Scientific Comp, University of Graz, Heinrichstrasse 36, 8010 Graz, Austria. *On strongly primary monoids and domains.*

A (commutative and cancellative) monoid H is strongly primary if for each non-invertible element $a \in H$ there is $n \in \mathbb{N}$ such that $\mathfrak{m}^n \subset aH$, where $\mathfrak{m}$ is the nonempty set of non-invertible elements of H. Numerical monoids are strongly primary and the same is true for large classes of Puiseux monoids (Puiseux monoids are additive submonoids of the nonnegative rational numbers). Furthermore, the multiplicative monoid of nonzero elements of a one-dimensional local noetherian domain is strongly primary.

We provide a survey on strongly primary monoids and their arithmetic (including their sets of lengths, sets of distances, and catenary degrees). New results stem from joint papers with Felix Gotti, Moshe Roitman, and Salvatore Tringali. (Received February 01, 2020)

1158-20-27 **Christopher M Drupieski*** (c.drupieski@depaul.edu) and **Jonathan R Kujawa** (kujawa@math.ou.edu). *Support varieties and modules of finite projective dimension for modular Lie superalgebras.*

Let $\mathfrak{g}$ be a finite-dimensional Lie superalgebra over an algebraically closed field k of characteristic $p \geq 3$. Using the cohomology ring of $\mathfrak{g}$, one can define for each finite-dimensional $\mathfrak{g}$-supermodule M a corresponding affine algebraic variety $X_{\mathfrak{g}}(M)$, called the cohomological support variety of M. In this talk I'll discuss recent work with Jonathan Kujawa in which we give an explicit 'rank variety' description for $X_{\mathfrak{g}}(M)$ as a subset of $X_{\mathfrak{g}} = \{x \in \mathfrak{g}_{\overline{1}} : [x, x] = 0\}$, the odd nullcone of $\mathfrak{g}$. A key tool in our approach is the Clifford filtration on $\mathfrak{g}$, which helps us to compare $X_{\mathfrak{g}}(M)$ to the support variety of a certain related unipotent finite supergroup scheme. As corollaries of our main result, I'll discuss a characterization of when a finite-dimensional $\mathfrak{g}$-supermodule is of finite projective dimension, and a positive characteristic analogue of a theorem of Bøgvad concerning a criterion for when the enveloping algebra of $\mathfrak{g}$ is of finite global dimension. (Received February 08, 2020)

1158-20-28 **Cris Negron*** (cnegron@email.unc.edu). *Cohomology for Drinfeld doubles of arbitrary finite group schemes.* Preliminary report.

I will discuss recent work concerning cohomology for Drinfeld doubles of finite group scheme. (The category of representations for the Drinfeld double of a given group scheme G is identified with the category of G-equivariant sheaves on G, under the adjoint action.) We show that the algebra of self-extensions of the unit, i.e. the trivial representation for the double, forms a finitely generated algebra, and that for any representation V of the double, extensions from the unit to V form a finitely generated module over this algebra. The project completes earlier work of E. M. Friedlander and the speaker, in which the given results were established for doubles of Frobenius kernels in smooth algebraic groups. (Received February 08, 2020)

1158-20-48 **Aaron N. Brookner*** (brookner@math.berkeley.edu). *On the Problem of Extension of Coalgebras to Bialgebras.*

Over a fixed base-field k, a bialgebra is a vector space which is an algebra, a coalgebra, and which satisfies some compatibility axioms. We present a sequence of coalgebras, all defined over the integers (as [co-]modules). And now varying k, we find the question of whether a coalgebra admits a bialgebra structure, and whether uniquely (even up to isomorphism), depends interestingly yet elementarily on the characteristic and algebraicity of k. We interpret one of the results both as a statement about trigonometry, and as a Morita equivalence between comodule categories. Finally we discuss ongoing work, involving the reinterpretation of the above results from an algebra-geometric perspective, and a new example of a bialgebra(-object) based on elementary trigonometry. (Received February 16, 2020)

1158-20-66 **Nicholas R Baeth*** (nicholas.baeth@fandm.edu), Department of Mathematics, Franklin & Marshall College, Lancaster, PA 17604. *Multiplicative factorization in numerical semigroups.* Preliminary report.

Though usually studied as an additive structure, each numerical semigroup is closed under multiplication and, with respect to this less-studied operation, is arithmetically interesting. With respect to multiplication, other than the trivial $\mathbb{N}$, numerical semigroups are non-Krull C-monoids. In this talk we classify the multiplicative atoms of numerical semigroups and give some characterization of their arithmetic. (Received February 19, 2020)

1158-20-121 **Noah White*** (noah@math.ucla.edu). *Schubert intersections and tableaux.*

The Littlewood-Richardson numbers famously count many different objects. Two classical examples are points in intersections of Schubert varieties in Grasmannians and certain tableaux. A natural question is whether one can label the points in the intersection of Schubert varieties by tableaux in some canonical way. I will explain exactly what this means, and a simple procedure to give such a labelling. I will also explain how this labelling arises from the representation theory of the symmetric group. (Received February 26, 2020)

1158-20-182 **Kurt D Herzinger***, 2354 Fairchild Dr Ste 6D234, Usaf Academy, CO 80840-6208. *Numerical Semigroups and the Game of Sylver Coinage.*

Sylver coinage is a two-player game played on the natural numbers. Players take turns naming numbers and eliminating from play any number that can be expressed as a sum of the numbers which have been named. The player who is forced to choose the number"1" loses the game. After a finite number of turns, the set of numbers available for play form the complement of a numerical semigroup. We will examine winning and losing positions in the game and how they relate to numerical semigroups. (Received March 01, 2020)

1158-20-204 **Robert Boltje*** (boltje@ucsc.edu). *Functorial structures in the representation theory of finite groups.*

About 25 years ago, building on the theory of Mackey functors, Serge Bouc developed a theory of so-called biset functors, noting that natural operations — as restriction, induction, inflation, deflation, and transport of structure via a group isomorphism — are all defined by tensoring with bimodules that arise as linearizations of bisets. This talk is meant as an introduction (with many examples) to Bouc's theory of biset functors and the notion of fibered biset functors (recently introduced together with Olcay Çoskun) which includes as operations the twist with one-dimensional modules. We conclude with some results on fibered biset functors. (Received March 01, 2020)

1158-20-206 **George J McNinch*** (george.mcninch@tufts.edu), Dept Mathmetics, Tufts University, 503 Boston Avenue, Medford, MA 02144. *Nilpotent elements and reductive subgroups over a local field.*

Let K be a *local field* – i.e. the field of fractions of a complete DVR A whose residue field k has characteristic $p > 0$ – and let G be a connected, absolutely simple algebraic K-group G which splits over an unramified

extension of K. We study the rational nilpotent orbits of G– i.e. the orbits of $G(K)$ in the nilpotent elements of $(G)(K)$ – under the assumption $p > 2h - 2$ where h is the Coxeter number of G.

A reductive group M over K is *unramified* if there is a reductive model $\mathcal{M}$ over A for which $M = \mathcal{M}_K$. We have shown for any nilpotent element $X_1 \in \mathrm{Lie}(G)$ that there is an unramified, reductive K-subgroup M which contains a maximal torus of G and for which $X_1 \in \mathrm{Lie}(M)$ is *geometrically distinguished*.

The talk will describe this result, and provide some context and examples. (Received March 01, 2020)

1158-20-274 **Subhadip Dey*** (sdey@math.ucdavis.edu) and **Michael Kapovich**. *Stein property of complex-hyperbolic Kleinian groups.*

Let M be a complex-hyperbolic n-manifold, i.e., a quotient of the complex-hyperbolic n-space $\mathbb{H}^n_{\mathbb{C}}$ by a torsion-free discrete group of isometries $\Gamma = \pi_1(M)$. We assume that M is *convex-cocompact*: This means that the convex core of M is a nonempty compact subset. In this talk, we will discuss a sufficient condition on Γ in terms of the growth-rate of its orbits in $\mathbb{H}^n_{\mathbb{C}}$ for which the complex manifold M is Stein. If time permits, we will also talk about some closely related questions. (Received March 03, 2020)

1158-20-285 **Cyane Gonzalez***, Department of Mathematics, 5245 North Backer Avenue M/S PB 108, California State University, Fresno, Fresno, CA 93740, and **Laura Marroquin**, **Maria S. Nogin** and **Taylor W. Yates**. *Twists and Turns: Learning Permutations through Rubik's Cubes.*

Did you know that a Rubik's cube and group theory have a lot in common? A Rubik's cube and its variations are interesting little toys that hide a complex world of cycles and permutations. In this talk we will share the secrets of Rubik's cubes and what it takes to solve them. Our Math Circle participants love sharing their algorithms for solving the cubes, including those that are not cubes at all, and creating various patterns. They ask questions, such as is it possible to just switch two corner pieces? What other configurations can or cannot be achieved? With a basic introduction to permutations, we can answer a lot of their questions. What a great incentive to learn about the parity of a permutation, cycles, inverses, commutators, and much more! (Received March 03, 2020)

1158-20-294 **James Cunningham Cameron*** (jcameron@math.ucla.edu). *Stratification for cochain algebras on Borel constructions of G-spaces.* Preliminary report.

For G a compact Lie group and X a finite G-CW complex I will discuss how the the Borel equivariant cohomology ring of X with $\mathbb{F}_p$ coefficients controls the structure of the derived category of the cochain algebra of the Borel construction. on X.

I will also indicate how generalizing from finite groups to compact Lie groups and G-spaces allows one to study some of the categories that appear in modular representation theory via categories that have structure that doesn't appear at the purely algebraic level. (Received March 03, 2020)

1158-20-305 **Scott Chapman*** (scott.chapman@shsu.edu), Department of Mathematics and Statistics, Huntsville, TX 77341, and **Felix Gotti** (felixgotti@berkeley.edu) and **Marly Gotti** (marlycormar@ufl.edu). *Factorization invariants of Puiseux monoids generated by geometric sequences.*

We study some of the factorization invariants of the class of Puiseux monoids generated by geometric sequences, and we compare and contrast them with the known results for numerical monoids generated by arithmetic sequences. The class we study here consists of all the atomic monoids of the form $S_r := \langle r^n \mid n \in \mathbb{N}_0 \rangle$, where r is a positive rational. As the atomic monoids S_r are nicely generated, we are able to give detailed descriptions of many of their factorization invariants. One distinguishing characteristic of S_r is that all its sets of lengths are arithmetic sequences of the same distance $|a - b|$, where $a, b \in \mathbb{N}$ are such that $r = a/b$ and $\gcd(a, b) = 1$. We prove this, and then use it to study the elasticity and tameness of S_r. (Received March 03, 2020)

1158-20-323 **David B Cohen**, **Chaim Goodman-Strauss** and **Yo'av Rieck*** (yoav@uark.edu), Department of Mathematical Sciences, University of Arkansas, Fayatteville, AR 72701. *Strongly aperiodic subshifts of finite type on hyperbolic groups.*

A *subshift of finite type* (SFT) on a finitely generated group is a set of local rules for distributing labels (out of a finite set) on the Cayley graph in a manner that is invariant under the group action. In his 1987 paper defining hyperbolic groups, Gromov constructed an SFT on any hyperbolic group and used it to study the dynamics of its action on the sphere at infinity.

An SFT is *strongly aperiodic* (SA) if no two group elements have identical neighborhoods of all sizes; in a way, the labels form an "address book" for the group. (This concept is similar to strongly aperiodic tiling on the group.) Many groups are known to admit an SA SFT, and the only known two obstructions are: undecidablity

of the word problem, and having more than one end. However, no infinite
hyperbolic
group was known to admit an SA SFT until Cohen and Goodman-Strauss (2017) showed existence for surface groups.

We show that any one-ended hyperbolic group admits a SA SFT. (Received March 03, 2020)

30 ► Functions of a complex variable

1158-30-55 **Summer Al-Hamdani*** (summera@mail.fresnostate.edu). *On Binomial Combinations of Chebyshev Polynomials.*

The sequence of Chebyshev polynomials of the second kind $\{U_m(z)\}_{m=0}^{\infty}$ is a well-known sequence of orthogonal polynomials whose zeros lie on the interval $(-1,1)$ and are dense there as $m \to \infty$. In this talk, we study the zero distribution of a sequence of polynomials $\{P_m(z)\}_{m=0}^{\infty}$ formed by linear combinations of Chebyshev polynomials whose coefficients are of binomial type. In particular, we prove that the probability the zeros of $P_m(z)$ lie on $(-1,1)$ approaches 100% as $m \to \infty$. (Received February 17, 2020)

1158-30-70 **Yury Grabovsky** and **Narek Hovsepyan*** (hovnarek@yahoo.com). *On feasibility of extrapolation of complex electromagnetic permittivity functions using Kramer-Kronig relations.*

As a function of frequency, the complex electromagnetic permittivity has analytic extension into the upper half-plane. It has further physical properties such as positive imaginary part and reflection symmetry about the imaginary axis. Such a function can be measured in a band of frequencies and one would like to extrapolate to a wider band of frequencies, using its analyticity. How reliable such extrapolation procedures can possibly be? We reformulate this question in terms of stability of analytic continuation of Hardy functions. The latter is then reduced to a solution of a linear integral equation of Fredholm type, which can be solved numerically, leading to a quantification of uncertainty of any extrapolation procedure. (Received February 20, 2020)

1158-30-82 **Marc Chamberland***, 1116 8th Ave., Grinnell, IA 50112. *When are a Polynomial's Zeros all Real and Distinct?*

There are many algebraic and analytic results concerning the location of zeros of single-variable polynomials. This note provides necessary and sufficient conditions to determine whether the zeros of a polynomial are real and distinct. The conditions are a finite set of inequalities that depend only on the given polynomial and its derivatives. After laying out these results, we explore extending these to entire functions. (Received February 21, 2020)

1158-30-123 **Khang Tran*** (khangt@mail.fresnostate.edu), Department of Mathematics, California State University, Fresno, 5245 North Backer Avenue M/S PB108, Fresno, CA 93740-0001. *Sequence of hyperbolic polynomials and its generating function.*

We study the reality of zeros of various sequences of polynomials generated by a bivariate generating function $G(t,z)$. In particular, if we expand a bivariate rational function $G(t,z)$ where $1/G(0,0) \neq 0$, as a power series in t, the coefficients could form a sequence of hyperbolic polynomials in z. In this case, we also look at the real interval where the union of all the sets of zeros of polynomials in this sequence is dense. (Received February 26, 2020)

1158-30-124 **Jack Quoc-Viet Luong*** (jackaham_luongcoln@mail.fresnostate.edu), 5245 North Backer Avenue M/S, Fresno, CA 93740, and **Khang Tran**. *Zeros of polynomials generated by the number of rook paths.*

Bivariate recurrence relations are of interest due to their application in various combinatorial problems. For example, they can be used to count the number of rook paths from one corner of an infinite chessboard to another corner. In this talk, we study a table of polynomials obtained from the bivariate recurrence relation $H_{m,n} + H_{m-1,n} + H_{m,n-1} + zH_{m-1,n-1} = 0$ with initial conditions $H_{0,0} = 1, H_{-1,n} = H_{m,-1} = 0$ for all natural numbers m and n. Equivalently, this table of polynomials is generated by $\sum_{m=0}^{\infty}\sum_{n=0}^{\infty} H_{m,n}t^m s^n = 1/(1+s+t+zst)$. We show that all zeros of any polynomial in this table are real. We also consider the more general class of recurrence relations generated by $R(s,t)/(1+s+t+zst)$ where $R(s,t)$ is a polynomial in s and t. In particular, we give an upper bound that depends only on $R(s,t)$ on the number of complex zeros of $H_{m,n}$. (Received February 26, 2020)

1158-30-142 **Zair Ibragimov*** (zibragimov@fullerton.edu), Fullerton, CA 92831. *Transfinite bi-metrics.*

It is known that one-point, hyperbolic-type metrics are not Gromov hyperbolic. These metrics are defined as a supremum of one-point metrics (i.e., metrics constructed using one boundary point) and the supremum is taken over all boundary points. In a recent joint work with Asuman Aksoy and Wesley Whiting we proved that taking the average of the one-point metrics instead of their supremum yields a Gromov hyperbolic metric. Moreover, the Gromov hyperbolicity constant of the resulting metric does not depend on the number of boundary points used in taking the average. In this talk we extend the idea of averaging metrics to the notion of trans finite bi-metrics. In some details we will discuss transfinite Apollonian metric and specific examples of domains where the transfi nite Apollonian metric can be computed explicitly. (Received February 27, 2020)

1158-30-171 **Rikard Bøgvad**, **Innocent Ndikubwayo*** (innocent@math.su.se) and **Boris Shapiro**. *Generalizing Tran's Conjecture.*

In 2014, Khang Tran conjectured that, for an arbitrary pair of polynomials $A(z)$ and $B(z)$, if $z_0 \in \mathbb{C} \setminus \{$ zeros of $A(z)\}$ is a zero of some polynomial in the sequence $\{P_n(z)\}_{n=1}^{\infty}$ generated by the three-term recurrence relation of length k

$$P_n(z) + B(z)P_{n-1}(z) + A(z)P_{n-k}(z) = 0$$

with standard initial conditions $P_0(z) = 1$, $P_{-1}(z) = \cdots = P_{-k+1}(z) = 0$, then z_0 lies on the real algebraic curve $\mathcal{C} \subset \mathbb{C}$ given by

$$\Im\left(\frac{B^k(z)}{A(z)}\right) = 0 \quad \text{and} \quad 0 \le (-1)^k \Re\left(\frac{B^k(z)}{A(z)}\right) \le \frac{k^k}{(k-1)^{k-1}}.$$

In this talk, I will generalize Tran's Conjecture and prove existence of such a real algebraic curve. This is based on a joint work with R. Bøgvad and B. Shapiro. (Received February 29, 2020)

1158-30-260 **Bogdan Ion***, Department of Mathematics, 301 Thackeray Hall, University of Pittsburgh, Pittsburgh, PA 15260. *Bernoulli polynomials and Dirichlet series.*

For a given sequence one can associate a power series and a Dirichlet series. We investigate the relationship between possible singularities that appear went we analytically continue both of these series. The most basic case, when the power series has a pole singularity at $z = 1$ is analyzed in detail by employing some (infinite order) discrete derivative operator (associated to the power series) that we call Bernoulli operator. Its main property is that it naturally acts on the vector space of analytic functions in the plane (with possible isolated singularities) that fall in the image of the Laplace-Mellin transform (for the variable in some half-plane). The action of the Bernoulli operator on the function t^s, provides the analytic continuation of the associated Dirichlet series and also detailed information about the location of poles, their resides, and special values. Using examples of arithmetic origin, I will attempt to illustrate what is reasonable to expect when the power series has a non-pole singularity at $z = 1$. (Received March 02, 2020)

1158-30-269 **Andrzej Piotrowski***, 11066 Auke Lake Way, Dept. of Natural Sciences, M/S SOB1, Juneau, AK 99801. *Quantifying Real Zeros of Polynomials via Curve Theorems.* Preliminary report.

In some cases, the real zeros of a polynomial (or a sequence of polynomials) can be ascertained as the intersection of a particular curve with a related real algebraic variety. In 1916, G. Pólya proved a result along these lines and provided a geometric link between the Hermite-Poulain Theorem, Schur's Theorem, and the transcendental characterization of multiplier sequences. In 2007, the author strengthened a result of G. Pólya's to obtain the first characterization of multiplier sequences for Hermite expansions of polynomials. The overarching ideas behind this curve theorem and its applications will be given, along with some conjectures on how to further extend G. Pólya's result. In addition, we will discuss how intersecting curves can be used to re-prove a result of K. Tran, which demonstrates that the zeros of the polynomials H_k must all be real, where the polynomials H_k are defined by $1/(P(t) + zt^r) = \sum H_k(z)t^k$ for some quadratic polynomial P and $r \in \{1, 2\}$. Furthermore, our approach will show that the zeros of H_k and H_{k+1} are interlacing for all k. The latter result came about as part of a collaboration with D. A. Cardon and T. Forgács. (Received March 02, 2020)

1158-30-304 **Rajesh Pereira*** (pereirar@uoguelph.ca), Department of Mathematics and Statistics, University of Guelph, 50 Stone Road E., Guelph, Ontario N1G 2W1, Canada. *Jensen's formula and the weak log majorization order.*

Jensen's formula is a key tool in the study of zero sets of entire functions. We will explore a connection between Jensen's formula and the weak log majorization order. In addition to exploring some new results on entire

functions, we will show that several historical results can be reinterpreted in terms of the weak log majorization order. (Received March 03, 2020)

1158-30-357 **Kourosh Tavakoli*** (ktavakoli@okcu.edu). *Some Infinite Compositions of Holomorphic Functions.*

In this research, we studied the geometric and analytic behavior of some compositions of holomorphic functions. We present Several interesting properties. (Received March 03, 2020)

33 ► Special functions

1158-33-129 **Karl Dilcher*** (dilcher@mathstat.dal.ca), Department of Mathematics and Statistics, Dalhousie University, Halifax, NS B3H4R2, Canada, and **Kenneth B. Stolarsky** and **Maciej Ulas**. *Some polynomial sequences whose zeros converge to algebraic curves.*

I will present three sequences of polynomials that arise naturally from questions in number theory and special functions. They all have in common that they have fairly easy explicit forms, are in some ways related to the Chebyshev polynomials, and have certain algebraic curves as limit curves of their zeros. In spite of these common features, the limit curves are quite different from each other. (Received February 27, 2020)

1158-33-138 **David A Cardon*** (cardon@mathematics.byu.edu), Department of Mathematics, TMCB 275, Brigham Young University, Provo, UT 84602. *Interlacing properties of coefficient polynomials in differential operator representations of real-root preserving linear transformations.*

We study linear transformations $T\colon \mathbb{R}[x] \to \mathbb{R}[x]$ of the form $T[x^n] = P_n(x)$ where $\{P_n(x)\}$ is a real orthogonal polynomial system. In particular, we are interested in linear transformations that map polynomials with all real zeros to polynomials with all real zeros. It is well known that any transformation $T\colon \mathbb{C}[x] \to \mathbb{C}[x]$ has a differential operator representation $T = \sum_{k=0}^{\infty} \frac{Q_k(x)}{k!} D^k$. We seek to understand the behavior of the transformation T by studying the roots of the $Q_k(x)$. We prove three main things. First, we show that the only case where the $Q_k(x)$ are constant and $\{P_n(x)\}$ are an orthogonal system is that when the orthogonal system is a shifted set of generalized probabilist Hermite polynomials. Second, we show that the coefficient polynomials $Q_k(x)$ have real roots when the $P_n(x)$ are the physicist Hermite polynomials or the Laguerre polynomials. Lastly, we show that in these cases, the roots of successive polynomials strictly interlace, a property that has not yet been studied for coefficient polynomials. We conclude by presenting several open problems. (Received February 27, 2020)

1158-33-377 **Jane McDougall***, Dept of Math and Computer Science, Colorado College, 14 E Cache La poudre, Colorado Springs, CO 80904. *Hypergeometric functions, their symmetries, and applications to harmonic mappings.*

We consider modifications of power functions z and z^{n-1} which include factors that are ${}_2F_1$ Gauss hypergometric functions. The particular pairs of functions studied here become "affine rotations" of one another and also possess reflectional symmetry. Moreover they can be utilized to form new examples of univalent harmonic functions (harmonic mappings). In this talk we explore these pairs of ${}_2F_1$ functions and describe conditions under which they lead to new harmonic mappings, and even new minimal surfaces. (Received March 04, 2020)

34 ► Ordinary differential equations

1158-34-13 **EKA OCHE OGBAJI*** (ogbajieka@yahoo.com) and **A Edu**. *Mathematical Model for Market Competition in the Telecommunication Industries in Nigeria.* Preliminary report.

This model was formulated by using some basic assumptions and the method of linearization was used to investigate the stability of the system and the data that has been used in this research has been obtained from national communication commission (NCC). Thus secondary data has been employed to achieve the objectives of the study from the year 2016 to 2018. The Lotka-Volterra competition model was used to analyze the competition for market supremacy among the three Nigerian telecommunication companies namely; MTN, AIRTEL and GLO. Firms market shares have been used to refer the competition among them. Estimation of market equilibrium and testing of stability has been performed in this study. The research result shows that MTN will be superior company in the long term and the study provides valuable information to the players for strategic planning and also for making informed decisions. (Received January 09, 2020)

1158-34-168 **Bhuvaneswari Sambandham*** (buna.sambandham@dixie.edu), **Aghalaya S Vatsala** and **Vinodh Chellamuthu**. *Linear Sequential Caputo Fractional Boundary Value Problems with Mixed Nonhomogeneous Boundary Conditions and the Numerical Results.* Preliminary report.

We recently developed the numerical results for linear sequential Caputo fractional boundary value problems with mixed homogeneous boundary conditions. We used Green's function representation to obtain the existence of the solution of the nonlinear problem. Also, we developed the numerical code for the boundary value problems with general nonhomogeneous terms with mixed nonhomogeneous boundary conditions. In this work, we plan to develop the accurate code for the linear nonhomogeneous sequential Caputo fractional boundary value problems for any nonhomogeneous terms with mixed boundary conditions. This will be useful in solving the nonlinear Caputo fractional boundary value problem with Caputo mixed boundary conditions by monotone method. (Received February 29, 2020)

1158-34-173 **Vinodh Kumar Chellamuthu***, Department of Mathematics, 225 S 700 E, Saint George, UT 84770, and **Jisun Otterson** and **Ammon Taylor**. *Modeling Dengue Outbreaks: Analyzing the Effects of Wolbachia Transinfection.* Preliminary report.

Dengue is a mosquito-borne viral infection that is usually found in tropical and subtropical regions around the world. The cycle of dengue transmission can be broken by infecting mosquitoes with Wolbachia bacterium, which reduces the level of dengue virus in the mosquito and shorten the host mosquito's lifespan. Several studies have shown that infecting mosquitos with Wolbachia and releasing them (a process called Transinfection) can spread this bacterium to the local mosquito population and mitigate the impact of the disease. We developed a mathematical model to investigate the efficacy of Wolbachia in blocking dengue virus transmission. Our model also incorporates the local temperature data, which can affect the procreation and growth of mosquitos. Furthermore, our simulation results provide new insights into the effectiveness of Wolbachia in reducing dengue at a population level. (Received February 29, 2020)

1158-34-219 **J Caleb Macdonald*** (joshua.macdonald1@louisiana.edu) and **Hayriye Gulbudak**. *Complex Dynamics of an NPZ Model Induced by Distinct Zooplankton Loss Terms with Applications to Algal Blooms.*

NPZ (Nutrient-Phytoplankton-Zooplankton) models are relatively simple frameworks for the interaction and population dynamics of the eponymous organisms. One of the central questions in plankton ecology is determining what factors drive these dynamics. We investigate this question in the context of ecological disturbances and algal blooms. Algal blooms can be harmful to aquatic ecosystems, and such blooms are increasing in frequency as the effects of anthropogenic climate change become more profound. Using a family of heuristic models, we suggest biological factors which affect qualitative bloom dynamics. These conclusions are based upon both threshold analysis and simulation studies, which we use to explore the rich dynamics of the family of models, including Hopf bifurcation and forward hysteresis. (Received March 02, 2020)

1158-34-284 **Susmita Sadhu*** (susmita.sadhu@gcsu.edu). *Analysis of long term transient dynamics in a three species predator-prey model with two timescales.*

Several ecosystems exhibit long lasting transient behaviors before experiencing regime shifts under seemingly constant environmental conditions. In this talk, I will present a singularly perturbed three-species model consisting of two predators competing for their common prey with explicit interference competition. In a neighborhood of the so-called "singular Hopf" bifurcation, the system exhibits long lasting transient dynamics, which may last for hundreds of generations, in form of chaotic mixed mode oscillations or relaxation oscillations before the system approaches its asymptotic state. To analyze the underlying mechanisms responsible for generations of such transient dynamics and transitions to its asymptotic states, the model will be reduced to a suitable normal form. A set of conditions will be obtained to determine whether a trajectory would exhibit another cycle of transient dynamics (leading to a population outbreak) before reaching its asymptotic state. These conditions can serve as early warning signs for sudden population shifts in an ecosystem. (Received March 03, 2020)

1158-34-361 **Lale Asik*** (lale.asik@ttu.edu), 1108 Memorial Circle, Lubbock, TN 79409, and **Ming Chen** (chenm688@dlmu.edu.cn) and **Angela Peace**. *The Effects of Excess Food Nutrient Content on a Tri-trophic Food Chain Model in the Aquatic Ecosystem.*

Ecological stoichiometry is an approach that focuses on the balance of energy and elements in environmental interactions, and it leads to new insights and a better understanding of ecological processes and outcomes. Modeling under this framework enables us to investigate the effects of nutrient content (i.e., food quality) on organisms whether the imbalance involves insufficient or excess nutrient content. This phenomenon is called

the "stoichiometric knife-edge." In this study, we develop and analyze a three-trophic-level food chain model that captures this phenomenon. The model tracks two essential elements, carbon, and phosphorus, in each species. We analyze the dynamics of the system, such as boundedness and positivity of the solutions, existence, and stability conditions of boundary and internal equilibria. Through numerical simulations and bifurcation analyses, we observe the switching of the dynamics of the system between periodic oscillations and chaos. Our findings also show that nutrient-rich food consumption causes (direct or indirect) adverse effects on species. (Received March 03, 2020)

35 ► *Partial differential equations*

1158-35-7 **Boya Liu*** (boyaliu1129@gmail.com), CA. *Increasing stability in a partial data inverse boundary value problem for biharmonic operators.*

We study the inverse boundary value problems of determining a potential in the Helmholtz type equation for the perturbed biharmonic operator from the knowledge of the partial Cauchy data set. Our geometric setting is that of a domain whose inaccessible portion of the boundary is contained in a hyperplane, and we are given the Cauchy data set on the complement. The uniqueness and logarithmic stability for this problem were established in Yang, Y., *Determining the first order perturbation of a bi-harmonic operator on bounded and unbounded domains from partial data*, J. Differential Equations **257** (2014), no. 10, 3607–3639 and Choudhury, A., Heck, H., *Stability of the inverse boundary value problem for the biharmonic operator: logarithmic estimates.* J. Inverse Ill-Posed Probl. **25** (2017), no. 2, 251–263 , respectively. We show, under mild regularity assumptions on the potentials, that the logarithmic stability estimate can be improved to the one of Hölder type, in the high frequency regime. (Received November 07, 2019)

1158-35-21 **Phoebe A Coy*** (pcoy@ucmerced.edu), **Arnold Kim** and **Chrysoula Tsogka**. *A Method of Fundamental Solutions for the One-Dimensional Wave Equation.* Preliminary report.

We develop a new method for approximating solutions to the one-dimensional wave equation with various boundary conditions, including moving boundaries. Our method is inspired by the Method of Fundamental Solutions for Laplace's equation. We find an approximate solution $u(x,t)$ that satisfies the wave equation and initial conditions exactly, and satisfies the boundary conditions at N points in time. Our method has significant advantages over the finite differences method, namely that the error produced by our method does not increase in time, and that it has the flexibility to be implemented for moving boundaries. This work will serve as a foundation for implementation in two and three dimensions. (Received February 04, 2020)

1158-35-29 **Igor Kukavica** and **Weinan Wang*** (wangwein@usc.edu). *Persistence of regularity and long time behavior of solutions for the Boussinesq equations.*

We address the global persistence of Sobolev regularity for the $2D$ fractional Boussinesq equations and then we consider the long time behavior of solutions for the $2D$ Boussinesq equations in different domains, including $\mathbb{R}^2$, $\mathbb{T}^2$, bounded domain with Lions or Dirichlet boundary conditions. (Received February 08, 2020)

1158-35-85 **Aghalaya S Vatsala*** (vatsala@louisiana.edu), Department of Mathematics, University of Louisiana at Lafayette, Lafayette, LA 70504, and **Yunxiang Bai**. *Sequential Versus Non sequential Caputo Dynamic Systems and Applications.* Preliminary report.

Caputo differential equations is closer to integer dynamic equations, since the initial and boundary conditions for both of them are the same. However, in general the solution of Caputo fractional differential equation does not yield the solution of the integer dynamic system as a special case unless the order of the fractional derivative is such that 0 <q <1. In order to take advantage of the global nature of the fractional derivative, study of sequential Caputo fractional dynamic systems is more advantageous. In this work, we demonstrate this both analytically as well as with applications. (Received February 22, 2020)

1158-35-177 **Pradeep G Chhetri***, pchhetri@rose.edu, and **Aghalaya Vatsala**, vatsala@louisiana.edu.
Existence of the Solution in the large for Caputo Fractional Reaction Diffusion Equation by Picard's Method and Generalized Quasilinearization Method.

Existence of the Solution in the Large for Caputo Fractional Reaction Diffusion Equation by Picard's Method and Generalized Quasilinearization Method

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We have developed Picard's iterative method to prove the existence and uniqueness of the solution of the nonlinear Caputo fractional reaction diffusion equation in one dimensional space. The order of the fractional time derivative q is such that $0.5 \leq q \leq 1$. The existence result has been proved by a priori assuming the solution is bounded. Thus, we refer to this method as existence of solution in the large. The method can be extended to the Caputo fractional reaction diffusion system also. Then we used this result in Generalized Quasilinearization Method for Caputo Fractional Reaction Diffusion Equation.

(Received February 29, 2020)

1158-35-255 **Subhash Subedi*** (ssubedi@coastal.edu) and **Aghalaya S Vatsala**. *Quenching behavior of Time Caputo Fractional Reaction-Diffusion Equation in a Circular Domain.*

We study the quenching problem for a time-dependent Caputo-fractional reaction-diffusion equation with a nonlinear source in a finite circular domain. We establish the existence and quenching of the solution. We also obtain the sufficient condition of quenching of the solution and the bounds for the quenching time of the solution.
(Received March 02, 2020)

1158-35-322 **Subekshya Bidari*** (subi3192@colorado.edu). *Effects of Spatial Structure of the Hive in the Recruitment Process of Honeybee Colonies.*

Honey bees make decisions regarding foraging and nest site selection in groups ranging from hundreds to thousands. To effectively make these decisions bees need to be able to communicate with each other within a spatially distributed group. Honeybees use a form of dance language to communicate information regarding the quality of alternatives they are deciding between. A dancing bee can communicate the quality, location and distance of the alternative they have discovered to recruit uncommitted bees in the group. We analyze this effect in the recruitment process due to the spatial structure of an enclosed hive. Our models range from treatment of the hive as a collection of discrete patches to a model of hive as a continuous unit where information is propagated through the group as they move diffusively. We report a combination of asymptotic results obtained using timescale separation techniques as well as numerical simulations that agree well with our theory. We find that the spatial structure of the hive and the statistics of bees' motion within the hive can strongly determine the rate of recruitment and consolidation of decision consensus. (Received March 03, 2020)

1158-35-346 **Fabian Santiago*** (fsantiago3@ucmerced.edu), Applied Mathematics, 5200 North Lake Road, Merced, CA 95343, and **Suzanne Sindi** (ssindi@ucmerced.edu), Applied Mathematics, 5200 North Lake Road, Merced, CA 95343. *A Likelihood Approach for Estimating Kinetic Rates of Prion Replication from a Structured Population Model.* Preliminary report.

Prion proteins cause a variety of fatal neurodegenerative diseases in mammals but are harmless to yeast, making it an ideal model organism for these diseases. Determining kinetic parameters of prion replication in yeast are complicated because experiments reflect both the disease and yeast population dynamics. We present a structured population model describing the distribution and replication of yeast prions in a population of cells. We then develop a likelihood based approach for estimating kinetic rates on simulated data and six different yeast prion strains. (Received March 03, 2020)

37 ► Dynamical systems and ergodic theory

1158-37-11 **Gangaram S. Ladde*** (gladde@usf.edu), 108 N. Greenfield Avenue, Temple Terrace, FL 33617. *An Innovative Dynamic Modeling of Binary State Processes.* Preliminary report.

By introducing the concepts of binary state potential energy and activation processes, we initiate a study of network systems in a coherent manner.. A dynamic algorithm for investigating binary digital state dynamic processes in a systematic and unified way. The modeling dynamic approach is based on historical development of classical models of McCulloch-Pitts, biological and social processes coupled with feedback process in neural network. This provides an alternate approach for studying binary state digital dynamic processes in neural network systems.Furthermore, the development of the work is initiated and guided by well-known illustration artificial neural network literature. The role and scope the approach exhibited (Received December 10, 2019)

1158-37-58 **Adam L. MacLean***, macleana@usc.edu, CA 90089. *Inference of stem cell fate and gene regulatory dynamics from single-cell data.*

Since single-cell RNA sequencing technologies have become widespread, great efforts have been made to develop appropriate computational methods to learn biological features from high dimensional datasets. Despite advances in many areas, such as cell clustering and trajectory inference, major gaps remain. We address two such gaps: how to infer cell-cell communication networks, and how to combine dynamical models with data-derived cell states. A prerequisite to such integration is a sufficient description of the cell states and their interactions: to accomplish this we introduce a method to learn cell states by nonnegative matrix factorization (SoptSC). We go on to infer cell-cell communication networks directly from data. Coupled to the inferred cell lineage relationships, SoptSC predicts both known and novel regulatory interactions. Finally, we describe how dynamical signaling models can be derived from data, and how their parameters can be identified via Bayesian inference, with application to hematopoiesis. (Received February 18, 2020)

1158-37-159 **Md Rafiul Islam*** (rafiul.islam@ttu.edu), 1612 Avenue Y Apt 205A, LUBBOCK, TX 79401, and **Angela Peace** and **Matt Gray**. *Mathematical model of an Emerging Fungal Pathogen Batrachochytrium Salamandrivorans transmission on the Eastern Newt with multiple stages of infections.*

Epidemic dynamics of infectious diseases with multiple routes of transmission are complex. Mathematical models can be used to determine invasion potential and identify which transmission pathway is dominant and can ultimately help identify appropriate intervention strategies. We developed compartmental host-pathogen models to examine the transmission dynamics of an emerging fungal pathogen on a North American salamander population. Multiple stages of infection are incorporated into the model, allowing disease-induced mortality and zoospore shedding rates to vary as the disease progresses. Parameter sensitivity analysis shows that the recovery and disease-induced mortality rates, the length of the incubation period, and environmental zoospore degradation rates are influential parameters. Calculation of the basic reproductive number highlights the invasion potential of this pathogen and was used to determine that direct transmission contact was the dominant transmission pathway for small population densities, while environmental transmission dominated in large populations. Collectively, these results suggest strategies that reduce host contacts at small densities or reduce the environmental persistence of zoospores at high host densities may be effective Bsal management strategies. (Received February 28, 2020)

40 ► Sequences, series, summability

1158-40-26 **SUKHDEV SINGH*** (singh.sukhdev01@gmail.com), Department of Mathematics, Lovely Professional University, Jalandhar-Delhi G.T Road (NH-1), Phagwara, Punjab 144411, India. *On C_2-sequence spaces in 2-normed spaces using ideal convergence and Orlicz function.*

In this paper, we introduced some C_2-sequence spaces that are defined using an Orlicz function and ideal convergence in 2-normed spaces and examined their topological and algebraic properties. (Received February 08, 2020)

41 ► Approximations and expansions

1158-41-302 **Christian Bueno*** (christianbueno@ucsb.edu) and **Alan G. Hylton** (alan.g.hylton@nasa.gov). *Permutation-Invariant Neural Networks.* Preliminary report.

Point clouds form an important data type in machine learning that inherently lack order. This makes them tricky inputs for traditional neural networks. As a result, interest in permutation-invariant and permutation-equivariant neural networks has grown and led to recent innovations in neural network design. Approaches for dealing with group actions in neural networks can range from using weight sharing with pooling, group averaging and using invariant polynomials. In this talk we explore some of these models, their theoretical capabilities, and their limitations when it comes to representing or approximating functions of interest. Along the way we will generalize the classical universal approximation theorem to general compact Hausdorff spaces, apply them to spaces of point clouds, and show that two popular permutation-invariant models have different representational power. (Received March 03, 2020)

42 ► Fourier analysis

1158-42-95 **Benjamin Cowen*** (benjamin.cowen.math@gmail.com). *Uncertainty Propagation through the Logarithmic Derivative of a Convolution.* Preliminary report.

During explosive experiments, acquired data must be *stored* far away from where they are *detected*. In the mid-20th-century, such detected signals were often transmitted to storage devices by very long (multi-kilometer) systems of coaxial cables. The signal of interest to us is the logarithmic derivative of the measured one, called *alpha*, making the total system from detection to alpha nonlinear. We seek to identify and characterize factors that could have distorted our data en route through the electromagnetic cable system. We present an analysis of how errors in a cable's parameters propagate to uncertainty in the alpha signal. (Received February 24, 2020)

46 ► Functional analysis

1158-46-119 **Igor V. Nikolaev*** (igor.v.nikolaev@gmail.com). *Arithmetic complexity of noncommutative tori.* Preliminary report.

The noncommutative torus $\mathcal{A}_\theta$ is a C^*-algebra generated by two unitary operators U and V satisfying the relation $VU = e^{2\pi I\theta}UV$ for a real constant θ. We study a special case of the algebras $\mathcal{A}_\theta$ when θ is the irrational root of a quadratic polynomial with integer coefficients. It is well known, that in this case θ unfolds into an infinite periodic continued fraction. With the help of this fraction we introduce an integer number $c(\mathcal{A}_\theta)$ called an arithmetic complexity of the noncommutative torus. It is shown, that the $c(\mathcal{A}_\theta)$ is linked to the ranks r of elliptic curves $\mathcal{E}$ by a simple formula $r(\mathcal{E}) = c(\mathcal{A}_\theta) - 1$. Reference: arXiv:2002.10854 (Received February 26, 2020)

1158-46-155 **Konrad Aguilar*** (konrad.aguilar@gmail.com). *Quantum metrics on the tensor product of a commutative C*-algebra and an AF C*-algebra.*

Given a compact metric space X and a unital AF algebra A equipped with a faithful tracial state, we place quantum metrics on the tensor product of C(X) and A given established quantum metrics on C(X) and A from work with Bice and Latremoliere. We prove the inductive limit of C(X) tensor A given by A is a metric limit in the Gromov-Hausdorff propinquity. We show that our quantum metric is compatible with the tensor product by producing a Leibniz rule on elementary tensors and showing the diameter of our quantum metric on the tensor product is bounded above the diameter of the Cartesian product of the quantum metric spaces. We provide continuous families of C(X) tensor A which extends our previous results with Latremoliere on UHF algebras. (Received February 28, 2020)

1158-46-166 **Hafedh Herichi**, Santa Monica College, Santa Monica, CA , and **Michel L. Lapidus*** (lapidus@math.ucr.edu), University of California, Riverside, CA 92521-0135. *Spectral Operators, Quantized Number Theory and the Riemann Hypothesis.*

We discuss results concerning the invertibility, quasi-invertibility and universality of the spectral operator obtained by the authors in the coming book [HerLap] (World Sci., in press, 470pp) and in a paper by the presenter [Lap1] (Philos. Trans, Royal Soc.,No. 2047,vol. 373,2015,22pp.). The spectral operator was introduced semi-heuristically by the presenter and M. van Frankenhuijsen in [Lap-vFr] (Springer Monographs in Math., 2nd, edn., Springer, 2013) as the operator sending the geometry onto the spectrum of fractal strings. We rigorously define

here the spectral operator as the Riemann zeta function applied to (in the sense of the functional calculus for unbounded normal operators) the infinitesimal shift of the real line (acting on.a suitable Hilbert space, indexed by a dimensional parameter c). We then obtain, in particular, several reformulations of the Riemann hypothesis (RH) providing natural operator-theoretic counterparts of the initial refomulation of RH by the presenter and H. Maier in [LapMai] (J. London Math. Soc.,vol. 52,1995,pp.16-34) in terms of inverse spectral problems for fractal strings, and expressed here in terms of the quasi-invertibility [HerLap] (resp., invertibilty, [Lap1]) of the spectral operator for all values of c in (0,1) (resp., (0,1/2)). (Received February 29, 2020)

1158-46-208 **Ievgen (Eugene) Bilokopytov*** (bilokopi@myumanitoba.ca), 10-166 River Avenue, Winnipeg, MB R3L 0A9, Canada. *Maximum modulus principle for multipliers.*

In this talk I will show that (not necessarily holomorphic) multipliers of a wide class of normed spaces of continuous functions over a connected Hausdorff topological space cannot attain their multiplier norms, unless they are constants. A corollary of this fact is that every non-trivial contractive multiplication operator has to be completely non-unitary. If the time allows I will present another proof of that fact using Birkhoff orthogonality. (Received March 01, 2020)

1158-46-240 **Andrea Arauza Rivera***, 25800 Carlos Bee Blvd, Hayward, CA 94542. *An Introduction to Noncommutative Fractal Geometry.*

Noncommutative geometry stems from the duality between the category of compact Hausdorff topological spaces and commutative unital C* algebras. When one lifts the commutativity requirement one is able to study "noncommutative" topological spaces. Led by Alain Connes, this field has developed many interesting tools for studying noncommutative spaces. We introduce how one may use these tools to study fractal sets. (Received March 02, 2020)

47 ► Operator theory

1158-47-12 **Sven-Ake Wegner*** (s.wegner@tees.ac.uk), Middlesbrough, TS13BX, United Kingdom. *Port-Hamiltonian PDEs on infinite networks.*

We consider the port-Hamiltonian partial differential equation

$$\frac{\partial x}{\partial t}(\xi,t) = P_1 \frac{\partial}{\partial \xi}(\mathcal{H}(\xi)x(\xi,t)) + P_0\,\mathcal{H}(\xi)x(\xi,t) \text{ for } t \geqslant 0 \text{ and } \xi \in \Gamma.$$

Here, $\Gamma = (V,E)$ is a graph in which each edge $e \in E$ comes with an interval $I_e \subseteq \mathbb{R}$ and $\xi \in \Gamma$ means $\xi = (\xi_e)_{e\in E}$ with $\xi_e \in I_e$. Under suitable conditions on the matrices $P_0,\ P_1 \in \mathbb{C}^{n\times n}$, the Hamiltonian $\mathcal{H} = \mathcal{H}(\xi)$ and the graph Γ we identify boundary conditions that, imposed at the vertices $v \in V$, turn the operator

$$A\colon D(A) \subseteq X_{\Gamma,\mathcal{H}} \to X_{\Gamma,\mathcal{H}},\ Ax = \Big(P_1\frac{\partial}{\partial\xi} + P_0\Big)(\mathcal{H}x)$$

into the generator of a C_0-semigroup on a suitable Hilbert space $X_{\Gamma,\mathcal{H}}$ associated with $\mathcal{H}$ and Γ. Results of this type are available for uniformly bounded $\mathcal{H}$, finite intervals and finite graphs. In the talk we discuss infinite intervals, infinite graphs and Hamiltonians that can be unbounded or tend to zero at infinity. The results are joint work with Birgit Jacob (Wuppertal) and Marcus Waurick (Glasgow). (Received December 23, 2019)

1158-47-24 **Sergey Belyi*** (sbelyi@troy.edu), Department of Mathematics, MSCX 235-B, Troy University, Troy, AL 36082, and **Eduard Tsekanovskii** (tsekanov@niagara.edu), Department of Mathematics, Niagara University, Niagara University, NY 14109. *On von Neumann's parameter of extremal Shrödinger operator.*

We study connections between extremal accretive quasi-self-adjoint dissipative extensions of a non-negative symmetric Shrödinger operator with deficiency indices $(1,1)$ on $L_2[\ell,+\infty)$ and the moduli of their von Neumann's parameters. It is shown that the modulus of the corresponding von Neumann's parameter belongs to the interval $[\kappa_0,1)$, where $\kappa_0 \geq 0$. For κ_0 we derive a new formula in terms of the values of the Weyl-Titchmarsh function $m_\infty(-0)$ and $m_\infty(i)$. An example that illustrates the obtained results is presented.

[1] S. Belyi, E. Tsekanovskiĭ, *On von Neumann's parameter of extremal Shrödinger operator*, Bulletin of Donetsk National University, Series A: Natural Sciences, vol. 2, (2019), pp. 91-97.

(Received February 06, 2020)

1158-47-74 **Asuman G. Aksoy*** (aaksoy@cmc.edu). *Factorization of Q-compact operators.* Preliminary report.

The theory of compact linear operators between two Banach spaces X and Y has a classical core and is familiar to many. Perhaps lesser known is the factorization of compact maps through a closed subspace of c_0 [2]. This

factorization theorem has a number of important connections and consequences analogous to how the ideals of continuous linear operators factoring compactly through ℓ^p-spaces ($1 \leq p < \infty$) were studied by many authors (see [1] and the references therein). In this talk, we consider a Banach space X and another Banach space $(Y, \{A_n\})$ where $\{A_n\}$ is a linear approximation scheme with $\dim A_n < \infty$ on Y. We study compactness with respect to an approximation scheme Q and get characterization and factorization theorems for Q-compact maps. Joint work with Yunied Puig De Dios.

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[2] T. Terzioğlu, *A characterization of compact linear mappings*, Arch. Math. **22** (1971), 76-78.

(Received February 20, 2020)

1158-47-141 **Marat V. Markin*** (mmarkin@csufresno.edu), Department of Mathematics, California State University, Fresno, 5245 N. Backer Avenue, M/S PB 108, Fresno, CA 93740-8001. *On a Spectral Mapping Theorem for Scalar Type Spectral Operators, the Weak Spectral Mapping Theorem for and Stability of C_0-Semigroups of Such Operators.* Preliminary report.

We prove a *spectral mapping theorem* for scalar type spectral operators and apply it to extend the *weak spectral mapping theorem* and the *generalized Lyapunov stability theorem*, known to hold for the C_0-semigroups of normal operators on complex Hilbert spaces, to the more general case of C_0-semigroups of scalar type spectral operators on complex Banach spaces. We also establish for the latter an analogue of the *Gearhart-Prüss-Greiner characterization of the uniform exponential stability* for C_0-semigroups on complex Hilbert spaces. (Received February 27, 2020)

1158-47-194 **Don W. Hadwin*** (operatorguy@gmail.com), 33 Academic Way, Math Dept UNH, Durham, NH 03824. *An Elementary Approach to Weak* Limits.* Preliminary report.

We begin, only using elementary techniques, to study and characterized the weak operator closure of composition operators induced by invertible measure-preserving transformations on the unit interval. We the apply this to studying the weak* operator closure of inner automorphisms of a von Neumann algebra. For example in a finite von Neumann algebra with no minimal projections (i.e., no finite-dimensional direct summands) the center-valued trace is a pointwise weak* limit of inner automorphisms. For such algebras on a separable Hilbert space this closure contains a conditional expectation onto every hyperfinite subalgebra containing the center. We also determine the weak* closure of the similarity orbit of every operator in a separable factor. (Received March 01, 2020)

49 ► Calculus of variations and optimal control; optimization

1158-49-225 **Radoslav G Vuchkov*** (rvuchkov@ucmerced.edu), 5200 N. Lake Road, Merced, CA 95343, **Cosmin Gheorghita Petra** (petra1@llnl.gov), 7000 East Avenue, Livermore, CA 94550, and **Noemi Petra** (npetra@ucmerced.edu), 5200 N. Lake Road, Merced, CA 95343. *Quasi-Newton Methods for Infinite-Dimensional Inverse Problems Governed by PDEs.*

Many engineering applications are modeled as optimization problems governed by differential equations. Newton's method is usually the method of choice to solve these problems due it's superior convergence properties compared to gradient-based or derivative-free optimization algorithms. However, deriving and computing second-order derivatives needed by Newton's method often is not trivial and sometimes not possible. In such cases quasi-Newton algorithms are a great alternative since they use only gradient information to build approximations to the second derivative. In this talk, we provide a new derivation of the Broyden-Fletcher-Goldfarb-Shanno (BFGS) quasi-Newton method update in an infinite-dimensional Hilbert space setting. The key component of this derivation is the formulation of variational problems related to the quasi-Newton methods within the space of self-adjoint Hilbert-Schmidt operators. Similarly by changing the constraints of the variational problem we obtain updates for other quasi-Newton methods such as Davidon–Fletcher–Powell, Symmetric Rank One, and Powell-Symmetric-Broyden. We show numerical results for an inverse problem governed by a partial differential equation and demonstrate the desired mesh-independent property of the quasi-Newton methods. (Received March 02, 2020)

51 ► Geometry

1158-51-330 **William Reimer*** (wreimer14@mail.fresnostate.edu) and **Michael A Bishop** (mibishop@csufresno.edu). *An Introduction to Fractal Dimensions.* Preliminary report.

This talk will explore what fractals are as well as how they can be constructed. Fractals are sets without a well-defined integer-valued dimension and require generalized notions of dimension. We define the box counting dimension of a set and verify that this gives the usual integer dimension for familiar sets such as intervals and boxes. We then provide examples of iterated function systems which can be used to construct self-similar fractals. We will end by discussing different ways to generate fractals using code in Python by calling functions recursively. This is an undergraduate research project. (Received March 03, 2020)

52 ► Convex and discrete geometry

1158-52-251 **Michael DiPasquale** (michael.dipasquale@colostate.edu), **Bryan R Gillespie*** (bryan.gillespie@colostate.edu) and **Chris Peterson** (peterson@math.colostate.edu). *Ehrhart Theory of Numerical Semigroup Structures.*

In this talk we will present enumerative results for the collection $S(n,k)$ of numerical semigroups with Frobenius number at most n and genus $n-k$. We introduce a discrete structure associated with numerical semigroups which allows the construction of a rational cone whose integer lattice points correspond with elements of $S(n,k)$. Using Ehrhart theory, we conclude that for fixed k, the number of elements $|S(n,k)|$ is a quasi-polynomial of degree k. If time permits, we will discuss a generalization of the construction to the higher dimensional context of affine semigroups. (Received March 02, 2020)

1158-52-368 **Nathan Thom*** (nathanthom@csufresno.edu) and **Oscar Vega** (ovega@csufresno.edu). *The Veldkamp Space of $W(q)$.* Preliminary report.

Given a partial linear space, S, the Veldkamp space of S, denoted $mathcalV(S)$, is defined using the geometric hyperplanes of S. In this talk, we will examine the forms the geometric hyperplanes of S can take when S is a finite generalized quadrangle and the cardinality of their pairwise intersections in order to understand how $mathcalV(S)$ behaves. In general, $mathcalV(S)$ does not possess desirable geometric attributes, so we focus on families of generalized quadrangles that impart some nice properties to their Veldkamp spaces. In particular, we consider the case when S is the generalized quadrangle $W(q)$, for q odd, and the case of its dual, $Q(4,q)$. Using a combinatorial approach, we will show that the Veldkamp space of $W(q)$ is isomorphic to $PG(3,q)$. (Received March 03, 2020)

53 ► Differential geometry

1158-53-367 **Jeffrey S. Meyer*** (jeffrey.meyer@csusb.edu) and **Brian Benson**. *Programming Sage to Compute Cheeger Constants.* Preliminary report.

The goal of this talk is to outline recent work with Brian Benson in which we use Sage to explicitly compute the Cheeger constants of certain principal congruence hyperbolic surfaces. Applying the Cheeger and Buser inequalities to Selberg's celebrated spectral gap theorem, it follows that the Cheeger constants of principal congruence surfaces are uniformly bounded from above and below. There has been significant work over the years to improve these bounds. However, explicit computations of these Cheeger constants have remained notoriously absent. In this talk, I will give a quick introduction to principal congruence surfaces. I will use Sage to produce certain fundamental domains called Farey domains. I will explain and demonstrate how our code searches for and finds the Cheeger constants for small n. There will be many colorful pictures. (Received March 03, 2020)

54 ► General topology

1158-54-125 **Pawel Grzegrzolka*** (pawelg@stanford.edu) and **Jeremy Siegert**. *Inductive dimension of coarse proximity spaces.*

In this talk, we define the asymptotic inductive dimension, asInd, of coarse proximity spaces. In the case of metric spaces equipped with their metric coarse proximity structure, this definition is equivalent to the definition of asInd given by Dranishnikov for proper metric spaces, which include finitely generated groups with the associated word

metric. We show that if the boundary of a coarse proximity space is completely traceable, then the asymptotic inductive dimension of the space is equal to the large inductive dimension of its boundary. (Received February 26, 2020)

1158-54-250 **Carmen Caprau**, **Nicolle Gonzalez*** (nicolle@math.ucla.edu), **Christine Ruey Shan Lee**, **Adam Lowrance**, **Radmilla Sazdanovic** and **Melissa Zhang**. *Ribbon concordance obstructions from Khovanov homology.*

I will describe how to extend Levine and Zemke's results on ribbon concordance obstruction from Khovanov homology to $\mathfrak{sl}_n$ homology for $n \geq 0$ and universal $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ homology theories. In particular, this shows that homology theories for all $n \geq 2$ give obstructions to ribbon concordance. (Received March 02, 2020)

1158-54-362 **Pawel Dlotko**, **Mustafa Hajij**, **Jesse Levitt*** (jslevitt@usc.edu) and **Radmila Sazdanovic**. *Finding Structure in the Knot Polynomials using Data Science.* Preliminary report.

Using Principal Component Analysis and Topological Data Analysis we analyze the distributions of the knot polynomials. These tools prove useful for both seeing correlations between different knot invariants and for distinguishing how well they separate the knots into distinct families. We focus on how the Ball Mapper of P. Dlotko, an exploratory data analysis tool that builds graphs from high dimensional clouds of data using just a radius measure, confirms and further illuminates substructure in this data. This includes some specific ways in which the s-invariant and signature differ through the distribution of the Alexander and Jones polynomials. (Received March 03, 2020)

55 ► Algebraic topology

1158-55-354 **Matthew Hogancamp*** (m.hogancamp@northeastern.edu), **Eugene Gorsky** and **Paul Wedrich**. *Categorifying the skein module of the solid torus.*

In this talk I will discuss recent joint work with E. Gorsky and P. Wedrich in which we describe a categorification of (the positive half of) the HOMFLY-PT skein module of the solid torus. Despite its rather sophisticated definition, we find that our category is equivalent to the category is equivalent to the category of modules over an explicit wreath product algebra. (Received March 03, 2020)

57 ► Manifolds and cell complexes

1158-57-23 **Blake Mellor*** (blake.mellor@lmu.edu). *Finite N-quandles of knots, links and graphs.* Preliminary report.

In this talk, we will investigate a generalization of the n-quandle for a knot, link or spatial graph. If a link or spatial graph has k components or edges, and given a k-tuple $N = (n_1, \ldots, n_k)$ of positive integers, we define the N-quandle by adding relations $x^{y^{n_i}} = x$ to the full quandle whenever y is an arc of edge i. The usual n-quandle is the case when $n_i = n$ for every i. In this case, Hoste and Shanahan provided a complete list of links where the n-quandle is finite, proving a conjecture of Przytycki. We extend the conjecture to spatial graphs and links where the entries of N are not all the same, and provide evidence for the extended conjecture by computing finite N-quandles associated to a number of links and spatial graphs. (Received February 05, 2020)

1158-57-35 **Rachel Marie Harris*** (rachel.harris@ttu.edu), 1108 Memorial Circle Drive, Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX 79410. *Developing a software for skein computations in knot complements.* Preliminary report.

Skein manipulations prove to be computationally intensive because of the exponential nature of the skein relations. The purpose of this work is to construct an automated tool that would generate a library of examples for testing conjectures in Chern-Simons theory. (Received February 11, 2020)

1158-57-36 **Razvan Gelca***, 1108 Memorial Circle Drive, Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX 79409, and **Shamonvenishiya Almeida**, 1108 Memorial Circle Dr., Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX 79409. *A linear map that computes colored Jones polynomials.* Preliminary report.

We will describe a linear map that acts on the skein module of a knot complement. This linear map gives rise to a recursive way to compute colored Jones polynomials. We will discuss the examples of the trefoil and figure 8 knots. (Received February 11, 2020)

1158-57-42 **Rhea Palak Bakshi*** (rhea_palak@gwu.edu). *A Counterexample to Marché's Generalization of Witten's Conjecture.*

In 2019, Julien Marché formulated a generalization Witten's finiteness conjecture for skein modules over the ring of Laurent polynomials in an invertible element:

Let M be a closed compact oriented 3-manifold. Then there exists an integer $d \geq 0$ and finitely generated $\mathbb{Z}[A^{\pm 1}]$-modules N_k so that

$$\mathcal{S}_{2,\infty}(M) = (\mathbb{Z}[A^{\pm 1}])^d \oplus \bigoplus_{k \geq 1} N_k,$$

where N_k is a $(A^k - A^{-k})$-torsion module for each k.

In this talk, I will discuss a counterexample to this conjecture, which I recently found. The counterexample is based on the Kauffman bracket skein module of the connected sum of two copies of the projective space. The counterexample is based on the work of Maciej Mroczkowski. (Received February 13, 2020)

1158-57-54 **M. Ishikawa**, **Thomas W. Mattman*** (tmattman@csuchico.edu), **K. Namiki** and **K. Shimokawa**. *Alternating knots with large boundary slope diameter.*

We show that, for an alternating knot, the ratio of the diameter of the set of boundary slopes to the crossing number can be arbitrarily large. (Received February 17, 2020)

1158-57-60 **Robert Owczarek*** (rowczare@unm.edu), 59 Coryphodon Ln, Jemez Springs, NM 87025. *Quantum Teichmüller spaces and Przytycki's invariant of rooted trees.* Preliminary report.

A continuation of exploration of the connection between quantum Teichmüller spaces and Przytycki's invariant of rooted trees will be presented. (Received February 18, 2020)

1158-57-61 **Charles Frohman*** (charles-frohman@uiowa.edu) and **Joanna Kania-Bartoszynska**. *Representation Theory of the Kauffman Bracket skein algebra of a surface at a root of unity.*

Let F be a closed surface of negative Euler characteristic, and ζ be a complex root of unity whose order is not divisible by 4. The Kauffman bracket skein algebra of F at ζ, $K_\zeta(F)$ is an algebra built out of framed links in $F \times [0,1]$ modulo the Kauffman bracket skein relation. The center $Z_\zeta(F)$ of $K_\zeta(F)$ is the coordinate ring of the $SL_2\mathbb{C}$ character variety of $\pi_1(F)$.

A representation $\rho : \pi_1(F) \to SL_2\mathbb{C}$ is **nonelementary** if it is irreducible, has infinite image and there is a loop on the surface α so that $|tr(\rho(\alpha))| > 2$. The skein algebra can be reduced at such a representation to yield a matrix algebra.

We show that if the nonelementary representation ρ extends over the fundamental group of a handlebody H whose boundary is F, then the skein module of the handlebody reduced at the extension of ρ is an irreducible representation of the skein algebra. (Received February 19, 2020)

1158-57-65 **Rhea Palak Bakshi** and **Dionne Ibarra***, 801 22nd street NW, Phillips 725, Washington, DC, and **Sujoy Mukherjee** and **Jozef Przytycki**. *Gram Determinant Type Mb.*

General formula's for the Gram determinant of Type A and Type B were formulated in the 1990's and 2000's, respectively. By using the relative skein module we generalize the definition of a Gram determinant motivated by knot theory, present the Gram determinant type Mb, and discuss our results that support Qi Chen's conjecture of a general closed formula for this determinant. (Received February 19, 2020)

1158-57-67 **Sarah Goodhill**, **Adam M Lowrance*** (adlowrance@vassar.edu), **Valeria Munoz Gonzales**, **Jessica Rattray** and **Amelia Zeh**. *Unknotting via region crossing changes.*

Region crossing change is an operation on a knot diagram where every crossing incident to a region or face of the diagram is changed. Shimizu proved that region crossing change is an unknotting operation. There are existing invariants that measure how difficult it is to unknot a knot using region crossing changes, such as the

region unknotting number and the region index. We define a similar new invariant called the multi-region index. We derive a lower bound for the multi-region index of a knot coming from Goeritz matrices and compute the multi-region index for many knots in the tables. (Received February 20, 2020)

1158-57-71 **Rhea Palak Bakshi** and **Dionne Ibarra*** (dfkunkel@gwu.edu), 801 22nd Street NW, Washington, DC 20052, and **Sujoy Mukherjee**, **Takefumi Nosaka** and **Jozef Przytycki**. *Schur multipliers and second quandle homology.*

For X, an Alexander quandle with invertible (1-t) , we express the second quandle homology of quasigroup Alexander quandles in terms of the exterior algebra of X and present a short self-contained proof of its structure and provide some computational examples. (Received February 20, 2020)

1158-57-72 **Jose Roman Aranda Cuevas*** (romanaranda123@gmail.com), 751 W Benton St Apt 27, Iowa City, IA 52246. *Thin position through trisections of 4-manifolds.*

In 1994, M. Scharlemann and A. Thompson introduced a complexity for handle decompositions of 3-manifolds called the width. In the last decade, taking a minimal width decomposition of a 3-manifold has been a useful technique. As an example, M. Scharlemann and J. Schultens used this to show that the connected sum of n knots in S^3 has tunnel number at least n. In this talk, we use ideas from the theory of trisections of 4-manifolds to define the width of a 4-dimensional handle decomposition and introduce the thin position of a compact smooth 4-manifold. (Received February 20, 2020)

1158-57-77 **Louis H Kauffman*** (kauffman@uic.edu), Louis H Kauffman, 5530 South Shore Drive, Apt 7C, Chicago, IL 60637-1946. *Affine Index and Sawollek Polynomials for Virtual Knots.*

This talk will discuss the relationships between the Affine Index Polynomial and the Sawollek Polynomial for virtual knots and links. (Received February 20, 2020)

1158-57-84 **Christine Ruey Shan Lee*** (crslee@southalabama.edu) and **Roland van der Veen**. *Cancellations in the degree of the colored Jones polynomial.*

The colored Jones polynomial is a link invariant generalized from the Jones polynomial using ideas from representation theory and physics. The coefficients and the degrees of the polynomial have been shown to relate to topological and geometric properties of the knot such as boundary slopes and volume. We propose an expansion of the colored Jones polynomial which recovers the degree formulas previously obtained by the authors and S. Garoufalidis for Montesinos knots. We will discuss how it can be used to obtain the degrees of the polynomial for many new examples, and we will also discuss the application of this expansion to computing the stable coefficients of the colored Jones polynomial. (Received February 22, 2020)

1158-57-86 **Micah Chrisman*** (chrisman.76@osu.edu). *Milnor's concordance invariants for knots on surfaces.*

Milnor's $\bar{\mu}$-invariants for links in the 3-sphere vanish on any link concordant to a boundary link. In particular, they are are trivial for any classical knot. Here we define an analogue of Milnor's concordance invariants for knots in thickened surfaces $\Sigma \times [0,1]$, where Σ is closed and oriented. These invariants vanish on any knot concordant to a homologically trivial knot in $\Sigma \times [0,1]$. We use them to give new examples of non-slice virtual knots having trivial Rasmussen invariant, graded genus, affine index polynomial, and generalized Alexander polynomial. Moreover, we complete the slice status classification of the 2564 virtual knots having at most five classical crossings and reduce to four (of 90235) the number of virtual knots with six classical crossings having unknown slice status. Furthermore, we prove that in contrast to the classical knot concordance group, the virtual knot concordance group is not abelian. (Received February 22, 2020)

1158-57-89 **Jozef H. Przytycki*** (przytyck@gwu.edu), l, DC , and **Xiao Wang**. *Computing Yang-Baxter homology.*

I will demonstrate how to compute the second Yang-Baxter homology for the Yang Baxter operators giving Jones and HOMFLYPT polynomials. In particular, I will prove that:

$$H_2(R_m) = k^{1+\binom{m}{2}} \oplus \left(k/(1-y^2)\right)^{\binom{m}{2}} \oplus \left(k/(1-y^4)\right)^{m-1}.$$

I will speculate on relations with Khovanov homology. (Received February 23, 2020)

1158-57-101 **Tejas Kalelkar*** (tejas@iiserpune.ac.in) and **Advait Phanse**. *Pachner moves on geometric triangulations of constant curvature manifolds.*

A geometric triangulation of a Riemannian manifold is a triangulation by totally geodesic simplexes. A Pachner move is a local combinatorial transformation of a triangulation. We show that up to derived subdivisions any

two geometric triangulations of a constant curvature compact manifold are related by Pachner moves through geometric triangulations. We also give an explicit bound on the number of Pachner moves needed to relate two geometric triangulations of such a manifold. This bound is in terms of the dimension of the manifold, the number of top dimensional simplexes and bounds on the lengths of edges. This gives an algorithm to check if two geometrically triangulated closed hyperbolic or low dimensional spherical manifolds are isometric or not. (Received February 25, 2020)

1158-57-109 **Emily Hamilton*** (mhamil09@calpoly.edu), California Polytechnic State University, 1 Grand Avenue, San Luis Obispo, CA 93407. *Separability of double cosets and conjugacy classes in n-manifold groups.*

In previous work with Wilton and Zalesskii we showed that if M is a hyperbolic 3-manifold of finite volume with fundamental group Γ, then double cosets of abelian subgroups of Γ are separable in Γ. As a consequence, we proved that fundamental groups of compact, orientable 3-manifolds are conjugacy separable. In this talk, we consider double coset separability of fundamental groups of hyperbolic n-manifolds for $n \geq 2$. We prove that if M is a closed hyperbolic n-manifold with fundamental group Γ, H and K are abelian subgroups of Γ, and $g \in \Gamma$, then the double coset HgK is separable in Γ. We generalize this result to cocompact lattices in linear, semisimple Lie groups of (real) rank one. (Received February 25, 2020)

1158-57-110 **Rhea Palak Bakshi**, **Dionne Ibarra**, **Gabriel Montoya-Vega**, **Jozef Przytycki** and **Deborah Weeks*** (deweeks@gwu.edu), The George Washington University, Washington, DC 20052. *Changing Framings of Links Requires Non-Separating S^2.*

Extending the work of McCullough and Chernov, the conditions under which the framing of a link can be changed via ambient isotopy are examined. A change in framing can only be accomplished via the Dirac trick in a manifold with a properly embedded non-separating S^2. Due to the invariance of spin structure and the parallelizability of every compact oriented 3-manifold the framing of a knot can only change by an even power. (Received February 25, 2020)

1158-57-111 **Rhea Palak Bakshi**, **Dionne Ibarra**, **Gabriel Montoya-Vega*** (gmontoyavega@gwu.edu), **Jozef H Przytycki** and **Deborah Weeks**. *Torsion in skein modules and framing changes of links.*

The framing of a knot K in an oriented 3-manifold remains unchanged unless there exists a properly embedded non-separating 2-sphere which intersects K exactly once; in which case, the change of framing is given by the Dirac trick. Since 1987, when Przytycki introduced skein modules, these have been extensively studied with the goal of building an algebraic topology based on knots. In this talk, we formulate the results on the framing of knots in the language of skein modules. (Received February 25, 2020)

1158-57-117 **Jozef H. Przytycki*** (przytyck@gwu.edu), George Washington University, Phillips Hall, Washington, DC 20052-0001, and **Marithania Silvero**. *Complexity of extreme Khovanov homology for 4-braids.*

We present here recent progress on polynomial complexity algorithm for computing extreme Khovanov homology for 4-braids. One of the results involves outerplanar circle graphs and their geometric realization (via independence simplicial complex). (Received February 25, 2020)

1158-57-126 **Sujoy Mukherjee*** (mukherjee.166@osu.edu). *On the homology of quasibands.* Preliminary report.

In general, rack and one-term homology are very useful tools for studying self-distributive algebraic structures like shelves, racks, and quandles. However, it turns out that when self-distributive algebraic structures are associative, their rack and one-term homology groups are not very interesting.

Associative shelves are examples of quasibands. In this talk, after introducing the homology of quasibands, I will show that it is more useful than rack or one-term homology for certain sub-families of associative shelves. Finally, I will describe my attempts to connect this homology theory to knot theory via Jones monoids. (Received February 26, 2020)

1158-57-130 **Elaina Aceves*** (elaina-aceves@uiowa.edu). *Sequence of Undetectable Nonquasipositive Braids.*

When K is a quasipositive transverse knot, Hedden and Plamenevskaya proved that $2\tau(K) - 1 = sl(K)$, where τ is the Ozsváth-Szabó concordance invariant and sl is the self-linking number of the knot. We construct a sequence of braids whose closures are nonquasipositive knots where we add more negative crossings as we progress through the sequence. Therefore, the knots formed from the sequence become more nonquasipositive as the sequence

progresses. Furthermore, we conjecture that the knots obtained from this sequence behave like quasipositive knots in that they uphold the equality $2\tau(K) - 1 = sl(K)$. Because of this property, the nonquasipositive knots obtained from our sequence are undetectable by transverse invariants like Ozsváth and Szabó's $\hat{\theta}(K)$ from Heegaard Floer homology and Plamenevskaya's $\psi(K)$ from Khovanov homology. (Received February 28, 2020)

1158-57-145 **Kanako Oshiro*** (oshirok@sophia.ac.jp) and **Natsumi Oyamaguchi**. *Dehn colorings and vertex-weight invariants of spatial graphs.*

we introduce Dehn colorings for spatial graphs, and give a family of spatial graph invariants that are called vertex-weight invariants. We give some examples of spatial graphs that can be distinguished by a vertex-weight invariant, whereas distinguished by neither their constituent links nor the number of Dehn colorings. (Received February 28, 2020)

1158-57-156 **Kristóf Huszár*** (kristof.huszar@ist.ac.at), **Jonathan Spreer** and **Uli Wagner**. *Combinatorial width parameters for 3-manifolds.*

Algorithms in computational 3-manifold topology typically take a triangulation as an input, and return topological information about the underlying 3-manifold. However, extracting the desired information from a triangulation (e.g., evaluating an invariant) is often computationally very expensive. In recent years this complexity barrier has been successfully tackled in some cases by importing ideas from the theory of parametrized algorithms into the realm of 3-manifolds. It was shown that various computationally hard problems about 3-manifolds can be efficiently solved for triangulations that are sufficiently "thin" or "tree-like."

In this talk we focus on the key combinatorial parameter in the above context, i.e., on the treewidth of a compact, connected 3-manifold. By building on the work of Scharlemann–Thompson and Scharlemann–Schultens–Saito on generalized Heegaard splittings, and on the work of Jaco–Rubinstein on layered triangulations, we relate the treewidth of a 3-manifold to classical invariants thereof. In particular, we show that the treewidth of a closed, orientable, irreducible, non-Haken manifold is always within a constant factor of its Heegaard genus. (Received February 28, 2020)

1158-57-158 **Tommaso Cremaschi***, cremasch@esc.edeu, and **Andrew Yarmola** and **Jose Andres Rodriguez-Migueles**. *On volumes and filling collections of multicurves.*

Let Γ be a hyperbolic link in a Seifert-fibered space N over a surface S of negative Euler characteristic. When Γ is carried by horizontal surfaces we show that the volume of $N \setminus \Gamma$ is quasi-isometric to expressions involving distances in the pants graph. We will build many examples with specific emphasis on canonical lifts in $PT^1(S)$. (Received February 28, 2020)

1158-57-162 **Migiwa Sakurai*** (migiwa@shibaura-it.ac.jp), 307 Fukasaku, Minuma-ku, Saitama-shi, Saitama 337-8570, Japan, and **Yoshiyuki Ohyama** (ohyama@lab.twcu.ac.jp), 2-6-1, Zempukuji, Suginami-ku, Tokyo 167-8585, Japan. *Virtual knots with properties of Kishino's knot.*

Satoh and Taniguchi defined a virtual knot invariant J_n called the n-writhe for each non-zero integer n. The n-writhes give the coefficients of some polynomial invariants for virtual knots including the index polynomial, the odd writhe polynomial and the affine index polynomial. It is obvious that the virtualization of a real crossing is an unknotting operation for virtual knots. The unknotting number by the virtualization is called the virtual unknotting number. Kishino found a virtual knot with special properties, called Kishino's Knot. Kishino's knot is a virtual unknotting number one knot whose n-writhes, the index polynomial, the odd writhe polynomial, the affine index polynomial, and the Jones polynomial are the same as those of the trivial knot. However, it is not known whether there exist many virtual knots with such properties.

In this talk, we construct infinitely many virtual knots with the same properties as Kishino's knot. By using the Miyazawa polynomial, we showed that these virtual knots are non-trivial and non-classical, and their knot types are all different. (Received February 29, 2020)

1158-57-169 **Andrew Yarmola***, Department of Mathematics, Fine Hall, Washington Road, Princeton, NJ 08544. *Searching for marked Kleinian groups.*

We will give a general overview of computational techniques involved in parameter space searches for marked Kleinian groups. These tools open the door to several difficult results in the study of hyperbolic 3-manifolds, including questions about volumes, tubes, Dehn surgery, and Margulis constants. An important aspect of this approach is that it can answer definitive questions about infinite families of Kleinian groups. The talk with focus on the necessary ideas and constructs for implementing such searches, as well as a brief review of applications and a discussion of future projects. (Received February 29, 2020)

1158-57-170 **Jim Hoste***, 1166 Cragmont Ave, Berkeley, CA 94708, and **Patrick D Shanahan** and **Cornelia Van Cott**. *Crosscap number of 2-bridge knots and the partial order on prime knots.* Preliminary report.

Suppose that K and J are distinct 2-bridge knots in the 3-sphere and that $K > J$, that is, there is an epimorphism of the knot group of K onto the knot group of J. Then $\gamma(K) \geq 3\gamma(J) - 4$, where $\gamma(K)$ is the *crosscap number*, or *non-orientable genus* of the knot K. (Received February 29, 2020)

1158-57-185 **Yanwen Luo*** (ywluo@ucdavis.edu), 200 atrium Way Apt 224, Davis, CA 95616. *The space of geodesic triangulations on surfaces.*

How to construct a straight-line embedding of any given planar graph is a fundamental problem in computational geometry. In 1963, Tutte provided a simple constructive method to produce a straight-line embedding of a 3-vertex-connected planar graph by solving a sparse linear system. In this talk, we will show that this idea can be applied to give a new proof of the Bloch-Connelly-Henderson theorem, which states that the space of geodesic triangulations of a convex polygon with a fixed combinatorial type is a contractible space. We will also prove that the space of geodesic triangulations on a flat torus with a fixed combinatorial type has the homotopy type of a torus. Finally, we will discuss the conjecture for the space of geodesic triangulations on hyperbolic surfaces. (Received March 01, 2020)

1158-57-187 **David Gabai**, **Robert Haraway**, **Robert Meyerhoff*** (meyerhog@bc.edu), **Nathaniel Thurston** and **Andrew Yarmola**. *The Complete Hyperbolic 3-Manifolds with a Maximal Cusp of Minimum Volume.*

We have proven that the figure-eight knot complement in the 3-sphere and its sibling are the complete hyperbolic 3-manifolds with a maximal cusp of minimum volume. This volume is the square root of 3. In this talk I'll discuss some of the history of this result and give some indication of the proof. (Received March 01, 2020)

1158-57-193 **Sujoy Mukherjee*** (mukherjee.166@osu.edu) and **Jozef H. Przytycki** (przytyck@gwu.edu). *On multi-term distributive homology.* Preliminary report.

Two-term distributive (rack) homology, introduced in 1990, and its enhancement, twisted quandle homology, are very useful tools for constructing link invariants and for studying self-distributive algebraic structures. The second author proposed the theory of multi-term distributive homology in 2011 to study the homology of self-distributive algebraic structures under a general framework.

After a brief discussion about the monoid of all binary operations on a set X and multi-shelves, I will introduce multi-term distributive homology. Finally, I will talk about our present work on the topic. (Received March 01, 2020)

1158-57-197 **Mark Brittenham*** (mbrittenham2@unl.edu), Department of Mathematics, Avery 203, University of Nebraska, Lincoln, NE 68588-0130, and **Susan Hermiller** (hermiller@unl.edu), Department of Mathematics, Avery 203, University of Nebraska, Lincoln, NE 68588-0130. *The further adventures of Bernhard and Jablan.*

In the 1990s Bernhard and Jablan independently proposed a method for computing the unknotting number of a knot, using minimal crossing number diagrams and crossing changes. This method has proved to have a remarkable track record of accurately predicting the unknotting numbers of knots, but in recent work we showed that it cannot have 100 percent accuracy. We will discuss this result, as well as what the data amassed in searching for counterexamples can say about other open problems on unknotting number. (Received March 01, 2020)

1158-57-200 **Susan Hermiller***, Department of Mathematics, University of Nebraska, Lincoln, NE 68588. *Fundamental groups of 3-manifolds and finite state automata.*

The word problem is solvable for all fundamental groups of compact 3-manifolds; that is, for each 3-manifold group there is an algorithm which, upon input of a word in the generators, determines whether the word represents the identity. In the case that the 3-manifold has no Nil or Sol prime factors, the word problem is solvable in quadratic time. In this talk I will discuss solutions the word problem for 3-manifold groups using finite state automata (computers with a finite amount of memory), and methods to build these algorithms. Based on joint projects with M. Brittenham and T. Susse, and with D. Holt, S. Rees, and T. Susse. (Received March 01, 2020)

1158-57-213 **Feng Luo**, **Matthias Goerner** and **Maria Trnkova*** (mtrnkova@math.ucdavis.edu), One Shields Ave., Davis, CA 95616. *Spun triangulations of closed hyperbolic 3-manifolds.* Preliminary report.

W. Thurston showed that every hyperbolic 3-manifold can be decomposed into a set of ideal tetrahedra and also simple closed geodesics (for closed manifolds). Unfortunately such a decomposition sometimes produces overlapping or flat tetrahedra, a so called non-geometric ideal triangulation. There is a conjecture that every cusped hyperbolic 3-manifold admits a geometric ideal triangulation. We are interested in spun (ideal) triangulations of closed hyperbolic 3-manifolds. SnapPy's census provides 11031 orientable closed hyperbolic 3-manifolds. According to HIKMOT all of them admit a geometric spun triangulation except one, it is known as Vol3. In this talk we show that Vol3 does not have any geometric ideal triangulation of small complexity up to 6. The main techniques that we use are Thurston's Dehn surgery theorem, Dehn parental test and a gluing variety. (Received March 02, 2020)

1158-57-216 **David B Ben*** (dmcreyno@purdue.edu), 150 N University St, W Lafayette, IN 47907-2067. *When are two manifolds the same?*

I will discuss the problem of deciding when two closed (hyperbolic) n-manifolds are isometric. The focus will be two fold. In one direction, one proves that certain invariants or collections of invariants can distinguish between two manifolds. In another direction, one constructs examples where these invariants fail to distinguish between two manifolds. (Received March 02, 2020)

1158-57-222 **Alexander Zupan*** (zupan@unl.edu). *Distance and intersection number in the reducing sphere complex.*

The Powell Conjecture proposes a generating set for the group of diffeomorphisms of the 3-sphere that preserve a Heegaard surface of genus g. In recent work, we showed that the Powell Conjecture is equivalent to the assertion that the reducing sphere complex is connected. Given a Heegaard surface Σ_g for S^3, the reducing sphere complex $\mathcal{R}(\Sigma_g)$ is the full subcomplex of the curve complex generated by those curves that bound disks in both handlebodies cut out by Σ_g. In this talk, we will prove that although any pair of reducing curves meeting in only four points is connected by a path in $\mathcal{R}(\Sigma_g)$, the distance between two such curves can be arbitrarily large, starkly contrasting the relationship between distance and intersection number in other related complexes. (Received March 02, 2020)

1158-57-226 **Matthias Goerner*** (enischte@gmail.com), 1 Kelton Ct, 8G, Oakland, CA 94611. *SnapPy, raytracing and verified computations.*

I will give a brief introduction to SnapPy, a software to study hyperbolic 3-manifolds, and demo some features I added recently, such as the raytraced inside view of a hyperbolic 3-manifold and verified computations of, e.g., the complex volume and the maximal cusp area matrix. (Received March 02, 2020)

1158-57-230 **Kevin Lamb***, 3601 Pacific Avenue, Stockton, CA 95211, and **Patrick Weed**. *A Distance for Circular Heegaard Splittings.*

For a knot $K \subset S^3$, its exterior $E(K) = S^3 \backslash \eta(K)$ has a singular foliation by Seifert surfaces of K derived from a circle-valued Morse function $f\colon E(K) \to S^1$. When f is self-indexing and has no critical points of index 0 or 3, the regular levels that separate the index-1 and index-2 critical points decompose $E(K)$ into a pair of compression bodies. We call such a decomposition a *circular Heegaard splitting* of $E(K)$. We define the notion of *circular distance* (similar to Hempel distance) for this class of Heegaard splitting and show that it can be bounded under certain circumstances. Specifically, if the circular distance of a circular Heegaard splitting is too large: (1) $E(K)$ can't contain low-genus incompressible surfaces, and (2) a minimal-genus Seifert surface for K is unique up to isotopy.

This is joint work with Patrick Weed. (Received March 02, 2020)

1158-57-231 **David Futer*** (dfuter@temple.edu), Mathematics Department, 1805 North Broad St., Suite 638, Philadelphia, PA 19122, and **Jessica S. Purcell** and **Saul Schleimer**. *Systoles and cosmetic surgeries.*

A pair of distinct slopes for a cusped hyperbolic manifold M is called a *cosmetic surgery pair* if the Dehn fillings along those slopes yield the same oriented 3–manifold. Gordon conjectured that such pairs do not exist.

I will describe a theorem that says any potential cosmetic surgery pairs on a hyperbolic manifold M belong to a finite list of slopes, whose size is determined by the *systole* (shortest closed geodesic) of M. If M is a knot complement in S^3, the result is particularly simple and there are typically no more than 10 pairs of slopes on the list. If M is a general cusped manifold, the list of slopes is larger but still tractable.

Using this result, we have checked the cosmetic surgery conjecture for the enumerated knots and the manifolds in the SnapPy census. I will describe the verification algorithm that results from the above theorem. (Received March 02, 2020)

1158-57-249 **D. Cooper**, **S. Tillmann** and **W. Worden*** (william.worden@rice.edu). *The Thurston norm via spun normal surfaces.*

The Thurston norm is a norm on the second homology of a hyperbolic 3-manifold which, given a homology class, returns the minimal complexity over all surfaces representing the class. The unit ball for this norm is a polyhedron symmetric about the origin, and by understanding the surfaces representing the vertices of this polyhedron one gets a wealth of information about the embedded surfaces in the manifold. In 2008, Cooper and Tillmann gave an algorithm for computing the Thurston norm ball of a closed manifold, using normal surfaces. I will discuss an adaptation of their techniques to the case of cusped hyperbolic 3-manifolds using spun normal surfaces, and give a demonstration of a computer program, available at https://pypi.org/project/tnorm/, that implements this algorithm. This is joint work with Daryl Cooper and Stephan Tillmann. (Received March 02, 2020)

1158-57-261 **Patrick Chu***, pychu@iu.edu. *Framed and Oriented 2 Dimensional Extended Topological Field Theories.* Preliminary report.

2 dimensional fully extended (2,1,0) topological field theories to the bicategory of algebras, bimodules, and intertwiners have already been classified in the oriented and framed extended cobordism categories. State sum models for computing these TFTs on closed oriented surfaces via triangulations/polygonalizations were described over nice families of fields. In this talk, we will discuss generalizing this state sum method to arbitrary fields using the classification results and also generalize this for framed cobordisms. (Received March 02, 2020)

1158-57-267 **Joshua A. Howie*** (jahowie@ucdavis.edu), Department of Mathematics, University of California, Davis, One Shields Ave, Davis, CA 95616. *An algorithm to decide if a link is alternating.*

In previous work, the author found a characterization of alternating link exteriors and used it to produce a normal surface algorithm which can decide if a knot is alternating. Similar results were obtained independently by Greene, Juhász, and Lackenby. The author's algorithm relied on a theorem of Jaco and Sedgwick which is not true for manifolds with multiple boundary tori. Here we overcome this problem by a careful study of normal spanning surfaces and their aggregate slopes. Consequently, we produce a normal surface algorithm which can decide if a link is alternating. (Received March 02, 2020)

1158-57-277 **Stanislav Jabuka*** (jabuka@unr.edu), Department of Mathematics and Statistics, University of Nevada, Reno, NV 89557. *Concordance crosscut number and rational Witt classes.*

The concordance crosscap number of a knot is the smallest crosscap number among all knots concordant to it, and it can be seen as interpolating between the classical 3-dimensional crosscap number and its 4-dimensional counterpart. To date only two results have appeared on computing this invariant. The first by Gengyu Zhang relies on the knot signature and determinant, while the second, due to Charles Livingston, uses the Alexander polynomial. We present in this talk a third way of thinking about the concordance crosscap number, one that uses the rational Witt class of a knot. We give examples to show that our methods give obstructions even when Zhang's and Livingston's do not. (Received March 02, 2020)

1158-57-278 **Puttipong Pongtanapaisan*** (puttipong-pongtanapaisan@uiowa.edu) and **Jason Joseph**. *Meridional rank and bridge number of knotted surfaces.* Preliminary report.

Meier and Zupan showed that every smoothly embedded surface in the 4-sphere can be put in bridge position. This is a 4-dimensional analog of bridge position for a knot in the 3-sphere, and gives rise to a complexity for knotted surfaces called the bridge number. In this talk, we discuss computations of the bridge numbers for surfaces that are not spun knots. The lower bound involves calculations of the meridional rank, which is the minimum number of meridional generators taken over all presentations of the fundamental group of the surface's exterior in the 4-sphere. (Received March 02, 2020)

1158-57-280 **Melissa Zhang*** (melissa.zhang@uga.edu) and **Matthew Stoffregen**. *Symmetries and Khovanov homology.*

Nearly a century ago, P. A. Smith studied symmetries of topological objects and their fixed-point sets under these actions. Nowadays, we can study symmetries through our homology-type invariants in low-dimensional

topology. In this talk, we'll explore the implications of a $\mathbb{Z}/p$ action on a link in its Khovanov homology, as well as some decategorified consequences. This is joint work with Matthew Stoffregen. (Received March 02, 2020)

1158-57-325 **Carson Rogers*** (carson.rogers@bc.edu). *Cosmetic twisting on fibered and hyperbolic knots.* Preliminary report.

The Cosmetic Crossing conjecture aims to classify all crossing changes on knots in S^3 which preserve knot type. In this talk, we examine an extension of this question to the operation of an *n-tangle replacement*. When n is even, this is known in the literature as a generalized crossing change. After summarizing previous work on that case, we discuss two results of the speaker which restrict cosmetic n-tangle replacements on fibered knots in rational homology spheres. In particular, we show that such knots do not admit cosmetic generalized crossing changes, extending a result of Kalfagianni.

Then, motivated by recent developments in hyperbolic Dehn surgery, we consider the case of hyperbolic knots. We explain how work of Cooper and Lackenby can be used to show that a hyperbolic knot only admits cosmetic generalized crossing changes of finitely many orders. To conclude, we pose questions about the potential to obtain an effective version of this result. (Received March 03, 2020)

1158-57-351 **David Bachman*** (bachman@pitzer.edu), **Ryan Derby-Talbot** (ryan.derby-talbot@fulbright.edu.vn) and **Eric Sedgwick** (esedgwick@cdm.depaul.edu). *Computing Heegaard Genus is NP-Hard.*

We show that Heegaard Genus <=g, the problem of deciding whether a triangulated 3-manifold admits a Heegaard splitting of genus less than or equal to g, is NP-hard. The result follows from a quadratic time reduction of the NP-complete problem CNF-SAT to Heegaard Genus <= g. This is joint work with Ryan Derby-Talbot and Eric Sedgwick (Received March 03, 2020)

58 ► Global analysis, analysis on manifolds

1158-58-328 **Brian Benson*** (bbenson@ucr.edu). *Algorithms for Computing the Cheeger Constants of Hyperbolic Manifolds.*

The Cheeger constant is an invariant defined on a Riemannian manifold, from which lower and upper bounds on Laplace eigenvalues of the manifold can be established. Further, the Cheeger constant is related to the isoperimetric problem for a given manifold, specifically, finding subsets of minimum perimeter amongst those have a fixed volume. I will outline work on and results from various recent projects with collaborators Grant Lakeland, Jeffrey Meyer, and Holger Then which directly compute and certify the Cheeger constants of hyperbolic surfaces. In addition, I hope to discuss challenges related to computing the Cheeger constant of hyperbolic 3-manifolds. (Received March 03, 2020)

60 ► Probability theory and stochastic processes

1158-60-140 **Michael I Ham***, LANL, LOS ALAMOS, NM 87545. *Two color ejecta measurement mystery in Sn.* Preliminary report.

Passing two colors of lasers through a cloud of ejecta can determine the size of particles. Here results are compared between a successful series of experiments in Sn where small ejecta were observed and a second unsuccessful set where no small ejecta results were observed. Comparisons with MIE scattering in both instances are provided. A discussion about possible reasons for the discrepancies follows. (Received February 27, 2020)

1158-60-288 **Edward C Keppelmann*** (keppelma@unr.edu), Department of Mathematics & Statistics MS 084, University of Nevada Reno, Reno, NV 89557. *Expert Games of Chance.* Preliminary report.

A technology based way to think about games in a math circle is to code them as an expert that the circle participants can play against. We have had the most success in this experience with games of chance. In the game of shut-up for example the approach adopted by the computer (after being told what to maximize and allowing it to look many moves ahead) is one that makes immediate sense probabilistically and turns out to be quite instructive. On the other hand, in games like Yhatzee, where optimal strategies are far more illusive, the issues are quite deep and varied. In this case it is not even clear what the computer should optimize. We

illustrate this on a small scale with a simplified version of this familiar dice game where such analysis is far more tenable. (Received March 03, 2020)

62 ► Statistics

1158-62-20 **Michael Reyes** and **Andrey Sarantsev***, 1664 N Virginia St, Davidson Math & Science Center 314, Reno, NV 89557. *Long-Term Stock Market Bayesian Modeling.*

We fit a dynamic factor model of the Standard & Poor Composite Index, with earnings, dividends, short- and long-term Treasury rates as factors. We use Bayesian inference. We evaluate the conventional wisdom that future long-term market returns are likely lower than the historical average of 6-7%. (Received February 03, 2020)

1158-62-78 **Hanna Makaruk*** (hanna_m@lanl.gov), MS H810 LANL, Los Alamos, NM 87545. *Inverse Problem of Recreating a Process from the Data Collected in One-of -a -Kind experiment: why Range of True Experimental Values and Measurement Uncertainty are Independent from each other.* Preliminary report.

A golden standard in science is to repeat an experiment a statistically significant number of times, recording data using the same set of detectors and the same data analysis methodology. In such case experimental error includes both the range of true values generated by repetitions of the experiment, and measurement uncertainty caused by the detector. They are independent. Range of true values depends on the physics of the experiment, while measurement uncertainty depends on the measurement method (properties of the detector not of the experiment). When only data from one–of –a kind experiment are available, only the measurement uncertainty is reported. It gives no information about the range, in which the true values of experiment would spread if the experiment was repeated. One needs to observe that reliability of a data point is independent from its measurement uncertainty. However, in practice reliable measurement methods frequently have high measurement uncertainty, while low reliability methods are applied to limit measurement uncertainty. Comparison of reliable data with high measurement uncertainty to not so reliable data measured with low uncertainty is discussed – in different scenarios different data analysis methods are applicable. (Received February 21, 2020)

1158-62-79 **Adriano Zanin Zambom*** (adriano.zambom@csun.edu), 6651 Neddy Ave, West Hills, CA 91307, and **Gregory Matthews**. *Sure Independence Screening in the Presence of Missing Data.*

Variable selection in ultra-high dimensional data sets is an increasingly prevalent issue with the readily available data arising from, for example, genome-wide associations studies or gene expression data. When the dimension of the feature space is exponentially larger than the sample size, it is desirable to screen out unimportant predictors in order to bring the dimension down to a moderate scale. In this paper we consider the case when observations of the predictors are missing at random. We propose performing screening using the marginal linear correlation coefficient between each predictor and the response variable accounting for the missing data using maximum likelihood estimation. This method is shown to have the sure screening property. Moreover, a novel method of screening that uses additional predictors when estimating the correlation coefficient is proposed. Simulations show that simply performing screening using pairwise complete observations is out-performed by both the proposed methods and is not recommended. Finally, the proposed methods are applied to a gene expression study on prostate cancer. (Received February 21, 2020)

1158-62-107 **Shishen Xie*** (xies@uhd.edu), Department of Mathematics and Statistics, University of Houston-Downtown, 1 Main Street, Houston, TX 77002. *Data Analysis to predict the effectiveness of a clinical treatment.* Preliminary report.

In this study clinical data is analyzed on the effectiveness of an anti-tuberculosis treatment, and the results at different stages of the treatment is predicted. The data is collected from monthly IFN-gamma levels of subjects who were diagnosed tuberculosis positive and participated in the treatment. Statistical methods are developed to compare the mean IFN-gamma levels on the different stages of the treatment and to seek the lowest mean IFN-gamma level in the duration of the treatment. Recommendations based on the data analysis are made to the modification and improvement of the treatment. (Received February 25, 2020)

1158-62-144 **Sky K. Sjue*** (sjue@lanl.gov). *Velocity and density reconstructions of detonation waves in conventional high explosives from proton radiographic images obtained at Los Alamos Neutron Science Center.*

We will discuss the imaging techniques used in proton radiography at Los Alamos, then discuss some approximations for the non-stationary chromatic blur due to non-constant energy loss of the protons through a dynamic experiment. A low order form for the density profile of a detonation wave in the high explosive will be assumed based on the ZND model. The effect of motion blur will be included due to integration over proton micro-pulses 5 nanoseconds wide for a duration of 50 nanoseconds in a single image. The time averaging will be performed consistently through a series of images. These assumptions will be subjected to a constant mass constraint to estimate the most likely density of the explosives in the reaction zone, the width of the reaction zone, and the velocity of the detonation wave, plus confidence intervals for these quantities of interest. The results from this forward analysis will be compared to an alternative analysis based on a more direct Abel inversion of the images. (Received February 28, 2020)

1158-62-212 **Türkü Özlüm Çelik**, **Asgar Jamneshan*** (jasgar@math.ucla.edu), **Guido Montúfar**, **Bernd Sturmfels** and **Lorenzo Venturello**. *Wasserstein Distance to Independence Models.*

An independence model for discrete random variables is a Segre-Veronese variety in a probability simplex. That simplex is a metric space with respect to the Wasserstein distance, a polyhedral norm whose unit ball is dual to the Lipschitz polytope. Given any empirical distribution, we seek to minimize its Wasserstein distance to the model. The solution to this optimization problem is a piecewise algebraic function of the given data. We compute this function explicitly in small instances, and we characterize its combinatorial structure and algebraic degrees in the general case. (Received March 02, 2020)

1158-62-281 **Andrew Kaplan*** (amkaplan3@gmail.com), Mathematical Sciences Building, 520 Portola Plaza, Room 8209, University of California, Los Angeles, Los Angeles, CA 90095. *Nonparametric Estimation of the Recursive Point Process Model with Application to Mumps in Pennsylvania.*

The self-exciting Hawkes point process model (Hawkes, 1971) has been used to describe and forecast communicable diseases. A variant of the Hawkes model, called the recursive model, was proposed by Schoenberg et al. (2018) and shown to fit well to cases of Rocky Mountain Spotted Fever in California. Unlike the Hawkes model, the recursive model allows the productivity to vary as the overall rate of incidence of the disease varies. Here, we extend the data-driven non-parametric E-M method of Marsan and Lengliné (2008) in order to fit the recursive model without assuming a particular functional form for the productivity. The non-parametric recursive model is trained to fit to weekly reported cases of mumps in Pennsylvania during the January 1970 - September 1990 time frame and then assessed using one week forecasts for the October 1990 - December 2001 time period. Both its training and predictive ability are evaluated compared to that of other candidate models, such as Hawkes and SEIR (Susceptible, Exposed, Infected, Recovered) compartmental models. (Received March 02, 2020)

1158-62-333 **Pallavi Basu** and **Luella Fu*** (luella@sfsu.edu), 1600 Holloway Ave, Mathematics Department, San Francisco, CA 94132, and **Wenguang Sun**. *Optimal Control of False Discovery Exceedance.* Preliminary report.

In multiple-testing, scientific findings usually come from experiments that have been performed only a few times or even just once, but researchers often use the false discovery rate (FDR), a Type I error control based on averages, in their testing procedures. It is more desirable to control the FDP for each experiment so that the results from a single operation of a testing procedure are reliable. From this reproducibility standpoint, the false discovery exceedance (FDX) is very relevant, because it does not rely on averages as FDR does. The FDX keeps a low probability on the event that the FDP exceeds an acceptable proportion each time the testing procedure is carried out. We propose an FDX procedure using an empirical Bayes framework and compare this method with other well-known Type I error criterion. (Received March 03, 2020)

1158-62-334 **Mohammad Rafiqul Islam*** (mislam02@student.ysu.edu), 830 Ohio Avenue, Apt 7, Youngstown, OH 44504, and **Nguyet Nguyen** (ntnguyen01@ysu.edu), 522 Lincoln Building, Department of Mathematics and Statistics, Youngstown, OH 44505. *Comparison of ARIMA, Artificial Neural Network (ANN), and Geometric Brownian Motion (GBM) models for stock price prediction.* Preliminary report.

Time series analysis of daily basis stock data and building predictive models are complicated because of the nonstationarity of the data. However, alternative methods could be used to avoid such complexity and to have a similar or better approximation. This paper presents a comparative study for stock price prediction using

three different methods, namely autoregressive integrated moving average (ARIMA), artificial neural network (ANN), and stochastic process-geometric Brownian motion (GBM). Each of the methods has been used to build predictive models using historical stock data collected from yahoo finance. Empirical results show that the neural network method gives a better approximation. (Received March 03, 2020)

1158-62-348 **Guangliang Chen*** (guangliang.chen@sjsu.edu), San Jose, CA 95192. *All data are "documents": A scalable spectral clustering framework based on landmark points and cosine similarity.*

We present a unified scalable computing framework for various versions of spectral clustering, such as the Ng-Jordan-Weiss algorithm (NIPS '01), Normalized Cut (Shi and Malik, 2000), and Diffusion Maps (Coifman and Lafon, 2006). We first consider the special setting of cosine similarity for clustering sparse data (e.g., documents under the bag-of-words model) or data with at most a few hundred dimensions (e.g., small images). We show that in such cases, spectral clustering can be implemented solely based on three kinds of efficient matrix operations on the data matrix – elementwise manipulation, matrix-vector multiplication and low-rank SVD. Next, for any given similarity (and for any kind of data), we introduce a landmark-based technique to convert the data (and the selected landmarks) into a "document-term" matrix and then apply the scalable implementation of spectral clustering with cosine similarity to cluster them. Lastly, we demonstrate the performance of our proposed algorithm by comparing it with a few existing methods on several benchmark data sets. (Received March 03, 2020)

1158-62-355 **Mark V Albert*** (mark.albert@unt.edu), **Riyad Bin Rafiq**, **François Modave** and **Shion Guha**. *Validation methods for machine learning to promote real-world applicability in medicine.*

The impact of AI on health care has been dramatic; however, there is a considerable degree of skepticism among clinicians about the real-world applicability of advanced predictive models; for this reason, it is particularly important to emphasize proper model validation in machine learning. Often model skepticism is well-placed as modelers may overclaim the real-world replicability for their models, understate the known limitations, or simply not be aware of the hidden limits of the modeling approach. Explanations limited to rigorous and thorough justification of all model design decisions may not be practical given model complexity. This also becomes more challenging as state-of-the-art models with the highest benchmark accuracy are becoming less interpretable, e.g. ensemble methods or deep learning. Here we provide an overview of the common limitations of model validation methods we have observed in medical predictive modeling, and we present solutions with a focus on strengthening the validity of predictive models in medical contexts. (Received March 03, 2020)

1158-62-373 **Md Sazib Hasan*** (mdsazib.hasan@dixie.edu), 368 S Mall Drive, I206, St George, UT 84790, and **Kalimuthu Krishnamoorthy**. *Inference for Delta-Lognormal Distribution.*

Inference based on delta-lognormal distribution will be addressed in this talk. There are practical situations where one encounters data that are skewed and contain a relatively high proportion of zeros. For example, the diagnostic test charges data in a study include zeroes because some patients had no diagnostic tests during the study and nonzero diagnostic testing charges were highly skewed to the right. In workplace exposure/pollution assessment studies, measurements collected from individual workers or locations within a working facility often include zeroes, and positive measurements are highly skewed. Inference based on log-normal distribution using fiducial inference will be discussed. Fiducial inference is practically very simple to implement and very accurate in result. (Received March 04, 2020)

65 ► Numerical analysis

1158-65-6 **Lewa' M Alzaleq*** (lewa.alzaleq@wsu.edu), Department of Mathematics and Statistics, Pullman, WA 99164, and **Manoranjan**. *Analysis of a Fish Population Model.*

Diffusion equations with nonlinear reactive terms are frequently used to describe dissipative dynamical systems in population modeling. In the talk, we will propose and study a Fish population model. A phase plane analysis of the model will be carried out. We will provide analytic solitary wave solutions when the wave speed is constant or an arbitrary function. The model will be studied using a finite difference method with exponential upwinding. The interactions of nonperiodic solitary waves will be investigated numerically. (Received October 18, 2019)

1158-65-40 **Mirjeta Pasha***, 1300 Lefton Esplanade, Kent, OH 44240, and **Alessandro Buccini** and **Lothar Reichel**, Kent, OH 44242. *Linearized Krylov Subspace Bregman Iteration with Nonnegativity Constraint.*

Bregman-type iterative methods have attracted considerable attention in recent years due to their ease of implementation and the high quality of the computed solutions they deliver. However, these iterative methods may require a large number of iterations and this reduces their attractiveness. This talk discusses a computationally attractive linearized Bregman algorithm by projecting the problem to be solved into an appropriately chosen low-dimensional Krylov subspace. The projection reduces the computational effort required for each iteration. A variant of this solution method, in which nonnegativity of each computed iterate is imposed, also is described. Convergence of the latter method is shown. Extensive numerical examples illustrate the performance of the proposed methods. (Received February 13, 2020)

1158-65-62 **Daniele Venturi*** (venturi@ucsc.edu), 1156 high street, Santa Cruz, CA 95064. *Dynamically orthogonal tensor methods for high-dimensional nonlinear PDEs.*

I will present a new method to compute the numerical solution of high-dimensional time-dependent PDEs on tensor manifolds with constant rank. The key idea relies on a hierarchical decomposition of the solution space in terms of a sequence of nested projections. This process can be visualized in terms of binary trees, and it yields series expansions of the solution that include classical Tensor-Train (TT) and Hierarchical Tucker (HT) representations. By enforcing suitable dynamic orthogonality conditions at each level of the binary tree, we obtain coupled evolution equations for the tensor modes spanning each subspace. This allows us to integrate time-dependent nonlinear PDEs on tensor manifolds with constant rank, with no need for computationally expensive rank reduction methods such as high-order SVD. Numerical examples are presented and discussed for high-dimensional linear hyperbolic and parabolic PDEs on separable periodic domains. (Received February 19, 2020)

1158-65-221 **Heyrim Cho*** (heyrimc@ucr.edu) and **Howard Elman**. *Adaptive Reduced Basis Method for Anisotropic High-dimensional Stochastic PDEs.*

The combination of reduced basis and collocation methods enables efficient and accurate evaluation of the solutions to parameterized PDEs. In this work, we study the stochastic collocation methods that can be combined with reduced basis methods to solve high-dimensional parameterized stochastic PDEs. We also propose an adaptive algorithm using a probabilistic collocation method (PCM) and ANOVA decomposition. This procedure involves two stages. First, the method employs an ANOVA decomposition to identify the effective dimensions, i.e., subspaces of the parameter space in which the contributions to the solution are larger, and sort the reduced basis solution in descending order of error. Then, the adaptive search refines the parametric space by increasing the order of polynomials until the algorithm is terminated by a saturation constraint. We demonstrate the effectiveness of the proposed algorithm for solving a stationary stochastic convection diffusion equation, a benchmark problem chosen because solutions contain steep boundary layers and anisotropic features. We show that two stages of adaptivity are critical in a benchmark problem with anisotropic stochasticity. (Received March 02, 2020)

1158-65-234 **Yifan Wang*** (qcutexu@berkeley.edu), UC berkeley, Berkeley, CA 94704, and **Suncica Canic**, berkeley, CA 94704. *Numerical simulation of the islet encapsulation device using fluid poroelastic structure interaction model.*

Artificial pancreas is a microelectromechanical (MEMS) device that encapsulates the transplanted islets by using the silicon nanopore membranes and allows islets to survive without immunosuppressive therapy. Our work is motivated by promoting the survival time of transplanted islets inside the device. We propose a Navier-Stokes-Biot model involving a poroelastic material and incompressible Newtonian fluid to mimic the fluid-poroelastic interaction between the islets compartment and blood flow. A monolithic approach is adopted to solve the proposed problem, with Nitsche's method to enforce the coupling conditions at the fluid-poroelastic structure interface. Numerical simulations are carried out carefully with promising findings in improving the design of the artificial pancreas. (Received March 02, 2020)

1158-65-238 **Jamie Haddock*** (jhaddock@math.ucla.edu) and **Anna Ma**. *Analyzing Hybrid Randomized and Greedy Projection Methods.*

Stochastic iterative algorithms have gained recent interest for solving large-scale systems of equations, Ax=y. One such example is the Randomized Kaczmarz (RK) algorithm, which acts only on single rows of the matrix A at a time. While RK randomly selects a row, Motzkin's algorithm employs a greedy row selection; the Sampling

Kaczmarz-Motzkin (SKM) algorithm combines these two strategies. In this talk, we present a convergence analysis for SKM which interpolates between RK and Motzkin's algorithm. (Received March 02, 2020)

1158-65-271 **Camille Carvalho***, Applied Math Deparment, 5200 N Lake Rd, Merced, CA. *Fourth order and energy stable FDTD method for Maxwell's equations in metamaterials.*

We present a novel fully fourth order in time and space finite difference method and its analysis for the time domain Maxwell's equations in metamaterials. The proposed method is based on the second order formulation of the time domain Maxwell's equations for the electric and magnetic fields along with the polarization and magnetization current densities. Our discretization employs fourth order staggering in space of different field components and the modified equation approach to obtain fourth order accuracy in time. The designed numerical schemes are conditionally stable as their associated discrete energy mimics the one from the continuous model under a CFL condition. Numerical simulations will be provided to illustrate fourth order convergence as well as compare with second order schemes. (Received March 02, 2020)

1158-65-338 **Malgorzata Peszynska*** (mpesz@math.oregonstate.edu), Oregon State University, Department of Mathematics, Corvallis, OR 97331. *Multiscale computational model of biofilm in porous media: from images to finite element analysis to the outlook beyond DNS.*

We discuss a project motivated by applications in reservoir engineering with data from micro-ct tomography of biofilm formation at the pore-scale. We motivate the model posed as a system of parabolic variational inequalities coupled by Monod growth terms. Next we present rigorous finite element analysis of the approximate solutions, and illustrate the model with simulations. We also discuss upscaling as well as outlook beyond Direct Numerical Simulations. This research is joint with many collaborators and students to be named in the talk. (Received March 03, 2020)

1158-65-339 **Jean M. Sexton*** (jmsexton@lbl.gov), Lawrence Berkeley National Laboratory, 1 Cyclotron Road Mailstop 50A3111, Berkeley, CA 94720. *A Deferred Correction Coupling Strategy for Cosmological Simulations.*

We present a strategy for coupling the equations of baryonic gas dynamics to those for radiative heating and cooling in the context of Nyx, an N-body + gas dynamics code for large-scale cosmological simulations. The radiative and advective processes have differing time scales. To avoid Strang splitting errors, we couple the processes with a simplified spectral deferred correction strategy. For certain physical regimes, this improved coupling strategy improves stability and reduces the cost of integrating the heating and cooling terms. Numerical results will be presented for relevant cosmological simulations, including our Lyman-alpha forest test case. (Received March 03, 2020)

68 ► Computer science

1158-68-45 **Arnaud de Mesmay*** (arnaud.demesmay@u-pem.fr), Laboratoire d'Informatique Gaspard Monge, 5 boulevard Descartes, CEDEX 2, 77452 Marne la Vallée, France, **Marcus Schaefer** (mschaefer@cdm.depaul.edu), dePaul University, School of Computing, Chicago, IL 60604, and **Eric Sedgwick** (esedgwick@cdm.depaul.edu), dePaul University, School of Computing, Chicago, IL 60604. *Link Crossing Number is NP-hard.*

We show that determining the crossing number of a link is NP-hard. For some weaker notions of link equivalence, we also show NP-completeness. (Received February 14, 2020)

1158-68-295 **Gobind S. Puniani*** (gobind_puniani@mail.fresnostate.edu), **Manuel Chacón** and **Mario Banuelos**. *The Role of Activation Functions in Deep Learning Architectures.* Preliminary report.

Artificial neural networks are computer models that "learn" to perform certain tasks (e.g. image classification) by processing vast amounts of labelled input data. Like the neural networks in our brain, these artificial neural networks learn how to perform a task by reinforcing pathways that lead to success and abandoning pathways that lead to failure. Activation functions play a key role in neural networks by introducing non-linearity into the system. However, activation functions are often overlooked by researchers in the field of deep learning, who prefer to rely on standard ones like ReLU and Tanh without considering whether they are the best for a given task. We investigate two new activation functions, TAct and mTAct, which interpolate between other activation functions, depending on the task. We explore the role of these functions on accuracy over a multitude of neural network architectures varying in complexity and learning rate. These activation functions were tested in the context of image classification, using an extensive image dataset known as CIFAR-10. Preliminary results show

that TAct and mTAct seem to perform better than standard activation functions on simple models with few parameters and low learning rates. (Received March 03, 2020)

1158-68-324 **Erica L Sawyer***, ericasawyer@mail.fresnostate.edu, and **Mario Banuelos**. *Deep Learning Methods for Detecting Structural Variants in Related Individuals.* Preliminary report.

Structural variants (SVs) are observed differences between the sequenced genome of an individual as compared to a reference genome for that species. These differences include deletions, inversions, insertions, and duplications. Since some genomic variations are associated with certain diseases, our work focuses on developing methods to detect such changes. Current methods to detect structural variation still suffer from high false positive rates, and we seek to develop machine learning and deep learning methods for identifying probable deletions in DNA based on genomic information of related individuals.

In this talk, we explain the implementation of neural networks, gradient descent, and logistic regression to predict SVs. We then introduce our preliminary model, which uses the observed genomic information of two parents and an offspring to predict locations of SVs in the genome of the child. Furthermore, we discuss the expansion of our model to simultaneously predict SVs for both the parents and the child and investigate the performance of this model under different neural network architectures using various performance metrics. (Received March 03, 2020)

1158-68-360 **Yu-Hang Tang*** (tang.maxin@gmail.com), 1 Cyclotron Rd, Berkeley, CA 94720. *Outlier Detection on Experimental Data Using Gaussian Process Regression with A Non-Stationary Noise Model.* Preliminary report.

Supervised machine learning performs optimally using accurately labeled ground truth as training data. However, many next-generation high-throughput chemical experiment techniques produce intrinsically noisy data. As such, it is very challenging to train machine learning models using this type of noisy data. We propose a non-stationary noise model on top of the Gaussian process regression framework. Together with a graph kernel that operates directly on graph representations of molecules, our method can infer the reliability of chromatography data on a sample-by-sample basis. This information can improve the trained machine learning model and prevents overfitting. More importantly, our method can identify outliers at a per-sample granularity and can thus serve as a feedback for experiment automation and manual checkup. (Received March 03, 2020)

1158-68-363 **Alexis Chung*** (nycrick@gmail.com), Cresskill, NJ , and **Richard Kyung**, Cresskill, NJ. *Mathematical and Physical Analysis of a Violin Note Using Computational Simulations.*

The amplitude of the sound as a function of time is well displayed on a spectrum, an ordered array of the components. We used such time spectrums to slice out the files or certain parts of a music note. In this paper, mathematical and physical knowledge were used to perform the spectral analysis of Violin. First, the wave forms and spectrums of Violin were obtained and they were compared to other instruments. Compared to Violin, it was interesting to see that several instruments, such as brass instruments have much more energy in their second, third, and/or third harmonics than in the first frequency. Most of them generate a fuzzy tone. Also, Compared to Violin, the flute shows the purest tones until 1000 Hz and most fuzzy mode after the 1000Hz. The brass and woodwind instruments show most of their energy in the second and third harmonics rather than the first frequency. Specifically, the string instrument such as Violin shows strong first and third harmonic components. Finally, psychoacoustics that includes various topics are studied how it is relevant to music psychology and music therapy. (Received March 03, 2020)

1158-68-365 **Alex Eugene Han*** (nycrick@gmail.com), Cresskill, NJ , and **Richard Kyung**, Cresskill, NJ. *A Study on the Prediction of Economic Indicators Using Mathematical and Computational Statistics.*

Economic indicators are fluctuating up-and-down during periods of such as recession and expansion. Due to the complexity of the data that include trend and irregular components, general approximation of the cyclic factors for the long-term behavior of economic data were studied for the presented analysis.

The sum of squares total, the sum of squares regression, and the sum of squares error were employed for modeling and predicting which were performed in an iterative way using a linear exponential smoothing technique with sequential updating equations. Smoothing parameters were used to estimate the trend and level.

In this paper major goals such as predicting, modeling, and characterization were performed in order to get a better strategy of forecasting the model.

As the last step in the model building, Analysis of Variance technique was used to assess the overall errors using the R-statistics and the least squares criterion was used to minimize the sum of square of vertical deviations. (Received March 03, 2020)

1158-68-372 **Jewon Yoo*** (nycrick@gmail.com), Cresskill, NJ , and **Richard Kyung**, Cresskill, NJ.
Study on the Forecasting of Stock Market Using Statistical and Computational Analysis.

Even though it is noted that the US fundamentals remain strong and that the central bank is not prepared to use additional stimulus measures aside from rate cuts, Wall Street closed deeply in the red on a volatile session, as an emergency rate cut by the Federal Reserve failed to ease concerns of slower growth ahead due to the coronavirus outbreak.

To find the underlying trend or fluctuation of an economic data, such as stock market, multiple types of data and other indices need to be collected over time. Because these data sets display change as time progresses, time series analysis is necessary to be able to analyze such sequential data by using math, statistics, and computation. For time series analysis and forecasting, this research used the a distribution method and the concept of return periods to map out how frequently certain data values appear.

In this paper, computational tools were used for the time series analysis. Since this data requires a proper statistical method in which the fitting model exactly matches the data, least squares method was used to minimize the sum of the squares of the deviations between the assumed model and the actual data. (Received March 04, 2020)

1158-68-374 **Dongeon Lee*** (nycrick@gmail.com), Cresskill, NJ , and **Richard Kyung**, Cresskill, NJ.
Study on the Efficient Algorithms to Reduce Noise Levels Using the Proposed Filters and Window Functions.

Noise reduction is essential to produce audible sound files and proceed precise communication; it is also of vital importance in creating digital images. Especially, when noise interferes with speech, the noise obscures the speech and results the speech in slightly degraded to completely unintelligible, depending on its amount and type.

Because we are relatively more sensitive to the higher frequency sounds than lower frequency sounds, we are vulnerable to noises, mainly consisted of high frequency sounds compared to main sound sources such as voice.

To emphasize and enhance the original signal, we diminish the noise and amplify the signal.

Numerous filters are designed to reduce the effects of noise and enhance the original sound. These noise removing systems can be improved through using a proper algorithms with various filters and windows.

We implicate the filter design at simple trigonometric functions and realistic samples to substantiate the noise removing efficiency of the filter design. We used MATLAB to analyze the audio files and execute Fast Fourier Transform.

The main objective for this research is to find the best-fit filter design in order to reach the highest efficiency in noise reduction. (Received March 04, 2020)

70 ► Mechanics of particles and systems

1158-70-380 **Yoonha Lee***, Kamuela, HI. *Study on the Spinning Maple Seed Inspired Turbine Blades Using Mechanical and Computational Analysis.*

The objective of this study is to efficiently find the best-fit model of turbine blade structures using mechanical and computational analysis. The research design was inspired from examining the falling of maple seeds. Maple seeds fall extremely slowly due to its unique geometrical structure and their rotation is fast enough to use its maximum air resistance creating high lift forces. To spread the seed widely, the maple seed auto-rotates as it falls reducing the dropping velocity even though the maple seed has a high mass to surface ratio. Not only does the maple seed have a low dropping velocity but also a high motion stability, the flight characteristics of the maple seed is an effective aerodynamic model. To find the best-fit model for various designs of efficient engine turbine structures, models of the National Advisory Committee for Aeronautics were used to describe a large number of airfoil sections. In this research, computer simulation was executed to examine the properties of the various types of blades. With the use of simulation tools, the fluid dynamic properties such as drag force, lift force, angle of attack and effectiveness of the maple seed inspired blades of engine turbine were examined. (Received March 04, 2020)

76 ► Fluid mechanics

1158-76-53 **Yonatan Dukler**, **Hangjie Ji*** (hangjie@math.ucla.edu), **Claudia Falcon** and **Andrea Bertozzi**. *A theory for undercompressive shocks in tears of wine.*

We revisit the tears of wine problem for thin films in water-ethanol mixtures and present a new model for the climbing dynamics. The new formulation includes a Marangoni stress balanced by both the normal and tangential components of gravity as well as surface tension which lead to distinctly different behavior. The prior literature did not address the wine tears but rather the behavior of the film at earlier stages and the behavior of the meniscus. In the lubrication limit we obtain an equation that is already well-known for rising films in the presence of thermal gradients. Such models can exhibit non-classical shocks that are undercompressive. We present basic theory that allows one to identify the signature of an undercompressive (UC) wave. We observe both compressive and undercompressive waves in new experiments and we argue that, in the case of a pre-swirled glass, the famous "wine tears" emerge from a reverse undercompressive shock originating at the meniscus. (Received February 17, 2020)

1158-76-252 **Francois Blanchette***, 5200 N. Lake Rd, Merced, CA 95343. *Modeling and simulations of porous marine aggregates.*

Settling marine aggregates plays an important role in transporting carbon from the surface ocean to the deep ocean. Investigations of settling rates is critical to understand the ecological impact of these particles. We present first a numerical study of the settling of a porous sphere in a density-stratified ambient fluid. We quantify the retention time, during which the sphere is nearly at rest as a function of various governing parameters. However, naturally occurring marine aggregates are very irregular, having in fact been shown to have a fractal structure. We therefore study numerically the fluid forces acting on aggregates formed by the diffusion-limited aggregation of cubic particles. We characterize the drag of translation flows, the torque of rotational flows, and the straining force of extensional flows acting on aggregates as a function of their size and mode of formation. Time permitting, we will revisit the formation mechanism of marine aggregates to incorporate the results obtained on their drag and torque. We study numerically the formation of aggregates using a Brownian dynamics model where the mobility tensor of each aggregate depends on its shape and size as computed by our boundary integral method. (Received March 02, 2020)

1158-76-290 **Aleksey S Telyakovskiy*** (alekseyt@unr.edu), Department of Math & Stat, University of Nevada, Reno, NV 89557. *Approximate method for the analysis of the groundwater flows.* Preliminary report.

The Boussinesq equation describes water flows in unconfined groundwater aquifers under the Dupuit assumption that the equipotential lines are vertical, making flow horizontal. It is a nonlinear diffusion equation with diffusivity depending linearly on water head. The generalized Boussinesq equation or porous medium equation is an equation where the diffusivity is a power law function of water head. For certain types of initial and boundary conditions similarity reduction is possible. Here we construct approximate analytical solutions that respect the scaling properties of the equations. (Received March 03, 2020)

78 ► Optics, electromagnetic theory

1158-78-127 **Michail D Todorov*** (mtod@tu-sofia.bg), 6111 Estelle St, San Diego, CA 92115. *Coupled System of Nonlinear Schroedinger equations: Nonlinearity and/or Integrability.*

For a coupled system of nonlinear Schroedinger equations (CSNSE) we investigate numerically the head-on collision dynamics of solitons. In the case of general elliptic polarization, exact solutions for the shapes of steadily propagating solitons are not available. We use a superposition of sech-like polarized solitons as the initial condition for investigating the soliton dynamics. We consider the interactions with and without cross-modulation, with stationary shapes and with breathers. For general nontrivial cross-modulation, a polarization shock takes place after the collision. For moderate and large alues of the nonlinear coupling parameter, additional solitons are created during the collision of the initial ones. Depending on the initial phases of the solitons, the polarizations of the system of solitons after the collision change, even for trivial cross-modulation. This sets the limits of practical validity of the celebrated Manakov solution. In the majority of cases the interaction is ostensibly inelastic: either one of the solitons virtually disappears, or additional solitons are born after the interaction. Since the Manakov system loses its full integrability when the nontrivial nonlinear coupling is present, the approach for its study is numerical. (Received February 26, 2020)

1158-78-291 **Omar DeGuchy*** (odeguchy@ucmerced.edu), **Arnold D Kim**, **Roummel F Marcia** and **Chrysoula Tsogka**. *Forward and Inverse Scattering In Synthetic Aperture Radar Using Machine Learning.*

In this work we consider machine learning for solving the forward and inverse scattering problems associated with the remote sensing application of synthetic aperture radar (SAR). Our approach consists of formulating the inverse problem as a linear system of the form $A\rho = d$ where d is the measured data, ρ the unknown reflectivity and A is the sensing matrix that depends on the characteristics of the imaging system. In all generality the sensing matrix is typically unknown and the reflectivity is reconstructed using an approximation of the sensing matrix. The quality of the reconstruction is critically dependent on the accuracy of the sensing matrix model. The idea that we pursue in this work, uses machine learning so as to recover the sensing matrix A or its inverse from a set of data corresponding to known reflectivities. To keep the algorithm simple and interpretable, we use a single, fully connected layer where the weights of the network correspond to the matrix A or its recovered inverse. We show with numerical simulations that the learned version of the sensing matrix is close to the actual sensing matrix while its learned inverse leads to improved reflectivity reconstructions. We provide evidence that the performance is robust over a range of SNR values. (Received March 03, 2020)

81 ► Quantum theory

1158-81-76 **Louis H Kauffman*** (kauffman@uic.edu), Louis H Kauffman, 5530 South Shore Drive, Apt 7C, Chicago, IL 60637-1946. *Majorana Fermions and the Majorana Dirac Equation.* Preliminary report.

This talk is joint work with Peter Rowlands. We examine how the Clifford algebra associated with Majorana Fermions is related to Majorana's original approach to real solutions to the Dirac equation. (Received February 20, 2020)

1158-81-100 **Rasulkhozha Sultonkhozhaevich Sharafiddinov*** (rasulkhozha@yahoo.com), Theoretical Physics Division, Institute of Nuclear Physics, Uzbekistan Academy of Sciences, Tashkent, Ulugbek 100214, Uzbekistan. *Set Criterion for Selected Quantum Theory Equations.*

One of differential comparison theorems for a set of objects from quantum theory may be formulated by the following manner. Given the availability of a sharp interconnection, the difference in sizes with an internal parts of current of flavor must constitute their linearly ordered set. This theorem relates the constancy of each type of flavor and the partially order of a set of cross sections of the interaction with a gauge boson of the structural components of its current. Thereby, it requires one to follow the mathematical logic of selected quantum theory equations, namely, of the Dirac, Klein-Gordon, Euler-Lagrange, and Schrödinger equations in the presence of mass, energy, and momentum matrices. From their point of view, nature itself separates Minkowski space into longitudinal and transversal spaces concerning a certain middle dynamical line. It characterizes herewith any particle both by longitudinal and transversal space-time coordinates. It is not excluded therefore that each of the dynamical laws of a set, presented in a given work for the first time, comes forward in it as a crucial proof of the existence of a longitudinal (transversal) Minkowski space-time. (Received February 24, 2020)

83 ► Relativity and gravitational theory

1158-83-246 **Michael A Bishop*** (mibishop@csufresno.edu) and **Gerardo Munoz**. *An Operator Algebraic Approach to the Unruh Effect.* Preliminary report.

The Unruh effect is an unusual physical phenomenon where an observer in a relativistic frame measures different temperatures than an observer in an inertial frame. This violates the fundamental physical principle that intrinsic properties of a system should be observer independent (or at least accounted for in some reasonable manner). Physics literature studies this idea through the lens of unbounded operators on Hilbert spaces. We present the Unruh effect through an algebraic approach which carefully handles issues with unbounded operators to show that this thermal effect is well-defined and demonstrate that the discrepancies between observers is due to their time evolution and space of states. (Received March 02, 2020)

1158-83-371 **Christopher Edward Alexander*** (christopher.alexander.ca@gmail.com), CA.
General relativistic self-similar waves that induce cosmic acceleration without a cosmological constant. Preliminary report.

A two-parameter family of exact, global, spherically symmetric, self-similar, shock-wave solutions of the Einstein field equations for a perfect fluid are constructed. The solutions are obtained by matching a two-parameter family of Friedmann-Lemaitre-Robertson-Walker (FLRW) type solutions to the one-parameter family of static solutions across a spherical shock surface. The families of FLRW type and static solutions will be assumed to have isothermal equations of state of the form $p = \sigma\rho$ and $\bar{p} = \bar{\sigma}\bar{\rho}$ respectively, with the constants σ and $\bar{\sigma}$ representing a single parameter for each family. The remaining parameter, denoted by a, is the *acceleration parameter* and represents a perturbation from the original FLRW solution, the current cosmological model for the Universe. Neither family of solutions incorporate a cosmological constant but cosmic acceleration is still present in the family of shock-wave solutions, with the acceleration parameterised by a. The conservation of mass and momentum across the shock surface imposes a constraint on the parameters, for which an implicit expression is obtained. (Received March 03, 2020)

85 ► Astronomy and astrophysics

1158-85-342 **Kohl James***, kjames3@csustan.edu, and **Victoria Morado** and **Andy Ponce**.
Numerical Image Analysis on Galaxies.

Through computer programming, we quantitatively define the center of other galaxies. We then develop a process to find the distance between other objects within these galaxies and their respective centers. (Received March 03, 2020)

90 ► Operations research, mathematical programming

1158-90-210 **Amelia E Musselman** (musselman5@llnl.gov), 7000 East Avenue, L-053, Livermore, CA 94550, **Robin L Wong** (wong133@llnl.gov), 7000 East Avenue, L-053, Livermore, CA 94550, and **Carol A Meyers*** (meyers14@llnl.gov), 7000 East Avenue, L-053, Livermore, CA 94550. *Laboratory Throughput Modeling: Using Operations Research to Optimize the Workplace.*

We discuss the efforts of our team at Lawrence Livermore National Laboratory to identify chokepoints in infrastructure, people, and processes across our workplace. We describe the variety of techniques that our group has used to study laboratory throughput, and we present an example using mathematical programming to identify capacity-limiting resources in the manufacturing and fielding of large experiments. We discuss the challenges with obtaining data to populate such models, and the iterative process required to make models useful. Ultimately the models built by our group influence tens of millions of dollars of infrastructure funding decisions at our site. (Received March 02, 2020)

1158-90-256 **Ali Mohammad Nezhad*** (mohamm42@purdue.edu), 150 N. University St., West Lafayette, IN 47906, and **Saugata Basu**. *On the central path of semidefinite optimization: degree and worst-case convergence rate.* Preliminary report.

Semidefinite optimization is the optimization of a linear objective function over the cone of positive semidefinite matrices intersected with an affine subspace. From an algorithmic point of view, primal-dual interior point methods have been comprehensively studied for an approximate optimal solution of semidefinite optimization problems. However, the worst-case convergence rate of the central path, which lies at the heart of primal-dual path-following interior point methods, is still unknown. In this talk, we address this question and derive upper bounds on the worst-case convergence rate of the central path and the degree of its Zariski closure. Since a primal-dual interior point method closely follows the central path, the worst-case convergence rate serves as a quantitative measure of hardness for numerically solving a semidefinite optimization problem. Joint work with Saugata Basu. (Received March 02, 2020)

1158-90-329 **Juliane Mueller*** (julianemueller@lbl.gov), 1 Cyclotron Rd., MS 50A-3111, Berkeley, CA 94720. *Efficient optimization of computationally expensive black-box problems with constraints.* Preliminary report.

In this talk, we describe an efficient method for solving optimization problems that have expensive-to-evaluate black-box objective and constraint functions. The objective and the constraints are independent, and thus we

first evaluate the constraint to assess the feasibility of a proposed solution before committing to spending more time on evaluating the objective function. We use separate Gaussian process models to approximate the objective and the constraints and we compare different adaptive sampling strategies for iteratively selecting new sample points. Finally, we show the performance of our method on an application problem that arises in the search for more efficient fuels by simulation. (Received March 03, 2020)

91 ► Game theory, economics, social and behavioral sciences

1158-91-326 **Khanh P. Nguyen***, 12218 North Palm Lake Dr., Houston, TX 77034, and **Zachary Kilpatrick** and **Kresimir Josic**. *Probabilistic feedback updates priors in sequential decisions with accumulated evidence.*

To make the best decisions organisms must flexibly accumulate information, accounting for what is relevant and ignoring what is not. Many decision-making studies focus on sequences of independent trials in which the evidence gathered to make a choice, as well as the resulting actions and feedback are irrelevant to future decisions. Normative theories have been developed for two-alternative forced choice tasks when rewards provide the sole evidence. Less is known about how observers should integrate probabilistic rewards interspersed with noisy evidence to inform their decisions in future, correlated trials. To understand decision-making under more natural conditions, we extend drift-diffusion models of evidence accumulation in serial correlations trials with and without feedback. Ideal observers integrate noisy evidence within a trial and optimize their reward rate across trials by adjusting their sequence of decision thresholds to deliberate longer on early decisions and respond more quickly in later trials. We show how conflicts between feedback and accumulated evidence are resolved and under which conditions one of them dominates. Our findings are consistent with experimentally observed response trends. (Received March 03, 2020)

1158-91-358 **Olivia J. Chu***, ochu@princeton.edu, and **Marc Wiedermann** and **Jonathan F. Donges**. *An Adaptive Voter Model in Heterogeneous Environments.* Preliminary report.

In human social systems, it is natural to assume that individuals' opinions influence and are influenced by their interactions. Mathematically, it is common to represent such systems as networks, where nodes are individuals and edges denote connections. Adaptive network models explore the dynamic relationship between node properties and network topology. For opinion dynamics, adaptive voter models provide two mechanisms through which changes can occur within the network. First, through homophily, an edge forms between two individuals who already agree; second, through social learning, an individual adopts one of their neighbor's opinions. Central to these models is assortative mixing: individuals more frequently attach to those who are similar to them, which facilitates the formation of sub-communities of like-minded individuals. However, it is not always the case that individuals want to cluster into homogeneous groups. Instead, they might attempt to surround themselves with individuals who both agree and disagree with them, in an effort to attain a balance of inclusion and distinctiveness in their social environments. In this work, we explore the effects that such preferences for heterogeneous environments have on the dynamics of the adaptive voter model. (Received March 03, 2020)

92 ► Biology and other natural sciences

1158-92-9 **Zhijun Wu*** (zhijun@iastate.edu), 462 Carver Hall, Ames, IA 50011. *Social Grouping as a Multi-Player Evolutionary Game.* Preliminary report.

Fish swim in schools and sometimes group tightly in ball shapes called fish bait balls. Birds fly in flocks and stay very close to each other and sometimes form a giant ball of birds as well. Such behaviors can be considered as a collective result from the attempt of each individual fish or bird to stay in touch with all others, i.e., always keep a close distance to all others but of course also not too close. Similar phenomena are also observed in some inorganic systems where non-bonded atoms form large molecular clusters with the so-called Van der Waals forces. We therefore model the grouping behaviors of fish or birds with a function similar to that used for modeling the Van der Waals forces in physics, and form a multi-player evolutionary game for the study of the grouping behaviors of these species. We show how grouping states of fish or birds at equilibrium and in particular, the ball shapes of fish or bird groups, can be simulated and predicted via this model. (Received December 07, 2019)

1158-92-10 **David M. Bortz*** (dmbortz@colorado.edu), Department of Applied Mathematics, University of Colorado, Boulder, CO 80309-0526. *Data-driven Model Selection for Spatio-temporal Models of Wound Healing.* Preliminary report.

Recently there has been a dramatic rise in interest in the data-driven creation of mathematical models governing biological phenomena. In this talk, we will provide an overview of our information-theoretic-based sparse regression approach for recovering parsimonious models from data. To illustrate the utility of this approach we consider the migration of keratinocytes during the re-epithelialization stage of wound healing. We will discuss the biological insights gained concerning physical vs. chemical signaling in the instigation of migration. We will also discuss some of the computational challenges in our approach as well as the framework we are developing for model distinguishability. (Received December 08, 2019)

1158-92-33 **Daniel B Cooney***, Fine Hall, Washington Road, Floor 2, Princeton, NJ 08540. *PDE Models of Multilevel Selection: The Evolution of Cooperation and The Shadow of Lower-Level Selection.*

In this presentation, we will discuss PDE models for multilevel selection, with an emphasis on studying the evolution of cooperation when there is reproductive competition both between individuals and between groups. We focus on the derivation and analysis of the long-time behavior of the two-level replicator dynamics, a nonlocal hyperbolic PDE describing deterministic birth-death competition for both individuals and groups. When interactions consist of the Prisoners' Dilemma, we show that whether the within-group advantage of defectors or the between-group advantage of groups with many cooperators wins out in the long run depends on the relative selection strength at the two levels. A notable finding is that lower-level selection casts a long shadow: if groups are best off with a mix of cooperators and defectors, then there will always be fewer cooperators than optimal at steady state, even in the limit of infinitely strong selection strength at the group level. (Received February 10, 2020)

1158-92-63 **Yang Kuang*** (kuang@asu.edu), Arizona State University, Tempe, AZ 85283. *Predictive Mathematical Models of Hormone Treatment for Prostate Cancer.*

Prostate cancer is commonly treated by a form of hormone therapy called androgen suppression. This form of treatment, while successful at reducing the cancer cell population, adversely affects quality of life and typically leads to a recurrence of the cancer in an androgen-independent form. Intermittent androgen suppression aims to alleviate some of these adverse affects by cycling the patient on and off treatment. We present several mathematical models of prostate cancer growth to study the dynamics of androgen suppression therapy and the production of prostate- specific antigen (PSA), a clinical marker for prostate cancer. Biologically crude models were based on the assumption of an androgen independent (AI) cell population with constant net growth rate. These models gave poor accuracy when fitting clinical data during simulation. More re fined models presented hypothesizes an AI population with increased sensitivity to low levels of androgen and these models generate high levels of accuracy in fitting clinical data. In general, we found that biologically more plausible models can forecast future PSA levels more accurately. These findings suggest that including more realistic mechanisms of resistance development may help predict the timing of androgen resistance. (Received February 19, 2020)

1158-92-64 **Yang Kuang*** (kuang@asu.edu), Arizona State University, Tempe, AZ 85283. *Oscillatory dynamics of an intravenous glucose tolerance test model with delay interval.*

Delay differential equations have been frequently used to study complex dynamics observed in nature. More recently, they are used to understand intriguing physiological phenomena such as those expressed by glucose and insulin interaction. We propose a simple set of delay differential equations to model an intravenous glucose tolerance test. This model uses two parameters to simulate not only both discrete time delay and distributed time delay in the past interval, but also the time delay distributed in a past sub-interval. We show that this relatively simple model provides good fit to fluctuating patient data sets and reveals some intriguing dynamics. Most importantly, our model may remove the defect in the well known Minimal Model which often overestimates the glucose effectiveness index. (Received February 19, 2020)

1158-92-91 **Jim M Cushing*** (cushing@math.arizona.edu), Department of Mathematics, 617 N Santa Rita Ave, Tucson, AZ 85721-0089. *Discrete time Darwinian models and an application to the evolution of post-reproduction survival.*

I will discuss the derivation and analysis of discrete time Darwinian dynamic models designed to address evolutionary trade-offs between reproductive effort and post-reproduction survival. A focus will be placed on the

role of nonlinear density effects in determining circumstances under which evolution selects low or high post-reproduction survival (semelparity or iteroparity). The mathematical analysis utilizes bifurcation and stability theory methods. (Received February 23, 2020)

1158-92-116 **Abba Gumel***, School of Mathematical and Statistical Sci., Tempe, AZ 85287. *Insecticide resistance and the quest for malaria eradication.*

Recent dramatic declines in global malaria burden and mortality can be largely attributed to the large-scale deployment of insecticidal-based measures, namely long-lasting insecticidal nets (LLINs) and indoor residual spraying (IRS). However, the sustainability of these gains, and the feasibility of global malaria eradication by 2040, may be affected by increasing insecticide resistance among the Anopheles malaria vector. This talk is based on the design and use of a new mechanistic model, which incorporates the full, weather-dependent mosquito lifecycle, to assess the population-level impact of the large-scale use of LLINs, under different levels of Anopheles pyrethroid insecticide resistance, on malaria transmission dynamics and control in a community (Received February 25, 2020)

1158-92-128 **Mikahl Banwarth-Kuhn*** (mbanwarth-kuhn@ucmerced.edu), **Jordan Collignon** and **Suzanne Sindi**. *Study of the Impact of Individual Cell Behaviors on Prion Protein Aggregation and Spatial Structuring of Yeast Colonies.* Preliminary report.

Prion proteins are most commonly associated with fatal neurodegenerative diseases in mammals, but they are also responsible for a number of harmless phenotypes in yeast. The diseased state (or phenotype) in yeast arises when a misfolded form of a protein, i.e. prion, appears and, rather than be removed by cellular quality control mechanisms, persists. In addition to gaining understanding the exact nature of the complex molecular processes and biochemical kinetics governing protein misfolding dynamics, connecting protein misfolding mechanisms *in vitro* to disease mechanism *in vivo* requires uncovering the non-intuitive relationships between protein aggregation mechanisms at multiple scales. We introduce a two-dimensional agent-based model of a budding yeast colony with detailed representation of cell-type specific biological processes. The model is used to study how individual cell behaviors drive the spatial structuring of yeast colonies and impact prion phenotypes. Spatial arrangement of cells is modeled using a center-based modeling approach and prion aggregation is modeled using simplified intracellular dynamics. The unified model may have the potential to predict mechanisms underlying spatial structuring of yeast and other microbial groups. (Received February 26, 2020)

1158-92-131 **Jesse Kreger*** (kregerj@uci.edu), **Natalia L Komarova** and **Dominik Wodarz**. *Effect of synaptic cell-to-cell transmission and recombination on the evolution of double mutants in HIV.*

Recombination in HIV infection impacts virus evolution in complex ways. The effect of free virus versus synaptic transmission on the evolution of double mutants, however, has not been investigated. Consistent with data, we assume spatial constraints for synaptic but not for free virus transmission. Two important effects are observed: (i) For disadvantageous mutants, synaptic transmission protects against detrimental effects of recombination on double mutant persistence. Under free virus transmission, recombination increases double mutant levels for negative epistasis, but reduces them for positive epistasis. (ii) The mode of virus spread also directly influences the evolutionary fate of double mutants. For disadvantageous mutants, double mutant production is the predominant driving force, and hence synaptic transmission leads to highest double mutant levels due to increased transmission efficiency. For advantageous mutants, double mutant spread is the most important force, and hence free virus transmission leads to fastest invasion. For neutral mutants, both production and spread of double mutants are important, and hence an optimal mixture of both transmission methods maximizes double mutant fractions. Implications for drug resistance are also discussed. (Received February 27, 2020)

1158-92-135 **Johannes G. Reiter*** (johannes.reiter@stanford.edu). *A mathematical model of ctDNA shedding predicts tumor detection size.*

Early cancer detection aims to find tumors before they progress to an incurable stage. We developed a stochastic mathematical model of tumor evolution and circulating tumor DNA (ctDNA) shedding to investigate the potential of ctDNA-based cancer early detection tests. We inferred ctDNA shedding rates of early stage lung cancers and calculated that a 15 mL blood sample contains on average only 1.5 genome equivalents of ctDNA for lung tumors with 1 billion cells (size of ≈ 1 cm^3). We considered two clinically different scenarios: cancer screening and cancer relapse detection. For monthly relapse testing with a sequencing panel covering 20 tumor specific mutations, we found a median detection size of 0.24 cm^3 corresponding to a lead time of 160 days compared to imaging based relapse detection. For annual screening, we found a median detection size of $2.8-4.8$ cm^3 depending on the sequencing panel size and on the mutation frequency. The expected detection sizes correspond to lead

times of $390 - 520$ days compared to current median lung tumor sizes at diagnosis. This quantitative framework provides a mechanistic interpretation of ctDNA based cancer detection approaches and helps to optimize cancer early detection strategies. (Received February 27, 2020)

1158-92-136 **Joseph C Watkins*** (jwatkins@math.arizona.edu), Department of Mathematics, Tucson, AZ 85721. *The role of genetic background in the clinical outcomes of a genetic disease.*

Here we use the example of Dravet syndrome, a rare, devastating form of childhood epilepsy often associated with mutations in the voltage-gated sodium channel gene, SCN1A to consider the dependence of clinical outcomes on variants in other genes associated with brain development. We consider three possible explanations 1) a single common variant that is enriched in one of the phenotypic groups, 2) sets of common or rare variants aggregated in and around genes associated with clinical outcome, and 3) rare variants in candidate genes associated with neuronal excitability. Our population are individuals with SCN1A truncation variants, a population with a nearly identical genetic cause for Dravet syndrome. The central question for evolution – are those with poor clinical outcomes unlucky and have poor genetic background or are those with mild clinical outcome lucky and have an optimal genetic background? (Received February 27, 2020)

1158-92-157 **Patrick D. Shipman***, shipman@math.colostate.edu. *Mechanism-enabled population balance modeling and applications to pattern formation.*

Nucleation and growth processes lead to the formation of nanoparticles. Protein aggregates, such as those involved in neurological diseases, are examples of nanoparticles in biological systems. An outstanding mystery in the chemisty of particle formation is how to understand and control particle size distributions. Through "mechanism-enabled population balance modeling" we suggest crucial characteristics of particle-formation systems that allow for narrow particle-size distributions. We apply ME-PBM to systems involving vaporchromatic experiments, in which cell vacuolar pigments called anthocyanins form colored patterns. (Received February 28, 2020)

1158-92-165 **Natalia L. Komarova***, Department of Mathematics, University of California Irvine, Irvine, CA 92617. *Mathematical methods in evolutionary dynamics.*

Evolutionary dynamics permeates life and life-like systems. Mathematical methods can be used to study evolutionary processes, such as selection, mutation, and drift, and to make sense of many phenomena in life sciences. I will present two very general types of evolutionary patterns, loss-of-function and gain-of-function mutations, and discuss scenarios of population dynamics – including stochastic tunneling and calculating the rate of evolution. I will also talk about evolution in random environments. The presence of temporal or spatial randomness significantly affects the competition dynamics in populations and gives rise to some counterintuitive observations. Applications include origins of cancer, passenger and driver mutations, and how aspirin might help prevent cancer. (Received February 29, 2020)

1158-92-179 **Chadi M Saad-Roy*** (csaadroy@princeton.edu). *Coupled epidemiological and evolutionary dynamics of pathogens.* Preliminary report.

Pathogens that cause infectious diseases can be strongly influenced by rapid evolutionary pressures, e.g. by host immune systems. Thus, pathogen evolutionary dynamics can play a substantial role in disease control strategies, underlying the importance of determining and understanding the qualitative behaviour of these dynamics. In this talk, I will discuss recent mathematical modelling work that considers both the epidemiology and the evolution of pathogens. I will also illustrate how these findings relate to disease control. (Received February 29, 2020)

1158-92-180 **Weitao Chen*** (weitaoc@ucr.edu). *Coupled Mechanochemical Multiscale Model to Study the Tissue Development.* Preliminary report.

Growth regulation is one of the main problems in developmental biology. During the growth, cells stop proliferation precisely when the intended size of the tissue or organ is achieved. Meanwhile, differential cell shapes in space are integrated to give rise to well-organized overall structure. Uncontrolled growth of the cells in tissues or organs will lead to abnormal development or fatal diseases such as cancer. Therefore, developing an extensible predictive mathematical model for exploring the mechanisms involved in the tissue development is significant for understanding the fundamental principles in developmental biology, with a broad range of applications from tissue engineering to biomanufacturing and biotech industry. Experimental data suggests that mechanical properties of cells and chemical signals in both intracellular and extracellular domains play critical roles in size control and shape formation. Here we develop a multiscale, mechochemical coupled model of tissue growth with subcellular details. This model is applied to test competing hypotheses in the field to resolve the highly debated

question of how tissues reach their final size, as well as how the tissue shape is determined simultaneously. (Received February 29, 2020)

1158-92-191 **Suzanne Sindi***, 5200 North Lake Road, Merced, CA 95340. *Ectoparasites and Malaria: Outcomes from Collaborative Workshops for Women in Mathematical Biology.*

I will describe two research projects that began at Collaborative Workshops for Women in Mathematical Biology.

First, we model the interactions between social organization and allogrooming efficiency to consider how parasitic threats may have shaped the evolution of social behaviors. We consider several dynamic models of social organization the varying levels of social grooming and find that allogrooming may interrupt parasitic epidemics in two meaningful ways: protection against increasing individual parasitic load via parasite reproduction and protection against spread of parasites among hosts.

Second, we consider the dynamics of Malaria, a vector-borne disease responsible for >400K deaths per year. Malaria control efforts are focused at the national level, but movement of individuals between countries may complicate elimination efforts. Here, we consider the case of neighboring countries Botswana and Zimbabwe, connected by human mobility. We use a two-patch Ross-MacDonald Model with Lagrangian human mobility to examine the coupled disease dynamics between these two countries. We find that dynamics and interventions in Zimbabwe dramatically influence pathways to elimination in Botswana, largely driven by Zimbabwe's population size and larger basic reproduction number. (Received March 01, 2020)

1158-92-192 **Suzanne Sindi***, 5200 North Lake Road, Merced, CA 95340. *A Minimal Model for Prion Aggregate Infectivity Shows Prion Strains Differ by Nucleus Size.*

Prion proteins are responsible for a variety of neurodegenerative diseases in mammals such as Creutzfeldt-Jakob disease in humans and "mad-cow" disease in cattle. While these diseases are fatal to mammals, a host of harmless phenotypes have been associated with prion proteins in *S. cerevisiae*, making yeast an ideal model organism for prion diseases.

The common assumption in prion phenotypes is that the only limiting event is the establishment of a stable prion aggregate of minimal size. We show this model is inconsistent with seeding experiments. We then develop a minimal model of prion phenotype appearance: the first successful amplification of an aggregate. Formally, we develop a chemical master equation of prion aggregate dynamics through conversion (polymerization) and fragmentation under the assumption of a minimal stable size. We frame amplification as a first-arrival time process that must occur on a time-scale consistent with the yeast cell cycle. This model, and subsequent experiments, then establish for the first time that two standard yeast prion strains have different minimally stable aggregate sizes. This suggests an novel approach (albeit entirely theoretical) for managing prion diseases, shifting prion strains towards larger nucleus sizes. (Received March 01, 2020)

1158-92-199 **Marek Kimmel*** (kimmel@rice.edu), **Khanh N Dinh** and **Roman Jaksik**, , Poland, **Amaury Lambert**, , France, and **Simon Tavare**. *Statistical inference for the evolutionary history of cancer genomes.*

Recent years have produced a large amount of work on inference about cancer evolution from mutations identified in cancer samples. Much of the modeling work has been based on classical models of population genetics, generalized to accommodate time-varying cell population size. Reverse-time genealogical views of such models, commonly known as coalescents, have been used to infer aspects of the past of growing populations. Another approach is to use branching processes, the simplest scenario being the linear birth-death process (lbdp), a binary fission Markov age-dependent branching process. A genealogical view of such models is also available. The two approaches lead to similar but not identical results. In this paper, we examine how the statistics based on birth-death processes differ from those based on the coalescent model. We also present a model of tumor evolution with selective sweeps, based on coalescence, and demonstrate how it may help in understanding the past history of tumor as well the influence of data pre-processing. We illustrate the theory with applications to several examples of The Cancer Genome Atlas tumors. (Received March 01, 2020)

1158-92-209 **Joceline Lega*** (lega@math.arizona.edu), Department of Mathematics, University of Arizona, Tucson, AZ 85721. *Forecasting the course of disease outbreaks with ICC curves.*

In this talk, I will present a new approach to forecast the course of a disease outbreak, which centers around the use of Incidence - Cumulative Cases (ICC) curves. I will first introduce ICC curves and give an exact formulation in the case of the SIR (Susceptible - Infected - Removed) epidemiological model. I will then discuss how ICC curves may be used for parameter identification and outbreak forecasting. Finally, I will show some examples of application, including forecasts for COVID-19 outbreaks in various locations. (Received March 02, 2020)

1158-92-232 **Alla Borisyuk*** (borisyuk@math.utah.edu) and **Gregory Handy**. *Effect of Astrocytes in Neuronal Networks.* Preliminary report.

Astrocytes are glial cells making up 50% of brain volume, with each one wrapping around thousands of synapses, and playing multiple important roles. However their exact effect in governing the dynamics of the synapse and neuronal networks is being debated. In particular, the tightness of their wrapping (or "degree of ensheathement") and the number of the synapses ensheathed varies by brain region and in certain disease states such as some forms of epilepsy. We set out to investigate whether the changes in ensheathment properties contribute to the disease state of the network or, conversely, play a protective role. First, we consider an individual synapse as a DiRT (Diffusion with Recharging Traps) model: diffusing particles can escape through absorbing parts of the boundary, or can be captured by traps on the boundary. We show that a synapse tightly ensheathed by an astrosyte makes neuronal connection faster, weaker, and less reliable. These influences can then be included in a neuronal network model by adding an "effective" astrocyte on every synapse. Depending on the number of synapses ensheathed, and the ensheathment strength, the astrocytes are able to push the network to synchron and to exhibiting strong spatial patterns, possibly contributing to epileptic disorder. (Received March 02, 2020)

1158-92-239 **J. Foo*** (jyfoo@umn.edu), **E. Gunnarsson**, **S. De** and **K. Leder**. *Understanding the role of phenotypic switching in cancer drug resistance.*

The emergence of acquired drug resistance in cancer represents a major barrier to treatment success. While research has traditionally focused on genetic sources of resistance, recent findings suggest that cancer cells can acquire transient resistant phenotypes via epigenetic modifications and other non-genetic mechanisms. Although these resistant phenotypes are eventually relinquished by individual cells, they can temporarily 'save' the tumor from extinction and enable the emergence of more permanent resistance mechanisms. These observations have generated interest in the potential of epigenetic therapies for long-term tumor control or eradication. In this work, we develop a mathematical model to study how phenotypic switching at the single-cell level affects resistance evolution in cancer. We highlight unique features of non-genetic resistance, probe the evolutionary consequences of epigenetic drugs and explore potential therapeutic strategies. (Received March 02, 2020)

1158-92-253 **Junling Ma***, Department of Mathematic and Statistics, University of Victoria, Victoria, BC V8W 2Y2, Canada. *The disease threshold condition for SIS dynamics on a random contact network.* Preliminary report.

t is know that, on a random network, the basic reproduction number of an SIS model is larger than that of an SIR model. It has also been shown that, there is no disease threshold on a scale free network with finite second moment of the degree distribution. These results contrast the prediction of the classical heterogeneous mixing models. We should that, for any positive transmission rate, as long as the network contains nodes with large enough degree, the disease persists on the network, independently to the degree distribution of the network. (Received March 02, 2020)

1158-92-262 **Anastasia Bizyaeva*** (bizyaeva@princeton.edu), **Sebastian Musslick** and **Naomi Ehrich Leonard**. *Normative and evolutionary perspectives on individual differences in cognitive flexibility.* Preliminary report.

Humans are often faced with antagonistic demands, such as maintaining focus on a single task (e.g. do taxes) while also being able to quickly switch to another task (e.g. respond to a fire alarm). In previous work, we studied how individuals balance the tradeoff between cognitive stability (the ability to focus on a task) and cognitive flexibility (the ability to switch between tasks) in a given environment. Leveraging dynamical systems theory, we showed that it is optimal for a single individual to sacrifice cognitive stability in the service of cognitive flexibility, especially if the environment requires frequent switches between tasks. Here, we explore the optimal balance between cognitive stability and cognitive flexibility at the group level. We investigate how heterogeneity in cognitive flexibility among individuals contributes to performance of the individuals in a collaborative task-switching environment. We further study how heterogeneity in cognitive flexibility emerges in a population with evolutionary interactions between individual traits. (Received March 02, 2020)

1158-92-265 **Azmy Ackleh**, LA, and **Karyn Sutton** and **Tingting Tang*** (ttang2@sdsu.edu), El Centro, CA 92243, and **Lihong Zhao**. *A Second Order Finite Difference Scheme for a Structured Model of Mycobacterium marinum Dynamics in Aquatic Animals.*

In this talk, we present a novel second order finite difference scheme for a modified version of a model, to study the transmission dynamics of a human TB-like bacterium, Mycobacterium marinum, affecting aquatic animals on a similar scale. Convergence of the finite difference approximation to the unique weak solution of the model is shown. Numerical results confirming the second order convergence of the scheme are presented.

The computational advantages of using this high-resolution scheme in comparison to the first order scheme is illustrated (Received March 02, 2020)

1158-92-275 **Chayu Yang*** (rgb855@mocs.utc.edu), **Lolika**, **Mushayabasa** and **Wang**. *Modeling the spatiotemporal variations in brucellosis transmission.*

We propose a nonlinear modeling framework to investigate the transmission dynamics of brucellosis, incorporating both the spatial and seasonal variations. The spatial modeling is based on a patch structure, and the seasonal impact is represented by utilizing time-periodic model parameters. We demonstrate this framework through a two-patch model and conduct detailed analysis, for the cases with and without seasonal oscillations, respectively. In particular, we establish the threshold dynamics results using the reproduction numbers defined under different scenarios. Our findings underscore the importance of including spatial and seasonal heterogeneities in the design of control strategies for brucellosis. (Received March 02, 2020)

1158-92-283 **H. Tomas Rube*** (trube@ucmerced.edu), UC Merced, School of Engineering, 5200 N. Lake Road, Merced, CA 95343, and **Chaitanya Rastogi**, **Judith F. Kribelbauer**, **Xiaoting Li** and **Harmen J. Bussemaker**. *Quantifying sequence readout by transcription factors through principled analysis of high-throughput SELEX data.* Preliminary report.

Transcription factors (TFs) are central to cellular information processing and control gene expression by binding specific DNA sequences in the genome. In recent years, the sequence specificity of hundreds of TFs have been characterized using high-throughput in vitro assays that utilize deep sequencing to quantify DNA binding. Variations of these assays can characterize binding by a single TF, multiple TFs forming a complex, RNA-binding proteins, or binding to chemically modified DNA. However, no unified mathematical method for analyzing all these data yet exists. Challenges in developing such a method include both the diversity and complexity of the experimental designs, and the sparsity of the data, where each DNA sequence is observed at most once. In this talk I will introduce a principled modeling framework, ProBound, that overcomes these challenges and infers quantitative models of TF binding specificity. The flexibility of ProBound allows it to work on all currently available data. ProBound uses multi-task learning to infer robust binding models from multiple experiments, a feature especially useful for analyzing the binding specificity of TF complexes. I also will discuss similarities and differences between ProBound and deep learning. (Received March 03, 2020)

1158-92-286 **Blerta Shtylla***, Berta Shtylla, CA. *Methods for model calibration with data: blood and solid tumor model case studies.*

In this talk, we discuss a framework by which two mathematical models are calibrated with synthetic data. We consider first an example of a model for multiple myeloma, where multiple global parameter sensitivity analysis methods are applied to reduce parameter space. Next, we develop a framework for assessing model fitting and model prediction uncertainity for solid tumor growth under various radiotherapy treatment protocols. In both cases, structural and practical identifiability analysis is used to assess what portion of parameter space can be reliably fitted to data. (Received March 03, 2020)

1158-92-287 **Mario Banuelos*** (mbanuelos22@csufresno.edu), **Omar DeGuchy** (odeguchy@ucmerced.edu), **Suzanne Sindi** (ssindi@ucmerced.edu) and **Roummel F. Marcia** (rmarcia@ucmerced.edu). *Related Inference: A Supervised Learning Approach to Detect Genomic Variation.*

The human genome is composed of 3 billion nucleotides, given by the letters A,C,G,T. Typically, organisms in the same species have similar genomes that differ by only a few sequences of varying lengths at varying positions. These differences may be observed in the form of regions where letters are inserted, deleted, or inverted. These anomalies, or novel rearrangements, are known as structural variants (SVs) and are often difficult to detect. Standard approaches for identifying such variation involves comparing fragments of DNA from the genome of interest to a reference genome. This process is usually complicated by errors produced in both the sequencing and mapping process which may result in an increase in false positive detections. In this work, we propose a gradient boosting and neural network approach for reducing the number of false positives. We focus our attention on the deletions detected by the popular SV tool delly. In particular, we consider the ability of simultaneously considering sequencing data from a parent and a child using a neural network and gradient boosting. We compare the performance of each method on simulated and real parent-child data and show that including related individuals in training data greatly improves the ability to detect true SVs. (Received March 03, 2020)

1158-92-289 **Ivana Bozic***, ibozic@uw.edu. *Evolutionary dynamics of cancer and its response to treatment.*

Cancer is the result of a stochastic evolutionary process characterized by the accumulation of mutations that are responsible for tumor growth, immune escape, and drug resistance, as well as mutations with no effect on the phenotype. Stochastic modeling can be used to describe the dynamics of tumor cell populations and to obtain insights into the hidden evolutionary processes leading to cancer. I will present recent approaches that use branching process models of cancer evolution to quantify intra-tumor heterogeneity and the development of drug resistance, and their implications for interpretation of cancer sequencing data and the design of optimal treatment strategies. (Received March 03, 2020)

1158-92-310 **Jordan Collignon***, 5200 North Lake Road, Merced, CA 95343, and **Mikahl Banwarth-Kuhn** and **Suzanne Sindi**. *An Investigation into the Impact of Cell-Budding on Colony Structure in Simulated Yeast Colonies.* Preliminary report.

High-performance computing has made it possible to generate biologically relevant populations of cells while tracking their position and shape in time. Such models have been used to investigate the shape of bacterial colonies, progression of cancer tumors, and development of tissues in plants and mammals. Most cell-based models were developed under the assumption that cell division progressed through fission, but the yeast extit(Saccharomyces cerevisiae) reproduces through budding, a biophysically and biomechanically different process where mother and daughter cells remain attached. The few cell-based models that have considered budding, typically have done so by approximating the budding process or only considering the development of a small colony. In this work, we implement a cell-based model of yeast incorporating budding. We do this by considering mother-daughter pairs to have strong adhesion and then consider the impact of budding on colony structure. We study the impact that budding has on quantitative features of colony structure including the sparseness, expanse and connectivity. We find that budding produces significant differences in these colony features and, as such, is an important biophysical feature to include when comparing to experimental yeast colony data. (Received March 03, 2020)

1158-92-317 **Mari Kawakatsu***, mari.kawakatsu@princeteon.edu, and **Simon A. Levin** and **Corina E. Tarnita**. *Evolutionary dynamics of polarization and cooperation in group-structured populations.* Preliminary report.

The emergence of stable social structures has long fascinated scholars from biology to political science. Political theorists as early as James Madison posited that a diversity of political interests would limit the formation of factions: if individuals have many interests, they must form cooperative alliances across differences to achieve common goals. But this idea runs counter to a result established by evolutionary models of cooperation: cooperation often promotes greater connectivity, which makes the system vulnerable to invasion, leading to fragmentation. Under what conditions, then, can cooperation and social cohesion evolve together? We study this question in the context of multiple, overlapping groups that represent various issues of interest. Individuals belong to an interest group if they have opinions about the corresponding issue, and they adopt strategies that are conditional on others' opinions. Individuals interact via an evolutionary game if they share a common interest, with both strategies and group memberships subject to evolutionary updating. We explore factors that promote social cohesion and their effects on population structure and diversity of strategies. Our work contributes to a growing literature on how individual heterogeneity shapes social structure. (Received March 03, 2020)

1158-92-332 **Krishnakant Dasika**, **Ksenia Zlobina**, **Jianhong Chen** and **Marcella Gomez***, 1156 High St., BSOE, SOE2, Santa Cruz, CA 95064. *Predictive models in wound healing for biological control.* Preliminary report.

Wound healing involves a symphony of biological processes wherein the sequence and timing of said processes play a critical role in maintaining a normal healthy trajectory to wound closure. In the United States alone, chronic wounds affect around 6.5 million patients. It is claimed that an excess of US $25 billion is spent annually on treatment of chronic wounds, and this number is expected to increase. In order to make advances in wound healing it is important to identify candidate biomarkers as leverage points for intervention as well as arrive at a predictive model for wound healing stage. Here, we present a data-driven method of to uncover underlying dynamics in wound healing. Machine learning methods are used to identify the dynamics present and dynamic models used to simulate the activity of key processes that uniquely identify wound healing stage. (Received March 03, 2020)

1158-92-335 **Azmy S. Ackleh**, **Md Istiaq Hossain**, **Amy Veprauskas*** (amy.veprauskas@louisiana.edu) and **Aijun Zhang**. *Evolution in a discrete-time predator-prey model.*

Interactions between species may change over time due to various reasons, including small evolutionary changes called microevolution. In this talk, we construct a discrete-time predator-prey model in which the prey is assumed to evolve in response to an environmental stressor. We model this evolution using a Darwinian dynamics approach which describes changes in a mean phenotypic trait. Here we assume this trait measures prey resistance to the environmental stressor. We provide thorough analysis of this model to understand how prey evolution may impact predator-prey dynamics. In particular, we establish conditions for stability of equilibria and system persistence and explore the existence of cycles and chaotic attractors. All together, we find that, when evolution is slow, the evolutionary dynamics are nothing but a continuous perturbation of the non-evolutionary system. However, when the speed of evolution is increased, evolution may introduce rich dynamics that are not observed when evolution is absent. (Received March 03, 2020)

1158-92-353 **Aqib Hasnain*** (aqib@ucsb.edu) and **Enoch Yeung** (eyeung@ucsb.edu). *Structured dynamic mode decomposition to quantify the impact of a genetic circuit on its host.*

Biophysical systems often exhibit emergent behavior when new elements are incorporated. Biophysical models typically fail to reconcile these emergent dynamics when confronted with data. Existing biophysical models do not account for the added interactions, arguably the leading obstacle in building robust synthetic genetic circuits. Often these models are created independent of data to describe only genetic circuit components; they lack a mechanism for incorporating whole cell data. We improve on these models through a learning process which takes advantage of the experimental process involved in collecting sequencing data. The learning process approximates the Koopman operator, K, an infinite-dimensional operator that acts on observables to produce linear dynamics. Sequencing measurements are not handled well by the state-of-the-art Dynamic Mode Decomposition (DMD) algorithms to approximate K. The data are high-dimensional, noisy, and typically contain a low number of timepoints. To overcome these difficulties, we develop structured DMD and integrate it with the learning algorithm for sparse DMD, which is a regularized DMD that can handle sparse time-series data. We demonstrate our method on a real dataset collected from a NAND gate built in Escherichia coli. (Received March 03, 2020)

1158-92-369 **Angelica Bloomquist*** (abloomquist@sdsu.edu) and **Naveen K. Vaidya**. *Modeling the risk of HIV infection for drug abusers.* Preliminary report.

Drugs of abuse, such as opiates, are one of the leading causes for transmission of HIV in many parts of the world. Drug abusers often face a higher risk of acquiring HIV because target cell (CD4+ T-cell) receptor expression differs in response to morphine, a metabolite of common opiates. In this study, we use a viral dynamics model that incorporates the T-cell expression difference to formulate the probability of infection among drug abusers. We quantify how the risk of infection is exacerbated in morphine conditioning, depending on the timings of morphine intake and virus exposure. With in-depth understanding of the viral dynamics and the increased risk for these individuals, we further evaluate how preventive therapies, including pre- and post-exposure prophylaxis, affect the infection risk in drug abusers. These results are useful to devise ideal treatment protocols to combat the several obstacles those under drugs of abuse face. (Received March 03, 2020)

1158-92-378 **Amanda N. Laubmeier*** (laubmeier@unl.edu), **Richard Rebarber** and **Brigitte Tenhumberg**. *Viability of reduced observations for tracking Astragalus abundance in the field.* Preliminary report.

Long-term studies monitoring ecological populations can be costly and time-consuming. To support their duration, it may be necessary to compromise the desired resolution of data. Here, we explore this tradeoff by first maximizing information gained from a detailed data set and then investigating whether we could sustain a similar study with fewer observations. We specifically consider the problem of modelling the abundance of a flowering plant, Astragalus scaphoides, which was previously monitored at four sites in a decades-long study. Using abundances from multiple life stages alongside individual-level demographic data, we can establish a discrete-time model for the plant populations. We then investigate the effect of only tracking large, flowering stages of the plant, reducing the burden of data collection but sacrificing rich demographic information. We compare two tools for accommodating these partial observations - dynamic observers and bayesian estimates - to assess the viability of a reduced protocol. (Received March 04, 2020)

94 ► Information and communication, circuits

1158-94-201 **Calvin Zhang-Molina*** (calvinz@math.arizona.edu), University of Arizona, Department of Mathematics, 617 N Santa Rita Ave, Tucson, AZ 85721. *An Integrated Theoretical Framework for Understanding Synaptic Transmission: The Functional Significance of Spontaneous Vesicle Release and Undocking.* Preliminary report.

Synaptic transmission is the mechanism of information transfer from one neuron to another. A crucial step in this process is the probabilistic release of neurotransmitter vesicles, which is the most significant source of randomness in the central nervous system. To quantify stochastic vesicle dynamics and evaluate its functional role, we develop an integrated theoretical framework that bridges dynamical systems, stochastic processes, optimal filtering, and control principles for understanding neuronal information processing at synaptic and circuit levels. Using this new framework, we investigate how the rate of spontaneous vesicle release and vesicle undocking influence neuronal computation at the synaptic level. Our results indicate that spontaneous vesicle release and undocking can play a functional role in synaptic transmission by enhancing the ability of a model synapse in estimating various desired signals. We suggest that, in addition to the rate of vesicle release, experimentalists should also pay attention to the rate of spontaneous vesicle release and undocking. (Received March 01, 2020)

1158-94-248 **Andrew G Benedek*** (benedek.andras@btk.mta.hu), Institute of Philosophy, Research Centre for the Humanities, 4. Toth Kalman st., Budapest, 1097, Hungary. *Models and Logics in the Practice of Mathematical Heuristics: Tools for concept development and discovery or just for retroactive reconstruction?*

I survey some models of Building Theories and History Based Logics of concept formation both from the point of view of conceptual and historical reconstructions of heuristics. Discussing Inquisitive Semantics and Interrogative Models of Inquiry in terms of Dynamic Epistemic Logic I raise three questions. 1) Is it necessary that "the owl of Minerva spreads its wings only with the falling of the dusk", i.e., we obtain logic models of reasoning only after the historical event of discovery, or 2) can we also use the Dynamic Epistemic Logical and Game Theoretical tools for concept development in mathematics education and actual scientific research? 3) Could we blend these approaches with the toolbox of Model Theory? My answer to the first question is "No"; "Yes" to the second; and "Should" for the third one. I argue my case w.r.t (1) by pointing out that reconstructions of thoughts overlap with the field of discovery; show (2) by logical formalizations of Polya-type and Lakatosian heuristics; and support (3) by use-cases of Model Theory and Reverse Mathematics. The main point of the talk is that the challenge of describing real-world practice of mathematical heuristics is to combine modeling mathematical structures with the dynamics of epistemic logics of reasoning about them. (Received March 03, 2020)

97 ► Mathematics education

1158-97-15 **Chanchal Dass*** (cdass01@gmail.com), F 201 Parth Avenue, Near ONGC office, Chandkheda, Ahmedabad, Gujarat 382424, India. *Activity-Based Mathematics Teaching with an examples of Kinematics.*

Over the years many mathematics teaching techniques have been evolved but with the advent of modern computing technology, the opportunity of new teaching techniques grew wider. There are many options available now. Now for teaching mathematics audios, videos, augmented reality, virtual reality, etc. are being used. An innovative technique of teaching mathematics has been developed taking advantage of the advancements of technology. It is an activity-based mathematics teaching technique that uses simple spreadsheets to explain all the mathematical concepts. The core of this innovation is a few simple techniques that can help in converting any mathematical or scientific concept into geometrical shapes without any programming knowledge. This is a great innovation in the field of mathematics education. With this technical explanation of all advanced and complex mathematical concepts becomes very simple. In this paper, an attempt is made to explain this teaching technique to teach kinematics. It will start with the theories available to draw simple curves and extend this knowledge to explain complex motions of machine parts including robotics. (Received January 31, 2020)

1158-97-22 **Peter Tannenbaum*** (petertannenbaum@gmail.com), Santa Barbara, CA. *The Shortest Distance Between Three Points.*

Given three random points on the plane, what is the most efficient way to connect them? This seemingly simple question turns out to be surprisingly deep and multi-layered, touching on a wide swath of secondary school mathematics. While neither a game nor a puzzle, the question (and a series of subsequent questions that potentially follow it) seems particularly well suited for Math Circle exploration and experimentation. In this

talk I will suggest a few of the many ways to pursue this. If time permits I will also perform a couple of magic tricks. (Received February 05, 2020)

1158-97-32 **Bogdan D. Suceavă*** (bsuceava@fullerton.edu), Department of Mathematics, Cal State Fullerton, 800 N. College Blvd., Fullerton, CA 92834. *From Gheorghe Tzitzeica's Contest of the 'Mathematical Gazette' to the 1980s Version of the Mini-Contest: Explorations of an Interesting Idea in Mathematical Education.*

The classical contest of the Bucharest-based 'Mathematical Gazette' was designed in the early 20th century to identify and motivate young mathematicians. The heart and soul of this extensive editorial and educational project was a doctoral student of Gaston Darboux (1842–1917), namely Gheorghe Tzitzeica (1873–1939), who was a full professor of geometry at the University of Bucharest (1903-1937), while he served as editor-in-chief of the 'Gazette'. Today, Tzitzeica is remembered as the esteemed founder of a research direction called affine differential geometry. When the 'Gazette"s tradition in organizing contests during the summer (different than the math olympiads) was revived in the 1980s, mainly through the vision of Nicolae Teodorescu (1908–2000), a new component was added to the summer school of the 'Mathematical Gazette': the Mini-Contest. We will describe this idea and its philosophy, and we will illustrate it with examples from that period. This Eastern European experience is of particular interest for its active learning component and the high level of student involvement in the overall creative process. (Received February 09, 2020)

1158-97-37 **Peter Tannenbaum*** (petertannenbaum@gmail.com), 1318 ANACAPA ST, Santa Barbara, CA 93101. *Polya's "Let us Teach Guessing" Revisited: Shortest Distance Problems.*

Given N random points on the plane, what is the shortest network connecting these points? This is a classic Polya-type question—easy to state but surprisingly deep and multi-layered. In this talk I will borrow from Polya's "teach by guessing" approach to pluck some of the low hanging fruit of this problem: N = 3, and a few special cases for larger values of N. (Received February 11, 2020)

1158-97-47 **David Patrick*** (patrick@aops.com). *Socks and Swish: card games with mathematical foundations.* Preliminary report.

The card game SET is widely used in math circles to introduce participants to finite affine geometry. We will discuss two other, lesser known card games whose gameplay is similar to SET: all involve finding groups of cards that have some sort of matching property. The first game, Socks, is quite similar to SET, but Socks models a finite projective (as opposed to affine) geometry. The second, Swish, is a bit different than the other two, and (aside from the usual combinatorial questions that one can ask about such games) can be used as an introduction to symmetry groups of polygons, and more generally to groups acting on sets. Besides discussing the games and their underlying mathematics, we will discuss our observations from a March 2020 session of the San Diego Math Teachers' Circle in which these games were explored by our participating teachers. (Received February 14, 2020)

1158-97-106 **Jennifer E Clinkenbeard*** (jclinkenbeard@csumb.edu). *Food For Thought: Exploring Euclidean and non-Euclidean Geometries.* Preliminary report.

We present a series of inquiry-oriented tasks to visualize and model different geometries using common food items. We explore: (1) Euclid's parallel postulate and its consequences in the first book of Elements with spaghetti; (2) spherical geometry on the surface of an orange; (3) hyperbolic geometry on Pringles chips; and (4) incidence geometry with marshmallows and spaghetti. This activity series has been implemented in an inquiry-oriented, upper-division geometry course at CSU Monterey Bay. We discuss samples of the tasks above that have been implemented in the course, as well as samples of student models and responses to metacognitive writing prompts. We discuss the strengths and limitations of this approach with respect to heuristic reasoning. As Pólya observed in *How to Solve It*, heuristic reasoning supports the development of provisional reasoning. This does not constitute formal proof, nor should it be presented as such. We conclude with a discussion of these considerations in the context of Euclidean and non-Euclidean geometries, and how students' exploration with the models may contribute to readiness for more formal argument. (Received February 25, 2020)

1158-97-132 **Péter Juhász*** (juhasz.peter@renyi.hu), Budapest, Hungary. *Unusual chessboard tasks for gifted learners.*

Lajos Pósa has been organizing and leading weekend math camps for highly gifted students for more than 30 years. A special form of gifted education is being implemented in these camps, in which the focus is laid not on preparing for competitions, but on intense and autonomous thinking.

In one of the long term sequences of tasks, the colors of the squares of a chessboard, or some numbers written on the squares can be changed, and various related questions are to be answered. Students are also encouraged to pose connected questions, which may expand the set of problems to be solved by the group. The presentation offers a sample of this sequence of tasks. (Received February 27, 2020)

1158-97-134 **Robert G Stein*** (bstein@csusb.edu), 4 Beacon Way, Apt 503, Jersey City, NJ 07304. *Let's Take Polya Back to School.*

Is it possible to design a K-12 curriculum that embodies Polya's philosophy of mathematics education? This talk explores what such a curriculum might look like as well as the practical issues involved in creating and implementing it.

How to build a curriculum based on active learning. a. Can you find suitable activities to cover the entire curriculum? b. Will students get enough practice to master basic skills? A Polya-style approach to some basic Preparing teachers Preparing administrators and the public (Received February 27, 2020)

1158-97-137 **Lance D Burger*** (lburger@mail.fresnostate.edu). *Towards a Lacanian Critique in Mathematics Education: The University Discourse and Current Work.* Preliminary report.

After introducing some Lacanian terminology and theory of discourses by example of the University discourse, this talk discusses an extension of Lacan's four discourses to a total of twenty four by allowing the usual symbols (defined as numbers) $S_1 \equiv 1$, $S_2 \equiv 2$, $a \equiv 3$ and $\$ \equiv 4$; to occupy any of the four possible positions in Lacan's matrix of Truth-Agent-Other-Production, as permutations of an identity element in a theorized isomorphic group-theoretic model, based on an 'Unconscious' discourse defined as the group identity element and the usual operation of cycle composition symbolically representing how two discourses compose through their interleaved mapping to another discourse. Results include a conjecture for an inverse 'Socialist' discourse, complementing Lacan's fifth Capitalist discourse, permutation subgroups suggesting different organizations for 'universes of discourse,' and applications to the field of mathematics education. A central motivation for this endeavor is to refine the applicability of Lacan's four discourses and thus widen the theory's scope through the addition of twenty more discourses, functioning within a model having interactions. (Received February 27, 2020)

1158-97-139 **Péter Juhász*** (juhasz.peter@renyi.hu), Budapest, Hungary. *Step out of your comfort zone – Wythoff's game is action.*

2-pile Nim is a game well-known to many people, which may be appropriate even for gifted 7th graders. K-pile Nim is a natural generalization of it. There is a less known possible follow-up to the 2-pile version, called Wythoff's Nim. We usually pose this problem to gifted students of 9-10th grades. Most of them discover the recursive method for finding the losing positions. Finding an explicit formula for the set of losing positions is essentially more challenging. It is indeed not evident even to conjecture, while proving appears to be absolutely difficult. For solving this problem, students need to step out of the comfort zone determined by the task. This step prepares the moment when we need to step out of the world of real numbers for solving cubic equations. This is an extraordinarily difficult idea, and there a non-existent world had to be created. We have a bit easier situation here, as we do not need to create a non-existent world, however, we need to step out of the world of integers, into the world of real numbers, needing irrationals too. This is quite surprising for students, who do not expect, before, this need for leaving the familiar and friendly world of integers in order for working out the solution. (Received February 27, 2020)

1158-97-154 **Emiliano Gomez*** (emgomez@berkeley.edu), 985 Evans Hall, Dept. of Mathematics, University of California, Berkeley, CA 94720-3840. *Mathematical thinking and problem solving through games and puzzles.*

We will present examples of mathematical games and puzzles that can be used in math circles with elementary through high school students, and that can be used to teach problem-solving strategies and to introduce important mathematical concepts. (Received February 28, 2020)

1158-97-195 **Ravi Damodar Vakil*** (rvakil@stanford.edu), Dept. of Mathematics, Stanford University, Stanford, CA 94305. *Learning through Problem-Solving.*

There is, I think, a rough consensus among us that one learns best by doing, so that in many situations, the best way to help people learn something is by setting up the right environment in which learners can play with ideas and investigate, and be inevitably be led to think of things in deeper and better ways. In this talk, I intend to share some things that I think have worked in varied situations, in the hopes of hearing many other ideas from other people. (Received March 01, 2020)

1158-97-218 **Adam Glesser*** (aglesser@fullerton.edu), 800 N. State College Blvd., Fullerton, CA 92831, and **Matt Rathbun** (mrathbun@fullerton.edu). *Micro Robots in the Classroom.*

At California State University, Fullerton, we have developed a course, "Strategies of Problem Solving", that teaches our incoming students about mathematical problem solving through games, logic puzzles, riddles, paradoxes, and contest problems. In this talk, we will focus on one particular game used in this class, Micro Robots, a game accessible and fun for all ages, and one which has mathematical connections to group theory, graph theory, and linear algebra. (Received March 02, 2020)

1158-97-235 **Aubrey Kemp*** (akemp2@csub.edu). *Promoting Reading Comprehension of Proofs in Undergraduate Mathematics.*

Since Fall 2016, a group of mathematicians and mathematics educators from two large universities in the United States have implemented educational-research-based methods for teaching and assessing learning of proofs in undergraduate mathematics. Using a framework proposed by Mejia Ramos et. al (2012) that describes an assessment model for proof comprehension, multiple activities, worksheets, and surveys were developed. These materials were given to students in introduction to proof courses and other selected upper level math courses such as Linear Algebra, Analysis I and II, etc. These activities and the responses from students were analyzed and some interviews were conducted with a subgroup of students. In this presentation, the framework that is used to assess proof comprehension is described, examples are provided of how this research group is using the framework to facilitate student learning, and some results and implications will be presented that have emerged from this collaboration. (Received March 02, 2020)

1158-97-245 **Agnes Tuska*** (agnest@csufresno.edu). *Playing with representations at the legs of Dienes.*

Zoltan Paul Dienes (1916-2014) advocated the use of mathematical games and appropriate learning aids (manipulatives) throughout his long and productive life as a teaching wizard mathematics educator. One of his remarkable contributions related to the use of manipulatives was the systematic use of arithmetic blocks with various bases (Multi-base Arithmetic Blocks), highlighting the structure of place-value systems. Inspired by the work of Dienes, I propose structured activities that can lead to the construction and generalization of divisibility rules in various bases. After exploratory-manipulative type of plays with Multi-base Arithmetic Blocks, Math Circle participants can be engaged in card games (rule-bound plays) that require them to communicate and justify their conjectures about the dependence of divisibility rules on the representations of numbers. Participants can create their own games, too. These internalized experiences make the participants ready for the invention and acceptance of concepts such as congruence, modular arithmetic, and equivalence classes (representational plays). (Received March 02, 2020)

1158-97-258 **Zoe C Ashton*** (ashton.95@osu.edu), 230 N Oval Mall OSU, Dept of Philosophy, Columbus, OH 43210. *Linking IBL and Research through Dialectical Heuristics.*

Hadamard, in "Thoughts on the Heuristic Method," wrote that "I tried, in other words to discern the rules, that all mathematicians unconsciously follow when they reason, at least the main ones ... They are on the other hand common sense rules – truisms so to speak." In this talk, I explore what exactly this 'truisms' claim amounts to and how it relates the way mathematicians solve problems and how students should solve problems. In part, I will argue that these truisms are reflected in a certain way that mathematicians conceive of their audiences. Namely that these audiences are universal – the way we solve problems in mathematics is not completely removed from the way we solve problems in dialectical milieus. By viewing our problem-solving practices through the dialectical lens, this can help the expert construct designed tasks to implement in IBL. The core idea is to reveal (a) how mathematical problem solving is dialectical, (b) recognize what the dialectical moves look like, (c) identify what kind of audience is involved in that dialectic, (d) recreate that dialectic in IBL tasks. (Received March 02, 2020)

1158-97-292 **Mario Banuelos*** (mbanuelos22@csufresno.edu). *Investigating Resampling Methods in Upper Division Math Courses.*

In the context of supervised learning, researchers rely on a variety of practices to validate models. Resampling methods provide a means to assess trained models' performance on unseen data, known as testing data. Rather than providing a set of procedures for students to follow after training their models, students will investigate the effects of different validation metrics with respect to model testing. Specifically, I will discuss two different collaborative approaches in upper division math courses to explore validation set splits and bootstrapping data. Students explore the fraction of testing and training splits in regression problems and how that affects the test mean squared error. For the bootstrap activity, students work in groups to answer specific questions about

calculating standard errors for linear regression estimates. Both activities are implemented in RStudio and overall student feedback will also be discussed. (Received March 03, 2020)

1158-97-298 **Jack Luong** and **Maria Diaz*** (mdiaz1824@mail.fresnostate.edu). *Chomp and Nim in K-12 Math Circles.* Preliminary report.

Playing and discussing mathematical games in K-12 math circles teaches students deep mathematics in a fun way. Through mathematical games, students can learn vital concepts such as symmetry, logic and abstraction. In this talk, we will describe a couple mathematical games such as Chomp and Nim that are used or can be used at the Fresno State Math Circle, and how they develop mathematical thinking in Math Circle participants. (Received March 03, 2020)

1158-97-306 **Megan E Selbach-Allen*** (mselbach@stanford.edu). *The What and Why Behind IBL Communities: Beginning to Build in CA & NV.* Preliminary report.

As evidence has mounted for the effectiveness of various active learning techniques, the use of Inquiry Based Learning (IBL) in college mathematics classes has grown rapidly. IBL is now a special interest group in the MAA and several national conferences have been held in recent years along with a series of practitioner workshops across multiple summers to help instructors change their practice and understand the teaching methodology. While these national efforts have sown the seeds, the initial funding streams are ending and the need for more localized, regional communities is emerging. These smaller regional groups have the potential to serve as communities of practice for mathematics faculty who desire to continuously improve their instruction and use of IBL techniques. In this talk I will share some of the short history of IBL communities and how they have been enacted in different regions around the United States. I will also share recent efforts to establish an IBL community in California and Nevada and solicit feedback from participants to think collaboratively about what this community might become for our region. (Received March 03, 2020)

1158-97-359 **Oleg Gleizer*** (prof1140g@math.ucla.edu). *The Josephus problem.*

A Jewish general turned Roman historian, Josephus Flavius lived in the first century AD. His book, The Judean War, describes a Jewish rebellion against the Roman Empire that ended up with the destruction of Jerusalem in AD 70.

During the war, Josephus and his forty soldiers were trapped in a cave, with Romans blocking the exit. The Jews chose suicide over capture. Since suicide is not allowed in Judaism, the Jews decided to proceed the following way. They formed a circle. Then a fighter killed his neighbor on the right with a blow of his sword and passed the sword further right. They kept doing so until there was one man left standing. That one had no other choice but to kill himself. The last one happened to be Josephus. Instead of committing a suicide, he turned himself over to the Romans. Josephus proceeded to become a friend of the Roman emperor Vespasian Flavius and a popular historian.

The above story gave rise to a family of problems all bearing the common name, the Josephus Problem. In this lecture, we will discuss the following:

Problem. Suppose that there are n soldiers, including Josephus, in the cave. What should the position of Josephus be in order for him to stay alive? (Received March 03, 2020)

1158-97-376 **Bridget Kinsella Druken*** (bdruken@fullerton.edu). *What P-12 Pre-service Teacher Coursework and Math Camps Tell Us about Low-floor, High-Ceiling Tasks.*

Engaging students in rich, worthwhile tasks that engender the mathematical problem solving lies at the heart of doing mathematics. Research in the teaching and learning of mathematics continues to reveal helpful insights into ways to support all doers of mathematics. In this talk, a mathematics teacher educator who teaches mathematics content courses to future P-12 teachers and co-coordinated a math camp for middle schoolers shares tasks and ideas about criteria for engaging students in rich mathematics, including procedural fluency, conceptual understanding, strategic problem solving, adaptive reasoning, and productive disposition. Routines, problems, and games developed for mathematics content courses for future P-12 teachers and math camp participants will be shared to highlight ways to develop and facilitate low-floor, high-ceiling tasks math circles. (Received March 04, 2020)

2010 MATHEMATICS SUBJECT CLASSIFICATION

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01 History and biography
03 Mathematical logic and foundations
05 Combinatorics
06 Order, lattices, ordered algebraic structures
08 General algebraic systems
11 Number theory
12 Field theory and polynomials
13 Commutative rings and algebras
14 Algebraic geometry
15 Linear and multilinear algebra; matrix theory
16 Associative rings and algebras
17 Nonassociative rings and algebras
18 Category theory; homological algebra
19 *K*-theory
20 Group theory and generalizations
22 Topological groups, Lie groups
26 Real functions
28 Measure and integration
30 Functions of a complex variable
31 Potential theory
32 Several complex variables and analytic spaces
33 Special functions
34 Ordinary differential equations
35 Partial differential equations
37 Dynamical systems and ergodic theory
39 Difference and functional equations
40 Sequences, series, summability
41 Approximations and expansions
42 Fourier analysis
43 Abstract harmonic analysis
44 Integral transforms, operational calculus
45 Integral equations
46 Functional analysis
47 Operator theory
49 Calculus of variations and optimal control; optimization
51 Geometry
52 Convex and discrete geometry
53 Differential geometry
54 General topology
55 Algebraic topology
57 Manifolds and cell complexes
58 Global analysis, analysis on manifolds
60 Probability theory and stochastic processes
62 Statistics
65 Numerical analysis
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70 Mechanics of particles and systems
74 Mechanics of deformable solids
76 Fluid mechanics
78 Optics, electromagnetic theory
80 Classical thermodynamics, heat transfer
81 Quantum theory
82 Statistical mechanics, structure of matter
83 Relativity and gravitational theory
85 Astronomy and astrophysics
86 Geophysics
90 Operations research, mathematical programming
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92 Biology and other natural sciences
93 Systems theory; control
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